Scientific Computing

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Preface

These notes started as a study guide for CS 111 at UC Santa Barbara and turned into a habit. Whenever I ran into an idea I couldn't explain from scratch, I wrote it down until I could. Feynman's test (if you can't explain something simply, you don't really understand it) turns out to be a reliable way to find gaps, and research kept turning up new ones in eigenvalue algorithms, iterative solvers, spectral graph methods, and Koopman theory. What you're reading is a working reference I wrote as a graduate student, mostly for my own use.

Computers were first built for numerical work. A "computer" was originally a person, someone paid to evaluate functions by hand. During the Second World War the U.S. Army employed rooms full of them to produce artillery firing tables, and the backlog was large enough to justify building ENIAC (1945). Among the machine's first jobs were calculations for thermonuclear weapons design, work in which John von Neumann was closely involved. His 1947 paper with Goldstine laid the foundations of numerical stability analysis. In 1950, Jule Charney's group used ENIAC to run the first computer weather forecast, a calculation Lewis Fry Richardson had once estimated would take 64,000 people working in concert. Later, the Apollo Guidance Computer navigated to the Moon with a few kilobytes of working memory, and the Cooley–Tukey FFT (1965) cut a core signal-processing computation from $O(n^2)$ to $O(n \log n)$.

Most of what follows is linear algebra. Nearly every numerical method works by reducing a nonlinear problem to a sequence of linear ones, because linear problems are the ones we understand best. We know when solutions exist, how sensitive they are to perturbations, and how to compute them reliably. Newton's method, finite element discretization, PCA, and gradient-based optimization all come down to repeated linearization. Once linear algebra feels familiar, most of the other methods in these notes are easier to follow.

For a more rigorous treatment, start with Principles of Scientific Computing by Bindel and Goodman and G. W. Stewart's Afternotes on Numerical Analysis. For a broader tour of the literature, the CS 6241 reading list at Cornell is a great place to look. Much of the perspective here comes from **CS 210A (Matrix Analysis)**, taught by Professor Shivkumar Chandrasekaran in the ECE department at UC Santa Barbara. If you get the chance to take it, do.

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1 Fundamentals

We begin with coordinates and linear maps, then introduce lengths, projections, and norms. These measurements let us state how large an error is and how much a matrix can amplify it. Floating-point arithmetic and operation counts complete the basic tools needed to compare numerical algorithms.

1.1 Vectors, Coordinates, and Linear Maps

Scientific computing begins with representation. A physical state, a polynomial, a sampled signal, and an image are different objects, yet each can be represented by a finite list of numbers. The list is useful only after we say what its entries mean. That distinction between an object and its coordinates will matter throughout these notes.

Throughout, $\mathbb{R}$, $\mathbb{Z}$, and $\mathbb{C}$ denote the real numbers, integers, and complex numbers. Most examples use real arithmetic. Complex numbers appear naturally when we study oscillations and eigenvalues.

Def 1.1 **(Scalar)** An element of the number system under discussion is a **scalar**. In these notes that number system is usually $\mathbb{R}$ and occasionally $\mathbb{C}$.

Def 1.2 **(Vector)** A **vector space** V over $\mathbb{R}$ is a set with addition and scalar multiplication satisfying the following laws. Addition is associative and commutative, has a zero element, and gives each vector an additive inverse. Scalar multiplication satisfies

$$\alpha(\beta \boldsymbol{x}) = (\alpha\beta)\boldsymbol{x}, \qquad 1\boldsymbol{x} = \boldsymbol{x},$$
$$\alpha(\boldsymbol{x} + \boldsymbol{y}) = \alpha\boldsymbol{x} + \alpha\boldsymbol{y}, \qquad (\alpha + \beta)\boldsymbol{x} = \alpha\boldsymbol{x} + \beta\boldsymbol{x}.$$

Elements of V are called **vectors**. The same definition over $\mathbb{C}$ permits complex scalars. A column of real numbers is one example of a vector. A polynomial is another: polynomial addition and scalar multiplication obey these laws.

Remark 1.1 A column of numbers therefore represents a vector only after we know what its coordinates refer to. The coordinates $(3, 2)^T$ might describe a point, a velocity, the coefficients of $3 + 2t$, or two temperature readings. Computation manipulates the coordinates, while interpretation tells us what vector they represent.

Def 1.3 **($\mathbb{R}^n$)** The vector space $\mathbb{R}^n$ consists of all ordered n-tuples of real numbers. We write vectors as columns unless stated otherwise:

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

Vectors are added entry by entry and multiplied by scalars entry by entry. The zero vector is denoted by $\mathbf{0}$.

Remark 1.2 NumPy normally stores a vector as an array of shape `(n,)`, which is neither a row nor a column. A shape `(n,1)` array is a genuine two-dimensional column. This implementation detail affects transposes and broadcasting, so shapes should be checked rather than guessed.

1.1.1 Span, Independence, and Basis

Def 1.4 **(Linear Combination, Span, and Subspace)** Given vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$, a **linear combination** is any vector of the form

$$c_1\boldsymbol{v}_1 + \cdots + c_k\boldsymbol{v}_k.$$

Their **span** is the set of all such combinations. A nonempty set $S \subseteq \mathbb{R}^n$ is a **subspace** if every linear combination of vectors in S remains in S.

Ex 1.1 Decide which of the following subsets of $\mathbb{R}^3$ are subspaces: the plane $x_1 + x_2 = 0$, the plane $x_1 + x_2 = 1$, the set of vectors with at least one zero entry, and the span of $(1, 1, 0)^T$ and $(0, 1, 1)^T$. For every negative answer, identify a subspace axiom that fails.

Example The vectors $(1, 0, 0)^T$ and $(0, 1, 0)^T$ span the coordinate plane in $\mathbb{R}^3$ on which the third coordinate is zero. A plane not passing through the origin is not a subspace: it does not contain $\mathbf{0}$ and is not closed under scalar multiplication.

A spanning list may contain redundant vectors. For example, $(1, 1)^T$ adds no new vectors to the span of $(1, 0)^T$ and $(0, 1)^T$. We need a criterion for detecting this redundancy before using the list as a coordinate system.

Def 1.5 **(Linear Independence and Basis)** Vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ are **linearly independent** if

$$c_1\boldsymbol{v}_1 + \cdots + c_k\boldsymbol{v}_k = \mathbf{0}$$

forces every coefficient c_i to be zero. A **basis** of a vector space is a linearly independent spanning set.

Thm 1.1 **(Exchange, Extension, and Dimension)** Let V be spanned by m vectors.

1. Every linearly independent list in V contains at most m vectors.
2. Any linearly independent list can be extended to a basis by adding vectors from a spanning list.
3. Any two bases of V contain the same number of vectors. This common number is the **dimension** of V.

Ex 1.2 Prove Thm. 1.1. Begin with a spanning list $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_m$ and a nonzero vector $\boldsymbol{v}_1$. Express $\boldsymbol{v}_1$ in the spanning list and replace one $\boldsymbol{w}_j$ having a nonzero coefficient by $\boldsymbol{v}_1$. Show that the new list still spans. Repeat with $\boldsymbol{v}_2, \ldots, \boldsymbol{v}_k$, using independence to show that a replaceable old vector always remains. Apply the result in both directions to two bases.

Remark 1.3 The preceding exercise supplies the replacement argument behind this theorem. This theorem justifies phrases used later such as “extend a basis of the null space to a basis of the domain.”

Thm 1.2 **(A Basis Gives Unique Coordinates)** Let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ be a basis of a vector space V. Every $\boldsymbol{x} \in V$ has exactly one representation

$$\boldsymbol{x} = x_1\boldsymbol{v}_1 + \cdots + x_n\boldsymbol{v}_n.$$

The scalars $x_1, \ldots, x_n$ are the coordinates of $\boldsymbol{x}$ in this basis.

Proof We must prove existence and uniqueness separately.

1. A basis is a spanning list. Consequently, the definition of span gives scalars $x_1, \dots, x_n$ such that $\boldsymbol{x} = \sum_i x_i \boldsymbol{v}_i$. This establishes at least one coordinate representation.

2. Suppose another list of scalars gives the same vector:

$$\boldsymbol{x} = \sum_{i=1}^{n} x_i \boldsymbol{v}_i, \qquad \boldsymbol{x} = \sum_{i=1}^{n} y_i \boldsymbol{v}_i.$$

   Subtract the second equality from the first. Distributivity of scalar multiplication gives

$$\mathbf{0} = \sum_{i=1}^{n} (x_i - y_i) \boldsymbol{v}_i.$$

3. The basis vectors are linearly independent. By the definition of independence, every coefficient in a vanishing linear combination must be zero. Thus $x_i - y_i = 0$ for each i, so the two coordinate lists coincide. Spanning supplied existence, whereas independence supplied uniqueness.

□

Ex 1.3 Let $p(t) = a + bt + ct^2$. Find its coordinates in the bases $1, t, t^2$ and $1, t-1, (t-1)^2$. Construct the matrix that converts the second coordinate vector into the first, and interpret each column.

Remark 1.4 Coordinates change when the basis changes, but the vector does not. For example, the polynomial $p(t) = 3 + 2t$ has coordinates $(3, 2)^T$ in the basis $1, t$ and coordinates $(5, 2)^T$ in the basis $1, t-1$. Confusing a vector with one coordinate list is a common source of conceptual errors in change-of-basis and eigenvalue arguments.

1.1.2 Matrices and Linear Maps

Def 1.6 **(Linear Map)** A map $T : V \to W$ is **linear** if, for all vectors $\boldsymbol{x}, \boldsymbol{y} \in V$ and scalars α, β,

$$T(\alpha \boldsymbol{x} + \beta \boldsymbol{y}) = \alpha T(\boldsymbol{x}) + \beta T(\boldsymbol{y}).$$

Linearity says that the response to a combination of inputs is the same combination of the individual responses.

Def 1.7 **(Matrix)** A **matrix** $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a rectangular array of scalars with m rows and n columns. It represents a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$ once bases have been selected in the domain and range.

Thm 1.3 **(The Columns Determine the Map)** Let $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$ be the standard basis of $\mathbb{R}^n$, and let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear. If the jth column of $\mathbf{A}$ is $\boldsymbol{a}_j = T(\boldsymbol{e}_j)$, then

$$T(\boldsymbol{x}) = \mathbf{A}\boldsymbol{x} = \sum_{j=1}^{n} x_j \boldsymbol{a}_j.$$

Thus a linear map is completely determined by what it does to a basis.

Proof Write an arbitrary input in the standard basis and apply linearity one operation at a time:

$$\begin{aligned} \boldsymbol{x} &= x_1\boldsymbol{e}_1 + \cdots + x_n\boldsymbol{e}_n, \\ T(\boldsymbol{x}) &= T(x_1\boldsymbol{e}_1 + \cdots + x_n\boldsymbol{e}_n) \\ &= T(x_1\boldsymbol{e}_1) + \cdots + T(x_n\boldsymbol{e}_n) \\ &= x_1T(\boldsymbol{e}_1) + \cdots + x_nT(\boldsymbol{e}_n) \\ &= x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n. \end{aligned}$$

The second line uses preservation of addition. The third uses preservation of scalar multiplication. The final line uses the definition $\boldsymbol{a}_j = T(\boldsymbol{e}_j)$.

To identify this vector with the usual matrix product, take its ith entry:

$$\left(\sum_{j=1}^{n} x_j\boldsymbol{a}_j\right)_i = \sum_{j=1}^{n} a_{ij}x_j = (\mathbf{A}\boldsymbol{x})_i.$$

The entries agree for every i, so $T(\boldsymbol{x}) = \mathbf{A}\boldsymbol{x}$. In particular, two linear maps that agree on all standard basis vectors agree on every input. □

Remark 1.5 This column view is more informative than memorizing the row-by-column rule. The set of all possible outputs $\mathbf{A}\boldsymbol{x}$ is the span of the columns. A zero output means that one linear combination of the columns vanishes. Existence and uniqueness for linear systems will grow directly from these two observations.

Example For

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \qquad \boldsymbol{x} = \begin{pmatrix} 4 \\ 5 \end{pmatrix},$$

the product is best read as

$$\mathbf{A}\boldsymbol{x} = 4\begin{pmatrix} 1 \\ 3 \end{pmatrix} + 5\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 14 \\ 7 \end{pmatrix}.$$

The same calculation can be read row by row: each output coordinate is a weighted sum of the input coordinates.

1.1.3 Matrix Products

Composition of linear maps defines the matrix product. If $\mathbf{B} \in \mathbb{R}^{n\times p}$ represents a map from $\mathbb{R}^p$ to $\mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{m\times n}$ represents a map from $\mathbb{R}^n$ to $\mathbb{R}^m$, then the composition sends $\boldsymbol{e}_j$ first to the jth column $\boldsymbol{b}_j$ of $\mathbf{B}$ and then to $\mathbf{A}\boldsymbol{b}_j$. By Thm 1.3, the matrix of the composition therefore has columns $\mathbf{A}\boldsymbol{b}_1, \ldots, \mathbf{A}\boldsymbol{b}_p$.

Def 1.8 **(Matrix Product)** For $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{n\times p}$, the product $\mathbf{AB} \in \mathbb{R}^{m\times p}$ is the matrix with entries

$$(\mathbf{AB})_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Equivalently, the jth column of $\mathbf{AB}$ is $\mathbf{A}\boldsymbol{b}_j$. The product is defined only when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$.

Ex 1.4 For $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix}$, compute $\mathbf{AB}$ and $\mathbf{BA}$ by entries and by columns. Find a matrix other than a multiple of $\mathbf{I}$ that commutes with $\mathbf{A}$.

Because the product represents composition, it is associative and distributes over addition. It is not commutative. For

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{AB} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = \mathbf{BA}.$$

Scaling the coordinates and then exchanging them is a different map from exchanging them and then scaling.

1.1.4 Transpose and Symmetry

Def 1.9 **(Matrix Transpose)** The **transpose** of $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the matrix $\mathbf{A}^T \in \mathbb{R}^{n \times m}$ defined by

$$(\mathbf{A}^T)_{ij} = a_{ji}.$$

It turns columns into rows and rows into columns.

Def 1.10 **(Symmetric and Skew-Symmetric Matrices)** A square matrix $\mathbf{A}$ is **symmetric** if $\mathbf{A}^T = \mathbf{A}$ and **skew-symmetric** if $\mathbf{A}^T = -\mathbf{A}$.

Remark 1.6 Every square matrix splits uniquely into symmetric and skew-symmetric parts:

$$\mathbf{A} = \frac{\mathbf{A} + \mathbf{A}^T}{2} + \frac{\mathbf{A} - \mathbf{A}^T}{2}.$$

Symmetric matrices will be especially important because they possess real eigenvalues and orthogonal eigenvectors.

Thm 1.4 **(Properties of the Transpose)** For conformable matrices and any scalar α,

$$(\mathbf{A}^T)^T = \mathbf{A}, \qquad (\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T,$$
$$(\alpha\mathbf{A})^T = \alpha\mathbf{A}^T, \qquad (\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T.$$

Ex 1.5 Prove the first three transpose identities entry by entry. For the product identity, write $((\mathbf{AB})^T)_{ij}$ as a sum and identify the same sum in $\mathbf{B}^T\mathbf{A}^T$. Explain why the order of the factors reverses.

Cor 1.5 **(Reversal Law for Multiple Factors)** Repeated use of the product rule gives

$$(\mathbf{A}_1\mathbf{A}_2 \cdots \mathbf{A}_k)^T = \mathbf{A}_k^T \cdots \mathbf{A}_2^T\mathbf{A}_1^T.$$

Cor 1.6 **(Outer Product is Symmetric of Rank One)** For every nonzero $\boldsymbol{x} \in \mathbb{R}^n$, the outer product $\boldsymbol{x}\boldsymbol{x}^T$ is symmetric and has rank one, meaning its column space has dimension one. Moreover,

$$\boldsymbol{x}^T\boldsymbol{x} = \sum_{i=1}^{n} x_i^2 > 0.$$

1.1.5 Inverses

A linear map from $\mathbb{R}^n$ to itself can be undone exactly when its matrix has an inverse. The inverse is a useful theoretical object, but, as the remark below explains, it is rarely computed.

Def 1.11 **(Identity Matrix)** The identity matrix $\mathbf{I}_n$ has ones on the diagonal and zeros elsewhere. It represents the map that leaves every vector unchanged: $\mathbf{I}_n\boldsymbol{x} = \boldsymbol{x}$. We omit the subscript when the dimension is clear. The zero matrix has every entry zero and sends every vector to the zero vector.

Ex 1.6 Use the column interpretation of multiplication to prove $\mathbf{I}_m\mathbf{A} = \mathbf{A} = \mathbf{A}\mathbf{I}_n$ for $\mathbf{A} \in \mathbb{R}^{m\times n}$. Why do the two identity matrices have different dimensions when $m \neq n$?

Def 1.12 **(Matrix Inverse)** A square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is **invertible**, or nonsingular, if there is a matrix $\mathbf{A}^{-1}$ with

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

The inverse is unique. If $\mathbf{B}$ and $\mathbf{C}$ both satisfy the definition, then $\mathbf{B} = \mathbf{B}(\mathbf{A}\mathbf{C}) = (\mathbf{B}\mathbf{A})\mathbf{C} = \mathbf{C}$.

Thm 1.7 **(Properties of the Inverse)** Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$ be invertible and let $\alpha \neq 0$. Then $\mathbf{A}^{-1}$, $\mathbf{A}\mathbf{B}$, $\alpha\mathbf{A}$, and $\mathbf{A}^T$ are invertible, with

$$(\mathbf{A}^{-1})^{-1} = \mathbf{A}, \qquad (\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}, \qquad (\alpha\mathbf{A})^{-1} = \alpha^{-1}\mathbf{A}^{-1}, \qquad (\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T.$$

Ex 1.7 Prove the inverse identities by multiplying each proposed inverse on both sides. For the transpose identity, transpose $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ and use the reversal law. Which assumption is needed for the scalar identity?

Remark 1.7 **(Factor instead of inverting)** Scientific codes almost never form $\mathbf{A}^{-1}$. To solve $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$, they factor $\mathbf{A}$ into triangular or orthogonal factors and solve with each factor in turn, which costs less and is more accurate than forming the inverse and multiplying. The reversal law $(\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ describes this procedure. To solve $\mathbf{A}\mathbf{B}\boldsymbol{x} = \boldsymbol{b}$, we first solve $\mathbf{A}\boldsymbol{y} = \boldsymbol{b}$ and then $\mathbf{B}\boldsymbol{x} = \boldsymbol{y}$.

1.1.6 Further Exercises

Ex 1.8 Prove that a list containing the zero vector is dependent. Prove that two nonzero vectors are dependent exactly when one is a scalar multiple of the other. Then explain why three vectors in $\mathbb{R}^2$ cannot be independent.

Ex 1.9 Suppose $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear and $T(\boldsymbol{e}_j) = \mathbf{0}$ for every standard basis vector. Prove that T is the zero map. Why is checking only one vector insufficient when $n > 1$? Would checking a redundant spanning list suffice?

Ex 1.10 Write a matrix as both a collection of columns and a collection of rows. Use the two views to explain, without entrywise multiplication, what $\mathbf{A}\boldsymbol{x}$ and $\boldsymbol{y}^T\mathbf{A}$ mean.

Ex 1.11 A square matrix is **involutory** if $\mathbf{A}^2 = \mathbf{I}$. Prove that $\mathbf{A}$ is involutory exactly when $\mathbf{A}^{-1} = \mathbf{A}$, and give an involutory 2×2 matrix other than $\pm\mathbf{I}$. Interpret your example geometrically.

Ex 1.12 Prove Cor. 1.6. In particular, show that every column of $\boldsymbol{x}\boldsymbol{x}^T$ is a multiple of $\boldsymbol{x}$, and identify one nonzero column.

1.2 Inner Products, Geometry, and Norms

To compare an approximation with an exact vector, we need to measure their difference. Euclidean length is one possible measurement. The largest coordinate error is another, and the two can rank approximations differently. We begin with the inner product, which defines lengths and perpendicular directions, and use it to find closest points on lines and subspaces. We then study norms that measure length without requiring an inner product.

1.2.1 Inner Products and Euclidean Geometry

Def 1.13 **(Real Inner Product)** An **inner product** on $\mathbb{R}^n$ is a function $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ satisfying, for all vectors $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}$ and scalars α, β,

1. **Symmetry:** $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle$.
2. **Linearity:** $\langle \alpha\boldsymbol{x} + \beta\boldsymbol{y}, \boldsymbol{z} \rangle = \alpha\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \beta\langle \boldsymbol{y}, \boldsymbol{z} \rangle$.
3. **Positive definiteness:** $\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$, with equality if and only if $\boldsymbol{x} = \boldsymbol{0}$.

The standard Euclidean inner product is

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_2 = \boldsymbol{x}^T \boldsymbol{y} = \sum_{i=1}^{n} x_i y_i.$$

Ex 1.13 For $\boldsymbol{x} = (1, 2)^T$ and $\boldsymbol{y} = (2, -1)^T$, compute $\langle \boldsymbol{x}, \boldsymbol{y} \rangle_2$. Verify the three axioms for the Euclidean inner product. Next replace it by $\langle \boldsymbol{x}, \boldsymbol{y} \rangle_W = 4x_1y_1 + x_2y_2$. Are these two vectors still orthogonal in the new inner product?

The Euclidean formula is a familiar example of these axioms. If $\boldsymbol{x} = (x_1, x_2)^T$, then $\langle \boldsymbol{x}, \boldsymbol{x} \rangle = x_1^2 + x_2^2$ is the square of its length. An inner product allows us to use the same arguments after changing the way directions are weighted. The inequalities below follow from the axioms, so they apply to every such choice.

Def 1.14 **(Norm Induced by an Inner Product)** Every inner product defines a norm by

$$\|\boldsymbol{x}\| = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}.$$

For the Euclidean inner product this gives

$$\|\boldsymbol{x}\|_2 = \sqrt{\boldsymbol{x}^T \boldsymbol{x}}.$$

The square root is chosen so that scaling a vector by α scales its length by $|\alpha|$. The triangle inequality is less immediate and follows from the Cauchy–Schwarz inequality.

Thm 1.8 **(Cauchy–Schwarz Inequality)** For every real inner product and all vectors $\boldsymbol{x}, \boldsymbol{y}$,

$$|\langle \boldsymbol{x}, \boldsymbol{y} \rangle| \leq \|\boldsymbol{x}\| \, \|\boldsymbol{y}\|.$$

Equality holds if and only if $\boldsymbol{x}$ and $\boldsymbol{y}$ are linearly dependent.

Proof If $\boldsymbol{y} = \mathbf{0}$, both $|\langle \boldsymbol{x}, \boldsymbol{y} \rangle|$ and $\|\boldsymbol{x}\|\|\boldsymbol{y}\|$ are zero. Hence the claim holds in that case. Suppose now that $\boldsymbol{y} \neq \mathbf{0}$, so $\langle \boldsymbol{y}, \boldsymbol{y} \rangle > 0$.

1. For every real scalar t, positive definiteness gives $\langle \boldsymbol{x} - t\boldsymbol{y}, \boldsymbol{x} - t\boldsymbol{y} \rangle \geq 0$. Expand using linearity and symmetry:

$$0 \leq \|\boldsymbol{x}\|^2 - 2t\langle \boldsymbol{x}, \boldsymbol{y} \rangle + t^2\|\boldsymbol{y}\|^2.$$

2. Set $t = \langle \boldsymbol{x}, \boldsymbol{y} \rangle / \|\boldsymbol{y}\|^2$. The denominator is positive. Substitution gives

$$0 \leq \|\boldsymbol{x}\|^2 - \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle^2}{\|\boldsymbol{y}\|^2}.$$

   Multiply by $\|\boldsymbol{y}\|^2$ and take nonnegative square roots. This yields $|\langle \boldsymbol{x}, \boldsymbol{y} \rangle| \leq \|\boldsymbol{x}\|\|\boldsymbol{y}\|$.

3. Equality holds precisely when $\|\boldsymbol{x} - t\boldsymbol{y}\|^2 = 0$. Positive definiteness makes this equivalent to $\boldsymbol{x} = t\boldsymbol{y}$. Thus equality occurs precisely for linearly dependent vectors, including the zero-vector cases.

□

Cor 1.9 **(Triangle Inequality for an Inner-Product Norm)** For all vectors $\boldsymbol{x}$ and $\boldsymbol{y}$,

$$\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|.$$

Ex 1.14 Expand $\|\boldsymbol{x} + \boldsymbol{y}\|^2$, bound the cross term using Cauchy–Schwarz, and take square roots. When does equality hold? Treat the case in which one vector is zero separately.

Def 1.15 **(Angle and Orthogonality)** For nonzero vectors $\boldsymbol{x}$ and $\boldsymbol{y}$, their angle $\theta \in [0, \pi]$ is defined by

$$\cos\theta = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{x}\|\,\|\boldsymbol{y}\|}.$$

Cauchy–Schwarz guarantees that the right-hand side lies in $[-1, 1]$. The vectors are **orthogonal**, written $\boldsymbol{x} \perp \boldsymbol{y}$, when $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$.

Thm 1.10 **(Pythagorean Theorem and Its Converse)** For vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ in a real inner-product space,

$$\boldsymbol{x} \perp \boldsymbol{y} \quad \Longleftrightarrow \quad \|\boldsymbol{x} + \boldsymbol{y}\|^2 = \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2.$$

Ex 1.15 Expand $\langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{x} + \boldsymbol{y} \rangle$ and prove both directions of the stated equivalence. Explain where the assumption of a real inner product is used.

1.2.2 Orthogonal Projection

The coefficient used in the proof of Cauchy–Schwarz has a geometric meaning. It selects the point on the line through $\boldsymbol{y}$ that is closest to $\boldsymbol{x}$.

Def 1.16 **(Projection onto a Nonzero Vector)** For $\boldsymbol{y} \neq \boldsymbol{0}$, the orthogonal projection of $\boldsymbol{x}$ onto the line spanned by $\boldsymbol{y}$ is

$$\operatorname{proj}_{\boldsymbol{y}}(\boldsymbol{x}) = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\langle \boldsymbol{y}, \boldsymbol{y} \rangle} \boldsymbol{y}.$$

The residual $\boldsymbol{x} - \operatorname{proj}_{\boldsymbol{y}}(\boldsymbol{x})$ is orthogonal to $\boldsymbol{y}$.

Thm 1.11 **(Closest Point on a Line)** Let $\boldsymbol{y} \neq \boldsymbol{0}$ and let

$$\alpha = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\langle \boldsymbol{y}, \boldsymbol{y} \rangle}.$$

Then $\alpha \boldsymbol{y}$ is the unique closest point to $\boldsymbol{x}$ among all scalar multiples of $\boldsymbol{y}$:

$$\|\boldsymbol{x} - \alpha \boldsymbol{y}\| \leq \|\boldsymbol{x} - \beta \boldsymbol{y}\| \qquad \text{for every } \beta \in \mathbb{R}.$$

Proof Let $\alpha = \langle \boldsymbol{x}, \boldsymbol{y} \rangle / \langle \boldsymbol{y}, \boldsymbol{y} \rangle$ and set $\boldsymbol{r} = \boldsymbol{x} - \alpha \boldsymbol{y}$. The assumption $\boldsymbol{y} \neq \boldsymbol{0}$ makes the denominator positive.

1. Compute

   $$\langle \boldsymbol{r}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle - \alpha \langle \boldsymbol{y}, \boldsymbol{y} \rangle = 0.$$

   Thus $\alpha \boldsymbol{y}$ has the required perpendicular residual.

2. For any real β,

   $$\boldsymbol{x} - \beta \boldsymbol{y} = \boldsymbol{r} + (\alpha - \beta) \boldsymbol{y}.$$

   Expand the squared norm. The cross term is zero by the preceding calculation:

   $$\begin{aligned} \|\boldsymbol{x} - \beta \boldsymbol{y}\|^2 &= \|\boldsymbol{r}\|^2 + 2(\alpha - \beta) \langle \boldsymbol{r}, \boldsymbol{y} \rangle + (\alpha - \beta)^2 \|\boldsymbol{y}\|^2 \\ &= \|\boldsymbol{r}\|^2 + (\alpha - \beta)^2 \|\boldsymbol{y}\|^2 \geq \|\boldsymbol{r}\|^2. \end{aligned}$$

3. Since $\|\boldsymbol{y}\|^2 > 0$, equality holds only when $\beta = \alpha$. Therefore $\alpha \boldsymbol{y}$ is the unique closest point on the line.

□

Def 1.17 **(Orthogonal Complement)** For a subspace $S \subseteq \mathbb{R}^n$, its orthogonal complement is

$$S^{\perp} = \{ \boldsymbol{z} \in \mathbb{R}^n : \langle \boldsymbol{z}, \boldsymbol{s} \rangle = 0 \text{ for every } \boldsymbol{s} \in S \}.$$

Thm 1.12 **(Projection onto a Subspace with an Orthonormal Basis)** Let the columns of $\mathbf{Q} \in \mathbb{R}^{n\times k}$ be an orthonormal basis for a subspace S. Then

$$\mathbf{P} = \mathbf{Q}\mathbf{Q}^T$$

is the orthogonal projector onto S. Every vector has the unique decomposition

$$\boldsymbol{x} = \mathbf{P}\boldsymbol{x} + (\mathbf{I} - \mathbf{P})\boldsymbol{x}, \qquad \mathbf{P}\boldsymbol{x} \in S, \qquad (\mathbf{I} - \mathbf{P})\boldsymbol{x} \in S^{\perp}.$$

Moreover, $\mathbf{P}\boldsymbol{x}$ is the unique point in S closest to $\boldsymbol{x}$.

Proof Let $\mathbf{Q}$ contain the orthonormal basis vectors of S, so $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$, and let $\mathbf{P} = \mathbf{Q}\mathbf{Q}^T$.

1. The vector $\mathbf{Q}^T\boldsymbol{x}$ is a list of coefficients. Therefore $\mathbf{P}\boldsymbol{x} = \mathbf{Q}(\mathbf{Q}^T\boldsymbol{x})$ is a linear combination of the columns of $\mathbf{Q}$ and lies in S.

2. First calculate

   $$\mathbf{Q}^T(\boldsymbol{x} - \mathbf{P}\boldsymbol{x}) = \mathbf{Q}^T\boldsymbol{x} - \mathbf{Q}^T\mathbf{Q}\mathbf{Q}^T\boldsymbol{x} = \mathbf{0}.$$

   Every $\boldsymbol{s} \in S$ has the form $\boldsymbol{s} = \mathbf{Q}\boldsymbol{c}$. Hence

   $$\boldsymbol{s}^T(\boldsymbol{x} - \mathbf{P}\boldsymbol{x}) = \boldsymbol{c}^T\mathbf{Q}^T(\boldsymbol{x} - \mathbf{P}\boldsymbol{x}) = 0.$$

3. For any $\boldsymbol{s} \in S$, write

   $$\boldsymbol{x} - \boldsymbol{s} = (\boldsymbol{x} - \mathbf{P}\boldsymbol{x}) + (\mathbf{P}\boldsymbol{x} - \boldsymbol{s}).$$

   The first term is perpendicular to S, and the second belongs to S. Pythagoras gives

   $$\|\boldsymbol{x} - \boldsymbol{s}\|_2^2 = \|\boldsymbol{x} - \mathbf{P}\boldsymbol{x}\|_2^2 + \|\mathbf{P}\boldsymbol{x} - \boldsymbol{s}\|_2^2.$$

   The second summand is nonnegative and vanishes only when $\boldsymbol{s} = \mathbf{P}\boldsymbol{x}$. This proves both minimality and uniqueness.

□

Ex 1.16 **Orthogonal versus oblique projection.** Let

$$\mathbf{P} = \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}.$$

Verify that $\mathbf{P}^2 = \mathbf{P}$ but $\mathbf{P}^T \neq \mathbf{P}$. Find its range and null space. Give a vector for which $\mathbf{P}\boldsymbol{x}$ is not the closest point in the range of $\mathbf{P}$.

Ex 1.17 **Projection and uniqueness.** Let S have orthonormal basis matrix $\mathbf{Q}$. Prove directly that $\mathbf{P} = \mathbf{Q}\mathbf{Q}^T$ satisfies $\mathbf{P}^T = \mathbf{P}$ and $\mathbf{P}^2 = \mathbf{P}$. Then prove that these two identities characterize orthogonal projectors.

Remark 1.8 For an orthonormal basis, the projection coefficients are simply the inner products $\mathbf{Q}^T\boldsymbol{x}$. If the basis columns are stored in a general full-rank matrix $\mathbf{V}$, the closest vector $\mathbf{V}\boldsymbol{c}$ instead satisfies

$$\mathbf{V}^T\mathbf{V}\boldsymbol{c} = \mathbf{V}^T\boldsymbol{x}.$$

The matrix $\mathbf{V}^T\mathbf{V}$ records all pairwise inner products between the basis vectors. Orthogonality reduces this matrix to the identity. This is the first reason orthogonal bases simplify computation.

1.2.3 Vector Norms

An inner product supplies both length and angle. A norm keeps only the notion of length. This weaker structure is enough for error bounds and convergence, even when no meaningful angle is available.

Def 1.18 **(Vector Norm)** A function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ is a **vector norm** if, for all vectors $\boldsymbol{x}, \boldsymbol{y}$ and scalars α,

1. $\|\boldsymbol{x}\| \geq 0$, with equality if and only if $\boldsymbol{x} = \boldsymbol{0}$.
2. $\|\alpha\boldsymbol{x}\| = |\alpha|\|\boldsymbol{x}\|$.
3. $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.

Def 1.19 **(The Standard 1-, 2-, and Infinity Norms)** For $\boldsymbol{x} \in \mathbb{R}^n$,

$$\|\boldsymbol{x}\|_1 = \sum_{i=1}^{n} |x_i|, \qquad \|\boldsymbol{x}\|_2 = \left(\sum_{i=1}^{n} x_i^2\right)^{1/2}, \qquad \|\boldsymbol{x}\|_\infty = \max_{1\leq i\leq n} |x_i|.$$

More generally, for $1 \leq p < \infty$,

$$\|\boldsymbol{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}.$$

For an exact vector $\boldsymbol{x}$ and an approximation $\hat{\boldsymbol{x}}$, the absolute error in a chosen norm is $\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|$. If $\boldsymbol{x} \neq \boldsymbol{0}$, its relative error is $\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|/\|\boldsymbol{x}\|$. The norm and scale must be stated together. An error of one unit may be small in one application and unacceptable in another.
Each norm answers a different question. The 1-norm measures total magnitude across coordinates. The 2-norm measures Euclidean energy. The infinity norm measures the largest coordinate error. A tolerance stated without a norm is therefore incomplete.

Example Let $\boldsymbol{x} = (1, 1, 1, 1)^T$ and $\boldsymbol{y} = (4, 0, 0, 0)^T$. Both vectors satisfy $\|\boldsymbol{x}\|_1 = \|\boldsymbol{y}\|_1 = 4$, but

$$\|\boldsymbol{x}\|_2 = 2, \qquad \|\boldsymbol{y}\|_2 = 4, \qquad \|\boldsymbol{x}\|_\infty = 1, \qquad \|\boldsymbol{y}\|_\infty = 4.$$

The vectors have the same total magnitude, but one distributes it evenly and the other concentrates it in one coordinate. No single norm measures every interpretation of size.

Thm 1.13 **(Sharp Comparisons of the Standard Norms)** For every $\boldsymbol{x} \in \mathbb{R}^n$,

$$\|\boldsymbol{x}\|_\infty \leq \|\boldsymbol{x}\|_2 \leq \|\boldsymbol{x}\|_1 \leq \sqrt{n}\,\|\boldsymbol{x}\|_2 \leq n\|\boldsymbol{x}\|_\infty.$$

The factors $\sqrt{n}$ and n cannot be replaced by smaller constants that work for every vector.

Proof We prove all four comparisons and show that their constants are attained.

1. Choose k with $|x_k| = \|\boldsymbol{x}\|_\infty$. Then

$$\|\boldsymbol{x}\|_\infty^2 = x_k^2 \leq \sum_i x_i^2 = \|\boldsymbol{x}\|_2^2.$$

Taking square roots gives $\|\boldsymbol{x}\|_\infty \leq \|\boldsymbol{x}\|_2$. Also $|x_i| \leq \|\boldsymbol{x}\|_\infty$ for every i, so

$$\|\boldsymbol{x}\|_2^2 = \sum_i |x_i|^2 \leq n\|\boldsymbol{x}\|_\infty^2.$$

2. To compare the 1- and 2-norms, expand

$$\|\boldsymbol{x}\|_1^2 = \sum_i |x_i|^2 + 2\sum_{i<j} |x_i||x_j| \geq \sum_i |x_i|^2 = \|\boldsymbol{x}\|_2^2.$$

For the reverse estimate, apply Cauchy–Schwarz to $(|x_1|, \ldots, |x_n|)^T$ and the all-ones vector:

$$\|\boldsymbol{x}\|_1 = \sum_i |x_i| \leq \sqrt{\sum_i |x_i|^2}\sqrt{\sum_i 1^2} = \sqrt{n}\|\boldsymbol{x}\|_2.$$

3. For a coordinate vector, all three norms equal one, so neither lower comparison can have a larger constant. For the all-ones vector, the three norms are 1, $\sqrt{n}$, and n. Thus neither factor $\sqrt{n}$ in the upper comparisons can be reduced.

□

Ex 1.18 Prove the remaining comparisons in Thm 1.13. For $\|\boldsymbol{x}\|_2 \leq \|\boldsymbol{x}\|_1$, expand the square of the right side. For $\|\boldsymbol{x}\|_\infty \leq \|\boldsymbol{x}\|_2$, select an entry of largest magnitude. Use a coordinate vector and the all-ones vector to show that the constants cannot be improved. Find all equality cases in $\|\boldsymbol{x}\|_1 \leq \sqrt{n}\|\boldsymbol{x}\|_2$.

Thm 1.14 **(Equivalence of Norms in Fixed Finite Dimension)** For any two norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ on $\mathbb{R}^n$, there are constants $c, C > 0$ such that

$$c\|\boldsymbol{x}\|_\beta \leq \|\boldsymbol{x}\|_\alpha \leq C\|\boldsymbol{x}\|_\beta \qquad \text{for every } \boldsymbol{x} \in \mathbb{R}^n.$$

Consequently, the two norms define the same convergent sequences on a fixed $\mathbb{R}^n$.

Proof We first compare one arbitrary norm with the Euclidean norm. Let $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$ be the coordinate vectors and define the finite constant

$$C_\alpha = \left(\sum_{i=1}^n \|\boldsymbol{e}_i\|_\alpha^2\right)^{1/2}.$$

Writing $\boldsymbol{x} = \sum_i x_i \boldsymbol{e}_i$, and applying the triangle inequality followed by scalar Cauchy–Schwarz, gives

$$\begin{aligned}\|\boldsymbol{x}\|_\alpha &\leq \sum_i \|x_i \boldsymbol{e}_i\|_\alpha = \sum_i |x_i| \|\boldsymbol{e}_i\|_\alpha \\ &\leq \left(\sum_i |x_i|^2\right)^{1/2} \left(\sum_i \|\boldsymbol{e}_i\|_\alpha^2\right)^{1/2} = C_\alpha \|\boldsymbol{x}\|_2.\end{aligned}$$

This provides an upper bound without assuming continuity of the arbitrary norm.

The triangle inequality also gives $\|\boldsymbol{x}\|_\alpha \leq \|\boldsymbol{y}\|_\alpha + \|\boldsymbol{x} - \boldsymbol{y}\|_\alpha$. Interchange $\boldsymbol{x}$ and $\boldsymbol{y}$ and combine the two inequalities:

$$|\|\boldsymbol{x}\|_\alpha - \|\boldsymbol{y}\|_\alpha| \leq \|\boldsymbol{x} - \boldsymbol{y}\|_\alpha \leq C_\alpha \|\boldsymbol{x} - \boldsymbol{y}\|_2.$$

Thus Euclidean-close vectors have close α-norms. This establishes continuity in the Euclidean coordinates.

The Euclidean unit sphere is closed and bounded, hence compact in finite dimensions. A continuous real function on a compact set attains its minimum and maximum. Let those values of $\|\cdot\|_\alpha$ be m_α and M_α. The attained minimum cannot be zero: a vector attaining it is on the unit sphere and is nonzero, while positive definiteness makes its α-norm positive. Hence $m_\alpha > 0$.

For $\boldsymbol{x} \neq \mathbf{0}$, the vector $\boldsymbol{x}/\|\boldsymbol{x}\|_2$ lies on that sphere. Homogeneity gives

$$m_\alpha \leq \frac{\|\boldsymbol{x}\|_\alpha}{\|\boldsymbol{x}\|_2} \leq M_\alpha.$$

Apply the same reasoning to the β-norm. The inequalities $\|\boldsymbol{x}\|_\beta \leq M_\beta \|\boldsymbol{x}\|_2$ and $m_\beta \|\boldsymbol{x}\|_2 \leq \|\boldsymbol{x}\|_\beta$ then give

$$\frac{m_\alpha}{M_\beta}\|\boldsymbol{x}\|_\beta \leq \|\boldsymbol{x}\|_\alpha \leq \frac{M_\alpha}{m_\beta}\|\boldsymbol{x}\|_\beta.$$

Both constants are positive and independent of $\boldsymbol{x}$. The zero vector satisfies the same inequalities because every term is zero. □

Remark 1.9 **(Equivalent does not mean interchangeable)** Norm equivalence says that convergence in one norm implies convergence in another when the dimension is fixed. It does not say that the numerical values or bounds are equally useful. The comparison constants may grow with n. A bound containing $\sqrt{n}$ may be harmless for ten unknowns and useless for ten million.
The 2-norm is induced by an inner product. The 1- and infinity norms are not. The difference is visible in the geometry of their unit balls and in a simple algebraic identity.

Thm 1.15 **(Parallelogram Identity)** Every norm induced by an inner product satisfies

$$\|\boldsymbol{x} + \boldsymbol{y}\|^2 + \|\boldsymbol{x} - \boldsymbol{y}\|^2 = 2\|\boldsymbol{x}\|^2 + 2\|\boldsymbol{y}\|^2.$$

Ex 1.19 **A norm without angles.** Use the parallelogram identity to prove that neither $\|\cdot\|_1$ nor $\|\cdot\|_\infty$ on $\mathbb{R}^2$ can be induced by an inner product. Explain which step of the projection argument fails when only a general norm is available.

Ex 1.20 Expand the two squared norms and add. Which cross terms cancel? Use the identity to recover $\langle \boldsymbol{x}, \boldsymbol{y} \rangle$ from the three lengths $\|\boldsymbol{x} + \boldsymbol{y}\|$, $\|\boldsymbol{x}\|$, and $\|\boldsymbol{y}\|$.

Example Take $\boldsymbol{x} = (1, 0)^T$ and $\boldsymbol{y} = (0, 1)^T$. For the 1-norm, the left side of the parallelogram identity equals $2^2 + 2^2 = 8$, while the right side equals $2 + 2 = 4$. Thus the 1-norm cannot come from any inner product. The same test shows that the infinity norm is not induced by an inner product.

Def 1.20 **(Weighted Inner Product)** Let $\mathbf{W} \in \mathbb{R}^{n \times n}$ be symmetric and satisfy $\boldsymbol{x}^T \mathbf{W} \boldsymbol{x} > 0$ for every nonzero $\boldsymbol{x}$. Then

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\mathbf{W}} = \boldsymbol{x}^T \mathbf{W} \boldsymbol{y}, \qquad \|\boldsymbol{x}\|_{\mathbf{W}} = \sqrt{\boldsymbol{x}^T \mathbf{W} \boldsymbol{x}}$$

define an inner product and its induced norm.

Ex 1.21 **Weighted geometry.** Let

$$\mathbf{W} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}.$$

Verify the inner-product axioms for $\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\mathbf{W}}$. Sketch the unit ball $\|\boldsymbol{x}\|_{\mathbf{W}} = 1$, find the $\mathbf{W}$-orthogonal complement of $(1,1)^T$, and compute the $\mathbf{W}$-orthogonal projection of $(1,0)^T$ onto the line spanned by $(1,1)^T$.
A weighted inner product changes the geometry in which lengths and angles are measured. Directions with large quadratic weight are expensive, and directions with small weight are cheap. In a finite-element method the weight may represent physical energy. In statistics it may account for unequal uncertainty. In optimization it may encode local curvature. The chapter on positive definite matrices develops this geometry further.

1.2.4 Further Exercises

Ex 1.22 **Cauchy–Schwarz as a one-variable minimization.** For fixed $\boldsymbol{x}$ and nonzero $\boldsymbol{y}$, minimize $\|\boldsymbol{x} - t\boldsymbol{y}\|_2^2$ over $t \in \mathbb{R}$. Derive the minimizing coefficient, the projection formula, and Cauchy–Schwarz from the non-negativity of the minimum.

Ex 1.23 **The norm is part of the question.** Two approximations have error vectors $\boldsymbol{e}_1 = (10^{-3}, \dots, 10^{-3})^T$ and $\boldsymbol{e}_2 = (10^{-2}, 0, \dots, 0)^T$ in $\mathbb{R}^{1000}$. Compare them in the 1-, 2-, and infinity norms. State which approximation is preferable under each measurement and explain why no ranking is meaningful until the application chooses a norm.

1.3 Matrix Norms and Condition Numbers

A matrix changes the size of a vector. To bound the effect of an error in the input, we need the largest possible ratio between output length and input length. This ratio defines the induced matrix norm. We distinguish it from the Frobenius norm, which measures the entries of a matrix as data.

An invertible matrix may stretch one direction and compress another. Solving a system reverses that action. The condition number compares the largest stretch with the smallest and will explain why a small residual can accompany a large solution error.

1.3.1 Induced Matrix Norms

Def 1.21 **(Induced Operator Norm)** Let $\|\cdot\|_{\text{in}}$ be a norm on $\mathbb{R}^n$ and let $\|\cdot\|_{\text{out}}$ be a norm on $\mathbb{R}^m$. For $\mathbf{A} \in \mathbb{R}^{m \times n}$, the induced operator norm is

$$\|\mathbf{A}\| = \sup_{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\mathbf{A}\boldsymbol{x}\|_{\text{out}}}{\|\boldsymbol{x}\|_{\text{in}}} = \max_{\|\boldsymbol{x}\|_{\text{in}}=1} \|\mathbf{A}\boldsymbol{x}\|_{\text{out}}.$$

When the same p-norm is used in the domain and range, we write $\|\mathbf{A}\|_p$.

Ex 1.24 For $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, compare $\|\mathbf{A}\boldsymbol{x}\|_2 / \|\boldsymbol{x}\|_2$ for $\boldsymbol{x} = (1,1)^T$ and $\boldsymbol{x} = (1,-1)^T$. Find the largest ratio by expanding $\|\mathbf{A}\boldsymbol{x}\|_2^2$. Explain why neither one test vector nor the largest matrix entry determines the induced norm.
The quotient is unchanged when $\boldsymbol{x}$ is rescaled, so only directions matter. Restricting to the unit sphere removes the irrelevant magnitude. In finite dimensions the unit sphere is compact and the objective is continuous, so the maximum is attained.

Thm 1.16 **(Fundamental Properties of Induced Norms)** For compatible matrices and vectors,

1. **Consistency:** $\|\mathbf{A}\boldsymbol{x}\| \le \|\mathbf{A}\|\,\|\boldsymbol{x}\|$.
2. **Submultiplicativity:** $\|\mathbf{AB}\| \le \|\mathbf{A}\|\,\|\mathbf{B}\|$.
3. **Identity:** $\|\mathbf{I}\| = 1$.
4. **Norm axioms:** the induced norm is positive definite, absolutely homogeneous, and satisfies the triangle inequality.

Proof We verify each assertion from the definition. All norms on intermediate vector spaces must be compatible, so the output norm for $\mathbf{B}$ is the input norm for $\mathbf{A}$.

1. If $\boldsymbol{x} = \mathbf{0}$, the inequality reads $0 \le 0$. Otherwise the quotient $\|\mathbf{A}\boldsymbol{x}\|/\|\boldsymbol{x}\|$ is one of the values bounded by the supremum defining $\|\mathbf{A}\|$. Multiplication by $\|\boldsymbol{x}\| > 0$ gives the result.
2. For any unit vector $\boldsymbol{x}$, apply consistency twice:
$$\|\mathbf{AB}\boldsymbol{x}\| \le \|\mathbf{A}\|\|\mathbf{B}\boldsymbol{x}\| \le \|\mathbf{A}\|\|\mathbf{B}\|\|\boldsymbol{x}\| = \|\mathbf{A}\|\|\mathbf{B}\|.$$
The right side does not depend on $\boldsymbol{x}$. Taking the maximum over unit inputs proves submultiplicativity.
3. When the same norm is used on input and output, $\|\mathbf{I}\boldsymbol{x}\|/\|\boldsymbol{x}\| = 1$ for every nonzero input. Its supremum is therefore one.
4. Every quotient is nonnegative. If $\mathbf{A} = \mathbf{0}$, all quotients vanish. Conversely, if $\|\mathbf{A}\| = 0$, consistency gives $\mathbf{A}\boldsymbol{e}_j = \mathbf{0}$ for each coordinate vector. These are exactly the columns of $\mathbf{A}$, so $\mathbf{A} = \mathbf{0}$.
5. For $\|\boldsymbol{x}\| = 1$, $\|\alpha\mathbf{A}\boldsymbol{x}\| = |\alpha|\|\mathbf{A}\boldsymbol{x}\|$. Taking maxima yields $\|\alpha\mathbf{A}\| = |\alpha|\|\mathbf{A}\|$.
6. For every unit input,
$$\|(\mathbf{A}+\mathbf{B})\boldsymbol{x}\| \le \|\mathbf{A}\boldsymbol{x}\| + \|\mathbf{B}\boldsymbol{x}\| \le \|\mathbf{A}\| + \|\mathbf{B}\|.$$
Taking the maximum gives $\|\mathbf{A}+\mathbf{B}\| \le \|\mathbf{A}\| + \|\mathbf{B}\|$.

□

Ex 1.25 **Submultiplicativity from amplification.** Prove $\|\mathbf{A}_1 \cdots \mathbf{A}_k\| \le \prod_{j=1}^{k} \|\mathbf{A}_j\|$. Explain why this bound may be conservative when the direction of greatest stretch for one factor is not mapped into the direction of greatest stretch for the next.

Example For the diagonal matrix
$$\mathbf{D} = \begin{pmatrix} 4 & 0 \\ 0 & 1/2 \end{pmatrix},$$
the Euclidean unit circle is sent to an ellipse with semiaxis lengths 4 and 1/2. Hence $\|\mathbf{D}\|_2 = 4$. The norm records the greatest stretch, not the average stretch and not the largest entry by convention. In this example those quantities happen to agree because the coordinate axes are the stretching directions.

Thm 1.17 **(The Induced 1- and Infinity Norms)** For $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{m\times n}$,

$$\|\mathbf{A}\|_1 = \max_{1\le j\le n} \sum_{i=1}^{m} |a_{ij}|, \qquad \|\mathbf{A}\|_\infty = \max_{1\le i\le m} \sum_{j=1}^{n} |a_{ij}|.$$

Thus the induced 1-norm is the largest absolute column sum, while the induced infinity norm is the largest absolute row sum.

Proof For each formula we need an upper bound for every input and an input that attains it.

Let $C = \max_j \sum_i |a_{ij}|$, the largest absolute column sum. Expanding each output entry and using the triangle inequality gives

$$\begin{aligned}\|\mathbf{A}\boldsymbol{x}\|_1 &= \sum_{i=1}^{m} \left| \sum_{j=1}^{n} a_{ij}x_j \right| \\ &\le \sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}||x_j| \\ &= \sum_{j=1}^{n} |x_j| \left(\sum_{i=1}^{m} |a_{ij}| \right) \le C \sum_{j=1}^{n} |x_j| = C\|\boldsymbol{x}\|_1.\end{aligned}$$

The interchange of sums is legitimate because both are finite. For unit inputs this proves $\|\mathbf{A}\|_1 \le C$.

Choose a column j_* attaining C. The input $\boldsymbol{e}_{j_*}$ has 1-norm one, and multiplying by it selects that column. Therefore

$$\|\mathbf{A}\boldsymbol{e}_{j_*}\|_1 = \sum_i |a_{ij_*}| = C.$$

The maximum is at least C, so the upper bound is exact.

For the infinity norm, put $R = \max_i \sum_j |a_{ij}|$. For any input,

$$|(\mathbf{A}\boldsymbol{x})_i| \le \sum_j |a_{ij}||x_j| \le \left(\sum_j |a_{ij}| \right) \|\boldsymbol{x}\|_\infty \le R\|\boldsymbol{x}\|_\infty.$$

This holds for every row, so taking their maximum gives $\|\mathbf{A}\|_\infty \le R$.

Choose a maximizing row i_*. Set $x_j = 1$ when $a_{i_*j} \ge 0$ and $x_j = -1$ when $a_{i_*j} < 0$. In particular choose 1 for a zero entry, so the input has infinity norm one even for the zero matrix. Each product in that row is $a_{i_*j}x_j = |a_{i_*j}|$. Thus $(\mathbf{A}\boldsymbol{x})_{i_*} = R$, proving attainment and the second formula. □

Ex 1.26 **Three induced norms.** For

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix},$$

compute $\|\mathbf{A}\|_1$ and $\|\mathbf{A}\|_\infty$. Estimate $\|\mathbf{A}\|_2$ from below by testing at least four unit vectors and from above using $\|\mathbf{A}\|_F$.

Def 1.22 **(The Induced Euclidean Norm)** The induced Euclidean norm is

$$\|\mathbf{A}\|_2 = \max_{\|\boldsymbol{x}\|_2=1} \|\mathbf{A}\boldsymbol{x}\|_2.$$

Equivalently,

$$\|\mathbf{A}\|_2^2 = \max_{\|\boldsymbol{x}\|_2=1} \boldsymbol{x}^T\mathbf{A}^T\mathbf{A}\boldsymbol{x}.$$

The second expression follows by squaring the output norm. It already shows why $\mathbf{A}^T\mathbf{A}$ appears in questions about Euclidean amplification. Once the singular value decomposition is available, Thm 4.19 proves the computable formula

$$\|\mathbf{A}\|_2 = \sqrt{\lambda_{\max}(\mathbf{A}^T\mathbf{A})} = \sigma_{\max}(\mathbf{A}).$$

Thm 1.18 **(Orthogonal Transformations Have Unit Euclidean Norm)** If $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$, then

$$\|\mathbf{Q}\boldsymbol{x}\|_2 = \|\boldsymbol{x}\|_2 \qquad \text{and} \qquad \|\mathbf{Q}\|_2 = 1.$$

Ex 1.27 Compute $\|\mathbf{Q}\boldsymbol{x}\|_2^2$ using $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$. Then take the maximum over unit vectors. Does the argument require $\mathbf{Q}$ to be square?

1.3.2 Dual Norms

An inner product turns a vector $\boldsymbol{y}$ into the linear measurement $\boldsymbol{x} \mapsto \boldsymbol{y}^T\boldsymbol{x}$. The dual norm measures the largest value of this measurement on the unit ball.

Def 1.23 **(Dual Norm)** For a norm $\|\cdot\|$ on $\mathbb{R}^n$, its dual norm is

$$\|\boldsymbol{y}\|_* = \max_{\|\boldsymbol{x}\|=1} |\boldsymbol{y}^T\boldsymbol{x}|.$$

Consequently,

$$|\boldsymbol{y}^T\boldsymbol{x}| \le \|\boldsymbol{y}\|_*\|\boldsymbol{x}\|.$$

Thm 1.19 **(Duals of the Standard Norms)** The Euclidean norm is self-dual. The 1- and infinity norms are dual to one another:

$$\|\boldsymbol{y}\|_{2,*} = \|\boldsymbol{y}\|_2, \qquad \|\boldsymbol{y}\|_{1,*} = \|\boldsymbol{y}\|_\infty, \qquad \|\boldsymbol{y}\|_{\infty,*} = \|\boldsymbol{y}\|_1.$$

Proof For a dual norm, an upper estimate alone is insufficient. We must also find a unit vector attaining that estimate. If $\boldsymbol{y} = \mathbf{0}$, every claimed norm is zero. Assume below that $\boldsymbol{y} \neq \mathbf{0}$.

1. For $\|\boldsymbol{x}\|_2 = 1$, Cauchy–Schwarz gives $|\boldsymbol{y}^T\boldsymbol{x}| \le \|\boldsymbol{y}\|_2$. Choose $\boldsymbol{x} = \boldsymbol{y}/\|\boldsymbol{y}\|_2$. It is a unit vector, and $\boldsymbol{y}^T\boldsymbol{x} = \|\boldsymbol{y}\|_2$. Hence the maximum is exactly $\|\boldsymbol{y}\|_2$.

2. For $\|\boldsymbol{x}\|_1 = 1$,

$$|\boldsymbol{y}^T\boldsymbol{x}| \le \sum_i |y_i||x_i| \le (\max_i |y_i|) \sum_i |x_i| = \|\boldsymbol{y}\|_\infty.$$

Choose an index j of largest $|y_j|$ and set $\boldsymbol{x} = \operatorname{sign}(y_j)\boldsymbol{e}_j$. Its 1-norm is one and $\boldsymbol{y}^T\boldsymbol{x} = |y_j| = \|\boldsymbol{y}\|_\infty$.

3. For $\|\boldsymbol{x}\|_\infty = 1$,

$$|\boldsymbol{y}^T\boldsymbol{x}| \le \sum_i |y_i||x_i| \le \sum_i |y_i| = \|\boldsymbol{y}\|_1.$$

Set $x_i = 1$ if $y_i \ge 0$ and $x_i = -1$ if $y_i < 0$. Then $\|\boldsymbol{x}\|_\infty = 1$ and every product $y_i x_i$ equals $|y_i|$. Thus equality is attained.

□

Dual norms explain why the norm used for a residual determines the norm that appears in the corresponding linear functional. They also explain the row-sum and column-sum formulas above. The transpose interchanges inputs with linear measurements, giving the useful identity

$$\|\mathbf{A}\|_1 = \|\mathbf{A}^T\|_\infty.$$

1.3.3 The Frobenius Norm

Matrices themselves form a vector space. When a matrix is being treated as data rather than as an operator, the natural Euclidean measurement is entrywise.

Def 1.24 **(Frobenius Inner Product and Norm)** For matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m\times n}$, the Frobenius inner product is

$$\langle \mathbf{A}, \mathbf{B}\rangle_F = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ij} = \operatorname{trace}(\mathbf{A}^T\mathbf{B}).$$

Its induced norm is

$$\|\mathbf{A}\|_F = \left(\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2\right)^{1/2}.$$

Ex 1.28 **Frobenius geometry.** Verify the inner-product axioms for $\langle \mathbf{A}, \mathbf{B}\rangle_F$. Prove the matrix Cauchy–Schwarz inequality $|\langle \mathbf{A}, \mathbf{B}\rangle_F| \le \|\mathbf{A}\|_F\|\mathbf{B}\|_F$ by treating the entries as vectors.

Remark 1.10 **(Two different questions)** The Frobenius norm measures the collective size of the entries. The induced 2-norm measures the largest amplification of a vector. Both are useful, but they answer different questions. Calling both a "matrix norm" does not make them interchangeable.

Thm 1.20 **(Bounds Relating Frobenius and Operator Norms)** For $\mathbf{A} \in \mathbb{R}^{m\times n}$,

$$\|\mathbf{A}\|_2 \le \|\mathbf{A}\|_F \le \sqrt{n}\,\|\mathbf{A}\|_2.$$

For compatible matrices $\mathbf{A}$ and $\mathbf{B}$,

$$\|\mathbf{AB}\|_F \le \|\mathbf{A}\|_2\|\mathbf{B}\|_F.$$

Proof Let $\boldsymbol{a}_i^T$ denote row i of $\mathbf{A}$. For $\|\boldsymbol{x}\|_2 = 1$, each output entry is $\boldsymbol{a}_i^T\boldsymbol{x}$. Cauchy–Schwarz applies separately in each row:

$$\|\mathbf{A}\boldsymbol{x}\|_2^2 = \sum_{i=1}^{m} |\boldsymbol{a}_i^T\boldsymbol{x}|^2 \le \sum_{i=1}^{m} \|\boldsymbol{a}_i\|_2^2\|\boldsymbol{x}\|_2^2 = \sum_{i,j} a_{ij}^2 = \|\mathbf{A}\|_F^2.$$

Taking the maximum over unit $\boldsymbol{x}$ and then square roots gives $\|\mathbf{A}\|_2 \le \|\mathbf{A}\|_F$.

For the opposite comparison, use columns instead of rows. Column j is $\mathbf{A}\boldsymbol{e}_j$, and $\|\boldsymbol{e}_j\|_2 = 1$. Consistency gives $\|\mathbf{A}\boldsymbol{e}_j\|_2 \le \|\mathbf{A}\|_2$. Summing the squared estimates over all n columns yields

$$\|\mathbf{A}\|_F^2 = \sum_{j=1}^{n} \|\mathbf{A}\boldsymbol{e}_j\|_2^2 \le \sum_{j=1}^{n} \|\mathbf{A}\|_2^2 = n\|\mathbf{A}\|_2^2.$$

The nonnegative square root gives the factor $\sqrt{n}$.

Finally let $\mathbf{B} \in \mathbb{R}^{n\times k}$ have columns $\boldsymbol{b}_1, \dots, \boldsymbol{b}_k$. Column j of the product is $\mathbf{A}\boldsymbol{b}_j$. Apply consistency to each such column before summing:

$$\begin{aligned}\|\mathbf{AB}\|_F^2 &= \sum_{j=1}^{k} \|\mathbf{A}\boldsymbol{b}_j\|_2^2 \\ &\le \sum_{j=1}^{k} \|\mathbf{A}\|_2^2\|\boldsymbol{b}_j\|_2^2 \\ &= \|\mathbf{A}\|_2^2\|\mathbf{B}\|_F^2.\end{aligned}$$

Taking square roots proves the product estimate. The dimensions n and k enter only through the corresponding sums over columns. □

1.3.4 Matrix Condition Numbers

An invertible matrix can stretch some directions and compress others. A solve with that matrix must undo both effects. The condition number measures the total distortion.

Def 1.25 **(Condition Number)** For a nonsingular square matrix $\mathbf{A}$ and an induced norm,

$$\kappa(\mathbf{A}) = \|\mathbf{A}\|\,\|\mathbf{A}^{-1}\|.$$

Thm 1.21 **(Two-Sided Stretching Bound)** For every nonzero vector $\boldsymbol{x}$,

$$\frac{1}{\|\mathbf{A}^{-1}\|} \le \frac{\|\mathbf{A}\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \le \|\mathbf{A}\|.$$

The ratio between the largest possible upper stretch and the guaranteed lower stretch is $\kappa(\mathbf{A})$.

Proof The assumption that $\mathbf{A}$ is nonsingular is needed for the lower estimate. Fix $\boldsymbol{x} \neq \mathbf{0}$.

1. By the definition of the induced norm, the ratio for this particular input cannot exceed the largest possible ratio:

$$\frac{\|\mathbf{A}\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq \sup_{\boldsymbol{z}\neq\mathbf{0}} \frac{\|\mathbf{A}\boldsymbol{z}\|}{\|\boldsymbol{z}\|} = \|\mathbf{A}\|.$$

2. Undo the action of $\mathbf{A}$ and apply consistency to its inverse:

$$\boldsymbol{x} = \mathbf{A}^{-1}(\mathbf{A}\boldsymbol{x}),$$
$$\|\boldsymbol{x}\| \leq \|\mathbf{A}^{-1}\|\|\mathbf{A}\boldsymbol{x}\|.$$

   Both $\|\boldsymbol{x}\|$ and $\|\mathbf{A}^{-1}\|$ are positive. Dividing by their product gives

$$\frac{1}{\|\mathbf{A}^{-1}\|} \leq \frac{\|\mathbf{A}\boldsymbol{x}\|}{\|\boldsymbol{x}\|}.$$

3. The upper stretch is $\|\mathbf{A}\|$ and the lower stretch is $1/\|\mathbf{A}^{-1}\|$. Dividing the former by the latter gives $\|\mathbf{A}\|\|\mathbf{A}^{-1}\| = \kappa(\mathbf{A})$.

□

Ex 1.29 **Conditioning as a thin ellipse.** Let $\mathbf{A} = \mathrm{diag}(1, \varepsilon)$ with $0 < \varepsilon < 1$. Describe the image of the Euclidean unit circle, compute $\|\mathbf{A}\|_2$, $\|\mathbf{A}^{-1}\|_2$, and $\kappa_2(\mathbf{A})$ directly from the definition, and explain which component of a solution is sensitive to perturbations.
If $\kappa(\mathbf{A})$ is near one, the matrix treats every direction comparably. If it is large, at least one direction is compressed relative to another. Recovering a component in a strongly compressed direction requires amplification by $\mathbf{A}^{-1}$. Consequently, ill-conditioned linear systems can turn small data errors into large solution errors. Theorem 2.7 makes this statement quantitative.

1.3.5 Further Exercises

Ex 1.30 **Entries do not determine action.** Compare

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}.$$

Both matrices have the same entry magnitudes. Find their ranges and null spaces, identify directions they annihilate, and explain why an entrywise description misses this structure.

Ex 1.31 **Duality and transpose.** Starting from the definition of a dual norm, prove $\|\mathbf{A}\|_1 = \|\mathbf{A}^T\|_\infty$. Then formulate the corresponding statement for an arbitrary norm and its dual.

Ex 1.32 **Changing units.** Suppose the two coordinates of $\boldsymbol{x}$ are measured in meters and millimeters. Introduce a diagonal scaling matrix that expresses both in meters. Derive the transformed matrix for the same linear map and compare its induced norm before and after scaling. Explain why a condition number is meaningful only after the input and output measurements have been specified.

1.4 Floating Point Arithmetic

Digital computers store only finitely many real numbers. Even when a formula is exact on paper, its implementation rounds intermediate results to this finite set. We need to determine the size of one rounding error before studying how an algorithm propagates many of them.

Floating-point arithmetic is usually accurate locally, since one elementary operation is rounded to a nearby representable number. Instability appears when an algorithm repeatedly amplifies these tiny local errors, or when it subtracts nearly equal quantities and exposes digits that were already contaminated by rounding.

1.4.1 Floating Point Representation

In binary floating-point arithmetic, a normalized number is stored in the form $x = (-1)^s (1.f)_2 \, 2^e$. The subscript 2 means that the digits after the point represent powers $2^{-1}, 2^{-2}, \ldots$. IEEE 754 specifies common formats and rounding rules. The fields have the following roles.

- **s:** Sign bit, determining whether the number is positive or negative.
- **f:** Fractional significand (mantissa). The leading 1 is implicit to maximize precision.
- **e:** Unbiased exponent. The stored exponent field adds a fixed bias to e; reserved fields encode zeros, subnormals, infinities, and NaNs.

The exponent controls scale and the significand controls precision. This means floating-point numbers have roughly fixed **relative** spacing, not fixed absolute spacing. There are many representable numbers near zero and far fewer, in absolute terms, near 10^{100}. The gap between adjacent numbers near x is proportional to $|x|$.

Remark 1.11 **(IEEE 754 Standards)**

- **Single (32-bit):** 1 sign, 8 exponent, 23 fraction bits.
- **Double (64-bit):** 1 sign, 11 exponent, 52 fraction bits.

Example A binary16 number has one sign bit, five exponent bits, and ten fraction bits. The exponent bias is 15. To store -3.25, we compute the fields as follows.

1. $s = 1$.
2. $3.25_{10} = 11.01_2 = 1.101_2 \times 2^1$.
3. $e = 1 + 15 = 16 = 10000_2$.
4. $f = 1010000000_2$.

Result: `1 10000 1010000000`.

Ex 1.33

1. Encode +5.5 in 16-bit format.
2. How many distinct normalized numbers for 5 exp bits, 10 frac bits?

Def 1.26 **(Machine Epsilon ($\varepsilon_{\text{mach}}$))** Machine epsilon is the distance from 1 to the next larger floating-point number. It is a spacing, rather than the threshold at which an increment first changes a rounded sum. It characterizes the spacing of representable numbers near 1 and is given by $\varepsilon_{\text{mach}} = 2^{1-p}$, where p is the number of bits in the significand.

- For double precision (64-bit), $\varepsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$.

Remark 1.12 **(Spacing rule)** Near a normalized number x, the distance to the next representable number is approximately $\varepsilon_{\text{mach}}|x|$. Near 10^{16}, the gap is on the order of 1, whereas near 1, the gap is on the order of 10^{-16}. This explains why adding a small number to a sufficiently large number has no effect.

Thm 1.22 **(Absorption Criterion)** Let x be a representable number with two finite neighbors. In round-to-nearest arithmetic, if $|y|$ is less than half the smaller of the two distances from x to its neighbors, then

$$\text{fl}(x + y) = x.$$

Equivalently, small increments disappear when they are below the local resolution of the floating-point grid.

Ex 1.34

1. Use Def 1.26 to predict whether `1.0 + 1e-16` and `1.0 + 2e-16` differ from `1.0`. Then test the prediction in Python.
2. Find $\varepsilon_{\text{mach}}$ empirically via loop.
3. Use Thm 1.22 to predict the smallest power of 10 for which `1e16 + 10**k` changes `1e16`.
4. Explain why the answer changes if `1e16` is replaced by `1.0`.

Def 1.27 **(Unit Roundoff (u))** Unit roundoff bounds the relative error caused by rounding a nonzero real number within the normalized range to the nearest floating-point number:

$$\frac{|\text{fl}(x) - x|}{|x|} \leq u.$$

For round-to-nearest arithmetic, $u = \varepsilon_{\text{mach}}/2 = 2^{-p}$.

Remark 1.13 **(Machine epsilon vs. unit roundoff)** $\varepsilon_{\text{mach}}$ is a spacing near 1. The unit roundoff u is an error bound for rounding to nearest. Many texts use "machine epsilon" for both quantities, so always check the convention. In these notes, elementary rounding errors are bounded by u.

Thm 1.23 **(Fundamental Axiom)** Suppose the exact result $a \circ b$ is finite, nonzero, and lies in the normalized floating-point range. In round-to-nearest arithmetic,

$$\text{fl}(a \circ b) = (a \circ b)(1 + \delta), \qquad |\delta| \leq u,$$

for $\circ \in \{+, -, \times, \div\}$. Overflow and gradual underflow require separate absolute-error models.

Proof Assume rounding to nearest, and exclude overflow and the subnormal range as in the theorem. It suffices to consider $|x|$, because the representable numbers and rounding rule are symmetric about zero.

1. Choose the integer e such that $2^e \le |x| < 2^{e+1}$. In this interval, changing the last stored significand bit changes the value by $s = 2^{e+1-p}$. Thus consecutive representable normalized values are separated by s.

2. Rounding to nearest selects one of the two representable values surrounding $|x|$. Its distance is at most half their separation. Therefore

$$|\,\mathrm{fl}(x) - x| \le \frac{s}{2} = 2^{e-p}.$$

   At an exactly representable value the error is zero, which also satisfies this bound.

3. Since $|x| \ge 2^e > 0$,

$$\frac{|\,\mathrm{fl}(x) - x|}{|x|} \le \frac{2^{e-p}}{2^e} = 2^{-p} = u.$$

   Define $\delta = (\mathrm{fl}(x) - x)/x$. Then $\mathrm{fl}(x) = x(1+\delta)$ and $|\delta| \le u$. For $x = 0$, take $\delta = 0$.

□

Remark 1.14 **(Accumulation of rounding errors)** A sequence of n benign operations often produces an error proportional to nu when $nu \ll 1$. The rule is a heuristic. Cancellation and ill-conditioning can amplify the perturbations well beyond nu. Operation order matters because every intermediate result is rounded.

Ex 1.35 Let $x = 1$ and $y = 1 - 10^{-12}$ be exact real data, and let $\hat{x}$ and $\hat{y}$ be their rounded values. Distinguish the error made when subtracting $\hat{x} - \hat{y}$ from the error caused by rounding x and y before subtraction. Which error does the fundamental rounding model bound? Estimate the amplification of the operand errors relative to $x - y$.

1.4.2 IEEE 754 Special Values

- **Signed Zero:** $+0, -0$. $1/(+0) = \infty$, $1/(-0) = -\infty$.
- **Subnormals:** Fill the gap between 0 and the smallest normalized number. Reduced precision, allows gradual underflow.
- **Infinities:** $\pm\infty$. Result of overflow or $x/0$.
- **NaN:** Not a Number ($0/0$, $\infty - \infty$). Unequal to everything, including itself.

Remark 1.15 **(NaN propagation)** A NaN usually contaminates all later arithmetic. This behavior helps with debugging, because the NaN marks an invalid operation that happened earlier even if the failure is observed much later.

Ex 1.36

1. Using NumPy scalars, compare `np.float64(1.0)/np.float64(0.0)` with division by `np.float64(-0.0)`. Explain the warnings and the signs of the results. (Python's built-in float division raises `ZeroDivisionError` instead.)

2. Evaluate `float('inf') - float('inf')` and `float('nan') == float('nan')`.

1.4.3 Arithmetic Pitfalls

Because floating-point numbers are discrete, the standard laws of algebra do not always hold. In particular, operations are neither associative nor distributive in general.

- **Non-Associativity:** The order of operations can change the result, i.e., $(a+b)+c \neq a+(b+c)$.
- **Non-Distributivity:** $a(b+c) \neq ab+ac$.
- **Absorption:** If $|a|$ is much larger than $|b|$, then $a+b$ may equal a exactly because b is smaller than the gap between representable numbers near a.

Example **(Absorption)** In double precision, `1e16 + 1.0 == 1e16` is true. The number 1 is smaller than the spacing between adjacent representable numbers near 10^{16}, so it is rounded away. Consequently `(1e16 + 1.0) - 1e16` returns `0.0`, not `1.0`.

Example **(Non-associativity)** Let $a = 10^{16}$, $b = -10^{16}$, and $c = 1$. Then

$$(a+b)+c = 1, \qquad a+(b+c) = 0$$

in double precision. The second expression first rounds $b+c$ back to -10^{16}, so the 1 is lost before the final addition.

Remark 1.16 **(Summation rule)** Add numbers of similar magnitude when possible. Summing from smallest to largest, pairwise summation, or compensated summation reduces the loss of small terms. Consequently, library reductions may give different answers than a hand-written loop and still be more accurate.

Ex 1.37

1. Before running Python, use Thm 1.22 to predict `(1e16 + 1.0) - 1e16`.
2. `a=1e16, b=1`. Find `(a+b)-a` in Python. Explain result.
3. Find a, b, c where `a*(b+c) != a*b + a*c`.
4. Before running Python, predict the results of summing $[10^{16}, 1, -10^{16}]$ in the listed order and in the order $[10^{16}, -10^{16}, 1]$.
5. Compare naive summation, pairwise summation, and Kahan summation on a list containing one large number and many small numbers. Which method best preserves the small terms?

1.5 Quantifying Approximation Error

1.5.1 Absolute and Relative Errors

- **Absolute Error:** $|\hat{x} - x|$.
- **Relative Error:** $\varepsilon = \frac{|\hat{x}-x|}{|x|}$. This is the standard measure for precision in numerical computing.

Absolute error measures the size of the mistake in the original units. Relative error measures the size of the mistake compared to the quantity being computed. In numerical computing, relative error is usually more informative. An error of 10^{-6} is excellent for a number of size 10^3 and terrible for a number of size 10^{-9}.

Remark 1.17 **(Comparing floating-point numbers)** Avoid testing computed real numbers with exact equality. Use a tolerance such as

$$|\hat{x} - x| \leq \text{atol} + \text{rtol}|x|.$$

This principle underlies `np.isclose`. The absolute tolerance bounds errors near zero, whereas the relative tolerance scales with magnitude.

Ex 1.38

1. Compute the absolute and relative errors of using 1.414 for $\sqrt{2}$.
2. Let $\hat{x} = x(1 + \delta_x)$ and $\hat{y} = y(1 + \delta_y)$ with $xy \neq 0$. Derive the exact relative perturbation of $\hat{x}\hat{y}$ before rounding the product. Bound it when $|\delta_x|, |\delta_y| \leq u$, and identify the first-order term.

Def 1.28 **(Catastrophic Cancellation) Catastrophic cancellation** occurs when two nearly equal approximations are subtracted. The leading digits cancel, so errors already present in the operands become large relative to the small difference.

- For $x \neq y$ and operand errors $|\hat{x}-x| \leq u|x|$, $|\hat{y}-y| \leq u|y|$, exact subtraction of the perturbed operands has relative error at most $u(|x|+|y|)/|x-y|$. Rounding that subtraction contributes a further local error.
- As $x \to y$, the amplification factor $(|x| + |y|)/|x - y|$ can become arbitrarily large.

Proof We distinguish errors in the operands from any additional rounding of the subtraction itself. Assume $x - y \neq 0$, since relative error with respect to a zero answer is undefined. Write the perturbed operands as

$$\hat{x} = x(1 + \varepsilon_x), \qquad \hat{y} = y(1 + \varepsilon_y), \qquad |\varepsilon_x|, |\varepsilon_y| \leq u.$$

First regard $\hat{x} - \hat{y}$ as an exact real subtraction of these perturbed data. Expand and cancel the unperturbed terms:

$$\begin{aligned}(\hat{x} - \hat{y}) - (x - y) &= x + x\varepsilon_x - y - y\varepsilon_y - x + y \\ &= x\varepsilon_x - y\varepsilon_y.\end{aligned}$$

The triangle inequality and the operand bounds give

$$\begin{aligned}|(\hat{x} - \hat{y}) - (x - y)| &\leq |x||\varepsilon_x| + |y||\varepsilon_y| \\ &\leq u(|x| + |y|).\end{aligned}$$

Dividing by $|x - y| > 0$ yields the stated amplification factor.

This factor can be large when the operands have comparable nonzero magnitude but their difference is small. It measures sensitivity to the operand errors. If the subtraction is also

rounded, the floating-point model adds a further error bounded by $u|\hat{x} - \hat{y}|$ in absolute value. That additional operation error is separate from the potentially amplified operand errors. The bound does not assert that every subtraction exhibits its worst possible error. □

Remark 1.18 **(Numerical stability principle)** Rounding error is an inevitable consequence of finite precision. However, catastrophic cancellation is an algorithmic flaw that can often be avoided by mathematically rearranging the problem.

Example **(Stabilizing Cancellation)** $f(x) = \sqrt{x+1} - \sqrt{x}$ for $x = 10^{16}$.

- **Direct:** In double precision, both square roots round to 10^8, so the computed difference is zero.
- **Stable:** $\frac{1}{\sqrt{x+1}+\sqrt{x}} \approx 5 \times 10^{-9}$.

Example **(Quadratic formula)** The roots of $ax^2 + bx + c = 0$ are

$$x_{\pm} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

If $b^2 \gg 4ac$, one numerator subtracts nearly equal numbers. A stable strategy computes the root with no cancellation first, then uses $x_1 x_2 = c/a$ to recover the other root.

Remark 1.19 **(Use stable library functions)** Functions such as `log1p(x)`, `expm1(x)`, and `hypot(x, y)` implement stable rearrangements for common cancellation and overflow patterns. Prefer them over hand-expanded formulas when the inputs may be small, large, or nearly cancelling.

Ex 1.39 Stabilize:

1. Use Def 1.28 to explain why $(1 - \cos x)/x^2$ is unstable for small x, then derive a stable form.
2. Use Def 1.28 to identify which root of $x^2 - 10^8 x + 1 = 0$ is unstable in the quadratic formula, then compute it using $x_1 x_2 = c/a$.
3. Use the Taylor series of e^x to explain why $e^x - 1$ loses relative accuracy for small x. Compare with `np.expm1`.

1.5.2 Error Analysis

A computed result can be judged in two ways. Forward error analysis bounds the distance between the computed output and the exact output. Backward error analysis instead seeks the smallest perturbation of the input for which the computed output is exact. An algorithm is **backward stable** if this perturbation is a modest multiple of the unit roundoff u for every input.

Backward error is often easier to bound than forward error, and it is directly comparable with the uncertainty in the data. If the input is known only to a relative accuracy larger than the backward error, the algorithm has introduced no error beyond what the data already contain. The condition number then converts a backward error into a forward error.

Def 1.29 **(Condition Number (κ))** The condition number measures the local sensitivity of a problem's output to perturbations in its input. For sufficiently small relative perturbations, the first-order relation has the form

$$\text{relative forward error} \lesssim \kappa \, \text{relative backward error},$$

with the precise definition of κ determined by the problem and the chosen norms. For the linear system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$, Section 2.2 shows that this role is played by the matrix condition number of Def 1.25.

Remark 1.20 **(Stability vs. conditioning)** Conditioning is a property of the mathematical problem. Stability is a property of the algorithm. A stable algorithm can still return a poor answer on an ill-conditioned problem, because the problem itself amplifies small perturbations.

Ex 1.40 A problem has condition number $\kappa = 10^8$, and an algorithm computes a solution with relative backward error 10^{-15}. Use Def 1.29 to estimate the largest relative forward error, and state how many correct significant digits the result can be expected to have.

1.6 Cost Models and Sparse Storage

To choose between algorithms, we compare both arithmetic and storage. A matrix may be small enough to store but too expensive to factor. A sparse matrix has sufficiently few nonzero entries that storing only those entries changes the cost of its basic operations.

We count additions and multiplications for dense products, then describe storage by coordinates and by rows. These formats also show how to assemble a sparse matrix from local contributions without first allocating a dense array.

1.6.1 Asymptotic Notation and Operation Counts

Def 1.30 **(Asymptotic Notation)** Let f and g be positive functions of an integer n. We write $f(n) = O(g(n))$ if there are constants $C > 0$ and n_0 such that $f(n) \le Cg(n)$ for every $n \ge n_0$. We write $f(n) = \Theta(g(n))$ if both $f(n) = O(g(n))$ and $g(n) = O(f(n))$ hold.

Ex 1.41 Show that $3n^2 + 100n + 5 = \Theta(n^2)$, and arrange $5n^3 + n$, $100n^2$, $n^2 \log n$, and $2^{10}n^2$ in order of growth.

Def 1.31 **(Flop)** A **flop** is one floating-point addition, subtraction, multiplication, division, or square root.

Thm 1.24 **(Costs of the Basic Dense Kernels)** Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{B} \in \mathbb{R}^{n\times p}$, and let $\mathbf{L} \in \mathbb{R}^{n\times n}$ be nonsingular and lower triangular. The standard algorithms require

1. $2n - 1$ flops for the inner product $\boldsymbol{x}^T\boldsymbol{y}$,
2. $m(2n - 1)$ flops for the product $\mathbf{A}\boldsymbol{x}$,
3. $mp(2n - 1)$ flops for the product $\mathbf{AB}$,
4. n^2 flops for the solution of $\mathbf{L}\boldsymbol{z} = \boldsymbol{y}$ by forward substitution.

Proof We count scalar arithmetic operations and exclude assignments and index calculations.

1. To form $\sum_{j=1}^{n} x_j y_j$, first compute its n products. Adding n numbers requires $n-1$ additions. The total is $n + (n - 1) = 2n - 1$ operations.
2. Each output entry is an inner product between one row of length n and the input vector. There are m output entries. The count is therefore $m(2n - 1)$, whose leading term is $2mn$.

3. If $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{n\times p}$, there are mp output entries. Each entry uses an inner product of length n. Hence the count is $mp(2n-1)$, with leading term $2mnp$.

4. In forward substitution, row i forms $i-1$ products, subtracts those $i-1$ terms from the right-hand side, and divides by the diagonal entry. The diagonal is nonzero by triangular nonsingularity. The count per row is $2(i-1)+1$. Therefore

$$\sum_{i=1}^{n}[2(i-1)+1] = 2\frac{n(n-1)}{2} + n = n^2.$$

For square matrices, put $m = n = p$ to obtain the familiar leading count $2n^3$. These counts describe the straightforward algorithms under the stated convention. □

The order of evaluation matters even when the result does not change. For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{x} \in \mathbb{R}^n$, forming $(\mathbf{AB})\boldsymbol{x}$ costs about $2n^3$ flops, whereas $\mathbf{A}(\mathbf{B}\boldsymbol{x})$ costs about $4n^2$ flops. The factorizations of later sections have costs of the same kind. Dense LU factorization requires about $\frac{2}{3}n^3$ flops (Thm 2.13), Cholesky about $\frac{1}{3}n^3$, and Householder QR of an $m \times n$ matrix about $2mn^2 - \frac{2}{3}n^3$.

Example **(The dense barrier)** At a sustained rate of 10^{10} flops per second, a dense LU factorization with $n = 10^4$ requires about $\frac{2}{3}\cdot 10^{12}$ flops, or roughly one minute. With $n = 10^5$ it requires about $\frac{2}{3}\cdot 10^{15}$ flops, or roughly nineteen hours, and storing the matrix in double precision requires $8\cdot 10^{10}$ bytes. A finite-difference discretization on a $300\times 300\times 300$ grid already has $n \approx 2.7\times 10^7$ unknowns. Dense methods are therefore out of reach for such problems, and the matrix must be stored and used in a form that exploits its zero entries.

Remark 1.21 **(Memory traffic)** Flop counts ignore the cost of moving data between memory and the processor. A dense matrix–matrix product reuses each loaded entry $O(n)$ times, and blocked implementations run close to the peak arithmetic rate. A sparse matrix–vector product performs two flops for each stored nonzero but reads at least one value and one index for it. Its running time is therefore set by memory bandwidth rather than by the flop count, and reducing the number of stored entries reduces the running time directly.

1.6.2 Sparse Matrix Storage

A matrix is **sparse** when the number of its nonzero entries, written $\mathrm{nnz}(\mathbf{A})$, is much smaller than n^2. The finite-difference Poisson matrix on an $N \times N$ grid has $n = N^2$ unknowns and fewer than $5n$ nonzero entries. Stored densely, it would occupy n^2 words, and a matrix–vector product would cost $2n^2$ flops. A sparse format stores only the nonzero entries and their positions.

Def 1.32 **(Coordinate and Compressed Sparse Row Formats)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ have $\mathrm{nnz}(\mathbf{A})$ nonzero entries.

1. The **coordinate** (COO) format stores three arrays of length $\mathrm{nnz}(\mathbf{A})$ that list each nonzero entry as a triplet (i, j, a_{ij}). The triplets may appear in any order, and repeated positions are understood to be summed.

2. The **compressed sparse row** (CSR) format stores the nonzero values row by row in an array `val`, their column indices in an array `col` of the same length, and an array `ptr` of length $n+1$. The entries of row i occupy positions $\texttt{ptr}_i, \ldots, \texttt{ptr}_{i+1} - 1$ of `val` and `col`.

The arrays below use mathematical indexing, beginning at one. In a Python implementation, column indices and positions begin at zero, and the first row pointer is zero. The difference $\texttt{ptr}_{i+1} - \texttt{ptr}_i$ counts stored entries in row i under either convention. An empty row has equal consecutive pointers.

Ex 1.42 Write the COO triplets and the CSR arrays of the 5×5 matrix $\operatorname{tridiag}(-1, 2, -1)$. Then compute $\mathbf{A}\boldsymbol{x}$ for $\boldsymbol{x} = (1, 2, 3, 4, 5)^T$ directly from the CSR arrays.

Example The matrix

$$\mathbf{A} = \begin{pmatrix} 4 & 0 & -1 & 0 \\ 0 & 3 & 0 & 0 \\ -1 & 0 & 5 & 2 \\ 0 & 0 & 2 & 6 \end{pmatrix}$$

has eight nonzero entries. Its CSR arrays are

$$\texttt{val} = (4, -1,\ 3,\ -1, 5, 2,\ 2, 6), \qquad \texttt{col} = (1, 3,\ 2,\ 1, 3, 4,\ 3, 4), \qquad \texttt{ptr} = (1, 3, 4, 7, 9).$$

Row 3 occupies positions $\texttt{ptr}_3 = 4$ through $\texttt{ptr}_4 - 1 = 6$.

Thm 1.25 **(Cost of a CSR Matrix–Vector Product)** For $\mathbf{A}$ stored in CSR format, the product $\boldsymbol{y} = \mathbf{A}\boldsymbol{x}$ computed by

$$y_i = \sum_{k=\texttt{ptr}_i}^{\texttt{ptr}_{i+1}-1} \texttt{val}_k \, x_{\texttt{col}_k}, \qquad i = 1, \ldots, n,$$

requires at most $2\,\mathrm{nnz}(\mathbf{A})$ flops. The format stores $2\,\mathrm{nnz}(\mathbf{A}) + n + 1$ numbers.

Proof Consider the computation row by row. For each stored entry a_{ij}, compute the product $a_{ij}x_j$ and add it to the accumulator for output entry i. This uses one multiplication and at most one addition per stored entry. Thus at most $2\,\mathrm{nnz}(\mathbf{A})$ arithmetic operations are needed for the contributions.

There are also n row accumulators to initialize or write. Traversing the row pointers takes $O(n)$ work. Consequently the complete count is $O(\mathrm{nnz}(\mathbf{A}) + n)$. In the common case of at least one stored entry in every row, $n \le \mathrm{nnz}(\mathbf{A})$ and the count simplifies to $O(\mathrm{nnz}(\mathbf{A}))$.

Storage consists of one value and one column index per stored entry, together with $n + 1$ row pointers. Its size is also $O(\mathrm{nnz}(\mathbf{A}) + n)$. Empty rows explain why the row-pointer term cannot be discarded without an assumption. □

The CSR format is efficient for computing $\mathbf{A}\boldsymbol{x}$ but inconvenient for constructing a matrix, because the position of an entry in $\texttt{val}$ depends on how many entries precede it. Matrices are therefore usually assembled in COO format, where a new entry is simply appended, and converted to CSR once assembly is complete.

Thm 1.26 **(Conversion from COO to CSR)** A matrix given by t triplets can be converted to CSR format in $O(t + n)$ operations.

Proof We describe an explicit algorithm. The matrix has n rows and columns, and t is the number of input triplets, including duplicates.

1. Initialize n counters to zero. For each triplet (i, j, a), increment the counter for row i. Initialization costs $O(n)$ and the pass costs $O(t)$.

2. Starting with $\texttt{ptr}_1 = 1$, add the row counts successively. This prefix sum costs $O(n)$ and gives the start and end positions of every row in an intermediate array of length t.

3. Keep a next-free-position counter for each row. Pass over the input again and place the column and value of each triplet in its row's next free slot. Each placement and increment takes constant time. This pass costs $O(t + n)$ including counter initialization.

4. Use arrays of length n called `stamp` and `pos`. When processing row i, a column j has already appeared in that row exactly when $\texttt{stamp}_j = i$. If it has appeared, add the value to output slot $\texttt{pos}_j$. Otherwise append a new output entry and set $\texttt{stamp}_j = i$ and $\texttt{pos}_j$ to its slot. The row stamp avoids clearing a length-n array for each row. Each intermediate entry is examined once.

5. There are a fixed number of passes of length t, and a fixed number of arrays initialized or scanned in $O(n)$ time. The total is $O(t + n)$. The resulting CSR columns need not be sorted within a row. Requiring sorted column indices is an additional operation and is not included in this bound.

□

Example **(Assembling the Poisson matrix)** For the one-dimensional Poisson matrix, we append for each $i = 1, \dots, n$ the triplet $(i, i, 2h^{-2})$, together with $(i, i-1, -h^{-2})$ when $i > 1$ and $(i, i+1, -h^{-2})$ when $i < n$. The result has $3n-2$ triplets and converts to CSR in $O(n)$ operations. Finite-element matrices are assembled in the same way by looping over elements, and each element contributes a small block of triplets that overlaps the contributions of its neighbors. The overlapping triplets are summed during the conversion.

Remark 1.22 **(Column storage and transposes)** The compressed sparse column (CSC) format stores $\mathbf{A}$ column by column and is the CSR format of $\mathbf{A}^T$. Column-oriented algorithms, such as sparse Cholesky factorization and column-oriented triangular solves, use CSC. A product $\mathbf{A}^T\boldsymbol{x}$ with a CSR matrix is computed by scattering $x_i\,\texttt{val}_k$ into position $\texttt{col}_k$ of the result. The scatter has the same flop count as the CSR product, but parallel threads that process different rows may then write to the same output entry.

1.6.3 Further Exercises

Ex 1.43 Count the flops and the memory reads of the CSR product in Thm 1.25, assuming 8-byte values and 4-byte indices. Compute the ratio of flops to bytes read, and explain why this ratio makes the product limited by memory bandwidth.

Ex 1.44 Write the scatter algorithm for $\mathbf{A}^T\boldsymbol{x}$ from CSR arrays, and prove that it costs at most $2\,\mathrm{nnz}(\mathbf{A})$ flops.

Ex 1.45 Assemble the five-point Poisson matrix on an $N \times N$ grid in COO format, convert it to CSR, and compare the time of a sparse and a dense matrix–vector product for $N = 50, 100, 200$. Report nnz, the storage of both formats, and the two times.

2 Direct Methods for Linear Systems

A linear system may have one solution, many solutions, or none. We first determine which case occurs and how perturbations change a solution. Elimination then constructs triangular factors that can be reused for different right-hand sides. Symmetry, positive definiteness, and a small number of nonzero entries can reduce the work.

2.1 Linearity and Fundamental Subspaces

A linear system asks for coefficients that combine the columns of a matrix to produce a prescribed vector. The answer depends on two different questions. Is the prescribed vector in the span of the columns? If it is, can two different coefficient vectors produce it? The range and null space express these questions directly.

2.1.1 Linear Systems and Geometric Interpretations

A matrix equation $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ expresses the requirement that a data vector $\boldsymbol{b}$ be produced by a linear combination of the columns of $\mathbf{A}$, weighted by the unknown coordinates $\boldsymbol{x}$.

Def 2.1 **(Linear System)** A linear system of m equations in n unknowns is written in matrix-vector form as

$$\mathbf{A}\boldsymbol{x} = \boldsymbol{b},$$

where $\mathbf{A} \in \mathbb{R}^{m\times n}$ is the coefficient matrix, $\boldsymbol{x} \in \mathbb{R}^n$ is the unknown solution vector, and $\boldsymbol{b} \in \mathbb{R}^m$ is the right-hand side vector.

Ex 2.1 Consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Find every solution for $\boldsymbol{b} = (1, 2)^T$, then show that no solution exists for $\boldsymbol{b} = (1, 3)^T$. Interpret the two systems as pairs of lines. Which multiples of the columns can the matrix produce, and which nonzero vectors does it send to zero?

Remark 2.1 **(Row versus column perspectives)** There are two complementary geometric interpretations of $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$:

1. **Row view:** Each nonzero row $\boldsymbol{a}_i^T\boldsymbol{x} = b_i$ defines an affine hyperplane in $\mathbb{R}^n$. A zero row gives either no restriction or an inconsistent equation, depending on b_i. Solving the system means locating the mutual intersection of these m hyperplanes.

2. **Column view:** Partitioning $\mathbf{A} = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n]$ into its columns, the product evaluates to $\mathbf{A}\boldsymbol{x} = \sum_{j=1}^{n} x_j\boldsymbol{a}_j$. Solving the system means expressing the vector $\boldsymbol{b}$ as a linear combination of the column vectors of $\mathbf{A}$.

Thm 2.1 **(Superposition Principle)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ define a linear operator. If $\mathbf{A}\boldsymbol{x}_1 = \boldsymbol{b}_1$ and $\mathbf{A}\boldsymbol{x}_2 = \boldsymbol{b}_2$, then for any scalars $\alpha, \beta \in \mathbb{R}$,

$$\mathbf{A}(\alpha\boldsymbol{x}_1 + \beta\boldsymbol{x}_2) = \alpha\boldsymbol{b}_1 + \beta\boldsymbol{b}_2.$$

Consequently, the solution to a linear system with a composite source term decomposes into a linear combination of individual response vectors.

Ex 2.2 Prove superposition by distributing $\mathbf{A}$ over the linear combination. If $\mathbf{A}$ is singular and the two systems are consistent, explain why the conclusion still holds even though their solutions may not be unique.

2.1.2 Invertibility and the Four Fundamental Subspaces

The solvability and uniqueness of solutions are governed by four fundamental subspaces associated with $\mathbf{A}$.

Def 2.2 **(The Four Fundamental Subspaces)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$.

1. The **Column Space (Range)** $\mathcal{R}(\mathbf{A}) = \{\mathbf{A}\boldsymbol{x} : \boldsymbol{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$.
2. The **Null Space (Kernel)** $\mathcal{N}(\mathbf{A}) = \{\boldsymbol{x} \in \mathbb{R}^n : \mathbf{A}\boldsymbol{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$.
3. The **Row Space** $\mathcal{R}(\mathbf{A}^T) = \{\mathbf{A}^T\boldsymbol{y} : \boldsymbol{y} \in \mathbb{R}^m\} \subseteq \mathbb{R}^n$.
4. The **Left Null Space** $\mathcal{N}(\mathbf{A}^T) = \{\boldsymbol{y} \in \mathbb{R}^m : \mathbf{A}^T\boldsymbol{y} = \mathbf{0}\} \subseteq \mathbb{R}^m$.

The rank $r = \operatorname{rank}(\mathbf{A})$ is the dimension of the column space $\mathcal{R}(\mathbf{A})$.

Thm 2.2 **(The Rank-Nullity Theorem)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ with rank $r = \dim \mathcal{R}(\mathbf{A})$. Then

$$\operatorname{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = n.$$

Proof Let $k = \dim \mathcal{N}(\mathbf{A})$, and choose a basis $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$ for the null space. Extend this list to a basis $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ of $\mathbb{R}^n$. The basis-extension theorem permits this choice because the null space is a subspace of $\mathbb{R}^n$. The first k basis vectors are precisely those on which the matrix vanishes. We will show that the images of the other $n - k$ vectors form a basis of the range.

For any $\boldsymbol{x} \in \mathbb{R}^n$, expand $\boldsymbol{x} = \sum_{i=1}^n c_i \boldsymbol{v}_i$. Linearity gives

$$\mathbf{A}\boldsymbol{x} = \sum_{i=1}^{k} c_i \mathbf{A}\boldsymbol{v}_i + \sum_{i=k+1}^{n} c_i \mathbf{A}\boldsymbol{v}_i = \mathbf{0} + \sum_{i=k+1}^{n} c_i \mathbf{A}\boldsymbol{v}_i = \sum_{i=k+1}^{n} c_i \mathbf{A}\boldsymbol{v}_i.$$

Thus the vectors $\{\mathbf{A}\boldsymbol{v}_{k+1}, \dots, \mathbf{A}\boldsymbol{v}_n\}$ span $\mathcal{R}(\mathbf{A})$. Spanning alone would only bound the rank from above. We must also rule out dependence among these output vectors before concluding that their number equals the rank.

To establish linear independence, suppose $\sum_{i=k+1}^n d_i \mathbf{A}\boldsymbol{v}_i = \mathbf{0}$. Linearity yields

$$\mathbf{A}\left(\sum_{i=k+1}^{n} d_i \boldsymbol{v}_i\right) = \mathbf{0} \quad \Longrightarrow \quad \boldsymbol{w} = \sum_{i=k+1}^{n} d_i \boldsymbol{v}_i \in \mathcal{N}(\mathbf{A}).$$

Because $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$ is a basis for $\mathcal{N}(\mathbf{A})$, there exist coefficients $a_1, \dots, a_k$ such that $\boldsymbol{w} = \sum_{j=1}^k a_j \boldsymbol{v}_j$. Equating the two expressions gives

$$\sum_{j=1}^{k} a_j \boldsymbol{v}_j - \sum_{i=k+1}^{n} d_i \boldsymbol{v}_i = \mathbf{0}.$$

Since the full set $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ is linearly independent, every coefficient vanishes:

$$a_1 = \cdots = a_k = d_{k+1} = \cdots = d_n = 0.$$

Therefore $\{\mathbf{A}\boldsymbol{v}_{k+1}, \ldots, \mathbf{A}\boldsymbol{v}_n\}$ is a basis for $\mathcal{R}(\mathbf{A})$, containing exactly $n-k$ vectors. Hence

$$r = n - k = n - \dim \mathcal{N}(\mathbf{A}) \quad \Longrightarrow \quad \operatorname{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = n.$$

□

Thm 2.3 **(Fundamental Theorem of Linear Algebra and Orthogonal Decompositions)** For any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$:

1. The row space and null space are orthogonal complements in $\mathbb{R}^n$:

$$\mathcal{R}(\mathbf{A}^T) \perp \mathcal{N}(\mathbf{A}) \qquad \text{and} \qquad \mathbb{R}^n = \mathcal{R}(\mathbf{A}^T) \oplus \mathcal{N}(\mathbf{A}).$$

2. The column space and left null space are orthogonal complements in $\mathbb{R}^m$:

$$\mathcal{R}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A}^T) \qquad \text{and} \qquad \mathbb{R}^m = \mathcal{R}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^T).$$

Consequently, $\dim \mathcal{R}(\mathbf{A}^T) = \dim \mathcal{R}(\mathbf{A}) = \operatorname{rank}(\mathbf{A})$.

Proof Let the rows of $\mathbf{A}$ be $\boldsymbol{a}_1^T, \ldots, \boldsymbol{a}_m^T$. The equation $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ is equivalent to the m scalar equations $\boldsymbol{a}_i^T \boldsymbol{x} = 0$. Thus a vector belongs to the null space exactly when it is perpendicular to every row.

A vector perpendicular to each row is perpendicular to every linear combination of rows, because

$$\left(\sum_i c_i \boldsymbol{a}_i\right)^T \boldsymbol{x} = \sum_i c_i \boldsymbol{a}_i^T \boldsymbol{x} = 0.$$

Conversely, being perpendicular to the row span implies being perpendicular to each individual row. Since this span is $\mathcal{R}(\mathbf{A}^T)$,

$$\mathcal{N}(\mathbf{A}) = \mathcal{R}(\mathbf{A}^T)^{\perp}.$$

In a finite-dimensional inner-product space, $S^{\perp\perp} = S$: extend an orthonormal basis of S to one of the entire space, and the remaining basis vectors span $S^{\perp}$. Taking the complement again returns the original basis vectors. Hence

$$\mathcal{N}(\mathbf{A})^{\perp} = \mathcal{R}(\mathbf{A}^T).$$

The same calculation applied to the matrix $\mathbf{A}^T$ gives $\mathcal{N}(\mathbf{A}^T) = \mathcal{R}(\mathbf{A})^{\perp}$ and its equivalent reversed-complement identity.

The orthonormal-basis construction also proves the direct-sum decompositions and the dimension rule $\dim S + \dim S^{\perp} = n$ in $\mathbb{R}^n$. Therefore

$$\dim \mathcal{R}(\mathbf{A}^T) = n - \dim \mathcal{N}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}),$$

where the last equality is rank–nullity. Thus row rank equals column rank. □

Thm 2.4 **(Solvability and the Left Null Space)** The linear system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has a solution if and only if $\boldsymbol{b} \in \mathcal{R}(\mathbf{A})$. Equivalently, $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ is solvable if and only if $\boldsymbol{b}$ is orthogonal to every vector in the left null space of $\mathbf{A}$:

$$\mathbf{A}\boldsymbol{x} = \boldsymbol{b} \text{ has a solution } \iff \boldsymbol{y}^T\boldsymbol{b} = 0 \quad \text{for all } \boldsymbol{y} \text{ such that } \mathbf{A}^T\boldsymbol{y} = \mathbf{0}.$$

If a solution exists and $\boldsymbol{x}_p$ is one particular solution, the complete set of solutions is the affine space

$$\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{z}, \qquad \boldsymbol{z} \in \mathcal{N}(\mathbf{A}).$$

Proof Solvability means that $\boldsymbol{b}$ can be produced as an output $\mathbf{A}\boldsymbol{x}$. By the definition of the range, this is exactly $\boldsymbol{b} \in \mathcal{R}(\mathbf{A})$. The preceding theorem identifies that range with $\mathcal{N}(\mathbf{A}^T)^\perp$. Consequently

$$\mathbf{A}\boldsymbol{x} = \boldsymbol{b} \text{ is solvable} \quad \iff \quad \boldsymbol{y}^T\boldsymbol{b} = 0 \text{ for every } \boldsymbol{y} \text{ with } \mathbf{A}^T\boldsymbol{y} = \mathbf{0}.$$

One direction can also be checked directly: if $\boldsymbol{b} = \mathbf{A}\boldsymbol{x}$, then $\boldsymbol{y}^T\boldsymbol{b} = (\mathbf{A}^T\boldsymbol{y})^T\boldsymbol{x} = 0$. The subspace theorem supplies the converse, which is not obtained by this calculation alone.

Now suppose a solution $\boldsymbol{x}_p$ has been found. For any $\boldsymbol{z} \in \mathcal{N}(\mathbf{A})$,

$$\mathbf{A}(\boldsymbol{x}_p + \boldsymbol{z}) = \mathbf{A}\boldsymbol{x}_p + \mathbf{A}\boldsymbol{z} = \boldsymbol{b} + \mathbf{0} = \boldsymbol{b}.$$

Thus every vector in $\boldsymbol{x}_p + \mathcal{N}(\mathbf{A})$ is a solution.

To prove that these are all the solutions, take any $\boldsymbol{x}$ satisfying the equations and subtract the equations for $\boldsymbol{x}_p$:

$$\mathbf{A}(\boldsymbol{x} - \boldsymbol{x}_p) = \mathbf{A}\boldsymbol{x} - \mathbf{A}\boldsymbol{x}_p = \boldsymbol{b} - \boldsymbol{b} = \mathbf{0}.$$

Hence $\boldsymbol{z} = \boldsymbol{x} - \boldsymbol{x}_p$ belongs to the null space and $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{z}$. A trivial null space therefore gives uniqueness. A nonzero null-space vector gives distinct solutions by adding its scalar multiples. □

Ex 2.3 **The Fredholm alternative and inconsistent systems.** Consider the 3×2 system

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 3 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 1 \\ 2 \\ b_3 \end{pmatrix}.$$

1. Compute a basis for the left null space $\mathcal{N}(\mathbf{A}^T)$.
2. Apply Thm 2.4 to find the exact value of b_3 that makes the system consistent.
3. For $b_3 = 5$, find the orthogonal projection of $\boldsymbol{b}$ onto $\mathcal{R}(\mathbf{A})$ and compute the minimum residual norm $\min_{\boldsymbol{x}} \|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2$.

Remark 2.2 This compatibility condition is often called the finite-dimensional Fredholm alternative. Each left-null vector describes one obstruction that a solvable right-hand side must avoid.

2.1.3 Further Exercises

Ex 2.4 **Dimensions and bases for the four fundamental subspaces.** Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 7 \end{pmatrix}.$$

1. Compute the row reduced echelon form of $\mathbf{A}$. Determine the rank $r = \operatorname{rank}(\mathbf{A})$.
2. Find bases for $\mathcal{R}(\mathbf{A})$, $\mathcal{N}(\mathbf{A})$, $\mathcal{R}(\mathbf{A}^T)$, and $\mathcal{N}(\mathbf{A}^T)$. After studying QR factorization, return and replace them by orthonormal bases.
3. Confirm that $\dim \mathcal{R}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}^T) = 3$ and $\dim \mathcal{R}(\mathbf{A}^T) + \dim \mathcal{N}(\mathbf{A}) = 4$.
4. Verify numerically that every vector in your basis of $\mathcal{R}(\mathbf{A}^T)$ is orthogonal to every vector in your basis of $\mathcal{N}(\mathbf{A})$.

Ex 2.5 **Decomposition into row space and null space components.** Let $\mathbf{A} \in \mathbb{R}^{m \times n}$.

1. Prove that every vector $\boldsymbol{x} \in \mathbb{R}^n$ has a unique orthogonal decomposition $\boldsymbol{x} = \boldsymbol{x}_{\text{row}} + \boldsymbol{x}_{\text{null}}$ with $\boldsymbol{x}_{\text{row}} \in \mathcal{R}(\mathbf{A}^T)$ and $\boldsymbol{x}_{\text{null}} \in \mathcal{N}(\mathbf{A})$.
2. Show that $\mathbf{A}\boldsymbol{x} = \mathbf{A}\boldsymbol{x}_{\text{row}}$, demonstrating that the null-space component contributes nothing to the output.
3. Prove that among all solutions to a consistent underdetermined system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$, the unique solution of minimal Euclidean norm $\|\boldsymbol{x}\|_2$ lies strictly in the row space: $\boldsymbol{x}_{\min} \in \mathcal{R}(\mathbf{A}^T)$.

2.2 Conditioning and Numerical Stability

A computed vector can satisfy the equations closely and still be far from the exact solution. We first relate these two errors using the condition number. We then ask whether a computed vector exactly solves a nearby problem. The size of the required change in the data is its backward error.

Conditioning concerns the sensitivity of the mathematical problem. Backward stability concerns the algorithm. Keeping the two questions separate lets us determine whether lost accuracy comes from the data or from the computation.

2.2.1 Residual, Forward Error, and Backward Error

Let $\mathbf{A}$ be nonsingular, let $\boldsymbol{x}^*$ denote the exact solution, and let $\hat{\boldsymbol{x}}$ be an approximation. We can evaluate $\boldsymbol{b} - \mathbf{A}\hat{\boldsymbol{x}}$ using the available data. We cannot evaluate $\hat{\boldsymbol{x}} - \boldsymbol{x}^*$ unless we already know the solution. This distinction is why residuals are used to assess a computation.

Def 2.3 **(Forward Error, Residual, and Backward Error)** For a computed approximation $\hat{\boldsymbol{x}}$:

1. The **forward error** is the vector difference $\Delta\boldsymbol{x} = \hat{\boldsymbol{x}} - \boldsymbol{x}^*$, with relative measure $\frac{\|\hat{\boldsymbol{x}} - \boldsymbol{x}^*\|}{\|\boldsymbol{x}^*\|}$.
2. The **residual** is the vector defect $\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\hat{\boldsymbol{x}}$, with relative measure $\frac{\|\boldsymbol{r}\|}{\|\mathbf{A}\|\|\hat{\boldsymbol{x}}\| + \|\boldsymbol{b}\|}$.

3. The **backward error** is the smallest perturbation $(\delta\mathbf{A}, \delta\boldsymbol{b})$ such that $\hat{\boldsymbol{x}}$ is the exact solution to the perturbed system $(\mathbf{A} + \delta\mathbf{A})\hat{\boldsymbol{x}} = \boldsymbol{b} + \delta\boldsymbol{b}$.

A backward error is a scalar size assigned to allowable changes in the data, rather than the pair of perturbations itself. We must specify which data may change and how their changes are measured. The theorem on normwise backward error below uses one relative bound for both $\delta\mathbf{A}$ and $\delta\boldsymbol{b}$. Relative forward error requires $\boldsymbol{x}^* \neq \mathbf{0}$. If a denominator in a residual test vanishes, use an absolute error test instead.

Thm 2.5 **(The Fundamental Residual-Error Bound)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be nonsingular, let $\boldsymbol{b} \neq \mathbf{0}$, and let $\boldsymbol{x}^*$ satisfy $\mathbf{A}\boldsymbol{x}^* = \boldsymbol{b}$. For any vector $\hat{\boldsymbol{x}}$ with residual $\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\hat{\boldsymbol{x}}$, the relative forward error satisfies

$$\frac{1}{\kappa(\mathbf{A})}\frac{\|\boldsymbol{r}\|}{\|\boldsymbol{b}\|} \leq \frac{\|\hat{\boldsymbol{x}} - \boldsymbol{x}^*\|}{\|\boldsymbol{x}^*\|} \leq \kappa(\mathbf{A})\frac{\|\boldsymbol{r}\|}{\|\boldsymbol{b}\|},$$

where $\kappa(\mathbf{A}) = \|\mathbf{A}\|\|\mathbf{A}^{-1}\|$ is the condition number of $\mathbf{A}$.

Proof Let $\boldsymbol{e} = \hat{\boldsymbol{x}} - \boldsymbol{x}^*$. Subtract the exact and approximate matrix products:

$$\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\hat{\boldsymbol{x}} = \mathbf{A}\boldsymbol{x}^* - \mathbf{A}\hat{\boldsymbol{x}} = -\mathbf{A}\boldsymbol{e}.$$

Thus $\boldsymbol{e} = -\mathbf{A}^{-1}\boldsymbol{r}$. The sign has no effect on a norm.

For the upper error bound, consistency applied to this last identity gives $\|\boldsymbol{e}\| \leq \|\mathbf{A}^{-1}\|\|\boldsymbol{r}\|$. We need to normalize by the unknown exact solution norm. Since $\boldsymbol{b} = \mathbf{A}\boldsymbol{x}^*$,

$$\|\boldsymbol{b}\| \leq \|\mathbf{A}\|\|\boldsymbol{x}^*\|, \qquad \frac{1}{\|\boldsymbol{x}^*\|} \leq \frac{\|\mathbf{A}\|}{\|\boldsymbol{b}\|}.$$

The divisions are legitimate because $\boldsymbol{b} \neq \mathbf{0}$ and invertibility makes $\boldsymbol{x}^* \neq \mathbf{0}$. Multiply the two estimates:

$$\frac{\|\boldsymbol{e}\|}{\|\boldsymbol{x}^*\|} \leq \|\mathbf{A}^{-1}\|\|\boldsymbol{r}\|\frac{\|\mathbf{A}\|}{\|\boldsymbol{b}\|} = \kappa(\mathbf{A})\frac{\|\boldsymbol{r}\|}{\|\boldsymbol{b}\|}.$$

For the lower bound, use $\boldsymbol{r} = -\mathbf{A}\boldsymbol{e}$ to obtain $\|\boldsymbol{e}\| \geq \|\boldsymbol{r}\|/\|\mathbf{A}\|$. Also $\boldsymbol{x}^* = \mathbf{A}^{-1}\boldsymbol{b}$ gives $\|\boldsymbol{x}^*\| \leq \|\mathbf{A}^{-1}\|\|\boldsymbol{b}\|$. A lower bound for the numerator and an upper bound for its positive denominator give

$$\frac{\|\boldsymbol{e}\|}{\|\boldsymbol{x}^*\|} \geq \frac{\|\boldsymbol{r}\|/\|\mathbf{A}\|}{\|\mathbf{A}^{-1}\|\|\boldsymbol{b}\|} = \frac{1}{\kappa(\mathbf{A})}\frac{\|\boldsymbol{r}\|}{\|\boldsymbol{b}\|}.$$

Combining the upper and lower inequalities proves the statement. They are separate estimates, rather than one being obtained by reversing the other. □

Example **(A small residual does not guarantee a small error)** Consider the diagonal system

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 10^{-8} \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 1 \\ 10^{-8} \end{pmatrix}.$$

The exact solution is $\boldsymbol{x}^* = (1,1)^T$. Suppose an algorithm produces the approximation $\hat{\boldsymbol{x}} = (1,0)^T$. The residual evaluates to

$$\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\hat{\boldsymbol{x}} = \begin{pmatrix} 1 \\ 10^{-8} \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 10^{-8} \end{pmatrix},$$

which satisfies $\|\boldsymbol{r}\|_2/\|\boldsymbol{b}\|_2 \approx 10^{-8}$. However, the forward error is $\hat{\boldsymbol{x}} - \boldsymbol{x}^* = (0,-1)^T$, yielding a relative forward error of $\|\hat{\boldsymbol{x}} - \boldsymbol{x}^*\|_2/\|\boldsymbol{x}^*\|_2 = 1/\sqrt{2} \approx 0.707$ (a seventy percent error). The matrix has condition number $\kappa_2(\mathbf{A}) = 10^8$, which amplifies the small residual into an $O(1)$ error in the solution.

2.2.2 Sensitivity to Perturbations in the Data

Theorem 2.5 measures the error of a given approximation through its residual. A related question asks how far the exact solution moves when the data $\mathbf{A}$ and $\boldsymbol{b}$ are perturbed. The answer is again governed by the condition number $\kappa(\mathbf{A}) = \|\mathbf{A}\|\|\mathbf{A}^{-1}\|$ of Def 1.25. Perturbing $\mathbf{A}$ requires one preliminary fact, namely that a sufficiently small perturbation of a nonsingular matrix remains nonsingular.

Lem 2.6 **(Perturbation of the Identity)** Let $\|\cdot\|$ be an induced matrix norm and let $\mathbf{M}$ be a square matrix with $\|\mathbf{M}\| < 1$. Then $\mathbf{I} + \mathbf{M}$ is nonsingular and

$$\|(\mathbf{I} + \mathbf{M})^{-1}\| \leq \frac{1}{1 - \|\mathbf{M}\|}.$$

Proof Put $q = \|\mathbf{M}\|$, so $0 \leq q < 1$.

1. If $(\mathbf{I} + \mathbf{M})\boldsymbol{z} = \mathbf{0}$, then $\boldsymbol{z} = -\mathbf{M}\boldsymbol{z}$. Taking norms gives $\|\boldsymbol{z}\| \leq q\|\boldsymbol{z}\|$, or $(1-q)\|\boldsymbol{z}\| \leq 0$. Since $1 - q > 0$, we obtain $\boldsymbol{z} = \mathbf{0}$. A square matrix with trivial null space is nonsingular.

2. Given any $\boldsymbol{y}$, let $\boldsymbol{z} = (\mathbf{I} + \mathbf{M})^{-1}\boldsymbol{y}$. Rearrange $\boldsymbol{y} = \boldsymbol{z} + \mathbf{M}\boldsymbol{z}$ to obtain $\boldsymbol{z} = \boldsymbol{y} - \mathbf{M}\boldsymbol{z}$. Therefore

$$\|\boldsymbol{z}\| \leq \|\boldsymbol{y}\| + q\|\boldsymbol{z}\|, \qquad (1-q)\|\boldsymbol{z}\| \leq \|\boldsymbol{y}\|.$$

   For nonzero $\boldsymbol{y}$, division gives

$$\frac{\|(\mathbf{I} + \mathbf{M})^{-1}\boldsymbol{y}\|}{\|\boldsymbol{y}\|} \leq \frac{1}{1-q}.$$

   Taking the supremum over inputs proves the inverse-norm bound.

□

Thm 2.7 **(Perturbation Bound for Linear Systems)** Let $\mathbf{A}$ be nonsingular, let $\boldsymbol{b} \neq \mathbf{0}$, and let $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$. Suppose the perturbations $\delta\mathbf{A}$ and $\delta\boldsymbol{b}$ satisfy $\kappa(\mathbf{A})\|\delta\mathbf{A}\|/\|\mathbf{A}\| < 1$. Then $\mathbf{A} + \delta\mathbf{A}$ is nonsingular, and the solution of $(\mathbf{A} + \delta\mathbf{A})(\boldsymbol{x} + \delta\boldsymbol{x}) = \boldsymbol{b} + \delta\boldsymbol{b}$ satisfies

$$\frac{\|\delta\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq \frac{\kappa(\mathbf{A})}{1 - \kappa(\mathbf{A})\|\delta\mathbf{A}\|/\|\mathbf{A}\|} \left(\frac{\|\delta\mathbf{A}\|}{\|\mathbf{A}\|} + \frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{b}\|} \right).$$

Proof Write $q = \|\mathbf{A}^{-1}\|\|\delta\mathbf{A}\|$. The hypothesis is $q < 1$, because $q = \kappa(\mathbf{A})\|\delta\mathbf{A}\|/\|\mathbf{A}\|$.

Factor the perturbed matrix:

$$\mathbf{A} + \delta\mathbf{A} = \mathbf{A}(\mathbf{I} + \mathbf{A}^{-1}\delta\mathbf{A}).$$

Consistency gives $\|\mathbf{A}^{-1}\delta\mathbf{A}\| \le q < 1$. The inverse lemma makes the second factor nonsingular and bounds its inverse by $1/(1-q)$. The first factor is nonsingular by assumption, so the product is nonsingular as well.

Expand the perturbed equation before subtracting $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$:

$$\begin{aligned}\mathbf{A}\boldsymbol{x} + \mathbf{A}\delta\boldsymbol{x} + \delta\mathbf{A}\boldsymbol{x} + \delta\mathbf{A}\delta\boldsymbol{x} &= \boldsymbol{b} + \delta\boldsymbol{b},\\ (\mathbf{A} + \delta\mathbf{A})\delta\boldsymbol{x} &= \delta\boldsymbol{b} - \delta\mathbf{A}\boldsymbol{x}.\end{aligned}$$

Use the factorization of the inverse to solve this equation:

$$\delta\boldsymbol{x} = (\mathbf{I} + \mathbf{A}^{-1}\delta\mathbf{A})^{-1}\mathbf{A}^{-1}(\delta\boldsymbol{b} - \delta\mathbf{A}\boldsymbol{x}).$$

Taking norms, applying consistency, and then the triangle inequality yields

$$\|\delta\boldsymbol{x}\| \le \frac{\|\mathbf{A}^{-1}\|}{1-q}(\|\delta\boldsymbol{b}\| + \|\delta\mathbf{A}\|\|\boldsymbol{x}\|).$$

Since $\boldsymbol{b} \neq \mathbf{0}$, its exact solution is nonzero. Divide by $\|\boldsymbol{x}\|$. From $\|\boldsymbol{b}\| = \|\mathbf{A}\boldsymbol{x}\| \le \|\mathbf{A}\|\|\boldsymbol{x}\|$ we obtain

$$\frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{x}\|} \le \|\mathbf{A}\|\frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{b}\|}.$$

Substitute this estimate, factor out $\|\mathbf{A}^{-1}\|\|\mathbf{A}\| = \kappa(\mathbf{A})$, and replace q by its relative expression. This produces precisely the bound in the statement. Every denominator used is positive by the hypotheses. □

To first order, relative perturbations of size ε in $\mathbf{A}$ and in $\boldsymbol{b}$ change the solution by a relative amount of at most about $2\kappa(\mathbf{A})\varepsilon$. The hypothesis $\kappa(\mathbf{A})\|\delta\mathbf{A}\|/\|\mathbf{A}\| < 1$ cannot be dropped, because a larger perturbation may make $\mathbf{A} + \delta\mathbf{A}$ singular. In the Euclidean norm, the smallest perturbation that does so has $\|\delta\mathbf{A}\|_2 = \sigma_{\min}(\mathbf{A}) = \|\mathbf{A}\|_2/\kappa_2(\mathbf{A})$ by Thm 4.21. The reciprocal $1/\kappa_2(\mathbf{A})$ is therefore the relative distance from $\mathbf{A}$ to the nearest singular matrix.

2.2.3 Backward Error and Backward Stability

The perturbation bound describes the problem and makes no reference to an algorithm. To apply it to a computed solution $\hat{\boldsymbol{x}}$, we need perturbations $\delta\mathbf{A}$ and $\delta\boldsymbol{b}$ for which $\hat{\boldsymbol{x}}$ is the exact solution. The smallest such perturbations can be computed from the residual.

Thm 2.8 **(Rigal–Gaches Theorem on Normwise Backward Error)** In the Euclidean vector norm and its induced matrix norm, let $\hat{\boldsymbol{x}} \neq \mathbf{0}$ have residual $\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\hat{\boldsymbol{x}}$. The smallest relative perturbation size satisfying $(\mathbf{A} + \delta\mathbf{A})\hat{\boldsymbol{x}} = \boldsymbol{b} + \delta\boldsymbol{b}$ with $\|\delta\mathbf{A}\|_2 \le \eta\|\mathbf{A}\|_2$ and $\|\delta\boldsymbol{b}\|_2 \le \eta\|\boldsymbol{b}\|_2$ is

$$\eta(\hat{\boldsymbol{x}}) = \frac{\|\boldsymbol{r}\|_2}{\|\mathbf{A}\|_2\|\hat{\boldsymbol{x}}\|_2 + \|\boldsymbol{b}\|_2}.$$

An algorithm is **backward stable** if, for every nonsingular $\mathbf{A}$ and every $\boldsymbol{b}$, the computed solution satisfies $\eta(\hat{\boldsymbol{x}}) \le c_n u$, where u is the unit roundoff and c_n grows at most polynomially with n.

Rigal and Gaches proved this formula for the normwise backward error (Rigal and Gaches, 1967). The proof below constructs the minimizing perturbations explicitly.

Proof Let $d = \|\mathbf{A}\|_2\|\hat{\boldsymbol{x}}\|_2 + \|\boldsymbol{b}\|_2 > 0$. We first bound any admissible perturbations, then construct ones attaining the bound.

Rearrange the perturbed equation to get

$$\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\hat{\boldsymbol{x}} = \delta\mathbf{A}\hat{\boldsymbol{x}} - \delta\boldsymbol{b}.$$

If both relative perturbations are bounded by η, then

$$\begin{aligned}\|\boldsymbol{r}\|_2 &\le \|\delta\mathbf{A}\|_2\|\hat{\boldsymbol{x}}\|_2 + \|\delta\boldsymbol{b}\|_2 \\ &\le \eta\|\mathbf{A}\|_2\|\hat{\boldsymbol{x}}\|_2 + \eta\|\boldsymbol{b}\|_2 = \eta d.\end{aligned}$$

Thus every admissible value satisfies $\eta \ge \|\boldsymbol{r}\|_2/d$.

To attain this value, define the nonnegative weights

$$a = \frac{\|\mathbf{A}\|_2\|\hat{\boldsymbol{x}}\|_2}{d}, \qquad b = \frac{\|\boldsymbol{b}\|_2}{d}, \qquad a + b = 1,$$

and choose

$$\delta\mathbf{A} = a\frac{\boldsymbol{r}\hat{\boldsymbol{x}}^T}{\|\hat{\boldsymbol{x}}\|_2^2}, \qquad \delta\boldsymbol{b} = -b\boldsymbol{r}.$$

The assumption $\hat{\boldsymbol{x}} \ne \mathbf{0}$ makes this matrix well defined. Multiplying it by $\hat{\boldsymbol{x}}$ cancels the squared norm in its denominator, so $\delta\mathbf{A}\hat{\boldsymbol{x}} = a\boldsymbol{r}$. Consequently

$$\delta\mathbf{A}\hat{\boldsymbol{x}} - \delta\boldsymbol{b} = (a + b)\boldsymbol{r} = \boldsymbol{r}.$$

This is exactly the rearranged perturbed equation. Thus the constructed perturbations really make $\hat{\boldsymbol{x}}$ an exact solution.

For a rank-one matrix $\boldsymbol{u}\boldsymbol{v}^T$, consistency and Cauchy–Schwarz give norm at most $\|\boldsymbol{u}\|_2\|\boldsymbol{v}\|_2$, and a unit input along $\boldsymbol{v}$ attains it when both vectors are nonzero. Therefore

$$\begin{aligned}\|\delta\mathbf{A}\|_2 &= a\frac{\|\boldsymbol{r}\|_2}{\|\hat{\boldsymbol{x}}\|_2} = \frac{\|\boldsymbol{r}\|_2}{d}\|\mathbf{A}\|_2, \\ \|\delta\boldsymbol{b}\|_2 &= b\|\boldsymbol{r}\|_2 = \frac{\|\boldsymbol{r}\|_2}{d}\|\boldsymbol{b}\|_2.\end{aligned}$$

The same equalities hold when the residual is zero, by taking zero perturbations. Both perturbation bounds are attained at the lower bound for η, proving minimality. This is the normwise backward-error formula of Rigal and Gaches (1967). □

Cor 2.9 **(Forward Error of a Backward Stable Solution)** In the Euclidean norm, let $\hat{\boldsymbol{x}}$ have backward error $\eta = \eta(\hat{\boldsymbol{x}})$ as in Thm 2.8, and suppose $\kappa_2(\mathbf{A})\eta < 1$. Then

$$\frac{\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2} \le \frac{2\kappa_2(\mathbf{A})\eta}{1 - \kappa_2(\mathbf{A})\eta}.$$

Proof The backward-error theorem provides perturbations satisfying

$$(\mathbf{A}+\delta\mathbf{A})\hat{\boldsymbol{x}}=\boldsymbol{b}+\delta\boldsymbol{b},$$
$$\frac{\|\delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2}\le\eta,\qquad\qquad\frac{\|\delta\boldsymbol{b}\|_2}{\|\boldsymbol{b}\|_2}\le\eta.$$

Set $\delta\boldsymbol{x}=\hat{\boldsymbol{x}}-\boldsymbol{x}$. Then $\boldsymbol{x}+\delta\boldsymbol{x}=\hat{\boldsymbol{x}}$, so these are precisely the perturbed data and solution in the perturbation theorem.

Let $\kappa=\kappa_2(\mathbf{A})$. Its smallness hypothesis holds because

$$\kappa\frac{\|\delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2}\le\kappa\eta<1.$$

Apply that theorem and then replace each perturbation ratio by its upper bound:

$$\begin{aligned}\frac{\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2}&\le\frac{\kappa}{1-\kappa\|\delta\mathbf{A}\|_2/\|\mathbf{A}\|_2}\left(\frac{\|\delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2}+\frac{\|\delta\boldsymbol{b}\|_2}{\|\boldsymbol{b}\|_2}\right)\\&\le\frac{\kappa}{1-\kappa\eta}(\eta+\eta)=\frac{2\kappa\eta}{1-\kappa\eta}.\end{aligned}$$

The second step uses a lower bound for the positive denominator and upper bounds for the nonnegative numerator terms. This explains the direction of the inequality. □

Ex 2.6

1. Let $\mathbf{A}=\begin{pmatrix}1 & 1\\ 1 & 1+10^{-10}\end{pmatrix}$. Compute $\mathbf{A}^{-1}$ explicitly and evaluate $\kappa_\infty(\mathbf{A})$.
2. Suppose $\mathbf{A}\boldsymbol{x}=\boldsymbol{b}$ is solved using a backward stable algorithm in arithmetic with unit roundoff $u\approx10^{-16}$. If $\kappa_2(\mathbf{A})=10^{11}$, estimate the number of correct decimal digits in the computed solution.
3. Prove that for any two nonsingular matrices $\mathbf{A}$ and $\mathbf{B}$, $\kappa(\mathbf{AB})\le\kappa(\mathbf{A})\kappa(\mathbf{B})$.
4. Show that for an orthogonal matrix $\mathbf{Q}$, $\kappa_2(\mathbf{QA})=\kappa_2(\mathbf{A})$ for any square matrix $\mathbf{A}$.

For a backward stable algorithm, η is a modest multiple of u, and Cor 2.9 bounds the relative forward error by about $2\kappa_2(\mathbf{A})u$. In double precision, with $u\approx1.1\times10^{-16}$, a system with $\kappa_2(\mathbf{A})=10^{10}$ can therefore be solved to about five correct digits by a stable method, and this is a worst-case accuracy estimate, not a limit attained for every right-hand side. A large forward error with a small backward error indicates an ill-conditioned problem, not an unstable algorithm.

2.3 LU Factorization

Triangular systems can be solved one unknown at a time. Gaussian elimination transforms a general system into that form by subtracting multiples of rows. Recording those multipliers produces the LU factorization. The factors separate the work of eliminating the matrix from the cheaper work of solving for a particular right-hand side.

The algebraic construction also exposes the numerical difficulty. A small pivot can produce large multipliers, so practical elimination interchanges rows. We study the factorization first, then examine what pivoting controls and what rounding errors remain.

2.3.1 Gaussian Elimination

A triangular system can be solved one unknown at a time. In an upper triangular system $\mathbf{U}\boldsymbol{x} = \boldsymbol{y}$, the final equation contains only x_n. Once x_n is known, the preceding equation contains only x_{n-1}, and backward substitution continues until all components are resolved. Gaussian elimination systematically transforms a general system into this upper triangular form by subtracting multiples of each row from the rows beneath it.

Example Consider the coefficient matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 2 & 2 & 3 \end{pmatrix}.$$

Subtracting twice the first row from the second and once the first row from the third produces

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

The first column now has the zeros needed for an upper triangular matrix. One more row subtraction completes elimination. To solve a system, we apply each subtraction to the right-hand side as well. To reuse the calculation, we store the multipliers instead.

Ex 2.7 Complete this elimination. Collect the multipliers in the positions below the diagonal of a matrix $\mathbf{L}$ with diagonal entries one, and call the final upper triangular matrix $\mathbf{U}$. Multiply the two factors to verify $\mathbf{A} = \mathbf{LU}$. Why must $\mathbf{L}$ record the multipliers with their original signs rather than the minus signs used in the row subtractions?

Def 2.4 **(LU Factorization)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$. An LU factorization of $\mathbf{A}$ is a decomposition

$$\mathbf{A} = \mathbf{LU},$$

where $\mathbf{L} \in \mathbb{R}^{n\times n}$ is unit lower triangular ($\ell_{ii} = 1$ and $\ell_{ij} = 0$ for $j > i$) and $\mathbf{U} \in \mathbb{R}^{n\times n}$ is upper triangular ($u_{ij} = 0$ for $i > j$).

Lem 2.10 **(Algebra of Unit Lower Triangular Matrices)** The set of $n \times n$ unit lower triangular matrices forms a group under matrix multiplication. In particular:

1. The product of two unit lower triangular matrices is unit lower triangular.
2. The inverse of a unit lower triangular matrix is unit lower triangular.
3. The determinant of every unit lower triangular matrix equals one.

Ex 2.8 For $i < j$, show that every term in $(\mathbf{L}_1\mathbf{L}_2)_{ij}$ vanishes. Compute the diagonal. Next solve $\mathbf{L}\boldsymbol{x}_j = \boldsymbol{e}_j$ by forward substitution and show that $x_{ij} = 0$ for $i < j$ and $x_{jj} = 1$. Deduce the assertions about the inverse and determinant.

Thm 2.11 **(Elimination Matrices and the Structure of L)** Let $\mathbf{A}^{(0)} = \mathbf{A}$. At step k of Gaussian elimination, assuming the pivot entry $a_{kk}^{(k-1)} \neq 0$, the subdiagonal entries in column k are annihilated by applying the elementary elimination matrix

$$\mathbf{M}_k = \mathbf{I} - \boldsymbol{\ell}_k \boldsymbol{e}_k^T,$$

where $\boldsymbol{\ell}_k = (0, \ldots, 0, \ell_{k+1,k}, \ldots, \ell_{n,k})^T$ contains the multipliers $\ell_{ik} = a_{ik}^{(k-1)}/a_{kk}^{(k-1)}$. After $n-1$ elimination steps, the upper triangular factor is $\mathbf{U} = \mathbf{M}_{n-1} \ldots \mathbf{M}_1 \mathbf{A}$. The lower triangular factor is the inverse of the accumulated elimination operations:

$$\mathbf{L} = \mathbf{M}_1^{-1} \mathbf{M}_2^{-1} \ldots \mathbf{M}_{n-1}^{-1} = \mathbf{I} + \sum_{k=1}^{n-1} \boldsymbol{\ell}_k \boldsymbol{e}_k^T.$$

Consequently, the entries of $\mathbf{L}$ strictly below the main diagonal are precisely the multipliers computed during Gaussian elimination.

Proof The inverse of an elementary elimination matrix $\mathbf{M}_k = \mathbf{I} - \boldsymbol{\ell}_k \boldsymbol{e}_k^T$ is $\mathbf{M}_k^{-1} = \mathbf{I} + \boldsymbol{\ell}_k \boldsymbol{e}_k^T$. This identity follows from the fact that the first k entries of $\boldsymbol{\ell}_k$ are zero, which implies $\boldsymbol{e}_k^T \boldsymbol{\ell}_k = 0$ and therefore

$$(\mathbf{I} - \boldsymbol{\ell}_k \boldsymbol{e}_k^T)(\mathbf{I} + \boldsymbol{\ell}_k \boldsymbol{e}_k^T) = \mathbf{I} - \boldsymbol{\ell}_k \boldsymbol{e}_k^T + \boldsymbol{\ell}_k \boldsymbol{e}_k^T - \boldsymbol{\ell}_k (\boldsymbol{e}_k^T \boldsymbol{\ell}_k) \boldsymbol{e}_k^T = \mathbf{I}.$$

Writing $\mathbf{N}_k = \boldsymbol{\ell}_k \boldsymbol{e}_k^T$, the condition $\boldsymbol{e}_k^T \boldsymbol{\ell}_k = 0$ says $\mathbf{N}_k^2 = \mathbf{0}$. Hence both $(\mathbf{I} - \mathbf{N}_k)(\mathbf{I} + \mathbf{N}_k)$ and the product in the reverse order equal the identity. This verifies both sides of the proposed inverse.

Next, consider the product of two successive inverse matrices $\mathbf{M}_j^{-1} \mathbf{M}_k^{-1}$ with $j < k$. Because $\boldsymbol{e}_j^T \boldsymbol{\ell}_k = 0$, the cross term vanishes:

$$(\mathbf{I} + \boldsymbol{\ell}_j \boldsymbol{e}_j^T)(\mathbf{I} + \boldsymbol{\ell}_k \boldsymbol{e}_k^T) = \mathbf{I} + \boldsymbol{\ell}_j \boldsymbol{e}_j^T + \boldsymbol{\ell}_k \boldsymbol{e}_k^T + \boldsymbol{\ell}_j (\boldsymbol{e}_j^T \boldsymbol{\ell}_k) \boldsymbol{e}_k^T = \mathbf{I} + \boldsymbol{\ell}_j \boldsymbol{e}_j^T + \boldsymbol{\ell}_k \boldsymbol{e}_k^T.$$

By induction, the product $\mathbf{M}_1^{-1} \ldots \mathbf{M}_{n-1}^{-1}$ simply places each column of multipliers into column k of the identity matrix without modifying any previously placed multipliers. Multiplying $\mathbf{U} = \mathbf{M}_{n-1} \ldots \mathbf{M}_1 \mathbf{A}$ on the left by $\mathbf{L}$ yields $\mathbf{A} = \mathbf{L}\mathbf{U}$.

The ordering matters. Multiplying the final upper matrix on the left successively by $\mathbf{M}_{n-1}^{-1}, \ldots, \mathbf{M}_1^{-1}$ undoes the row operations in reverse order. Grouping the inverses as $\mathbf{L} = \mathbf{M}_1^{-1} \cdots \mathbf{M}_{n-1}^{-1}$ therefore gives $\mathbf{L}\mathbf{U} = \mathbf{A}$. The vanishing cross terms show that these inverse operations record precisely the multipliers in the lower triangle. □

Thm 2.12 **(Existence and Uniqueness of the LU Factorization)** A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ has a unique LU factorization with unit lower triangular $\mathbf{L}$ and nonsingular upper triangular $\mathbf{U}$ if and only if every leading principal submatrix $\mathbf{A}_k = \mathbf{A}[1:k, 1:k]$ is nonsingular for each $k = 1, \ldots, n$.

Proof We proceed by induction on the matrix dimension n. For $n = 1$, $\mathbf{A} = (a_{11})$. Because $\det(\mathbf{A}_1) = a_{11} \neq 0$, the unique factorization is $\mathbf{L} = (1)$ and $\mathbf{U} = (a_{11})$. Assume the result holds for matrices of dimension $n-1$. Partition the $n \times n$ matrix $\mathbf{A}$ into blocks:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{n-1} & \boldsymbol{c} \\ \boldsymbol{d}^T & a_{nn} \end{pmatrix},$$

where $\mathbf{A}_{n-1} \in \mathbb{R}^{(n-1)\times(n-1)}$, $\boldsymbol{c}, \boldsymbol{d} \in \mathbb{R}^{n-1}$, and $a_{nn} \in \mathbb{R}$. Every leading principal block of $\mathbf{A}_{n-1}$ is also a leading principal block of $\mathbf{A}$, so it satisfies the theorem's hypotheses. This verifies the condition needed to invoke the induction hypothesis on the smaller matrix.

By the induction hypothesis, $\mathbf{A}_{n-1}$ has a unique factorization $\mathbf{A}_{n-1} = \mathbf{L}_{n-1}\mathbf{U}_{n-1}$ with $\mathbf{L}_{n-1}$ unit lower triangular and $\mathbf{U}_{n-1}$ nonsingular. We seek block factors

$$\mathbf{L} = \begin{pmatrix} \mathbf{L}_{n-1} & \mathbf{0} \\ \boldsymbol{\ell}^T & 1 \end{pmatrix}, \qquad \mathbf{U} = \begin{pmatrix} \mathbf{U}_{n-1} & \boldsymbol{u} \\ \mathbf{0}^T & u_{nn} \end{pmatrix}.$$

The zero lower-left block of $\mathbf{U}$ and zero upper-right block of $\mathbf{L}$ are required by triangularity. The diagonal one in $\mathbf{L}$ is required by the unit-diagonal convention. Only the new vectors and the final scalar remain undetermined.

Multiplying $\mathbf{L}$ and $\mathbf{U}$ and equating block by block gives:

$$\mathbf{L}_{n-1}\boldsymbol{u} = \boldsymbol{c}, \qquad \boldsymbol{\ell}^T\mathbf{U}_{n-1} = \boldsymbol{d}^T, \qquad \boldsymbol{\ell}^T\boldsymbol{u} + u_{nn} = a_{nn}.$$

Because $\mathbf{L}_{n-1}$ is unit lower triangular (hence invertible), $\boldsymbol{u} = \mathbf{L}_{n-1}^{-1}\boldsymbol{c}$ is uniquely determined. Because $\mathbf{U}_{n-1}$ is nonsingular, $\boldsymbol{\ell}^T = \boldsymbol{d}^T\mathbf{U}_{n-1}^{-1}$ is uniquely determined. Finally, the scalar diagonal entry is uniquely determined by $u_{nn} = a_{nn} - \boldsymbol{\ell}^T\boldsymbol{u}$. Taking determinants of the block decomposition gives

$$\det(\mathbf{A}) = \det(\mathbf{L})\det(\mathbf{U}) = (1 \cdot \det(\mathbf{L}_{n-1}))(\det(\mathbf{U}_{n-1}) \cdot u_{nn}) = \det(\mathbf{A}_{n-1})u_{nn}.$$

Because $\det(\mathbf{A}) \neq 0$ and $\det(\mathbf{A}_{n-1}) \neq 0$, we have $u_{nn} = \det(\mathbf{A})/\det(\mathbf{A}_{n-1}) \neq 0$, which ensures that $\mathbf{U}$ is nonsingular. This completes the induction.

For necessity, if $\mathbf{A} = \mathbf{LU}$ with $\mathbf{U}$ nonsingular, its leading block satisfies $\mathbf{A}_k = \mathbf{L}_k\mathbf{U}_k$. Both triangular factors are nonsingular, so every $\mathbf{A}_k$ is nonsingular. This proves the converse as well.

For the converse, triangularity ensures that the first k rows and columns of the product involve only the leading factors: no omitted trailing index contributes. The diagonal of $\mathbf{U}$ is nonzero because its determinant is their product and $\mathbf{U}$ is nonsingular. Therefore every leading triangular block $\mathbf{U}_k$ is nonsingular. The unit lower block $\mathbf{L}_k$ is nonsingular as well, proving every leading block of $\mathbf{A}$ is nonsingular. □

2.3.2 Triangular Solves and Computational Complexity

Def 2.5 **(Forward and Backward Substitution)** Let $\mathbf{L} \in \mathbb{R}^{n\times n}$ be unit lower triangular and let $\mathbf{U} \in \mathbb{R}^{n\times n}$ be nonsingular upper triangular. Solving $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ decouples into two triangular steps:

1. **Forward substitution ($\mathbf{L}\boldsymbol{y} = \boldsymbol{b}$):**

$$y_1 = b_1, \qquad y_i = b_i - \sum_{j=1}^{i-1} \ell_{ij}y_j \quad (i = 2, \ldots, n).$$

2. **Backward substitution ($\mathbf{U}\boldsymbol{x} = \boldsymbol{y}$):**

$$x_n = \frac{y_n}{u_{nn}}, \qquad x_i = \frac{1}{u_{ii}}\left(y_i - \sum_{j=i+1}^{n} u_{ij}x_j\right) \quad (i = n-1, \ldots, 1).$$

Thm 2.13 **(Computational Operation Count of LU Factorization)** For a general dense matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, computing the LU factorization requires asymptotically $\frac{2}{3}n^3$ floating-point operations. Evaluating the forward and backward triangular substitution steps requires $2n^2 - n$ flops for each right-hand side vector $\boldsymbol{b}$.

Proof At step k of elimination, calculating the $n-k$ multipliers $\ell_{ik} = a_{ik}/a_{kk}$ requires $n-k$ divisions. Updating the active $(n-k)\times(n-k)$ submatrix via $a_{ij} \leftarrow a_{ij} - \ell_{ik}a_{kj}$ requires one multiplication

and one subtraction for each entry, totaling $2(n-k)^2$ operations. Summing over all $n-1$ elimination steps gives

$$\begin{aligned}\sum_{k=1}^{n-1}\Big[(n-k)+2(n-k)^2\Big] &= 2\sum_{m=1}^{n-1} m^2 + \sum_{m=1}^{n-1} m \\ &= \frac{(n-1)n(2n-1)}{3} + \frac{(n-1)n}{2} \\ &= \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n.\end{aligned}$$

The summation index $m = n-k$ is the active trailing size. Expanding $(n-1)n(2n-1)/3$ gives $(2n^3-3n^2+n)/3$, and adding $(n^2-n)/2$ gives the stated polynomial. Its leading coefficient is $2/3$, while the remaining terms have smaller order.

For forward substitution with unit diagonal, row i requires $i-1$ multiplications and $i-1$ subtractions, totaling $\sum_{i=1}^{n} 2(i-1) = n^2 - n$ flops. For backward substitution, row i requires $n-i$ multiplications, $n-i$ subtractions, and one division, totaling $\sum_{i=1}^{n}(2(n-i)+1) = n^2$ flops. The combined solve phase requires $2n^2 - n \approx 2n^2$ flops.

The factors can be reused for every new right-hand side. The quadratic solve count is therefore paid once per right-hand side, whereas the cubic factorization count is paid only when the coefficient matrix must be factored again. This is the distinction behind the complexity claim. □

2.3.3 Partial Pivoting and the Growth Factor

In floating-point arithmetic, unpivoted Gaussian elimination fails if a pivot is zero, and it can be unstable when a pivot is small relative to the subdiagonal entries. Dividing by a small pivot creates large multipliers and can amplify rounding errors. Partial pivoting interchanges rows so that the pivot has maximal absolute value in its column. It keeps every multiplier at most one in magnitude and is usually stable, but it does not guarantee a small growth factor. The Wilkinson example below shows the exceptional worst case.

Def 2.6 **(PA = LU Factorization with Partial Pivoting)** Gaussian elimination with partial pivoting produces the decomposition

$$\mathbf{PA} = \mathbf{LU},$$

where $\mathbf{P} \in \mathbb{R}^{n\times n}$ is a permutation matrix recording row swaps, $\mathbf{L} \in \mathbb{R}^{n\times n}$ is unit lower triangular with all subdiagonal entries bounded by $|\ell_{ij}| \le 1$, and $\mathbf{U} \in \mathbb{R}^{n\times n}$ is upper triangular.

Ex 2.9 **Hand execution of unpivoted versus pivoted LU.** Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 10^{-4} & 1 \\ 1 & 1 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

1. Compute the unpivoted LU factorization $\mathbf{A} = \mathbf{LU}$ in exact arithmetic, and evaluate the multiplier ℓ_{21}.

2. Simulate standard 3-digit floating-point arithmetic with rounding. Show that unpivoted elimination computes the erroneous solution $\hat{\boldsymbol{x}} = (0,1)^T$.

3. Compute the pivoted factorization $\mathbf{PA} = \mathbf{LU}$ using partial pivoting. Verify that in 3-digit arithmetic, the computed solution recovers $(1,1)^T$ with negligible error.

Thm 2.14 **(Wilkinson Growth Factor Bound for Partial Pivoting)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$. The growth factor ρ of Gaussian elimination is defined by

$$\rho = \frac{\max_{i,j,k} |a_{ij}^{(k)}|}{\max_{i,j} |a_{ij}|},$$

where $a_{ij}^{(k)}$ denotes the entries of the active submatrix after step k. For Gaussian elimination with partial pivoting, the growth factor satisfies

$$\rho \leq 2^{n-1}.$$

Proof Let $M_k = \max_{i,j} |a_{ij}^{(k)}|$ denote the maximum absolute entry of the active matrix after k elimination steps, with $M_0 = \max_{i,j} |a_{ij}|$. At step k, partial pivoting guarantees that the pivot $|a_{kk}^{(k-1)}| = \max_{i\geq k} |a_{ik}^{(k-1)}|$, which ensures that all multipliers satisfy

$$|\ell_{ik}| = \left| \frac{a_{ik}^{(k-1)}}{a_{kk}^{(k-1)}} \right| \leq 1 \qquad \text{for all } i = k+1, \ldots, n.$$

Swapping rows changes no entry magnitudes. Thus the only possible growth in this step comes from the subtraction update, and the pivot comparison bounds the multiplier used in each such subtraction.

The elimination update rule is $a_{ij}^{(k)} = a_{ij}^{(k-1)} - \ell_{ik} a_{kj}^{(k-1)}$. Applying the triangle inequality yields

$$|a_{ij}^{(k)}| \leq |a_{ij}^{(k-1)}| + |\ell_{ik}||a_{kj}^{(k-1)}| \leq M_{k-1} + (1)M_{k-1} = 2M_{k-1}.$$

Taking the maximum over all active indices gives $M_k \leq 2M_{k-1}$. By induction, $M_{n-1} \leq 2^{n-1} M_0$. Dividing by M_0 establishes $\rho \leq 2^{n-1}$.

For every intermediate step k, the same induction gives $M_k \leq 2^k M_0 \leq 2^{n-1} M_0$. Pivot rows already retained in $\mathbf{U}$ were bounded when they left the active matrix. Hence the bound applies to the maximum across all stages, which is the maximum in the growth-factor definition, not only to the final active entry. □

Ex 2.10 **The Wilkinson counterexample and exponential growth.** Consider the $n \times n$ matrix

$$\mathbf{W}_n = \begin{pmatrix} 1 & 0 & 0 & \ldots & 0 & 1 \\ -1 & 1 & 0 & \ldots & 0 & 1 \\ -1 & -1 & 1 & \ldots & 0 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -1 & -1 & \ldots & 1 & 1 \\ -1 & -1 & -1 & \ldots & -1 & 1 \end{pmatrix}.$$

1. Assume that ties in pivot selection keep the current row. Show that partial pivoting performs no row swaps on $\mathbf{W}_n$ because every diagonal entry already has absolute value 1, matching the maximum in its column.

2. Compute the entries of the final column after each elimination step $k = 1, 2, \ldots, n-1$.

3. Prove that the final bottom-right entry grows as $u_{nn} = 2^{n-1}$, confirming that the bound in Thm 2.14 is tight.

The growth factor matters because it controls the rounding errors committed during elimination. The next theorem expresses these errors as a perturbation of $\mathbf{A}$. It uses the unit roundoff u of Def 1.27 and the constant $\gamma_k = ku/(1-ku)$, defined whenever $ku < 1$. Absolute values and inequalities between matrices are taken entrywise.

Thm 2.15 **(Backward Error of Gaussian Elimination)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be nonsingular, and suppose Gaussian elimination with partial pivoting runs to completion in floating-point arithmetic. Assume $3nu < 1$, where u is the unit roundoff, and exclude overflow and underflow from the rounding model. Let $\mathbf{P}$ be the computed permutation, let $\hat{\mathbf{L}}$ and $\hat{\mathbf{U}}$ be the computed factors, and let $\hat{\boldsymbol{x}}$ be the solution of $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ computed by forward and backward substitution. Then

$$(\mathbf{A} + \Delta\mathbf{A})\hat{\boldsymbol{x}} = \boldsymbol{b}, \qquad |\mathbf{P}\,\Delta\mathbf{A}| \le \gamma_{3n}|\hat{\mathbf{L}}||\hat{\mathbf{U}}|.$$

Proof We use the floating-point model and assume that underflow and overflow do not invalidate it. Absolute values and inequalities between matrices below are entrywise. The standard componentwise analysis is developed in Higham (2002).

We first record how repeated scalar rounding is bounded. If $|\delta_j| \le u$ and $\rho_j = \pm 1$, each factor $(1+\delta_j)^{\rho_j}$ lies between $1-u$ and $(1-u)^{-1}$. A product of at most s factors therefore lies between $(1-u)^s$ and $(1-u)^{-s}$. Bernoulli's inequality gives $(1-u)^s \ge 1-su$. If $su < 1$, its reciprocal is at most $(1-su)^{-1} = 1+\gamma_s$. Thus the product can be written $1+\theta_s$ with $|\theta_s| \le \gamma_s$.

To apply this bound to a row calculation, consider computing

$$x = \frac{a - \sum_{k=1}^{j-1} b_k c_k}{d}.$$

Let a multiplication have error μ_k, each successive subtraction error δ_k, and the final division error δ_d. Expanding the successive computed subtractions and moving their factors to the other side gives an exact identity of the form

$$a = d\hat{x}(1+\theta_0) + \sum_{k=1}^{j-1} b_k c_k (1+\theta_k), \qquad |\theta_k| \le \gamma_j.$$

Here the coefficient of $d\hat{x}$ is $[(1+\delta_d)\prod_{l=1}^{j-1}(1+\delta_l)]^{-1}$. The coefficient of $b_k c_k$ is $(1+\mu_k)/\prod_{l=1}^{k-1}(1+\delta_l)$. Each involves at most j rounding factors or inverse factors, so the preceding product bound applies. Omitting the division gives the same bound for a computed upper-factor entry.

For elimination, these are exactly the scalar recurrences determining entries of the computed factors. Applying the identity to entry (i,j) of their product, with at most n factors, gives

$$\hat{\mathbf{L}}\hat{\mathbf{U}} = \mathbf{P}\mathbf{A} + \mathbf{E}, \qquad |\mathbf{E}| \le \gamma_n |\hat{\mathbf{L}}||\hat{\mathbf{U}}|.$$

To see the entrywise estimate, subtract the unweighted sum $\sum_k \hat{\ell}_{ik}\hat{u}_{kj}$ from the weighted exact row identity. The error is a sum of these same products times coefficients of magnitude at most γ_n. The triangle inequality gives the displayed nonnegative matrix product. Row permutations only relabel entries and introduce no rounding.

Apply the same scalar calculation to forward and backward substitution. It produces triangular perturbations satisfying

$$(\hat{\mathbf{L}} + \Delta\mathbf{L})\hat{\boldsymbol{y}} = \mathbf{P}\boldsymbol{b}, \qquad |\Delta\mathbf{L}| \le \gamma_n|\hat{\mathbf{L}}|,$$
$$(\hat{\mathbf{U}} + \Delta\mathbf{U})\hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}, \qquad |\Delta\mathbf{U}| \le \gamma_n|\hat{\mathbf{U}}|.$$

Eliminate $\hat{\boldsymbol{y}}$ by substituting the second equality into the first. The coefficient matrix of the resulting system is

$$\mathbf{PA} + \mathbf{E} + \hat{\mathbf{L}}\Delta\mathbf{U} + \Delta\mathbf{L}\hat{\mathbf{U}} + \Delta\mathbf{L}\Delta\mathbf{U}.$$

Define $\mathbf{P}\Delta\mathbf{A}$ to be the sum of the four perturbation terms. Then $(\mathbf{A} + \Delta\mathbf{A})\hat{\boldsymbol{x}} = \boldsymbol{b}$, and the triangle inequality gives

$$|\mathbf{P}\Delta\mathbf{A}| \le (3\gamma_n + \gamma_n^2)|\hat{\mathbf{L}}||\hat{\mathbf{U}}|.$$

For $3nu < 1$, $1 + 3\gamma_n + \gamma_n^2 \le (1 + \gamma_n)^3 = (1 - nu)^{-3} \le (1 - 3nu)^{-1} = 1 + \gamma_{3n}$. The penultimate inequality follows from $(1 - nu)^3 \ge 1 - 3nu$. Subtracting one proves the stated bound with γ_{3n}. □

Cor 2.16 **(Normwise Backward Error and the Growth Factor)** Under the hypotheses of Thm 2.15, let ρ be the growth factor of the computed elimination. Then

$$\frac{\|\Delta\mathbf{A}\|_\infty}{\|\mathbf{A}\|_\infty} \le n^2\gamma_{3n}\,\rho.$$

Proof Let $a_{\max} = \max_{i,j}|a_{ij}|$. Partial pivoting bounds each computed multiplier in magnitude by one under the usual round-to-nearest model. The diagonal entries of $\hat{\mathbf{L}}$ are one, so every entry of $|\hat{\mathbf{L}}|$ is at most one. By the definition of growth, every entry of $|\hat{\mathbf{U}}|$ is at most $\rho a_{\max}$.

Entry (i, j) of their product is a sum of n nonnegative terms. Each term is at most $\rho a_{\max}$, giving

$$(|\hat{\mathbf{L}}||\hat{\mathbf{U}}|)_{ij} = \sum_{k=1}^{n} |\hat{\ell}_{ik}||\hat{u}_{kj}| \le n\rho a_{\max}.$$

The preceding componentwise theorem therefore bounds each entry of $|\mathbf{P}\Delta\mathbf{A}|$ by $\gamma_{3n}n\rho a_{\max}$.

The infinity norm is a maximum row sum. Each row contains n entries, so

$$\|\mathbf{P}\Delta\mathbf{A}\|_\infty \le n(\gamma_{3n}n\rho a_{\max}) = n^2\gamma_{3n}\rho a_{\max}.$$

A row permutation changes only the order of row sums, not their maximum. Thus $\|\mathbf{P}\Delta\mathbf{A}\|_\infty = \|\Delta\mathbf{A}\|_\infty$. Finally $a_{\max} \le \|\mathbf{A}\|_\infty$, because the row containing a largest entry has a row sum at least that large. Divide by $\|\mathbf{A}\|_\infty > 0$ to obtain the claimed relative estimate. □

Corollary 2.16 shows that partial pivoting is backward stable whenever the growth factor is modest, and Cor 2.9 then bounds the forward error by the condition number. The worst-case bound $\rho \le 2^{n-1}$ of Thm 2.14 is attained by special matrices such as the Wilkinson matrix in the exercises below, but such growth is rarely observed. For random matrices, Trefethen and Schreiber measured average growth factors of about $n^{2/3}$ for partial pivoting (Trefethen and Schreiber, 1990). The factor n^2 in the corollary is also pessimistic, since the bound assumes that every rounding error takes its worst possible value and sign.

2.3.4 Iterative Refinement

Theorem 2.15 bounds the backward error of the computed solution, but the forward error can still be as large as $\kappa(\mathbf{A})$ times that backward error. Iterative refinement reuses the computed factors to reduce the forward error. Each step costs one residual evaluation and one pair of triangular solves, which is small compared with the factorization.

Def 2.7 **(Iterative Refinement)** Let $\hat{\boldsymbol{x}}^{(0)}$ be computed from the factorization $\mathbf{PA} \approx \hat{\mathbf{L}}\hat{\mathbf{U}}$. For $k = 0, 1, 2, \ldots$:

1. Compute the residual $\boldsymbol{r}^{(k)} = \boldsymbol{b} - \mathbf{A}\hat{\boldsymbol{x}}^{(k)}$, if possible in higher precision.
2. Solve the correction equation $\mathbf{A}\boldsymbol{d}^{(k)} = \boldsymbol{r}^{(k)}$ with the existing factors.
3. Update $\hat{\boldsymbol{x}}^{(k+1)} = \hat{\boldsymbol{x}}^{(k)} + \boldsymbol{d}^{(k)}$.

Refinement improves the forward error when the factors are accurate enough to give a useful correction and the residual is computed more accurately than the factorization. The precise convergence conditions depend on $\kappa(\mathbf{A})$ and on the precisions used for the factorization, the residual, and the update (Higham, 2002).

2.3.5 Further Exercises

Ex 2.11 **Banded matrices and fill-in.** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be a tridiagonal matrix ($a_{ij} = 0$ for $|i - j| > 1$).

1. Prove that without pivoting, the factors $\mathbf{L}$ and $\mathbf{U}$ retain the tridiagonal sparsity structure, meaning $\mathbf{L}$ has bandwidth 1 on the subdiagonal and $\mathbf{U}$ has bandwidth 1 on the superdiagonal.
2. Calculate the flop count for unpivoted tridiagonal LU factorization and forward/backward substitution. Show that the total work is $O(n)$ instead of $O(n^3)$.
3. Explain what happens to the upper bandwidth of $\mathbf{U}$ when partial pivoting performs row interchanges.

Ex 2.12 **Block LU decomposition and the Schur complement.** Partition a nonsingular matrix $\mathbf{A}$ into blocks

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix},$$

where $\mathbf{A}_{11} \in \mathbb{R}^{k\times k}$ is nonsingular.

1. Show that block Gaussian elimination factors $\mathbf{A}$ as

$$\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21}\mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{S} \end{pmatrix},$$

   where $\mathbf{S} = \mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}$ is the Schur complement of $\mathbf{A}_{11}$ in $\mathbf{A}$.
2. Prove that $\det(\mathbf{A}) = \det(\mathbf{A}_{11})\det(\mathbf{S})$.
3. Relate the Schur complement $\mathbf{S}$ to the active submatrix remaining after k steps of standard Gaussian elimination.

Ex 2.13 **Determinants and multiple right-hand sides.** Let $\mathbf{PA} = \mathbf{LU}$ be the computed LU factorization with partial pivoting.

1. Prove that $\det(\mathbf{A}) = (-1)^s \prod_{i=1}^{n} u_{ii}$, where s is the number of row interchanges recorded by $\mathbf{P}$.
2. Explain why computing the determinant via $\mathbf{PA} = \mathbf{LU}$ requires $O(n^3)$ flops, whereas evaluating the Leibniz expansion formula requires $O(n!)$ flops.
3. For m distinct right-hand side vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_m$, analyze the computational cost of solving $\mathbf{A}\boldsymbol{x}_j = \boldsymbol{b}_j$ by reusing the single factorization $\mathbf{PA} = \mathbf{LU}$ versus computing m independent solves from scratch.

2.4 Symmetric Positive Definite Matrices

For some symmetric matrices, the scalar $\boldsymbol{x}^T\mathbf{A}\boldsymbol{x}$ is positive for every nonzero vector. This property will permit a factorization with only one triangular factor and no row interchanges. We first examine what the scalar measures, then construct the factorization by eliminating one variable at a time.

2.4.1 Positive Quadratic Energy

The expression $\boldsymbol{x}^T\mathbf{A}\boldsymbol{x}$ is called a *quadratic form*. For a symmetric matrix of order two, it is

$$\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2.$$

Positive diagonal entries alone do not ensure positivity of this expression, because the mixed term can have either sign. We require positivity in every direction.

Ex 2.14 Expand the quadratic forms for $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. For the first, express the result as a sum of positive multiples of squares. For the second, find a vector that makes the result negative. Both matrices have positive diagonal entries. What distinguishes them?

Def 2.8 **(Symmetric Positive Definite Matrix)** A real matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is **symmetric positive definite**, abbreviated SPD, if $\mathbf{A} = \mathbf{A}^T$ and

$$\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} > 0 \qquad \text{for every } \boldsymbol{x} \neq \mathbf{0}.$$

It is **symmetric positive semidefinite** if the strict inequality is replaced by $\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} \geq 0$.

Thm 2.17 **(First Consequences of Positive Definiteness)** If $\mathbf{A}$ is SPD, then every diagonal entry is positive, every principal submatrix is SPD, and $\mathbf{A}$ is nonsingular.

Proof We test positive definiteness with vectors chosen to isolate the desired entries.

1. The coordinate vector $\boldsymbol{e}_i$ is nonzero. Therefore

$$a_{ii} = \boldsymbol{e}_i^T\mathbf{A}\boldsymbol{e}_i > 0.$$

2. Let $I = (i_1, \dots, i_k)$ be a set of selected indices, and let $\mathbf{B} = \mathbf{A}_{I,I}$. Given nonzero $\boldsymbol{y} \in \mathbb{R}^k$, form $\boldsymbol{x} \in \mathbb{R}^n$ by placing y_j in position i_j and zeros in all other positions. This $\boldsymbol{x}$ is nonzero. Expanding both quadratic forms shows

$$\boldsymbol{y}^T \mathbf{B} \boldsymbol{y} = \sum_{p,q=1}^{k} y_p a_{i_p i_q} y_q = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} > 0.$$

   The submatrix is symmetric because $a_{i_p i_q} = a_{i_q i_p}$. It is therefore SPD.

3. If $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ for some nonzero $\boldsymbol{x}$, then $\boldsymbol{x}^T \mathbf{A} \boldsymbol{x} = 0$, contradicting positive definiteness. Thus the null space is trivial, and the square matrix is nonsingular.

□

Ex 2.15 Let $\mathbf{A}$ be SPD and let $\mathbf{P}$ be a permutation matrix. Prove directly that $\mathbf{PAP}^T$ is SPD. Then consider $\mathbf{DAD}$ for a nonsingular diagonal matrix $\mathbf{D}$. Explain why changing the order or scale of the unknowns in this way preserves positive definiteness.

Def 2.9 **(The A-Inner Product and Energy Norm)** If $\mathbf{A}$ is SPD, then

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\mathbf{A}} = \boldsymbol{x}^T \mathbf{A} \boldsymbol{y}, \qquad \|\boldsymbol{x}\|_{\mathbf{A}} = \sqrt{\boldsymbol{x}^T \mathbf{A} \boldsymbol{x}}$$

define an inner product and its induced norm. The scalar $\boldsymbol{x}^T \mathbf{A} \boldsymbol{x}$ is often called the **energy** of $\boldsymbol{x}$.

Ex 2.16 Verify the inner-product axioms directly, identifying the use of symmetry and of positive definiteness. Then obtain the triangle inequality from the inner-product result proved earlier. What fails if $\mathbf{A}$ is only positive semidefinite?

Example For

$$\mathbf{A} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix},$$

the unit-energy curve is $4x_1^2 + x_2^2 = 1$. A displacement in the first coordinate costs four times as much energy as the same displacement in the second. A norm chosen this way also expresses which errors matter in the model.

2.4.2 Cholesky Factorization

Thm 2.18 **(Cholesky Factorization)** Every SPD matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ has a unique factorization

$$\mathbf{A} = \mathbf{L}\mathbf{L}^T,$$

where $\mathbf{L}$ is lower triangular with positive diagonal entries.

Proof We construct the factor one block at a time. In addition to existence, the positive choice of diagonal will determine each new block uniquely.

We proceed by induction on n. For $n = 1$, $\mathbf{A} = (\alpha)$ with $\alpha > 0$, so $\mathbf{L} = (\sqrt{\alpha})$ is the unique positive square root. For the induction step, partition $\mathbf{A} \in \mathbb{R}^{n \times n}$ as

$$\mathbf{A} = \begin{pmatrix} \alpha & \boldsymbol{r}^T \\ \boldsymbol{r} & \mathbf{B} \end{pmatrix}.$$

Since $\mathbf{A}$ is SPD, $\alpha = \boldsymbol{e}_1^T \mathbf{A} \boldsymbol{e}_1 > 0$. Set $\ell_{11} = \sqrt{\alpha}$ and $\boldsymbol{\ell} = \boldsymbol{r}/\ell_{11}$. The first diagonal entry α is positive by testing the quadratic form with the first coordinate vector. Thus its positive square root and all divisions by it are defined.

Form the Schur complement

$$\mathbf{S} = \mathbf{B} - \boldsymbol{\ell}\boldsymbol{\ell}^T = \mathbf{B} - \frac{\boldsymbol{r}\boldsymbol{r}^T}{\alpha}.$$

For any nonzero $\boldsymbol{y} \in \mathbb{R}^{n-1}$, choose $\boldsymbol{x} = (-\boldsymbol{\ell}^T\boldsymbol{y}/\ell_{11}, \boldsymbol{y}^T)^T \in \mathbb{R}^n$. Expanding the quadratic form:

$$\begin{aligned}
\boldsymbol{x}^T \mathbf{A} \boldsymbol{x} &= \begin{pmatrix} -\frac{\boldsymbol{\ell}^T\boldsymbol{y}}{\ell_{11}} \\ \boldsymbol{y} \end{pmatrix}^T \begin{pmatrix} \alpha & \boldsymbol{r}^T \\ \boldsymbol{r} & \mathbf{B} \end{pmatrix} \begin{pmatrix} -\frac{\boldsymbol{\ell}^T\boldsymbol{y}}{\ell_{11}} \\ \boldsymbol{y} \end{pmatrix} \\
&= \alpha \left(\frac{\boldsymbol{\ell}^T\boldsymbol{y}}{\ell_{11}} \right)^2 - 2\frac{(\boldsymbol{\ell}^T\boldsymbol{y})(\boldsymbol{r}^T\boldsymbol{y})}{\ell_{11}} + \boldsymbol{y}^T\mathbf{B}\boldsymbol{y} \\
&= (\boldsymbol{\ell}^T\boldsymbol{y})^2 - 2(\boldsymbol{\ell}^T\boldsymbol{y})^2 + \boldsymbol{y}^T\mathbf{B}\boldsymbol{y} \\
&= \boldsymbol{y}^T(\mathbf{B} - \boldsymbol{\ell}\boldsymbol{\ell}^T)\boldsymbol{y} = \boldsymbol{y}^T\mathbf{S}\boldsymbol{y}.
\end{aligned}$$

Since $\boldsymbol{y} \neq \mathbf{0}$, the vector $\boldsymbol{x}$ is nonzero, so $\boldsymbol{y}^T\mathbf{S}\boldsymbol{y} = \boldsymbol{x}^T\mathbf{A}\boldsymbol{x} > 0$. Hence $\mathbf{S}$ is SPD. By induction, $\mathbf{S} = \mathbf{K}\mathbf{K}^T$ for a unique lower triangular $\mathbf{K}$ with positive diagonal entries. Block multiplication gives

$$\mathbf{A} = \begin{pmatrix} \ell_{11} & \mathbf{0}^T \\ \boldsymbol{\ell} & \mathbf{K} \end{pmatrix} \begin{pmatrix} \ell_{11} & \boldsymbol{\ell}^T \\ \mathbf{0} & \mathbf{K}^T \end{pmatrix} = \begin{pmatrix} \ell_{11}^2 & \ell_{11}\boldsymbol{\ell}^T \\ \ell_{11}\boldsymbol{\ell} & \boldsymbol{\ell}\boldsymbol{\ell}^T + \mathbf{K}\mathbf{K}^T \end{pmatrix} = \begin{pmatrix} \alpha & \boldsymbol{r}^T \\ \boldsymbol{r} & \mathbf{B} \end{pmatrix}.$$

The positivity of the diagonal requires $\ell_{11} = \sqrt{\alpha}$, which uniquely fixes $\boldsymbol{\ell} = \boldsymbol{r}/\ell_{11}$, and uniqueness of $\mathbf{K}$ makes the entire factor $\mathbf{L}$ unique. □

Ex 2.17 Complete the block calculation in the proof of Thm. 2.18: verify exactly that $\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} = \boldsymbol{y}^T\mathbf{S}\boldsymbol{y}$. Then prove the converse statement that if $\alpha > 0$ and $\mathbf{S}$ is SPD, the block matrix $\mathbf{A}$ is SPD.

Thm 2.19 **(Equivalent Characterizations of Positive Definiteness)** For a real symmetric matrix $\mathbf{A}$, the following statements are equivalent.

1. $\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} > 0$ for every nonzero $\boldsymbol{x}$.
2. $\mathbf{A} = \mathbf{L}\mathbf{L}^T$ for a lower triangular $\mathbf{L}$ with positive diagonal.
3. $\mathbf{A} = \mathbf{W}^T\mathbf{W}$ for a nonsingular matrix $\mathbf{W}$.
4. Every leading principal minor is positive.

Proof We prove the first three equivalences, then connect them to the leading principal minors.

If $\mathbf{A}$ is SPD, the Cholesky theorem supplies $\mathbf{A} = \mathbf{L}\mathbf{L}^T$ with positive diagonal entries. A triangular matrix with nonzero diagonal is nonsingular. Taking $\mathbf{W} = \mathbf{L}^T$ gives the third representation, $\mathbf{A} = \mathbf{W}^T\mathbf{W}$.

Conversely, suppose $\mathbf{A} = \mathbf{W}^T\mathbf{W}$ with $\mathbf{W}$ nonsingular. A nonzero $\boldsymbol{x}$ cannot have $\mathbf{W}\boldsymbol{x} = \mathbf{0}$. Therefore

$$\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} = (\mathbf{W}\boldsymbol{x})^T(\mathbf{W}\boldsymbol{x}) = \|\mathbf{W}\boldsymbol{x}\|_2^2 > 0.$$

This proves positive definiteness and completes the equivalence of statements (1), (2), and (3).

Next suppose $\mathbf{A}$ is SPD. Every leading principal block $\mathbf{A}_j$ is SPD by Thm 2.17. Its Cholesky factor has positive diagonal, so

$$\det(\mathbf{A}_j) = \det(\mathbf{L}_j)\det(\mathbf{L}_j^T) = \prod_{i=1}^{j} \ell_{ii}^2 > 0.$$

Thus (1) implies (4).

For the converse, assume every leading principal minor is positive. We use induction on the dimension. For dimension one, this is exactly the positive-definiteness condition. In a larger dimension, write

$$\mathbf{A} = \begin{pmatrix} \alpha & \boldsymbol{r}^T \\ \boldsymbol{r} & \mathbf{B} \end{pmatrix}, \qquad \mathbf{S} = \mathbf{B} - \boldsymbol{r}\boldsymbol{r}^T/\alpha.$$

The first leading minor is $\alpha > 0$. For a leading block of size j in $\mathbf{S}$, subtracting r_i/α times the first row from each lower row of $\mathbf{A}_{j+1}$ gives a block upper triangular matrix with diagonal blocks α and $\mathbf{S}_j$. These row subtractions leave the determinant unchanged, so

$$\det(\mathbf{S}_j) = \frac{\det(\mathbf{A}_{j+1})}{\alpha} > 0.$$

The Schur complement is symmetric and has positive leading minors. The induction hypothesis makes it SPD.

Finally complete the square for an arbitrary vector $(s, \boldsymbol{y}^T)^T$:

$$\begin{aligned} \begin{pmatrix} s \\ \boldsymbol{y} \end{pmatrix}^T \mathbf{A} \begin{pmatrix} s \\ \boldsymbol{y} \end{pmatrix} &= \alpha s^2 + 2s\boldsymbol{r}^T\boldsymbol{y} + \boldsymbol{y}^T\mathbf{B}\boldsymbol{y} \\ &= \alpha\left(s + \frac{\boldsymbol{r}^T\boldsymbol{y}}{\alpha}\right)^2 + \boldsymbol{y}^T\mathbf{S}\boldsymbol{y}. \end{aligned}$$

If $\boldsymbol{y} \neq \mathbf{0}$, the second term is positive. If $\boldsymbol{y} = \mathbf{0}$ and the full vector is nonzero, then $s \neq 0$ and the first term is positive. Thus $\mathbf{A}$ is SPD, proving (4) implies (1). This is Sylvester's criterion (Meyer, 2000; Horn and Johnson, 2012). □

Ex 2.18 Let $\mathbf{X} \in \mathbb{R}^{m\times n}$ with $m \geq n$. Prove that $\mathbf{X}^T\mathbf{X}$ is SPD exactly when the columns of $\mathbf{X}$ are linearly independent. State what remains true when they are dependent.

Remark 2.3 After the spectral theorem is available, one more characterization follows directly. A real symmetric matrix is SPD exactly when all its eigenvalues are positive. The Cholesky argument does not need this fact, so we postpone it and keep the present argument self-contained.

2.4.3 The Cholesky Algorithm

Thm 2.20 **(The Cholesky Algorithm)** The entries of the factor in $\mathbf{A} = \mathbf{L}\mathbf{L}^T$ are computed column by column:

$$\begin{aligned} \ell_{jj} &= \sqrt{a_{jj} - \sum_{k=1}^{j-1} \ell_{jk}^2}, \\ \ell_{ij} &= \frac{a_{ij} - \sum_{k=1}^{j-1} \ell_{ik}\ell_{jk}}{\ell_{jj}}, \qquad i = j+1, \ldots, n. \end{aligned}$$

For an SPD matrix every expression under a square root is positive. A nonpositive pivot signals that the input is not positive definite, apart from the effects of rounding.

Proof We derive one column at a time from the identity $\mathbf{A} = \mathbf{L}\mathbf{L}^T$.

1. For $i \geq j$, triangularity gives

$$a_{ij} = (\mathbf{L}\mathbf{L}^T)_{ij} = \sum_{k=1}^{n} \ell_{ik}\ell_{jk} = \sum_{k=1}^{j} \ell_{ik}\ell_{jk}.$$

   Terms with $k > j$ vanish because $\ell_{jk} = 0$.

2. Put $i = j$ and separate the last term:

$$a_{jj} = \sum_{k=1}^{j-1} \ell_{jk}^2 + \ell_{jj}^2.$$

   The Cholesky convention chooses the positive square root, so

$$\ell_{jj} = \sqrt{a_{jj} - \sum_{k=1}^{j-1} \ell_{jk}^2}.$$

   Existence of the Cholesky factor for an SPD matrix ensures that the radicand is strictly positive in exact arithmetic.

3. For $i > j$, move the known terms to the other side:

$$a_{ij} - \sum_{k=1}^{j-1} \ell_{ik}\ell_{jk} = \ell_{ij}\ell_{jj}.$$

   Division by $\ell_{jj} > 0$ gives the stated update. All quantities on its right side come from preceding columns or from the diagonal entry just computed.

□

Example For

$$\mathbf{A} = \begin{pmatrix} 4 & 2 & -2 \\ 2 & 10 & 2 \\ -2 & 2 & 6 \end{pmatrix},$$

the algorithm gives

$$\begin{aligned} \ell_{11} &= 2, & \ell_{21} &= 1, & \ell_{31} &= -1, \\ \ell_{22} &= 3, & \ell_{32} &= 1, & \ell_{33} &= 2. \end{aligned}$$

Thus

$$\mathbf{L} = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 1 & 2 \end{pmatrix}, \qquad \mathbf{L}\mathbf{L}^T = \mathbf{A}.$$

To solve $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$, first solve $\mathbf{L}\boldsymbol{y} = \boldsymbol{b}$ and then $\mathbf{L}^T\boldsymbol{x} = \boldsymbol{y}$.

2.4.4 Cost and Stability

Thm 2.21 **(Flop Count and Storage)** Dense Cholesky factorization requires $n^3/3 + O(n^2)$ floating-point operations. Only one triangular half of the matrix and factor need be stored. The two triangular solves for one right-hand side require $2n^2 + O(n)$ further operations.

Proof In column j, there are $n - j + 1$ entries on and below the diagonal. Each uses a sum with $j - 1$ products. Counting a multiplication and addition as two operations, the leading inner-product work is

$$2\sum_{j=1}^{n}(n-j+1)(j-1).$$

Set $k = j - 1$. Then $n - j + 1 = n - k$, so the sum becomes

$$\begin{aligned} 2\sum_{k=0}^{n-1}(n-k)k &= 2n\sum_{k=0}^{n-1}k - 2\sum_{k=0}^{n-1}k^2 \\ &= 2n\frac{n(n-1)}{2} - 2\frac{(n-1)n(2n-1)}{6} \\ &= \frac{n^3-n}{3}. \end{aligned}$$

There are at most $n(n-1)/2$ divisions, n square roots, and $O(n^2)$ additional scalar operations. They do not change the cubic leading term. Hence the total is $n^3/3 + O(n^2)$ flops under the usual convention of counting each scalar operation once.

Only the lower triangle of the factor needs to be stored. It contains $1 + 2 + \cdots + n = n(n+1)/2$ entries. □

Remark 2.4 Cholesky needs no pivoting for an SPD matrix. Moreover, the identity

$$a_{ii} = \sum_{k=1}^{i}\ell_{ik}^2$$

implies $|\ell_{ik}| \le \sqrt{a_{ii}}$, so the factor cannot exhibit the element growth possible in general elimination. In standard floating-point arithmetic, and absent overflow or underflow, the computed factor is the exact Cholesky factor of a nearby symmetric matrix. This is the appropriate backward-stability statement. It does not say that an ill-conditioned system has an accurate forward solution.

Remark 2.5 Once singular values are available, Cor 4.20 gives

$$\kappa_2(\mathbf{A}) = \frac{\lambda_{\max}(\mathbf{A})}{\lambda_{\min}(\mathbf{A})}, \qquad \kappa_2(\mathbf{L}) = \sqrt{\kappa_2(\mathbf{A})}.$$

These are spectral interpretations of structure already visible in the energy and the factorization.

2.4.5 Sparse Cholesky and the Elimination Graph

For a sparse SPD matrix, the Cholesky factor can have many more nonzero entries than the matrix itself. These new entries are called **fill**. Fill determines the memory and the arithmetic of a sparse factorization, and it depends strongly on the order in which the unknowns are eliminated. A graph associated with the matrix predicts the fill before any arithmetic is done.

Def 2.10 **(Graph of a Symmetric Matrix)** The graph of a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ has vertices $1, \dots, n$ and an edge $\{i, j\}$ for each pair $i \neq j$ with $a_{ij} \neq 0$. The neighbors of a vertex v are the vertices joined to v by an edge, and the degree of v is the number of its neighbors.

Thm 2.22 **(Elimination as a Graph Operation)** Let $\mathbf{A}$ be SPD with graph G, and partition $\mathbf{A}$ as in the proof of Thm 2.18, with first column $(\alpha, \boldsymbol{r}^T)^T$ and trailing block $\mathbf{B}$. The first column of $\mathbf{L}$ is nonzero exactly in the rows of the neighbors of vertex 1. Apart from exact numerical cancellation, the graph of the Schur complement $\mathbf{S} = \mathbf{B} - \boldsymbol{r}\boldsymbol{r}^T/\alpha$ is obtained from G by deleting vertex 1 and joining every pair of its neighbors by an edge.

Proof Partition the matrix around the first variable as $\mathbf{A} = \begin{pmatrix} \alpha & \boldsymbol{r}^T \\ \boldsymbol{r} & \mathbf{B} \end{pmatrix}$. In its Cholesky factor, the first entry is $\sqrt{\alpha}$, and the remaining first-column entries satisfy $\ell_{i1}\sqrt{\alpha} = r_i$. Hence

$$\ell_{i1} = r_i/\sqrt{\alpha}.$$

Since $\alpha > 0$, this entry is nonzero precisely when r_i is nonzero. Those positions are the neighbors of the eliminated vertex in the matrix graph.

The remaining factorization acts on the Schur complement

$$\mathbf{S} = \mathbf{B} - \frac{\boldsymbol{r}\boldsymbol{r}^T}{\alpha}, \qquad s_{ij} = b_{ij} - \frac{r_i r_j}{\alpha}.$$

If i, j are both neighbors, then $r_i r_j \neq 0$. Even when $b_{ij} = 0$, the update introduces a nonzero entry at that position. Thus elimination can connect every pair of neighbors, forming a clique in the structural graph.

If one of the two vertices is not a neighbor, the corresponding factor r_i or r_j is zero and no entry is introduced by this elimination step. Existing nonzero entries may occasionally cancel numerically, so the graph statement concerns the pattern predicted in the absence of cancellation. Repeating this local rule explains fill in the Cholesky factor (Davis, 2006). □

Ex 2.19 Consider an SPD matrix of order five whose only off-diagonal nonzeros lie in its first row and column. Its graph is a star. Draw the elimination graphs with the center first and with the center last; count the nonzeros of $\mathbf{L}$ in each case, assuming no cancellation.

Applying Thm 2.22 repeatedly describes the whole factorization. Eliminating a vertex turns its remaining neighbors into a clique, and the nonzero pattern of column k of $\mathbf{L}$ is the set of neighbors of vertex k at the moment it is eliminated. Reordering the unknowns by a permutation $\mathbf{P}$ and factoring $\mathbf{P}\mathbf{A}\mathbf{P}^T$ changes the elimination order and therefore the fill, without changing the solution.

Example **(An arrow matrix)** Let $\mathbf{A}$ be SPD with nonzero entries only on the diagonal and in the first row and column. Its graph is a star with center 1. If the center is eliminated first, its $n-1$ neighbors become a clique, the Schur complement is full, and $\mathbf{L}$ has about $n^2/2$ nonzero entries. If the center is eliminated last, every leaf has the center as its only neighbor, no edges are added, and $\mathbf{L}$ has $2n-1$ nonzero entries. The two orders give the same solution at costs of $O(n^3)$ and $O(n)$ flops.

Remark 2.6 **(Fill-reducing orderings)** Finding the ordering with the least fill is NP-complete (Yannakakis, 1981), so practical codes use heuristics. The **minimum degree** heuristic always eliminates a

vertex of smallest current degree, which bounds the fill created at each step by the square of that degree. The degrees change after every elimination, and an efficient implementation stores the vertices in a bucket queue, with one doubly linked list of vertices for each degree. Moving a vertex to a new bucket then takes constant time, and the next vertex is found by scanning upward from the lowest nonempty bucket (Davis, 2006). **Nested dissection** instead removes a small set of vertices that separates the graph into two parts, orders the separator last, and recurses on the parts.

Nested dissection determines the cost of sparse Cholesky for the Poisson problem. On an $N \times N$ grid with $n = N^2$ unknowns, the natural row-by-row order gives a factor with bandwidth N, so $\mathbf{L}$ has about $n^{3/2}$ nonzero entries and the factorization costs about n^2 flops. Nested dissection with grid lines as separators reduces these to $O(n \log n)$ nonzero entries and $O(n^{3/2})$ flops (George, 1973). On an $N \times N \times N$ grid, separators are planes with $n^{2/3}$ vertices, and nested dissection leaves $O(n^{4/3})$ nonzero entries and $O(n^2)$ flops. The growth of the factor with n is the reason that large three-dimensional problems are solved by the iterative methods of Section 6.2.

2.4.6 Further Exercises

Ex 2.20 Compute the Cholesky factor of

$$\mathbf{A} = \begin{pmatrix} 9 & 3 & -6 \\ 3 & 26 & -7 \\ -6 & -7 & 9 \end{pmatrix}$$

by hand, then use it to solve $\mathbf{A}\boldsymbol{x} = (6, 44, 2)^T$.

Ex 2.21 Prove that the largest entry of an SPD matrix in absolute value lies on the main diagonal. Begin with a two by two principal submatrix and choose a useful vector, or use its positive determinant.

Ex 2.22 Use Thm 2.22 to prove that, apart from cancellation, $\ell_{ij} \neq 0$ for $i > j$ exactly when G contains a path from i to j whose interior vertices all precede j in the elimination order.

Ex 2.23 Assemble the five-point Poisson matrix on an $N \times N$ grid for $N = 20, 40, 80$. Factor it with a sparse direct solver using the natural order and a minimum degree order, and report the number of nonzero entries in the factors and the factorization times.

Ex 2.24 Show that $\|\boldsymbol{x}\|_{\mathbf{A}} = \|\mathbf{L}^T \boldsymbol{x}\|_2$. Use this identity to interpret the $\mathbf{A}$-unit sphere as the inverse image of the Euclidean unit sphere, without mentioning eigenvalues.

2.5 Factorization Updates

A sequence of linear systems often changes only slightly from one solve to the next. If only the right-hand side changes, the existing factors can be used directly. If the matrix changes, we ask whether the new solve can be reduced to solves with the old matrix and a much smaller correction problem.

We derive this reduction for changes of rank one and then for changes of rank r. A separate construction uses neighboring block interactions to keep elimination local. In both cases, the amount of work saved depends on which entries change and on the conditioning of the remaining solves.

2.5.1 Rank-One Changes and the Sherman–Morrison Formula

Let a factorization of a nonsingular matrix $\mathbf{A}$ already be available. Suppose the matrix changes to $\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T$. A fresh dense factorization costs $O(n^3)$ operations. The rank-one formula replaces it by solves with the old factors and a few inner products.
For a solution of $(\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T)\boldsymbol{x} = \boldsymbol{b}$, let $\gamma = \boldsymbol{v}^T\boldsymbol{x}$. The system then reads $\mathbf{A}\boldsymbol{x} = \boldsymbol{b} - \gamma\boldsymbol{u}$. If $\boldsymbol{y}$ and $\boldsymbol{z}$ solve $\mathbf{A}\boldsymbol{y} = \boldsymbol{b}$ and $\mathbf{A}\boldsymbol{z} = \boldsymbol{u}$, respectively, then $\boldsymbol{x} = \boldsymbol{y} - \gamma\boldsymbol{z}$. Taking the inner product with $\boldsymbol{v}$ gives a scalar equation,

$$(1 + \boldsymbol{v}^T\boldsymbol{z})\gamma = \boldsymbol{v}^T\boldsymbol{y}.$$

Thus two solves with the old matrix and one scalar division determine the correction, provided the coefficient of γ is nonzero. The following theorem states the same calculation as an identity for the inverse.

Thm 2.23 **(Sherman–Morrison Formula)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be nonsingular, and let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$. The updated matrix $\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T$ is nonsingular if and only if

$$1 + \boldsymbol{v}^T\mathbf{A}^{-1}\boldsymbol{u} \neq 0.$$

When this condition holds,

$$(\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\boldsymbol{u}\boldsymbol{v}^T\mathbf{A}^{-1}}{1 + \boldsymbol{v}^T\mathbf{A}^{-1}\boldsymbol{u}}.$$

Proof Set $\boldsymbol{z} = \mathbf{A}^{-1}\boldsymbol{u}$ and $\alpha = 1 + \boldsymbol{v}^T\boldsymbol{z}$. We must handle the singular and nonsingular cases separately.

Suppose $\alpha = 0$. If $\boldsymbol{u}$ were zero, then $\boldsymbol{z}$ would be zero and α would be one. Hence $\boldsymbol{u} \neq \mathbf{0}$, and invertibility of $\mathbf{A}$ gives $\boldsymbol{z} \neq \mathbf{0}$. Now calculate

$$\begin{aligned}(\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T)\boldsymbol{z} &= \mathbf{A}\boldsymbol{z} + \boldsymbol{u}(\boldsymbol{v}^T\boldsymbol{z}) \\ &= \boldsymbol{u} + \boldsymbol{u}(\boldsymbol{v}^T\boldsymbol{z}) = \boldsymbol{u}(1 + \boldsymbol{v}^T\boldsymbol{z}) = \mathbf{0}.\end{aligned}$$

The updated matrix has a nonzero null vector, so it is singular.

Suppose instead that $\alpha \neq 0$ and define $\mathbf{B} = \mathbf{A}^{-1} - \boldsymbol{z}\boldsymbol{v}^T\mathbf{A}^{-1}/\alpha$. The first product needed is

$$(\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T)\mathbf{A}^{-1} = \mathbf{I} + \boldsymbol{u}\boldsymbol{v}^T\mathbf{A}^{-1}.$$

The calculation in the first case, now without setting α to zero, gives $(\mathbf{A}+\boldsymbol{u}\boldsymbol{v}^T)\boldsymbol{z} = \alpha\boldsymbol{u}$. Therefore

$$(\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T)\mathbf{B} = \mathbf{I} + \boldsymbol{u}\boldsymbol{v}^T\mathbf{A}^{-1} - \frac{\alpha\boldsymbol{u}\boldsymbol{v}^T\mathbf{A}^{-1}}{\alpha} = \mathbf{I}.$$

A square matrix possessing a right inverse is nonsingular, so $\mathbf{B}$ is its inverse. Replacing $\boldsymbol{z}$ by $\mathbf{A}^{-1}\boldsymbol{u}$ gives the stated formula. This identity is due to Sherman and Morrison (1950). □

Ex 2.25 **A nearly singular correction.** Let $\mathbf{A} = \mathbf{I}$, $\boldsymbol{u} = -(1 - \delta)\boldsymbol{e}_1$, and $\boldsymbol{v} = \boldsymbol{e}_1$. Compute the Sherman–Morrison denominator and the updated inverse. Explain why the correction amplifies roundoff as $\delta \to 0$, even though solving with the base matrix is perfectly conditioned.

Def 2.11 **(Sherman–Morrison Solve)** To solve $(\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T)\boldsymbol{x} = \boldsymbol{b}$ from stored factors of $\mathbf{A}$:

1. Solve $\mathbf{A}\boldsymbol{z} = \boldsymbol{u}$ once for the update.
2. Compute $\alpha = 1 + \boldsymbol{v}^T\boldsymbol{z}$. Refactor the updated matrix if α is zero or so small that division by it would magnify roundoff.
3. For each right-hand side, solve $\mathbf{A}\boldsymbol{y} = \boldsymbol{b}$.
4. Compute

$$\boldsymbol{x} = \boldsymbol{y} - \boldsymbol{z}\frac{\boldsymbol{v}^T\boldsymbol{y}}{\alpha}.$$

The algorithm never forms $\mathbf{A}^{-1}$.

Ex 2.26 **Sherman–Morrison by hand.** Repeat the worked example with $\boldsymbol{b} = (0, 1)^T$. Compute the corrected solution from Def 2.11, and verify it by multiplying with $\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T$.

Example Let

$$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \qquad \boldsymbol{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \boldsymbol{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The two solves with $\mathbf{A}$ give $\boldsymbol{z} = (1/2, 1/3)^T$ and $\boldsymbol{y} = (1/2, 0)^T$. Since $\alpha = 1 + 1/2 - 1/3 = 7/6$,

$$\boldsymbol{x} = \boldsymbol{y} - \boldsymbol{z}\frac{1/2}{7/6} = \begin{pmatrix} 2/7 \\ -1/7 \end{pmatrix}.$$

Multiplication by $\mathbf{A} + \boldsymbol{u}\boldsymbol{v}^T = \begin{pmatrix} 3 & -1 \\ 1 & 2 \end{pmatrix}$ verifies the solution.

Remark 2.7 **(Cost of a rank-one change)** With a dense LU factorization, preparing the correction requires one $O(n^2)$ solve. Each new right-hand side requires one further solve and $O(n)$ correction work. A fresh $O(n^3)$ factorization is avoided. For a sparse factorization, the same comparison uses the measured cost of a triangular solve rather than the dense estimate.

2.5.2 Low-Rank Updates and the Woodbury Formula

A change described by r pairs of vectors has the form $\mathbf{A} + \mathbf{U}\mathbf{V}^T$, where $\mathbf{U}$ and $\mathbf{V}$ have r columns and $r \ll n$. The correction requires an $r \times r$ system. This small system determines both the cost and the reliability of the update.

Thm 2.24 **(Woodbury Formula)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be nonsingular, and let $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{n\times r}$. Define

$$\mathbf{G} = \mathbf{I}_r + \mathbf{V}^T\mathbf{A}^{-1}\mathbf{U}.$$

The matrix $\mathbf{A} + \mathbf{U}\mathbf{V}^T$ is nonsingular if and only if $\mathbf{G}$ is nonsingular. In that case,

$$(\mathbf{A} + \mathbf{U}\mathbf{V}^T)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\mathbf{G}^{-1}\mathbf{V}^T\mathbf{A}^{-1}.$$

Proof Suppose $\mathbf{G}\boldsymbol{c} = \mathbf{0}$ for some nonzero $\boldsymbol{c}$. Then $\mathbf{U}\boldsymbol{c} \neq \mathbf{0}$, since otherwise $\mathbf{G}\boldsymbol{c} = \boldsymbol{c}$. Consequently, $\boldsymbol{x} = \mathbf{A}^{-1}\mathbf{U}\boldsymbol{c}$ is nonzero and

$$(\mathbf{A} + \mathbf{U}\mathbf{V}^T)\boldsymbol{x} = \mathbf{U}\boldsymbol{c} + \mathbf{U}\mathbf{V}^T\mathbf{A}^{-1}\mathbf{U}\boldsymbol{c} = \mathbf{U}\mathbf{G}\boldsymbol{c} = \mathbf{0}.$$

Thus a singular $\mathbf{G}$ makes the updated matrix singular.

The nonzero null vector has been transferred from the small matrix to the updated full matrix. We now transfer in the other direction to prove the equivalence, rather than only one implication.

Conversely, suppose $(\mathbf{A} + \mathbf{U}\mathbf{V}^T)\boldsymbol{x} = \mathbf{0}$ for some nonzero $\boldsymbol{x}$. Let $\boldsymbol{c} = \mathbf{V}^T\boldsymbol{x}$. The vector $\boldsymbol{c}$ cannot be zero, because that would imply $\mathbf{A}\boldsymbol{x} = \mathbf{0}$. From $\boldsymbol{x} = -\mathbf{A}^{-1}\mathbf{U}\boldsymbol{c}$, we obtain

$$\mathbf{G}\boldsymbol{c} = \boldsymbol{c} + \mathbf{V}^T\mathbf{A}^{-1}\mathbf{U}\boldsymbol{c} = \mathbf{V}^T\boldsymbol{x} - \mathbf{V}^T\boldsymbol{x} = \mathbf{0}.$$

Hence the two matrices are nonsingular together.

Both matrices are square. A square matrix is nonsingular precisely when its null space is trivial. The two constructions therefore prove that either both are singular or both are nonsingular.

When $\mathbf{G}$ is nonsingular, direct multiplication verifies the inverse:

$$\begin{aligned}
&(\mathbf{A} + \mathbf{U}\mathbf{V}^T)\left(\mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\mathbf{G}^{-1}\mathbf{V}^T\mathbf{A}^{-1}\right) \\
&\quad = \mathbf{A}\mathbf{A}^{-1} - \mathbf{A}\mathbf{A}^{-1}\mathbf{U}\mathbf{G}^{-1}\mathbf{V}^T\mathbf{A}^{-1} + \mathbf{U}\mathbf{V}^T\mathbf{A}^{-1} - \mathbf{U}\mathbf{V}^T\mathbf{A}^{-1}\mathbf{U}\mathbf{G}^{-1}\mathbf{V}^T\mathbf{A}^{-1} \\
&\quad = \mathbf{I}_n - \mathbf{U}\mathbf{G}^{-1}\mathbf{V}^T\mathbf{A}^{-1} + \mathbf{U}\mathbf{V}^T\mathbf{A}^{-1} - \mathbf{U}(\mathbf{V}^T\mathbf{A}^{-1}\mathbf{U})\mathbf{G}^{-1}\mathbf{V}^T\mathbf{A}^{-1} \\
&\quad = \mathbf{I}_n + \mathbf{U}\left(\mathbf{I}_r - (\mathbf{I}_r + \mathbf{V}^T\mathbf{A}^{-1}\mathbf{U})\mathbf{G}^{-1}\right)\mathbf{V}^T\mathbf{A}^{-1} \\
&\quad = \mathbf{I}_n + \mathbf{U}\left(\mathbf{I}_r - \mathbf{G}\mathbf{G}^{-1}\right)\mathbf{V}^T\mathbf{A}^{-1} \\
&\quad = \mathbf{I}_n + \mathbf{U}(\mathbf{I}_r - \mathbf{I}_r)\mathbf{V}^T\mathbf{A}^{-1} = \mathbf{I}_n.
\end{aligned}$$

In the inverse verification, the bracket becomes zero because its sum $\mathbf{I}_r + \mathbf{V}^T\mathbf{A}^{-1}\mathbf{U}$ is exactly $\mathbf{G}$. A right inverse of a square matrix is its inverse, so checking this one product suffices after nonsingularity has been established. See Woodbury (1950) for the original identity. □

Ex 2.27 **Matrix determinant lemma.** Under the hypotheses of Thm 2.24, prove

$$\det(\mathbf{A} + \mathbf{U}\mathbf{V}^T) = \det(\mathbf{A})\det(\mathbf{I}_r + \mathbf{V}^T\mathbf{A}^{-1}\mathbf{U}).$$

Begin with $\mathbf{A}+\mathbf{U}\mathbf{V}^T = \mathbf{A}(\mathbf{I}+\mathbf{A}^{-1}\mathbf{U}\mathbf{V}^T)$ and prove the needed identity $\det(\mathbf{I}+\mathbf{X}\mathbf{Y}) = \det(\mathbf{I}+\mathbf{Y}\mathbf{X})$ using a block determinant in two different orders.

Def 2.12 **(Woodbury Solve)** Let stored factors of $\mathbf{A}$ be available.

1. Solve the block system $\mathbf{A}\mathbf{W} = \mathbf{U}$.
2. Form $\mathbf{G} = \mathbf{I}_r + \mathbf{V}^T\mathbf{W}$ and factor the small matrix $\mathbf{G}$.
3. For a right-hand side matrix $\mathbf{B}$, solve $\mathbf{A}\mathbf{Y} = \mathbf{B}$.
4. Solve $\mathbf{G}\mathbf{C} = \mathbf{V}^T\mathbf{Y}$ and return

$$\mathbf{X} = \mathbf{Y} - \mathbf{W}\mathbf{C}.$$

When several right-hand sides are available, solve them together. This is usually faster than solving one column at a time.

The setup requires r solves with the old factors, the product $\mathbf{V}^T\mathbf{W}$, and one factorization of order r. Each later right-hand side requires one old solve, two products involving an $n \times r$ matrix, and one small solve. For dense factors, the setup costs $O(rn^2 + nr^2 + r^3)$ rather than $O(n^3)$. This comparison favors Woodbury only when r remains small.

Remark 2.8 **(Transpose solves)** The same retained factors also solve the transposed updated system. Compute $\mathbf{Z} = \mathbf{A}^{-T}\mathbf{V}$ and use

$$\mathbf{G}^T = \mathbf{I}_r + \mathbf{U}^T\mathbf{Z}.$$

The transpose solve then has the same four steps with $\mathbf{U}$ and $\mathbf{V}$ interchanged. No factorization of the full transpose is needed when the base factor already supports a transpose solve.

Remark 2.9 **(Updating only the inverse diagonal)** Suppose the diagonal of $\mathbf{A}^{-1}$ is already known. The corrected diagonal follows from

$$\operatorname{diag}((\mathbf{A} + \mathbf{U}\mathbf{V}^T)^{-1}) = \operatorname{diag}(\mathbf{A}^{-1}) - \operatorname{diag}(\mathbf{W}\mathbf{G}^{-1}\mathbf{Z}^T),$$

where $\mathbf{W} = \mathbf{A}^{-1}\mathbf{U}$ and $\mathbf{Z} = \mathbf{A}^{-T}\mathbf{V}$. Only the diagonal of the final product is needed. The full inverse should not be formed. When no diagonal is known in advance, Section 6.7 estimates the whole diagonal of $\mathbf{A}^{-1}$ from a few random probes and one factorization.

Prop 2.25 **(Changes in a Few Rows and Columns)** Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices that differ only in rows and columns indexed by a set of size k. Then $\mathbf{B} - \mathbf{A}$ has rank at most $2k$.

Proof Let I be the set of k selected indices. Form $\mathbf{E}$ by taking the corresponding columns of the identity matrix, and put $\mathbf{P} = \mathbf{E}\mathbf{E}^T$. Multiplication by $\mathbf{P}$ keeps the selected coordinates and replaces the others by zero.

Set $\mathbf{D} = \mathbf{B} - \mathbf{A}$. Only selected rows or selected columns can contain nonzero entries of $\mathbf{D}$. Thus a submatrix using neither selected rows nor selected columns is zero. In matrix form this is

$$(\mathbf{I} - \mathbf{P})\mathbf{D}(\mathbf{I} - \mathbf{P}) = \mathbf{0}.$$

Split the rows first, and then split the columns of the remaining term:

$$\begin{aligned}\mathbf{D} &= \mathbf{P}\mathbf{D} + (\mathbf{I} - \mathbf{P})\mathbf{D} \\ &= \mathbf{P}\mathbf{D} + (\mathbf{I} - \mathbf{P})\mathbf{D}\mathbf{P} + (\mathbf{I} - \mathbf{P})\mathbf{D}(\mathbf{I} - \mathbf{P}) \\ &= \mathbf{E}(\mathbf{E}^T\mathbf{D}) + (\mathbf{I} - \mathbf{E}\mathbf{E}^T)\mathbf{D}\mathbf{E}\mathbf{E}^T.\end{aligned}$$

The first term factors through a k-dimensional space, so its rank is at most k. The second also factors through the k columns of $\mathbf{E}$, so its rank is at most k. The range of their sum lies in the sum of their ranges, whose dimension is at most $2k$. Hence $\operatorname{rank}(\mathbf{B} - \mathbf{A}) \le 2k$. □

This rank bound is useful when a model changes by adding, removing, or modifying a few states. The changed rows are stored once, and the changed columns are stored after their overlap with those rows has been removed. Woodbury then corrects the retained factorization with rank at most $2k$.

Remark 2.10 **(When to refactor)** The small matrix $\mathbf{G}$ can be sensitive to roundoff. A reliable implementation estimates that sensitivity, checks the residual of the corrected solution, and computes a fresh factorization when either check fails. No fixed update-count limit or sensitivity threshold applies to all problems. Choose such thresholds in relation to the requested accuracy, and compare against a fresh solve when validating an implementation.

2.5.3 Tridiagonal Systems

Low rank describes how a matrix changes. Banded structure describes which unknowns interact. A tridiagonal matrix couples each unknown only to its two neighbors. Gaussian elimination preserves this pattern when no row interchange is required, so the solve takes linear rather than cubic work.

Def 2.13 **(Thomas Algorithm)** Consider the equations

$$a_i x_{i-1} + d_i x_i + c_i x_{i+1} = r_i,$$

with the missing endpoint terms omitted. Let $\widetilde{d}_1 = d_1$ and $\widetilde{r}_1 = r_1$. For $i = 2, \ldots, n$, compute

$$\begin{aligned} m_i &= \frac{a_i}{\widetilde{d}_{i-1}}, \\ \widetilde{d}_i &= d_i - m_i c_{i-1}, \\ \widetilde{r}_i &= r_i - m_i \widetilde{r}_{i-1}. \end{aligned}$$

Backward substitution gives

$$x_n = \frac{\widetilde{r}_n}{\widetilde{d}_n}, \qquad x_i = \frac{\widetilde{r}_i - c_i x_{i+1}}{\widetilde{d}_i}, \quad i = n-1, \ldots, 1.$$

Prop 2.26 **(Cost of Tridiagonal Elimination)** If every modified pivot $\widetilde{d}_i$ is nonzero, the Thomas algorithm solves an $n \times n$ tridiagonal system in $O(n)$ operations and $O(n)$ storage. The diagonal and right-hand side may be overwritten, so the additional storage can be constant when the input need not be retained.

Proof The forward elimination loop visits $i = 2, \ldots, n$. At each position it forms one multiplier, updates one diagonal entry, and updates one right-hand-side entry. Each operation involves a fixed number of scalars, independent of n. Thus the entire forward loop costs at most a constant times $n - 1$ operations.

Back substitution starts with one division for the final unknown. Each preceding unknown is computed from one upper-diagonal coefficient, the next solution component, its right-hand-side entry, and its modified diagonal. Again the work per position is bounded by a constant. The backward loop therefore costs $O(n)$ as well.

The data comprise n diagonal entries, $n - 1$ entries on each adjacent diagonal, and n right-hand-side entries. Their total count is $4n - 2$. Modified diagonal and right-hand-side entries can overwrite their old values because those old values are never needed by a later step. The input and output storage is therefore $O(n)$, with only a constant number of extra scalar temporaries if overwriting is allowed.

These counts assume every required pivot is nonzero. The cost argument establishes neither the existence of those pivots nor numerical stability. □

Remark 2.11 **(Pivoting)** The Thomas algorithm is unpivoted elimination. It is safe for important classes such as symmetric positive definite tridiagonal matrices, but a general tridiagonal matrix can have a zero or dangerously small pivot. In that case, use a tridiagonal solver with pivoting or a general banded factorization, even though pivoting may enlarge the band.

2.5.4 Block-Tridiagonal Systems

Many models place several unknowns at each location or time level. Grouping those unknowns produces a block-tridiagonal matrix. The scalar divisions in the Thomas algorithm become solves with dense diagonal blocks, and the scalar updates become matrix products.

Def 2.14 **(Block-Tridiagonal Matrix)** A block-tridiagonal matrix has the form

$$\mathbf{A} = \begin{pmatrix} \mathbf{D}_1 & \mathbf{U}_1 & & \\ \mathbf{L}_1 & \mathbf{D}_2 & \mathbf{U}_2 & \\ & \ddots & \ddots & \ddots \\ & & \mathbf{L}_{s-1} & \mathbf{D}_s \end{pmatrix}.$$

The diagonal blocks $\mathbf{D}_k$ are square. Neighboring blocks may have different sizes, so $\mathbf{U}_k$ and $\mathbf{L}_k$ may be rectangular.

Thm 2.27 **(Block-Tridiagonal LU Factorization)** Let $\mathbf{S}_1 = \mathbf{D}_1$. For $k = 1, \dots, s-1$, suppose $\mathbf{S}_k$ is nonsingular and compute

$$\mathbf{M}_k \mathbf{S}_k = \mathbf{L}_k, \qquad \mathbf{S}_{k+1} = \mathbf{D}_{k+1} - \mathbf{M}_k \mathbf{U}_k.$$

If the final block $\mathbf{S}_s$ is also nonsingular, then $\mathbf{A} = \mathcal{L}\mathcal{U}$, where $\mathcal{L}$ is block unit lower bidiagonal with blocks $\mathbf{M}_k$, and $\mathcal{U}$ is block upper bidiagonal with diagonal blocks $\mathbf{S}_k$ and upper blocks $\mathbf{U}_k$.

Proof We check every possible nonzero block of $\mathcal{L}\mathcal{U}$. The lower factor has identity diagonal blocks and subdiagonal blocks $\mathbf{M}_k$. The upper factor has diagonal blocks $\mathbf{S}_k$ and superdiagonal blocks $\mathbf{U}_k$.

In position $(k, k+1)$, the only contributing product is $\mathbf{I}\mathbf{U}_k$, which gives the required upper block. In position $(k+1, k)$, the only contributing product is $\mathbf{M}_k \mathbf{S}_k$. The construction solves $\mathbf{M}_k \mathbf{S}_k = \mathbf{L}_k$, so this is the required lower block.

For the first diagonal position, the product is $\mathbf{I}\mathbf{S}_1 = \mathbf{D}_1$. At a later diagonal position k, two terms contribute:

$$(\mathcal{L}\mathcal{U})_{kk} = \mathbf{S}_k + \mathbf{M}_{k-1}\mathbf{U}_{k-1}.$$

The defining update $\mathbf{S}_k = \mathbf{D}_k - \mathbf{M}_{k-1}\mathbf{U}_{k-1}$ makes this sum equal to $\mathbf{D}_k$.

A block farther from the diagonal cannot receive a contribution. In row i, $\mathcal{L}$ can have nonzero blocks only in columns $i-1$ and i. The corresponding rows of $\mathcal{U}$ can reach only columns $i-1$, i, and $i+1$. Hence their product is block tridiagonal. We have checked all its potentially nonzero blocks and found exactly those of $\mathbf{A}$. This proves the factorization. □

Ex 2.28 **Thomas as scalar block elimination.** Take every block in Thm 2.27 to have size one. Derive the three forward recurrences in Def 2.13 and the backward substitution formula. Identify the modified scalar pivot with the scalar Schur complement.

The equation $\mathbf{M}_k\mathbf{S}_k = \mathbf{L}_k$ should not be evaluated with an inverse. Instead, solve the transposed system

$$\mathbf{S}_k^T\mathbf{M}_k^T = \mathbf{L}_k^T$$

using the stored factorization of $\mathbf{S}_k$.

Def 2.15 **(Block Forward and Backward Sweeps)** Partition the right-hand side into blocks $\boldsymbol{b}_k$. First compute

$$\begin{aligned}\boldsymbol{y}_1 &= \boldsymbol{b}_1,\\ \boldsymbol{y}_k &= \boldsymbol{b}_k - \mathbf{M}_{k-1}\boldsymbol{y}_{k-1}, \qquad k = 2, \dots, s.\end{aligned}$$

Then solve backward:

$$\begin{aligned}\mathbf{S}_s\boldsymbol{x}_s &= \boldsymbol{y}_s,\\ \mathbf{S}_k\boldsymbol{x}_k &= \boldsymbol{y}_k - \mathbf{U}_k\boldsymbol{x}_{k+1}, \qquad k = s-1, \dots, 1.\end{aligned}$$

The same recurrences accept several right-hand sides at once.

Ex 2.29 **Transpose block solve.** Starting from $\mathbf{A} = \mathcal{L}\mathcal{U}$ in Thm 2.27, derive the two sweeps for $\mathbf{A}^T\boldsymbol{x} = \boldsymbol{b}$. State the direction of each sweep and identify which stored off-diagonal block it uses.

Remark 2.12 **(Work and storage)** If there are s blocks of comparable size p, the factorization costs $O(sp^3)$ operations and stores $O(sp^2)$ numbers. A solve with q right-hand sides costs $O(sp^2q)$. Since the full order is $n = sp$, the factorization cost is $O(n^3/s^2)$. Unequal neighboring block sizes add rectangular products, so the largest block often controls the cost.

Remark 2.13 **(Transpose and multiple right-hand sides)** The factors also solve $\mathbf{A}^T\boldsymbol{x} = \boldsymbol{b}$. The transpose reverses the order of the two sweeps and interchanges the stored upper and lower blocks. Keep several right-hand sides together throughout the sweeps. This reuses memory and is usually faster than processing the columns separately.

2.5.5 The Symmetric Positive Definite Case

When the block-tridiagonal matrix is symmetric positive definite, block Cholesky removes the need for pivoting and stores only one off-diagonal side. The key fact is that every diagonal Schur complement remains positive definite.

Thm 2.28 **(Positive Definite Schur Complement)** Let

$$\mathbf{A} = \begin{pmatrix}\mathbf{D} & \mathbf{B}^T\\ \mathbf{B} & \mathbf{C}\end{pmatrix}$$

be symmetric positive definite. Then $\mathbf{D}$ is positive definite, and the Schur complement

$$\mathbf{C} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^T$$

is positive definite.

Proof For every nonzero $\boldsymbol{x}$, testing with $(\boldsymbol{x}, \mathbf{0})^T$ gives

$$\begin{pmatrix}\boldsymbol{x}\\ \mathbf{0}\end{pmatrix}^T \mathbf{A} \begin{pmatrix}\boldsymbol{x}\\ \mathbf{0}\end{pmatrix} = \boldsymbol{x}^T\mathbf{D}\boldsymbol{x} > 0,$$

so $\mathbf{D}$ is positive definite and $\mathbf{D}^{-1}$ exists. The leading block $\mathbf{D}$ is symmetric because it is a principal block of the symmetric matrix. Thus $\mathbf{D}^{-1}$ is symmetric too, by transposing $\mathbf{D}\mathbf{D}^{-1} = \mathbf{I}$. This fact is used when transposing the chosen vector in the next expansion.

Now let $\boldsymbol{y} \neq \mathbf{0}$, and choose $\boldsymbol{x} = -\mathbf{D}^{-1}\mathbf{B}^T\boldsymbol{y}$. Expanding the block quadratic form:

$$\begin{aligned}\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{y}\end{pmatrix}^T \begin{pmatrix}\mathbf{D} & \mathbf{B}^T\\ \mathbf{B} & \mathbf{C}\end{pmatrix}\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{y}\end{pmatrix} &= \boldsymbol{x}^T\mathbf{D}\boldsymbol{x} + \boldsymbol{x}^T\mathbf{B}^T\boldsymbol{y} + \boldsymbol{y}^T\mathbf{B}\boldsymbol{x} + \boldsymbol{y}^T\mathbf{C}\boldsymbol{y}\\ &= (-\mathbf{D}^{-1}\mathbf{B}^T\boldsymbol{y})^T\mathbf{D}(-\mathbf{D}^{-1}\mathbf{B}^T\boldsymbol{y}) + 2(-\mathbf{D}^{-1}\mathbf{B}^T\boldsymbol{y})^T\mathbf{B}^T\boldsymbol{y} + \boldsymbol{y}^T\mathbf{C}\boldsymbol{y}\\ &= \boldsymbol{y}^T\mathbf{B}\mathbf{D}^{-1}\mathbf{B}^T\boldsymbol{y} - 2\boldsymbol{y}^T\mathbf{B}\mathbf{D}^{-1}\mathbf{B}^T\boldsymbol{y} + \boldsymbol{y}^T\mathbf{C}\boldsymbol{y}\\ &= \boldsymbol{y}^T(\mathbf{C} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^T)\boldsymbol{y}.\end{aligned}$$

Because $\boldsymbol{y} \neq \mathbf{0}$, the compound vector $(\boldsymbol{x}, \boldsymbol{y})^T$ is nonzero. The positive definiteness of $\mathbf{A}$ implies that this quadratic form is strictly positive:

$$\boldsymbol{y}^T(\mathbf{C} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^T)\boldsymbol{y} > 0.$$

The Schur complement $\mathbf{C} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^T$ is therefore positive definite.

The Schur complement is symmetric because both $\mathbf{C}$ and $\mathbf{B}\mathbf{D}^{-1}\mathbf{B}^T$ are symmetric. The quadratic-form calculation proves strict positivity for every nonzero $\boldsymbol{y}$. These two facts together prove that it is SPD and justify repeating the same elimination step on the trailing block. □

Block Cholesky follows the same order as block LU. Factor $\mathbf{S}_k = \mathbf{F}_k\mathbf{F}_k^T$, solve

$$\mathbf{C}_k\mathbf{F}_k^T = \mathbf{L}_k,$$

and update

$$\mathbf{S}_{k+1} = \mathbf{D}_{k+1} - \mathbf{C}_k\mathbf{C}_k^T.$$

The theorem above shows that every $\mathbf{S}_k$ remains positive definite. The stored factors form a block lower-bidiagonal Cholesky factor of $\mathbf{A}$.

2.5.6 Block Structure and Factor Reuse

A block order need not be visible in the original row numbering. Make an undirected graph with one vertex for each row and one edge for each nonzero connection. Choose one or more starting vertices and label every vertex by its distance from them. The labels at the ends of an edge differ by at most one. Grouping equal labels therefore produces a block-tridiagonal matrix.

Remark 2.14 **(Block width)** The sizes of the levels matter as much as their number. A poor starting set can place many rows in one level and create an expensive dense block. A useful ordering keeps the largest level small and avoids abrupt changes between neighboring level sizes. Before factorization, check that every nonzero lies in a diagonal block or one of the two neighboring block bands.

Local changes permit a second form of reuse. Suppose block j is the first one that changes. The factors before it remain valid, but the connection between blocks $j-1$ and j touches the changed part. Keep the earlier factors, recompute this connection, and factor blocks j through the end. Keeping the old connection would give the wrong Schur complement.

Remark 2.15 **(Choosing the reuse method)** Use the stored factor directly when only the right-hand side changes. Use Sherman–Morrison or Woodbury when a few rows and columns change throughout the matrix. Recompute from the first changed block when the change begins late in the block order. Compute a fresh factorization when the change can no longer be described by a few vectors, the small Woodbury matrix becomes too sensitive to roundoff, or the block order no longer fits the matrix.

Remark 2.16 **(Implementation details)** A reusable solver should provide ordinary, transpose, and multiple-right-hand-side solves through the same interface. It should retain work arrays instead of allocating them inside every solve. It should compare the corrected solution with the original equations by a residual. Finally, it should report whether it reused a factorization or computed a new one. These details determine whether the algebraic shortcut reduces actual running time.

2.5.7 Further Exercises

Ex 2.30 **Changed rows and columns.** Work through Prop 2.25 for two changed indices. Construct the two factors with at most four columns, show explicitly how the overlap of the changed rows and columns is counted once, and identify a case in which the actual rank is smaller than four.

Ex 2.31 **Low-rank solve versus refactorization.** Let $\mathbf{A}$ be a diagonally dominant dense matrix and change k rows and columns. Compare a fresh LU factorization with a Woodbury correction for several values of n and k. Report the initial cost, the time per right-hand side, the relative residual, and the difference from a fresh solve. Find how many right-hand sides are needed before reuse saves time overall.

Ex 2.32 **A two-block Cholesky solve.** Consider

$$\mathbf{D}_1 = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \qquad \mathbf{L}_1 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathbf{D}_2 = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix},$$

with upper block $\mathbf{L}_1^T$. Compute the first Cholesky block, the scaled lower block, and the second Schur complement. Then solve the system with right-hand side $(1, 0, 0, 1)^T$ and check the residual.

Ex 2.33 **Graph-distance blocks.** For the five-point grid graph on a 3×3 lattice, label every vertex by its distance from a corner. List the block sizes and verify that every edge joins equal or adjacent levels. Repeat with the center vertex as the seed and compare the largest block.

Ex 2.34 **Woodbury or partial refactorization.** Construct a sequence of block-tridiagonal matrices in which the final two blocks change at every step. Implement both a low-rank correction and a refactorization that begins at the first changed block. Compare their cost and residual as the block size and update rank vary. Use the operation counts in this chapter to explain when one method becomes faster than the other.

Read **Further reading.** Golub and Van Loan give the standard dense matrix treatment, while Davis develops the sparse direct viewpoint (Golub and Van Loan, 2013; Davis, 2006). Demmel and Higham are useful guides to deciding when an algebraically correct update may still be unsafe in floating-point arithmetic (Demmel, 1997; Higham, 2002).

3 Orthogonality and Least Squares

Orthogonal transformations preserve Euclidean lengths. We use them to construct bases and to solve systems whose equations cannot all be satisfied. The connection is projection: a least-squares fit is the point in a column space closest to the data.

3.1 Orthogonal Transformations and the QR Factorization

Changing to an orthonormal basis preserves Euclidean distances. This lets us simplify a matrix without magnifying the errors already present in a vector. We construct such changes of basis by reflections and by subtracting projections.

The resulting QR factorization records each original column in an orthonormal basis. Its triangular factor records the coefficients of those columns. The same construction will turn least squares into a triangular solve.

3.1.1 Orthogonality, Isometry, and Invariance

Def 3.1 **(Orthogonal Matrix)** A square matrix $\mathbf{Q} \in \mathbb{R}^{n\times n}$ is termed **orthogonal** if its columns form an orthonormal basis for $\mathbb{R}^n$. Equivalently,

$$\mathbf{Q}^T\mathbf{Q} = \mathbf{Q}\mathbf{Q}^T = \mathbf{I}_n \iff \mathbf{Q}^{-1} = \mathbf{Q}^T.$$

For rectangular matrices $\mathbf{Q} \in \mathbb{R}^{m\times n}$ with $m \geq n$, we say that $\mathbf{Q}$ has orthonormal columns if $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_n$.

Thm 3.1 **(Geometric Invariance Properties of Orthogonal Transformations)** Let $\mathbf{Q} \in \mathbb{R}^{n\times n}$ be an orthogonal matrix. For all vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, the transformation $\boldsymbol{x} \mapsto \mathbf{Q}\boldsymbol{x}$ satisfies the following properties:

1. **Inner Product Preservation:** $\langle \mathbf{Q}\boldsymbol{x}, \mathbf{Q}\boldsymbol{y}\rangle = (\mathbf{Q}\boldsymbol{x})^T(\mathbf{Q}\boldsymbol{y}) = \boldsymbol{x}^T\boldsymbol{y} = \langle \boldsymbol{x}, \boldsymbol{y}\rangle$.
2. **Isometry (Length Preservation):** $\|\mathbf{Q}\boldsymbol{x}\|_2 = \|\boldsymbol{x}\|_2$.
3. **Matrix 2-Norm and Conditioning:** $\|\mathbf{Q}\|_2 = 1$ and $\kappa_2(\mathbf{Q}) = \|\mathbf{Q}\|_2\|\mathbf{Q}^{-1}\|_2 = 1$.
4. **Determinant:** $\det(\mathbf{Q}) = \pm 1$, where the sign records whether orientation is preserved or reversed. In higher dimensions an orientation-reversing orthogonal map need not be a single hyperplane reflection.

Proof We establish the claims in the order in which they depend on one another.

1. The transpose of $\mathbf{Q}\boldsymbol{x}$ is $\boldsymbol{x}^T\mathbf{Q}^T$. Hence

   $$(\mathbf{Q}\boldsymbol{x})^T(\mathbf{Q}\boldsymbol{y}) = \boldsymbol{x}^T(\mathbf{Q}^T\mathbf{Q})\boldsymbol{y} = \boldsymbol{x}^T\mathbf{I}\boldsymbol{y} = \boldsymbol{x}^T\boldsymbol{y}.$$

   The middle equality is exactly where orthogonality is used.

2. Set $\boldsymbol{y} = \boldsymbol{x}$. Then $\|\mathbf{Q}\boldsymbol{x}\|_2^2 = \|\boldsymbol{x}\|_2^2$. Norms are nonnegative, so taking square roots gives equality of lengths.

3. Every Euclidean unit input has a Euclidean unit output. The maximum output length is therefore one. Since $\mathbf{Q}^{-1} = \mathbf{Q}^T$ and $\mathbf{Q}^T$ is also orthogonal, both factors in $\kappa_2(\mathbf{Q}) = \|\mathbf{Q}\|_2\|\mathbf{Q}^{-1}\|_2$ equal one.

4. Take determinants in $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$. Multiplicativity and invariance under transpose give

$$1 = \det(\mathbf{I}) = \det(\mathbf{Q}^T)\det(\mathbf{Q}) = (\det \mathbf{Q})^2.$$

The determinant is real, so it is 1 or -1. These signs distinguish preservation and reversal of orientation.

□

3.1.2 The QR Factorization

For two independent columns, begin with $\boldsymbol{q}_1 = \boldsymbol{a}_1/\|\boldsymbol{a}_1\|_2$. Subtract the projection of $\boldsymbol{a}_2$ onto $\boldsymbol{q}_1$ and normalize what remains to obtain $\boldsymbol{q}_2$. Thus

$$\boldsymbol{a}_1 = r_{11}\boldsymbol{q}_1, \qquad \boldsymbol{a}_2 = r_{12}\boldsymbol{q}_1 + r_{22}\boldsymbol{q}_2.$$

Writing these two relations side by side gives a matrix product with an upper triangular factor. Each additional independent column adds one new basis vector in the same way.

Def 3.2 **(Full and Thin QR Factorizations)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $m \geq n$.

1. The **Full QR Factorization** is given by

$$\mathbf{A} = \mathbf{QR} = \begin{pmatrix}\mathbf{Q}_1 & \mathbf{Q}_2\end{pmatrix}\begin{pmatrix}\mathbf{R}_1 \\ \mathbf{0}\end{pmatrix},$$

where $\mathbf{Q} \in \mathbb{R}^{m\times m}$ is orthogonal, $\mathbf{Q}_1 \in \mathbb{R}^{m\times n}$, $\mathbf{Q}_2 \in \mathbb{R}^{m\times(m-n)}$, and $\mathbf{R}_1 \in \mathbb{R}^{n\times n}$ is upper triangular.

2. The **Thin (Economy) QR Factorization** retains only the first n columns:

$$\mathbf{A} = \mathbf{Q}_1\mathbf{R}_1,$$

where $\mathbf{Q}_1 \in \mathbb{R}^{m\times n}$ satisfies $\mathbf{Q}_1^T\mathbf{Q}_1 = \mathbf{I}_n$ and $\mathbf{R}_1 \in \mathbb{R}^{n\times n}$ is upper triangular.

If $\mathbf{A}$ has full column rank, the thin QR factorization is unique when the diagonal entries of $\mathbf{R}_1$ are chosen to be strictly positive.

Ex 3.1 Suppose $\mathbf{A} = \mathbf{Q}_1\mathbf{R}_1 = \widehat{\mathbf{Q}}_1\widehat{\mathbf{R}}_1$ has full column rank and both triangular factors have positive diagonal. Show that $\mathbf{Q}_1^T\widehat{\mathbf{Q}}_1 = \mathbf{R}_1\widehat{\mathbf{R}}_1^{-1}$ is orthogonal and upper triangular. Prove that such a matrix with positive diagonal must be the identity, and deduce uniqueness.

Remark 3.1 **(Subspace identification via QR)** When $\mathbf{A}$ has full column rank, the columns of $\mathbf{Q}_1$ form an orthonormal basis for the column space (range) $\mathcal{R}(\mathbf{A})$, while the columns of $\mathbf{Q}_2$ form an orthonormal basis for the orthogonal complement $\mathcal{R}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^T)$. For overdetermined least-squares problems where $m \gg n$, the thin factorization requires less storage ($O(mn)$ vs $O(m^2)$) and is used in practical implementations.

3.1.3 Householder Triangularization

The standard method for computing the QR factorization of a dense matrix is the Householder reflection technique. At each step, a Householder reflector is constructed to zero out all subdiagonal entries of a column simultaneously (Householder, 1958).

A reflection leaves the component parallel to a mirror unchanged and reverses the perpendicular component. If $\boldsymbol{v}$ is normal to the mirror, that perpendicular component of $\boldsymbol{y}$ is $\boldsymbol{v}(\boldsymbol{v}^T\boldsymbol{y})/(\boldsymbol{v}^T\boldsymbol{v})$. Subtracting twice this component gives the formula below.

Def 3.3 **(Householder Reflector)** Let $\boldsymbol{v} \in \mathbb{R}^m$ be a nonzero vector. The Householder reflector associated with $\boldsymbol{v}$ is the matrix

$$\mathbf{H} = \mathbf{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}}.$$

Geometrically, $\mathbf{H}$ reflects any vector across the hyperplane orthogonal to $\boldsymbol{v}$. Every Householder reflector is symmetric ($\mathbf{H}^T = \mathbf{H}$) and orthogonal ($\mathbf{H}^T\mathbf{H} = \mathbf{H}^2 = \mathbf{I}$).

Ex 3.2 Let $\mathbf{P} = \boldsymbol{v}\boldsymbol{v}^T/(\boldsymbol{v}^T\boldsymbol{v})$. Show that $\mathbf{P}^T = \mathbf{P}$ and $\mathbf{P}^2 = \mathbf{P}$, and deduce that $(\mathbf{I} - 2\mathbf{P})^2 = \mathbf{I}$. Determine the action of the reflector on $\operatorname{span}\{\boldsymbol{v}\}$ and its orthogonal complement. If the column to be reduced is zero, explain why no reflector is needed.

Thm 3.2 **(Householder Vector Construction)** Given a nonzero vector $\boldsymbol{x} \in \mathbb{R}^m$, we wish to find a reflector $\mathbf{H}$ that aligns $\boldsymbol{x}$ with the first canonical coordinate axis: $\mathbf{H}\boldsymbol{x} = \pm\|\boldsymbol{x}\|_2\boldsymbol{e}_1$. The vector $\boldsymbol{v}$ is chosen as

$$\boldsymbol{v} = \boldsymbol{x} + \operatorname{sign}(x_1)\|\boldsymbol{x}\|_2\boldsymbol{e}_1,$$

where $\operatorname{sign}(x_1) = 1$ if $x_1 \geq 0$ and -1 if $x_1 < 0$. Choosing the sign of the first component ensures that the two added terms have identical signs and avoids cancellation when the first component of $\boldsymbol{v}$ is formed.

Proof We want $\mathbf{H}\boldsymbol{x} = \alpha\boldsymbol{e}_1$ with $\mathbf{H}$ orthogonal. Orthogonality preserves the norm, so any possible α must satisfy $|\alpha| = \|\boldsymbol{x}\|_2$. Choose $\alpha = -\operatorname{sign}(x_1)\|\boldsymbol{x}\|_2$ with the convention in the statement, and set $\boldsymbol{v} = \boldsymbol{x} - \alpha\boldsymbol{e}_1$.

This vector is nonzero. Its first entry is $x_1 + \operatorname{sign}(x_1)\|\boldsymbol{x}\|_2$, whose magnitude is $|x_1| + \|\boldsymbol{x}\|_2 > 0$. The reflector defined from $\boldsymbol{v}$ is therefore well defined.

Calculate its denominator before applying it:

$$\begin{aligned}\boldsymbol{v}^T\boldsymbol{v} &= (\boldsymbol{x} - \alpha\boldsymbol{e}_1)^T(\boldsymbol{x} - \alpha\boldsymbol{e}_1)\\ &= \|\boldsymbol{x}\|_2^2 - 2\alpha x_1 + \alpha^2\\ &= 2\|\boldsymbol{x}\|_2^2 - 2\alpha x_1 = 2(\boldsymbol{x} - \alpha\boldsymbol{e}_1)^T\boldsymbol{x} = 2\boldsymbol{v}^T\boldsymbol{x}.\end{aligned}$$

The third line uses $\alpha^2 = \|\boldsymbol{x}\|_2^2$. Since $\boldsymbol{v} \neq 0$, the denominator is positive. Substitute the identity into the reflector action:

$$\mathbf{H}\boldsymbol{x} = \boldsymbol{x} - 2\boldsymbol{v}\frac{\boldsymbol{v}^T\boldsymbol{x}}{\boldsymbol{v}^T\boldsymbol{v}} = \boldsymbol{x} - \boldsymbol{v} = \alpha\boldsymbol{e}_1.$$

Thus all entries except the first are zero.

The chosen sign also makes the two terms forming v_1 have the same sign. It avoids subtracting two nearly equal magnitudes, which could lose relative accuracy in the vector defining the reflector. The opposite choice of α can work algebraically when its vector is nonzero, but lacks this numerical protection (Higham, 2002; Golub and Van Loan, 2013). □

Ex 3.3 Construct the Householder reflector $\mathbf{H}$ that transforms the vector $\boldsymbol{x} = (3, 4)^T$ into $\alpha \boldsymbol{e}_1 = (-5, 0)^T$. Explicitly write out the 2×2 matrix $\mathbf{H}$ and verify that $\mathbf{H} = \mathbf{H}^T$ and $\mathbf{H}^2 = \mathbf{I}$.

Remark 3.2 **(Matrix-free application and complexity)** In numerical algorithms, a Householder matrix $\mathbf{H}$ is usually stored through its defining vector rather than as an explicit $m \times m$ array. To apply $\mathbf{H}$ to a vector $\boldsymbol{y}$, we compute

$$\mathbf{H}\boldsymbol{y} = \boldsymbol{y} - \left(\frac{2\boldsymbol{v}^T \boldsymbol{y}}{\boldsymbol{v}^T \boldsymbol{v}} \right) \boldsymbol{v},$$

which requires only an inner product and a vector update costing $O(m)$ flops. For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, Householder QR requires $2mn^2 - \frac{2}{3}n^3$ flops.

3.1.4 Gram–Schmidt Orthogonalization

The Gram–Schmidt process constructs the thin QR factorization column by column by projecting each column of $\mathbf{A}$ onto the orthogonal complement of the previously computed basis vectors.

Thm 3.3 **(Classical and Modified Gram–Schmidt Algorithms)** Let $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n \in \mathbb{R}^m$ be linearly independent columns of $\mathbf{A}$. This assumption ensures that every vector to be normalized below is nonzero in exact arithmetic.

1. **Classical Gram–Schmidt (CGS):** For each column $k = 1, \ldots, n$, compute all projections against the original vector simultaneously:

$$r_{ik} = \boldsymbol{q}_i^T \boldsymbol{a}_k \quad (i = 1, \ldots, k-1), \qquad \boldsymbol{v}_k = \boldsymbol{a}_k - \sum_{i=1}^{k-1} r_{ik} \boldsymbol{q}_i, \qquad r_{kk} = \|\boldsymbol{v}_k\|_2, \qquad \boldsymbol{q}_k = \frac{\boldsymbol{v}_k}{r_{kk}}.$$

2. **Modified Gram–Schmidt (MGS):** For each column k, subtract projections sequentially from the partially updated vector:

$$\boldsymbol{v}_k^{(1)} = \boldsymbol{a}_k, \qquad \boldsymbol{v}_k^{(i+1)} = \boldsymbol{v}_k^{(i)} - (\boldsymbol{q}_i^T \boldsymbol{v}_k^{(i)}) \boldsymbol{q}_i \quad (i = 1, \ldots, k-1), \qquad \boldsymbol{q}_k = \frac{\boldsymbol{v}_k^{(k)}}{\|\boldsymbol{v}_k^{(k)}\|_2}.$$

Ex 3.4 Rearrange the classical Gram–Schmidt update to express $\boldsymbol{a}_k$ in terms of $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$. Assemble these relations to prove $\mathbf{A} = \mathbf{Q}_1 \mathbf{R}_1$. Explain what a zero diagonal entry would tell you about the original columns. In modified Gram–Schmidt, which inner products must be saved to construct $\mathbf{R}_1$?

Ex 3.5 Compute the thin QR factorization of $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$ using Modified Gram-Schmidt.

Remark 3.3 **(Stability hierarchy among orthogonalization methods)** In exact arithmetic, CGS and MGS produce the same orthonormal basis, up to signs. In floating-point arithmetic their organization matters:

1. Classical Gram–Schmidt can lose orthogonality when the columns are nearly dependent.
2. Modified Gram–Schmidt retains more orthogonality because each projection is taken from the current residual rather than from the original column.
3. Householder QR is backward stable and normally produces a computed $\mathbf{Q}$ that is orthogonal to working precision.

Precise bounds depend on dimension, scaling, and the condition of the column space. Reorthogonalization can make Gram–Schmidt reliable when an explicit basis is required.

3.1.5 Orthogonal Projectors

Def 3.4 **(Orthogonal Projector)** Let $S \subset \mathbb{R}^m$ be a subspace with orthonormal basis columns stored in $\mathbf{Q}_1 \in \mathbb{R}^{m\times n}$. The orthogonal projector onto S is the matrix

$$\mathbf{P} = \mathbf{Q}_1\mathbf{Q}_1^T.$$

Every orthogonal projector is symmetric ($\mathbf{P}^T = \mathbf{P}$) and idempotent ($\mathbf{P}^2 = \mathbf{P}$). The complementary projector $\mathbf{P}^\perp = \mathbf{I} - \mathbf{P}$ projects onto the orthogonal complement $S^\perp$.

Thm 3.4 **(Symmetric Idempotents are Orthogonal Projectors)** A square matrix $\mathbf{P}$ is the orthogonal projector onto its range if and only if

$$\mathbf{P}^T = \mathbf{P}, \qquad \mathbf{P}^2 = \mathbf{P}.$$

For such a matrix, every vector has the orthogonal decomposition

$$\boldsymbol{x} = \mathbf{P}\boldsymbol{x} + (\mathbf{I} - \mathbf{P})\boldsymbol{x},$$

with the first term in $\mathcal{R}(\mathbf{P})$ and the second in $\mathcal{N}(\mathbf{P})$.

Proof We prove both directions.

1. Assume $\mathbf{P}^T = \mathbf{P}$ and $\mathbf{P}^2 = \mathbf{P}$. For any $\boldsymbol{x}$,

   $$\boldsymbol{x} = \mathbf{P}\boldsymbol{x} + (\mathbf{I} - \mathbf{P})\boldsymbol{x}.$$

   The first term belongs to the range by definition. The second belongs to the null space because

   $$\mathbf{P}(\mathbf{I} - \mathbf{P})\boldsymbol{x} = (\mathbf{P} - \mathbf{P}^2)\boldsymbol{x} = \mathbf{0}.$$

   If $\boldsymbol{y} = \mathbf{P}\boldsymbol{u}$ is any range vector and $\boldsymbol{z}$ is any null-space vector, then

   $$\boldsymbol{y}^T\boldsymbol{z} = \boldsymbol{u}^T\mathbf{P}^T\boldsymbol{z} = \boldsymbol{u}^T\mathbf{P}\boldsymbol{z} = 0.$$

   Symmetry was needed to replace $\mathbf{P}^T$ by $\mathbf{P}$. Thus the two components are perpendicular.

2. For any $\boldsymbol{s}$ in the range, $\mathbf{P}\boldsymbol{x} - \boldsymbol{s}$ also lies in the range. Consequently

   $$\|\boldsymbol{x} - \boldsymbol{s}\|_2^2 = \|(\mathbf{I} - \mathbf{P})\boldsymbol{x}\|_2^2 + \|\mathbf{P}\boldsymbol{x} - \boldsymbol{s}\|_2^2.$$

   The unique minimum occurs at $\boldsymbol{s} = \mathbf{P}\boldsymbol{x}$.

3. If $\mathbf{P}$ is the orthogonal projector onto a subspace with orthonormal basis $\mathbf{Q}_1$, then $\mathbf{P} = \mathbf{Q}_1\mathbf{Q}_1^T$. Direct multiplication gives

$$\mathbf{P}^T = (\mathbf{Q}_1\mathbf{Q}_1^T)^T = \mathbf{Q}_1\mathbf{Q}_1^T = \mathbf{P},$$
$$\mathbf{P}^2 = \mathbf{Q}_1(\mathbf{Q}_1^T\mathbf{Q}_1)\mathbf{Q}_1^T = \mathbf{Q}_1\mathbf{Q}_1^T = \mathbf{P}.$$

□

Ex 3.6 Let $\mathbf{Q}$ and $\widehat{\mathbf{Q}}$ have orthonormal columns spanning the same subspace. Prove that $\mathbf{Q}\mathbf{Q}^T = \widehat{\mathbf{Q}}\widehat{\mathbf{Q}}^T$. Thus projection depends on the subspace, not on the chosen orthonormal basis.

3.1.6 Further Exercises

Ex 3.7 An idempotent matrix need not be an orthogonal projector. For $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, identify the range and null space, sketch its action, and explain the role of symmetry in Thm. 3.4.

Ex 3.8 Explain why the condition number of any orthogonal matrix is equal to 1. What does this imply about the propagation of errors when applying orthogonal transformations?

3.2 Linear Least Squares

Suppose the columns of $\mathbf{A}$ describe the responses available to a model. Every prediction has the form $\mathbf{A}\boldsymbol{x}$ and therefore lies in the column space of $\mathbf{A}$. If an observation $\boldsymbol{b}$ lies outside that space, no choice of parameters can fit it exactly. Least squares asks for the point in the column space closest to the observation.

The Euclidean criterion adds the squares of the individual errors. Squaring prevents positive and negative errors from cancelling and gives greater weight to larger errors. Other norms express different choices about how errors should be measured.

3.2.1 The Geometric Problem

Def 3.5 **(Linear Least-Squares Problem)** For $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$, the linear least-squares problem is

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 = \min_{\boldsymbol{x}\in\mathbb{R}^n} \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij}x_j - b_i \right)^2 .$$

The residual associated with $\boldsymbol{x}$ is $\boldsymbol{r}(\boldsymbol{x}) = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}$.

Example Suppose three measurements of one constant are 1, 2, and 6. A fitted constant c predicts $(c, c, c)^T$, so the squared error is

$$(c-1)^2 + (c-2)^2 + (c-6)^2 = 3(c-3)^2 + 14.$$

It is smallest at $c = 3$. The residual $(-2, -1, 3)^T$ is perpendicular to $(1, 1, 1)^T$, the direction in which the predictions can move. The residual itself is nonzero because the observations disagree.

Ex 3.9 Replace the three observations by $b_1, \dots, b_m$. Complete the square to show that their mean gives the best fitted constant. Show that the residual entries sum to zero. Why would minimizing the sum of the errors without squaring them fail to define the same problem?

For a general matrix, changing coefficient x_j moves the fitted vector along column $\boldsymbol{a}_j$. At a closest point, movement in either direction along that column cannot decrease the squared distance. This requires the residual to be perpendicular to every column. The word *normal* in the next theorem means perpendicular.

Thm 3.5 **(Orthogonal Characterization and Normal Equations)** A vector $\widehat{\boldsymbol{x}}$ is a least-squares minimizer if and only if its residual is orthogonal to every column of $\mathbf{A}$:

$$\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\widehat{\boldsymbol{x}} \perp \mathcal{R}(\mathbf{A}).$$

Equivalently,

$$\mathbf{A}^T\boldsymbol{r} = \mathbf{0}, \qquad \mathbf{A}^T\mathbf{A}\widehat{\boldsymbol{x}} = \mathbf{A}^T\boldsymbol{b}.$$

These are the **normal equations**. A minimizer is unique exactly when the columns of $\mathbf{A}$ are linearly independent.

Proof Decompose $\boldsymbol{b} - \mathbf{A}\boldsymbol{x}$ by inserting the candidate fitted vector $\mathbf{A}\widehat{\boldsymbol{x}}$:

$$\boldsymbol{b} - \mathbf{A}\boldsymbol{x} = (\boldsymbol{b} - \mathbf{A}\widehat{\boldsymbol{x}}) + \mathbf{A}(\widehat{\boldsymbol{x}} - \boldsymbol{x}).$$

If $\boldsymbol{b} - \mathbf{A}\widehat{\boldsymbol{x}} \perp \mathcal{R}(\mathbf{A})$, then since $\mathbf{A}(\widehat{\boldsymbol{x}} - \boldsymbol{x}) \in \mathcal{R}(\mathbf{A})$, the two summands are orthogonal:

$$(\boldsymbol{b} - \mathbf{A}\widehat{\boldsymbol{x}})^T\mathbf{A}(\widehat{\boldsymbol{x}} - \boldsymbol{x}) = 0.$$

The Pythagorean identity therefore yields

$$\begin{aligned}\|\boldsymbol{b} - \mathbf{A}\boldsymbol{x}\|_2^2 &= \|(\boldsymbol{b} - \mathbf{A}\widehat{\boldsymbol{x}}) + \mathbf{A}(\widehat{\boldsymbol{x}} - \boldsymbol{x})\|_2^2 \\ &= \|\boldsymbol{b} - \mathbf{A}\widehat{\boldsymbol{x}}\|_2^2 + \|\mathbf{A}(\widehat{\boldsymbol{x}} - \boldsymbol{x})\|_2^2 \geq \|\boldsymbol{b} - \mathbf{A}\widehat{\boldsymbol{x}}\|_2^2.\end{aligned}$$

Equality holds if and only if $\|\mathbf{A}(\widehat{\boldsymbol{x}} - \boldsymbol{x})\|_2^2 = 0$, meaning $\mathbf{A}(\widehat{\boldsymbol{x}} - \boldsymbol{x}) = \mathbf{0}$. Thus $\widehat{\boldsymbol{x}}$ minimizes the residual norm.

This proves sufficiency: a candidate with perpendicular residual gives a residual no larger than that of any competitor. We now prove necessity without assuming beforehand that the candidate has this orthogonality.

Conversely, if $\widehat{\boldsymbol{x}}$ is a minimizer, define for any $\boldsymbol{v} \in \mathbb{R}^n$ and $t \in \mathbb{R}$ the scalar function

$$\begin{aligned}\phi(t) &= \|\boldsymbol{b} - \mathbf{A}(\widehat{\boldsymbol{x}} + t\boldsymbol{v})\|_2^2 \\ &= \|\boldsymbol{r} - t\mathbf{A}\boldsymbol{v}\|_2^2 \\ &= \|\boldsymbol{r}\|_2^2 - 2t\,\boldsymbol{v}^T\mathbf{A}^T\boldsymbol{r} + t^2\|\mathbf{A}\boldsymbol{v}\|_2^2,\end{aligned}$$

where $\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\widehat{\boldsymbol{x}}$. Because ϕ has a global minimum at $t = 0$,

$$\phi'(0) = -2\boldsymbol{v}^T\mathbf{A}^T\boldsymbol{r} = 0 \quad \text{for all } \boldsymbol{v} \in \mathbb{R}^n.$$

Choosing $\boldsymbol{v} = \boldsymbol{e}_j$ for $j = 1, \ldots, n$ gives $\mathbf{A}^T\boldsymbol{r} = \mathbf{0}$, which means $\boldsymbol{r} \perp \mathcal{R}(\mathbf{A})$. Expanding $\mathbf{A}^T(\boldsymbol{b} - \mathbf{A}\widehat{\boldsymbol{x}}) = \mathbf{0}$ yields the normal equations

$$\mathbf{A}^T\mathbf{A}\widehat{\boldsymbol{x}} = \mathbf{A}^T\boldsymbol{b}.$$

A minimizer exists because the column space is a finite-dimensional subspace, so the orthogonal projection of $\boldsymbol{b}$ onto it exists. By definition of that column space, some coefficient vector produces the projected point. The orthogonality argument just proved makes that vector a minimizer. Existence of a fitted point does not yet imply uniqueness of its coefficients.

For uniqueness, suppose $\widehat{\boldsymbol{x}}_1$ and $\widehat{\boldsymbol{x}}_2$ are both minimizers. Then $\mathbf{A}(\widehat{\boldsymbol{x}}_1 - \widehat{\boldsymbol{x}}_2) = \mathbf{0}$, so $\widehat{\boldsymbol{x}}_1 - \widehat{\boldsymbol{x}}_2 \in \mathcal{N}(\mathbf{A})$. When $\mathbf{A}$ has full column rank, $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$, forcing $\widehat{\boldsymbol{x}}_1 = \widehat{\boldsymbol{x}}_2$. If $\boldsymbol{z} \in \mathcal{N}(\mathbf{A}) \setminus \{\mathbf{0}\}$, then $\mathbf{A}(\widehat{\boldsymbol{x}} + \boldsymbol{z}) = \mathbf{A}\widehat{\boldsymbol{x}}$, so $\widehat{\boldsymbol{x}} + \boldsymbol{z}$ is also a minimizer.

The fitted vector is unique even when the coefficient vector is not. Indeed two minimizers have residuals perpendicular to the range, and their fitted-vector difference lies in that range and its orthogonal complement, so it is zero. When a nonzero null vector exists, adding any scalar multiple preserves this common fitted vector and produces distinct coefficients. □

Remark 3.4 The same condition follows by differentiating one coordinate at a time. For $f(\boldsymbol{x}) = \frac{1}{2}\|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2$, hold every coordinate except x_j fixed. Then

$$\frac{\partial f}{\partial x_j} = \boldsymbol{a}_j^T(\mathbf{A}\boldsymbol{x} - \boldsymbol{b}).$$

The vector of these partial derivatives is called the gradient. Its vanishing gives the normal equations. The proof above also establishes sufficiency, which does not follow merely from setting derivatives to zero for an arbitrary function.

Remark 3.5 **(Conditioning of Normal Equations)** When $\mathbf{A}$ has full column rank, $\mathbf{A}^T\mathbf{A}$ is SPD because

$$\boldsymbol{x}^T\mathbf{A}^T\mathbf{A}\boldsymbol{x} = \|\mathbf{A}\boldsymbol{x}\|_2^2 > 0$$

for nonzero $\boldsymbol{x}$. This proves that the normal equations have a unique solution.

The normal equations replace the rectangular problem by a square one, but forming $\mathbf{A}^T\mathbf{A}$ hides the geometry and worsens sensitivity. QR instead chooses orthonormal coordinates adapted to the column space.

Thm 3.6 **(Least-Squares Solution by QR)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have full column rank and thin QR factorization $\mathbf{A} = \mathbf{Q}_1\mathbf{R}_1$. Then the unique least-squares solution is obtained from

$$\mathbf{R}_1\widehat{\boldsymbol{x}} = \mathbf{Q}_1^T\boldsymbol{b}.$$

The fitted vector and residual are

$$\widehat{\boldsymbol{b}} = \mathbf{Q}_1\mathbf{Q}_1^T\boldsymbol{b}, \qquad \boldsymbol{r} = (\mathbf{I} - \mathbf{Q}_1\mathbf{Q}_1^T)\boldsymbol{b}.$$

Proof Let $\mathbf{Q} = (\mathbf{Q}_1 \;\; \mathbf{Q}_2) \in \mathbb{R}^{m\times m}$ be orthogonal, where $\mathbf{Q}_1 \in \mathbb{R}^{m\times n}$ and $\mathbf{Q}_2 \in \mathbb{R}^{m\times(m-n)}$. Since

$\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$, the Euclidean norm is orthogonally invariant:

$$\begin{aligned}\|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 &= \|\mathbf{Q}^T(\mathbf{A}\boldsymbol{x} - \boldsymbol{b})\|_2^2 \\ &= \left\| \begin{pmatrix} \mathbf{Q}_1^T \\ \mathbf{Q}_2^T \end{pmatrix} (\mathbf{Q}_1\mathbf{R}_1\boldsymbol{x} - \boldsymbol{b}) \right\|_2^2 \\ &= \left\| \begin{pmatrix} \mathbf{R}_1\boldsymbol{x} - \mathbf{Q}_1^T\boldsymbol{b} \\ -\mathbf{Q}_2^T\boldsymbol{b} \end{pmatrix} \right\|_2^2 \\ &= \|\mathbf{R}_1\boldsymbol{x} - \mathbf{Q}_1^T\boldsymbol{b}\|_2^2 + \|\mathbf{Q}_2^T\boldsymbol{b}\|_2^2.\end{aligned}$$

The columns $\mathbf{Q}_1$ span the fitted subspace. The columns $\mathbf{Q}_2$ span its perpendicular complement. In these coordinates, only the first block can change when $\boldsymbol{x}$ changes, because $\mathbf{Q}_2^T\mathbf{A} = \mathbf{0}$.

The second term $\|\mathbf{Q}_2^T\boldsymbol{b}\|_2^2$ is fixed and independent of $\boldsymbol{x}$. Because $\mathbf{A}$ has full column rank, $\mathbf{R}_1$ is non-singular, and setting $\mathbf{R}_1\widehat{\boldsymbol{x}} = \mathbf{Q}_1^T\boldsymbol{b}$ reduces the first term to zero, attaining the minimum value $\|\mathbf{Q}_2^T\boldsymbol{b}\|_2^2$. The fitted vector and residual follow:

$$\begin{aligned}\widehat{\boldsymbol{b}} &= \mathbf{A}\widehat{\boldsymbol{x}} = \mathbf{Q}_1\mathbf{R}_1(\mathbf{R}_1^{-1}\mathbf{Q}_1^T\boldsymbol{b}) = \mathbf{Q}_1\mathbf{Q}_1^T\boldsymbol{b}, \\ \boldsymbol{r} &= \boldsymbol{b} - \widehat{\boldsymbol{b}} = (\mathbf{I} - \mathbf{Q}_1\mathbf{Q}_1^T)\boldsymbol{b}.\end{aligned}$$

The triangular matrix $\mathbf{R}_1$ is nonsingular because $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ would otherwise hold for a nonzero vector in its null space. Thus the triangular system has one solution and sets the adjustable block exactly to zero. This proves uniqueness without forming $\mathbf{A}^T\mathbf{A}$. It is also the algebraic reason for the QR method's use in least squares (Golub and Van Loan, 2013). □

Ex 3.10 Let $\mathbf{W}$ be SPD. Weighted least squares minimizes $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_{\mathbf{W}}^2$. Prove that the residual is $\mathbf{W}$-orthogonal to $\mathcal{R}(\mathbf{A})$ and derive $\mathbf{A}^T\mathbf{W}\mathbf{A}\boldsymbol{x} = \mathbf{A}^T\mathbf{W}\boldsymbol{b}$. Explain what statistical or physical information the weight matrix expresses.

Remark 3.6 **(Geometry vs. Conditioning in Householder QR)** For a matrix with full column rank, Thm 4.19 proves $\kappa_2(\mathbf{A}^T\mathbf{A}) = \kappa_2(\mathbf{A})^2$. Householder QR avoids forming $\mathbf{A}^T\mathbf{A}$ and is the safer dense default. The normal equations remain useful when special structure, cost, or repeated updates make the trade worthwhile.

Example Fitting $y = c_0 + c_1 t$ to $(0, 1)$, $(1, 2)$, and $(2, 4)$ gives

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}.$$

The solution is $c_0 = 5/6$ and $c_1 = 3/2$. The residual is orthogonal to both the constant column and the time column. Hence its entries sum to zero and have zero first moment with respect to the sample times.

3.2.2 Rank-Deficient Problems

Thm 3.7 **(All Minimizers and the Minimum-Norm Choice)** Let $\widehat{\boldsymbol{x}}$ be any least-squares minimizer. Then every minimizer has the form

$$\widehat{\boldsymbol{x}} + \boldsymbol{z}, \qquad \boldsymbol{z} \in \mathcal{N}(\mathbf{A}).$$

Among them there is exactly one vector of smallest Euclidean norm. It is characterized by

$$\widehat{\boldsymbol{x}}_{\min} \perp \mathcal{N}(\mathbf{A}), \qquad \text{equivalently} \qquad \widehat{\boldsymbol{x}}_{\min} \in \mathcal{R}(\mathbf{A}^T).$$

Proof Let $\widehat{\boldsymbol{x}}$ be any minimizer. Any two minimizers have the same fitted vector by the preceding characterization. Their difference is therefore in the null space. This justifies describing every minimizer by adding a null-space vector to one chosen minimizer.

For any $\boldsymbol{z} \in \mathcal{N}(\mathbf{A})$, $\mathbf{A}(\widehat{\boldsymbol{x}}+\boldsymbol{z}) = \mathbf{A}\widehat{\boldsymbol{x}}$, so $\widehat{\boldsymbol{x}}+\boldsymbol{z}$ is also a minimizer. Conversely, if $\boldsymbol{x}'$ is any minimizer, then $\mathbf{A}\boldsymbol{x}' = \mathbf{A}\widehat{\boldsymbol{x}}$ by uniqueness of the orthogonal projection $\widehat{\boldsymbol{b}}$, which gives $\mathbf{A}(\boldsymbol{x}' - \widehat{\boldsymbol{x}}) = \mathbf{0}$, so $\boldsymbol{x}' - \widehat{\boldsymbol{x}} \in \mathcal{N}(\mathbf{A})$.

Every vector decomposes uniquely as $\boldsymbol{x} = \boldsymbol{x}_\perp + \boldsymbol{x}_0$ with $\boldsymbol{x}_0 \in \mathcal{N}(\mathbf{A})$ and $\boldsymbol{x}_\perp \in \mathcal{N}(\mathbf{A})^\perp$. For any minimizer $\widehat{\boldsymbol{x}} + \boldsymbol{z}$, we have $\boldsymbol{z} \in \mathcal{N}(\mathbf{A})$, so

$$\begin{aligned}\|\widehat{\boldsymbol{x}} + \boldsymbol{z}\|_2^2 &= \|(\widehat{\boldsymbol{x}}_\perp + \widehat{\boldsymbol{x}}_0) + \boldsymbol{z}\|_2^2 \\ &= \|\widehat{\boldsymbol{x}}_\perp + (\widehat{\boldsymbol{x}}_0 + \boldsymbol{z})\|_2^2 \\ &= \|\widehat{\boldsymbol{x}}_\perp\|_2^2 + \|\widehat{\boldsymbol{x}}_0 + \boldsymbol{z}\|_2^2 \geq \|\widehat{\boldsymbol{x}}_\perp\|_2^2.\end{aligned}$$

The minimum is attained uniquely when $\boldsymbol{z} = -\widehat{\boldsymbol{x}}_0$, which yields $\widehat{\boldsymbol{x}}_{\min} = \widehat{\boldsymbol{x}}_\perp \in \mathcal{N}(\mathbf{A})^\perp$. By the fundamental theorem of linear algebra, $\mathcal{N}(\mathbf{A})^\perp = \mathcal{R}(\mathbf{A}^T)$.

The decomposition into $\mathcal{N}(\mathbf{A})$ and $\mathcal{N}(\mathbf{A})^\perp$ is unique. The perpendicular component is therefore shared by all minimizers, and only the null-space component can vary. The displayed sum of squared norms shows directly why removing that component minimizes the norm, rather than merely asserting it from orthogonality. □

Ex 3.11 Suppose $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ is consistent but underdetermined. Prove that the minimum-norm solution is orthogonal to the null space. Explain why adding any invisible null-space component increases the norm without changing the data fit.

If a column combination is nearly zero, a large change in its coefficients produces little change in the predictions. Noisy observations can then determine those coefficients poorly. One way to control this sensitivity is to penalize large coefficient vectors, while allowing a larger residual. The parameter in the following definition specifies the strength of that preference.

Def 3.6 **(Tikhonov Regularization)** For $\lambda > 0$, Tikhonov or ridge regularization solves

$$\min_{\boldsymbol{x}} \left(\|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \lambda \|\boldsymbol{x}\|_2^2 \right).$$

It trades residual fit against parameter size. The problem is equivalently the augmented least-squares problem

$$\min_{\boldsymbol{x}} \left\| \begin{pmatrix} \mathbf{A} \\ \sqrt{\lambda}\mathbf{I} \end{pmatrix} \boldsymbol{x} - \begin{pmatrix} \boldsymbol{b} \\ \mathbf{0} \end{pmatrix} \right\|_2^2,$$

and its normal equations are

$$(\mathbf{A}^T\mathbf{A} + \lambda\mathbf{I})\boldsymbol{x} = \mathbf{A}^T\boldsymbol{b}.$$

Ex 3.12 For the scalar problem $\min_x\{(ax-b)^2+\lambda x^2\}$, derive $x = ab/(a^2+\lambda)$. Compare its sensitivity to b with that of $x = b/a$ when a is small. For a general matrix, prove $\mathbf{A}^T\mathbf{A}+\lambda\mathbf{I}$ is SPD whenever $\lambda > 0$, even if $\mathbf{A}$ has dependent columns.

Remark 3.7 **(Interpretation of Regularization)** Regularization also adds a modeling assumption. Among models with similar fit, it prefers smaller parameter vectors. Whether that assumption is scientifically appropriate depends on the meaning and scaling of the parameters. Theorem 4.23 identifies the directions that regularization damps and the amount by which each is damped.

3.2.3 Further Exercises

Ex 3.13 Starting only from the orthogonal decomposition theorem, rederive the normal equations and prove both directions of Thm. 3.5. Do not use derivatives.

Ex 3.14 Fit a parabola $y = c_0 + c_1 t + c_2 t^2$ to $(-1,1)$, $(0,0)$, $(1,1)$, and $(2,5)$ using thin QR. Verify orthogonality of the residual against each column to working precision.

Ex 3.15 Construct two almost parallel columns in $\mathbb{R}^3$. Compare the condition and numerical orthogonality of solutions obtained from normal equations, modified Gram–Schmidt QR, and Householder QR as the angle between the columns decreases.

Ex 3.16 Rescale one column of a data matrix by 10^6. The set of fitted vectors does not change if its coefficient is rescaled inversely, but the ridge penalty does. Explain why regularization cannot be interpreted independently of units.

Read **Further reading.** Meyer develops the geometry and algebra with unusual clarity (Meyer, 2000). Golub and Van Loan give the algorithmic treatment, while Higham explains what finite precision changes (Golub and Van Loan, 2013; Higham, 2002). Trefethen and Bau provide a concise bridge between the geometry and the computation (Trefethen and Bau, 1997).

4 Eigenvalues and the SVD

We study directions and subspaces on which a matrix acts simply. Eigenvectors describe invariant directions of a square matrix. Singular vectors describe stretching by a rectangular matrix. These decompositions explain sensitivity and approximation, and they suggest algorithms that compute either an entire decomposition or a few selected components.

4.1 Eigenvalues, Invariant Subspaces, and Their Sensitivity

Repeated multiplication by a matrix can change both the length and the direction of a vector. Some directions are simpler: the matrix only multiplies them by a scalar. These are the eigenvector directions. A basis made from them would reduce repeated matrix multiplication to repeated scalar multiplication.

Such a basis need not exist. We examine that obstruction, then construct a triangular representation that always exists over the complex numbers. Symmetry gives the stronger conclusion of an orthonormal eigenvector basis. We use this basis to study optimization and sensitivity.

4.1.1 Invariant Directions and Change of Basis

Def 4.1 **(Eigenvalues and Eigenvectors)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$. A nonzero vector $\boldsymbol{v} \in \mathbb{C}^n$ is an eigenvector of $\mathbf{A}$ associated with eigenvalue $\lambda \in \mathbb{C}$ if

$$\mathbf{A}\boldsymbol{v} = \lambda \boldsymbol{v}.$$

The set of eigenvalues is the spectrum $\sigma(\mathbf{A})$. When eigenvalues are listed as $\lambda_1, \ldots, \lambda_n$, repetitions are included according to algebraic multiplicity.

Ex 4.1 For $\mathbf{A} = \operatorname{diag}(2, -1)$, compute $\mathbf{A}^k\boldsymbol{x}$ for $\boldsymbol{x} = (x_1, x_2)^T$ and identify its invariant coordinate lines. Next solve the eigenvalue equation for $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Why are complex scalars needed for the rotation even though all matrix entries are real?

Def 4.2 **(Invariant Subspace)** A subspace $S \subseteq \mathbb{C}^n$ is **invariant** under $\mathbf{A}$ if $\mathbf{A}\boldsymbol{x} \in S$ whenever $\boldsymbol{x} \in S$. An eigenvector spans a one-dimensional invariant subspace. Higher-dimensional invariant subspaces remain meaningful even when no basis of eigenvectors exists.

Example Three two-dimensional examples show what the definition can and cannot promise.

1. A diagonal matrix scales the coordinate axes, leaving both axes invariant.
2. The shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only one independent eigenvector. Repeated application keeps sliding vectors in the same direction, so a repeated eigenvalue need not supply a full eigenbasis.
3. A planar rotation through an angle that is not an integer multiple of π has no real invariant line, but it has complex eigenvectors. The complex eigenvalues encode the rotation as multiplication by a complex phase.

Def 4.3 **(Similarity and a Change of Basis)** Matrices $\mathbf{A}$ and $\mathbf{B}$ are **similar** if

$$\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$$

for some nonsingular $\mathbf{S}$. They represent the same linear map in different bases: the columns of $\mathbf{S}$ are the new basis vectors written in the old coordinates.

Thm 4.1 **(Similarity Preserves Eigenvalues)** If $\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$, then $\mathbf{A}$ and $\mathbf{B}$ have the same eigenvalues. More precisely, $\mathbf{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ if and only if

$$\mathbf{B}(\mathbf{S}^{-1}\boldsymbol{v}) = \lambda(\mathbf{S}^{-1}\boldsymbol{v}).$$

Ex 4.2 Multiply $\mathbf{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ by $\mathbf{S}^{-1}$ and insert $\mathbf{S}\mathbf{S}^{-1}$ to prove the forward implication. Prove the reverse implication and explain why the transformed eigenvector is nonzero. Use determinants to show that algebraic multiplicities are preserved as well.

Thm 4.2 **(Diagonalization Means an Eigenvector Basis)** A matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is similar to a diagonal matrix if and only if it has a basis of eigenvectors. If $\mathbf{S} = (\boldsymbol{v}_1 \ \cdots \ \boldsymbol{v}_n)$ contains such a basis, then

$$\mathbf{A}\mathbf{S} = \mathbf{S}\boldsymbol{\Lambda}, \qquad \mathbf{A} = \mathbf{S}\boldsymbol{\Lambda}\mathbf{S}^{-1}.$$

Ex 4.3 Compare the jth columns of $\mathbf{AS}$ and $\mathbf{S\Lambda}$ to prove both implications. Explain why linear independence of the eigenvectors is essential.

4.1.2 Multiplicity and Schur Form

Def 4.4 **(Algebraic and Geometric Multiplicity)** Let λ be an eigenvalue of $\mathbf{A}$.

1. The algebraic multiplicity $\mu_a(\lambda)$ is the multiplicity of λ as a root of $\det(\mathbf{A} - z\mathbf{I}) = 0$.
2. The geometric multiplicity $\mu_g(\lambda) = \dim \mathcal{N}(\mathbf{A} - \lambda\mathbf{I})$ is the dimension of the corresponding eigenspace.

Every eigenvalue satisfies $1 \leq \mu_g(\lambda) \leq \mu_a(\lambda)$. When $\mu_g(\lambda) < \mu_a(\lambda)$ for any eigenvalue, $\mathbf{A}$ is defective and does not possess a complete basis of eigenvectors.

Ex 4.4 **(A defective shear)** For $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, find the algebraic and geometric multiplicities of the eigenvalue. Compute $\mathbf{A}^k$ and explain how its growth is invisible if one records only the eigenvalue 1.

For complex vectors, $\mathbf{Q}^*$ denotes the conjugate transpose: transpose the matrix and conjugate its entries. The inner product is $\boldsymbol{x}^*\boldsymbol{y}$, and a unitary matrix satisfies $\mathbf{Q}^*\mathbf{Q} = \mathbf{I}$. These extend the real inner product and orthogonal matrices. The conjugation is necessary because $\boldsymbol{x}^*\boldsymbol{x} = \sum_i |x_i|^2$ must be positive for nonzero $\boldsymbol{x}$.

Ex 4.5 Prove that eigenvectors associated with distinct eigenvalues are linearly independent. For an induction, apply $\mathbf{A} - \lambda_k\mathbf{I}$ to a proposed dependence among k eigenvectors. Conclude that a matrix with n distinct eigenvalues has an eigenvector basis over $\mathbb{C}$.

Repeated eigenvalues can prevent diagonalization, as the shear shows. We can still begin a basis with one eigenvector. The entries below the first diagonal position then vanish. Repeating the construction on the remaining block produces a triangular matrix, without requiring the remaining basis vectors to be eigenvectors of the original matrix.

Thm 4.3 **(Schur Triangularization Decomposition)** Every square matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is unitarily similar to an upper triangular matrix:

$$\mathbf{A} = \mathbf{QTQ}^*,$$

where $\mathbf{Q} \in \mathbb{C}^{n\times n}$ is unitary ($\mathbf{Q}^*\mathbf{Q} = \mathbf{I}$) and $\mathbf{T} \in \mathbb{C}^{n\times n}$ is upper triangular. The diagonal entries of $\mathbf{T}$ are the eigenvalues of $\mathbf{A}$.

Proof We proceed by induction on the matrix dimension n. For $n = 1$, the statement holds trivially with $\mathbf{Q} = [1]$ and $\mathbf{T} = \mathbf{A}$.

The induction will isolate one eigenvector and then apply the result only to a matrix of order $n - 1$. We do not assume that $\mathbf{A}$ has a complete eigenvector basis.

Assume the theorem holds for all square matrices of order $n - 1$. By the Fundamental Theorem of Algebra, the characteristic polynomial $\det(\mathbf{A} - z\mathbf{I}) = 0$ possesses at least one complex root λ_1. Let $\boldsymbol{v}_1 \in \mathbb{C}^n$ be an eigenvector associated with λ_1, normalized so that $\|\boldsymbol{v}_1\|_2 = 1$. Using the

Gram–Schmidt process or Householder reflectors, construct an orthonormal basis for $\mathbb{C}^n$ whose first column is $\boldsymbol{v}_1$, forming a unitary matrix $\mathbf{U}_1 = [\boldsymbol{v}_1, \mathbf{W}] \in \mathbb{C}^{n\times n}$ with $\mathbf{W} \in \mathbb{C}^{n\times(n-1)}$.

The completed basis satisfies $\boldsymbol{v}_1^*\boldsymbol{v}_1 = 1$, $\mathbf{W}^*\boldsymbol{v}_1 = \mathbf{0}$, and $\mathbf{W}^*\mathbf{W} = \mathbf{I}_{n-1}$. These identities will determine the first column of the transformed matrix.

Applying this unitary similarity transformation to $\mathbf{A}$ gives

$$\mathbf{U}_1^*\mathbf{A}\mathbf{U}_1 = \begin{pmatrix} \boldsymbol{v}_1^* \\ \mathbf{W}^* \end{pmatrix} \mathbf{A} \begin{pmatrix} \boldsymbol{v}_1 & \mathbf{W} \end{pmatrix} = \begin{pmatrix} \boldsymbol{v}_1^*\mathbf{A}\boldsymbol{v}_1 & \boldsymbol{v}_1^*\mathbf{A}\mathbf{W} \\ \mathbf{W}^*\mathbf{A}\boldsymbol{v}_1 & \mathbf{W}^*\mathbf{A}\mathbf{W} \end{pmatrix}.$$

Because $\mathbf{A}\boldsymbol{v}_1 = \lambda_1\boldsymbol{v}_1$, the leading entry is $\boldsymbol{v}_1^*(\lambda_1\boldsymbol{v}_1) = \lambda_1\|\boldsymbol{v}_1\|_2^2 = \lambda_1$. The lower-left block is $\mathbf{W}^*(\lambda_1\boldsymbol{v}_1) = \lambda_1(\mathbf{W}^*\boldsymbol{v}_1) = \mathbf{0}$, because the columns of $\mathbf{W}$ are orthogonal to $\boldsymbol{v}_1$. Thus

$$\mathbf{U}_1^*\mathbf{A}\mathbf{U}_1 = \begin{pmatrix} \lambda_1 & \boldsymbol{w}^* \\ \mathbf{0} & \mathbf{A}_1 \end{pmatrix},$$

where $\mathbf{A}_1 = \mathbf{W}^*\mathbf{A}\mathbf{W} \in \mathbb{C}^{(n-1)\times(n-1)}$ and $\boldsymbol{w}^* = \boldsymbol{v}_1^*\mathbf{A}\mathbf{W}$.

By the induction hypothesis, there exists a unitary matrix $\mathbf{Q}_1 \in \mathbb{C}^{(n-1)\times(n-1)}$ such that $\mathbf{Q}_1^*\mathbf{A}_1\mathbf{Q}_1 = \mathbf{T}_1$ is upper triangular. Define the $n \times n$ block-diagonal unitary matrix

$$\mathbf{V} = \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{Q}_1 \end{pmatrix}.$$

Setting $\mathbf{Q} = \mathbf{U}_1\mathbf{V}$, which is unitary as the product of unitary matrices, we obtain

$$\begin{aligned} \mathbf{Q}^*\mathbf{A}\mathbf{Q} &= \mathbf{V}^*(\mathbf{U}_1^*\mathbf{A}\mathbf{U}_1)\mathbf{V} \\ &= \begin{pmatrix} \lambda_1 & \boldsymbol{w}^*\mathbf{Q}_1 \\ \mathbf{0} & \mathbf{Q}_1^*\mathbf{A}_1\mathbf{Q}_1 \end{pmatrix} = \begin{pmatrix} \lambda_1 & \boldsymbol{w}^*\mathbf{Q}_1 \\ \mathbf{0} & \mathbf{T}_1 \end{pmatrix}. \end{aligned}$$

To verify that the new matrix is unitary, multiply

$$\mathbf{Q}^*\mathbf{Q} = \mathbf{V}^*\mathbf{U}_1^*\mathbf{U}_1\mathbf{V} = \mathbf{V}^*\mathbf{V} = \mathbf{I}.$$

The block $\mathbf{Q}_1$ acts only on the trailing coordinates, so it cannot introduce entries below the zero lower-left block.

Because $\mathbf{T}_1$ is upper triangular, the entire block matrix $\mathbf{T} = \mathbf{Q}^*\mathbf{A}\mathbf{Q}$ is upper triangular. The diagonal entries of $\mathbf{T}$ are λ_1 and the eigenvalues of $\mathbf{A}_1$, which collectively form the eigenvalues of $\mathbf{A}$.

Finally, similarity preserves the characteristic polynomial, and the determinant of a triangular matrix is the product of its diagonal entries. Therefore $\det(\mathbf{T} - z\mathbf{I}) = \prod_i (t_{ii} - z)$, which proves that the diagonal entries list all eigenvalues with algebraic multiplicities. See Horn and Johnson (2012) for Schur form. □

Thm 4.4 **(Spectral Theorem for Real Symmetric Matrices)** If $\mathbf{A} \in \mathbb{R}^{n\times n}$ is real symmetric ($\mathbf{A} = \mathbf{A}^T$), then all eigenvalues of $\mathbf{A}$ are real, and $\mathbf{A}$ is orthogonally diagonalizable:

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T = \sum_{i=1}^{n} \lambda_i \boldsymbol{q}_i \boldsymbol{q}_i^T,$$

where $\mathbf{Q} \in \mathbb{R}^{n\times n}$ is real orthogonal ($\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$) and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ is real diagonal.

Proof First, let $\lambda \in \mathbb{C}$ and $\boldsymbol{v} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$ satisfy $\mathbf{A}\boldsymbol{v} = \lambda\boldsymbol{v}$. Taking the inner product with $\boldsymbol{v}$,

$$\boldsymbol{v}^*\mathbf{A}\boldsymbol{v} = \boldsymbol{v}^*(\lambda\boldsymbol{v}) = \lambda\|\boldsymbol{v}\|_2^2.$$

Taking the conjugate transpose of this scalar and using $\mathbf{A} = \mathbf{A}^T = \mathbf{A}^*$,

$$\boldsymbol{v}^*\mathbf{A}\boldsymbol{v} = (\boldsymbol{v}^*\mathbf{A}\boldsymbol{v})^* = (\lambda\|\boldsymbol{v}\|_2^2)^* = \bar{\lambda}\|\boldsymbol{v}\|_2^2.$$

Subtracting these two scalar expressions gives

$$(\lambda - \bar{\lambda})\|\boldsymbol{v}\|_2^2 = 0.$$

Because $\boldsymbol{v} \neq \mathbf{0}$, $\|\boldsymbol{v}\|_2^2 > 0$, forcing $\lambda = \bar{\lambda} \in \mathbb{R}$. Thus all eigenvalues are real.

Real eigenvalues alone do not yet give an orthogonal basis. We now construct that basis by reducing the dimension, retaining symmetry at every reduction.

Because $\lambda_1 \in \mathbb{R}$ and $\mathbf{A}$ is real, writing $\boldsymbol{v} = \boldsymbol{x} + i\boldsymbol{y}$ gives $\mathbf{A}\boldsymbol{x} = \lambda_1\boldsymbol{x}$ and $\mathbf{A}\boldsymbol{y} = \lambda_1\boldsymbol{y}$. At least one of $\boldsymbol{x}, \boldsymbol{y}$ is nonzero, yielding a real unit eigenvector $\boldsymbol{q}_1$. Extend $\boldsymbol{q}_1$ to an orthogonal matrix $\mathbf{Q}_1 = [\boldsymbol{q}_1, \mathbf{W}] \in \mathbb{R}^{n\times n}$ with $\mathbf{W} \in \mathbb{R}^{n\times(n-1)}$. Because $\mathbf{A}\boldsymbol{q}_1 = \lambda_1\boldsymbol{q}_1$ and $\mathbf{W}^T\boldsymbol{q}_1 = \mathbf{0}$:

$$\mathbf{Q}_1^T\mathbf{A}\mathbf{Q}_1 = \begin{pmatrix} \boldsymbol{q}_1^T \\ \mathbf{W}^T \end{pmatrix} \mathbf{A} \begin{pmatrix} \boldsymbol{q}_1 & \mathbf{W} \end{pmatrix} = \begin{pmatrix} \boldsymbol{q}_1^T\mathbf{A}\boldsymbol{q}_1 & \boldsymbol{q}_1^T\mathbf{A}\mathbf{W} \\ \mathbf{W}^T\mathbf{A}\boldsymbol{q}_1 & \mathbf{W}^T\mathbf{A}\mathbf{W} \end{pmatrix} = \begin{pmatrix} \lambda_1 & (\mathbf{W}^T\mathbf{A}\boldsymbol{q}_1)^T \\ \mathbf{0} & \mathbf{A}_1 \end{pmatrix} = \begin{pmatrix} \lambda_1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{A}_1 \end{pmatrix},$$

The upper-right block vanishes for a specific reason: $\boldsymbol{q}_1^T\mathbf{A}\mathbf{W} = (\mathbf{W}^T\mathbf{A}\boldsymbol{q}_1)^T$, and the lower-left block is zero. This equality uses symmetry. Without it, Schur form would leave a possibly nonzero upper-right block.

where $\mathbf{A}_1 = \mathbf{W}^T\mathbf{A}\mathbf{W} \in \mathbb{R}^{(n-1)\times(n-1)}$ is symmetric. Indeed, $\mathbf{A}_1^T = \mathbf{W}^T\mathbf{A}^T\mathbf{W} = \mathbf{A}_1$. For dimension one the theorem is immediate. At each larger dimension we may therefore apply the induction hypothesis to this real symmetric trailing block.

By induction on dimension, there exists an orthogonal $\mathbf{Q}_2 \in \mathbb{R}^{(n-1)\times(n-1)}$ such that $\mathbf{Q}_2^T\mathbf{A}_1\mathbf{Q}_2 = \operatorname{diag}(\lambda_2, \ldots, \lambda_n)$. The matrix

$$\mathbf{Q} = \mathbf{Q}_1 \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{Q}_2 \end{pmatrix}$$

is orthogonal and satisfies $\mathbf{Q}^T\mathbf{A}\mathbf{Q} = \mathbf{\Lambda}$, so $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$.

Reading the columns of $\mathbf{A}\mathbf{Q} = \mathbf{Q}\mathbf{\Lambda}$ shows $\mathbf{A}\boldsymbol{q}_i = \lambda_i\boldsymbol{q}_i$ for every column. Orthogonality of $\mathbf{Q}$ makes these vectors an orthonormal basis. This verifies the eigenvector assertion as well as the displayed factorization. □

For a nonzero vector, define its Rayleigh quotient by $R(\boldsymbol{x}) = \boldsymbol{x}^T\mathbf{A}\boldsymbol{x}/(\boldsymbol{x}^T\boldsymbol{x})$. In an orthonormal eigenvector basis, $\boldsymbol{x} = \sum_i c_i\boldsymbol{q}_i$ gives

$$R(\boldsymbol{x}) = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2}.$$

This is a weighted average of the eigenvalues, so its minimum and maximum are the smallest and largest eigenvalues. To isolate an intermediate eigenvalue, we restrict the quotient to subspaces of a specified dimension.

Ex 4.6 For $\mathbf{A} = \operatorname{diag}(1, 3, 7)$, find the extreme Rayleigh quotients on $\operatorname{span}\{\boldsymbol{e}_1, \boldsymbol{e}_2\}$ and on $\operatorname{span}\{\boldsymbol{e}_2, \boldsymbol{e}_3\}$. Explain why every plane through the origin intersects $\operatorname{span}\{\boldsymbol{e}_2, \boldsymbol{e}_3\}$ in a nonzero vector. Use that fact to bound the largest quotient on any plane from below by 3.

Thm 4.5 **(Courant–Fischer Minimax Theorem)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be real symmetric with eigenvalues ordered as $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$. Then for each $k \in \{1, \dots, n\}$,

$$\lambda_k = \min_{\dim(S)=k} \max_{\boldsymbol{x}\in S,\, \boldsymbol{x}\neq \mathbf{0}} \frac{\boldsymbol{x}^T \mathbf{A}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}} = \max_{\dim(S)=n-k+1} \min_{\boldsymbol{x}\in S,\, \boldsymbol{x}\neq \mathbf{0}} \frac{\boldsymbol{x}^T \mathbf{A}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}}.$$

Proof Let $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ be an orthonormal basis of eigenvectors of $\mathbf{A}$ with $\mathbf{A}\boldsymbol{q}_i = \lambda_i \boldsymbol{q}_i$. For any nonzero $\boldsymbol{x} = \sum_{i=1}^n c_i \boldsymbol{q}_i$, the Rayleigh quotient satisfies

$$R(\boldsymbol{x}) = \frac{\boldsymbol{x}^T \mathbf{A}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}} = \frac{\sum_{i=1}^n \lambda_i c_i^2}{\sum_{i=1}^n c_i^2}.$$

The quotient is unchanged under rescaling, so each maximum can equivalently be taken on the unit sphere of its subspace. That sphere is compact, which ensures the stated extrema are attained.

Define the subspace $V_k = \operatorname{span}\{\boldsymbol{q}_1, \dots, \boldsymbol{q}_k\}$, which has dimension k. For any nonzero $\boldsymbol{x} \in V_k$, $c_i = 0$ for all $i > k$, so

$$R(\boldsymbol{x}) = \frac{\sum_{i=1}^k \lambda_i c_i^2}{\sum_{i=1}^k c_i^2} \le \lambda_k \frac{\sum_{i=1}^k c_i^2}{\sum_{i=1}^k c_i^2} = \lambda_k.$$

Because $\boldsymbol{q}_k \in V_k$ achieves $R(\boldsymbol{q}_k) = \lambda_k$, we have $\max_{\boldsymbol{x}\in V_k,\, \boldsymbol{x}\neq \mathbf{0}} R(\boldsymbol{x}) = \lambda_k$.

Now let S be any arbitrary subspace of $\mathbb{R}^n$ with $\dim(S) = k$. Define $W_{k-1} = \operatorname{span}\{\boldsymbol{q}_k, \dots, \boldsymbol{q}_n\}$, which has dimension $n - k + 1$. By the dimension formula,

$$\dim(S \cap W_{k-1}) = \dim(S) + \dim(W_{k-1}) - \dim(S + W_{k-1}) \ge k + (n - k + 1) - n = 1.$$

Therefore, there exists a nonzero vector $\boldsymbol{y} \in S \cap W_{k-1}$. Because $\boldsymbol{y} \in W_{k-1}$, its expansion $\boldsymbol{y} = \sum_{i=k}^n d_i \boldsymbol{q}_i$ contains only modes $i \ge k$, so

$$R(\boldsymbol{y}) = \frac{\sum_{i=k}^n \lambda_i d_i^2}{\sum_{i=k}^n d_i^2} \ge \lambda_k \frac{\sum_{i=k}^n d_i^2}{\sum_{i=k}^n d_i^2} = \lambda_k.$$

Because $\boldsymbol{y} \in S$, the maximum over S satisfies $\max_{\boldsymbol{x}\in S,\, \boldsymbol{x}\neq \mathbf{0}} R(\boldsymbol{x}) \ge R(\boldsymbol{y}) \ge \lambda_k$. Taking the minimum over all k-dimensional subspaces S proves the first equality. For the second characterization, the eigenvalues of $-\mathbf{A}$ in increasing order are $-\lambda_n, \dots, -\lambda_1$. Apply the first characterization to its index $n - k + 1$:

$$-\lambda_k = \min_{\dim S = n-k+1} \max_{\boldsymbol{x}\in S\setminus\{0\}} [-R(\boldsymbol{x})].$$

For each fixed S, $\max(-R) = -\min R$. Negating the outer minimum changes it into a maximum. Thus

$$\lambda_k = \max_{\dim S = n-k+1} \min_{\boldsymbol{x}\in S\setminus\{0\}} R(\boldsymbol{x}),$$

which proves the second equality without leaving an additional argument implicit. □

Thm 4.6 **(Cauchy Interlacing Theorem)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be symmetric, and let $\mathbf{B} \in \mathbb{R}^{(n-1)\times(n-1)}$ be a principal submatrix. If

$$\lambda_1 \le \cdots \le \lambda_n, \qquad \mu_1 \le \cdots \le \mu_{n-1}$$

are the eigenvalues of $\mathbf{A}$ and $\mathbf{B}$, respectively, then

$$\lambda_k \le \mu_k \le \lambda_{k+1}, \qquad k = 1, \dots, n-1.$$

Proof After a simultaneous permutation of rows and columns, we may assume that $\mathbf{B}$ is obtained by deleting the last row and column. Identify $\mathbb{R}^{n-1}$ with the coordinate hyperplane

$$H = \{\boldsymbol{x} \in \mathbb{R}^n : x_n = 0\}.$$

For $\boldsymbol{y} \in \mathbb{R}^{n-1}$, insert a final zero to get $\boldsymbol{x} = (\boldsymbol{y}, 0)^T$. Block multiplication gives $\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} = \boldsymbol{y}^T\mathbf{B}\boldsymbol{y}$ and $\boldsymbol{x}^T\boldsymbol{x} = \boldsymbol{y}^T\boldsymbol{y}$. Thus the identification preserves both the dimension of a subspace and its Rayleigh quotient.

For every nonzero $\boldsymbol{x} \in H$, the Rayleigh quotients of $\mathbf{A}$ and $\mathbf{B}$ agree.

Apply the first Courant–Fischer characterization from Thm 4.5. The minimization defining μ_k ranges only over k-dimensional subspaces of H, whereas the minimization defining λ_k ranges over all k-dimensional subspaces of $\mathbb{R}^n$. Restricting the admissible set cannot decrease the minimum, so $\lambda_k \le \mu_k$.

For the other inequality, use the dual characterization. Since $\dim(H) = n-1$,

$$\mu_k = \max_{\substack{S\subset H\\ \dim(S)=n-k}} \min_{\boldsymbol{x}\in S,\, \boldsymbol{x}\neq\mathbf{0}} R_{\mathbf{A}}(\boldsymbol{x}) \le \max_{\substack{S\subset \mathbb{R}^n\\ \dim(S)=n-k}} \min_{\boldsymbol{x}\in S,\, \boldsymbol{x}\neq\mathbf{0}} R_{\mathbf{A}}(\boldsymbol{x}) = \lambda_{k+1}.$$

Together the two bounds give the interlacing inequalities.

In the dual formula, the subspaces for $\mathbf{B}$ have dimension $(n-1)-k+1 = n-k$. The same dimension in the formula for $\mathbf{A}$ corresponds to its eigenvalue with index $k+1$. This index calculation explains the one-step shift in the upper bound. □

Ex 4.7 **(Cauchy interlacing on a concrete matrix)** For

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix},$$

compute the eigenvalues of $\mathbf{A}$ and of the leading 2×2 principal submatrix. Verify each inequality in Thm 4.6. Repeat after deleting the middle row and column.

4.1.3 Sensitivity of Eigenvalues

Every eigenvalue algorithm computes the eigenvalues of a matrix that differs from $\mathbf{A}$ by rounding errors, and every iterative eigensolver stops with an approximate eigenpair whose residual is small but nonzero. We therefore need to know how far eigenvalues move when the matrix is perturbed.

For symmetric matrices the answer follows from Thm 4.5, and eigenvalues move no farther than the norm of the perturbation. For a general diagonalizable matrix the answer also depends on the condition number of the eigenvector basis.

Thm 4.7 **(Weyl's Perturbation Theorem)** Let $\mathbf{A}$ and $\mathbf{E}$ be real symmetric $n \times n$ matrices. Let $\lambda_1 \le \cdots \le \lambda_n$ be the eigenvalues of $\mathbf{A}$ and $\mu_1 \le \cdots \le \mu_n$ the eigenvalues of $\mathbf{A} + \mathbf{E}$. Then

$$|\mu_k - \lambda_k| \le \|\mathbf{E}\|_2, \qquad k = 1, \dots, n.$$

Proof Write $R_{\mathbf{A}}(\boldsymbol{x}) = \boldsymbol{x}^T\mathbf{A}\boldsymbol{x}/(\boldsymbol{x}^T\boldsymbol{x})$ for a nonzero vector. Subtract the two quotients:

$$R_{\mathbf{A}+\mathbf{E}}(\boldsymbol{x}) - R_{\mathbf{A}}(\boldsymbol{x}) = \frac{\boldsymbol{x}^T\mathbf{E}\boldsymbol{x}}{\|\boldsymbol{x}\|_2^2}.$$

Cauchy–Schwarz and consistency bound the numerator:

$$|\boldsymbol{x}^T\mathbf{E}\boldsymbol{x}| \le \|\boldsymbol{x}\|_2\|\mathbf{E}\boldsymbol{x}\|_2 \le \|\mathbf{E}\|_2\|\boldsymbol{x}\|_2^2.$$

After division, this gives the pointwise inequalities

$$R_{\mathbf{A}}(\boldsymbol{x}) - \|\mathbf{E}\|_2 \le R_{\mathbf{A}+\mathbf{E}}(\boldsymbol{x}) \le R_{\mathbf{A}}(\boldsymbol{x}) + \|\mathbf{E}\|_2.$$

Fix a subspace S of dimension k. Each inequality holds for every nonzero vector of S, so it also holds after taking the maximum over those vectors. A constant independent of the vector passes unchanged through the maximum.

Now take the minimum over all such subspaces. The same constant is independent of S, so it again passes unchanged through this operation. Courant–Fischer identifies the resulting middle value with μ_k and the other two with $\lambda_k \pm \|\mathbf{E}\|_2$. Both matrices are symmetric, as required to use that theorem. Hence

$$\lambda_k - \|\mathbf{E}\|_2 \le \mu_k \le \lambda_k + \|\mathbf{E}\|_2,$$

which is equivalent to $|\mu_k - \lambda_k| \le \|\mathbf{E}\|_2$. This classical perturbation theorem is treated in Horn and Johnson (2012). □

Cor 4.8 **(Residual Bounds for Symmetric Matrices)** Let $\mathbf{A}$ be real symmetric, let $\boldsymbol{x}$ be a unit vector, let θ be real, and let $\boldsymbol{r} = \mathbf{A}\boldsymbol{x} - \theta\boldsymbol{x}$.

1. Some eigenvalue λ of $\mathbf{A}$ satisfies $|\lambda - \theta| \le \|\boldsymbol{r}\|_2$.
2. Suppose $\theta = \boldsymbol{x}^T\mathbf{A}\boldsymbol{x}$, let λ be an eigenvalue of $\mathbf{A}$ nearest to θ, and let $\delta > 0$ be the distance from θ to the remaining eigenvalues. Then

$$|\lambda - \theta| \le \frac{\|\boldsymbol{r}\|_2^2}{\delta}.$$

Proof Expand $\boldsymbol{x} = \sum_i c_i\boldsymbol{q}_i$ in an orthonormal eigenvector basis, so that $\sum_i c_i^2 = 1$ and $\boldsymbol{r} = \sum_i(\lambda_i - \theta)c_i\boldsymbol{q}_i$. Then

$$\|\boldsymbol{r}\|_2^2 = \sum_{i=1}^{n}(\lambda_i - \theta)^2 c_i^2 \ge \min_i(\lambda_i - \theta)^2,$$

because each squared distance is at least its minimum and the nonnegative weights c_i^2 sum to one. Taking square roots gives $\min_i |\lambda_i - \theta| \le \|\boldsymbol{r}\|_2$. The minimum is attained at some eigenvalue because there are finitely many of them.

For the second claim, the Rayleigh quotient condition is $\sum_i (\lambda_i - \theta) c_i^2 = 0$. Suppose $\lambda \le \theta$, since the case $\lambda \ge \theta$ follows by applying the argument to $-\mathbf{A}$. Let $\beta = \theta + \delta$.

An eigenvalue $\lambda_i \ne \lambda$ cannot lie in $(\lambda, \theta]$, because it would then be closer to θ than λ. Such an eigenvalue also cannot lie in (θ, β), by the definition of δ. Hence every such eigenvalue satisfies $\lambda_i < \lambda$ or $\lambda_i \ge \beta$, and in both cases $(\lambda_i - \lambda)(\lambda_i - \beta) \ge 0$. The product vanishes when $\lambda_i = \lambda$.

Writing $\lambda_i - \lambda = (\lambda_i - \theta) + (\theta - \lambda)$ and $\lambda_i - \beta = (\lambda_i - \theta) - \delta$ and expanding gives

$$0 \le \sum_{i=1}^{n} (\lambda_i - \lambda)(\lambda_i - \beta) c_i^2 = \|\boldsymbol{r}\|_2^2 + (\theta - \lambda - \delta) \sum_{i=1}^{n} (\lambda_i - \theta) c_i^2 - (\theta - \lambda)\delta = \|\boldsymbol{r}\|_2^2 - (\theta - \lambda)\delta.$$

For clarity, the expansion in the last display uses $a = \lambda_i - \theta$ and $d = \theta - \lambda$:

$$(a + d)(a - \delta) = a^2 + (d - \delta)a - d\delta.$$

Summing the first term gives the squared residual norm. The middle term vanishes because $\theta = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x}$. The last term is $-d\delta$ because the weights sum to one. These are the three contributions in the displayed inequality.

Therefore $(\theta - \lambda)\delta \le \|\boldsymbol{r}\|_2^2$. □

The first bound states that a small residual locates an eigenvalue of a symmetric matrix to within the residual norm. The second bound shows that the Rayleigh quotient is much more accurate than its residual when the eigenvalue is well separated from the rest of the spectrum. An eigenvector residual of 10^{-4} and a gap of order one, for example, give an eigenvalue error of order 10^{-8}.

Thm 4.9 **(Bauer–Fike Theorem)** Let $\mathbf{A} = \mathbf{X} \mathbf{\Lambda} \mathbf{X}^{-1}$ be diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_n$, and let $\mathbf{E}$ be any $n \times n$ matrix. Every eigenvalue μ of $\mathbf{A} + \mathbf{E}$ satisfies

$$\min_{1 \le i \le n} |\lambda_i - \mu| \le \kappa_2(\mathbf{X}) \|\mathbf{E}\|_2,$$

where $\kappa_2(\mathbf{X}) = \|\mathbf{X}\|_2 \|\mathbf{X}^{-1}\|_2$ is computed in the Euclidean norm on $\mathbb{C}^n$.

Proof Let μ be an eigenvalue of $\mathbf{A} + \mathbf{E}$ and choose a nonzero eigenvector $\boldsymbol{y}$. If μ is already an eigenvalue of $\mathbf{A}$, its distance to that spectrum is zero and the estimate follows. Assume it is not.

Write $\mathbf{A} = \mathbf{X} \mathbf{\Lambda} \mathbf{X}^{-1}$ and put $\boldsymbol{z} = \mathbf{X}^{-1} \boldsymbol{y}$. This vector is nonzero because $\mathbf{X}$ is invertible. Multiply $(\mathbf{A} + \mathbf{E})\boldsymbol{y} = \mu \boldsymbol{y}$ by $\mathbf{X}^{-1}$ and substitute $\boldsymbol{y} = \mathbf{X} \boldsymbol{z}$:

$$(\mathbf{\Lambda} - \mu \mathbf{I}) \boldsymbol{z} = -\mathbf{X}^{-1} \mathbf{E} \mathbf{X} \boldsymbol{z}.$$

Every diagonal entry $\lambda_i - \mu$ is nonzero, so the diagonal matrix is invertible. Therefore

$$\boldsymbol{z} = -(\mathbf{\Lambda} - \mu \mathbf{I})^{-1} \mathbf{X}^{-1} \mathbf{E} \mathbf{X} \boldsymbol{z}.$$

Take Euclidean norms and divide by $\|\boldsymbol{z}\|_2 > 0$:

$$1 \le \|(\mathbf{\Lambda} - \mu \mathbf{I})^{-1}\|_2 \|\mathbf{X}^{-1}\|_2 \|\mathbf{E}\|_2 \|\mathbf{X}\|_2.$$

For a diagonal matrix the Euclidean norm is the largest diagonal magnitude. Here it is

$$\max_i \frac{1}{|\lambda_i - \mu|} = \frac{1}{\min_i |\lambda_i - \mu|}.$$

Multiply the norm inequality by the positive denominator. The result is $\min_i |\lambda_i - \mu| \le \kappa_2(\mathbf{X})\|\mathbf{E}\|_2$. This is the Bauer–Fike theorem (Horn and Johnson, 2012; Golub and Van Loan, 2013). □

Cor 4.10 **(Residual Bound for Diagonalizable Matrices)** Let $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$, let $\boldsymbol{x}$ be a unit vector, let $\theta \in \mathbb{C}$, and let $\boldsymbol{r} = \mathbf{A}\boldsymbol{x} - \theta\boldsymbol{x}$. Then some eigenvalue λ of $\mathbf{A}$ satisfies $|\lambda - \theta| \le \kappa_2(\mathbf{X})\|\boldsymbol{r}\|_2$.

Proof The unit-norm hypothesis gives $\boldsymbol{x}^*\boldsymbol{x} = 1$, where the star denotes conjugate transpose. Define the rank-one perturbation $\mathbf{E} = -\boldsymbol{r}\boldsymbol{x}^*$. Its action on this particular vector is

$$\mathbf{E}\boldsymbol{x} = -\boldsymbol{r}(\boldsymbol{x}^*\boldsymbol{x}) = -\boldsymbol{r}.$$

Since $\boldsymbol{r} = \mathbf{A}\boldsymbol{x} - \theta\boldsymbol{x}$, we obtain

$$(\mathbf{A} + \mathbf{E})\boldsymbol{x} = \mathbf{A}\boldsymbol{x} - \boldsymbol{r} = \theta\boldsymbol{x}.$$

The vector is nonzero, so θ is an exact eigenvalue of the perturbed matrix.

For any unit vector $\boldsymbol{y}$, Cauchy–Schwarz gives

$$\|\mathbf{E}\boldsymbol{y}\|_2 = \|\boldsymbol{r}\|_2 |\boldsymbol{x}^*\boldsymbol{y}| \le \|\boldsymbol{r}\|_2.$$

Equality is attained at $\boldsymbol{y} = \boldsymbol{x}$, because $|\boldsymbol{x}^*\boldsymbol{x}| = 1$. Thus $\|\mathbf{E}\|_2 = \|\boldsymbol{r}\|_2$. Applying Bauer–Fike to this explicitly constructed perturbation gives the stated residual bound. The argument concerns eigenvalue location, not the distance to a particular eigenvector. □

For a symmetric matrix, Thm 4.4 provides an orthogonal $\mathbf{X}$ with $\kappa_2(\mathbf{X}) = 1$, and Cor 4.10 reduces to the first part of Cor 4.8. For a nonnormal matrix, $\kappa_2(\mathbf{X})$ can be large, and a small residual then locates an eigenvalue only loosely. Near a defective matrix the eigenvalues can move much farther than the perturbation, as the following example shows.

Example **(Eigenvalues of a nearly defective matrix)** The matrix

$$\mathbf{A}_\varepsilon = \begin{pmatrix} 0 & 1 \\ \varepsilon & 0 \end{pmatrix}$$

differs from the defective matrix $\mathbf{A}_0$ by a perturbation of norm ε, but its eigenvalues are $\pm\sqrt{\varepsilon}$. For $\varepsilon = 10^{-10}$, a perturbation at the level of 10^{-10} moves the double eigenvalue zero by 10^{-5}. The eigenvectors $(1, \pm\sqrt{\varepsilon})^T$ of $\mathbf{A}_\varepsilon$ become parallel as $\varepsilon \to 0$, so $\kappa_2(\mathbf{X})$ grows without bound, in agreement with Thm 4.9.

4.1.4 Further Exercises

Ex 4.8 **(Rayleigh quotient stationary points)** For a real symmetric matrix $\mathbf{A}$, define the Rayleigh quotient $R(\boldsymbol{x}) = \frac{\boldsymbol{x}^T\mathbf{A}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}}$ on $\mathbb{R}^n \setminus \{\mathbf{0}\}$.

1. Compute the gradient $\nabla R(\boldsymbol{x})$ directly.
2. Prove that $\nabla R(\boldsymbol{x}) = \mathbf{0}$ if and only if $\boldsymbol{x}$ is an eigenvector of $\mathbf{A}$.

3. Restrict the Hessian $\nabla^2 R(\boldsymbol{q}_i)$ to directions orthogonal to $\boldsymbol{q}_i$. Show that its eigenvalues on this tangent space are $\frac{2}{\|\boldsymbol{q}_i\|_2^2}(\lambda_j - \lambda_i)$ for $j \neq i$. Explain why the radial direction has zero curvature, and conclude that the smallest and largest eigenvalues are the global extrema of R.

Ex 4.9 **(Similarity is a coordinate change)** Choose a nonorthogonal basis of $\mathbb{R}^2$, form its basis matrix $\mathbf{S}$, and compute $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ for a simple diagonal matrix $\mathbf{A}$. Verify directly that eigenvalues are preserved although lengths and angles generally are not.

4.2 Vector Iterations and the QR Algorithm

To compute an eigenpair approximately, we seek a unit vector $\boldsymbol{v}$ and scalar θ for which $\mathbf{A}\boldsymbol{v} - \theta\boldsymbol{v}$ is small. The residual bounds of the preceding section explain what such a calculation establishes for symmetric matrices.

We first multiply a vector repeatedly by the matrix, then replace multiplication by the solution of a shifted linear system to select other eigenvalues. These methods approximate one eigenpair at a time. The QR algorithm instead changes the whole matrix by orthogonal similarity transformations to expose all its eigenvalues.

4.2.1 Power Iteration and Shifted Inverse Iteration

Def 4.5 **(Power Iteration)** Let $\mathbf{A}$ be real symmetric. Given a unit vector $\boldsymbol{v}^{(0)}$, power iteration constructs

$$\boldsymbol{w}^{(k)} = \mathbf{A}\boldsymbol{v}^{(k-1)}, \qquad \boldsymbol{v}^{(k)} = \frac{\boldsymbol{w}^{(k)}}{\|\boldsymbol{w}^{(k)}\|_2}, \qquad \lambda^{(k)} = (\boldsymbol{v}^{(k)})^T \mathbf{A}\boldsymbol{v}^{(k)}.$$

Order the eigenvalues by magnitude. If $|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n|$ and $\boldsymbol{q}_1^T \boldsymbol{v}^{(0)} \neq 0$, the direction of the iterate converges linearly:

$$\|\boldsymbol{v}^{(k)} - (\pm\boldsymbol{q}_1)\|_2 = O\left(\left|\frac{\lambda_2}{\lambda_1}\right|^k\right).$$

For a general nonnormal matrix, an eigenvector basis may be poorly conditioned or may not exist, and this simple orthogonal expansion no longer justifies the claim.

The convergence claim comes from expanding the starting vector in an orthonormal eigenbasis. Write $\boldsymbol{v}^{(0)} = \sum_i c_i \boldsymbol{q}_i$ with $c_1 \neq 0$. Then

$$\mathbf{A}^k \boldsymbol{v}^{(0)} = \lambda_1^k \left(c_1 \boldsymbol{q}_1 + \sum_{i=2}^{n} c_i (\lambda_i/\lambda_1)^k \boldsymbol{q}_i \right).$$

Each unwanted component decreases relative to the first at least as fast as $|\lambda_2/\lambda_1|^k$. Normalizing removes the common scale. A negative dominant eigenvalue can alternate the sign of the vector, so convergence is to a direction.

Ex 4.10 For $\mathbf{A} = \operatorname{diag}(4, 2, 1)$, write the normalized power iterate explicitly when $\boldsymbol{v}^{(0)} = (1, 1, 1)^T/\sqrt{3}$. Bound the angle to $\boldsymbol{e}_1$. What changes if the starting vector is $(0, 1, 1)^T/\sqrt{2}$? What fails if the first diagonal entry is replaced by -2?

Def 4.6 **(Shifted Inverse Iteration)** For a real symmetric matrix, an eigenvalue near a target σ can be isolated by applying power iteration implicitly to $(\mathbf{A}-\sigma\mathbf{I})^{-1}$:

$$(\mathbf{A}-\sigma\mathbf{I})\boldsymbol{w}^{(k)}=\boldsymbol{v}^{(k-1)},\qquad \boldsymbol{v}^{(k)}=\frac{\boldsymbol{w}^{(k)}}{\|\boldsymbol{w}^{(k)}\|_2}.$$

The eigenvalues of $(\mathbf{A}-\sigma\mathbf{I})^{-1}$ are $(\lambda_i-\sigma)^{-1}$. If exactly one eigenvalue is closest to σ, the shift is not an eigenvalue, and the starting vector has a component in its eigenspace, that eigenspace dominates the iteration. The matrix inverse is never formed. Each step solves a shifted linear system instead.

Def 4.7 **(Rayleigh Quotient Iteration (RQI))** For symmetric matrices, updating the shift dynamically to the current Rayleigh estimate $\sigma_k=(\boldsymbol{v}^{(k)})^T\mathbf{A}\boldsymbol{v}^{(k)}$ defines Rayleigh Quotient Iteration:

$$(\mathbf{A}-\sigma_k\mathbf{I})\boldsymbol{w}^{(k+1)}=\boldsymbol{v}^{(k)},\qquad \boldsymbol{v}^{(k+1)}=\frac{\boldsymbol{w}^{(k+1)}}{\|\boldsymbol{w}^{(k+1)}\|_2},\qquad \sigma_{k+1}=(\boldsymbol{v}^{(k+1)})^T\mathbf{A}\boldsymbol{v}^{(k+1)}.$$

The vectors used in the shift formula must have unit norm. Before solving the next shifted system, test the eigenpair residual. At an exact eigenpair, the shift makes the matrix singular and the calculation should stop. A singular shift can also occur away from an eigenvector, so the formula alone is not a globally defined iteration. The local theorem below concerns steps for which the shifted solve exists, with exact convergence treated as termination.

Thm 4.11 **(Local Cubic Convergence of Rayleigh Quotient Iteration)** Let $\mathbf{A}$ be real symmetric, and let λ_* be a simple eigenvalue with unit eigenvector $\boldsymbol{q}_*$. If a unit RQI iterate $\boldsymbol{v}^{(k)}$ is sufficiently close to $\boldsymbol{q}_*$, then

$$\sin\angle(\boldsymbol{v}^{(k+1)},\boldsymbol{q}_*)\le C\sin^3\angle(\boldsymbol{v}^{(k)},\boldsymbol{q}_*).$$

If λ_* is an extremal eigenvalue, the Rayleigh shifts also satisfy

$$|\sigma_{k+1}-\lambda_*|\le C'|\sigma_k-\lambda_*|^3$$

for iterates sufficiently close to $\boldsymbol{q}_*$.

Proof Write the current unit vector as

$$\boldsymbol{v}=\cos\theta\,\boldsymbol{q}_*+\sin\theta\,\boldsymbol{z},\qquad \boldsymbol{z}\perp\boldsymbol{q}_*,\qquad \|\boldsymbol{z}\|_2=1.$$

Its Rayleigh quotient $\sigma=\boldsymbol{v}^T\mathbf{A}\boldsymbol{v}$ obeys

$$\begin{aligned}\sigma&=(\cos\theta\,\boldsymbol{q}_*+\sin\theta\,\boldsymbol{z})^T\mathbf{A}(\cos\theta\,\boldsymbol{q}_*+\sin\theta\,\boldsymbol{z})\\&=\cos^2\theta\,\lambda_*+2\cos\theta\sin\theta\,\boldsymbol{q}_*^T\mathbf{A}\boldsymbol{z}+\sin^2\theta\,\boldsymbol{z}^T\mathbf{A}\boldsymbol{z}.\end{aligned}$$

Because $\mathbf{A}\boldsymbol{q}_*=\lambda_*\boldsymbol{q}_*$ and $\boldsymbol{q}_*^T\boldsymbol{z}=0$, the cross-term vanishes: $\boldsymbol{q}_*^T\mathbf{A}\boldsymbol{z}=\lambda_*\boldsymbol{q}_*^T\boldsymbol{z}=0$. Substituting $\cos^2\theta=1-\sin^2\theta$,

$$\sigma-\lambda_*=(1-\sin^2\theta)\lambda_*+\sin^2\theta\,\boldsymbol{z}^T\mathbf{A}\boldsymbol{z}-\lambda_*=\sin^2\theta\,(\boldsymbol{z}^T\mathbf{A}\boldsymbol{z}-\lambda_*),$$

so $|\sigma-\lambda_*|\le C_0\sin^2\theta$.

For example, the first constant can be taken as $C_0 = \|\mathbf{A} - \lambda_*\mathbf{I}\|_2$, because $|\boldsymbol{z}^T(\mathbf{A} - \lambda_*\mathbf{I})\boldsymbol{z}| \le C_0$ for unit $\boldsymbol{z}$. Thus it does not depend on the current angle.

Let $\gamma = \min_{\lambda_j \neq \lambda_*} |\lambda_j - \lambda_*| > 0$. Once θ is small enough, $|\lambda_j - \sigma| \ge \gamma/2$ for every other eigenvalue. Expanding the inverse-iteration vector $\boldsymbol{w} = (\mathbf{A} - \sigma\mathbf{I})^{-1}\boldsymbol{v}$ in an orthonormal eigenbasis, its component along $\boldsymbol{q}_*$ has magnitude

$$\frac{|\cos\theta|}{|\lambda_* - \sigma|},$$

whereas the norm of its orthogonal component is at most $2|\sin\theta|/\gamma$. More explicitly, expand $\boldsymbol{z} = \sum_{j\neq *} a_j \boldsymbol{q}_j$ with $\sum a_j^2 = 1$. The squared perpendicular-component norm is

$$\sin^2\theta \sum_{j\neq *} \frac{a_j^2}{|\lambda_j - \sigma|^2} \le \frac{4\sin^2\theta}{\gamma^2}.$$

Normalization does not change the ratio of perpendicular and parallel component lengths, which is the tangent of the acute angle. Therefore,

$$\tan\angle(\boldsymbol{v}^+, \boldsymbol{q}_*) \le \frac{2|\lambda_* - \sigma|}{\gamma|\cos\theta|}|\sin\theta| \le C_1|\sin\theta|^3.$$

For small angles, $\sin\theta$ and $\tan\theta$ are comparable, which proves cubic convergence of the eigenvector direction.

For angles at most $\pi/4$, $|\cos\theta| \ge 1/\sqrt{2}$, so the denominator stays uniformly away from zero. The cubic bound therefore has a constant independent of sufficiently close iterates. We consider only nonsingular shifted solves, as stipulated before the theorem.

If λ_* is extremal, the spectral gap also gives two-sided local bounds $c\sin^2\theta \le |\sigma - \lambda_*| \le C\sin^2\theta$. Applying the Rayleigh-quotient identity to the next iterate and using the cubic angle bound yields

$$|\sigma^+ - \lambda_*| \le C_2\sin^2\angle(\boldsymbol{v}^+, \boldsymbol{q}_*) \le C_3\sin^6\theta \le C_4|\sigma - \lambda_*|^3.$$

For an extremal simple eigenvalue, all differences $\lambda_j - \lambda_*$ have one sign and magnitudes at least γ. Hence their weighted average in the Rayleigh error cannot cancel, giving the lower bound $\gamma\sin^2\theta \le |\sigma - \lambda_*|$. This is the additional fact used to convert the angle estimate to a cubic eigenvalue estimate. □

4.2.2 The Practical QR Algorithm

Def 4.8 **(Basic QR Iteration)** Given $\mathbf{A}_0 = \mathbf{A}$, the unshifted QR iteration generates orthogonally similar matrices via

$$\mathbf{A}_k = \mathbf{Q}_k\mathbf{R}_k, \qquad \mathbf{A}_{k+1} = \mathbf{R}_k\mathbf{Q}_k = \mathbf{Q}_k^T\mathbf{A}_k\mathbf{Q}_k.$$

All iterates share the same eigenvalues. Under suitable separation conditions the subdiagonal entries tend to zero and the iterates reveal a Schur form. Shifts and deflation are essential for a reliable practical method.

Lem 4.12 **(Invariance of Hessenberg Structure under QR Iteration)** Let $\mathbf{H} \in \mathbb{R}^{n\times n}$ be an upper Hessenberg matrix ($H_{ij} = 0$ for all $i > j+1$). If $\mathbf{H} = \mathbf{QR}$ is computed by successive adjacent-row Givens rotations, then the reversed product $\mathbf{H}_+ = \mathbf{RQ}$ is also upper Hessenberg.

Proof Use the QR factorization obtained by eliminating successive subdiagonal entries with rotations of adjacent rows. This is the structure-preserving QR factorization used by the practical algorithm.

At column j, the current matrix has no entries below row $j+1$ in that column. A rotation of rows $j, j+1$ eliminates that one entry. Previously treated columns already have zeros in both of those rows, so this rotation does not spoil their zeros. Continuing gives an upper triangular matrix

$$\mathbf{R} = \mathbf{G}_{n-1} \cdots \mathbf{G}_1 \mathbf{H}, \qquad \mathbf{Q} = \mathbf{G}_1^T \cdots \mathbf{G}_{n-1}^T.$$

A zero entry requires no rotation, so this construction also covers singular matrices.

We check that $\mathbf{Q}$ is upper Hessenberg. For column $j < n$, the rightmost factors $\mathbf{G}_{n-1}^T, \ldots, \mathbf{G}_{j+1}^T$ leave $\boldsymbol{e}_j$ unchanged. The next factor $\mathbf{G}_j^T$ can introduce only coordinate $j+1$. Subsequent factors have smaller row indices and cannot introduce any coordinate beyond $j+1$. Thus $Q_{ij} = 0$ for $i > j+1$.

For an entry (i, j) of $\mathbf{RQ}$ below that subdiagonal, $i > j+1$, write

$$(\mathbf{RQ})_{ij} = \sum_{l=1}^{n} R_{il} Q_{lj}.$$

If $l < i$, triangularity makes $R_{il} = 0$. If $l \geq i$, then $l > j+1$, and the Hessenberg property makes $Q_{lj} = 0$. Every summand is therefore zero. This proves that the reversed product is upper Hessenberg without assuming an inverse of $\mathbf{R}$. □

Thm 4.13 **(Practical Shifted QR Framework)** A standard dense nonsymmetric eigenvalue computation has two phases:

1. **Hessenberg reduction:** Compute $\mathbf{H}_0 = \mathbf{U}_0^T \mathbf{A} \mathbf{U}_0$ with Householder reflectors. This one-time reduction costs $O(n^3)$ operations and preserves the eigenvalues.

2. **Implicit shifted QR steps:** Apply shifts while preserving Hessenberg form. A schematic single-shift step is

$$\mathbf{H}_k - \mu_k \mathbf{I} = \mathbf{Q}_k \mathbf{R}_k, \qquad \mathbf{H}_{k+1} = \mathbf{R}_k \mathbf{Q}_k + \mu_k \mathbf{I}.$$

   Lem. 4.12 explains why each step costs $O(n^2)$ rather than $O(n^3)$. Real nonsymmetric codes use implicit single or double shifts so that complex conjugate pairs can remain in real arithmetic. A small subdiagonal entry permits deflation and splits the problem into smaller blocks.

For a real symmetric matrix, Hessenberg form reduces to tridiagonal form. Wilkinson shifts from the trailing two by two block and implicit bulge chasing then give a particularly efficient symmetric algorithm.

4.2.3 The Jacobi Eigenvalue Algorithm

For a symmetric matrix, an alternative to the QR algorithm drives the off-diagonal entries to zero directly. Each step applies an orthogonal similarity by a plane rotation chosen to annihilate one off-diagonal entry. The method, due to Jacobi in 1846, is slower than shifted QR for a general dense matrix. It remains useful because it can compute the small eigenvalues of graded positive definite matrices to high relative accuracy (Demmel, 1997), and because rotations acting on disjoint pairs of indices can be applied in parallel.

Def 4.9 **(Givens Plane Rotation Matrix)** The matrix $\mathbf{G}(p, q, \theta) \in \mathbb{R}^{n\times n}$ is an identity matrix except for the four entries at the intersections of rows and columns p and q:

$$g_{pp} = c, \qquad g_{pq} = s, \qquad g_{qp} = -s, \qquad g_{qq} = c,$$

where $c = \cos\theta$ and $s = \sin\theta$. The transformation $\mathbf{A}' = \mathbf{G}^T\mathbf{A}\mathbf{G}$ modifies only rows and columns p and q of $\mathbf{A}$.

Thm 4.14 **(Derivation of the Optimal Rotation Angle)** To zero the transformed off-diagonal entry $a'_{pq} = 0$, we consider the 2×2 orthogonal similarity transformation on the submatrix at indices $\{p, q\}$:

$$\begin{aligned}\begin{pmatrix} a'_{pp} & a'_{pq} \\ a'_{pq} & a'_{qq} \end{pmatrix} &= \begin{pmatrix} c & -s \\ s & c \end{pmatrix} \begin{pmatrix} a_{pp} & a_{pq} \\ a_{pq} & a_{qq} \end{pmatrix} \begin{pmatrix} c & s \\ -s & c \end{pmatrix} \\ &= \begin{pmatrix} c^2 a_{pp} - 2csa_{pq} + s^2 a_{qq} & (c^2 - s^2)a_{pq} + cs(a_{pp} - a_{qq}) \\ (c^2 - s^2)a_{pq} + cs(a_{pp} - a_{qq}) & s^2 a_{pp} + 2csa_{pq} + c^2 a_{qq} \end{pmatrix}.\end{aligned}$$

Setting $a'_{pq} = 0$ yields the trigonometric condition

$$(c^2 - s^2)a_{pq} + cs(a_{pp} - a_{qq}) = 0 \implies \frac{\cos(2\theta)}{\sin(2\theta)} = \frac{a_{qq} - a_{pp}}{2a_{pq}}.$$

Defining $\tau = \cot(2\theta) = \frac{a_{qq}-a_{pp}}{2a_{pq}}$ and using $\cot(2\theta) = \frac{1-\tan^2\theta}{2\tan\theta}$, setting $t = \tan\theta$ produces

$$t^2 + 2\tau t - 1 = 0.$$

If $a_{pq} = 0$, no rotation is needed. Otherwise, to choose a rotation of magnitude at most $\pi/4$ and avoid cancellation, select the root of smaller absolute value, with $\operatorname{sign}(0) = 1$:

$$t = \frac{\operatorname{sign}(\tau)}{|\tau| + \sqrt{1 + \tau^2}}, \qquad c = \frac{1}{\sqrt{1 + t^2}}, \qquad s = ct.$$

Proof The scalar equation is $t^2 + 2\tau t - 1 = 0$. Its roots are $-\tau \pm \sqrt{1 + \tau^2}$, and their product is -1. Thus, unless both have magnitude one, one root has magnitude less than one and the other greater than one.

If $\tau > 0$, the smaller-magnitude root is $\sqrt{1 + \tau^2} - \tau$. Rationalize it:

$$\sqrt{1 + \tau^2} - \tau = \frac{(1 + \tau^2) - \tau^2}{\sqrt{1 + \tau^2} + \tau} = \frac{1}{\sqrt{1 + \tau^2} + \tau}.$$

If $\tau < 0$, put $a = -\tau > 0$. The smaller root is $a - \sqrt{1 + a^2} = -1/(a + \sqrt{1 + a^2})$. Combining these cases gives

$$t = \frac{\operatorname{sign}(\tau)}{|\tau| + \sqrt{1 + \tau^2}}.$$

For $\tau = 0$, the roots are 1 and -1. With the stated convention $\operatorname{sign}(0) = 1$, the formula selects 1.

The denominator is at least one, so $|t| \le 1$. Set $\theta = \arctan t$, giving $|\theta| \le \pi/4$, and define $c = 1/\sqrt{1 + t^2}$, $s = ct$. Then $c^2 + s^2 = 1$ and $s/c = t$. Substitution into the off-diagonal formula

gives zero because t satisfies the quadratic. The rationalized expression avoids subtracting two nearly equal positive numbers. □

Ex 4.11 Perform one Jacobi rotation by hand to zero out the off-diagonal entry of

$$\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 2 & 6 \end{pmatrix}.$$

Explicitly compute τ, t, c, s, and the resulting diagonal matrix $\mathbf{A}'$.

A rotation that annihilates a_{pq} may make a previously annihilated entry nonzero again, so the method does not terminate after $n(n-1)/2$ steps. Convergence follows instead from a quantity that decreases at every step.

Thm 4.15 **(Monotone Decrease of the Off-Diagonal Frobenius Norm)** Let the off-diagonal squared norm be defined by

$$\operatorname{off}(\mathbf{A})^2 = \sum_{i \neq j} a_{ij}^2 = \|\mathbf{A}\|_F^2 - \sum_{i=1}^{n} a_{ii}^2.$$

Applying the Jacobi rotation $\mathbf{A}' = \mathbf{G}(p, q, \theta)^T \mathbf{A} \mathbf{G}(p, q, \theta)$ with the optimal angle strictly reduces the off-diagonal norm:

$$\operatorname{off}(\mathbf{A}')^2 = \operatorname{off}(\mathbf{A})^2 - 2a_{pq}^2.$$

Proof Orthogonal multiplication preserves the Frobenius norm because it preserves the Euclidean norm of each column on the left, or each row on the right. Applying this twice gives the invariance used below.

Because $\mathbf{G}$ is orthogonal, the total Frobenius norm is preserved: $\|\mathbf{A}'\|_F^2 = \|\mathbf{A}\|_F^2$. The rotation affects only entries in rows and columns p and q. For the 2×2 submatrix, the sum of squares is invariant under orthogonal similarity:

$$(a'_{pp})^2 + (a'_{qq})^2 + 2(a'_{pq})^2 = a_{pp}^2 + a_{qq}^2 + 2a_{pq}^2.$$

Because the rotation enforces $a'_{pq} = 0$, the sum of squared diagonal entries increases:

$$(a'_{pp})^2 + (a'_{qq})^2 = a_{pp}^2 + a_{qq}^2 + 2a_{pq}^2.$$

Subtract the old squared diagonal sum from the new one in this block. The difference is exactly $2a_{pq}^2$. Thus diagonal energy has increased by that amount while total squared norm stays fixed.

All other diagonal entries a_{ii} $(i \neq p, q)$ are untouched. Substituting this into $\operatorname{off}(\mathbf{A}')^2 = \|\mathbf{A}'\|_F^2 - \sum (a'_{ii})^2$ proves that $\operatorname{off}(\mathbf{A}')^2 = \operatorname{off}(\mathbf{A})^2 - 2a_{pq}^2$.

In particular, a nonzero selected entry gives a strict decrease. If it was already zero and no rotation is performed, the formula gives equality. A single decreasing quantity does not by itself specify a global rate until the choice of pairs is considered. □

Ex 4.12 Prove that for an $n \times n$ matrix, if the largest off-diagonal entry is selected at each step (Classical Jacobi), then $\operatorname{off}(\mathbf{A}^{(k+1)})^2 \leq \left(1 - \frac{2}{n(n-1)}\right) \operatorname{off}(\mathbf{A}^{(k)})^2$.

Def 4.10 **(Cyclic Jacobi Algorithm)** Let $\mathbf{A}^{(0)} = \mathbf{A}$ and $\mathbf{V} = \mathbf{I}_n$. While $\operatorname{off}(\mathbf{A}) > \texttt{tol}\,\|\mathbf{A}^{(0)}\|_F$, sweep through the pairs $1 \le p < q \le n$ in a fixed order. For each pair, compute c and s from Thm 4.14, replace $\mathbf{A}$ by $\mathbf{G}(p,q,\theta)^T\mathbf{A}\mathbf{G}(p,q,\theta)$, and replace $\mathbf{V}$ by $\mathbf{V}\mathbf{G}(p,q,\theta)$. On termination, the diagonal of $\mathbf{A}$ approximates the eigenvalues and the columns of $\mathbf{V}$ the eigenvectors.

Theorem 4.15 shows that $\operatorname{off}(\mathbf{A})^2$ decreases by $2a_{pq}^2$ at every rotation. Choosing the largest off-diagonal entry at each step gives linear convergence, as the exercises show. The cyclic order avoids that search, and once the off-diagonal entries are small compared with the gaps between the eigenvalues, cyclic Jacobi converges quadratically, measured sweep by sweep (Golub and Van Loan, 2013). Each sweep costs $O(n^3)$ flops.

4.2.4 Shifted Hessenberg Systems

Several calculations require the solution of $(s\mathbf{I} - \mathbf{A})\boldsymbol{x} = \boldsymbol{b}$ for many complex shifts s with the same $\mathbf{A}$ and $\boldsymbol{b}$. The vector $(s\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{b}$ is the resolvent of $\mathbf{A}$ applied to $\boldsymbol{b}$. Resolvents appear in Laplace transforms of solutions of linear differential equations, in frequency responses, and in contour-integral formulas for matrix functions. Factoring $s\mathbf{I} - \mathbf{A}$ separately for each of N shifts costs about $\frac{2}{3}n^3$ flops per shift. The Hessenberg reduction of Thm 4.13 removes this repetition, because adding a multiple of the identity does not change Hessenberg structure.

Thm 4.16 **(Shifted Solves with a Hessenberg Matrix)** Let $\mathbf{A} = \mathbf{Q}\mathbf{H}\mathbf{Q}^T$ with $\mathbf{Q}$ orthogonal and $\mathbf{H}$ upper Hessenberg, and let s not be an eigenvalue of $\mathbf{A}$. Then $\boldsymbol{x} = (s\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{b}$ is given by

$$(s\mathbf{I} - \mathbf{H})\boldsymbol{y} = \mathbf{Q}^T\boldsymbol{b}, \qquad \boldsymbol{x} = \mathbf{Q}\boldsymbol{y}.$$

Gaussian elimination with partial pivoting solves the Hessenberg system in $O(n^2)$ operations. At each step, the only possible row interchange is between the pivot row and the row directly below it.

Proof Since $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$, we have $s\mathbf{I} - \mathbf{A} = \mathbf{Q}(s\mathbf{I} - \mathbf{H})\mathbf{Q}^T$, so $\boldsymbol{x} = \mathbf{Q}(s\mathbf{I} - \mathbf{H})^{-1}\mathbf{Q}^T\boldsymbol{b}$. This reduction can also be obtained without forming an inverse. Set $\boldsymbol{x} = \mathbf{Q}\boldsymbol{y}$ in $(s\mathbf{I} - \mathbf{A})\boldsymbol{x} = \boldsymbol{b}$, then multiply by $\mathbf{Q}^T$. Orthogonality gives exactly $(s\mathbf{I} - \mathbf{H})\boldsymbol{y} = \mathbf{Q}^T\boldsymbol{b}$. Nonsingularity follows because s is not an eigenvalue.

The matrix $s\mathbf{I} - \mathbf{H}$ differs from $-\mathbf{H}$ only on the diagonal, so it is upper Hessenberg. Suppose that after $k-1$ elimination steps the active submatrix is upper Hessenberg. Its first column then has a single nonzero entry below the diagonal, in row $k+1$. Partial pivoting compares only rows k and $k+1$, computes one multiplier, and updates one row of length at most $n-k+1$. Rows $k+2, \dots, n$ are untouched, so the next active submatrix is again upper Hessenberg. Summing the work over $k = 1, \dots, n-1$ gives $O(n^2)$ operations, and the triangular solve that follows costs another $O(n^2)$.

The total row-update length is bounded by a constant times $\sum_{k=1}^{n-1}(n-k+1)$. This sum is $(n-1)(n+2)/2 = O(n^2)$. Pivot comparisons and swaps cost at most the same order. An upper triangular solve also has a quadratic count, so adding it leaves the overall order unchanged. $\square$

Reducing $\mathbf{A}$ to Hessenberg form with Householder reflectors costs about $\frac{10}{3}n^3$ flops when $\mathbf{Q}$ is kept as a product of reflectors, and this reduction is done once (Golub and Van Loan, 2013). Applying $\mathbf{Q}^T$ and $\mathbf{Q}$ to a vector costs $O(n^2)$ flops. For N shifts, the total cost is therefore about $\frac{10}{3}n^3 + O(Nn^2)$ flops, compared with about $\frac{2}{3}Nn^3$ for separate factorizations. The reduction pays for itself after five shifts, and with one hundred shifts it reduces the cost by a factor of about twenty.

Remark 4.1 **(Contour integrals for the matrix exponential)** The Cauchy integral formula for matrix functions gives

$$e^{t\mathbf{A}}\boldsymbol{b} = \frac{1}{2\pi \mathrm{i}} \int_{\Gamma} e^{ts}(s\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{b}\, ds,$$

where Γ is a contour that encloses the spectrum of $\mathbf{A}$ (Higham, 2008). When the spectrum lies in the left half-plane, as for Markov generators and discretized diffusion operators, Γ can be chosen to wrap around the negative real axis so that e^{ts} decays along it. After a parametrization of Γ, the trapezoid rule with N nodes approximates the integral by a sum of N resolvent applications, each computed by Thm 4.16. For suitable Talbot-type contours the error decreases geometrically in N, and a few tens of nodes commonly reach double-precision accuracy (Weideman and Trefethen, 2007). The geometric convergence is the same mechanism as the spectral accuracy of the trapezoid rule for periodic analytic integrands in Section 5.3. For large sparse matrices, the Krylov approximations of Section 4.4 are usually preferable, because they avoid the dense Hessenberg reduction.

4.2.5 Further Exercises

Ex 4.13 **(Observed cubic convergence of RQI)** Apply RQI to a symmetric matrix with distinct eigenvalues, starting from several unit vectors close to the same eigenvector.

1. Record the angle errors $\theta_k = \angle(\boldsymbol{v}^{(k)}, \boldsymbol{q}_*)$ and the ratios $\sin\theta_{k+1}/\sin^3\theta_k$.
2. Record the Rayleigh errors and verify the cubic relation in Thm 4.11 before rounding error dominates.
3. Repeat with two nearly repeated eigenvalues and explain how the spectral gap enters the constants in the proof.

Ex 4.14 **(Failure of unshifted QR on an exchange matrix)** Let $\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

1. Compute the QR factorization $\mathbf{P} = \mathbf{Q}_0\mathbf{R}_0$.
2. Compute $\mathbf{P}_1 = \mathbf{R}_0\mathbf{Q}_0$ and show that $\mathbf{P}_1 = \mathbf{P}$.
3. Repeat the step with a nonzero shift. Determine which shifts break the cycle, and explain how an eigenvalue shift from the trailing two by two block permits immediate deflation in this example.

Ex 4.15 **(Givens rotation Hessenberg QR)** Write a Python function that executes one step of shifted QR on an upper Hessenberg matrix $\mathbf{H} \in \mathbb{R}^{n\times n}$ using $n-1$ Givens rotations. Verify that your implementation requires $O(n^2)$ flops rather than $O(n^3)$.

Ex 4.16 Verify that the eigenvalues computed on the diagonal of $\mathbf{A}'$ match the exact roots of $\det(\mathbf{A}-\lambda\mathbf{I}) = 0$.

Ex 4.17 **(Many shifted solves)** For a random 200×200 matrix, compare the time to solve $(s_k\mathbf{I}-\mathbf{A})\boldsymbol{x}_k = \boldsymbol{b}$ for $k = 1, \dots, N$ by separate LU factorizations and by one Hessenberg reduction followed by Thm 4.16. Find the value of N at which the two approaches cost the same, and compare it with the flop estimate in the text.

4.3 The Singular Value Decomposition

An eigenvector compares input and output in the same space. A rectangular matrix may have different input and output dimensions, so we instead seek orthonormal input directions whose images are mutually orthogonal. In those coordinates, the map merely scales each direction and may send some directions to zero.

The singular value decomposition makes this construction precise. It will evaluate the matrix norm, expose directions that are poorly determined by a least-squares problem, and identify which components can be discarded in an approximation of smaller rank.

4.3.1 The SVD Theorem and Subspace Characterization

Suppose unit input vectors $\boldsymbol{v}_i$ have images $\mathbf{A}\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$ with orthonormal $\boldsymbol{u}_i$ and nonnegative σ_i. Applying $\mathbf{A}$ then consists of expanding an input in the $\boldsymbol{v}_i$ basis, multiplying its coefficients by σ_i, and expressing the output in the $\boldsymbol{u}_i$ basis. To find such input vectors, observe that

$$(\mathbf{A}\boldsymbol{v}_i)^T(\mathbf{A}\boldsymbol{v}_j) = \boldsymbol{v}_i^T(\mathbf{A}^T\mathbf{A})\boldsymbol{v}_j.$$

An orthonormal eigenbasis of the symmetric matrix $\mathbf{A}^T\mathbf{A}$ therefore makes the images orthogonal. This observation supplies the proof.

Thm 4.17 **(The Singular Value Decomposition Theorem)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $m \geq n$. There exist orthogonal matrices $\mathbf{U} \in \mathbb{R}^{m\times m}$ and $\mathbf{V} \in \mathbb{R}^{n\times n}$, and a rectangular diagonal matrix $\mathbf{\Sigma} \in \mathbb{R}^{m\times n}$, such that

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T,$$

where the diagonal entries of $\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \sigma_2, \ldots, \sigma_n)$ are real, non-negative, and sorted in non-increasing order:

$$\sigma_1 \geq \cdots \geq \sigma_n \geq 0, \qquad \sigma_i > 0 \ (1 \leq i \leq r), \qquad \sigma_i = 0 \ (r < i \leq n).$$

The number of strictly positive singular values $r = \operatorname{rank}(\mathbf{A})$ equals the rank of $\mathbf{A}$. In dyadic outer-product form,

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T.$$

Proof Consider the Gram matrix $\mathbf{A}^T\mathbf{A} \in \mathbb{R}^{n\times n}$. Because $(\mathbf{A}^T\mathbf{A})^T = \mathbf{A}^T\mathbf{A}$, it is symmetric, and for every $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{x}^T(\mathbf{A}^T\mathbf{A})\boldsymbol{x} = \|\mathbf{A}\boldsymbol{x}\|_2^2 \geq 0$, so $\mathbf{A}^T\mathbf{A}$ is positive semidefinite.

By the Spectral Theorem (Thm 4.4), $\mathbf{A}^T\mathbf{A}$ has an orthonormal basis of eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{R}^n$ with corresponding real eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$. Define the singular values $\sigma_i = \sqrt{\lambda_i} \geq 0$ for $i = 1, \ldots, n$, and let r denote the number of strictly positive singular values, with positivity for $1 \leq i \leq r$ and zeros for $r < i \leq n$. Either index range may be empty.

For each index $i \in \{1, \ldots, r\}$, define the normalized vectors in $\mathbb{R}^m$:

$$\boldsymbol{u}_i = \frac{1}{\sigma_i}\mathbf{A}\boldsymbol{v}_i.$$

We verify that $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_r\}$ form an orthonormal set in $\mathbb{R}^m$:

$$\boldsymbol{u}_i^T \boldsymbol{u}_j = \frac{1}{\sigma_i \sigma_j} (\mathbf{A}\boldsymbol{v}_i)^T (\mathbf{A}\boldsymbol{v}_j) = \frac{1}{\sigma_i \sigma_j} \boldsymbol{v}_i^T (\mathbf{A}^T \mathbf{A} \boldsymbol{v}_j) = \frac{\lambda_j}{\sigma_i \sigma_j} \boldsymbol{v}_i^T \boldsymbol{v}_j = \frac{\sigma_j^2}{\sigma_i \sigma_j} \delta_{ij} = \delta_{ij}.$$

When $i = j$, the displayed inner product is $\sigma_i^2/\sigma_i^2 = 1$. When $i \neq j$, it is zero because $\boldsymbol{v}_i^T \boldsymbol{v}_j = 0$. All divisions used only positive singular values, so no zero singular value was divided by.

Using Gram–Schmidt orthonormalization, extend $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_r\}$ to a complete orthonormal basis $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_m\}$ for $\mathbb{R}^m$. Assemble the orthogonal matrices $\mathbf{U} = [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{R}^{m \times m}$ and $\mathbf{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{R}^{n \times n}$.

Evaluating the (i, j) entry of $\mathbf{U}^T \mathbf{A} \mathbf{V}$:

$$(\mathbf{U}^T \mathbf{A} \mathbf{V})_{ij} = \boldsymbol{u}_i^T \mathbf{A} \boldsymbol{v}_j.$$

For $j \leq r$, $\mathbf{A}\boldsymbol{v}_j = \sigma_j \boldsymbol{u}_j$, so $\boldsymbol{u}_i^T \mathbf{A} \boldsymbol{v}_j = \sigma_j \boldsymbol{u}_i^T \boldsymbol{u}_j = \sigma_j \delta_{ij}$. For $j > r$, $\lambda_j = 0$, so $\|\mathbf{A}\boldsymbol{v}_j\|_2^2 = \boldsymbol{v}_j^T \mathbf{A}^T \mathbf{A} \boldsymbol{v}_j = 0$, which implies $\mathbf{A}\boldsymbol{v}_j = \mathbf{0}$ and $\boldsymbol{u}_i^T \mathbf{A} \boldsymbol{v}_j = 0$. Thus $\mathbf{U}^T \mathbf{A} \mathbf{V} = \mathbf{\Sigma}$, which gives $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$.

The images of the first r right singular vectors are nonzero multiples of the independent vectors $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$. The remaining right singular vectors map to zero. Since these vectors form a basis of the domain, the range has dimension exactly r. This proves the rank assertion in addition to the factorization. □

If $m < n$, apply the theorem to $\mathbf{A}^T$ and transpose the factorization. The left and right factors exchange roles, and the rectangular diagonal factor is transposed. Thus the decomposition exists for every rectangular matrix. When $r = 0$, the outer-product sum is empty and $\mathbf{A} = \mathbf{0}$.

Ex 4.18 Construct an SVD of $\begin{pmatrix} 3 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$. Describe the image of the unit circle, then identify the range and the null space of the transpose. Repeat after replacing the entry 1 by zero. Which input direction has become invisible?

Thm 4.18 **(Thin SVD and the Four Fundamental Subspaces)** Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have rank $r \leq n \leq m$. The thin SVD is

$$\mathbf{A} = \mathbf{U}_1 \mathbf{\Sigma}_1 \mathbf{V}_1^T,$$

where $\mathbf{U}_1 \in \mathbb{R}^{m \times r}$ and $\mathbf{V}_1 \in \mathbb{R}^{n \times r}$ have orthonormal columns, and $\mathbf{\Sigma}_1 = \operatorname{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}^{r \times r}$ has strictly positive entries. The complete singular vectors partition the domain and codomain into the four fundamental subspaces:

1. **Column space (range):** $\mathcal{R}(\mathbf{A}) = \operatorname{span}\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_r\} \subset \mathbb{R}^m$.
2. **Left null space:** $\mathcal{N}(\mathbf{A}^T) = \operatorname{span}\{\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m\} \subset \mathbb{R}^m$.
3. **Row space:** $\mathcal{R}(\mathbf{A}^T) = \operatorname{span}\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_r\} \subset \mathbb{R}^n$.
4. **Null space (kernel):** $\mathcal{N}(\mathbf{A}) = \operatorname{span}\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n\} \subset \mathbb{R}^n$.

Proof Expand $\boldsymbol{x}$ in the complete orthonormal right singular basis: $\boldsymbol{x} = \sum_{i=1}^{n} c_i \boldsymbol{v}_i$, where $c_i = \boldsymbol{v}_i^T \boldsymbol{x}$. The SVD gives

$$\mathbf{A}\boldsymbol{x} = \sum_{i=1}^{r} \sigma_i c_i \boldsymbol{u}_i.$$

Thus every output lies in $\operatorname{span}\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$. Conversely, for $i \le r$, $\boldsymbol{u}_i = \mathbf{A}(\boldsymbol{v}_i/\sigma_i)$ is an output. Hence that span is exactly the range, not merely a space containing it.

For the null space, take the squared Euclidean norm of the output. Orthonormality of the left singular vectors gives

$$\|\mathbf{A}\boldsymbol{x}\|_2^2 = \sum_{i=1}^{r} \sigma_i^2 c_i^2.$$

Every σ_i in this sum is positive. The sum vanishes if and only if $c_i = 0$ for every $i \le r$. The remaining coefficients are arbitrary, so the null space is exactly $\operatorname{span}\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\}$.

For the transpose, expand $\boldsymbol{y} = \sum_{i=1}^{m} d_i \boldsymbol{u}_i$. Transposing the SVD gives

$$\mathbf{A}^T\boldsymbol{y} = \sum_{i=1}^{r} \sigma_i d_i \boldsymbol{v}_i.$$

Every output is in the first r right singular directions, and each $\boldsymbol{v}_i = \mathbf{A}^T(\boldsymbol{u}_i/\sigma_i)$ is an output. This proves the row-space claim. Likewise $\|\mathbf{A}^T\boldsymbol{y}\|_2^2 = \sum_{i=1}^{r} \sigma_i^2 d_i^2$, which vanishes precisely when the first r coefficients are zero. The left null space is therefore the span of the remaining $\boldsymbol{u}_i$. □

4.3.2 Norms and Condition Numbers

The induced Euclidean norm of Def 1.22 is the largest amplification of a unit vector, and the condition number of Def 1.25 compares the largest and the smallest amplification. Both were defined as optimization problems over the unit sphere. Because orthogonal matrices preserve Euclidean length, the SVD reduces these problems to a diagonal matrix, where they can be solved by inspection.

Thm 4.19 **(Norms and Condition Numbers from the SVD)** Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ with $m \ge n$ have singular values $\sigma_1 \ge \cdots \ge \sigma_n \ge 0$. Then

$$\|\mathbf{A}\|_2 = \sigma_1, \qquad \min_{\|\boldsymbol{x}\|_2 = 1} \|\mathbf{A}\boldsymbol{x}\|_2 = \sigma_n, \qquad \|\mathbf{A}\|_F = \Big(\sum_{i=1}^{n} \sigma_i^2\Big)^{1/2}.$$

If $\mathbf{A}$ has full column rank, we define $\kappa_2(\mathbf{A}) = \sigma_1/\sigma_n$. The condition number satisfies

$$\kappa_2(\mathbf{A}^T\mathbf{A}) = \kappa_2(\mathbf{A})^2.$$

If $\mathbf{A}$ is square and nonsingular, then $\|\mathbf{A}^{-1}\|_2 = 1/\sigma_n$, so $\kappa_2(\mathbf{A}) = \|\mathbf{A}\|_2\|\mathbf{A}^{-1}\|_2$ agrees with Def 1.25.

Proof Let $\boldsymbol{x}$ be a unit vector and let $\boldsymbol{y} = \mathbf{V}^T\boldsymbol{x}$, which is also a unit vector. Since $\mathbf{U}$ preserves Euclidean length,

$$\|\mathbf{A}\boldsymbol{x}\|_2^2 = \|\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\boldsymbol{x}\|_2^2 = \|\mathbf{\Sigma}\boldsymbol{y}\|_2^2 = \sum_{i=1}^{n} \sigma_i^2 y_i^2.$$

The sum is a weighted average of the σ_i^2, so it lies between σ_n^2 and σ_1^2. The upper value is attained at $\boldsymbol{y} = \boldsymbol{e}_1$, that is, at $\boldsymbol{x} = \boldsymbol{v}_1$, and the lower value is attained at $\boldsymbol{x} = \boldsymbol{v}_n$. These two cases prove the first two identities.

For any unit vector, $\sum_i y_i^2 = 1$. Therefore $\sum_i \sigma_i^2 y_i^2 \le \sigma_1^2 \sum_i y_i^2 = \sigma_1^2$, and the analogous lower inequality uses σ_n^2. These inequalities become equalities for the indicated coordinate vectors. Thus they give extrema, not just bounds.

For the Frobenius norm, $\|\mathbf{A}\|_F^2 = \operatorname{trace}(\mathbf{A}^T\mathbf{A})$ and $\mathbf{A}^T\mathbf{A} = \mathbf{V}\boldsymbol{\Sigma}^T\boldsymbol{\Sigma}\mathbf{V}^T$. The identity $\operatorname{tr}(\mathbf{BC}) = \operatorname{tr}(\mathbf{CB})$ follows by interchanging the two finite sums over entries. Applying it to this product gives

$$\operatorname{tr}(\mathbf{V}\boldsymbol{\Sigma}^T\boldsymbol{\Sigma}\mathbf{V}^T) = \operatorname{tr}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma}\mathbf{V}^T\mathbf{V}) = \operatorname{tr}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma}).$$

Thus the trace is invariant under the similarity by $\mathbf{V}$, so $\|\mathbf{A}\|_F^2 = \operatorname{trace}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma}) = \sum_i \sigma_i^2$.

The factorization $\mathbf{A}^T\mathbf{A} = \mathbf{V}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma})\mathbf{V}^T$ is itself an SVD of $\mathbf{A}^T\mathbf{A}$ with singular values $\sigma_1^2 \ge \cdots \ge \sigma_n^2$. When $\mathbf{A}$ has full column rank these are positive, and $\kappa_2(\mathbf{A}^T\mathbf{A}) = \sigma_1^2/\sigma_n^2 = \kappa_2(\mathbf{A})^2$. Finally, if $\mathbf{A}$ is square and nonsingular, then $\mathbf{A}^{-1} = \mathbf{V}\boldsymbol{\Sigma}^{-1}\mathbf{U}^T$ is an SVD whose largest singular value is $1/\sigma_n$. □

Cor 4.20 **(Symmetric Positive Definite Matrices)** If $\mathbf{A}$ is symmetric positive definite, its singular values are its eigenvalues, and $\kappa_2(\mathbf{A}) = \lambda_{\max}(\mathbf{A})/\lambda_{\min}(\mathbf{A})$. If $\mathbf{A} = \mathbf{L}\mathbf{L}^T$ is the Cholesky factorization of Thm 2.18, then $\kappa_2(\mathbf{L}) = \sqrt{\kappa_2(\mathbf{A})}$.

Proof For an SPD matrix, the spectral theorem gives $\mathbf{A} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^T$ with positive eigenvalues. Order them in decreasing order. Both outer factors are orthogonal, and the middle factor is diagonal with nonnegative decreasing entries. It therefore satisfies every condition of an SVD. Its singular values are the positive eigenvalues themselves.

Now suppose $\mathbf{A} = \mathbf{L}\mathbf{L}^T$ is the Cholesky factorization. To find singular values of $\mathbf{L}^T$, form its Gram matrix:

$$(\mathbf{L}^T)^T\mathbf{L}^T = \mathbf{L}\mathbf{L}^T = \mathbf{A}.$$

The SVD construction shows that squared singular values of a matrix are the eigenvalues of its Gram matrix. Thus $\sigma_i(\mathbf{L}^T) = \sqrt{\lambda_i(\mathbf{A})}$. Transposing an SVD exchanges its two orthogonal factors and transposes its diagonal factor, preserving the singular values. Hence the same formula holds for $\mathbf{L}$.

The smallest eigenvalue is positive, so both extreme singular values are nonzero. Consequently

$$\kappa_2(\mathbf{L}) = \frac{\sigma_{\max}(\mathbf{L})}{\sigma_{\min}(\mathbf{L})} = \frac{\sqrt{\lambda_{\max}(\mathbf{A})}}{\sqrt{\lambda_{\min}(\mathbf{A})}} = \sqrt{\kappa_2(\mathbf{A})}.$$

□

The identity $\kappa_2(\mathbf{A}^T\mathbf{A}) = \kappa_2(\mathbf{A})^2$ explains the preference for QR in Section 3.2. When the normal equations are formed and solved by a backward stable method, the forward error can be as large as a modest multiple of $\kappa_2(\mathbf{A})^2 u$, where u is the unit roundoff. Householder QR works with $\mathbf{A}$ directly, and when the least-squares residual is small its forward error is a modest multiple of $\kappa_2(\mathbf{A})u$ (Higham, 2002). For $\kappa_2(\mathbf{A}) = 10^6$ in double precision, these two estimates are of order 10^{-4} and 10^{-10}.

4.3.3 Low-Rank Matrix Approximation

Thm 4.21 **(Eckart–Young–Mirsky Low-Rank Approximation Theorem)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have singular value decomposition $\mathbf{A} = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$. For any $k < r$, let the rank-k truncated SVD be

$$\mathbf{A}_k = \sum_{i=1}^{k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T.$$

Then $\mathbf{A}_k$ is the best rank-k approximation to $\mathbf{A}$ in the spectral norm:

$$\min_{\operatorname{rank}(\mathbf{B})\le k} \|\mathbf{A}-\mathbf{B}\|_2 = \|\mathbf{A}-\mathbf{A}_k\|_2 = \sigma_{k+1},$$

and in the Frobenius norm:

$$\min_{\operatorname{rank}(\mathbf{B})\le k} \|\mathbf{A}-\mathbf{B}\|_F = \|\mathbf{A}-\mathbf{A}_k\|_F = \sqrt{\sum_{j=k+1}^{r} \sigma_j^2}.$$

Proof First, evaluating the spectral norm of $\mathbf{A}-\mathbf{A}_k$:

$$\mathbf{A}-\mathbf{A}_k = \sum_{i=k+1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T = \mathbf{U}\,\operatorname{diag}(0,\dots,0,\sigma_{k+1},\dots,\sigma_r,0,\dots,0)\mathbf{V}^T.$$

Because the 2-norm is unitarily invariant, $\|\mathbf{A}-\mathbf{A}_k\|_2 = \sigma_{k+1}$.

Now let $\mathbf{B} \in \mathbb{R}^{m\times n}$ be any arbitrary matrix with $\operatorname{rank}(\mathbf{B}) \le k$. By the Rank-Nullity Theorem, $\dim \mathcal{N}(\mathbf{B}) = n - \operatorname{rank}(\mathbf{B}) \ge n-k$. Define the $(k+1)$-dimensional subspace $W_{k+1} = \operatorname{span}\{\boldsymbol{v}_1,\dots,\boldsymbol{v}_{k+1}\} \subset \mathbb{R}^n$. By the dimension formula,

$$\begin{aligned}\dim(\mathcal{N}(\mathbf{B}) \cap W_{k+1}) &= \dim \mathcal{N}(\mathbf{B}) + \dim W_{k+1} \\ &\quad - \dim(\mathcal{N}(\mathbf{B}) + W_{k+1}) \\ &\ge (n-k) + (k+1) - n = 1.\end{aligned}$$

Therefore, there exists a nonzero vector $\boldsymbol{w} \in \mathcal{N}(\mathbf{B}) \cap W_{k+1}$, normalized so that $\|\boldsymbol{w}\|_2 = 1$.

Because $\boldsymbol{w} \in \mathcal{N}(\mathbf{B})$, $\mathbf{B}\boldsymbol{w} = \mathbf{0}$. Expanding $\boldsymbol{w} = \sum_{j=1}^{k+1} c_j \boldsymbol{v}_j$ with $\sum_{j=1}^{k+1} c_j^2 = 1$,

$$\begin{aligned}\|(\mathbf{A}-\mathbf{B})\boldsymbol{w}\|_2^2 &= \|\mathbf{A}\boldsymbol{w} - \mathbf{B}\boldsymbol{w}\|_2^2 = \|\mathbf{A}\boldsymbol{w}\|_2^2 \\ &= \left\|\mathbf{A}\sum_{j=1}^{k+1} c_j \boldsymbol{v}_j\right\|_2^2 = \left\|\sum_{j=1}^{k+1} c_j \sigma_j \boldsymbol{u}_j\right\|_2^2 \\ &= \sum_{j=1}^{k+1} c_j^2 \sigma_j^2 \ge \sigma_{k+1}^2 \sum_{j=1}^{k+1} c_j^2 = \sigma_{k+1}^2.\end{aligned}$$

Consequently, the operator norm satisfies

$$\|\mathbf{A}-\mathbf{B}\|_2 = \max_{\|\boldsymbol{x}\|_2=1} \|(\mathbf{A}-\mathbf{B})\boldsymbol{x}\|_2 \ge \|(\mathbf{A}-\mathbf{B})\boldsymbol{w}\|_2 \ge \sigma_{k+1}.$$

This proves that $\|\mathbf{A}-\mathbf{B}\|_2 \ge \sigma_{k+1}$ for every matrix $\mathbf{B}$ of rank at most k.

For the Frobenius norm, let $\mathbf{P}$ be the orthogonal projector onto $\mathcal{N}(\mathbf{B})$. Since $\operatorname{rank}(\mathbf{B}) \le k$, this null space has dimension at least $n-k$. Hence

$$\|\mathbf{A}-\mathbf{B}\|_F^2 \ge \|(\mathbf{A}-\mathbf{B})\mathbf{P}\|_F^2 = \|\mathbf{A}\mathbf{P}\|_F^2.$$

Choose an $(n-k)$-dimensional subspace of $\mathcal{N}(\mathbf{B})$ if the null space is larger, and let $\mathbf{P}$ project onto that subspace. Expanding in the right singular-vector basis gives

$$\begin{aligned}\|\mathbf{A}\mathbf{P}\|_F^2 &= \operatorname{tr}(\mathbf{P}^T\mathbf{A}^T\mathbf{A}\mathbf{P}) = \operatorname{tr}\left(\mathbf{P}\left(\sum_{i=1}^{n}\sigma_i^2\boldsymbol{v}_i\boldsymbol{v}_i^T\right)\mathbf{P}\right)\\ &= \sum_{i=1}^{n}\sigma_i^2\operatorname{tr}(\mathbf{P}\boldsymbol{v}_i\boldsymbol{v}_i^T\mathbf{P}) = \sum_{i=1}^{n}\sigma_i^2\|\mathbf{P}\boldsymbol{v}_i\|_2^2.\end{aligned}$$

The weights $d_i = \|\mathbf{P}\boldsymbol{v}_i\|_2^2$ satisfy $0 \le d_i \le 1$ and $\sum_i d_i = \operatorname{tr}(\mathbf{P}) = n-k$. To justify the minimum of this weighted sum, put $s_i = \sigma_i^2$ in decreasing order. From $\sum_i d_i = n-k$ we have

$$\sum_{i\le k} d_i = \sum_{i>k}(1-d_i) = t \ge 0.$$

Now subtract the desired tail sum:

$$\begin{aligned}\sum_i s_i d_i - \sum_{i>k} s_i &= \sum_{i\le k} s_i d_i - \sum_{i>k} s_i(1-d_i)\\ &\ge s_{k+1}t - s_{k+1}t = 0.\end{aligned}$$

The inequality uses $s_i \ge s_{k+1}$ in the first sum and $s_i \le s_{k+1}$ in the second. This proves the weighted minimum explicitly. Therefore

$$\|\mathbf{A}-\mathbf{B}\|_F^2 \ge \sum_{i=k+1}^{r}\sigma_i^2.$$

The truncated SVD attains this bound, completing the proof.

For the proposed truncation, its error has precisely the discarded singular values, so its squared Frobenius norm is their squared sum. For $k \ge r$, the full matrix itself is admissible and both minimum errors are zero. The approximation theorem in these two norms is discussed in Golub and Van Loan (2013). □

Ex 4.19 **(Low-rank error accounting)** For a diagonal matrix $\mathbf{A} = \operatorname{diag}(9,4,1)$, compute the best rank-one and rank-two approximations. Verify both the spectral- and Frobenius-norm errors predicted by Thm 4.21.

Remark 4.2 **(Numerical rank and noise filtering)** In measured or discretized data, singular values rarely vanish exactly. Given a tolerance $\tau > 0$, define the numerical rank by

$$r_\tau = \#\{i : \sigma_i > \tau\}.$$

A common floating-point default is $\tau = \varepsilon_{\text{mach}}\max(m,n)\sigma_1$, but a scientifically meaningful threshold should reflect the noise and discretization scales. Truncation removes components below the chosen threshold. Calling those components noise requires a model of the data.

4.3.4 Least Squares and Regularization

The least-squares theory of Section 3.2 characterized every minimizer and the minimum-norm minimizer, but it computed a solution only when $\mathbf{A}$ has full column rank. The SVD computes all minimizers in every case. It also shows which components of the solution are sensitive to errors in $\boldsymbol{b}$, and how truncation and Tikhonov regularization modify those components.

Def 4.11 **(Moore–Penrose Pseudoinverse)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have rank r and thin SVD $\mathbf{A} = \mathbf{U}_1\boldsymbol{\Sigma}_1\mathbf{V}_1^T$. The Moore–Penrose pseudoinverse of $\mathbf{A}$ is

$$\mathbf{A}^\dagger = \mathbf{V}_1\boldsymbol{\Sigma}_1^{-1}\mathbf{U}_1^T = \sum_{i=1}^{r} \frac{1}{\sigma_i}\boldsymbol{v}_i\boldsymbol{u}_i^T \in \mathbb{R}^{n\times m}.$$

Ex 4.20 **(Verification of the four Moore–Penrose conditions)** Using the thin SVD $\mathbf{A} = \mathbf{U}_1\boldsymbol{\Sigma}_1\mathbf{V}_1^T$ and $\mathbf{A}^\dagger = \mathbf{V}_1\boldsymbol{\Sigma}_1^{-1}\mathbf{U}_1^T$, prove algebraically:

1. $\mathbf{A}\mathbf{A}^\dagger\mathbf{A} = \mathbf{A}$.
2. $\mathbf{A}^\dagger\mathbf{A}\mathbf{A}^\dagger = \mathbf{A}^\dagger$.
3. $(\mathbf{A}\mathbf{A}^\dagger)^T = \mathbf{A}\mathbf{A}^\dagger$ (orthogonal projector onto $\mathcal{R}(\mathbf{A})$).
4. $(\mathbf{A}^\dagger\mathbf{A})^T = \mathbf{A}^\dagger\mathbf{A}$ (orthogonal projector onto $\mathcal{R}(\mathbf{A}^T)$).

Thm 4.22 **(Least Squares by the SVD)** Let $\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$ have rank r. A vector $\boldsymbol{x}$ minimizes $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2$ if and only if

$$\boldsymbol{v}_i^T\boldsymbol{x} = \frac{\boldsymbol{u}_i^T\boldsymbol{b}}{\sigma_i}, \qquad i = 1, \dots, r.$$

The minimum residual norm is $\left(\sum_{i=r+1}^{m}(\boldsymbol{u}_i^T\boldsymbol{b})^2\right)^{1/2}$, and the unique minimizer of smallest Euclidean norm is

$$\mathbf{A}^\dagger\boldsymbol{b} = \sum_{i=1}^{r} \frac{\boldsymbol{u}_i^T\boldsymbol{b}}{\sigma_i}\,\boldsymbol{v}_i.$$

Proof Introduce orthogonal coordinates $\boldsymbol{y} = \mathbf{V}^T\boldsymbol{x}$ and $\boldsymbol{c} = \mathbf{U}^T\boldsymbol{b}$. These changes of coordinates are bijective and preserve Euclidean lengths. In particular, minimizing over $\boldsymbol{x}$ is equivalent to minimizing over every $\boldsymbol{y} \in \mathbb{R}^n$.

Multiply the residual by $\mathbf{U}^T$ and insert the SVD:

$$\begin{aligned}\|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 &= \|\mathbf{U}^T(\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T\boldsymbol{x} - \boldsymbol{b})\|_2^2 \\ &= \|\boldsymbol{\Sigma}\boldsymbol{y} - \boldsymbol{c}\|_2^2 \\ &= \sum_{i=1}^{r}(\sigma_i y_i - c_i)^2 + \sum_{i=r+1}^{m} c_i^2.\end{aligned}$$

The final sum is fixed by the data and cannot be changed by the solution. Each term in the first sum is nonnegative. Since $\sigma_i > 0$ for $i \le r$, choose $y_i = c_i/\sigma_i$ to make every such term zero. This proves the minimum residual value and identifies all necessary constrained coordinates.

The coordinates $y_{r+1}, \dots, y_n$ never appear in the residual because the corresponding columns of $\mathbf{\Sigma}$ vanish. They can therefore be arbitrary in a least-squares minimizer. To choose the minimizer of smallest norm, write

$$\|\boldsymbol{x}\|_2^2 = \|\boldsymbol{y}\|_2^2 = \sum_{i=1}^{r} \frac{c_i^2}{\sigma_i^2} + \sum_{i=r+1}^{n} y_i^2.$$

The first sum is already fixed. The second is minimized uniquely by setting every free coordinate to zero. Transforming back gives

$$\boldsymbol{x}_{\min} = \sum_{i=1}^{r} \frac{\boldsymbol{u}_i^T \boldsymbol{b}}{\sigma_i} \boldsymbol{v}_i = \mathbf{A}^{\dagger} \boldsymbol{b}.$$

Empty index ranges, including rank zero, are interpreted as empty sums. □

The coefficient $\boldsymbol{u}_i^T \boldsymbol{b}/\sigma_i$ shows how errors in the data enter the solution. If $\boldsymbol{b}$ is replaced by $\boldsymbol{b} + \boldsymbol{e}$, the component of the minimum-norm solution along $\boldsymbol{v}_i$ changes by $\boldsymbol{u}_i^T \boldsymbol{e}/\sigma_i$. Directions with small singular values amplify the noise by $1/\sigma_i$, which is the mechanism behind the ill-conditioning measured by $\kappa_2(\mathbf{A})$. Regularization reduces this amplification by modifying the coefficients for small σ_i.

Thm 4.23 **(Filter Factors for Truncation and Tikhonov Regularization)** Let $\mathbf{A}$ have rank r and SVD $\mathbf{A} = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$. For $k \le r$, the minimum-norm least-squares solution for the truncated matrix $\mathbf{A}_k$ of Thm 4.21 is

$$\boldsymbol{x}_k = \mathbf{A}_k^{\dagger} \boldsymbol{b} = \sum_{i=1}^{k} \frac{\boldsymbol{u}_i^T \boldsymbol{b}}{\sigma_i} \boldsymbol{v}_i.$$

For $\lambda > 0$, the Tikhonov solution of Def 3.6 is

$$\boldsymbol{x}_\lambda = \sum_{i=1}^{r} \frac{\sigma_i^2}{\sigma_i^2 + \lambda} \frac{\boldsymbol{u}_i^T \boldsymbol{b}}{\sigma_i} \boldsymbol{v}_i.$$

Proof For the truncated matrix, only the first k positive singular triplets are retained. Applying the minimum-norm least-squares formula to that matrix gives

$$\boldsymbol{x}_k = \sum_{i=1}^{k} \frac{\boldsymbol{u}_i^T \boldsymbol{b}}{\sigma_i} \boldsymbol{v}_i.$$

Relative to the untruncated pseudoinverse, the first k terms have multiplier one and every discarded term has multiplier zero. These are the truncation filters.

For Tikhonov regularization, let $\boldsymbol{y} = \mathbf{V}^T \boldsymbol{x}$ and $\boldsymbol{c} = \mathbf{U}^T \boldsymbol{b}$. Substitute the SVD into the normal equations and multiply by $\mathbf{V}^T$ on the left:

$$(\mathbf{\Sigma}^T \mathbf{\Sigma} + \lambda \mathbf{I}) \boldsymbol{y} = \mathbf{\Sigma}^T \boldsymbol{c}.$$

This is a diagonal system. For $i \le r$, its ith equation is $(\sigma_i^2 + \lambda) y_i = \sigma_i c_i$, so

$$y_i = \frac{\sigma_i c_i}{\sigma_i^2 + \lambda} = \frac{\sigma_i^2}{\sigma_i^2 + \lambda} \frac{c_i}{\sigma_i}.$$

For $i > r$, the equation is $\lambda y_i = 0$, giving $y_i = 0$. The assumption $\lambda > 0$ makes all diagonal coefficients strictly positive, including those in null-space directions.

Multiplication by $\mathbf{V}$ gives the stated filtered sum. The filter $\sigma_i^2/(\sigma_i^2 + \lambda)$ lies strictly between zero and one. It is small for singular values much smaller than $\sqrt{\lambda}$, whereas it approaches one when σ_i^2 is large relative to λ. □

Both regularized solutions multiply the unregularized coefficient $\boldsymbol{u}_i^T\boldsymbol{b}/\sigma_i$ by a filter factor. Truncation uses the factor one for $i \le k$ and zero for $i > k$, so it removes the weak directions entirely. Tikhonov regularization uses $\sigma_i^2/(\sigma_i^2 + \lambda)$, which is close to one when $\sigma_i^2 \gg \lambda$ and close to σ_i^2/λ when $\sigma_i^2 \ll \lambda$. The amplification $1/\sigma_i$ is therefore replaced by at most $1/(2\sqrt{\lambda})$, and the transition between kept and damped directions occurs near $\sigma_i = \sqrt{\lambda}$. The choice between the two methods, and the choice of k or λ, depends on the noise level in $\boldsymbol{b}$ and on the assumption about the solution discussed after Def 3.6.

Remark 4.3 **(Computing the SVD)** The proof of Thm 4.17 constructs the SVD from the eigendecomposition of $\mathbf{A}^T\mathbf{A}$, but practical codes do not compute it that way. A backward stable eigensolver applied to $\mathbf{A}^T\mathbf{A}$ computes its eigenvalues with absolute errors of order $u\sigma_1^2$, so singular values below about $\sqrt{u}\,\sigma_1$ lose all relative accuracy. Golub and Kahan instead reduce $\mathbf{A}$ to an upper bidiagonal matrix $\mathbf{B} = \mathbf{U}_0^T\mathbf{A}\mathbf{V}_0$ by applying Householder reflections alternately from the left and from the right (Golub and Kahan, 1965). A shifted QR iteration applied implicitly to $\mathbf{B}$ then computes the singular values, and standard dense libraries follow this two-phase approach. For a large sparse matrix, the bidiagonal matrix can instead be built one column at a time by a recurrence that uses only products with $\mathbf{A}$ and $\mathbf{A}^T$. The recurrence is closely related to the Lanczos iteration of Section 4.5 applied to $\mathbf{A}^T\mathbf{A}$.

4.3.5 Further Exercises

Ex 4.21 **(Hand computation of full and thin SVD)** For $\mathbf{A} = \begin{pmatrix} 3 & 3 \\ 4 & 4 \end{pmatrix}$:

1. Compute $\mathbf{A}^T\mathbf{A}$ and find its eigenvalues λ_1, λ_2 and orthonormal eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2$.
2. Compute σ_1, σ_2 and the left singular vector $\boldsymbol{u}_1 = \frac{1}{\sigma_1}\mathbf{A}\boldsymbol{v}_1$.
3. Complete $\{\boldsymbol{u}_1\}$ to an orthonormal basis of $\mathbb{R}^2$ to find $\boldsymbol{u}_2$.
4. Write down the full SVD $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ and verify that $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \mathbf{A}$.

Ex 4.22 **(Truncation versus Tikhonov regularization)** Let $\mathbf{A} = \operatorname{diag}(1, 10^{-2}, 10^{-4})$, let $\boldsymbol{x}_* = (1, 1, 1)^T$, and let $\boldsymbol{b} = \mathbf{A}\boldsymbol{x}_* + 10^{-3}(1, -1, 1)^T$.

1. Compute the least-squares solution $\mathbf{A}^{-1}\boldsymbol{b}$ and its error $\|\mathbf{A}^{-1}\boldsymbol{b} - \boldsymbol{x}_*\|_2$. Identify the component responsible for the error using the coefficients $\boldsymbol{u}_i^T\boldsymbol{b}/\sigma_i$.
2. Compute the truncated solutions $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3$ of Thm 4.23 and their errors.
3. Compute the Tikhonov solutions for $\lambda = 10^{-2}, 10^{-4}, 10^{-6}$. For each λ, list the three filter factors and explain which components are kept and which are damped.

Ex 4.23 **(SVD image compression and energy decay)** In Python, load a grayscale image as a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$.

1. Compute the full SVD using `scipy.linalg.svd`.
2. Plot the singular value spectrum σ_i on a semilog scale and the cumulative energy ratio $E(k) = \frac{\sum_{i=1}^{k} \sigma_i^2}{\sum_{i=1}^{r} \sigma_i^2}$ as a function of k.
3. Reconstruct the image at ranks $k \in \{5, 20, 50\}$. Compare memory storage against the original image.

4.4 Krylov Subspaces and the Arnoldi Process

The QR algorithm of Thm 4.13 computes every eigenvalue of a dense matrix in $O(n^3)$ operations. Many applications instead require a few eigenvalues of a large sparse matrix, and the matrix is often available only through the product $\boldsymbol{v} \mapsto \mathbf{A}\boldsymbol{v}$. Power iteration (Def 4.5) uses only such products, but at step k it retains only the last vector $\mathbf{A}^{k-1}\boldsymbol{v}$ and discards the earlier ones.

The Arnoldi process retains the span of $\boldsymbol{v}, \mathbf{A}\boldsymbol{v}, \dots, \mathbf{A}^{k-1}\boldsymbol{v}$ through an orthonormal basis. We first define the subspace spanned by this sequence and explain why its natural basis is unusable in floating-point arithmetic. We then construct an orthonormal basis one vector at a time and derive the Arnoldi relation. The same relation underlies the Lanczos algorithm in the next section, the GMRES method for linear systems, and the Krylov approximation of the matrix exponential.

4.4.1 Krylov Subspaces

Def 4.12 **(Krylov Subspace)** Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ and let $\boldsymbol{v} \in \mathbb{R}^n$ be nonzero. The kth Krylov subspace generated by $\mathbf{A}$ and $\boldsymbol{v}$ is

$$\mathcal{K}_k(\mathbf{A}, \boldsymbol{v}) = \operatorname{span}\{\boldsymbol{v}, \mathbf{A}\boldsymbol{v}, \dots, \mathbf{A}^{k-1}\boldsymbol{v}\}.$$

Equivalently, $\mathcal{K}_k(\mathbf{A}, \boldsymbol{v}) = \{p(\mathbf{A})\boldsymbol{v} : p \in \mathcal{P}_{k-1}\}$, where $\mathcal{P}_{k-1}$ denotes the polynomials of degree at most $k-1$.

Ex 4.24 For $\mathbf{A} = \operatorname{diag}(1, 2, 3)$ and $\boldsymbol{v} = (1, 1, 0)^T$, find the dimensions of $\mathcal{K}_1$, $\mathcal{K}_2$, and $\mathcal{K}_3$. Find a polynomial of degree two for which $p(\mathbf{A})\boldsymbol{v} = \mathbf{0}$. Explain why no number of matrix products can recover the third coordinate direction from this starting vector.

Ex 4.25 **(Shift invariance)** Show that $\mathcal{K}_k(\mathbf{A} - \sigma\mathbf{I}, \boldsymbol{v}) = \mathcal{K}_k(\mathbf{A}, \boldsymbol{v})$ for every scalar σ. Conclude that the Arnoldi iteration applied to $\mathbf{A} - \sigma\mathbf{I}$ produces the same basis $\mathbf{V}_k$ and the Hessenberg matrix $\mathbf{H}_k - \sigma\mathbf{I}$.

The kth power iterate is a multiple of $\mathbf{A}^{k-1}\boldsymbol{v}$, so it lies in $\mathcal{K}_k(\mathbf{A}, \boldsymbol{v})$. Power iteration therefore selects one particular vector from the Krylov subspace, namely the one produced by the polynomial $p(t) = t^{k-1}$. Any method that searches the whole subspace can do at least as well with the same number of matrix–vector products. Section 4.5 makes this comparison precise for symmetric matrices.

Lem 4.24 **(Krylov Subspaces Stop Growing at an Invariant Subspace)** If $\mathcal{K}_{k+1}(\mathbf{A}, \boldsymbol{v}) = \mathcal{K}_k(\mathbf{A}, \boldsymbol{v})$, then $\mathcal{K}_k(\mathbf{A}, \boldsymbol{v})$ is invariant under $\mathbf{A}$ in the sense of Def 4.2, and $\mathcal{K}_j(\mathbf{A}, \boldsymbol{v}) = \mathcal{K}_k(\mathbf{A}, \boldsymbol{v})$ for every $j \geq k$.

Proof Write $\mathcal{K}_j = \mathcal{K}_j(\mathbf{A}, \boldsymbol{v})$.

1. Every $\boldsymbol{x} \in \mathcal{K}_k$ can be written as $\boldsymbol{x} = \sum_{i=0}^{k-1} c_i \mathbf{A}^i \boldsymbol{v}$. Multiplying gives

$$\mathbf{A}\boldsymbol{x} = \sum_{i=0}^{k-1} c_i \mathbf{A}^{i+1} \boldsymbol{v} \in \mathcal{K}_{k+1}.$$

   If $\mathcal{K}_{k+1} = \mathcal{K}_k$, this image belongs to $\mathcal{K}_k$. This is exactly the definition of invariance.

2. The equality of the two spaces implies $\mathbf{A}^k \boldsymbol{v} \in \mathcal{K}_k$. By invariance, applying $\mathbf{A}$ again gives $\mathbf{A}^{k+1}\boldsymbol{v} \in \mathcal{K}_k$. Repeating this reasoning gives $\mathbf{A}^j \boldsymbol{v} \in \mathcal{K}_k$ for every $j \geq k$.

3. For $j \geq k$, all generators of $\mathcal{K}_j$ now lie in $\mathcal{K}_k$, so $\mathcal{K}_j \subseteq \mathcal{K}_k$. The reverse inclusion follows because the generators of $\mathcal{K}_k$ are among those of $\mathcal{K}_j$. Therefore the spaces are equal.

□

Remark 4.4 **(Ill-conditioned monomial basis)** The vectors $\boldsymbol{v}, \mathbf{A}\boldsymbol{v}, \ldots, \mathbf{A}^{k-1}\boldsymbol{v}$ form a basis of $\mathcal{K}_k$ whenever the subspace has dimension k, but this basis can be severely ill-conditioned. When a unique dominant eigendirection is present in the starting vector, the power-iteration argument shows that the normalized vectors $\mathbf{A}^j\boldsymbol{v}/\|\mathbf{A}^j\boldsymbol{v}\|_2$ converge to the dominant eigenvector, so the later columns of the basis matrix become nearly parallel. The resulting near dependence makes the basis sensitive to rounding errors, which can destroy the information carried by the smaller components.

4.4.2 The Arnoldi Process

The remedy is to orthonormalize the Krylov sequence as it is generated. If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j$ is an orthonormal basis of $\mathcal{K}_j$, we obtain a new direction by multiplying the latest basis vector by $\mathbf{A}$ instead of forming $\mathbf{A}^j\boldsymbol{v}$. We then remove its components along the existing basis by Gram–Schmidt orthogonalization (Thm 3.3).

Def 4.13 **(The Arnoldi Iteration)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ and let $\boldsymbol{v} \in \mathbb{R}^n$ be nonzero. Set $\boldsymbol{v}_1 = \boldsymbol{v}/\|\boldsymbol{v}\|_2$. For $j = 1, 2, \ldots, k$:

1. Compute $\boldsymbol{w} = \mathbf{A}\boldsymbol{v}_j$.
2. For $i = 1, \ldots, j$, compute $h_{ij} = \boldsymbol{v}_i^T \boldsymbol{w}$ and update $\boldsymbol{w} \leftarrow \boldsymbol{w} - h_{ij}\boldsymbol{v}_i$.
3. Compute $h_{j+1,j} = \|\boldsymbol{w}\|_2$.
4. If $h_{j+1,j} = 0$, stop. Otherwise, set $\boldsymbol{v}_{j+1} = \boldsymbol{w}/h_{j+1,j}$.

Step 2 is modified Gram–Schmidt. The coefficients h_{ij} are collected in the $k \times k$ upper Hessenberg matrix $\mathbf{H}_k$ and in the $(k+1) \times k$ matrix $\bar{\mathbf{H}}_k$, which is $\mathbf{H}_k$ with the additional row $h_{k+1,k}\boldsymbol{e}_k^T$.

Thm 4.25 **(The Arnoldi Relation)** Suppose the Arnoldi iteration completes k steps with $h_{j+1,j} \neq 0$ for $j = 1, \ldots, k$, and let $\mathbf{V}_k = (\boldsymbol{v}_1 \ \cdots \ \boldsymbol{v}_k)$. Then:

1. The columns of $\mathbf{V}_k$ are orthonormal and span $\mathcal{K}_k(\mathbf{A}, \boldsymbol{v})$.
2. The matrices satisfy

$$\mathbf{A}\mathbf{V}_k = \mathbf{V}_k\mathbf{H}_k + h_{k+1,k}\boldsymbol{v}_{k+1}\boldsymbol{e}_k^T = \mathbf{V}_{k+1}\bar{\mathbf{H}}_k. \tag{1}$$

3. The Hessenberg matrix is the projection of $\mathbf{A}$ onto the Krylov subspace, $\mathbf{H}_k = \mathbf{V}_k^T\mathbf{A}\mathbf{V}_k$.

Proof All assertions concern exact arithmetic. We verify orthogonality, the matrix identity, the span, and the projected matrix in that order.

1. Assume $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j$ are orthonormal. At inner-loop position i, replace the current $\boldsymbol{w}$ by $\boldsymbol{w} - h_{ij}\boldsymbol{v}_i$, where $h_{ij} = \boldsymbol{v}_i^T \boldsymbol{w}$. Its inner product with $\boldsymbol{v}_i$ becomes $h_{ij} - h_{ij}\boldsymbol{v}_i^T \boldsymbol{v}_i = 0$. For an earlier vector $\boldsymbol{v}_l$, $l < i$, the inner product remains zero because $\boldsymbol{v}_l^T \boldsymbol{v}_i = 0$. Thus the final $\boldsymbol{w}$ is perpendicular to every previous basis vector. Dividing by its nonzero norm gives a new unit vector. The starting vector is unit by construction, so induction establishes orthonormality at every step.

2. The inner loop subtracts $\sum_{i=1}^{j} h_{ij}\boldsymbol{v}_i$ from the original vector $\mathbf{A}\boldsymbol{v}_j$. Its remaining vector is $h_{j+1,j}\boldsymbol{v}_{j+1}$. Consequently

$$\mathbf{A}\boldsymbol{v}_j = \sum_{i=1}^{j+1} h_{ij}\boldsymbol{v}_i.$$

   Putting the first k such equations side by side gives $\mathbf{A}\mathbf{V}_k = \mathbf{V}_{k+1}\bar{\mathbf{H}}_k$. The coefficients in column j have no indices beyond $j+1$, which is why the matrix is upper Hessenberg. Separating its final row gives the other form of the Arnoldi relation.

3. The first vector spans $\mathcal{K}_1$. Suppose the first j vectors span $\mathcal{K}_j$. The column identity expresses $\boldsymbol{v}_{j+1}$ as a combination of $\mathbf{A}\boldsymbol{v}_j$ and previous basis vectors. Each belongs to $\mathcal{K}_{j+1}$, so $\boldsymbol{v}_{j+1}$ does too. There are $j+1$ independent orthonormal vectors in this space, whose definition supplies at most $j+1$ generators. Its dimension is therefore exactly $j+1$, and these vectors span it.

4. Multiply the Arnoldi relation by $\mathbf{V}_k^T$ on the left:

$$\begin{aligned}\mathbf{V}_k^T \mathbf{A}\mathbf{V}_k &= \mathbf{V}_k^T \mathbf{V}_k \mathbf{H}_k + h_{k+1,k}\mathbf{V}_k^T \boldsymbol{v}_{k+1}\boldsymbol{e}_k^T \\ &= \mathbf{I}_k \mathbf{H}_k + \mathbf{0} = \mathbf{H}_k.\end{aligned}$$

   The two simplifications follow from orthonormality proved in the first step.

For the standard Arnoldi construction and its use in Krylov methods, see Saad (2003). □

Remark 4.5 **(Arnoldi cost)** Step j requires one product with $\mathbf{A}$ and j inner products and vector updates of length n. After k steps the iteration has used k matrix–vector products and $O(k^2 n)$ additional operations, and it stores $k+1$ vectors of length n. The orthogonalization cost and storage both grow with k. For a symmetric matrix, Section 4.5 shows that most of the coefficients h_{ij} vanish and that the growth disappears.

4.4.3 Ritz Pairs and Breakdown

The matrix $\mathbf{H}_k$ is small, so its eigenvalues can be computed by the dense QR algorithm. Its eigenpairs, lifted back to $\mathbb{R}^n$ through $\mathbf{V}_k$, serve as approximate eigenpairs of $\mathbf{A}$. The Arnoldi relation measures their quality without any further products with $\mathbf{A}$.

Def 4.14 **(Ritz Pairs)** Let $\mathbf{H}_k \boldsymbol{s} = \theta \boldsymbol{s}$ with $\|\boldsymbol{s}\|_2 = 1$. The scalar θ is a **Ritz value** and the unit vector $\tilde{\boldsymbol{u}} = \mathbf{V}_k \boldsymbol{s} \in \mathcal{K}_k(\mathbf{A}, \boldsymbol{v})$ is the corresponding **Ritz vector** of $\mathbf{A}$ with respect to $\mathcal{K}_k(\mathbf{A}, \boldsymbol{v})$.

Thm 4.26 **(Residual of a Ritz Pair)** Let $(\theta, \tilde{\boldsymbol{u}})$ be a Ritz pair with $\tilde{\boldsymbol{u}} = \mathbf{V}_k \boldsymbol{s}$, and let s_k denote the last entry of $\boldsymbol{s}$. Then

$$\mathbf{A}\tilde{\boldsymbol{u}} - \theta\tilde{\boldsymbol{u}} = h_{k+1,k} s_k \, \boldsymbol{v}_{k+1}, \qquad \|\mathbf{A}\tilde{\boldsymbol{u}} - \theta\tilde{\boldsymbol{u}}\|_2 = h_{k+1,k}|s_k|.$$

Ex 4.26 Multiply the Arnoldi relation by $\boldsymbol{s}$ and use $\mathbf{H}_k \boldsymbol{s} = \theta \boldsymbol{s}$. Identify the single remaining term, and take its norm. Explain why this residual evaluation needs no new product with $\mathbf{A}$.

The residual norm depends only on $h_{k+1,k}$ and the last entry of a k-dimensional eigenvector. We can therefore monitor convergence of every Ritz pair at each step and form $\tilde{\boldsymbol{u}} = \mathbf{V}_k \boldsymbol{s}$ only for the pairs that have converged. A small residual shows that $(\theta, \tilde{\boldsymbol{u}})$ is an exact eigenpair of a nearby matrix. For a symmetric matrix, Cor 4.8 then places an eigenvalue within the residual norm of θ. For a nonnormal diagonalizable matrix, Cor 4.10 multiplies this distance by the condition number of the eigenvector basis, which can be large.

Cor 4.27 **(Breakdown Gives Exact Eigenvalues)** If the Arnoldi iteration stops at step k because $h_{k+1,k} = 0$, then $\mathcal{K}_k(\mathbf{A}, \boldsymbol{v})$ is invariant under $\mathbf{A}$, and every Ritz pair is an exact eigenpair of $\mathbf{A}$.

Proof At breakdown, the unnormalized vector remaining after orthogonalization is zero. A vector $\boldsymbol{v}_{k+1}$ is therefore not defined, but it is also unnecessary: the column equations give the shortened identity

$$\mathbf{A}\mathbf{V}_k = \mathbf{V}_k \mathbf{H}_k.$$

1. An arbitrary vector in the Krylov space has the form $\boldsymbol{x} = \mathbf{V}_k \boldsymbol{y}$. Applying $\mathbf{A}$ gives $\mathbf{A}\boldsymbol{x} = \mathbf{V}_k \mathbf{H}_k \boldsymbol{y}$, which is another linear combination of the columns of $\mathbf{V}_k$. Thus the space is invariant.

2. Let $\mathbf{H}_k \boldsymbol{s} = \theta \boldsymbol{s}$ with $\boldsymbol{s} \neq \mathbf{0}$. Multiply by $\mathbf{V}_k$ and use the shortened identity:

   $$\mathbf{A}(\mathbf{V}_k \boldsymbol{s}) = \mathbf{V}_k \mathbf{H}_k \boldsymbol{s} = \theta(\mathbf{V}_k \boldsymbol{s}).$$

   The lifted vector is nonzero because $\|\mathbf{V}_k \boldsymbol{s}\|_2 = \|\boldsymbol{s}\|_2 > 0$. It is therefore an eigenvector, not merely a zero vector satisfying the equation. Every Ritz pair is consequently exact.

□

Ex 4.27 **(Exact termination)** Suppose $\boldsymbol{v}$ is a combination of exactly m eigenvectors of a diagonalizable matrix $\mathbf{A}$, with nonzero coefficients and distinct eigenvalues. Prove that $\dim \mathcal{K}_k(\mathbf{A}, \boldsymbol{v}) = \min(k, m)$, and use Lem 4.24 and Cor 4.27 to conclude that the Arnoldi iteration stops at step m with those m eigenvalues as its Ritz values.

4.4.4 Further Exercises

Ex 4.28 **(Monomial versus Arnoldi basis)** Let $\mathbf{A} = \operatorname{diag}(1, 2, \ldots, 100)$ and $\boldsymbol{v} = (1, \ldots, 1)^T$. For $k = 5, 10, 15, 20$, compute the condition number of the matrix with columns $\boldsymbol{v}, \mathbf{A}\boldsymbol{v}, \ldots, \mathbf{A}^{k-1}\boldsymbol{v}$, and the quantity $\|\mathbf{V}_k^T \mathbf{V}_k - \mathbf{I}\|_2$ for the Arnoldi basis. Explain the growth of the first quantity using power iteration.

Ex 4.29 **(Implementing Arnoldi)** Implement Def 4.13 for a function that applies $\mathbf{A}$ to a vector. Test it on a random 200×200 nonsymmetric matrix by checking $\|\mathbf{A}\mathbf{V}_k - \mathbf{V}_{k+1}\bar{\mathbf{H}}_k\|_2$ and by comparing the residuals of Thm 4.26 with directly computed residuals $\|\mathbf{A}\tilde{\boldsymbol{u}} - \theta\tilde{\boldsymbol{u}}\|_2$.

4.5 The Lanczos Algorithm

When $\mathbf{A}$ is real symmetric, the Arnoldi process of Section 4.4 simplifies in two ways. First, the projected matrix $\mathbf{H}_k = \mathbf{V}_k^T \mathbf{A} \mathbf{V}_k$ is symmetric as well as Hessenberg, so it is tridiagonal. Each new basis vector then needs to be orthogonalized only against the two preceding ones, and the cost per step no longer grows with k. Second, the Rayleigh quotient characterization of Thm 4.5 applies to the projected matrix. The characterization shows that the extreme Ritz values are the best estimates of the extreme eigenvalues available from the Krylov subspace.

We first derive the three-term recurrence that defines the Lanczos algorithm (Lanczos, 1950). We then show that the extreme Ritz values move monotonically toward the extreme eigenvalues and that they are never worse than the power-iteration estimate obtained with the same number of products. A numerical example compares the two methods. The section ends with the loss of orthogonality that occurs in floating-point arithmetic.

4.5.1 The Three-Term Recurrence

Thm 4.28 **(Tridiagonal Projection)** Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be symmetric, and suppose the Arnoldi iteration (Def 4.13) completes k steps. Then $\mathbf{H}_k$ is symmetric tridiagonal. Writing $\alpha_j = h_{jj}$ and $\beta_j = h_{j+1,j}$, we have

$$\mathbf{H}_k = \mathbf{T}_k = \begin{pmatrix} \alpha_1 & \beta_1 & & \\ \beta_1 & \alpha_2 & \ddots & \\ & \ddots & \ddots & \beta_{k-1} \\ & & \beta_{k-1} & \alpha_k \end{pmatrix},$$

and the Arnoldi coefficients satisfy $h_{ij} = 0$ for $i < j - 1$ and $h_{j-1,j} = \beta_{j-1}$.

Proof The Arnoldi matrix has two structural properties.

1. Since $\mathbf{H}_k = \mathbf{V}_k^T \mathbf{A} \mathbf{V}_k$ and $\mathbf{A}^T = \mathbf{A}$,

   $$\mathbf{H}_k^T = (\mathbf{V}_k^T \mathbf{A} \mathbf{V}_k)^T = \mathbf{V}_k^T \mathbf{A}^T \mathbf{V}_k = \mathbf{H}_k.$$

   Thus $h_{ij} = h_{ji}$ for every pair of indices.

2. Arnoldi constructs an upper Hessenberg matrix, so $h_{ij} = 0$ whenever $i > j + 1$.

3. If $i < j - 1$, then $j > i + 1$. The Hessenberg property applied to entry (j, i) gives $h_{ji} = 0$. Symmetry gives $h_{ij} = h_{ji} = 0$. Hence only the main diagonal and its two immediate neighbors can be nonzero.

4. The diagonal is $\alpha_j = h_{jj}$. Arnoldi defines $h_{j+1,j} = \beta_j$. Symmetry makes $h_{j,j+1} = \beta_j$ as well. These are exactly the displayed tridiagonal entries, and $h_{j-1,j} = \beta_{j-1}$.

□

Ex 4.30 Use symmetry directly to explain why $\boldsymbol{v}_i^T \mathbf{A} \boldsymbol{v}_j = 0$ for $i < j - 1$: move $\mathbf{A}$ to $\boldsymbol{v}_i$ and locate $\mathbf{A}\boldsymbol{v}_i$ in the earlier Krylov subspace. This gives a second explanation of why the recurrence needs only two previous directions in exact arithmetic.

In the Arnoldi iteration, the coefficient h_{ij} is the component of $\mathbf{A}\boldsymbol{v}_j$ along $\boldsymbol{v}_i$. Theorem 4.28 shows that these components vanish for $i < j-1$ in exact arithmetic, so the inner loop of Step 2 reduces to two subtractions. The coefficient for $i = j-1$ is already known from the previous step. Substituting these facts into Def 4.13 gives the Lanczos iteration.

Def 4.15 **(The Lanczos Iteration)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be symmetric and let $\boldsymbol{v}_1$ be a unit vector. Set $\boldsymbol{v}_0 = \mathbf{0}$ and $\beta_0 = 0$. For $j = 1, 2, \ldots, k$:

1. Compute $\boldsymbol{w} = \mathbf{A}\boldsymbol{v}_j - \beta_{j-1}\boldsymbol{v}_{j-1}$.
2. Compute $\alpha_j = \boldsymbol{v}_j^T\boldsymbol{w}$ and update $\boldsymbol{w} \leftarrow \boldsymbol{w} - \alpha_j\boldsymbol{v}_j$.
3. Compute $\beta_j = \|\boldsymbol{w}\|_2$.
4. If $\beta_j = 0$, stop. Otherwise, set $\boldsymbol{v}_{j+1} = \boldsymbol{w}/\beta_j$.

Cor 4.29 **(The Lanczos Relation)** After k steps of the Lanczos iteration without breakdown, the matrix $\mathbf{V}_k = (\boldsymbol{v}_1 \ \cdots \ \boldsymbol{v}_k)$ has orthonormal columns spanning $\mathcal{K}_k(\mathbf{A}, \boldsymbol{v}_1)$, and

$$\mathbf{A}\mathbf{V}_k = \mathbf{V}_k\mathbf{T}_k + \beta_k\boldsymbol{v}_{k+1}\boldsymbol{e}_k^T, \qquad \mathbf{T}_k = \mathbf{V}_k^T\mathbf{A}\mathbf{V}_k.$$

Proof Compare one Lanczos step with the corresponding Arnoldi column equation. For a symmetric matrix, the preceding theorem makes that equation

$$\mathbf{A}\boldsymbol{v}_j = \beta_{j-1}\boldsymbol{v}_{j-1} + \alpha_j\boldsymbol{v}_j + \beta_j\boldsymbol{v}_{j+1}.$$

Every earlier coefficient is zero in exact arithmetic. Subtracting the first two terms from $\mathbf{A}\boldsymbol{v}_j$, then normalizing the remainder, is exactly the Lanczos recurrence. Thus, from the same starting vector and positive normalization by a norm, the two processes generate the same basis vectors until termination.

The Arnoldi theorem consequently supplies orthonormality and the Krylov span. To see the matrix identity directly, place the first k three-term equations side by side. The first k components form $\mathbf{V}_k\mathbf{T}_k$. The only term outside these columns comes from the last equation, giving

$$\mathbf{A}\mathbf{V}_k = \mathbf{V}_k\mathbf{T}_k + \beta_k\boldsymbol{v}_{k+1}\boldsymbol{e}_k^T.$$

Multiplication by $\mathbf{V}_k^T$ makes the last term vanish and gives $\mathbf{V}_k^T\mathbf{A}\mathbf{V}_k = \mathbf{T}_k$. The use of exact arithmetic matters: omitted coefficients may become nonzero under rounding. □

Ex 4.31 Carry out two steps of the Lanczos iteration by hand for $\mathbf{A} = \operatorname{diag}(4, 2, 1)$ and $\boldsymbol{v}_1 = (1, 1, 1)^T/\sqrt{3}$. Write down α_1, β_1, α_2, and $\mathbf{T}_2$, and compute β_2.

Remark 4.6 **(Lanczos cost)** In exact arithmetic, each step requires one product with $\mathbf{A}$ and $O(n)$ further operations, independent of k. Reorthogonalization in floating-point arithmetic adds work and may require storing the earlier vectors. If only eigenvalues are wanted, the iteration needs to keep just $\boldsymbol{v}_{j-1}$, $\boldsymbol{v}_j$, and the scalars α_j, β_j. If Ritz vectors are also wanted, we either store $\mathbf{V}_k$ or run the recurrence a second time to form $\mathbf{V}_k\boldsymbol{s}$.

4.5.2 Rayleigh–Ritz Approximation

The Ritz pairs of Def 4.14 are now the eigenpairs of the symmetric tridiagonal matrix $\mathbf{T}_k$, so the Ritz values are real. Theorem 4.26 gives their residuals as $\beta_k|s_k|$. Symmetry also lets us say how good the Ritz values are. Throughout, let $\lambda_1 \le \cdots \le \lambda_n$ be the eigenvalues of $\mathbf{A}$ and let $\theta_1^{(k)} \le \cdots \le \theta_k^{(k)}$ be the eigenvalues of $\mathbf{T}_k$.

Thm 4.30 **(Rayleigh–Ritz Optimality)** Let $R(\boldsymbol{x}) = \boldsymbol{x}^T\mathbf{A}\boldsymbol{x}/\boldsymbol{x}^T\boldsymbol{x}$. The extreme Ritz values are the extreme values of the Rayleigh quotient over the Krylov subspace,

$$\theta_k^{(k)} = \max_{\boldsymbol{x}\in\mathcal{K}_k,\,\boldsymbol{x}\neq\mathbf{0}} R(\boldsymbol{x}), \qquad \theta_1^{(k)} = \min_{\boldsymbol{x}\in\mathcal{K}_k,\,\boldsymbol{x}\neq\mathbf{0}} R(\boldsymbol{x}).$$

Consequently, $\lambda_1 \le \theta_1^{(k)}$ and $\theta_k^{(k)} \le \lambda_n$.

Proof We compare two Rayleigh quotients by writing the same vector in two coordinate systems.

1. Since the columns of $\mathbf{V}_k$ are an orthonormal basis of $\mathcal{K}_k$, each $\boldsymbol{x} \in \mathcal{K}_k$ has a unique representation $\boldsymbol{x} = \mathbf{V}_k\boldsymbol{y}$. It is nonzero precisely when $\boldsymbol{y} \neq \mathbf{0}$, because $\|\mathbf{V}_k\boldsymbol{y}\|_2 = \|\boldsymbol{y}\|_2$.

2. Direct substitution gives

$$\boldsymbol{x}^T\boldsymbol{x} = \boldsymbol{y}^T\mathbf{V}_k^T\mathbf{V}_k\boldsymbol{y} = \boldsymbol{y}^T\boldsymbol{y},$$
$$\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} = \boldsymbol{y}^T(\mathbf{V}_k^T\mathbf{A}\mathbf{V}_k)\boldsymbol{y} = \boldsymbol{y}^T\mathbf{T}_k\boldsymbol{y}.$$

   Hence $R(\boldsymbol{x}) = \boldsymbol{y}^T\mathbf{T}_k\boldsymbol{y}/(\boldsymbol{y}^T\boldsymbol{y})$.

3. Expand $\boldsymbol{y}$ in an orthonormal eigenbasis of the symmetric matrix $\mathbf{T}_k$. Its quotient is a weighted average of the Ritz values. Its minimum is $\theta_1^{(k)}$ and maximum is $\theta_k^{(k)}$, attained at the corresponding eigenvectors. The parametrization in the first step ranges over the whole Krylov space, so these are exactly the extrema claimed there.

4. The Rayleigh quotient of $\mathbf{A}$ is between λ_1 and λ_n on every nonzero vector. Restricting the vectors to a subspace cannot move the minimum below λ_1 or the maximum above λ_n. Thus $\lambda_1 \le \theta_1^{(k)}$ and $\theta_k^{(k)} \le \lambda_n$.

□

Cor 4.31 **(Monotone Convergence of the Extreme Ritz Values)** The Ritz values at consecutive steps interlace,

$$\theta_i^{(k+1)} \le \theta_i^{(k)} \le \theta_{i+1}^{(k+1)}, \qquad i = 1, \ldots, k.$$

In particular, $\theta_k^{(k)}$ is nondecreasing and bounded above by λ_n, and $\theta_1^{(k)}$ is nonincreasing and bounded below by λ_1. They reach the extreme eigenvalues represented in the starting vector when the corresponding invariant Krylov subspace is exhausted. Reaching λ_n or λ_1 requires a nonzero starting component in its eigenspace.

Ex 4.32 Apply Cauchy interlacing to $\mathbf{T}_k$ as a principal submatrix of $\mathbf{T}_{k+1}$. Deduce monotonicity of the two extreme Ritz values. Does monotonicity alone show that their limits are the extreme eigenvalues of $\mathbf{A}$? Examine a starting vector orthogonal to an extreme eigenspace.

Ex 4.33 For $\mathbf{T}_2$ from the previous exercise, compute both Ritz values and their residuals using Thm 4.26. Verify the interlacing inequalities of Cor 4.31 between the Ritz value $\theta_1^{(1)} = \alpha_1$, the two eigenvalues of $\mathbf{T}_2$, and the eigenvalues of $\mathbf{A}$.

Cor 4.32 **(Lanczos Improves on Power Iteration)** Let $\mathbf{A}$ be symmetric positive definite, and let $\lambda^{(k-1)}$ be the Rayleigh quotient estimate of Def 4.5 after $k-1$ power steps from $\boldsymbol{v}^{(0)} = \boldsymbol{v}_1$. Both $\lambda^{(k-1)}$ and $\theta_k^{(k)}$ require k products with $\mathbf{A}$, and

$$\lambda^{(k-1)} \le \theta_k^{(k)} \le \lambda_n.$$

Proof After $k-1$ normalized power steps, the vector $\boldsymbol{v}^{(k-1)}$ is a nonzero scalar multiple of $\mathbf{A}^{k-1}\boldsymbol{v}_1$. This follows by induction, since normalizing after each multiplication changes only a scalar factor. Therefore it belongs to $\mathcal{K}_k$.

The Rayleigh–Ritz theorem maximizes the Rayleigh quotient over every nonzero vector in this space. The power iterate is one candidate, so

$$\lambda^{(k-1)} = R(\boldsymbol{v}^{(k-1)}) \le \max_{\boldsymbol{x} \in \mathcal{K}_k \setminus \{\mathbf{0}\}} R(\boldsymbol{x}) = \theta_k^{(k)} \le \lambda_n.$$

Each comparison concerns the same initial vector and the same matrix.

For the product count, $k-1$ multiplications form the power iterate. Computing its Rayleigh quotient requires one more product with $\mathbf{A}$. Thus the quoted power estimate uses k products. Lanczos uses one product in each of k steps and obtains $\mathbf{T}_k$ from those products. Its largest eigenvalue needs no further product with $\mathbf{A}$.

Positive definiteness ensures all eigenvalues are positive. Hence the eigenvalue of largest magnitude targeted by power iteration is the largest algebraic eigenvalue λ_n, the same endpoint being estimated by the largest Ritz value. □

The polynomial form of Def 4.12 explains the size of the improvement. Every vector in $\mathcal{K}_k$ is $p(\mathbf{A})\boldsymbol{v}_1$ for some $p \in \mathcal{P}_{k-1}$. Expanding $\boldsymbol{v}_1 = \sum_i c_i \boldsymbol{q}_i$ in the eigenvectors of $\mathbf{A}$ gives

$$\theta_k^{(k)} = \max_{p \in \mathcal{P}_{k-1}} R(p(\mathbf{A})\boldsymbol{v}_1) = \max_{p \in \mathcal{P}_{k-1}} \frac{\sum_{i=1}^n \lambda_i \, p(\lambda_i)^2 c_i^2}{\sum_{i=1}^n p(\lambda_i)^2 c_i^2}.$$

The quotient is close to λ_n when p is large at λ_n and small on the rest of the spectrum. Power iteration is restricted to $p(t) = t^{k-1}$, which separates λ_n from λ_{n-1} only by the factor $(\lambda_{n-1}/\lambda_n)^{k-1}$. Lanczos implicitly selects the best polynomial. A shifted Chebyshev polynomial, which is bounded by one on $[\lambda_1, \lambda_{n-1}]$ and grows rapidly outside it, gives the following bound.

Remark 4.7 **(Kaniel–Paige bound)** Assume $\lambda_n > \lambda_{n-1} > \lambda_1$ and $\boldsymbol{v}_1^T \boldsymbol{q}_n \neq 0$. Let $\phi \in [0, \pi/2)$ be the acute angle between $\boldsymbol{v}_1$ and the eigenvector $\boldsymbol{q}_n$, let $\gamma = (\lambda_n - \lambda_{n-1})/(\lambda_{n-1} - \lambda_1)$, and let $T_m(x) = \cosh(m \operatorname{arccosh} x)$ for $x \ge 1$. Then

$$0 \le \lambda_n - \theta_k^{(k)} \le (\lambda_n - \lambda_1) \left(\frac{\tan \phi}{T_{k-1}(1 + 2\gamma)} \right)^2.$$

A proof is given by Golub and Van Loan (Golub and Van Loan, 2013). Since $T_m(1 + 2\gamma)$ grows roughly like $(1 + 2\sqrt{\gamma})^m$ when γ is small, the error decreases geometrically at a rate set by the

square root of the relative gap. A square root of the same kind appears in the conjugate gradient bound of Thm 6.13. The analogous bound for $\theta_1^{(k)}$ follows by applying the result to $-\mathbf{A}$. Interior eigenvalues admit weaker bounds and converge more slowly, because no low-degree polynomial can be large at an interior point and small on both sides of it.

Example **(Lanczos and power iteration on a model spectrum)** Let $\mathbf{A} = \operatorname{diag}(1, 2, \dots, 100)$ and $\boldsymbol{v}_1 = (1, \dots, 1)^T/10$. We run Lanczos with full reorthogonalization, so that the computed Ritz values agree with exact arithmetic to rounding level, and compare the error in the largest Ritz value with the error in the power-iteration estimate after the same number of products.

k	$\lambda_n - \theta_k^{(k)}$	$\lambda_n - \lambda^{(k-1)}$	Kaniel–Paige bound
10	8.6×10^{-1}	4.5	9.9×10^{2}
20	5.0×10^{-2}	2.0	1.8×10^{1}
30	6.4×10^{-4}	1.2	3.3×10^{-1}
40	7.2×10^{-7}	8.2×10^{-1}	5.8×10^{-3}

The relative gap is $\gamma = 1/98$, and the power-iteration ratio is $\lambda_{99}/\lambda_{100} = 0.99$. After 40 products the power-iteration estimate still has an absolute error of about 0.82, whereas the largest Ritz value is correct to about eight significant digits. The bound is valid but conservative at every step. By the symmetry of this spectrum about 50.5 and of the starting vector, the error in $\theta_1^{(k)}$ is the same as the error in $\theta_k^{(k)}$.

At $k = 40$ the residual of Thm 4.26 for the largest Ritz pair is 3.7×10^{-3}, about four orders of magnitude larger than the eigenvalue error. Corollary 4.8 explains the difference. The corollary bounds the distance from the Ritz value to the nearest eigenvalue by the squared residual divided by the distance δ from the Ritz value to the other eigenvalues. Here δ is about one, and the bound 1.4×10^{-5} is consistent with the observed error.

Remark 4.8 **(Lanczos and conjugate gradients)** The conjugate gradient method of Section 6.4 builds the same Krylov subspace $\mathcal{K}_k(\mathbf{A}, \boldsymbol{r}^{(0)})$ for a symmetric positive definite system. Its residuals are multiples of the Lanczos vectors started from $\boldsymbol{v}_1 = \boldsymbol{r}^{(0)}/\|\boldsymbol{r}^{(0)}\|_2$, and its coefficients determine the entries of $\mathbf{T}_k$. Section 6.4 derives this correspondence and uses it to estimate the extreme eigenvalues during a CG solve.

4.5.3 Loss of Orthogonality and Ghost Eigenvalues

Theorem 4.28 relies on the exact orthogonality of $\boldsymbol{v}_{j+1}$ to $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-2}$. The Lanczos iteration never enforces this orthogonality explicitly, so rounding errors are free to destroy it.

Remark 4.9 **(Paige's analysis)** In floating-point arithmetic, the computed Lanczos vectors gradually lose mutual orthogonality. Paige showed that the loss is not random (Paige, 1976). Let $(\theta_i, \tilde{\boldsymbol{u}}_i)$ be a Ritz pair computed from $\mathbf{T}_k$, and let u denote the unit roundoff. Paige's analysis gives

$$\beta_k |s_{k,i}|\, |\tilde{\boldsymbol{u}}_i^T \boldsymbol{v}_{k+1}| \text{ has a rounding-error scale of order } u\|\mathbf{A}\|_2.$$

The denominator is the residual of Thm 4.26. This relation describes a rounding-level scale rather than an equality or a universal estimate for each run. The new Lanczos vector can therefore acquire a component along $\tilde{\boldsymbol{u}}_i$ precisely as that Ritz pair converges, and orthogonality is lost in the directions of converged Ritz vectors. Once orthogonality to a converged Ritz vector is lost, the iteration begins to approximate the same eigenvalue again and produces repeated copies of it, known as **ghost eigenvalues**. The number of such copies and the steps at which they

appear depend on rounding and implementation. The experiment below asks you to observe their appearance rather than predict a fixed count.

Remark 4.10 **(Remedies for ghost eigenvalues)** Practical implementations use one of three strategies.

1. **Full reorthogonalization.** We orthogonalize each new vector against all previous Lanczos vectors. This restores the behavior of exact arithmetic at the Arnoldi cost of $O(k^2 n)$ operations and $O(kn)$ storage.

2. **Selective reorthogonalization.** We orthogonalize only against converged Ritz vectors, and only when Paige's analysis indicates that orthogonality to them has been lost. The cost is proportional to the number of converged Ritz vectors.

3. **Cullum–Willoughby identification.** We run the iteration without reorthogonalization and discard spurious Ritz values afterward by comparing the eigenvalues of $\mathbf{T}_k$ with those of the matrix obtained by deleting its first row and column.

4.5.4 Further Exercises

Ex 4.34 Using Thm 4.30, show that $\theta_{k-1}^{(k)} = \max_S \min_{\boldsymbol{x}\in S,\, \boldsymbol{x}\neq \mathbf{0}} R(\boldsymbol{x})$, where S ranges over two-dimensional subspaces of $\mathcal{K}_k$. Conclude that $\theta_{k-1}^{(k)} \le \lambda_{n-1}$.

Ex 4.35 Let $\mathbf{A}$ be symmetric positive definite. Show that the power-iteration estimate satisfies $\lambda_n - \lambda^{(k-1)} \le (\lambda_n - \lambda_1)\tan^2\phi\,(\lambda_{n-1}/\lambda_n)^{2(k-1)}$, where ϕ is the angle between $\boldsymbol{v}_1$ and $\boldsymbol{q}_n$. Compare the rate with the Kaniel–Paige bound when γ is small.

Ex 4.36 Implement Def 4.15 with an option for full reorthogonalization. Reproduce the table in the example above, and plot all Ritz values against k for $k \le 200$ with and without reorthogonalization. Identify the steps at which ghost copies of λ_{100} appear, and relate them to the step at which the largest Ritz value converged.

Ex 4.37 Repeat the previous computation with $\mathbf{A} = \operatorname{diag}(1, 2, \dots, 99, 110)$. Explain, using the polynomial characterization, why the largest Ritz value now converges much faster while the convergence of $\theta_1^{(k)}$ is essentially unchanged.

Read **Further reading.** Horn and Johnson are a reference for the matrix structure behind the spectral results (Horn and Johnson, 2012). The computational treatment in Golub and Van Loan includes symmetric eigensolvers and the SVD (Golub and Van Loan, 2013). The original Lanczos and Golub–Kahan papers show how two algorithms in these chapters first appeared (Lanczos, 1950; Golub and Kahan, 1965).

5 Approximation and Integration

Function values are often available only at a finite set of points. We construct approximations from those values and ask how the choice of points affects the error. Integrating an interpolant produces a quadrature rule. Trigonometric approximation leads to Fourier transforms and fast calculations on periodic grids.

5.1 Polynomial and Spline Interpolation

Suppose a function is known only through measured or computed values at a few points. To estimate a value between those points, we can construct a simpler function that agrees with the available values. Interpolation imposes exact agreement at the data points. It differs from least squares, which permits a nonzero residual there.

Two distinct points determine a line. We ask whether $n+1$ distinct points similarly determine a polynomial of degree at most n, how to evaluate it, and how far it can lie from the function that supplied the data. When a single polynomial is unsuitable, we use a different cubic polynomial on each subinterval and join the pieces smoothly.

5.1.1 Existence, Uniqueness, and the Vandermonde Matrix

Thm 5.1 **(Existence and Uniqueness of the Interpolating Polynomial)** Let $x_0, x_1, \ldots, x_n$ be $n+1$ pairwise distinct real numbers, and let $y_0, y_1, \ldots, y_n$ be arbitrary real numbers. There exists a unique polynomial $P_n \in \mathcal{P}_n$ of degree at most n such that $P_n(x_i) = y_i$ for all $i = 0, 1, \ldots, n$.

Proof We first prove that two candidate interpolants cannot differ. Suppose $P_n(x_i) = Q_n(x_i) = y_i$ for every node. Their difference $R = P_n - Q_n$ has degree at most n and vanishes at all $n+1$ distinct nodes. A nonzero polynomial of degree at most n has at most n distinct roots. Hence R must be the zero polynomial, so $P_n = Q_n$.

This proves at most one interpolant exists. To prove that one actually exists for arbitrary data, write

$$P_n(x) = c_0 + c_1 x + \cdots + c_n x^n.$$

The interpolation condition at node x_i is $\sum_{j=0}^{n} c_j x_i^j = y_i$. These $n+1$ equations form a square system $\mathbf{V}\boldsymbol{c} = \boldsymbol{y}$, where row i of $\mathbf{V}$ is $(1, x_i, \ldots, x_i^n)$.

Suppose $\mathbf{V}\boldsymbol{c} = \mathbf{0}$. Then the polynomial with coefficients $\boldsymbol{c}$ vanishes at every node. By the same root-count argument it is the zero polynomial, meaning every c_j is zero. Thus $\mathbf{V}$ has a trivial null space. Rank–nullity makes its rank $n+1$, so its range is all of $\mathbb{R}^{n+1}$. It therefore has a solution for every data vector $\boldsymbol{y}$.

The coefficients of that solution define an interpolating polynomial, proving existence. The uniqueness argument applies to every pair of such polynomials and completes the theorem. Distinctness of the nodes was essential to the root count. □

Remark 5.1 Although the Vandermonde matrix provides a constructive proof of existence, solving $\mathbf{V}\boldsymbol{c} = \boldsymbol{y}$ in the monomial basis is often a poor numerical representation. Its condition number depends on the nodes and their scaling and can grow rapidly with degree. A large coefficient error does not imply that the interpolation problem itself is ill-posed. It may instead reflect an ill-conditioned choice of basis. The Lagrange and barycentric forms avoid solving for monomial coefficients.

5.1.2 The Lagrange Form

The Lagrange formulation circumvents the ill-conditioned Vandermonde linear system by constructing a basis of polynomials tailored to the specific nodes x_i.

Def 5.1 **(Lagrange Basis Polynomials)** Given $n+1$ distinct nodes $x_0, x_1, \ldots, x_n$, the i-th Lagrange

basis polynomial $L_i \in \mathcal{P}_n$ is defined by

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j} = \frac{(x - x_0) \cdots (x - x_{i-1})(x - x_{i+1}) \cdots (x - x_n)}{(x_i - x_0) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)}.$$

The Lagrange basis polynomials satisfy the cardinal Kronecker delta property:

$$L_i(x_j) = \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

The interpolating polynomial in Lagrange form is

$$P_n(x) = \sum_{i=0}^{n} y_i L_i(x).$$

Ex 5.1 Verify $L_i(x_j) = \delta_{ij}$. Use uniqueness of interpolation to prove $\sum_i L_i(x) = 1$ and $\sum_i x_i L_i(x) = x$ when $n \geq 1$. For the Vandermonde determinant, regard the last row as a polynomial in the last node. Identify its roots and leading coefficient, then complete an induction on the number of nodes.

Ex 5.2 For the data $(0, 1)$, $(1, 3)$, and $(3, 2)$, construct the interpolant first by solving the Vandermonde system and then by constructing the three Lagrange polynomials. Verify agreement by comparing coefficients. Once the barycentric formula has been derived below, use it to evaluate the same polynomial at $x = 2$.

Example To interpolate the three data points $(-1, 1)$, $(0, 0)$, and $(1, 1)$, we construct the degree-2 Lagrange basis polynomials on $x_0 = -1, x_1 = 0, x_2 = 1$:

$$\begin{aligned} L_0(x) &= \frac{(x - 0)(x - 1)}{(-1 - 0)(-1 - 1)} = \frac{x(x - 1)}{2} = \frac{x^2 - x}{2}, \\ L_1(x) &= \frac{(x - (-1))(x - 1)}{(0 - (-1))(0 - 1)} = \frac{(x + 1)(x - 1)}{-1} = 1 - x^2, \\ L_2(x) &= \frac{(x - (-1))(x - 0)}{(1 - (-1))(1 - 0)} = \frac{x(x + 1)}{2} = \frac{x^2 + x}{2}. \end{aligned}$$

Multiplying by the data values $y_0 = 1, y_1 = 0, y_2 = 1$ gives

$$P_2(x) = 1 \cdot \left(\frac{x^2 - x}{2} \right) + 0 \cdot (1 - x^2) + 1 \cdot \left(\frac{x^2 + x}{2} \right) = \frac{x^2 - x + x^2 + x}{2} = x^2.$$

Remark 5.2 The standard Lagrange form has two practical shortcomings. First, evaluating $P_n(x)$ at a single point x requires $O(n^2)$ floating-point operations. Second, adding a new node (x_{n+1}, y_{n+1}) alters every single basis polynomial $L_i(x)$, requiring a complete recomputation from scratch.

5.1.3 The Barycentric Interpolation Formula

The products in the Lagrange basis repeat much of the same work. Factoring out their common nodal polynomial gives a formula that costs $O(n)$ operations per evaluation after the weights have been computed. Its sensitivity still depends on the nodes.

Thm 5.2 **(Barycentric Interpolation Formulas)** Let $\ell(x) = \prod_{j=0}^{n}(x - x_j)$ denote the nodal polynomial. For each node x_i, define the barycentric weights

$$w_i = \frac{1}{\prod_{j\neq i}(x_i - x_j)} = \frac{1}{\ell'(x_i)}, \qquad i = 0, 1, \ldots, n.$$

Then the interpolating polynomial satisfies:

1. **First Barycentric Form:**

$$P_n(x) = \ell(x)\sum_{i=0}^{n} \frac{w_i}{x - x_i} y_i.$$

2. **Second (True) Barycentric Form:**

$$P_n(x) = \frac{\sum_{i=0}^{n} \frac{w_i}{x-x_i} y_i}{\sum_{i=0}^{n} \frac{w_i}{x-x_i}}.$$

Both displayed formulas use division by $x - x_i$ and are applied only away from the nodes. If $x = x_j$, return y_j directly. The singularities in the formulas are removable, but should not be evaluated as divisions by zero.

Proof Assume first that x is not any of the nodes. The nodal polynomial contains every factor $x - x_j$. Removing the ith factor gives

$$\frac{\ell(x)}{x - x_i} = \prod_{j\neq i}(x - x_j).$$

Multiplication by the weight $w_i = 1/\prod_{j\neq i}(x_i - x_j)$ gives precisely $L_i(x)$. Substitute into the Lagrange interpolant:

$$P_n(x) = \sum_i y_i L_i(x) = \ell(x)\sum_i \frac{w_i y_i}{x - x_i}.$$

This proves the first formula.

Now apply it to data $y_i = 1$ for every i. The constant polynomial 1 interpolates those data. By uniqueness it is the interpolant, so

$$1 = \ell(x)\sum_i \frac{w_i}{x - x_i}.$$

Because x is away from the nodes, $\ell(x) \neq 0$, and the sum in this identity is $1/\ell(x)$, also nonzero. Divide the first formula by this identity and cancel $\ell(x)$. The resulting quotient is the second formula.

At $x = x_j$, each displayed expression contains a division by zero and is not evaluated as written. In the second formula, multiply its numerator and denominator by $x - x_j$ and let $x \to x_j$. All other terms vanish in this limit, leaving $w_j y_j / w_j = y_j$. The weight is nonzero because all nodes are distinct. Thus the singularity is removable and the direct prescription to return y_j is consistent. □

Ex 5.3 Interpolate the data $y_i = 1$ to prove $1 = \ell(x)\sum_i w_i/(x - x_i)$. Divide the first formula by this identity to obtain the second. Why does multiplying every weight by the same nonzero constant leave the second formula unchanged? Take the limit as $x \to x_j$ and verify that it recovers y_j.

Remark 5.3 The two barycentric formulas have different rounding-error properties. The modified Lagrange (first barycentric) formula admits a backward-error analysis; the second formula has favorable forward-error bounds when the Lebesgue function is modest. Neither representation removes the sensitivity caused by poorly chosen nodes. Furthermore, common multiplicative factors in the weights w_i cancel out in the numerator and denominator, allowing simple analytical expressions for standard node families (such as Chebyshev nodes).

5.1.4 The Interpolation Error Formula

Exact agreement at the nodes says nothing by itself about the error between them. To estimate that error, we must assume something about the function beyond its sampled values. The following theorem assumes a bounded derivative of order $n + 1$. Its nodal factor vanishes at each data point, as the error must.

Thm 5.3 **(Cauchy Remainder Error Theorem)** Let $f \in C^{n+1}[a, b]$, and let $P_n \in \mathcal{P}_n$ interpolate f at $n+1$ distinct nodes $x_0, x_1, \ldots, x_n \in [a, b]$. For each point $x \in [a, b]$, there exists a point $\xi_x \in (a, b)$ such that

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!}\prod_{j=0}^{n}(x - x_j) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!}\omega_{n+1}(x),$$

where $\omega_{n+1}(x) = \prod_{j=0}^{n}(x - x_j)$ is the nodal error polynomial.

Proof If $x = x_i$ for some node x_i, both sides vanish and the formula holds trivially for any ξ_x. This case must be separated before dividing by the nodal polynomial, because that polynomial is zero at every node.

Now fix $x \in [a, b]$ with $x \neq x_i$ for all i. Define the constant

$$\lambda = \frac{f(x) - P_n(x)}{\omega_{n+1}(x)},$$

and consider the auxiliary function $g : [a, b] \to \mathbb{R}$ defined by

$$g(t) = f(t) - P_n(t) - \lambda\omega_{n+1}(t).$$

Because $f \in C^{n+1}[a, b]$ and polynomials are infinitely differentiable, $g \in C^{n+1}[a, b]$. We evaluate g at the $n + 2$ distinct points $x_0, x_1, \ldots, x_n$ and x:

$$g(x_i) = f(x_i) - P_n(x_i) - \lambda\omega_{n+1}(x_i) = y_i - y_i - 0 = 0, \qquad i = 0, \ldots, n,$$
$$g(x) = f(x) - P_n(x) - \left(\frac{f(x) - P_n(x)}{\omega_{n+1}(x)}\right)\omega_{n+1}(x) = 0.$$

Thus $g(t)$ vanishes at $n + 2$ distinct points in $[a, b]$.

The extra zero at the evaluation point is the reason for choosing λ in this way. The interpolation error alone has $n+1$ known zeros. Subtracting the scaled nodal polynomial creates one additional zero while preserving all the original zeros.

By Rolle's Theorem, between any two adjacent zeros of $g(t)$, its derivative $g'(t)$ has at least one zero. Therefore, $g'(t)$ has at least $n+1$ distinct zeros in (a,b). Applying Rolle's Theorem successively:

- $g''(t)$ has at least n distinct zeros in (a,b),
- $g'''(t)$ has at least $n-1$ distinct zeros in (a,b),
- Continuing inductively, $g^{(n+1)}(t)$ has at least one zero in (a,b), which we denote by ξ_x.

We now differentiate $g(t)$ exactly $n+1$ times with respect to t:

$$g^{(n+1)}(t) = f^{(n+1)}(t) - P_n^{(n+1)}(t) - \lambda \frac{d^{n+1}}{dt^{n+1}}[\omega_{n+1}(t)].$$

Since P_n has degree at most n, its $(n+1)$-th derivative is identically zero: $P_n^{(n+1)}(t) \equiv 0$. The nodal polynomial $\omega_{n+1}(t) = t^{n+1} + \dots$ is a monic polynomial of degree $n+1$, so its $(n+1)$-th derivative is $(n+1)!$. Evaluating at $t = \xi_x$ gives

$$0 = g^{(n+1)}(\xi_x) = f^{(n+1)}(\xi_x) - 0 - \lambda(n+1)!.$$

Solving for λ produces

$$\lambda = \frac{f^{(n+1)}(\xi_x)}{(n+1)!}.$$

Equating this to our definition of $\lambda = (f(x) - P_n(x))/\omega_{n+1}(x)$ completes the proof.

The point supplied by repeated Rolle arguments depends on the chosen evaluation point x. There is no assertion that a single ξ works for every x in the interval. The differentiability assumption is used exactly when Rolle's theorem is repeated through order $n+1$. □

5.1.5 Runge's Phenomenon and Chebyshev Nodes

The error bound contains two decoupled components:

$$\|f - P_n\|_\infty \le \frac{\|f^{(n+1)}\|_\infty}{(n+1)!} \|\omega_{n+1}\|_\infty.$$

The derivative term $\|f^{(n+1)}\|_\infty$ depends entirely on the analytical smoothness of f, while $\|\omega_{n+1}\|_\infty = \max_{x\in[a,b]} |\prod_{j=0}^{n}(x - x_j)|$ depends exclusively on the geometric placement of the interpolation nodes.

Remark 5.4 **(Runge phenomenon)** Analyticity alone does not make equally spaced interpolation converge. For Runge's function $f(x) = 1/(1+25x^2)$, the interpolants on equally spaced nodes develop increasing endpoint oscillations and satisfy

$$\lim_{n\to\infty} \|f - P_n\|_\infty = \infty,$$

even though f is analytic on the real interval. The growing Lebesgue constant amplifies the approximation error, while the nearby complex poles at $\pm i/5$ limit the rate of polynomial approximation. The derivative remainder bound above warns that convergence is not automatic, but a divergent upper bound by itself would not prove divergence.

The nodal factor can be made small by placing the nodes well. On $[-1,1]$, the best choice is the set of Chebyshev points $x_k = \cos\left((2k+1)\pi/(2n+2)\right)$, $k = 0,\dots,n$, the roots of the Chebyshev polynomial T_{n+1}. For these nodes $\|\omega_{n+1}\|_\infty = 2^{-n}$, the smallest value attained by any monic polynomial of degree $n+1$. Section 5.2 proves this minimax property and shows that interpolation at Chebyshev points converges geometrically for functions that are analytic in a neighborhood of $[-1,1]$.

Remark 5.5 The Lebesgue constant $\Lambda_n = \max_{x\in[a,b]} \sum_{i=0}^{n} |L_i(x)|$ bounds the stability of polynomial interpolation: $\|f - P_n\|_\infty \le (1+\Lambda_n)\min_{Q\in\mathcal{P}_n}\|f-Q\|_\infty$.

- For equally spaced nodes, $\Lambda_n = \Theta\left(\frac{2^n}{n\log n}\right)$ grows exponentially.
- For Chebyshev nodes, $\Lambda_n = \frac{2}{\pi}\log(n+1) + O(1)$ grows only logarithmically, remaining near-optimal.

5.1.6 Piecewise Polynomials: Cubic Splines

A single polynomial makes every coefficient depend on all the data. An alternative is to use a polynomial of low degree on each interval between consecutive nodes. Continuity of derivatives at the joins prevents corners. Cubic splines impose continuity through the second derivative, leaving two additional boundary conditions to specify.

Def 5.2 **(Cubic Spline Interpolant)** Given $n+1$ distinct points $a = x_0 < x_1 < \cdots < x_n = b$ and values $y_0, y_1, \dots, y_n$, a cubic spline $S(x)$ is a function such that:

1. On each subinterval $[x_i, x_{i+1}]$, $S(x) = S_i(x)$ is a cubic polynomial:
$$S_i(x) = a_i + b_i(x-x_i) + c_i(x-x_i)^2 + d_i(x-x_i)^3, \qquad i = 0,\dots,n-1.$$
2. Interpolation conditions: $S(x_i) = y_i$ for all $i = 0,\dots,n$.
3. C^2 continuity: $S(x)$, $S'(x)$, and $S''(x)$ are continuous on the entire interval $[a,b]$.

Thm 5.4 **(The Tridiagonal Spline System)** Let $h_i = x_{i+1} - x_i$ and let $M_i = S''(x_i)$ denote the second derivative of the spline at node x_i. The condition that $S'(x)$ is continuous at each interior node x_i $(i = 1,\dots,n-1)$ leads to the linear system
$$h_{i-1}M_{i-1} + 2(h_{i-1}+h_i)M_i + h_iM_{i+1} = 6\left(\frac{y_{i+1}-y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}\right).$$

To close the system of $n-1$ equations in $n+1$ unknowns, two boundary conditions are imposed:

1. **Natural (Free) Spline:** $M_0 = M_n = 0$.
2. **Clamped (Complete) Spline:** $S'(x_0) = f'(a)$ and $S'(x_n) = f'(b)$.
3. **Periodic Spline:** Require $y_0 = y_n$ and impose $S'(x_0) = S'(x_n)$ and $M_0 = M_n$.

For natural conditions, eliminate the prescribed endpoint values $M_0 = M_n = 0$. The resulting interior system is strictly diagonally dominant and symmetric positive definite. For clamped conditions, the endpoint equations obtained from the prescribed first derivatives give the same properties. It can be solved in $O(n)$ time using the Thomas algorithm.

Proof Since $S_i(x)$ is a cubic polynomial on $[x_i, x_{i+1}]$, its second derivative $S_i''(x)$ is a linear function interpolating $M_i = S''(x_i)$ and $M_{i+1} = S''(x_{i+1})$:

$$S_i''(x) = M_i \frac{x_{i+1} - x}{h_i} + M_{i+1} \frac{x - x_i}{h_i}.$$

The formula for the second derivative is the unique line taking values M_i and M_{i+1} at the two endpoints. It is therefore convenient to integrate it before imposing the function values.

Integrating twice with respect to x gives

$$S_i(x) = \frac{M_i(x_{i+1} - x)^3}{6h_i} + \frac{M_{i+1}(x - x_i)^3}{6h_i} + C_i(x_{i+1} - x) + D_i(x - x_i).$$

Applying the interpolation conditions $S_i(x_i) = y_i$ and $S_i(x_{i+1}) = y_{i+1}$ determines the constants C_i and D_i:

$$C_i = \frac{y_i}{h_i} - \frac{M_i h_i}{6}, \qquad D_i = \frac{y_{i+1}}{h_i} - \frac{M_{i+1} h_i}{6}.$$

Differentiating $S_i(x)$ to obtain $S_i'(x)$ and evaluating at the endpoints of $[x_i, x_{i+1}]$ yields

$$S_i'(x_i) = -\frac{M_i h_i}{3} - \frac{M_{i+1} h_i}{6} + \frac{y_{i+1} - y_i}{h_i},$$
$$S_{i-1}'(x_i) = \frac{M_{i-1} h_{i-1}}{6} + \frac{M_i h_{i-1}}{3} + \frac{y_i - y_{i-1}}{h_{i-1}}.$$

Each interior knot has two candidate first derivatives, one from the interval on its left and one from the interval on its right. The function and second derivative already agree there by construction. Only equality of these first derivatives remains to impose.

Equating $S_{i-1}'(x_i) = S_i'(x_i)$ for continuity of the first derivative and gathering terms produces the stated tridiagonal equation. Strict diagonal dominance follows from $2(h_{i-1} + h_i) > h_{i-1} + h_i$, which gives nonsingularity. Symmetry, positive diagonal entries, and strict diagonal dominance give positive definiteness, so tridiagonal Cholesky can be used.

For natural boundary conditions, eliminating M_0 and M_n leaves the interior tridiagonal system. The off-diagonal entry between equations i and $i+1$ is h_i in both positions, proving symmetry. In each row, the positive diagonal $2(h_{i-1} + h_i)$ exceeds the sum of remaining off-diagonal magnitudes. Every eigenvalue is therefore positive by the symmetric diagonal-dominance bound. This also proves uniqueness of the natural spline coefficients. Clamped endpoint equations arise by setting the displayed endpoint derivative formulas equal to the prescribed slopes. □

Ex 5.4 Set up the 2×2 system for the natural cubic spline through the data

$$(0, 0), \qquad (1, 1), \qquad (2, 0), \qquad (3, 1).$$

Solve for M_1 and M_2, and write down the explicit cubic polynomial for each subinterval.

Thm 5.5 **(Minimum Curvature (Energy Minimization) Property)** Let $S(x)$ be the natural cubic spline interpolating $f(x)$ at $x_0, x_1, \ldots, x_n$. Let $g \in C^2[a, b]$ be any other function satisfying $g(x_i) = y_i$ for all $i = 0, \ldots, n$. Then

$$\int_a^b [S''(x)]^2 \, dx \leq \int_a^b [g''(x)]^2 \, dx.$$

Natural cubic splines are the smoothest possible interpolants in the sense of minimizing total bending energy.

Proof Let $h(x) = g(x) - S(x)$. Since both g and S interpolate f at the knots $x_0, x_1, \ldots, x_n$, we have

$$h(x_i) = g(x_i) - S(x_i) = y_i - y_i = 0, \qquad i = 0, 1, \ldots, n.$$

Expanding the squared second derivative of $g = S + h$:

$$\begin{aligned}\int_a^b [g''(x)]^2\,dx &= \int_a^b [S''(x) + h''(x)]^2\,dx \\ &= \int_a^b [S''(x)]^2\,dx + 2\int_a^b S''(x)h''(x)\,dx + \int_a^b [h''(x)]^2\,dx.\end{aligned}$$

If the cross term is zero, the final term is a square integral and can only increase the energy. We therefore calculate that cross term rather than attempting to compare the two energies directly.

We evaluate the cross term by integrating by parts on each subinterval $[x_i, x_{i+1}]$:

$$\int_{x_i}^{x_{i+1}} S''(x)h''(x)\,dx = \big[S''(x)h'(x)\big]_{x_i}^{x_{i+1}} - \int_{x_i}^{x_{i+1}} S'''(x)h'(x)\,dx.$$

Because $S(x)$ is a cubic polynomial on $[x_i, x_{i+1}]$, its third derivative $S_i'''(x)$ is constant on (x_i, x_{i+1}). Let $S_i'''(x) = c_i$. Then

$$\int_{x_i}^{x_{i+1}} S'''(x)h'(x)\,dx = c_i \int_{x_i}^{x_{i+1}} h'(x)\,dx = c_i[h(x_{i+1}) - h(x_i)] = 0,$$

since $h(x_{i+1}) = h(x_i) = 0$. Summing the boundary terms over all subintervals:

$$\begin{aligned}\int_a^b S''(x)h''(x)\,dx &= \sum_{i=0}^{n-1} \Big(S''(x_{i+1}^-)h'(x_{i+1}) - S''(x_i^+)h'(x_i)\Big) \\ &= S''(b)h'(b) - S''(a)h'(a) + \sum_{i=1}^{n-1} \Big(S''(x_i^-) - S''(x_i^+)\Big)\,h'(x_i).\end{aligned}$$

At each interior knot, the contribution from the interval on the left has a plus sign and the contribution from the interval on the right a minus sign. They involve the same $h'(x_i)$ because h is continuously differentiable. This explains the cancellation structure in the sum.

Because $S \in C^2[a, b]$, the second derivative is continuous across interior knots: $S''(x_i^-) - S''(x_i^+) = 0$. For a natural cubic spline, $S''(a) = S''(b) = 0$, so the boundary terms vanish as well. Therefore

$$\int_a^b S''(x)h''(x)\,dx = 0.$$

Substituting this back yields

$$\int_a^b [g''(x)]^2\,dx = \int_a^b [S''(x)]^2\,dx + \int_a^b [h''(x)]^2\,dx \geq \int_a^b [S''(x)]^2\,dx,$$

with equality if and only if $h''(x) \equiv 0$, meaning h is linear and hence identically zero because $h(x_0) = h(x_n) = 0$.

If the remaining integral of $(h'')^2$ is zero, continuity of h'' forces it to vanish at every point: a nonzero value would give a neighborhood with a positive contribution. Then h is linear. Its zero values at the two distinct endpoints force both linear coefficients to be zero. Thus equality holds only for the spline itself. □

5.1.7 Further Exercises

Ex 5.5 Compute the barycentric weights $w_0, \dots, w_n$ explicitly for $n+1$ Chebyshev nodes on $[-1,1]$. Show that, up to a common nonzero factor, $w_k = (-1)^k \sin\left(\frac{2k+1}{2n+2}\pi\right)$ (or $(-1)^k\delta_k$ with $\delta_k = 1/2$ at boundary nodes for Chebyshev-Lobatto points).

Ex 5.6 Use Thm 5.3 to bound the maximum error when $f(x) = \sin(x)$ is interpolated by a polynomial of degree 4 on $[0, \pi/2]$ using equally spaced nodes.

Ex 5.7 For the natural spline through $(0,0)$, $(1,1)$, $(2,0)$, and $(3,1)$, compute its bending energy $\int_0^3 (S'')^2\,dx$ and compare it with the energy of one other twice-differentiable interpolant.

5.2 Chebyshev Polynomials and Minimax Approximation

The interpolation error contains a product of distances to the nodes. We can reduce its largest value by changing the nodes. Chebyshev polynomials solve this particular optimization problem. Their roots give nodes that cluster near the endpoints, where equally spaced interpolation can oscillate badly.

The construction begins with equally spaced angles and the substitution $x = \cos\theta$. We derive the polynomial recurrence from a cosine identity, then use alternating extreme values to prove the minimax property. The final convergence result requires complex analysis and can be read after the polynomial arguments.

5.2.1 Interpolation Error and Conditioning

Approximating a continuous function $f \in C([-1,1])$ by polynomial interpolation encounters two distinct failure modes. The first is algebraic. Vandermonde systems in the monomial basis are often ill-conditioned, so a coefficient calculation may lose accuracy even when the interpolating polynomial is well defined. The second is analytical. For functions such as Runge's example, interpolation at equally spaced nodes diverges near the endpoints even in exact arithmetic. These are different failures and require different remedies.

To study the effect of the nodes, begin with Thm 5.3. With n nodes $x_1, \dots, x_n$ and the monic nodal polynomial $\omega_n(x) = \prod_{j=1}^{n}(x - x_j)$, the theorem bounds the interpolation error by

$$\|f - p_{n-1}\|_\infty \le \frac{\|f^{(n)}\|_\infty}{n!}\,\|\omega_n\|_\infty.$$

The first factor depends only on f, and the second only on the nodes. Choosing the nodes to minimize $\|\omega_n\|_\infty$ is therefore a minimax problem over monic polynomials of degree n, which this section solves.

5.2.2 Chebyshev Polynomials and Fundamental Identities

Def 5.3 **(Chebyshev Polynomials of the First Kind)** For $x \in [-1,1]$, the Chebyshev polynomial of degree k is defined by

$$T_k(x) = \cos(k \arccos x).$$

Under the substitution $x = \cos\theta$ with $\theta \in [0, \pi]$,

$$T_k(\cos\theta) = \cos(k\theta).$$

Lem 5.6 **(Chebyshev Three-Term Recurrence)** The functions $T_k(x)$ are algebraic polynomials satisfying the three-term recurrence relation

$$T_0(x) = 1, \qquad T_1(x) = x, \qquad T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x) \quad \text{for } k \geq 1.$$

Furthermore, for every $k \geq 1$, the leading coefficient of $T_k(x)$ is 2^{k-1}.

Proof The cosine addition formulas give

$$\cos((k+1)\theta) = \cos(k\theta)\cos\theta - \sin(k\theta)\sin\theta,$$
$$\cos((k-1)\theta) = \cos(k\theta)\cos\theta + \sin(k\theta)\sin\theta.$$

Adding cancels the sine terms. Substitute $x = \cos\theta$ and $T_j(x) = \cos(j\theta)$ to obtain

$$T_{k+1}(x) + T_{k-1}(x) = 2xT_k(x).$$

Also $T_0(x) = 1$ and $T_1(x) = x$. Hence the recurrence constructs polynomials from polynomial starting values, although the original definition used trigonometric functions.

For the leading coefficient, first compute $T_2 = 2x^2 - 1$, whose leading coefficient is 2. The formula is also true for T_1, with leading coefficient $1 = 2^0$. Suppose T_k has degree k and leading coefficient 2^{k-1}. The product $2xT_k$ has degree $k+1$ and leading coefficient 2^k. Subtracting T_{k-1}, of smaller degree, cannot cancel that leading term. Thus T_{k+1} has the claimed degree and coefficient. This completes the induction. □

Ex 5.8 Compute T_2, T_3, T_4 from the recurrence. Prove by induction that T_k has degree k and leading coefficient 2^{k-1} for $k \geq 1$. Which term in the recurrence determines the leading coefficient? After proving the minimax theorem below, map $[a, b]$ to $[-1, 1]$ and show that the monic minimax polynomial of degree $n \geq 1$ on $[a, b]$ has uniform norm $(b-a)^n/2^{2n-1}$.

Thm 5.7 **(Orthogonality of Chebyshev Polynomials)** Chebyshev polynomials are orthogonal on $[-1, 1]$ with respect to the singular weight function $w(x) = (1-x^2)^{-1/2}$:

$$\int_{-1}^{1} T_j(x)T_k(x)\frac{dx}{\sqrt{1-x^2}} = \begin{cases} 0, & j \neq k, \\ \pi, & j = k = 0, \\ \frac{\pi}{2}, & j = k \geq 1. \end{cases}$$

Proof Use $x = \cos\theta$ with $0 \leq \theta \leq \pi$. Then $dx = -\sin\theta\, d\theta$ and $\sqrt{1-x^2} = \sin\theta$ on the interior. Reversing the endpoints cancels the minus sign, giving

$$\int_{-1}^{1} \frac{T_j(x)T_k(x)}{\sqrt{1-x^2}}\, dx = \int_0^{\pi} \cos(j\theta)\cos(k\theta)\, d\theta.$$

The original endpoint singularities are integrable under this substitution.

If $j \neq k$, use the product identity

$$\cos(j\theta)\cos(k\theta) = \tfrac{1}{2}\cos((j-k)\theta) + \tfrac{1}{2}\cos((j+k)\theta).$$

Both frequencies are nonzero integers, including when one of j, k is zero. For any nonzero integer l,

$$\int_0^\pi \cos(l\theta)\, d\theta = [\sin(l\theta)/l]_0^\pi = 0.$$

Thus the cross integral vanishes.

If $j = k = 0$, the integrand is one and its integral is π. If $j = k \geq 1$, use $\cos^2(k\theta) = (1+\cos(2k\theta))/2$. The constant term integrates to $\pi/2$ and the oscillating term to zero. These three cases give every value in the statement. □

5.2.3 Roots, Extrema, and the Nodal Error Polynomial

Thm 5.8 **(Zeros and Extreme Points of T_n)** For each $n \geq 1$, the polynomial $T_n(x)$ has:

1. Exactly n distinct real zeros in $(-1, 1)$, given by

$$x_j = \cos\left(\frac{2j-1}{2n}\pi\right), \qquad j = 1, \ldots, n.$$

2. Exactly $n+1$ alternating extreme points on $[-1, 1]$, given by

$$y_j = \cos\left(\frac{j\pi}{n}\right), \qquad j = 0, \ldots, n,$$

at which $T_n(y_j) = (-1)^j$.

Proof On $[-1, 1]$, the substitution $x = \cos\theta$ with $\theta \in [0, \pi]$ is one-to-one. Thus we can find all roots by solving $\cos(n\theta) = 0$ on that angle interval.

The cosine is zero at $n\theta = (j - 1/2)\pi$. The choices in $[0, n\pi]$ are exactly $j = 1, \ldots, n$. Therefore

$$\theta_j = \frac{2j-1}{2n}\pi, \qquad x_j = \cos\theta_j.$$

All angles are strictly between zero and π, and distinct angles give distinct points because cosine is strictly decreasing there. These are n distinct roots of a degree-n polynomial, so there are no additional roots.

For extreme values, $|T_n(x)| = |\cos(n\theta)| \leq 1$. Equality occurs exactly when $n\theta = j\pi$ for $j = 0, \ldots, n$. The corresponding points are $y_j = \cos(j\pi/n)$, and

$$T_n(y_j) = \cos(j\pi) = (-1)^j.$$

They are therefore alternating points attaining the global upper and lower values, including the two endpoints. In the interior, differentiating gives $T_n'(\cos\theta) = n\sin(n\theta)/\sin\theta$, so no other interior extremum is possible. □

Ex 5.9 List all angles in $[0, \pi]$ that satisfy each condition in the proof. Apply $x = \cos\theta$ and show that the resulting nodes are distinct. Why are there no additional zeros? Sketch T_3 and mark both its zeros and its alternating extreme values.

Def 5.4 **(Chebyshev Nodes)** The n Chebyshev nodes are the roots of $T_n(x)$. They interpolate polynomials of degree at most $n-1$:

$$x_j = \cos\left(\frac{2j-1}{2n}\pi\right), \qquad j = 1, \ldots, n.$$

These nodes cluster quadratically near the interval endpoints ± 1, with spacing magnitude $|x_{j+1} - x_j| = O(n^{-2})$ near the boundaries and $O(1/n)$ in the interior.

Thm 5.9 **(Representation of the Chebyshev Nodal Polynomial)** Let $\{x_j\}_{j=1}^n$ be the Chebyshev nodes from Def 5.4. The monic nodal polynomial satisfies

$$\omega_n(x) = \prod_{j=1}^{n}(x - x_j) = 2^{1-n}T_n(x).$$

Consequently, the uniform norm of the nodal polynomial is

$$\|\omega_n\|_\infty = 2^{1-n}\|T_n\|_\infty = 2^{1-n}.$$

Proof Both $\omega_n(x) = \prod_{j=1}^n (x - x_j)$ and $T_n(x)$ have the same n distinct zeros. Repeated use of the factor theorem makes T_n divisible by this product. Since both degrees are n, the quotient is a nonzero constant, say C.

The leading coefficient of the product ω_n is one: choosing x from every factor gives x^n. The leading coefficient of T_n is 2^{n-1} by the recurrence theorem. Comparing the coefficients of x^n in $T_n = C\omega_n$ gives $C = 2^{n-1}$. Hence

$$\omega_n(x) = 2^{1-n}T_n(x).$$

For every $x \in [-1, 1]$, write $x = \cos\theta$. Then $|T_n(x)| = |\cos(n\theta)| \leq 1$. At $x = 1$, its value is exactly one. Thus $\|T_n\|_\infty = 1$, and homogeneity of the uniform norm gives

$$\|\omega_n\|_\infty = 2^{1-n}\|T_n\|_\infty = 2^{1-n}.$$

This proves both the polynomial identity and its precise norm. □

5.2.4 The Minimax Property and Equioscillation

Thm 5.10 **(Chebyshev Minimax Theorem)** Let $\widetilde{\mathcal{P}}_n$ denote the set of all monic polynomials of degree n with real coefficients. Then $2^{1-n}T_n$ is the unique solution to the minimax problem on $[-1, 1]$:

$$\min_{q \in \widetilde{\mathcal{P}}_n} \|q\|_\infty = \|2^{1-n}T_n\|_\infty = 2^{1-n}.$$

Proof Let $m(x) = 2^{1-n}T_n(x) \in \widetilde{\mathcal{P}}_n$. By Thm 5.9, $\|m\|_\infty = 2^{1-n}$. At the $n+1$ extreme points $y_j = \cos(j\pi/n)$ for $j = 0, \ldots, n$,

$$m(y_j) = 2^{1-n}(-1)^j.$$

Suppose there exists a monic polynomial $q \in \widetilde{\mathcal{P}}_n$ such that $\|q\|_\infty < 2^{1-n}$. Consider the difference polynomial

$$r(x) = m(x) - q(x).$$

Because both m and q are monic polynomials of degree n, the leading terms x^n cancel. Thus $r \in \mathcal{P}_{n-1}$ has degree at most $n-1$.

Evaluating $r(x)$ at each extreme point y_j:

$$r(y_j) = m(y_j) - q(y_j) = 2^{1-n}(-1)^j - q(y_j).$$

Because $|q(y_j)| \le \|q\|_\infty < 2^{1-n}$, the sign of $r(y_j)$ is dictated entirely by $m(y_j)$:

- When j is even, $m(y_j) = 2^{1-n}$, so $r(y_j) = 2^{1-n} - q(y_j) > 0$.
- When j is odd, $m(y_j) = -2^{1-n}$, so $r(y_j) = -2^{1-n} - q(y_j) < 0$.

Thus $r(x)$ alternates sign across the $n+1$ ordered points $1 = y_0 > y_1 > \cdots > y_n = -1$. By the Intermediate Value Theorem, $r(x)$ must vanish at least once in each of the n disjoint open intervals (y_{j+1}, y_j) for $j = 0, \ldots, n-1$.

Consequently, $r(x)$ has at least n distinct real zeros in $(-1, 1)$. However, a nonzero polynomial of degree at most $n-1$ cannot have n roots. Therefore, $r(x)$ must be identically zero, which contradicts the strict inequality $\|q\|_\infty < 2^{1-n}$. This proves that $\|q\|_\infty \ge 2^{1-n}$ for all monic $q \in \widetilde{\mathcal{P}}_n$.

The strict sign-change argument proves minimality, but it does not by itself prove uniqueness: with a competing norm equal to the minimum, some signs may be zero. We now handle these weak inequalities separately.

To establish uniqueness, suppose $\|q\|_\infty = 2^{1-n}$ and retain $r = m - q \in \mathcal{P}_{n-1}$. The weak inequalities

$$(-1)^j r(y_j) \ge 0, \qquad j = 0, \ldots, n,$$

follow from $m(y_j) = 2^{1-n}(-1)^j$ and $|q(y_j)| \le 2^{1-n}$. The difference has degree at most $n-1$ because the leading coefficients of the two monic degree-n polynomials cancel.

Interpolate r at $y_0, \ldots, y_n$ using the Lagrange formula. Its coefficient of x^n is zero because $\deg r \le n-1$. Comparing that coefficient gives

$$0 = \sum_{j=0}^{n} \frac{r(y_j)}{\prod_{k \ne j}(y_j - y_k)}.$$

Because the nodes are ordered $1 = y_0 > y_1 > \cdots > y_n = -1$, the product in the denominator has exactly j negative factors:

$$\operatorname{sign}\left(\prod_{k \ne j}(y_j - y_k)\right) = \operatorname{sign}\left(\prod_{k<j}(y_j - y_k)\prod_{k>j}(y_j - y_k)\right) = (-1)^j.$$

Therefore

$$\frac{r(y_j)}{\prod_{k \ne j}(y_j - y_k)} = \frac{(-1)^j r(y_j)}{\left|\prod_{k \ne j}(y_j - y_k)\right|} \ge 0.$$

A sum of non-negative terms can vanish only if each numerator $(-1)^j r(y_j)$ is zero. Thus $r(y_j) = 0$ for all $j = 0, \dots, n$. Since $r \in \mathcal{P}_{n-1}$ has $n+1$ distinct roots, $r \equiv 0$, proving $q \equiv m$.

All terms in the zero sum have been shown nonnegative individually. If one were positive, the entire sum would be positive. Hence each must be zero. Since every denominator is nonzero, this forces $r(y_j) = 0$ at all nodes, completing the uniqueness argument without assuming strict sign changes. □

Ex 5.10 **(Equioscillation on a concrete polynomial)** Let $m(x) = 2^{1-4} T_4(x)$.

1. Expand m in the monomial basis and verify that it is monic.
2. Evaluate m at $y_j = \cos(j\pi/4)$ for $j = 0, \dots, 4$ and verify the alternating values $\pm 1/8$.
3. Choose any monic quartic $q \neq m$, evaluate it on a fine grid, and confirm numerically that $\|q\|_\infty > 1/8$, as predicted by Thm 5.10.

Remark 5.6 **(Lebesgue constant growth)** For any grid of n interpolation nodes X_n, the interpolation projection $I_n : C([-1,1]) \to \mathcal{P}_{n-1}$ satisfies the Lebesgue inequality $\|f - I_n f\|_\infty \leq (1+\Lambda_n) \min_{p \in \mathcal{P}_{n-1}} \|f - p\|_\infty$, where Λ_n is the Lebesgue constant. For equally spaced nodes, $\Lambda_n \sim \frac{2^n}{en \log n}$, which diverges exponentially and induces Runge's phenomenon. For Chebyshev nodes, $\Lambda_n \leq \frac{2}{\pi} \log n + 1$, guaranteeing that Chebyshev interpolation converges at a rate within a logarithmic factor of the optimal polynomial approximation.

5.2.5 Chebyshev Series and Spectral Accuracy

Def 5.5 **(Chebyshev Series Expansion)** A function with a convergent Chebyshev expansion is written using the primed-sum convention

$$f(x) = \sum_{k=0}^{\infty}{}' c_k T_k(x),$$

where the prime on the summation indicates that the initial term $k = 0$ carries a factor of $1/2$:

$$c_k = \frac{2}{\pi} \int_{-1}^{1} f(x) T_k(x) \frac{dx}{\sqrt{1-x^2}} = \frac{2}{\pi} \int_0^\pi f(\cos\theta) \cos(k\theta)\, d\theta.$$

Thm 5.11 **(Geometric Convergence for Analytic Functions)** Let f be analytic on and inside a Bernstein ellipse whose boundary is

$$\mathcal{E}_\rho = \left\{ z \in \mathbb{C} : z = \frac{w + w^{-1}}{2},\ |w| = \rho \right\}$$

with $\rho > 1$, semi-axes $a = \frac{\rho + \rho^{-1}}{2}$ and $b = \frac{\rho - \rho^{-1}}{2}$. Suppose $|f(z)| \leq M$ on this boundary. Then the Chebyshev coefficients satisfy

$$|c_k| \leq \frac{2M}{\rho^k} \qquad \text{for all } k \geq 0.$$

Consequently, the truncated polynomial approximation $p_n(x) = \sum_{k=0}^{n}{}' c_k T_k(x)$ converges geometrically:

$$\|f - p_n\|_\infty \leq \frac{2M}{\rho - 1} \rho^{-n}.$$

Proof Under the conformal map $z = \frac{w+w^{-1}}{2}$, the circle $|w| = r$ for $1 < r < \rho$ maps to the ellipse $\mathcal{E}_r$ in the complex z-plane. The transformed function $F(w) = f\left(\frac{w+w^{-1}}{2}\right)$ is analytic in the annulus $r^{-1} < |w| < r$ and satisfies $F(w) = F(w^{-1})$.

Here an annulus is the region between two circles. A Laurent series is a power series allowing negative as well as nonnegative integer powers, which is the appropriate expansion on such a region. The symmetry under $w \mapsto w^{-1}$ pairs its positive and negative coefficients.

Expanding $F(w)$ in its Laurent series $F(w) = \sum_{k=-\infty}^{\infty} a_k w^k$, the symmetry $F(w) = F(w^{-1})$ implies $a_k = a_{-k}$. On the unit circle $w = e^{i\theta}$, $F(e^{i\theta}) = f(\cos\theta) = a_0 + 2\sum_{k=1}^{\infty} a_k \cos(k\theta)$, so the Chebyshev coefficients are related to the Laurent coefficients by $c_k = 2a_k$ for $k \geq 1$. By Cauchy's Integral Formula, for any $1 < r < \rho$,

$$a_k = \frac{1}{2\pi i} \oint_{|w|=r} \frac{F(w)}{w^{k+1}}\, dw.$$

On the contour $|w| = r$, the factor $|w^{-(k+1)}|$ equals $r^{-(k+1)}$ and the contour length is $2\pi r$. Consequently the absolute value of the integral is bounded by $(2\pi)^{-1}(2\pi r)Mr^{-(k+1)} = Mr^{-k}$. The bound M also holds on the smaller ellipses by the maximum modulus principle.

Taking absolute values and using $|F(w)| \leq M$ gives $|a_k| \leq Mr^{-k}$. Taking the infimum over $r \to \rho^-$ yields $|c_k| \leq 2M\rho^{-k}$. Summing the tail of the series gives the uniform error bound:

$$\|f - p_n\|_\infty \leq \sum_{k=n+1}^{\infty} |c_k| \leq 2M \sum_{k=n+1}^{\infty} \rho^{-k} = \frac{2M\rho^{-n}}{\rho - 1}.$$

For the primed constant term, $c_0/2 = a_0$, so $|c_0| \leq 2M$ as well. In the tail estimate, $|T_k(x)| \leq 1$ for real $x \in [-1, 1]$. Thus the coefficient tail dominates the uniform function error. This connects the complex coefficient estimate to approximation on the real interval (Trefethen, 2013). □

5.2.6 Further Exercises

Ex 5.11 **(Clenshaw's evaluation algorithm)** Consider the unprimed sum $p(x) = \sum_{k=0}^{n} c_k T_k(x)$. The forward three-term recurrence already evaluates the basis in $O(n)$ operations; Clenshaw's recurrence evaluates the sum directly without storing the basis values or computing trigonometric functions.

1. Define the backward sequence $b_{n+2} = b_{n+1} = 0$, and for $k = n, n-1, \ldots, 1$,

   $$b_k = 2xb_{k+1} - b_{k+2} + c_k.$$

   Show that $p(x) = c_0 + xb_1 - b_2$.

2. Prove that this backward recurrence is algebraically equivalent to evaluating $p(x)$ by substituting the three-term relation $T_{k+1} = 2xT_k - T_{k-1}$.

3. Implement Clenshaw's algorithm in Python and compare its runtime against explicit monomial expansion for $n = 50$.

Ex 5.12 **(Lebesgue constant computation and Runge comparison)** The Lebesgue function is $\lambda_n(x) = \sum_{j=1}^{n} |\ell_j(x)|$, where $\ell_j(x)$ are the Lagrange cardinal polynomials.

1. Write a script to evaluate $\lambda_n(x)$ on a fine grid of 10^4 points in $[-1,1]$ for uniform nodes and Chebyshev nodes for $n \in \{5, 10, 15, 20\}$.
2. Plot the maximum $\Lambda_n = \max_{x\in[-1,1]} \lambda_n(x)$ versus n on a semilog scale. Verify that Λ_n grows exponentially for uniform nodes and logarithmically for Chebyshev nodes.
3. Interpolate Runge's function $f(x) = 1/(1+25x^2)$ with $n = 20$ using both node sets. Explain the boundary behavior in terms of the Bernstein ellipse parameter ρ.

Ex 5.13 **(Chebyshev differentiation matrix)** Let $\boldsymbol{x} = (x_0, x_1, \ldots, x_n)^T$ be the Chebyshev extreme nodes $x_j = \cos(j\pi/n)$.

1. Derive the entries of the spectral differentiation matrix $\mathbf{D}_n \in \mathbb{R}^{(n+1)\times(n+1)}$ that maps the nodal values $\mathbf{f} = (f(x_0), \ldots, f(x_n))^T$ to the exact derivative values of the polynomial interpolant $\mathbf{f}' = \mathbf{D}_n\mathbf{f}$.
2. Show that the diagonal entries satisfy $D_{00} = \frac{2n^2+1}{6}$ and $D_{nn} = -\frac{2n^2+1}{6}$.
3. Verify that the row sums of $\mathbf{D}_n$ vanish identically: $\mathbf{D}_n\mathbf{1} = \mathbf{0}$.

Ex 5.14 **(Bernstein ellipse parameter for Runge's function)** Consider $f(x) = \frac{1}{1+25x^2}$ on $[-1,1]$.

1. Find the complex poles of $f(z)$ in $\mathbb{C}$.
2. Determine the largest value of ρ such that the Bernstein ellipse $\mathcal{E}_\rho$ does not contain either pole.
3. Using Thm 5.11, compute the asymptotic error decay factor ρ^{-1} and predict how many Chebyshev modes are needed to achieve machine precision $\|f - p_n\|_\infty \le 10^{-16}$.

5.3 Numerical Quadrature

Numerical quadrature approximates the definite integral of a continuous function $f : [a,b] \to \mathbb{R}$ by a finite linear combination of sampled function values:

$$I(f) = \int_a^b f(x)\,dx \approx Q_n(f) = \sum_{i=0}^{n} w_i f(x_i).$$

The points $x_i \in [a,b]$ are the quadrature nodes, and the scalar coefficients $w_i \in \mathbb{R}$ are the quadrature weights. The design of quadrature rules balances algebraic precision, computational cost, and numerical stability.

5.3.1 Degree of Exactness and Interpolatory Quadrature

The standard criterion for evaluating a quadrature rule is its algebraic precision, defined by the highest degree of polynomials it integrates without error.

Example On $[0,1]$, suppose we use just the two endpoint values and seek $Q(f) = w_0 f(0) + w_1 f(1)$. Exact integration of 1 requires $w_0 + w_1 = 1$. Exact integration of x requires $w_1 = 1/2$. Thus both weights are $1/2$, giving the trapezoidal rule.

Ex 5.15 Test this rule on x^2. Is the quadrature value above or below the integral? Derive the endpoint weights on $[a, b]$ in the same way. Relate the result to the area under the line joining the two endpoint values.

Def 5.6 **(Degree of Exactness (Algebraic Precision))** A quadrature rule Q_n has degree of exactness m if

$$Q_n(p) = \int_a^b p(x)\,dx \qquad \text{for all } p \in \mathcal{P}_m,$$

and there exists at least one polynomial $q \in \mathcal{P}_{m+1}$ for which $Q_n(q) \neq \int_a^b q(x)\,dx$.

Thm 5.12 **(Interpolatory Quadrature Rules)** Let $x_0, x_1, \ldots, x_n$ be $n+1$ distinct nodes in $[a, b]$. The interpolatory quadrature rule approximates $I(f)$ by integrating the polynomial interpolant $P_n(x) = \sum_{i=0}^n f(x_i)L_i(x)$ in Lagrange form:

$$Q_n(f) = \int_a^b P_n(x)\,dx = \sum_{i=0}^n w_i f(x_i), \qquad w_i = \int_a^b L_i(x)\,dx.$$

Every interpolatory quadrature rule with $n+1$ distinct nodes has degree of exactness at least n.

Proof The Lagrange polynomials satisfy $L_i(x_j) = \delta_{ij}$. Therefore for any data values $f(x_i)$, the polynomial

$$P_n(x) = \sum_{i=0}^n f(x_i)L_i(x)$$

interpolates those values. Its integral is a finite sum, so linearity of integration gives

$$\int_a^b P_n(x)\,dx = \sum_{i=0}^n f(x_i) \int_a^b L_i(x)\,dx = \sum_{i=0}^n w_i f(x_i).$$

This is the proposed quadrature rule.

To establish exactness, take a polynomial p of degree at most n. It already satisfies the interpolation conditions for its own data. Uniqueness of interpolation implies that the constructed P_n equals p everywhere, not only at the nodes. Consequently

$$Q_n(p) = \int_a^b P_n(x)\,dx = \int_a^b p(x)\,dx.$$

The argument applies to every polynomial in $\mathcal{P}_n$. Thus the rule has degree of exactness at least n. It may be exact for higher degrees because additional cancellation can occur, so this argument establishes a lower bound for the degree rather than its exact value. □

Ex 5.16 Integrate $p(x) = \sum_i p(x_i)L_i(x)$ term by term to prove the theorem. Deduce $\sum_i w_i = b - a$ by choosing $p = 1$. For the nodes $-1, 0, 1$, integrate the three Lagrange basis polynomials to obtain Simpson's weights. Use symmetry to explain why the rule is exact for cubics as well as quadratics.

5.3.2 Newton–Cotes Formulas and Composite Rules

Newton–Cotes rules are interpolatory quadrature formulas built on equally spaced nodes. Because high-order single-panel Newton–Cotes rules suffer from negative weights and Runge's phenomenon, practical computations use composite rules on small subintervals of uniform width $h = (b-a)/N$.

Def 5.7 **(Composite Trapezoidal and Simpson's Rules)** Dividing $[a, b]$ into N subintervals with uniform width $h = (b - a)/N$ and grid points $x_j = a + jh$:

1. The **Composite Trapezoidal Rule** is

$$T_N(f) = h\left[\frac{f(a) + f(b)}{2} + \sum_{j=1}^{N-1} f(x_j)\right].$$

2. For an even number of panels N, the **Composite Simpson's Rule** is

$$S_N(f) = \frac{h}{3}\left[f(x_0) + 4\sum_{\substack{j=1\\ j\text{ odd}}}^{N-1} f(x_j) + 2\sum_{\substack{j=2\\ j\text{ even}}}^{N-2} f(x_j) + f(x_N)\right].$$

Thm 5.13 **(Composite Error Formulas)** Let $f \in C^2[a, b]$ for the trapezoidal rule, and let $f \in C^4[a, b]$ for Simpson's rule.

1. The composite trapezoidal error satisfies

$$I(f) - T_N(f) = -\frac{b - a}{12}h^2 f''(\xi_T), \qquad \xi_T \in (a, b).$$

2. The composite Simpson error satisfies

$$I(f) - S_N(f) = -\frac{b - a}{180}h^4 f^{(4)}(\xi_S), \qquad \xi_S \in (a, b).$$

Proof For the trapezoidal rule on panel $[x_j, x_{j+1}]$, the Cauchy interpolation error formula (Thm 5.3) gives

$$f(x) - P_1(x) = \frac{f''(\xi_j(x))}{2}(x - x_j)(x - x_{j+1}).$$

Because $(x - x_j)(x - x_{j+1}) \le 0$ on this panel, the Mean Value Theorem for Integrals implies there exists $\mu_j \in (x_j, x_{j+1})$ such that

$$\int_{x_j}^{x_{j+1}} [f(x) - P_1(x)]\,dx = \frac{f''(\mu_j)}{2}\int_{x_j}^{x_{j+1}} (x - x_j)(x - x_{j+1})\,dx = -\frac{h^3}{12}f''(\mu_j).$$

Summing across all N panels and applying the Intermediate Value Theorem to the average $\frac{1}{N}\sum_{j=0}^{N-1} f''(\mu_j) = f''(\xi_T)$ yields the trapezoidal error formula:

$$I(f) - T_N(f) = -\frac{h^3}{12}\sum_{j=0}^{N-1} f''(\mu_j) = -\frac{Nh^3}{12}f''(\xi_T) = -\frac{b - a}{12}h^2 f''(\xi_T).$$

The intermediate mean-value expression for the trapezoidal rule can be justified without assuming that a choice of $\xi_j(x)$ is continuous: the minimum and maximum of f'' bound the pointwise interpolation error multiplied by its fixed-sign nodal factor. Integrating these bounds puts the

scaled panel error between the minimum and maximum values of f''. Continuity then supplies the stated μ_j.

For Simpson's rule use the integral form of Taylor's theorem, which requires only $f \in C^4$. On the reference interval $[-1, 1]$, the error functional annihilates cubics. Its fourth-derivative kernel is

$$\begin{aligned} K(t) &= \frac{1}{6} E_x((x-t)_+^3) \\ &= \frac{(1-t)^4}{24} - \frac{2}{9}(-t)_+^3 - \frac{(1-t)^3}{18} \\ &= -\frac{(1-|t|)^3(1+3|t|)}{72}, \qquad -1 \le t \le 1. \end{aligned}$$

The notation E_x means that the quadrature-error operator acts on the variable x, with t held fixed. The factor $1/6$ comes from the third power in Taylor's integral remainder. On $[-1, 1]$, integrating $(x-t)_+^3/6$ gives $(1-t)^4/24$. Evaluating it at the Simpson nodes gives exactly the two nonzero sampled terms displayed above.

Thus $K \le 0$ and $\int_{-1}^{1} K(t)\,dt = -1/90$. Taylor's integral remainder gives $E(f) = \int_{-1}^{1} K(t) f^{(4)}(t)\,dt$; the next theorem proves this general kernel representation. A weighted mean-value argument and scaling $x = c_j + ht$ give the double-panel error $-h^5 f^{(4)}(\mu_j)/90$. Summing the $N/2$ double panels and using continuity of $f^{(4)}$ proves the composite formula.

On a double panel, $x = c_j + ht$ has Jacobian h and its fourth derivative with respect to t is $h^4 f^{(4)}(c_j + ht)$. These give the combined factor h^5. There are $N/2$ such panels, so $-(N/2)h^5/90 = -(b-a)h^4/180$, which is the composite constant. □

Ex 5.17 Verify the Simpson kernel formula separately for $t \ge 0$ and $t \le 0$, and integrate it. Apply the single-panel formulas to $f(x) = x^2$ for the trapezoidal rule and $f(x) = x^4$ for Simpson's rule. Check both the sign and the power of h.

5.3.3 Peano Kernel Error Representation

A quadrature error $E(f)$ is linear in f. If the rule integrates every polynomial of degree at most m exactly, those polynomial terms contribute no error. Taylor's formula then lets us express the whole error using $f^{(m+1)}$. The function multiplying this derivative under the integral is called the Peano kernel. Once we know that function, we can estimate the error for many different integrands without repeating the derivation.

Thm 5.14 **(Peano Kernel Theorem)** Let Q be a quadrature rule on $[a, b]$ with error functional $E(f) = \int_a^b f(x)\,dx - Q(f)$. If $E(p) = 0$ for all $p \in \mathcal{P}_m$, then for every $f \in C^{m+1}[a, b]$,

$$E(f) = \int_a^b f^{(m+1)}(t) K(t)\,dt,$$

where the Peano kernel $K(t)$ is defined by applying the error functional to the truncated power function:

$$K(t) = \frac{1}{m!} E_x\left((x-t)_+^m\right), \qquad (x-t)_+ = \max(0, x-t).$$

For $m = 0$, interpret $(x-t)_+^0$ as 1 when $x > t$ and 0 when $x \le t$.

If $K(t)$ does not change sign on $[a, b]$, then there exists $\xi \in (a, b)$ such that

$$E(f) = \frac{f^{(m+1)}(\xi)}{(m+1)!} E(x^{m+1}).$$

Proof By Taylor's theorem with integral remainder expanded about a:

$$f(x) = \sum_{k=0}^{m} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{1}{m!}\int_a^x f^{(m+1)}(t)(x-t)^m\,dt.$$

Extending the upper integration limit using the truncated power notation $(x-t)_+^m$ allows writing the remainder as $\frac{1}{m!}\int_a^b f^{(m+1)}(t)(x-t)_+^m\,dt$. The truncated power makes the integrand zero for $t > x$, so extending the upper limit adds only zeros. It is useful because both the quadrature sum and the integral can then act on functions of x over a fixed interval.

Applying the linear functional E to both sides, the polynomial terms vanish because $E(p) = 0$ for all $p \in \mathcal{P}_m$. Exchanging the order of integration and the functional gives

$$\begin{aligned} E(f) &= E_x\left(\frac{1}{m!}\int_a^b f^{(m+1)}(t)(x-t)_+^m\,dt\right) \\ &= \int_a^b f^{(m+1)}(t)\left[\frac{1}{m!}E_x((x-t)_+^m)\right]dt \\ &= \int_a^b f^{(m+1)}(t)K(t)\,dt. \end{aligned}$$

Interchanging a finite quadrature sum with integration follows from linearity. Interchanging the two ordinary integrals follows from continuity and integrability on the compact triangle. Thus no convergence of an infinite sum is needed in this step.

If K has constant sign, the weighted mean-value theorem gives

$$E(f) = f^{(m+1)}(\xi)\int_a^b K(t)\,dt.$$

Testing the identity with $f(x) = x^{m+1}$, we have $f^{(m+1)}(x) = (m+1)!$, which yields $E(x^{m+1}) = (m+1)!\int_a^b K(t)\,dt$. Substituting this integral proves the claim.

If K is identically zero, the error is zero and the final identity is immediate. Otherwise use $|K|$ as a nonnegative weight: the integral average of the continuous derivative lies between its minimum and maximum, so the intermediate value theorem supplies a point attaining that average. This is the weighted mean-value step used above. □

Ex 5.18 **Peano kernel of the trapezoidal rule.** Consider the single-panel trapezoidal rule on $[0, h]$ with error functional $E(f) = \int_0^h f(x)\,dx - \frac{h}{2}[f(0) + f(h)]$.

1. Show that $E(1) = 0$ and $E(x) = 0$, so $m = 1$.
2. Evaluate the Peano kernel $K(t) = E_x((x-t)_+) = \int_0^h (x-t)_+\,dx - \frac{h}{2}[(0-t)_+ + (h-t)_+]$.
3. Show that $K(t) = \frac{1}{2}t(t-h) \le 0$ for all $t \in [0, h]$.
4. Apply Thm 5.14 to compute $\int_0^h K(t)\,dt = -\frac{h^3}{12}$ and recover the exact error term $-\frac{h^3}{12}f''(\xi)$.

5.3.4 Gauss–Legendre Quadrature

Newton–Cotes rules fix equispaced nodes and optimize only the weights. Gaussian quadrature optimizes both the nodes $x_1, \dots, x_p$ and the weights $w_1, \dots, w_p$, providing $2p$ degrees of freedom to maximize the degree of exactness.

We need polynomials orthogonal for the ordinary integral on $[-1, 1]$, rather than the Chebyshev weight. Applying Gram–Schmidt to $1, x, x^2, \dots$ with inner product $\int_{-1}^{1} p(x)q(x)\,dx$ constructs them. Up to normalization these are the Legendre polynomials. Orthogonality will make part of the quadrature error vanish without adding more nodes.

Ex 5.19 Construct the first three polynomials in this process. You may leave their norms arbitrary. Show that the polynomial of degree two is proportional to $x^2 - 1/3$. Use its roots and exactness for 1 and x to derive the two-point Gaussian rule.

Lem 5.15 **(Roots of Orthogonal Polynomials)** Let $P_p(x)$ be the degree-p Legendre polynomial orthogonal on $[-1, 1]$ with respect to weight $w(x) \equiv 1$:

$$\int_{-1}^{1} P_p(x)q(x)\,dx = 0 \qquad \text{for all } q \in \mathcal{P}_{p-1}.$$

Then $P_p(x)$ has exactly p distinct, real roots, and all roots lie in the open interval $(-1, 1)$.

Proof Let $x_1, \dots, x_k$ denote the points in $(-1, 1)$ where $P_p(x)$ changes sign. Because P_p is continuous and orthogonal to the constant polynomial $q(x) = 1$, we have $\int_{-1}^{1} P_p(x)\,dx = 0$, so P_p must change sign at least once in $(-1, 1)$, meaning $k \geq 1$. We count sign-changing roots rather than all roots. A root of even multiplicity does not reverse the sign, whereas a root of odd multiplicity does. The polynomial is nonzero, so it cannot vanish on an interval.

Suppose for contradiction that $k < p$. Construct the polynomial $q(x) = \prod_{j=1}^{k}(x - x_j) \in \mathcal{P}_k$. By construction, $P_p(x)$ and $q(x)$ change sign at identical points in $(-1, 1)$, so their product $P_p(x)q(x)$ has constant sign on $[-1, 1]$ and is not identically zero. Consequently,

$$\int_{-1}^{1} P_p(x)q(x)\,dx \neq 0.$$

Across each listed point, both factors change sign and their product does not. At every other root, P_p has even multiplicity and again the product does not change sign. Away from finitely many roots the product is nonzero. Its integral over the interval therefore cannot vanish.

However, because $\deg(q) = k < p$, orthogonality requires $\int_{-1}^{1} P_p(x)q(x)\,dx = 0$, producing a contradiction. Therefore, $k = p$, proving that P_p has p distinct real roots in $(-1, 1)$.

A degree-p polynomial cannot have more than p distinct roots. The argument has found p distinct sign-changing roots inside the interval, exhausting the degree. Thus they are simple and there are no other roots, real or complex. □

Thm 5.16 **(Fundamental Theorem of Gaussian Quadrature)** Let $x_1, \dots, x_p$ be the roots of the degree-p Legendre polynomial $P_p(x)$ on $[-1, 1]$, and let $w_i = \int_{-1}^{1} L_i(x)\,dx$. The p-point Gauss–Legendre quadrature rule

$$\int_{-1}^{1} f(x)\,dx \approx \sum_{i=1}^{p} w_i f(x_i)$$

has degree of exactness $2p-1$. No p-point quadrature rule can achieve a degree of exactness higher than $2p-1$. Furthermore, all weights are strictly positive: $w_i > 0$ for all $i = 1, \ldots, p$.

Proof Let $f \in \mathcal{P}_{2p-1}$. Polynomial long division of $f(x)$ by $P_p(x)$ yields

$$f(x) = q(x)P_p(x) + r(x), \qquad \deg(q) \leq p-1, \quad \deg(r) \leq p-1.$$

Integrating both sides over $[-1, 1]$:

$$\int_{-1}^{1} f(x)\,dx = \int_{-1}^{1} q(x)P_p(x)\,dx + \int_{-1}^{1} r(x)\,dx.$$

Because $\deg(q) \leq p-1$, orthogonality of P_p against $\mathcal{P}_{p-1}$ implies $\int_{-1}^{1} q(x)P_p(x)\,dx = 0$, giving

$$\int_{-1}^{1} f(x)\,dx = \int_{-1}^{1} r(x)\,dx.$$

Applying the p-point quadrature sum at the Legendre roots $\{x_i\}_{i=1}^{p}$ where $P_p(x_i) = 0$:

$$\begin{aligned}\sum_{i=1}^{p} w_i f(x_i) &= \sum_{i=1}^{p} w_i \left[q(x_i)P_p(x_i) + r(x_i)\right] \\ &= \sum_{i=1}^{p} w_i \left[0 + r(x_i)\right] = \sum_{i=1}^{p} w_i r(x_i).\end{aligned}$$

Because $\deg(r) \leq p-1$, interpolatory quadrature on p distinct nodes integrates $r(x)$ exactly by Thm 5.12:

$$\sum_{i=1}^{p} w_i r(x_i) = \int_{-1}^{1} r(x)\,dx = \int_{-1}^{1} f(x)\,dx.$$

Thus $\sum_{i=1}^{p} w_i f(x_i) = \int_{-1}^{1} f(x)\,dx$ for all $f \in \mathcal{P}_{2p-1}$. Both the exact integral and the quadrature value have now been reduced to the same remainder polynomial r of degree at most $p-1$. This proves exactness through degree $2p-1$, rather than merely through the interpolation degree.

To prove that $2p-1$ is the maximal degree, define the nodal polynomial

$$g(x) = \prod_{i=1}^{p} (x - x_i)^2 \in \mathcal{P}_{2p}.$$

Because $g(x) \geq 0$ and vanishes only at isolated points, its integral is strictly positive:

$$\int_{-1}^{1} g(x)\,dx > 0.$$

However, evaluating the quadrature sum yields

$$\sum_{i=1}^{p} w_i g(x_i) = \sum_{i=1}^{p} w_i \cdot 0 = 0 \neq \int_{-1}^{1} g(x)\,dx.$$

Therefore no p-point quadrature rule can integrate all degree-$2p$ polynomials. This obstruction applies to any proposed p nodes, regardless of their weights. The squared nodal polynomial is tailored to those nodes, has a positive integral, and is invisible to the quadrature rule.

For weight positivity, let $L_i \in \mathcal{P}_{p-1}$ be the i-th Lagrange basis polynomial. The square satisfies $\deg(L_i^2) = 2p-2 \le 2p-1$. The rule integrates L_i^2 exactly:

$$0 < \int_{-1}^{1} L_i^2(x)\,dx = \sum_{j=1}^{p} w_j L_i^2(x_j) = \sum_{j=1}^{p} w_j \delta_{ij} = w_i.$$

Every weight w_i is strictly positive.

The positivity integral is strict because $L_i(x_i) = 1$, so L_i is not the zero polynomial and its square is positive on a neighborhood. The equality with the single weight uses $L_i(x_j)^2 = \delta_{ij}$. Gaussian quadrature and its error arguments are developed in Stoer and Bulirsch (2002). □

5.3.5 Richardson Extrapolation and Romberg Integration

When the error of a numerical method admits an asymptotic expansion in powers of h, Richardson extrapolation algebraically cancels the leading error terms.

Thm 5.17 **(Euler–Maclaurin Summation and Romberg Integration)** Let $f \in C^{2m+2}[a,b]$. The composite trapezoidal rule satisfies the asymptotic expansion

$$T(h) = I(f) + c_2 h^2 + c_4 h^4 + c_6 h^6 + \cdots + c_{2m} h^{2m} + O(h^{2m+2}),$$

where $c_{2k} = \frac{B_{2k}}{(2k)!}[f^{(2k-1)}(b) - f^{(2k-1)}(a)]$ depend on the Bernoulli numbers B_{2k}. Romberg integration systematically eliminates powers of h^2 via the tableau recurrence

$$R(k,0) = T(h_k), \qquad R(k,j) = \frac{4^j R(k,j-1) - R(k-1,j-1)}{4^j - 1} \quad (1 \le j \le k),$$

where $h_k = (b-a)/2^k$. Column j achieves order $O(h^{2j+2})$.

Proof Consider two consecutive grid widths, h_k and $h_{k-1} = 2h_k$. The first trapezoidal expansion gives

$$R(k,0) = I + c_2 h_k^2 + c_4 h_k^4 + O(h_k^6),$$
$$R(k-1,0) = I + 4c_2 h_k^2 + 16 c_4 h_k^4 + O(h_k^6).$$

Multiply the first by four and subtract the second. The h_k^2 terms cancel, while the constant term becomes $3I$. Dividing by three therefore gives

$$R(k,1) = \frac{4R(k,0) - R(k-1,0)}{3} = I + O(h_k^4).$$

This proves the first extrapolation step, including the reason for its two weights.

For the general step, suppose column $j-1$ has an expansion whose first possible error term has order $2j$:

$$R(k,j-1) = I + d_{2j} h_k^{2j} + O(h_k^{2j+2}).$$

The coefficients here are the same when the entire construction is carried out with width $2h_k$, because the expansion coefficients are independent of the width. Hence

$$R(k-1,j-1) = I + 2^{2j} d_{2j} h_k^{2j} + O(h_k^{2j+2}).$$

Since $2^{2j} = 4^j$, the combination in the stated recurrence cancels this leading term and divides its constant term by $4^j - 1$. Thus $R(k,j) = I + O(h_k^{2j+2})$. Induction proves the order of each fixed column for which the assumed smoothness supplies the required expansion. The starting Euler–Maclaurin expansion itself is a standard result of quadrature theory (Stoer and Bulirsch, 2002). □

5.3.6 Periodic Functions and Spectral Convergence

Thm 5.18 **(Spectral Accuracy of the Trapezoidal Rule on Periodic Domains)** Let f be a smooth, 2π-periodic function on $\mathbb{R}$. All endpoint derivatives in the Euler–Maclaurin expansion cancel identically:

$$f^{(2k-1)}(2\pi) - f^{(2k-1)}(0) = 0 \qquad \text{for all } k \geq 1.$$

Consequently, all algebraic error coefficients $c_2, c_4, c_6, \ldots$ vanish. If f is analytic and bounded in a strip $\{z \in \mathbb{C} : |\operatorname{Im}(z)| < d\}$, the composite trapezoidal rule converges exponentially fast:

$$|I(f) - T_N(f)| \leq Ce^{-dN}.$$

On smooth periodic domains, the composite trapezoidal rule achieves spectral accuracy.

Proof Because f is periodic and smooth, $f^{(m)}(2\pi) = f^{(m)}(0)$ for all $m \geq 0$. Every term c_{2k} in the Euler–Maclaurin formula is proportional to $f^{(2k-1)}(2\pi) - f^{(2k-1)}(0) = 0$, so the error decays faster than any power of $h = 2\pi/N$. For any chosen algebraic power, apply Euler–Maclaurin with enough derivatives. Its endpoint terms vanish and only a remainder of higher order remains. This is the meaning of faster-than-algebraic convergence for a smooth periodic function. Exponential convergence requires the additional analyticity assumption.

To quantify the exponential rate, expand $f(x)$ in its Fourier series:

$$f(x) = \sum_{k=-\infty}^{\infty} \hat{f}_k e^{ikx}, \qquad \hat{f}_k = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-ikx}\, dx.$$

The exact integral is $I(f) = \int_0^{2\pi} f(x)\, dx = 2\pi \hat{f}_0$. The composite trapezoidal rule with $h = 2\pi/N$ and nodes $x_j = 2\pi j/N$ for $j = 0, \ldots, N-1$ gives

$$\begin{aligned} T_N(f) = \frac{2\pi}{N} \sum_{j=0}^{N-1} f(x_j) &= \frac{2\pi}{N} \sum_{j=0}^{N-1} \sum_{k=-\infty}^{\infty} \hat{f}_k e^{2\pi ijk/N} \\ &= 2\pi \sum_{k=-\infty}^{\infty} \hat{f}_k \left(\frac{1}{N} \sum_{j=0}^{N-1} e^{2\pi ijk/N} \right). \end{aligned}$$

The inner sum satisfies discrete orthogonality:

$$\frac{1}{N} \sum_{j=0}^{N-1} e^{2\pi ijk/N} = \begin{cases} 1, & k = mN \text{ for some } m \in \mathbb{Z}, \\ 0, & k \not\equiv 0 \pmod{N}. \end{cases}$$

For a frequency not divisible by N, the inner sum is a geometric sum with ratio $r = e^{2\pi ik/N} \neq 1$ and $r^N = 1$. Its value is $(1 - r^N)/(1 - r) = 0$. If k is divisible by N, every term is one and the normalized sum is one.

Substituting this indicator isolates the aliased modes:

$$T_N(f) = 2\pi \sum_{m=-\infty}^{\infty} \hat{f}_{mN} = 2\pi \hat{f}_0 + 2\pi \sum_{m \neq 0} \hat{f}_{mN}.$$

The quadrature error is therefore the tail sum:

$$I(f) - T_N(f) = -2\pi \sum_{m\neq 0} \hat{f}_{mN}.$$

Because f is analytic and bounded by M_0 in the strip $|\operatorname{Im}(z)| < d$, shifting the integration path in the definition of $\hat{f}_k$ gives the geometric decay $|\hat{f}_k| \leq Me^{-d|k|}$. Applying this bound:

$$|I(f) - T_N(f)| \leq 2\pi M \sum_{m\neq 0} e^{-d|m|N} = 4\pi M \sum_{m=1}^{\infty} (e^{-dN})^m = 4\pi M \frac{e^{-dN}}{1-e^{-dN}} = O(e^{-dN}).$$

Because $d > 0$ and $N \geq 1$, $1 - e^{-dN} \geq 1 - e^{-d} > 0$. Hence the displayed fraction is bounded by a fixed constant times e^{-dN}. This supplies a constant independent of N for the claimed geometric bound. □

5.3.7 Further Exercises

Ex 5.20 **Method of undetermined coefficients.** Find the nodes and weights for a symmetric 3-point rule on $[-1, 1]$ of the form $\int_{-1}^{1} f(x)\,dx \approx w_0 f(-\alpha) + w_1 f(0) + w_2 f(\alpha)$.

1. For fixed $\alpha = 1$, enforce exactness for $f(x) = 1, x, x^2$ to derive the weights of Simpson's rule: $w_0 = 1/3, w_1 = 4/3, w_2 = 1/3$.
2. Let α be a free parameter. Enforce exactness for $f(x) = 1, x^2, x^4$ to obtain the three-point Gauss–Legendre rule: $\alpha = \sqrt{3/5}$, $w_0 = w_2 = 5/9$, and $w_1 = 8/9$. Use symmetry to establish exactness for odd powers, then test x^6 to determine the degree of exactness.
3. Compare the degree of exactness of both rules.

Ex 5.21 **Romberg tableau construction.** Approximate the integral $I = \int_0^1 e^{-x^2}\,dx$.

1. Compute $T(h)$ for $h = 1, 1/2, 1/4$ using the composite trapezoidal rule.
2. Construct the first three rows of the Romberg tableau to evaluate $R(1,1), R(2,1)$, and $R(2,2)$.
3. Confirm that $R(1,1)$ matches composite Simpson's rule with $N = 2$.

Ex 5.22 **Change of interval and Gaussian quadrature.** Let $\int_a^b f(t)\,dt$ be an integral on an arbitrary interval $[a, b]$.

1. Derive the affine substitution $t(x) = \frac{b-a}{2}x + \frac{a+b}{2}$ mapping $x \in [-1, 1]$ to $t \in [a, b]$.
2. Transform the 2-point Gauss–Legendre rule to $[0, 3]$ and approximate $\int_0^3 \sqrt{1+t^3}\,dt$.
3. Compare the result with Simpson's rule, which uses three evaluations rather than two. Report both the error and the evaluation count.

Ex 5.23 **Spectral accuracy on periodic integrals.** Consider the integral $I = \int_0^{2\pi} \frac{dx}{2+\cos x} = \frac{2\pi}{\sqrt{3}}$.

1. Identify the complex poles of $f(z) = \frac{1}{2+\cos z}$ by solving $\cos z = -2$.
2. Determine the half-width d of the strip of analyticity around the real axis.

3. Evaluate $T_N(f)$ for $N = 4, 8, 16$. Confirm that $|I - T_N(f)|$ decays as e^{-dN}.

Read **Further reading.** Stoer and Bulirsch give a classical numerical-analysis treatment of interpolation and quadrature (Stoer and Bulirsch, 2002). Trefethen is especially useful for polynomial approximation, and de Boor for splines (Trefethen, 2013; de Boor, 2001). The short paper by Clenshaw and Curtis is a good example of an algorithmic quadrature paper that remains approachable (Clenshaw and Curtis, 1960).

5.4 Fourier Analysis

A periodic vector can be described by the amplitude of each oscillation it contains. The discrete Fourier transform is a change from its sampled values to these amplitudes. We construct the oscillating basis vectors, prove their orthogonality, and use the basis to simplify shifts and circular convolution.

The fast Fourier transform computes the same change of basis by dividing one transform into smaller transforms. Finally, differentiating each oscillation gives an algorithm for differentiating smooth periodic functions from their sampled values.

5.4.1 The Discrete Fourier Transform

We consider discrete signals defined on a uniform periodic grid of n points. The Discrete Fourier Transform (DFT) expands a vector in the orthonormal basis of discrete complex exponentials.

Def 5.8 **(Periodic Grid Signal)** Let $t_j = 2\pi j/n$ for $j = 0, \dots, n-1$ be grid points in one period. A vector $\boldsymbol{x} \in \mathbb{C}^n$ stores samples $x_j = f(t_j)$. Extend its entries periodically by $x_{j+n} = x_j$. The grid coordinates themselves satisfy $t_{j+n} = t_j + 2\pi$, so it is the sampled values that repeat.

For integer frequency k, the vector with entries $e^{2\pi ijk/n}$ samples one complex oscillation. Frequency zero is constant. Increasing k increases the number of turns made by the complex phase over one period. Taking inner products with these vectors measures the corresponding amplitudes. The minus sign in the forward transform comes from conjugating the basis vectors.

Def 5.9 **(DFT Matrix)** Let $\omega_n = e^{-2\pi i/n}$ denote the primitive n-th root of unity. The Discrete Fourier Transform matrix $\mathbf{F} \in \mathbb{C}^{n\times n}$ has entries

$$F_{jk} = \frac{1}{\sqrt{n}}\omega_n^{jk} = \frac{1}{\sqrt{n}}e^{-2\pi ijk/n}, \qquad 0 \le j, k \le n-1.$$

The forward transform maps a spatial signal $\boldsymbol{x} \in \mathbb{C}^n$ to its frequency representation $\hat{\boldsymbol{x}} = \mathbf{F}\boldsymbol{x}$.

Lem 5.19 **(Unitarity of the DFT Matrix)** The normalized DFT matrix $\mathbf{F}$ is unitary:

$$\mathbf{F}^*\mathbf{F} = \mathbf{F}\mathbf{F}^* = \mathbf{I}_n.$$

Consequently, the inverse transform is given by the conjugate transpose: $\mathbf{F}^{-1} = \mathbf{F}^*$.

Proof We evaluate the (j, ℓ) entry of the Gram matrix product $\mathbf{F}^*\mathbf{F}$. Because $(\mathbf{F}^*)_{jk} = \overline{F_{kj}} = \frac{1}{\sqrt{n}}\omega_n^{-jk}$, we have

$$(\mathbf{F}^*\mathbf{F})_{j\ell} = \sum_{k=0}^{n-1}(\mathbf{F}^*)_{jk}F_{k\ell} = \frac{1}{n}\sum_{k=0}^{n-1}\omega_n^{-jk}\omega_n^{k\ell} = \frac{1}{n}\sum_{k=0}^{n-1}\left(\omega_n^{\ell-j}\right)^k.$$

The complex conjugate reverses the sign in the exponent, so the product depends only on the difference of column indices. This is why the geometric sum appears, rather than a general sum with unrelated terms.

If $j = \ell$, each term in the summation equals $\omega_n^0 = 1$, yielding $\frac{1}{n}\sum_{k=0}^{n-1} 1 = 1$. If $j \neq \ell$, let $r = \omega_n^{\ell-j}$. Because $-(n-1) \leq \ell - j \leq n-1$ and $j \neq \ell$, the base $r \neq 1$, while $r^n = (\omega_n^n)^{\ell-j} = 1^{\ell-j} = 1$. The finite geometric series sums to

$$\sum_{k=0}^{n-1} r^k = \frac{1-r^n}{1-r} = \frac{1-1}{1-r} = 0.$$

Therefore, $(\mathbf{F}^*\mathbf{F})_{j\ell} = \delta_{j\ell}$, which establishes $\mathbf{F}^*\mathbf{F} = \mathbf{I}_n$.

The Gram matrix is therefore the identity. Since $\mathbf{F}$ is square, $\mathbf{F}^*\mathbf{F} = \mathbf{I}$ makes $\mathbf{F}$ nonsingular and gives $\mathbf{F}^{-1} = \mathbf{F}^*$. Multiplication in the reverse order then gives $\mathbf{F}\mathbf{F}^* = \mathbf{I}$ as well. This explains both asserted unitary identities. □

Ex 5.24 **Unitary coordinates and geometric sums.** Let $\mathbf{F} \in \mathbb{C}^{4\times 4}$ be the 4-point DFT matrix. Write out its sixteen numerical entries explicitly using powers of $\omega_4 = -i$. Verify directly that columns 0 and 2 are orthogonal, and confirm that $\mathbf{F}^*\mathbf{F} = \mathbf{I}_4$.

Thm 5.20 **(Parseval's Identity and Isometry)** The Discrete Fourier Transform is an isometry on $\mathbb{C}^n$ with respect to the Euclidean norm:

$$\|\mathbf{F}\boldsymbol{x}\|_2 = \|\boldsymbol{x}\|_2 \qquad \text{for all } \boldsymbol{x} \in \mathbb{C}^n.$$

The inner product is likewise preserved: $\langle \mathbf{F}\boldsymbol{x}, \mathbf{F}\boldsymbol{y}\rangle = \langle \boldsymbol{x}, \boldsymbol{y}\rangle$.

Ex 5.25 Use $\mathbf{F}^*\mathbf{F} = \mathbf{I}$ to prove preservation of inner products, then set the two vectors equal to obtain Parseval's identity. What factor appears if the forward transform is left unnormalized?

Def 5.10 **(Frequency Bins and Nyquist Limit)** For a signal sampled on n points with uniform spacing $h = 2\pi/n$, integer frequency index k corresponds to the continuous Fourier mode e^{ikx}. The distinct resolvable frequencies are ordered as

$$0, 1, 2, \ldots, \lfloor n/2 \rfloor, -\lfloor (n-1)/2 \rfloor, \ldots, -1.$$

The maximum resolvable frequency is the Nyquist frequency $k_{\text{Nyq}} = \lfloor n/2 \rfloor$. Any continuous mode with frequency $|k| > k_{\text{Nyq}}$ aliases to a lower grid frequency satisfying $k' \equiv k \pmod{n}$.

Ex 5.26 On a grid of four points, write the sampled vectors for frequencies 0, 1, 2, and 5. Show that frequencies 1 and 5 give identical vectors. More generally, prove that adding n to an integer frequency leaves its samples unchanged. Why can samples alone not distinguish these frequencies?

Remark 5.7 **(Normalization convention)** Numerical libraries such as NumPy's `np.fft.fft` define the forward transform without the factor $1/\sqrt{n}$, so that $\hat{\boldsymbol{x}}_{\text{numpy}} = \sum_{j=0}^{n-1} x_j \omega_n^{jk}$, and place the full factor $1/n$ in the inverse transform `np.fft.ifft`. This choice scales the entries by $\sqrt{n}$ relative to the unitary operator $\mathbf{F}$ defined here.

5.4.2 Circulant Matrices and the Convolution Theorem

A linear system or filtering operation that commutes with spatial translation is represented by a circulant matrix. The Discrete Fourier Transform diagonalizes every circulant matrix.

Def 5.11 **(Circulant Matrix)** Let $\boldsymbol{c} = (c_0, c_1, \ldots, c_{n-1})^T \in \mathbb{C}^n$. The circulant matrix $\mathbf{C} = \operatorname{circ}(\boldsymbol{c}) \in \mathbb{C}^{n\times n}$ has entries $C_{jk} = c_{(j-k) \pmod n}$:

$$\mathbf{C} = \begin{pmatrix} c_0 & c_{n-1} & c_{n-2} & \cdots & c_1 \\ c_1 & c_0 & c_{n-1} & \cdots & c_2 \\ c_2 & c_1 & c_0 & \cdots & c_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_{n-1} & c_{n-2} & c_{n-3} & \cdots & c_0 \end{pmatrix}.$$

Thm 5.21 **(Spectral Diagonalization of Circulant Matrices)** Every circulant matrix $\mathbf{C} = \operatorname{circ}(\boldsymbol{c})$ is diagonalized by the unitary DFT matrix:

$$\mathbf{C} = \mathbf{F}^* \mathbf{\Lambda} \mathbf{F},$$

where the diagonal eigenvalue matrix $\mathbf{\Lambda} = \operatorname{diag}(\lambda_0, \ldots, \lambda_{n-1})$ has entries equal to the unnormalized DFT coefficients of the first column:

$$\lambda_k = \sqrt{n}(\mathbf{F}\boldsymbol{c})_k = \sum_{j=0}^{n-1} c_j \omega_n^{jk}.$$

Proof Let $\mathbf{S} \in \mathbb{R}^{n\times n}$ have first column $(0, 1, 0, \ldots, 0)^T$ and be circulant (for $n \geq 2$) be the cyclic permutation matrix that shifts coordinate indices forward by one: $\mathbf{S}\boldsymbol{x} = (x_{n-1}, x_0, x_1, \ldots, x_{n-2})^T$. Powers of $\mathbf{S}$ shift indices by j positions, satisfying $\mathbf{S}^n = \mathbf{I}$. Every circulant matrix is a polynomial in $\mathbf{S}$:

$$\mathbf{C} = \sum_{j=0}^{n-1} c_j \mathbf{S}^j.$$

The first column of $\mathbf{S}^j$ is the coordinate vector with its one in cyclic position j. Thus the first column of $\sum_j c_j \mathbf{S}^j$ is $\boldsymbol{c}$. Both sides are circulant and a circulant is determined by its first column, proving the polynomial representation.

Let $\boldsymbol{v}_k$ denote the k-th column of $\mathbf{F}^*$, with entries $(\boldsymbol{v}_k)_j = \frac{1}{\sqrt{n}} \omega_n^{-jk}$. Multiplying by the shift matrix gives

$$(\mathbf{S}\boldsymbol{v}_k)_j = (\boldsymbol{v}_k)_{(j-1) \pmod n} = \frac{1}{\sqrt{n}} \omega_n^{-(j-1)k} = \omega_n^k \left(\frac{1}{\sqrt{n}} \omega_n^{-jk} \right) = \omega_n^k (\boldsymbol{v}_k)_j.$$

Thus $\boldsymbol{v}_k$ is an eigenvector of $\mathbf{S}$ with eigenvalue ω_n^k. Applying the polynomial representation of $\mathbf{C}$ to $\boldsymbol{v}_k$ yields

$$\mathbf{C}\boldsymbol{v}_k = \sum_{j=0}^{n-1} c_j \mathbf{S}^j \boldsymbol{v}_k = \sum_{j=0}^{n-1} c_j (\omega_n^k)^j \boldsymbol{v}_k = \left(\sum_{j=0}^{n-1} c_j \omega_n^{jk} \right) \boldsymbol{v}_k = \lambda_k \boldsymbol{v}_k.$$

Collecting all n eigenvector equations in matrix form gives $\mathbf{C}\mathbf{F}^* = \mathbf{F}^* \mathbf{\Lambda}$. Multiplying on the right by $\mathbf{F}$ yields $\mathbf{C} = \mathbf{F}^* \mathbf{\Lambda} \mathbf{F}$.

The case $n = 1$ is the scalar identity and does not require a shift matrix. For larger n, the factor $\sqrt{n}$ in the eigenvalue formula appears because the chosen DFT is unitary, while the polynomial eigenvalue sum is unnormalized. It must not be dropped when using this normalization. □

Ex 5.27 **Circulant spectral inversion for the periodic Poisson problem.** Consider the discrete periodic Poisson equation $-\frac{u_{j-1}-2u_j+u_{j+1}}{h^2} = f_j$ on n points with periodic boundary conditions.

1. Express the coefficient matrix as a circulant matrix $\mathbf{C} = \operatorname{circ}(\boldsymbol{c})$ and identify the entries of $\boldsymbol{c}$.
2. Compute the eigenvalues λ_k of $\mathbf{C}$ using Thm 5.21. Show that $\lambda_0 = 0$ and $\lambda_k > 0$ for $k = 1, \dots, n-1$.
3. Explain why the system is solvable if and only if $\sum_{j=0}^{n-1} f_j = 0$, and formulate an $O(n \log n)$ solver using the FFT.

Thm 5.22 **(Discrete Convolution Theorem)** Let $\boldsymbol{c}, \boldsymbol{x} \in \mathbb{C}^n$. The circular convolution $\boldsymbol{z} = \boldsymbol{c} * \boldsymbol{x}$ has components

$$z_j = \sum_{k=0}^{n-1} c_k x_{(j-k) \pmod n} = (\mathbf{C}\boldsymbol{x})_j, \qquad \mathbf{C} = \operatorname{circ}(\boldsymbol{c}).$$

Applying the Discrete Fourier Transform converts convolution to entrywise multiplication:

$$\mathbf{F}(\boldsymbol{c} * \boldsymbol{x}) = \sqrt{n}(\mathbf{F}\boldsymbol{c}) \odot (\mathbf{F}\boldsymbol{x}),$$

where $\odot$ denotes the Hadamard (entrywise) vector product.

Ex 5.28 Write the convolution as $\mathbf{C}\boldsymbol{x}$ and substitute $\mathbf{C} = \mathbf{F}^*\mathbf{\Lambda}\mathbf{F}$. Apply $\mathbf{F}$ and use the formula for λ_k. Account explicitly for the factor $\sqrt{n}$.

5.4.3 The Fast Fourier Transform

Computing a matrix-vector product with a general dense $n \times n$ matrix requires $2n^2$ flops. The Cooley–Tukey Fast Fourier Transform (FFT) factors the DFT matrix into sparse structured factors, reducing the complexity to $O(n \log n)$ operations.

Thm 5.23 **(Radix-2 Cooley–Tukey Decomposition)** Let n be an even integer, and let $\boldsymbol{x} \in \mathbb{C}^n$. Partition $\boldsymbol{x}$ into even-indexed and odd-indexed subvectors $\boldsymbol{x}_E, \boldsymbol{x}_O \in \mathbb{C}^{n/2}$ defined by $(\boldsymbol{x}_E)_j = x_{2j}$ and $(\boldsymbol{x}_O)_j = x_{2j+1}$ for $j = 0, \dots, n/2-1$. The n-point unnormalized DFT $\hat{x}_k = \sum_{j=0}^{n-1} x_j \omega_n^{jk}$ satisfies the butterfly identity

$$\begin{aligned} \hat{x}_k &= (\hat{\boldsymbol{x}}_E)_k + \omega_n^k (\hat{\boldsymbol{x}}_O)_k, \\ \hat{x}_{k+n/2} &= (\hat{\boldsymbol{x}}_E)_k - \omega_n^k (\hat{\boldsymbol{x}}_O)_k, \end{aligned}$$

for each $k = 0, 1, \dots, n/2-1$, where $\hat{\boldsymbol{x}}_E, \hat{\boldsymbol{x}}_O$ are the $(n/2)$-point DFTs of $\boldsymbol{x}_E$ and $\boldsymbol{x}_O$. When $n = 2^p$, applying this reduction recursively requires $O(n \log n)$ floating-point operations.

Proof Splitting the DFT sum into even and odd indices gives

$$\hat{x}_k = \sum_{m=0}^{n/2-1} x_{2m} \omega_n^{2mk} + \sum_{m=0}^{n/2-1} x_{2m+1} \omega_n^{(2m+1)k}.$$

Because $\omega_n^2 = e^{-4\pi i/n} = e^{-2\pi i/(n/2)} = \omega_{n/2}$, the first sum is the $(n/2)$-point transform of $\boldsymbol{x}_E$:

$$\sum_{m=0}^{n/2-1} x_{2m} \omega_{n/2}^{mk} = (\hat{\boldsymbol{x}}_E)_k.$$

Factoring ω_n^k out of the second sum gives $\omega_n^k \sum_{m=0}^{n/2-1} x_{2m+1}\omega_{n/2}^{mk} = \omega_n^k(\hat{\boldsymbol{x}}_O)_k$. This establishes the first identity:

$$\hat{x}_k = (\hat{\boldsymbol{x}}_E)_k + \omega_n^k(\hat{\boldsymbol{x}}_O)_k.$$

For the second half of the spectrum $k + n/2$, the periodicity $\omega_{n/2}^{m(k+n/2)} = \omega_{n/2}^{mk}(\omega_{n/2}^{n/2})^m = \omega_{n/2}^{mk}$ and the half-period phase shift

$$\omega_n^{k+n/2} = \omega_n^k \omega_n^{n/2} = \omega_n^k e^{-\pi i} = -\omega_n^k$$

give the complementary butterfly equation:

$$\hat{x}_{k+n/2} = (\hat{\boldsymbol{x}}_E)_k - \omega_n^k(\hat{\boldsymbol{x}}_O)_k.$$

The second half of the output reuses the same two smaller transforms. Thus the reduction needs two transforms of length $n/2$, rather than four. The minus sign distinguishes the second half without requiring additional recursive calls.

In matrix form, introducing the diagonal twiddle matrix $\boldsymbol{\Omega}_{n/2} = \operatorname{diag}(1, \omega_n, \ldots, \omega_n^{n/2-1})$ and the even–odd permutation matrix $\mathbf{P}_n$:

$$\hat{\boldsymbol{x}} = \begin{pmatrix} \mathbf{I}_{n/2} & \boldsymbol{\Omega}_{n/2} \\ \mathbf{I}_{n/2} & -\boldsymbol{\Omega}_{n/2} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{x}}_E \\ \hat{\boldsymbol{x}}_O \end{pmatrix} = \begin{pmatrix} \mathbf{I}_{n/2} & \boldsymbol{\Omega}_{n/2} \\ \mathbf{I}_{n/2} & -\boldsymbol{\Omega}_{n/2} \end{pmatrix} \begin{pmatrix} \mathbf{W}_{n/2} & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_{n/2} \end{pmatrix} \mathbf{P}_n \boldsymbol{x}.$$

At each recursion stage, the butterfly recombination performs $n/2$ complex multiplications and n additions. The arithmetic complexity satisfies the recurrence

$$T(n) = 2T(n/2) + cn.$$

For $n = 2^p$, the depth of the recursion tree is $\log_2 n$, yielding total cost $T(n) = cn \log_2 n = O(n \log n)$.

To solve the work recurrence explicitly, divide it by n and put $n = 2^p$. Then $T(2^p)/2^p = T(2^{p-1})/2^{p-1} + c$. Repeating this equality p times gives $T(n)/n = T(1) + cp$. Since $p = \log_2 n$, the work is $nT(1) + cn \log_2 n$. This supplies the stated order including the base-case contribution. □

5.4.4 Spectral Differentiation

For smooth periodic functions, differentiation in the Fourier domain replaces finite-difference stencils with exact multiplication in frequency.

Thm 5.24 **(Spectral Differentiation of Periodic Functions)** Let $u : \mathbb{R} \to \mathbb{C}$ be a continuously differentiable, 2π-periodic function. Representing u by its Fourier series $u(x) = \sum_{k=-\infty}^{\infty} \hat{u}_k e^{ikx}$, the derivative has Fourier coefficients

$$\widehat{(u')}_k = ik\hat{u}_k.$$

If u is analytic and bounded in a strip $\{z \in \mathbb{C} : |\operatorname{Im}(z)| < d\}$, the truncation error of the discrete spectral derivative converges geometrically to zero:

$$\max_{x_j} |u'(x_j) - (D_n \boldsymbol{u})_j| = O(ne^{-dn/2}).$$

Proof For the coefficient of the derivative, integrate by parts over one period:

$$\begin{aligned}\widehat{(u')}_k &= \frac{1}{2\pi}\int_0^{2\pi} u'(x)e^{-ikx}\,dx \\ &= \frac{1}{2\pi}[u(x)e^{-ikx}]_0^{2\pi} + \frac{ik}{2\pi}\int_0^{2\pi} u(x)e^{-ikx}\,dx = ik\hat{u}_k.\end{aligned}$$

The boundary term is zero because $u(2\pi) = u(0)$ and $e^{-2\pi ik} = 1$ for integer k. This argument requires only the stated continuous differentiability, rather than an unjustified differentiation of a conditionally convergent series.

For the error bound, assume the additional analyticity and boundedness in the strip, with bound M. For $k > 0$, shift the coefficient integral to the line with imaginary part $-d'$, where $0 < d' < d$. Periodicity cancels the two vertical sides of the contour. On the shifted line, $|e^{-ik(x-id')}| = e^{-kd'}$, so $|\hat{u}_k| \le Me^{-kd'}$. For $k < 0$, shift upwards instead. Letting $d' \uparrow d$ gives $|\hat{u}_k| \le Me^{-d|k|}$.

Differentiation multiplies coefficient magnitude by $|k|$. If K is of order $n/2$, the differentiated tail is bounded by

$$2M\sum_{k=K}^{\infty} ke^{-dk}.$$

Set $q = e^{-d} < 1$ and $k = K + j$. The sum is

$$\sum_{k=K}^{\infty} kq^k = q^K\left(K\sum_{j=0}^{\infty} q^j + \sum_{j=0}^{\infty} jq^j\right) = q^K\left(\frac{K}{1-q} + \frac{q}{(1-q)^2}\right).$$

It is therefore $O(Ke^{-dK})$.

Sampling replaces a high mode by its aliased grid frequency. The difference between the true derivative multiplier and the aliased multiplier is at most $|k| + K$, which is at most a constant times $|k|$ for an omitted mode. Summing their coefficient bounds therefore gives the same order for aliasing as for the tail. Taking K of order $n/2$ yields $O(ne^{-dn/2})$.

For even n and real data, the Nyquist samples are those of a cosine whose derivative vanishes on the grid. Setting that first-derivative component to zero gives the derivative of the real trigonometric interpolant, as stipulated in the theorem. □

5.4.5 Further Exercises

Ex 5.29 **Fast linear convolution via zero padding.** Let $\boldsymbol{x} \in \mathbb{R}^p$ and $\boldsymbol{y} \in \mathbb{R}^q$ be two discrete sequences. Their linear convolution has length $m = p + q - 1$.

1. Explain how zero-padding $\boldsymbol{x}$ and $\boldsymbol{y}$ to length $N \ge p + q - 1$ prevents periodic wraparound in circular convolution.
2. Compare the arithmetic operation count of direct linear convolution ($2pq$ flops) against FFT-based circular convolution ($O(N \log N)$ flops).
3. For $p = q = 10^5$, estimate the speedup obtained by the FFT algorithm.

Ex 5.30 **Cooley–Tukey butterfly and bit-reversal permutation.** Consider the radix-2 FFT algorithm with $n = 8$.

1. Write the indices $0, 1, \ldots, 7$ in 3-bit binary notation $(b_2 b_1 b_0)$.
2. Reverse the bits $(b_0 b_1 b_2)$ to construct the bit-reversal permutation vector $\mathbf{p}$.
3. Show that sorting the input by $\mathbf{p}$ allows the recursive Cooley–Tukey butterflies to be evaluated in place without allocating auxiliary arrays.

Ex 5.31 **Spectral differentiation and the Runge phenomenon.** Implement the Fourier spectral differentiation matrix for $n = 16, 32, 64$ points on $[0, 2\pi)$.

1. Apply the spectral derivative to $u(x) = \sin(x)$ and measure the maximum pointwise error against $\cos(x)$. Confirm that error reaches machine precision for small n.
2. Apply the spectral derivative to $v(x) = \frac{1}{2+\cos(x)}$. Estimate the decay constant d from Thm 5.24 and confirm geometric convergence.
3. Apply the method to $w(x) = x$ on $[0, 2\pi)$. Identify the jump in its periodic extension. Plot the derivative error away from and near that jump, and explain why the uniform geometric-convergence theorem no longer applies.

Read **Further reading.** The Cooley–Tukey paper is short and still worth reading as an example of how reorganizing a calculation changes its cost (Cooley and Tukey, 1965). Trefethen's approximation text gives a gentle route from polynomial approximation to spectral accuracy (Trefethen, 2013).

6 Sparse Systems and Iterative Methods

A discretized differential equation can have millions of unknowns but only a few nonzero matrix entries per row. We construct such a system from the Poisson equation and solve it by successive corrections. The methods differ in how they choose a correction and in which parts of the error they reduce.

6.1 Linear Operators and the Poisson Equation

To approximate a differential equation numerically, we replace its unknown function by values on a grid. A local difference formula then relates each value to its neighbors and produces a sparse linear system. We derive this system for $-u'' = f$ with prescribed endpoint values, then relate its eigenvalues to those of the differential operator.

The boundary conditions affect both solvability and the spectrum. Keeping them in the derivation will explain why this matrix is positive definite and why its condition number grows when the grid is refined.

6.1.1 Differential Operators and Discretization

Def 6.1 **(Linear Operator)** A map $\mathcal{A} : X \to Y$ between vector spaces is linear if $\mathcal{A}(\alpha u + \beta v) = \alpha \mathcal{A}u + \beta \mathcal{A}v$ for all $u, v \in X$ and scalars α, β. Here the vectors may be functions. Differentiation is a linear map from continuously differentiable functions to continuous functions. A matrix is the corresponding construction for finite lists of coordinates.

Def 6.2 **(Dirichlet Poisson Problem)** The one-dimensional Poisson problem with homogeneous Dirichlet boundary data is

$$-u''(x) = f(x), \qquad 0 < x < 1, \qquad u(0) = u(1) = 0.$$

The boundary conditions are part of the problem data.

Def 6.3 **(Grid Function)** For n interior grid points, let $h = 1/(n+1)$ and $x_i = ih$, $i = 1, \dots, n$. A grid function is the vector

$$\boldsymbol{u} = (u_1, \dots, u_n)^T, \qquad u_i \approx u(x_i).$$

The boundary values are fixed separately by the boundary conditions.

Def 6.4 **(Finite Difference Stencil)** Approximating $u''(x)$ on a grid with spacing h:

$$u''(x_k) \approx \frac{u(x_{k-1}) - 2u(x_k) + u(x_{k+1})}{h^2}.$$

For $u \in C^4$, Taylor expansion gives a truncation error bounded by $h^2 \|u^{(4)}\|_\infty / 12$.

Ex 6.1

1. Taylor expand $u(x+h)$ and $u(x-h)$ about x.
2. Add the two expansions.
3. Solve for $u''(x)$.
4. Identify the leading error term and prove the $O(h^2)$ claim in Def 6.4.

At an interior node, the difference formula gives $-u_{i-1} + 2u_i - u_{i+1} = h^2 f(x_i)$. The endpoint values $u_0 = u_{n+1} = 0$ are known, so only $u_1, \dots, u_n$ are unknowns. Writing one such equation per interior node gives the matrix below.

Ex 6.2 Write the three equations when there are three interior nodes. Next prescribe $u(0) = \alpha$ and $u(1) = \beta$. Show that the coefficient matrix stays the same and determine the changes in the first and last entries of the right-hand side.

Def 6.5 **(1-D Discrete Laplacian)** Discretizing $-d^2/dx^2$ on $[0,1]$ with $u(0) = u(1) = 0$ yields the tridiagonal system $\mathbf{Tu} = \mathbf{f}$:

$$\mathbf{T} = \frac{1}{h^2}\text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n \times n}.$$

Remark 6.1 **(Sparsity)** Each row of $\mathbf{T}$ uses only the grid point and its two neighbors. This locality is why finite difference matrices are sparse.

Thm 6.1 **(Spectrum Convergence)** Eigenvalues of $\mathbf{T}$ are $\lambda_k = \frac{4}{h^2}\sin^2(\frac{k\pi}{2(n+1)})$. As $n \to \infty$, $\lambda_k \to k^2\pi^2$ (eigenvalues of continuous $-d^2/dx^2$).

Ex 6.3

1. Let $(\boldsymbol{v}_k)_j = \sin(jk\pi h)$ with $h = 1/(n+1)$.
2. Compute the j-th entry of $\mathbf{T}\boldsymbol{v}_k$.
3. Use $\sin(A-B) + \sin(A+B) = 2\sin A \cos B$.
4. Use $1 - \cos\theta = 2\sin^2(\theta/2)$.
5. Derive the eigenvalue formula in Thm 6.1.
6. Let $h \to 0$ with fixed k and recover $k^2\pi^2$.

Thm 6.2 **(Discrete Poisson Structure)** The matrix $\mathbf{T}$ in Def 6.5 is sparse, symmetric positive definite, and has condition number

$$\kappa_2(\mathbf{T}) = O(h^{-2}).$$

Proof Sparsity and symmetry follow directly from the tridiagonal stencil $\mathbf{T} = \frac{1}{h^2}\,\mathrm{tridiag}(-1, 2, -1)$. For any grid vector $\boldsymbol{u} \in \mathbb{R}^n$ with boundary conditions $u_0 = u_{n+1} = 0$, the quadratic form expands as a sum of squared forward differences:

$$\begin{aligned}
\boldsymbol{u}^T\mathbf{T}\boldsymbol{u} &= \frac{1}{h^2}\sum_{i=1}^{n} u_i(2u_i - u_{i-1} - u_{i+1}) \\
&= \frac{1}{h^2}\sum_{i=1}^{n}(u_i^2 - u_iu_{i-1} + u_i^2 - u_iu_{i+1}) \\
&= \frac{1}{h^2}\sum_{i=0}^{n}(u_{i+1} - u_i)^2.
\end{aligned}$$

In the sum of squared differences, each interior u_i^2 occurs twice, and each neighbor product occurs with coefficient -2. The endpoint squares vanish because both boundary values are zero. This reproduces exactly the expanded quadratic form and justifies the rearrangement.

Because $u_0 = 0$, if $\boldsymbol{u}^T\mathbf{T}\boldsymbol{u} = 0$, then $u_{i+1} - u_i = 0$ for all i, forcing $u_1 = \cdots = u_n = 0$. Hence $\boldsymbol{u}^T\mathbf{T}\boldsymbol{u} > 0$ for all $\boldsymbol{u} \neq \mathbf{0}$, proving that $\mathbf{T}$ is SPD.

By Thm 6.1, the extremal eigenvalues are

$$\begin{aligned}
\lambda_{\min} = \lambda_1 &= \frac{4}{h^2}\sin^2\left(\frac{\pi h}{2}\right) = \frac{4}{h^2}\left(\frac{\pi h}{2} + O(h^3)\right)^2 = \pi^2 + O(h^2), \\
\lambda_{\max} = \lambda_n &= \frac{4}{h^2}\sin^2\left(\frac{n\pi h}{2}\right) = \frac{4}{h^2}\cos^2\left(\frac{\pi h}{2}\right) = \frac{4}{h^2}(1 - O(h^2)).
\end{aligned}$$

The largest-mode identity uses $n\pi h = \pi - \pi h$, so $\sin(n\pi h/2) = \cos(\pi h/2)$. The smallest mode instead has angle tending to zero, where $\sin z = z + O(z^3)$. These two different limits explain why one eigenvalue stays of constant size while the other grows like h^{-2}.

The 2-norm condition number is therefore

$$\kappa_2(\mathbf{T}) = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{4/h^2 + O(1)}{\pi^2 + O(h^2)} = \frac{4}{\pi^2 h^2} + O(1) = O(h^{-2}).$$

□

Ex 6.4 Count the nonzero entries of $\mathbf{T}$. Then replace the Dirichlet boundary conditions by periodic conditions and form the analogous matrix for $n \geq 3$. Find a nonzero vector in its null space. Which step in the positive-definiteness proof fails, and what compatibility condition must the right-hand side satisfy?

6.1.2 Operator Norms and Conditioning

Def 6.6 **(Boundedness)** An operator is **bounded** if $\|\mathcal{A}u\|_Y \leq C\|u\|_X$ for some $C < \infty$.

- Matrices are always bounded.
- Boundedness of a differential operator depends on the domain and the norms. For a function on $[0, 1]$, the L^2 norm is $\|u\|_{L^2} = (\int_0^1 |u(x)|^2\, dx)^{1/2}$. The functions $u_k(x) = \sin(2\pi kx)$ all have norm $1/\sqrt{2}$, whereas their derivatives have norm $2\pi k/\sqrt{2}$. Thus no fixed constant C can bound $\|u_k'\|_{L^2}$ by $C\|u_k\|_{L^2}$ for every k.

  If instead we measure the input by $\|u\|_{H^1} = (\|u\|_{L^2}^2 + \|u'\|_{L^2}^2)^{1/2}$, then $\|u'\|_{L^2} \leq \|u\|_{H^1}$ directly. The space H^1 consists of square-integrable functions whose weak first derivative is also square-integrable. For the smooth functions considered here, that derivative is the ordinary derivative. Including it in the input norm makes differentiation bounded.

Remark 6.2 The condition-number estimate uses two separate facts: the smallest eigenvalue approaches the positive constant π^2, whereas the largest grows like $4h^{-2}$. Thus halving h approximately quadruples the condition number. This is consistent with the unboundedness of the differential operator in the L^2 norm, but unboundedness alone is not a proof of the estimate for a discretization.

Remark 6.3 **(Solver consequence)** Dense Gaussian elimination ignores sparsity. For Poisson matrices, sparse Cholesky, CG with preconditioning, multigrid, or FFT-based solvers are the natural algorithms, depending on geometry and boundary conditions.

6.1.3 Further Exercises

Ex 6.5 Build $\mathbf{T}$ for $n = 50$ and plot its eigenvalues vs. $k^2\pi^2$ using Thm 6.1.

Ex 6.6 Show that $\kappa(\mathbf{T}) \to \infty$ as $n \to \infty$ using Thm 6.2.

Ex 6.7 Assemble the two-dimensional Poisson matrix on an $m \times m$ interior grid, with $N = m^2$ unknowns. Compare sparse direct solves under natural and fill-reducing orderings. Relate the factorization times and factor nonzero counts to the estimates in the sparse Cholesky section; explain why timings alone cannot prove an asymptotic bound.

Ex 6.8 Prove the differentiation operator is unbounded on $L^2[0, 1]$ using $u_k(x) = \sin(k\pi x)$.

Read **Further reading.** LeVeque develops finite differences with close attention to consistency, stability, and boundary conditions (LeVeque, 2007). Strang connects the same model problems to broader ideas in applied mathematics and computation (Strang, 2007).

6.2 Iterative Methods for Linear Systems

A sparse matrix can have a factorization that is too expensive to store. Iterative methods instead improve an approximate solution through matrix–vector products and simpler solves. Their cost

depends both on the cost of one correction and on how many corrections are needed.

For an approximation $\boldsymbol{x}$, compute the residual $\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}$. The exact correction satisfies $\mathbf{A}\boldsymbol{\delta} = \boldsymbol{r}$. Solving that equation exactly would be the original task again. We instead solve $\mathbf{M}\boldsymbol{\delta} = \boldsymbol{r}$ with a matrix $\mathbf{M}$ that is cheap to use, and add the correction to $\boldsymbol{x}$.

6.2.1 Matrix Splittings and Stationary Iterations

The correction update is $\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \mathbf{M}^{-1}(\boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(k)})$. The inverse here denotes a solve with $\mathbf{M}$, rather than a matrix to form. Rearranging this expression gives the traditional splitting $\mathbf{A} = \mathbf{M} - \mathbf{N}$ and the recurrence below. The same $\mathbf{M}$ is used at every step, which explains the word *stationary*.

Def 6.7 **(Stationary Iteration)** Given a splitting $\mathbf{A} = \mathbf{M} - \mathbf{N}$, the corresponding stationary iteration computes successive approximations via the fixed-point recurrence

$$\mathbf{M}\boldsymbol{x}^{(k+1)} = \mathbf{N}\boldsymbol{x}^{(k)} + \boldsymbol{b} \iff \boldsymbol{x}^{(k+1)} = \mathbf{T}\boldsymbol{x}^{(k)} + \boldsymbol{c},$$

where $\mathbf{T} = \mathbf{M}^{-1}\mathbf{N} = \mathbf{I} - \mathbf{M}^{-1}\mathbf{A}$ is the **iteration matrix** and $\boldsymbol{c} = \mathbf{M}^{-1}\boldsymbol{b}$ is the modified right-hand side vector.

Ex 6.9 Consider the matrix $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ and the splitting $\mathbf{M} = 4\mathbf{I}$. Compute the iteration matrix $\mathbf{T} = \mathbf{I} - \mathbf{M}^{-1}\mathbf{A}$, find its eigenvalues, and evaluate its spectral radius $\rho(\mathbf{T})$.

Remark 6.4 The exact solution is a fixed point because its residual is zero. That fact establishes consistency of the iteration, but does not imply convergence to the fixed point. The correction must reduce every possible error direction. The following spectral-radius criterion states exactly when this occurs.

6.2.2 Convergence Theory and the Spectral Radius

The convergence of a linear stationary iteration is determined entirely by the spectral properties of the iteration matrix $\mathbf{T}$.

Ex 6.10 For the scalar equation $ax = b$ with $a > 0$, use the correction $\delta = (b - ax)/m$ with $m > 0$. Derive the error relation $e^{(k+1)} = (1 - a/m)e^{(k)}$. Determine the values of a/m for which the errors decrease, alternate in sign, remain unchanged in magnitude, or grow. The matrix criterion extends this calculation to all eigenvector directions.

Def 6.8 **(Spectral Radius)** Let $\mathbf{T} \in \mathbb{R}^{n\times n}$ with spectrum $\sigma(\mathbf{T}) \subset \mathbb{C}$. The spectral radius of $\mathbf{T}$ is defined as the maximum modulus among all its eigenvalues:

$$\rho(\mathbf{T}) = \max\{|\lambda| : \lambda \in \sigma(\mathbf{T})\}.$$

Thm 6.3 **(Gelfand's Spectral Radius Theorem)** For any matrix norm $\|\cdot\|$ induced by a vector norm on $\mathbb{C}^n$ and any matrix $\mathbf{T} \in \mathbb{R}^{n\times n}$,

$$\rho(\mathbf{T}) = \lim_{k\to\infty} \|\mathbf{T}^k\|^{1/k}.$$

In particular, for any $\varepsilon > 0$, there exists an induced matrix norm $\|\cdot\|_*$ such that $\|\mathbf{T}\|_* \le \rho(\mathbf{T}) + \varepsilon$.

Ex 6.11 The following construction proves the norm assertion and the limit. Use Schur form $\mathbf{T} = \mathbf{Q}\mathbf{U}\mathbf{Q}^*$ and set $\mathbf{D} = \operatorname{diag}(1, \eta, \dots, \eta^{n-1})$. Show that every strictly upper triangular entry of $\mathbf{D}^{-1}\mathbf{U}\mathbf{D}$ tends to zero as $\eta \to 0$. Choose η so that its infinity norm is at most $\rho(\mathbf{T}) + \varepsilon$, and define $\|\boldsymbol{x}\|_* = \|\mathbf{D}^{-1}\mathbf{Q}^*\boldsymbol{x}\|_\infty$. Finally use equivalence of norms to obtain $\limsup_k \|\mathbf{T}^k\|^{1/k} \le \rho(\mathbf{T}) + \varepsilon$ in any fixed norm, and an eigenvector to obtain the reverse bound. Let $\varepsilon \downarrow 0$.

Thm 6.4 **(Fundamental Convergence Criterion for Stationary Iterations)** The stationary iteration $\boldsymbol{x}^{(k+1)} = \mathbf{T}\boldsymbol{x}^{(k)} + \boldsymbol{c}$ converges to the unique solution $\boldsymbol{x}^* = \mathbf{A}^{-1}\boldsymbol{b}$ for every initial guess $\boldsymbol{x}^{(0)} \in \mathbb{R}^n$ if and only if

$$\rho(\mathbf{T}) < 1.$$

Proof Let $\boldsymbol{e}^{(k)} = \boldsymbol{x}^{(k)} - \boldsymbol{x}^*$ denote the error vector at step k. Subtracting the fixed-point identity $\boldsymbol{x}^* = \mathbf{T}\boldsymbol{x}^* + \boldsymbol{c}$ from the iteration formula $\boldsymbol{x}^{(k+1)} = \mathbf{T}\boldsymbol{x}^{(k)} + \boldsymbol{c}$:

$$\begin{aligned} \boldsymbol{e}^{(k+1)} &= \boldsymbol{x}^{(k+1)} - \boldsymbol{x}^* \\ &= (\mathbf{T}\boldsymbol{x}^{(k)} + \boldsymbol{c}) - (\mathbf{T}\boldsymbol{x}^* + \boldsymbol{c}) \\ &= \mathbf{T}(\boldsymbol{x}^{(k)} - \boldsymbol{x}^*) = \mathbf{T}\boldsymbol{e}^{(k)}. \end{aligned}$$

Applying this recurrence recursively yields

$$\boldsymbol{e}^{(k)} = \mathbf{T}^k \boldsymbol{e}^{(0)}.$$

The recurrence shows why the error, rather than the iterate itself, is the convenient object: the fixed right-hand side has cancelled. In particular, testing all initial guesses is equivalent to testing all initial errors, since any error vector can be realized by choosing $\boldsymbol{x}^{(0)} = \boldsymbol{x}^* + \boldsymbol{e}^{(0)}$.

The error $\boldsymbol{e}^{(k)} \to \mathbf{0}$ for all initial errors $\boldsymbol{e}^{(0)} \in \mathbb{R}^n$ if and only if $\lim_{k\to\infty} \mathbf{T}^k = \mathbf{0}$.

If $\rho(\mathbf{T}) < 1$, choose $\varepsilon = \frac{1-\rho(\mathbf{T})}{2} > 0$. By Gelfand's theorem (Thm 6.3), there exists an induced matrix norm $\|\cdot\|_*$ such that

$$\|\mathbf{T}\|_* \le \rho(\mathbf{T}) + \varepsilon = \frac{1 + \rho(\mathbf{T})}{2} < 1.$$

Then $\|\mathbf{T}^k\|_* \le \|\mathbf{T}\|_*^k \to 0$ as $k \to \infty$, so $\mathbf{T}^k \to \mathbf{0}$ and $\|\boldsymbol{e}^{(k)}\|_* \le \|\mathbf{T}\|_*^k \|\boldsymbol{e}^{(0)}\|_* \to 0$.

Conversely, suppose $\mathbf{T}^k \to \mathbf{0}$. If $\lambda \in \sigma(\mathbf{T})$ with eigenvector $\boldsymbol{v} \neq \mathbf{0}$, then $\mathbf{T}^k\boldsymbol{v} = \lambda^k\boldsymbol{v}$. For this sequence to converge to $\mathbf{0}$, we must have $|\lambda|^k \to 0$, forcing $|\lambda| < 1$. If an eigenvector is complex while the matrix is real, write it as $\boldsymbol{v} = \boldsymbol{a} + i\boldsymbol{b}$. Convergence of $\mathbf{T}^k$ on every real vector makes both $\mathbf{T}^k\boldsymbol{a}$ and $\mathbf{T}^k\boldsymbol{b}$ tend to zero, so convergence also holds for their complex combination. Thus the eigenvector test covers complex eigenvalues as well.

Because this holds for all eigenvalues, $\rho(\mathbf{T}) < 1$. □

Remark 6.5 **(Asymptotic convergence rate and iteration count)** The quantity $-\log_{10} \rho(\mathbf{T})$ represents the asymptotic rate of convergence, measuring the number of decimal digits of accuracy gained

per iteration. To reduce the initial error by a relative tolerance factor ε, the required number of iterations satisfies

$$k \approx \frac{\ln(1/\varepsilon)}{-\ln \rho(\mathbf{T})}.$$

If $\rho(\mathbf{T}) = 0.99$, we have $-\ln(0.99) \approx 0.01005$, which requires approximately 230 iterations to reduce the error by a factor of ten. As $\rho(\mathbf{T}) \to 1$, convergence slows drastically.

6.2.3 Krylov Subspaces and Matrix-Free Solvers

A stationary method applies the same splitting at every step. A Krylov method instead chooses the iterate $\boldsymbol{x}^{(k)}$ from the affine space $\boldsymbol{x}^{(0)} + \mathcal{K}_k(\mathbf{A}, \boldsymbol{r}^{(0)})$, where $\mathcal{K}_k$ is the Krylov subspace of Def 4.12 generated by the initial residual. The basis for this space is built by the Arnoldi process of Section 4.4, or by the Lanczos recurrence when $\mathbf{A}$ is symmetric. The conjugate gradient and GMRES methods differ in the optimality criterion used to select $\boldsymbol{x}^{(k)}$.

Thm 6.5 **(Krylov Polynomial Representation)** Let $\boldsymbol{x}^{(0)}$ be an initial guess with residual $\boldsymbol{r}^{(0)} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(0)}$. Any iterate $\boldsymbol{x}^{(k)}$ belonging to the affine Krylov space $\boldsymbol{x}^{(0)} + \mathcal{K}_k(\mathbf{A}, \boldsymbol{r}^{(0)})$ can be represented as

$$\boldsymbol{x}^{(k)} = \boldsymbol{x}^{(0)} + p_{k-1}(\mathbf{A})\boldsymbol{r}^{(0)},$$

where p_{k-1} is a polynomial of degree at most $k-1$. The corresponding residual vector satisfies

$$\boldsymbol{r}^{(k)} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(k)} = q_k(\mathbf{A})\boldsymbol{r}^{(0)},$$

where $q_k(t) = 1 - t p_{k-1}(t)$ is a polynomial of degree at most k satisfying the normalization constraint $q_k(0) = 1$.

Ex 6.12 Write a general correction as $\sum_{j=0}^{k-1} c_j \mathbf{A}^j \boldsymbol{r}^{(0)}$ and subtract its image from $\boldsymbol{r}^{(0)}$. Prove the polynomial representation. Conversely, if $q_k(0) = 1$, show that $(1 - q_k(t))/t$ is a polynomial of degree at most $k-1$.

Remark 6.6 **(Matrix-free computation)** Krylov subspace methods never require explicit access to the individual entries of $\mathbf{A}$. They interact with the linear operator exclusively by computing matrix–vector products $\boldsymbol{v} \mapsto \mathbf{A}\boldsymbol{v}$. Consequently, linear systems can be solved efficiently even when the matrix is defined implicitly as a differential operator or a numerical simulation routine.

Def 6.9 **(Stopping Criteria for Iterative Solvers)** Because the true error $\boldsymbol{e}^{(k)} = \boldsymbol{x}^{(k)} - \boldsymbol{x}^{*}$ is unobservable, termination is determined using the computable residual vector $\boldsymbol{r}^{(k)} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(k)}$. Common stopping criteria include:

1. **Relative residual test:** $\dfrac{\|\boldsymbol{r}^{(k)}\|_2}{\|\boldsymbol{b}\|_2} \le \texttt{tol}$.
2. **Preconditioned relative residual test:** $\dfrac{\|\mathbf{M}^{-1}\boldsymbol{r}^{(k)}\|_2}{\|\mathbf{M}^{-1}\boldsymbol{b}\|_2} \le \texttt{tol}$.
3. **Step difference test:** $\dfrac{\|\boldsymbol{x}^{(k+1)} - \boldsymbol{x}^{(k)}\|_2}{\|\boldsymbol{x}^{(k+1)}\|_2} \le \texttt{tol}$.

Ex 6.13 Prove that if $\|\mathbf{T}\| < 1$ for some induced matrix norm, then the error satisfies $\|\boldsymbol{e}^{(k)}\| \le \frac{\|\mathbf{T}\|}{1-\|\mathbf{T}\|}\|\boldsymbol{x}^{(k)} - \boldsymbol{x}^{(k-1)}\|$.

Relative tests require a nonzero denominator; for $\boldsymbol{b} = \mathbf{0}$, use an absolute residual tolerance. A small step alone can also signal stagnation. Recompute the true residual before accepting convergence, and use the residual–error bound to interpret its size. The normwise backward-error denominator $\|\mathbf{A}\|\|\boldsymbol{x}^{(k)}\| + \|\boldsymbol{b}\|$ often gives a more useful scale than $\|\boldsymbol{b}\|$ alone.

6.2.4 Further Exercises

Ex 6.14 For $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ and $\mathbf{M} = 4\mathbf{I}$, determine how many iterations guarantee $\|\boldsymbol{e}^{(k)}\|_\infty \le 10^{-6}\|\boldsymbol{e}^{(0)}\|_\infty$. Compare a bound from $\|\mathbf{T}\|_\infty$ with the asymptotic estimate from $\rho(\mathbf{T})$. Why are these different kinds of estimates?

Ex 6.15 Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ and let $\boldsymbol{r}^{(0)} \in \mathbb{R}^n$. Lem 4.24 shows that $\dim \mathcal{K}_k(\mathbf{A}, \boldsymbol{r}^{(0)})$ stops growing once the Krylov subspace is invariant. Prove the complementary statement: before that step, the dimension increases by exactly one at each step.

6.3 Jacobi, Gauss-Seidel, and SOR Methods

Each row of a linear system can be solved for its diagonal unknown when that diagonal entry is nonzero. Jacobi uses the old values of every other unknown. Gauss–Seidel uses new values as soon as they are available. Relaxation changes how far each such update is taken. We derive these choices and then determine sufficient conditions for convergence.

For the matrix formulas we use $\mathbf{A} = \mathbf{D} - \mathbf{L}_A - \mathbf{U}_A$, where $\mathbf{D}$ is diagonal and $-\mathbf{L}_A$, $-\mathbf{U}_A$ are the strictly lower and upper parts of $\mathbf{A}$. The minus signs are a convention. For example, if $a_{21} = 1$, then $(L_A)_{21} = -1$.

6.3.1 The Jacobi Method

Jacobi uses the diagonal as an easily solved approximation to the matrix. Strict diagonal dominance is a sufficient convergence condition, but is not part of the definition. It computes the $(k+1)$-th estimate of each unknown x_i exclusively from the k-th estimates of all other variables.

Def 6.10 **(Jacobi Iteration)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ with nonzero diagonal entries $a_{ii} \neq 0$. The Jacobi method corresponds to the splitting $\mathbf{M} = \mathbf{D}$ and $\mathbf{N} = \mathbf{L}_A + \mathbf{U}_A$. In component form, the update is

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right), \qquad i = 1, 2, \ldots, n.$$

In matrix notation, the iteration is given by

$$\boldsymbol{x}^{(k+1)} = \mathbf{T}_J \boldsymbol{x}^{(k)} + \boldsymbol{c}_J,$$

where $\mathbf{T}_J = \mathbf{D}^{-1}(\mathbf{L}_A + \mathbf{U}_A) = \mathbf{I} - \mathbf{D}^{-1}\mathbf{A}$ is the Jacobi iteration matrix and $\boldsymbol{c}_J = \mathbf{D}^{-1}\boldsymbol{b}$.

Remark 6.7 **(Parallel execution and memory)** Because each component $x_i^{(k+1)}$ depends only on values from the previous iteration vector $\boldsymbol{x}^{(k)}$, all n components can be computed concurrently. This fine-grained parallelism makes Jacobi attractive on vector and GPU hardware. However, it requires double buffering in memory to store $\boldsymbol{x}^{(k)}$ and $\boldsymbol{x}^{(k+1)}$ simultaneously.

6.3.2 The Gauss–Seidel Method

The Gauss–Seidel method uses updated variable values as soon as they become available within a sweep. This changes the iteration matrix, so its convergence must be examined separately.

Def 6.11 **(Gauss–Seidel Iteration)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ with nonzero diagonal entries. The Gauss–Seidel method corresponds to the triangular splitting $\mathbf{M} = \mathbf{D} - \mathbf{L}_A$ and $\mathbf{N} = \mathbf{U}_A$. In component form, the update is

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right), \qquad i = 1, 2, \ldots, n.$$

In matrix notation, the iteration is given by

$$(\mathbf{D} - \mathbf{L}_A)\boldsymbol{x}^{(k+1)} = \mathbf{U}_A \boldsymbol{x}^{(k)} + \boldsymbol{b} \iff \boldsymbol{x}^{(k+1)} = \mathbf{T}_{GS}\boldsymbol{x}^{(k)} + \boldsymbol{c}_{GS},$$

where $\mathbf{T}_{GS} = (\mathbf{D} - \mathbf{L}_A)^{-1}\mathbf{U}_A$ is the Gauss–Seidel iteration matrix and $\boldsymbol{c}_{GS} = (\mathbf{D} - \mathbf{L}_A)^{-1}\boldsymbol{b}$.

Ex 6.16 Compute the Jacobi and Gauss-Seidel iteration matrices for $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Find their spectral radii and verify that $\rho(\mathbf{T}_{GS}) = \rho(\mathbf{T}_J)^2$.

Remark 6.8 **(Sequential dependencies and in-place updates)** Because the calculation of $x_i^{(k+1)}$ requires the newly computed values $x_1^{(k+1)}, \ldots, x_{i-1}^{(k+1)}$, the standard Gauss–Seidel sweep is inherently sequential. However, it permits in-place updates, in which new values immediately overwrite old values in a single vector. For the standard Poisson problem in the natural ordering, the Gauss–Seidel spectral radius is the square of the Jacobi spectral radius. Such a comparison needs structure and does not hold for arbitrary matrices.

Example **(Comparison of one Jacobi and Gauss–Seidel step)** Consider the linear system

$$\begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ 5 \end{pmatrix},$$

whose exact solution is $\boldsymbol{x}^* = (1, 1, 1)^T$. Let the initial guess be $\boldsymbol{x}^{(0)} = (0, 0, 0)^T$.

- **Jacobi Step:**

$$\begin{aligned} x_1^{(1)} &= \frac{1}{4}(5 - 0 - 0) = 1.25, \\ x_2^{(1)} &= \frac{1}{4}(6 - 0 - 0) = 1.50, \\ x_3^{(1)} &= \frac{1}{4}(5 - 0 - 0) = 1.25, \qquad \Longrightarrow \boldsymbol{x}_J^{(1)} = \begin{pmatrix} 1.25 \\ 1.50 \\ 1.25 \end{pmatrix}. \end{aligned}$$

- **Gauss–Seidel Step:**

$$
\begin{aligned}
x_1^{(1)} &= \frac{1}{4}(5 - 0 - 0) = 1.25, \\
x_2^{(1)} &= \frac{1}{4}(6 - (1.25) - 0) = \frac{4.75}{4} = 1.1875, \\
x_3^{(1)} &= \frac{1}{4}(5 - (1.1875) - 0) = \frac{3.8125}{4} \approx 0.9531, \qquad \Longrightarrow \ \boldsymbol{x}_{GS}^{(1)} \approx \begin{pmatrix} 1.2500 \\ 1.1875 \\ 0.9531 \end{pmatrix}.
\end{aligned}
$$

The Gauss–Seidel iterate is closer to the true solution $(1, 1, 1)^T$ because the update for x_2 immediately used the new value of x_1, and the update for x_3 used the new value of x_2.

6.3.3 Convergence Theorems

Thm 6.6 **(Convergence Under Strict Diagonal Dominance)** A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is strictly diagonally dominant if

$$
|a_{ii}| > \sum_{j \neq i} |a_{ij}| \qquad \text{for all } i = 1, 2, \ldots, n.
$$

If $\mathbf{A}$ is strictly diagonally dominant, then both the Jacobi iteration and the Gauss–Seidel iteration converge for any initial guess $\boldsymbol{x}^{(0)} \in \mathbb{R}^n$.

Proof We show that the matrix ∞-norm of the iteration matrix $\mathbf{T}_J = \mathbf{D}^{-1}(\mathbf{L}_A + \mathbf{U}_A)$ is strictly bounded by one. The entries of $\mathbf{T}_J$ are

$$
(\mathbf{T}_J)_{ij} = \begin{cases} -\frac{a_{ij}}{a_{ii}}, & i \neq j, \\ 0, & i = j. \end{cases}
$$

The induced ∞-norm is the maximum absolute row sum:

$$
\|\mathbf{T}_J\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^{n} |(\mathbf{T}_J)_{ij}| = \max_{1 \leq i \leq n} \sum_{j \neq i} \frac{|a_{ij}|}{|a_{ii}|}.
$$

Because $\mathbf{A}$ is strictly diagonally dominant, $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for each $i \in \{1, \ldots, n\}$, so

$$
\sum_{j \neq i} \frac{|a_{ij}|}{|a_{ii}|} < 1 \implies \|\mathbf{T}_J\|_\infty < 1.
$$

By Gelfand's spectral radius theorem, $\rho(\mathbf{T}_J) \leq \|\mathbf{T}_J\|_\infty < 1$, which establishes convergence of the Jacobi iteration.

The estimate is uniform over all rows. Hence its maximum is a fixed number smaller than one, rather than a different bound for each iterate. Repeated multiplication by the iteration matrix then makes every Jacobi error tend to zero.

For Gauss–Seidel, let λ be an eigenvalue of $\mathbf{T}_{GS} = (\mathbf{D} - \mathbf{L}_A)^{-1}\mathbf{U}_A$ with eigenvector $\boldsymbol{v} \neq \mathbf{0}$, normalized so that $\|\boldsymbol{v}\|_\infty = |v_p| = 1$. From $\mathbf{U}_A \boldsymbol{v} = \lambda(\mathbf{D} - \mathbf{L}_A)\boldsymbol{v}$, row p reads

$$
-\sum_{j > p} a_{pj} v_j = \lambda a_{pp} v_p + \lambda \sum_{j < p} a_{pj} v_j.
$$

Rearranging and applying the triangle inequality with $|v_j| \le 1$ yields

$$|\lambda||a_{pp}| \le \sum_{j>p} |a_{pj}| + |\lambda| \sum_{j<p} |a_{pj}|.$$

If $|\lambda| \ge 1$, then

$$|a_{pp}| \le \frac{1}{|\lambda|} \sum_{j>p} |a_{pj}| + \sum_{j<p} |a_{pj}| \le \sum_{j \ne p} |a_{pj}|,$$

which contradicts strict diagonal dominance. The contradiction came from choosing an entry of maximal eigenvector magnitude. Such an entry is nonzero because the eigenvector is nonzero, so the normalization and divisions were legitimate. The index need not be the first or last row.

Hence $|\lambda| < 1$, proving $\rho(\mathbf{T}_{GS}) < 1$. □

Thm 6.7 **(Convergence of Gauss–Seidel for Symmetric Positive Definite Systems)** If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric positive definite, then the Gauss–Seidel method converges for every initial vector $\boldsymbol{x}^{(0)} \in \mathbb{R}^n$. The Jacobi method, however, is not guaranteed to converge for arbitrary SPD matrices.

Proof Let $\boldsymbol{x}^*$ be the exact solution and define $\phi(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} - \boldsymbol{b}^T\boldsymbol{x}$. During a sweep, let $\boldsymbol{y}$ denote the vector after any preceding coordinate updates, and let $r_i = b_i - (\mathbf{A}\boldsymbol{y})_i$ be its current residual entry.

Changing only coordinate i by d gives

$$\phi(\boldsymbol{y} + d\boldsymbol{e}_i) - \phi(\boldsymbol{y}) = -dr_i + \tfrac{1}{2}a_{ii}d^2.$$

The diagonal entry is positive because $\mathbf{A}$ is SPD. Completing the square gives

$$-dr_i + \tfrac{1}{2}a_{ii}d^2 = \tfrac{1}{2}a_{ii}(d - r_i/a_{ii})^2 - \frac{r_i^2}{2a_{ii}}.$$

The Gauss–Seidel update is exactly $d = r_i/a_{ii}$, using all coordinates at their current values. It minimizes this expression and decreases ϕ by $a_{ii}d^2/2$.

Sum the decreases over the sweep. They are all nonnegative. Their sum vanishes only when every coordinate change is zero. If no coordinate changes, the vector remains the same throughout the sweep, and $r_i = 0$ when each row is visited. All rows are therefore satisfied, so the vector is $\boldsymbol{x}^*$.

The energy identity gives $\phi(\boldsymbol{y}) - \phi(\boldsymbol{x}^*) = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{x}^*\|_{\mathbf{A}}^2$. Thus one sweep strictly decreases the energy norm of every nonzero error. Let $\mathbf{T}_{GS}$ be its linear error map. The function $\boldsymbol{e} \mapsto \|\mathbf{T}_{GS}\boldsymbol{e}\|_{\mathbf{A}}$ is continuous on the compact energy-unit sphere. Its attained maximum is a number $q < 1$, because every value there is strictly less than one. Homogeneity yields $\|\mathbf{T}_{GS}\boldsymbol{e}\|_{\mathbf{A}} \le q\|\boldsymbol{e}\|_{\mathbf{A}}$ for every error. Repeating sweeps gives $\|\boldsymbol{e}^{(k)}\|_{\mathbf{A}} \le q^k\|\boldsymbol{e}^{(0)}\|_{\mathbf{A}} \to 0$. This proves convergence. See also Saad (2003) for stationary-iteration theory. □

Ex 6.17 Complete the square for a coordinate update and derive the decrease $a_{ii}d_i^2/2$. To see why the same conclusion need not hold for Jacobi, take the 3×3 matrix with diagonal entries 1 and all off-diagonal entries $3/4$. Compute its eigenvalues and those of its Jacobi iteration matrix.

6.3.4 Successive Over-Relaxation (SOR)

In an SOR sweep, compute the value proposed by a Gauss–Seidel coordinate update and take a fraction ω of the change. Values $0 < \omega < 1$ shorten the change. Values $\omega > 1$ extend it. This can accelerate convergence for a suitable matrix and parameter, but can also cause divergence.

Def 6.12 **(Successive Over-Relaxation Scheme)** Let $\omega > 0$ be a relaxation parameter. The SOR method computes the updated components via

$$x_i^{(k+1)} = (1-\omega)x_i^{(k)} + \frac{\omega}{a_{ii}}\left(b_i - \sum_{j<i} a_{ij}x_j^{(k+1)} - \sum_{j>i} a_{ij}x_j^{(k)}\right).$$

In matrix notation, this corresponds to the splitting $\mathbf{M}_\omega = \frac{1}{\omega}\mathbf{D} - \mathbf{L}_A$ and $\mathbf{N}_\omega = \left(\frac{1}{\omega} - 1\right)\mathbf{D} + \mathbf{U}_A$. The iteration matrix is

$$\mathbf{T}_\omega = (\mathbf{D} - \omega\mathbf{L}_A)^{-1}\left((1-\omega)\mathbf{D} + \omega\mathbf{U}_A\right).$$

Ex 6.18 Prove the Kahan theorem: for any matrix $\mathbf{A}$ with nonzero diagonal entries, $\rho(\mathbf{T}_\omega) \geq |\omega - 1|$, which implies that SOR can only converge if $0 < \omega < 2$.

Thm 6.8 **(Ostrowski–Reich Theorem and the Optimal Parameter)** If $\mathbf{A}$ is symmetric positive definite, then the SOR iteration converges if and only if the relaxation parameter satisfies

$$0 < \omega < 2.$$

For the one-dimensional Dirichlet Poisson matrix in its natural ordering, with Jacobi spectral radius $\rho(\mathbf{T}_J) = \cos(\pi h) < 1$, the optimal relaxation parameter that minimizes $\rho(\mathbf{T}_\omega)$ is

$$\omega^* = \frac{2}{1 + \sqrt{1 - \rho(\mathbf{T}_J)^2}}.$$

With this optimal choice, the spectral radius satisfies $\rho(\mathbf{T}_{\omega^*}) = \omega^* - 1$.

Let $\phi(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} - \boldsymbol{b}^T\boldsymbol{x}$, and consider the current vector just before coordinate i is updated. Write $g_i = (\mathbf{A}\boldsymbol{x} - \boldsymbol{b})_i$. Holding the other coordinates fixed gives

$$\phi(\boldsymbol{x} + d_i\boldsymbol{e}_i) - \phi(\boldsymbol{x}) = g_i d_i + \tfrac{1}{2}a_{ii}d_i^2.$$

The Gauss–Seidel change is $-g_i/a_{ii}$. SOR takes $d_i = -\omega g_i/a_{ii}$, so $g_i = -a_{ii}d_i/\omega$. Substitution gives

$$\phi(\boldsymbol{x}) - \phi(\boldsymbol{x} + d_i\boldsymbol{e}_i) = \frac{a_{ii}(2-\omega)}{2\omega}d_i^2.$$

For an SPD matrix, $a_{ii} > 0$. Thus every nonzero coordinate change decreases the energy when $0 < \omega < 2$.

Let $\boldsymbol{x}_*$ solve the system and let $\boldsymbol{e} = \boldsymbol{x} - \boldsymbol{x}_*$. Expanding the quadratic and using $\mathbf{A}\boldsymbol{x}_* = \boldsymbol{b}$ gives $\phi(\boldsymbol{x}) - \phi(\boldsymbol{x}_*) = \frac{1}{2}\|\boldsymbol{e}\|_{\mathbf{A}}^2$. A complete sweep therefore strictly reduces the error norm unless every coordinate change is zero. If all changes are zero, the vector remains fixed throughout the sweep, and $g_i = 0$ for every i. It is then already the solution.

The error after a sweep is $\mathbf{T}_\omega \boldsymbol{e}$. The continuous function $\boldsymbol{e} \mapsto \|\mathbf{T}_\omega \boldsymbol{e}\|_{\mathbf{A}}$ attains a maximum q on the compact unit sphere $\|\boldsymbol{e}\|_{\mathbf{A}} = 1$. The strict decrease just proved holds at every point on that sphere, including a maximizing point, so $q < 1$. Scaling gives $\|\mathbf{T}_\omega \boldsymbol{e}\|_{\mathbf{A}} \leq q\|\boldsymbol{e}\|_{\mathbf{A}}$ for every error. After k sweeps the norm is at most q^k times its initial value, which tends to zero.
For necessity, the determinant exercise above gives $\rho(\mathbf{T}_\omega) \geq |\omega - 1|$. Convergence requires spectral radius less than one, hence $|\omega - 1| < 1$, or $0 < \omega < 2$. The optimal-parameter formula for the Poisson problem is stated without proof; see the treatment of SOR in (Saad, 2003).

Remark 6.9 **(Acceleration for discrete Poisson equations)** For the 1D discrete Laplacian on a grid with spacing h, the Jacobi spectral radius is $\rho(\mathbf{T}_J) = \cos(\pi h) \approx 1 - \frac{1}{2}\pi^2 h^2$. Consequently, Jacobi and Gauss–Seidel require $O(h^{-2}) = O(n^2)$ iterations to converge. With optimal SOR, $\omega^* \approx 2 - 2\pi h$, yielding $\rho(\mathbf{T}_{\omega^*}) \approx 1 - 2\pi h$. This reduces the required iteration count to $O(h^{-1}) = O(n)$, providing an order-of-magnitude computational speedup.

6.3.5 Further Exercises

Ex 6.19 Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. Show that $\mathbf{A}$ is symmetric but indefinite, and prove that both Jacobi and Gauss-Seidel diverge.

Ex 6.20 Write a Python script to solve the 1D discrete Poisson equation with $n = 50$ grid points using Jacobi, Gauss-Seidel, and optimal SOR. Plot the residual norm versus iteration number on a semilogarithmic scale.

6.4 The Conjugate Gradient Method

Suppose $\mathbf{A}$ is symmetric positive definite and we want to solve $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$. A factorization is one way to find $\boldsymbol{x}$. For a large sparse matrix, we would like to use the cheaper operation $\boldsymbol{v} \mapsto \mathbf{A}\boldsymbol{v}$ instead. Starting with a guess, how should we choose a correction that brings us closer to the solution?

Conjugate gradients answers this question by associating a scalar function with the system. The solution is the point where this function is smallest. We first establish that correspondence, then learn how to minimize the function along one direction. The final ingredient is to choose successive directions so that a new correction preserves the progress made along earlier ones. The resulting method uses one matrix–vector product per step and stores only a few vectors.

6.4.1 Quadratic Energy

Consider first the scalar equation $ax = b$, with $a > 0$. The function

$$q(x) = \frac{a}{2}x^2 - bx$$

has derivative $q'(x) = ax - b$. Thus the equation we want to solve appears when we set this derivative to zero. More is true: completing the square gives

$$q(x) = \frac{a}{2}\left(x - \frac{b}{a}\right)^2 - \frac{b^2}{2a}.$$

Because $a > 0$, the squared term is nonnegative and vanishes only at $x = b/a$. Solving the equation and minimizing q therefore give the same answer.

Ex 6.21 What changes when $a < 0$? When $a = 0$, consider $b = 0$ and $b \neq 0$ separately. Explain why setting a derivative to zero, by itself, does not establish that a point is a minimum.

For a symmetric matrix, the analogous function is

$$\phi(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\mathbf{A}\boldsymbol{x} - \boldsymbol{b}^T\boldsymbol{x}.$$

It takes a vector as input and returns one real number. The word *quadratic* refers to the products $x_i x_j$ in the first term. We call ϕ an *energy*. In some applications it is a physical potential energy. Here its role is to help us solve the equations. Its value need not be positive.

Example For the system

$$2x_1 + x_2 = 4, \qquad x_1 + 2x_2 = 5,$$

we have

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 4 \\ 5 \end{pmatrix},$$

and

$$\phi(x_1, x_2) = x_1^2 + x_1x_2 + x_2^2 - 4x_1 - 5x_2.$$

The solution is $(1, 2)^T$. With $e_1 = x_1 - 1$ and $e_2 = x_2 - 2$, completing the square in two variables gives

$$\begin{aligned} \phi(x_1, x_2) + 7 &= e_1^2 + e_1e_2 + e_2^2 \\ &= \frac{1}{2}((e_1 + e_2)^2 + e_1^2 + e_2^2). \end{aligned}$$

Every square vanishes at the solution, and their sum is positive everywhere else. Thus the smallest energy is -7, attained only at $(1, 2)^T$.

Ex 6.22 Expand the two translated expressions to verify the identity above. Next differentiate $\phi(x_1, x_2)$ with respect to each variable, holding the other fixed. Show that setting the two partial derivatives to zero recovers the two original equations. Where does symmetry of $\mathbf{A}$ enter the coefficient of x_1x_2?

The same completion-of-the-square argument works in any dimension. Recall that for an SPD matrix, $\|\boldsymbol{e}\|_{\mathbf{A}}^2 = \boldsymbol{e}^T\mathbf{A}\boldsymbol{e}$ is positive for every nonzero $\boldsymbol{e}$.

Thm 6.9 **(Energy and Solution Error)** Let $\mathbf{A}$ be SPD and let $\boldsymbol{x}^*$ solve $\mathbf{A}\boldsymbol{x}^* = \boldsymbol{b}$. For every $\boldsymbol{x}$,

$$\phi(\boldsymbol{x}) - \phi(\boldsymbol{x}^*) = \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^*)^T\mathbf{A}(\boldsymbol{x} - \boldsymbol{x}^*) = \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{x}^*\|_{\mathbf{A}}^2.$$

Consequently, $\boldsymbol{x}^*$ is the unique minimizer of ϕ.

Proof Let $\boldsymbol{e} = \boldsymbol{x} - \boldsymbol{x}^*$, so $\boldsymbol{x} = \boldsymbol{x}^* + \boldsymbol{e}$. We compare the two energies by substituting this expression into the definition.

1. Distribute the matrix product before collecting terms:

$$(\boldsymbol{x}^* + \boldsymbol{e})^T \mathbf{A}(\boldsymbol{x}^* + \boldsymbol{e}) = \boldsymbol{x}^{*T}\mathbf{A}\boldsymbol{x}^* + \boldsymbol{x}^{*T}\mathbf{A}\boldsymbol{e} + \boldsymbol{e}^T\mathbf{A}\boldsymbol{x}^* + \boldsymbol{e}^T\mathbf{A}\boldsymbol{e}.$$

   Also $\boldsymbol{b}^T(\boldsymbol{x}^* + \boldsymbol{e}) = \boldsymbol{b}^T\boldsymbol{x}^* + \boldsymbol{b}^T\boldsymbol{e}$. Therefore

$$\phi(\boldsymbol{x}^* + \boldsymbol{e}) - \phi(\boldsymbol{x}^*) = \tfrac{1}{2}\boldsymbol{x}^{*T}\mathbf{A}\boldsymbol{e} + \tfrac{1}{2}\boldsymbol{e}^T\mathbf{A}\boldsymbol{x}^* + \tfrac{1}{2}\boldsymbol{e}^T\mathbf{A}\boldsymbol{e} - \boldsymbol{b}^T\boldsymbol{e}.$$

   The terms containing only $\boldsymbol{x}^*$ have cancelled because they make up $\phi(\boldsymbol{x}^*)$.

2. A scalar equals its transpose. Symmetry of $\mathbf{A}$ consequently gives

$$\boldsymbol{x}^{*T}\mathbf{A}\boldsymbol{e} = (\boldsymbol{x}^{*T}\mathbf{A}\boldsymbol{e})^T = \boldsymbol{e}^T\mathbf{A}^T\boldsymbol{x}^* = \boldsymbol{e}^T\mathbf{A}\boldsymbol{x}^*.$$

   The two half-weighted cross terms sum to $\boldsymbol{e}^T\mathbf{A}\boldsymbol{x}^*$. Since $\mathbf{A}\boldsymbol{x}^* = \boldsymbol{b}$, this is $\boldsymbol{e}^T\boldsymbol{b} = \boldsymbol{b}^T\boldsymbol{e}$, exactly the term that is subtracted. What remains is $\frac{1}{2}\boldsymbol{e}^T\mathbf{A}\boldsymbol{e}$.

3. Positive definiteness says $\boldsymbol{e}^T\mathbf{A}\boldsymbol{e} > 0$ for every $\boldsymbol{e} \neq \mathbf{0}$, and the value is zero for $\boldsymbol{e} = \mathbf{0}$. Thus $\phi(\boldsymbol{x}) > \phi(\boldsymbol{x}^*)$ whenever $\boldsymbol{x} \neq \boldsymbol{x}^*$. This proves uniqueness as well as minimality.

□

This identity explains the choice of energy. Lowering ϕ lowers the squared solution error measured in the $\mathbf{A}$-norm by exactly the same amount, apart from the factor 1/2. We do not know $\boldsymbol{x}^*$, so we cannot evaluate that error directly. We can evaluate $\phi(\boldsymbol{x})$ and, more usefully, its change under a proposed correction using only $\mathbf{A}$, $\boldsymbol{b}$, and the current guess.

6.4.2 Line Search and Steepest Descent

Let $\boldsymbol{x}$ be the current guess, and choose a nonzero direction $\boldsymbol{p}$. Points on the line through $\boldsymbol{x}$ in this direction have the form $\boldsymbol{x} + \alpha\boldsymbol{p}$, where the scalar α is the step length. Choosing the best point on this line is a one-variable problem, even when $\boldsymbol{x}$ has millions of entries.
Write $\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}$ for the current residual. Expanding as in the preceding proof gives

$$\phi(\boldsymbol{x} + \alpha\boldsymbol{p}) = \phi(\boldsymbol{x}) - \alpha\boldsymbol{p}^T\boldsymbol{r} + \frac{\alpha^2}{2}\boldsymbol{p}^T\mathbf{A}\boldsymbol{p}.$$

The coefficient $\boldsymbol{p}^T\mathbf{A}\boldsymbol{p}$ is positive. This is an upward-opening quadratic in α, just like the scalar function with which we began. Differentiating with respect to α gives

$$\frac{d}{d\alpha}\phi(\boldsymbol{x} + \alpha\boldsymbol{p}) = -\boldsymbol{p}^T\boldsymbol{r} + \alpha\boldsymbol{p}^T\mathbf{A}\boldsymbol{p}.$$

Hence the unique minimizing step is

$$\alpha = \frac{\boldsymbol{p}^T\boldsymbol{r}}{\boldsymbol{p}^T\mathbf{A}\boldsymbol{p}}.$$

This is called an *exact line search*: exact refers to minimizing along the chosen line, not to solving the whole linear system in one step.

Ex 6.23 Substitute this step length into the quadratic to show that the energy decrease is

$$\phi(\boldsymbol{x}) - \phi(\boldsymbol{x} + \alpha\boldsymbol{p}) = \frac{(\boldsymbol{p}^T\boldsymbol{r})^2}{2\boldsymbol{p}^T\mathbf{A}\boldsymbol{p}}.$$

When is there no improvement? Relate your answer to the slope at $\alpha = 0$. Explain why replacing $\boldsymbol{p}$ by a nonzero scalar multiple changes the step length but leaves the new point unchanged.

How should we choose $\boldsymbol{p}$? The term linear in α is $-\alpha \boldsymbol{p}^T \boldsymbol{r}$. Among directions of Euclidean length one, Cauchy–Schwarz shows that the largest initial decrease occurs at $\boldsymbol{p} = \boldsymbol{r}/\|\boldsymbol{r}\|_2$, provided $\boldsymbol{r} \neq \boldsymbol{0}$. Thus the residual points in the locally steepest downhill direction. In the language of multivariable calculus, $\nabla\phi(\boldsymbol{x}) = \mathbf{A}\boldsymbol{x} - \boldsymbol{b} = -\boldsymbol{r}$ is the gradient, and the residual is its negative.

Since scaling the direction does not change the line, we can simply take $\boldsymbol{p} = \boldsymbol{r}$. Repeating the line minimization in the current residual direction gives *steepest descent*:

$$\alpha_k = \frac{\boldsymbol{r}^{(k)T}\boldsymbol{r}^{(k)}}{\boldsymbol{r}^{(k)T}\mathbf{A}\boldsymbol{r}^{(k)}},$$
$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \alpha_k \boldsymbol{r}^{(k)}.$$

If the residual is zero, the system is already solved. Otherwise $\alpha_k > 0$ and the energy strictly decreases.

Ex 6.24 Derive $\boldsymbol{r}^{(k+1)} = \boldsymbol{r}^{(k)} - \alpha_k \mathbf{A}\boldsymbol{r}^{(k)}$. Use the step-length formula to prove $\boldsymbol{r}^{(k+1)T}\boldsymbol{r}^{(k)} = 0$. Interpret this as saying that the energy has zero slope along the line just minimized. Does it imply orthogonality to every earlier residual?

Steepest descent makes the best local choice of direction, but a good local slope need not give rapid progress over many steps. If the energy level sets are elongated ellipses, the residual can point across the narrow part of the ellipse instead of along its long direction toward the minimum. Successive line searches can then alternate across this narrow valley. The following calculation shows this behavior without relying on a picture.

Ex 6.25 Let $\mathbf{A} = \operatorname{diag}(1, 9)$, $\boldsymbol{b} = \boldsymbol{0}$, and $\boldsymbol{x}^{(0)} = (9, 1)^T$, so the solution is the origin.

1. Compute $\boldsymbol{r}^{(0)}$ and show that $\alpha_0 = 1/5$. Find $\boldsymbol{x}^{(1)}$ and $\boldsymbol{r}^{(1)}$.
2. Take a second steepest-descent step. Show that $\alpha_1 = 1/5$ and $\boldsymbol{x}^{(2)} = (16/25)\boldsymbol{x}^{(0)}$.
3. Compare the directions of the two steps. Why did two exact line searches fail to reach the solution of this two-variable system?

6.4.3 Conjugate Directions

Suppose we have minimized along a direction $\boldsymbol{p}$, arriving at $\boldsymbol{y}$ with residual $\boldsymbol{r}_{\boldsymbol{y}}$. The zero-slope condition is $\boldsymbol{p}^T\boldsymbol{r}_{\boldsymbol{y}} = 0$. Now take a step of length β along another direction $\boldsymbol{q}$. The new residual is $\boldsymbol{r}_{\boldsymbol{y}} - \beta\mathbf{A}\boldsymbol{q}$, so

$$\boldsymbol{p}^T(\boldsymbol{r}_{\boldsymbol{y}} - \beta\mathbf{A}\boldsymbol{q}) = -\beta\boldsymbol{p}^T\mathbf{A}\boldsymbol{q}.$$

If $\boldsymbol{p}^T\mathbf{A}\boldsymbol{q} = 0$, the slope along $\boldsymbol{p}$ remains zero after the new step. We have found the condition that lets us retain an earlier line minimization. It is orthogonality in the energy inner product.

Def 6.13 **(A-Conjugate Vectors)** Nonzero vectors $\boldsymbol{p}_0, \ldots, \boldsymbol{p}_{k-1}$ are **A-conjugate** if

$$\boldsymbol{p}_i^T\mathbf{A}\boldsymbol{p}_j = 0 \qquad (i \neq j).$$

Equivalently, they are orthogonal in the inner product $\langle \boldsymbol{u}, \boldsymbol{v} \rangle_{\mathbf{A}} = \boldsymbol{u}^T\mathbf{A}\boldsymbol{v}$.

Ex 6.26 Return to $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $\boldsymbol{b} = (4, 5)^T$.

1. Take $\boldsymbol{p}_0 = (1,0)^T$. Find c so that $\boldsymbol{p}_1 = (c,1)^T$ is $\mathbf{A}$-conjugate to $\boldsymbol{p}_0$. Are the two directions Euclidean-orthogonal?

2. Starting at the origin, minimize first along $\boldsymbol{p}_0$ and then along $\boldsymbol{p}_1$. Verify that you arrive at $(1,2)^T$ and that the second step preserves the zero slope along $\boldsymbol{p}_0$.

3. Prove that any collection of nonzero $\mathbf{A}$-conjugate vectors is linearly independent. Take the $\mathbf{A}$-inner product of a vanishing linear combination with each direction.

Thm 6.10 **(Minimization along Conjugate Directions)** Let $\boldsymbol{p}_0, \dots, \boldsymbol{p}_{n-1}$ be an $\mathbf{A}$-conjugate basis. Starting from $\boldsymbol{x}^{(0)}$, minimize successively along these directions. With $\boldsymbol{r}^{(k)} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(k)}$, the steps are

$$\alpha_k = \frac{\boldsymbol{p}_k^T \boldsymbol{r}^{(k)}}{\boldsymbol{p}_k^T \mathbf{A} \boldsymbol{p}_k} = \frac{\boldsymbol{p}_k^T \boldsymbol{r}^{(0)}}{\boldsymbol{p}_k^T \mathbf{A} \boldsymbol{p}_k},$$
$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \alpha_k \boldsymbol{p}_k.$$

After k steps, the iterate minimizes ϕ, and hence the $\mathbf{A}$-norm of the solution error, among all points

$$\boldsymbol{x}^{(0)} + c_0 \boldsymbol{p}_0 + \cdots + c_{k-1} \boldsymbol{p}_{k-1}.$$

After n steps it is the exact solution.

Proof The essential point is that conjugacy separates a problem in several coefficients into independent scalar quadratics.

1. Put $\boldsymbol{d} = \sum_{i=0}^{k-1} c_i \boldsymbol{p}_i$ and $\boldsymbol{r}^{(0)} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(0)}$. The preceding energy expansion gives

   $$\phi(\boldsymbol{x}^{(0)} + \boldsymbol{d}) = \phi(\boldsymbol{x}^{(0)}) - \boldsymbol{d}^T \boldsymbol{r}^{(0)} + \tfrac{1}{2} \boldsymbol{d}^T \mathbf{A} \boldsymbol{d}.$$

   Substituting the sum for $\boldsymbol{d}$ produces a double sum in the last term:

   $$\boldsymbol{d}^T \mathbf{A} \boldsymbol{d} = \sum_{i,j=0}^{k-1} c_i c_j \boldsymbol{p}_i^T \mathbf{A} \boldsymbol{p}_j = \sum_{i=0}^{k-1} c_i^2 \boldsymbol{p}_i^T \mathbf{A} \boldsymbol{p}_i.$$

   The terms with $i \neq j$ vanish by conjugacy.

2. Define $d_i = \boldsymbol{p}_i^T \mathbf{A} \boldsymbol{p}_i > 0$ and $g_i = \boldsymbol{p}_i^T \boldsymbol{r}^{(0)}$. The terms depending on c_i are

   $$\tfrac{1}{2} d_i c_i^2 - g_i c_i = \tfrac{1}{2} d_i (c_i - g_i/d_i)^2 - \frac{g_i^2}{2 d_i}.$$

   Their unique minimum occurs at $c_i = g_i/d_i$. No other coefficient occurs in this expression, so optimizing a later coefficient cannot change an earlier minimizing value.

3. After the first k updates,

   $$\boldsymbol{r}^{(k)} = \boldsymbol{r}^{(0)} - \sum_{i=0}^{k-1} \alpha_i \mathbf{A} \boldsymbol{p}_i.$$

   Multiply by $\boldsymbol{p}_k^T$. Every term in the sum vanishes, giving $\boldsymbol{p}_k^T \boldsymbol{r}^{(k)} = \boldsymbol{p}_k^T \boldsymbol{r}^{(0)}$. Hence the line-search coefficient computed from the current residual is exactly g_k/d_k.

4. After k steps, every available coefficient has its unique minimizing value. After n steps, the basis spans $\mathbb{R}^n$, so the admissible points include the unique global minimizer $\boldsymbol{x}^*$. Thus the final point is the exact solution.

□

6.4.4 The Conjugate Gradient Algorithm

The theorem assumes that a conjugate basis is already available. Constructing and storing an entire basis before solving would be too expensive. CG instead builds one new direction per step from the current residual and the preceding direction. The proof below explains why this short recurrence makes the new direction conjugate to *all* earlier ones.
The first k directions span $\mathcal{K}_k(\mathbf{A}, \boldsymbol{r}^{(0)})$. Thus the points available after k steps form $\boldsymbol{x}^{(0)} + \mathcal{K}_k$, the initial guess plus a Krylov subspace. The preceding theorem explains what CG optimizes within that set: it minimizes the energy, or equivalently the error in the $\mathbf{A}$-norm.

Thm 6.11 **(The Conjugate Gradient Algorithm)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be symmetric positive definite and let $\boldsymbol{b} \in \mathbb{R}^n$. Given an initial estimate $\boldsymbol{x}^{(0)}$, compute $\boldsymbol{r}^{(0)} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(0)}$ and set $\boldsymbol{p}^{(0)} = \boldsymbol{r}^{(0)}$. If the initial residual is zero, return immediately. Otherwise, for $k = 0, 1, 2, \ldots$:

1. Compute the matrix–vector product $\boldsymbol{w}^{(k)} = \mathbf{A}\boldsymbol{p}^{(k)}$.
2. Compute the step length:
$$\alpha_k = \frac{\boldsymbol{r}^{(k)T}\boldsymbol{r}^{(k)}}{\boldsymbol{p}^{(k)T}\boldsymbol{w}^{(k)}}.$$
3. Update the solution iterate:
$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \alpha_k \boldsymbol{p}^{(k)}.$$
4. Update the residual vector:
$$\boldsymbol{r}^{(k+1)} = \boldsymbol{r}^{(k)} - \alpha_k \boldsymbol{w}^{(k)}.$$
5. Check termination: if $\|\boldsymbol{r}^{(k+1)}\|_2 \le \max(\texttt{atol}, \texttt{tol}\|\boldsymbol{b}\|_2)$, terminate. Here `tol` is a relative tolerance and `atol` is an absolute tolerance, used also when $\boldsymbol{b} = \mathbf{0}$.
6. Compute the orthogonalization coefficient:
$$\beta_k = \frac{\boldsymbol{r}^{(k+1)T}\boldsymbol{r}^{(k+1)}}{\boldsymbol{r}^{(k)T}\boldsymbol{r}^{(k)}}.$$
7. Compute the next search direction:
$$\boldsymbol{p}^{(k+1)} = \boldsymbol{r}^{(k+1)} + \beta_k \boldsymbol{p}^{(k)}.$$

Thm 6.12 **(Orthogonality Properties of CG)** In exact arithmetic, the sequences generated by the Conjugate Gradient algorithm satisfy the following orthogonality relations for all $i \neq j$:
$$\boldsymbol{r}^{(i)T}\boldsymbol{r}^{(j)} = 0 \qquad \text{and} \qquad \boldsymbol{p}^{(i)T}\mathbf{A}\boldsymbol{p}^{(j)} = 0.$$

Moreover, the residual vectors and search directions span the Krylov subspace:

$$\operatorname{span}\{\boldsymbol{r}^{(0)}, \boldsymbol{r}^{(1)}, \ldots, \boldsymbol{r}^{(k)}\} = \operatorname{span}\{\boldsymbol{p}^{(0)}, \boldsymbol{p}^{(1)}, \ldots, \boldsymbol{p}^{(k)}\} = \mathcal{K}_{k+1}(\mathbf{A}, \boldsymbol{r}^{(0)}).$$

Consequently, the method is guaranteed to reach the exact solution in at most n iterations in exact arithmetic.

Proof We use induction to establish the structure that makes the short recurrence valid. Write $\mathcal{K}_k = \mathcal{K}_k(\mathbf{A}, \boldsymbol{r}^{(0)})$ and set $\mathcal{K}_0 = \{\mathbf{0}\}$. All calculations are in exact arithmetic and stop when a residual is zero.

1. At the beginning of step k, we claim that the previous directions $\boldsymbol{p}^{(0)}, \ldots, \boldsymbol{p}^{(k-1)}$ are conjugate and span $\mathcal{K}_k$, while $\boldsymbol{r}^{(k)} \in \mathcal{K}_{k+1}$ and $\boldsymbol{r}^{(k)} \perp \mathcal{K}_k$. For $k = 0$, there are no previous directions, and $\boldsymbol{r}^{(0)} \in \mathcal{K}_1$ by definition. Its orthogonality to $\mathcal{K}_0$ is automatic. Set $\boldsymbol{p}^{(0)} = \boldsymbol{r}^{(0)}$.

2. Suppose $k \geq 1$ and the claims hold through the preceding step. Rearrange each previous residual update:

$$\mathbf{A}\boldsymbol{p}^{(i)} = \frac{\boldsymbol{r}^{(i)} - \boldsymbol{r}^{(i+1)}}{\alpha_i}.$$

   For $i < k - 1$, both residuals on the right belong to $\mathcal{K}_k$. Therefore

$$\boldsymbol{r}^{(k)T}\mathbf{A}\boldsymbol{p}^{(i)} = \frac{\boldsymbol{r}^{(k)T}\boldsymbol{r}^{(i)} - \boldsymbol{r}^{(k)T}\boldsymbol{r}^{(i+1)}}{\alpha_i} = 0.$$

   Only the direction with index $k - 1$ can require a correction.

3. In $\boldsymbol{p}^{(k)} = \boldsymbol{r}^{(k)} + \beta\boldsymbol{p}^{(k-1)}$, require conjugacy with $\boldsymbol{p}^{(k-1)}$. This gives

$$\beta = -\frac{\boldsymbol{r}^{(k)T}\mathbf{A}\boldsymbol{p}^{(k-1)}}{\boldsymbol{p}^{(k-1)T}\mathbf{A}\boldsymbol{p}^{(k-1)}}.$$

   The numerator is evaluated with the previous residual update. Since $\boldsymbol{r}^{(k)T}\boldsymbol{r}^{(k-1)} = 0$,

$$\boldsymbol{r}^{(k)T}\mathbf{A}\boldsymbol{p}^{(k-1)} = -\frac{\|\boldsymbol{r}^{(k)}\|_2^2}{\alpha_{k-1}}.$$

   The previous step length gives $\alpha_{k-1}\boldsymbol{p}^{(k-1)T}\mathbf{A}\boldsymbol{p}^{(k-1)} = \|\boldsymbol{r}^{(k-1)}\|_2^2$. Substitution yields exactly $\beta_{k-1} = \|\boldsymbol{r}^{(k)}\|_2^2/\|\boldsymbol{r}^{(k-1)}\|_2^2$. For every $i < k-1$, the new direction is conjugate to $\boldsymbol{p}^{(i)}$ because both its residual term and its previous-direction term have zero $\mathbf{A}$-inner product with that direction.

4. A nonzero $\boldsymbol{r}^{(k)}$ is perpendicular to $\mathcal{K}_k$, so it cannot belong to that space. It supplies one independent direction in $\mathcal{K}_{k+1}$. Adding a multiple of $\boldsymbol{p}^{(k-1)} \in \mathcal{K}_k$ does not change the expanded span. Therefore the first $k + 1$ directions span $\mathcal{K}_{k+1}$. For $k = 0$, the identity $\boldsymbol{p}^{(0)T}\boldsymbol{r}^{(0)} = \|\boldsymbol{r}^{(0)}\|_2^2$ follows from $\boldsymbol{p}^{(0)} = \boldsymbol{r}^{(0)}$. For $k \geq 1$, orthogonality gives

$$\boldsymbol{p}^{(k)T}\boldsymbol{r}^{(k)} = \|\boldsymbol{r}^{(k)}\|_2^2 + \beta_{k-1}\boldsymbol{p}^{(k-1)T}\boldsymbol{r}^{(k)} = \|\boldsymbol{r}^{(k)}\|_2^2.$$

   Thus exact line minimization gives the algorithm's stated α_k. Its denominator is positive because $\mathbf{A}$ is SPD and $\boldsymbol{p}^{(k)} \neq \mathbf{0}$.

5. For $i < k$, multiply the new residual update by $\boldsymbol{p}^{(i)T}$. The old residual is perpendicular to $\boldsymbol{p}^{(i)}$, and conjugacy removes the update term. For $i = k$, the step-length formula gives zero directly. Hence $\boldsymbol{r}^{(k+1)}$ is perpendicular to every direction spanning $\mathcal{K}_{k+1}$. The update also puts it in $\mathcal{K}_{k+2}$. This proves the induction assertions at the next step. The preceding conjugate-direction theorem proves energy minimality.

6. Earlier residuals lie in the previous Krylov spaces, so the orthogonality just proved makes all nonzero residuals mutually perpendicular. The direction recurrence gives equality of the residual and direction spans. There cannot be more than n independent directions in $\mathbb{R}^n$. If n nonzero directions have been generated, they span the whole space, and the next residual, perpendicular to that space, is zero. Thus termination occurs in at most n steps.

The standard CG analysis is developed in Saad (2003). □

Remark 6.10 **(Connection with Lanczos)** By Thm 6.12, the residuals $\boldsymbol{r}^{(0)}, \dots, \boldsymbol{r}^{(k-1)}$ are mutually orthogonal and the first j of them span $\mathcal{K}_j(\mathbf{A}, \boldsymbol{r}^{(0)})$. The Lanczos iteration of Def 4.15, started from $\boldsymbol{v}_1 = \boldsymbol{r}^{(0)}/\|\boldsymbol{r}^{(0)}\|_2$, produces an orthonormal basis with the same nested spans. Such a basis is unique up to the signs of its vectors, and the signs work out to

$$\boldsymbol{v}_{j+1} = (-1)^j \frac{\boldsymbol{r}^{(j)}}{\|\boldsymbol{r}^{(j)}\|_2}, \qquad j = 0, \dots, k-1.$$

The CG coefficients also determine the Lanczos matrix $\mathbf{T}_k$ of Cor 4.29. Its entries are

$$(\mathbf{T}_k)_{11} = \frac{1}{\alpha_0}, \qquad (\mathbf{T}_k)_{j+1,j+1} = \frac{1}{\alpha_j} + \frac{\beta_{j-1}}{\alpha_{j-1}}, \qquad (\mathbf{T}_k)_{j+1,j} = (\mathbf{T}_k)_{j,j+1} = \frac{\sqrt{\beta_{j-1}}}{\alpha_{j-1}},$$

where α_j and β_j are the CG step length and direction coefficient. Conjugate gradients may therefore be read as the Lanczos algorithm applied to a linear system.
The correspondence also describes the CG iterate directly. Write $\boldsymbol{x}^{(k)} = \boldsymbol{x}^{(0)} + \mathbf{V}_k \boldsymbol{y}_k$. Since $\boldsymbol{r}^{(k)}$ is orthogonal to $\mathcal{K}_k$, and since $\mathbf{V}_k^T \boldsymbol{r}^{(0)} = \|\boldsymbol{r}^{(0)}\|_2 \boldsymbol{e}_1$ and $\mathbf{V}_k^T \mathbf{A} \mathbf{V}_k = \mathbf{T}_k$, we obtain

$$\mathbf{0} = \mathbf{V}_k^T \boldsymbol{r}^{(k)} = \mathbf{V}_k^T (\boldsymbol{r}^{(0)} - \mathbf{A}\mathbf{V}_k \boldsymbol{y}_k) = \|\boldsymbol{r}^{(0)}\|_2 \boldsymbol{e}_1 - \mathbf{T}_k \boldsymbol{y}_k.$$

The CG iterate is thus obtained by solving the projected system $\mathbf{T}_k \boldsymbol{y}_k = \|\boldsymbol{r}^{(0)}\|_2 \boldsymbol{e}_1$. In addition, the extreme eigenvalues of $\mathbf{T}_k$ approach $\lambda_{\min}$ and $\lambda_{\max}$ by Thm 4.30, so the CG coefficients give a running estimate of $\kappa_2(\mathbf{A})$ at no extra cost in products with $\mathbf{A}$.

Example For $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\boldsymbol{b} = (4, 5)^T$, and $\boldsymbol{x}^{(0)} = \mathbf{0}$, the initial direction is $\boldsymbol{p}^{(0)} = \boldsymbol{r}^{(0)} = (4, 5)^T$. Thus

$$\mathbf{A}\boldsymbol{p}^{(0)} = (13, 14)^T, \qquad \alpha_0 = \frac{41}{122},$$
$$\boldsymbol{x}^{(1)} = \frac{1}{122}(164, 205)^T, \qquad \boldsymbol{r}^{(1)} = \frac{1}{122}(-45, 36)^T.$$

The new residual is perpendicular to the old residual because $-45 \cdot 4 + 36 \cdot 5 = 0$. Continuing along $\boldsymbol{r}^{(1)}$ alone would be steepest descent. CG instead adds a multiple of the old search direction to preserve the preceding minimization.

Ex 6.27 Continue this calculation with exact fractions. Show that $\beta_0 = 81/14884$ and $\boldsymbol{p}^{(1)} = (-5166, 4797)^T/14884$. Verify conjugacy with $\boldsymbol{p}^{(0)}$, then compute α_1 and recover $\boldsymbol{x}^{(2)} = (1, 2)^T$. Compare the second CG direction with the second steepest-descent direction. Which earlier line-minimization condition does each preserve?

6.4.5 Convergence Rate and Preconditioning

Although CG terminates in n steps in exact arithmetic, its primary utility in scientific computing is as a rapid iterative solver for which $k \ll n$.

Thm 6.13 **(Chebyshev Error Bound for Conjugate Gradient)** Let $\mathbf{A}$ be symmetric positive definite with 2-norm condition number $\kappa = \lambda_{\max}/\lambda_{\min}$. The error at step k satisfies

$$\frac{\|\boldsymbol{x}^{(k)} - \boldsymbol{x}^*\|_{\mathbf{A}}}{\|\boldsymbol{x}^{(0)} - \boldsymbol{x}^*\|_{\mathbf{A}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k.$$

Proof Assume first that the initial error $\boldsymbol{e}^{(0)} = \boldsymbol{x}^{(0)} - \boldsymbol{x}^*$ is nonzero. A zero initial error needs no iteration.

1. A Krylov correction is $s(\mathbf{A})\boldsymbol{r}^{(0)}$, with $\deg s \leq k-1$. Since $\boldsymbol{r}^{(0)} = -\mathbf{A}\boldsymbol{e}^{(0)}$, the corrected error is

   $$\boldsymbol{e}^{(0)} + s(\mathbf{A})\boldsymbol{r}^{(0)} = [\mathbf{I} - \mathbf{A}s(\mathbf{A})]\boldsymbol{e}^{(0)} = p(\mathbf{A})\boldsymbol{e}^{(0)},$$

   where $p(t) = 1 - ts(t)$ has degree at most k and $p(0) = 1$. Conversely, every polynomial with these properties has this form because $1 - p(t)$ is divisible by t.

2. Expand $\boldsymbol{e}^{(0)} = \sum_i c_i \boldsymbol{q}_i$ in an orthonormal eigenbasis. Then

   $$\begin{aligned}\|p(\mathbf{A})\boldsymbol{e}^{(0)}\|_{\mathbf{A}}^2 &= \sum_i \lambda_i p(\lambda_i)^2 c_i^2 \\ &\leq \left(\max_i |p(\lambda_i)|^2\right)\sum_i \lambda_i c_i^2.\end{aligned}$$

   CG minimizes the energy-norm error over these corrections. Taking square roots therefore bounds its relative error by $\max_i |p(\lambda_i)|$ for every admissible p.

3. Let $a = \lambda_{\min}$ and $b = \lambda_{\max}$. If $a < b$, the affine map $(b + a - 2t)/(b - a)$ takes $[a, b]$ to $[-1, 1]$. Choose

   $$p(t) = \frac{T_k((b + a - 2t)/(b - a))}{T_k((b + a)/(b - a))}.$$

   At $t = 0$ the numerator equals the denominator, so $p(0) = 1$. On $[a, b]$, the numerator has magnitude at most one. Therefore the relative error is at most $1/T_k(c)$, where $c = (b + a)/(b - a) > 1$.

4. Put $\zeta = c + \sqrt{c^2 - 1} > 1$. The Chebyshev recurrence, checked first at $k = 0, 1$, gives

   $$T_k(c) = \tfrac{1}{2}(\zeta^k + \zeta^{-k}) \geq \tfrac{1}{2}\zeta^k.$$

   Writing $\kappa = b/a$ and simplifying gives

   $$\zeta = \frac{b + a + 2\sqrt{ab}}{b - a} = \frac{\sqrt{b} + \sqrt{a}}{\sqrt{b} - \sqrt{a}} = \frac{\sqrt{\kappa} + 1}{\sqrt{\kappa} - 1}.$$

   Hence $1/T_k(c) \leq 2\zeta^{-k}$, which is the claimed bound.

5. If $a = b$, the spectral theorem gives $\mathbf{A} = a\mathbf{I}$. The first CG correction has step length $1/a$ and gives the exact solution. Thus the stated conclusion also holds for every $k \geq 1$ when $\kappa = 1$.

This is the classical Chebyshev bound for CG (Saad, 2003; Greenbaum, 1997). □

Ex 6.28 Derive the expression for $T_k(c)$ by checking its first two values and three-term recurrence. Then simplify $c+\sqrt{c^2-1}$ in terms of b/a. Explain why a polynomial that is small only at the eigenvalues can give a better CG estimate than one bounded on the whole interval.

Cor 6.14 **(Termination with Distinct Eigenvalues)** If $\mathbf{A}$ has only m distinct eigenvalues, CG reaches the exact solution in at most m iterations in exact arithmetic.

Proof Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues. Each is positive because $\mathbf{A}$ is SPD.

1. Define
$$p_m(t) = \prod_{j=1}^{m} (1 - t/\lambda_j).$$
It has degree m, value one at zero, and value zero at every eigenvalue. The divisions are legitimate because no eigenvalue is zero.

2. In an orthonormal eigenbasis $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$, powers of $\mathbf{A}$ give $p_m(\mathbf{A}) = \mathbf{Q}p_m(\mathbf{\Lambda})\mathbf{Q}^T$. Every diagonal entry of $p_m(\mathbf{\Lambda})$ is zero. Therefore $p_m(\mathbf{A}) = \mathbf{0}$.

3. As shown in the preceding proof, after m steps the admissible errors include every $p(\mathbf{A})\boldsymbol{e}^{(0)}$ with $\deg p \leq m$ and $p(0) = 1$. In particular they include $p_m(\mathbf{A})\boldsymbol{e}^{(0)} = \mathbf{0}$. CG selects the smallest energy norm among these errors. A norm is nonnegative, so a zero candidate forces the minimum to be zero. Thus the iterate is the exact solution, or the algorithm has already terminated earlier.

□

Ex 6.29 For a diagonal SPD matrix with three distinct eigenvalues, construct the polynomial in Cor 6.14 and verify that it annihilates the matrix.

Remark 6.11 **(The square root condition number advantage)** The CG bound depends on $\sqrt{\kappa}$. For comparison, the worst-case energy contraction factor for steepest descent with exact line searches is $(\kappa-1)/(\kappa+1)$. General stationary iterations depend on their iteration matrices, not solely on $\kappa(\mathbf{A})$. For an ill-conditioned system with $\kappa = 10^4$, the reduction factor per iteration for steepest descent is approximately $(\kappa-1)/(\kappa+1) \approx 0.9998$, whereas for CG it is $(\sqrt{\kappa}-1)/(\sqrt{\kappa}+1) = 99/101 \approx 0.9802$, giving roughly one hundred times as large an asymptotic logarithmic reduction per step in these bounds. Actual rates also depend on the spectrum and initial error.

Remark 6.12 **(Superlinear convergence)** The Chebyshev bound is a worst-case guarantee for an arbitrary spectrum. When the eigenvalues of $\mathbf{A}$ are grouped into tight clusters with only a few isolated outliers, CG eliminates the outlying eigenmodes in the first few iterations and then converges at a rate governed by the clustered eigenvalues. This behavior is termed superlinear convergence.

The convergence bound suggests changing coordinates so that different directions have more comparable scales. An SPD matrix $\mathbf{M}$ supplies such a change through its factorization $\mathbf{M} = \mathbf{C}\mathbf{C}^T$.

We analyze CG in the transformed coordinates, but implement it using solves with $\mathbf{M}$. The construction below explains why those solves preserve the required symmetry.

Def 6.14 **(Preconditioned Conjugate Gradient (PCG))** Let $\mathbf{M}$ be a symmetric positive definite preconditioner. Write $\mathbf{M} = \mathbf{C}\mathbf{C}^T$. PCG is ordinary CG applied to the SPD system $\mathbf{C}^{-1}\mathbf{A}\mathbf{C}^{-T}\boldsymbol{y} = \mathbf{C}^{-1}\boldsymbol{b}$, with $\boldsymbol{x} = \mathbf{C}^{-T}\boldsymbol{y}$. This is essential because $\mathbf{M}^{-1}\mathbf{A}$ need not be symmetric in the Euclidean inner product. Initialize $\boldsymbol{z}^{(0)}$ by solving $\mathbf{M}\boldsymbol{z}^{(0)} = \boldsymbol{r}^{(0)}$ and set $\boldsymbol{p}^{(0)} = \boldsymbol{z}^{(0)}$. Use the usual solution and residual updates, then solve for the new $\boldsymbol{z}$ and set $\boldsymbol{p}^{(k+1)} = \boldsymbol{z}^{(k+1)} + \beta_k \boldsymbol{p}^{(k)}$. At each step, a linear system $\mathbf{M}\boldsymbol{z}^{(k)} = \boldsymbol{r}^{(k)}$ is solved, and the step lengths and direction updates become

$$\alpha_k = \frac{\boldsymbol{r}^{(k)T}\boldsymbol{z}^{(k)}}{\boldsymbol{p}^{(k)T}\mathbf{A}\boldsymbol{p}^{(k)}}, \qquad \beta_k = \frac{\boldsymbol{r}^{(k+1)T}\boldsymbol{z}^{(k+1)}}{\boldsymbol{r}^{(k)T}\boldsymbol{z}^{(k)}}.$$

The convergence bound applies to the SPD matrix $\mathbf{C}^{-1}\mathbf{A}\mathbf{C}^{-T}$. A useful preconditioner improves its eigenvalue distribution while remaining inexpensive to solve with. The cost of constructing and applying the preconditioner must be included in the comparison.

6.4.6 Further Exercises

Ex 6.30 Implement the Preconditioned Conjugate Gradient algorithm in Python using an incomplete Cholesky preconditioner, and test it on a 2D discrete Laplacian of size 100×100.

6.5 The Generalized Minimal Residual (GMRES) Method

For a nonsymmetric matrix, the SPD energy argument no longer supplies an objective to minimize. The residual norm remains meaningful: we can choose the available correction that best satisfies the equations. GMRES applies this choice to an expanding Krylov subspace (Saad and Schultz, 1986).

Arnoldi provides orthonormal coordinates for the corrections and their images under $\mathbf{A}$. In those coordinates, the minimization becomes a small least-squares problem. We derive that reduction, update its solution with plane rotations, and examine what is lost when the stored basis is discarded at a restart.

6.5.1 Reduction to Hessenberg Least Squares

Let $\boldsymbol{r}^{(0)} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(0)}$ be nonzero and let $\beta = \|\boldsymbol{r}^{(0)}\|_2$. We run the Arnoldi iteration of Def 4.13 from $\boldsymbol{v}_1 = \boldsymbol{r}^{(0)}/\beta$. By Thm 4.25, after k steps the columns of $\mathbf{V}_k$ form an orthonormal basis of $\mathcal{K}_k(\mathbf{A}, \boldsymbol{r}^{(0)})$ and $\mathbf{A}\mathbf{V}_k = \mathbf{V}_{k+1}\bar{\mathbf{H}}_k$. These two facts reduce the residual minimization in $\mathbb{R}^n$ to a small least-squares problem.

At the first step, the correction has the form $\alpha\boldsymbol{r}^{(0)}$. We therefore minimize $\|\boldsymbol{r}^{(0)} - \alpha\mathbf{A}\boldsymbol{r}^{(0)}\|_2$. The next exercise solves this scalar problem. Later steps allow more correction directions, but apply the same least-squares principle.

Ex 6.31 Assume $\mathbf{A}\boldsymbol{r}^{(0)} \neq \mathbf{0}$. Expand the squared residual and derive

$$\alpha = \frac{(\mathbf{A}\boldsymbol{r}^{(0)})^T\boldsymbol{r}^{(0)}}{\|\mathbf{A}\boldsymbol{r}^{(0)}\|_2^2}.$$

Compare this with the first CG step for an SPD matrix. Which quantity is each method minimizing? Find an invertible matrix and a nonzero residual for which this first GMRES correction is zero.

Thm 6.15 **(The GMRES Projected Least-Squares Problem)** Assume the Arnoldi process has completed k steps without breakdown. Let any vector in the affine space $\boldsymbol{x}^{(0)} + \mathcal{K}_k(\mathbf{A}, \boldsymbol{r}^{(0)})$ be parameterized as $\boldsymbol{x} = \boldsymbol{x}^{(0)} + \mathbf{V}_k \boldsymbol{y}$ for some vector $\boldsymbol{y} \in \mathbb{R}^k$. The residual vector satisfies

$$\boldsymbol{r}(\boldsymbol{x}) = \boldsymbol{b} - \mathbf{A}(\boldsymbol{x}^{(0)} + \mathbf{V}_k \boldsymbol{y}) = \boldsymbol{r}^{(0)} - \mathbf{A}\mathbf{V}_k \boldsymbol{y} = \beta \boldsymbol{v}_1 - \mathbf{V}_{k+1} \bar{\mathbf{H}}_k \boldsymbol{y} = \mathbf{V}_{k+1}(\beta \boldsymbol{e}_1 - \bar{\mathbf{H}}_k \boldsymbol{y}).$$

Because the columns of $\mathbf{V}_{k+1}$ are orthonormal, $\mathbf{V}_{k+1}$ is an isometry in the Euclidean norm:

$$\|\boldsymbol{b} - \mathbf{A}\boldsymbol{x}\|_2 = \|\beta \boldsymbol{e}_1 - \bar{\mathbf{H}}_k \boldsymbol{y}\|_2,$$

where $\boldsymbol{e}_1 = (1, 0, \ldots, 0)^T \in \mathbb{R}^{k+1}$. The GMRES iterate is $\boldsymbol{x}^{(k)} = \boldsymbol{x}^{(0)} + \mathbf{V}_k \boldsymbol{y}_k$, where $\boldsymbol{y}_k$ uniquely solves the $(k+1) \times k$ upper Hessenberg least-squares problem

$$\boldsymbol{y}_k = \arg\min_{\boldsymbol{y} \in \mathbb{R}^k} \|\beta \boldsymbol{e}_1 - \bar{\mathbf{H}}_k \boldsymbol{y}\|_2.$$

Proof The unknown in the reduced problem is the coefficient vector $\boldsymbol{y}$, not the full vector $\boldsymbol{x}$. Since $\mathbf{V}_k$ is a basis of the correction space, every admissible approximation has a unique representation $\boldsymbol{x} = \boldsymbol{x}^{(0)} + \mathbf{V}_k \boldsymbol{y}$.

Substitute it into the residual and keep track of the dimensions. The Arnoldi relation uses $\bar{\mathbf{H}}_k \in \mathbb{R}^{(k+1)\times k}$ and $\mathbf{V}_{k+1} \in \mathbb{R}^{n \times (k+1)}$:

$$\begin{aligned} \boldsymbol{b} - \mathbf{A}\boldsymbol{x} &= \boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(0)} - \mathbf{A}\mathbf{V}_k \boldsymbol{y} \\ &= \boldsymbol{r}^{(0)} - \mathbf{V}_{k+1} \bar{\mathbf{H}}_k \boldsymbol{y} \\ &= \beta \boldsymbol{v}_1 - \mathbf{V}_{k+1} \bar{\mathbf{H}}_k \boldsymbol{y} \\ &= \mathbf{V}_{k+1}(\beta \boldsymbol{e}_1 - \bar{\mathbf{H}}_k \boldsymbol{y}). \end{aligned}$$

The last equality uses $\mathbf{V}_{k+1} \boldsymbol{e}_1 = \boldsymbol{v}_1$.

Put $\boldsymbol{z} = \beta \boldsymbol{e}_1 - \bar{\mathbf{H}}_k \boldsymbol{y}$. Orthonormality gives

$$\|\mathbf{V}_{k+1} \boldsymbol{z}\|_2^2 = \boldsymbol{z}^T \mathbf{V}_{k+1}^T \mathbf{V}_{k+1} \boldsymbol{z} = \boldsymbol{z}^T \boldsymbol{z} = \|\boldsymbol{z}\|_2^2.$$

Taking square roots proves equality of the full and reduced residual norms for every coefficient vector. Hence minimizing one is exactly equivalent to minimizing the other.

For uniqueness, suppose $\bar{\mathbf{H}}_k \boldsymbol{y} = \mathbf{0}$. Its last row gives $h_{k+1,k} y_k = 0$. Before breakdown this subdiagonal entry is nonzero, so $y_k = 0$. The preceding row then gives $h_{k,k-1} y_{k-1} = 0$, so $y_{k-1} = 0$. Continue upwards to obtain all entries zero. Thus the reduced matrix has full column rank and its least-squares minimizer is unique. This is the standard GMRES reduction (Saad and Schultz, 1986; Saad, 2003). □

Ex 6.32 Before Arnoldi breakdown, show that $\bar{\mathbf{H}}_k$ has full column rank. Start with its last row and use the nonzero subdiagonal entries to show that $\bar{\mathbf{H}}_k \boldsymbol{y} = \mathbf{0}$ implies $\boldsymbol{y} = \mathbf{0}$. Then use least-squares orthogonality to prove $\boldsymbol{r}^{(k)} \perp \mathbf{A}\mathcal{K}_k$. Explain why enlarging the subspace cannot increase the minimum residual norm.

Thm 6.16 **(Polynomial Residual Characterization and Spectral Bound)** The k-th GMRES residual satisfies the polynomial minimization property

$$\|\boldsymbol{r}^{(k)}\|_2 = \min_{p\in\mathcal{P}_k,\, p(0)=1} \|p(\mathbf{A})\boldsymbol{r}^{(0)}\|_2.$$

If $\mathbf{A}$ is diagonalizable with $\mathbf{A} = \mathbf{X}\boldsymbol{\Lambda}\mathbf{X}^{-1}$, where $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then the relative residual is bounded by

$$\frac{\|\boldsymbol{r}^{(k)}\|_2}{\|\boldsymbol{r}^{(0)}\|_2} \le \kappa_2(\mathbf{X}) \min_{p\in\mathcal{P}_k,\, p(0)=1} \max_{\lambda\in\sigma(\mathbf{A})} |p(\lambda)|.$$

Proof Write a general Krylov correction using its generating powers:

$$\boldsymbol{x} - \boldsymbol{x}^{(0)} = \sum_{j=0}^{k-1} c_j \mathbf{A}^j \boldsymbol{r}^{(0)} = q(\mathbf{A})\boldsymbol{r}^{(0)}, \qquad q(t) = \sum_{j=0}^{k-1} c_j t^j.$$

Multiplication by $\mathbf{A}$ and subtraction from the initial residual gives

$$\begin{aligned} \boldsymbol{r}(\boldsymbol{x}) &= \boldsymbol{r}^{(0)} - \mathbf{A}q(\mathbf{A})\boldsymbol{r}^{(0)} \\ &= [\mathbf{I} - \mathbf{A}q(\mathbf{A})]\boldsymbol{r}^{(0)} = p(\mathbf{A})\boldsymbol{r}^{(0)}, \end{aligned}$$

where $p(t) = 1 - tq(t)$ has degree at most k and satisfies $p(0) = 1$.

Conversely, let $p(t) = 1 + a_1 t + \cdots + a_k t^k$. Then $q(t) = -a_1 - a_2 t - \cdots - a_k t^{k-1}$ is a polynomial and $p(t) = 1 - tq(t)$. Thus every normalized polynomial in the statement corresponds to an admissible correction. GMRES minimizes over exactly this set, proving the first equality.

Now assume diagonalizability, $\mathbf{A} = \mathbf{X}\boldsymbol{\Lambda}\mathbf{X}^{-1}$. Induction gives $\mathbf{A}^j = \mathbf{X}\boldsymbol{\Lambda}^j\mathbf{X}^{-1}$, and substituting these powers into a polynomial gives $p(\mathbf{A}) = \mathbf{X}p(\boldsymbol{\Lambda})\mathbf{X}^{-1}$. For any admissible p,

$$\begin{aligned} \|p(\mathbf{A})\boldsymbol{r}^{(0)}\|_2 &\le \|\mathbf{X}\|_2 \|p(\boldsymbol{\Lambda})\|_2 \|\mathbf{X}^{-1}\|_2 \|\boldsymbol{r}^{(0)}\|_2 \\ &= \kappa_2(\mathbf{X}) \max_i |p(\lambda_i)| \|\boldsymbol{r}^{(0)}\|_2. \end{aligned}$$

The Euclidean norm of a diagonal matrix is its largest diagonal magnitude, which explains the last equality. GMRES has residual no larger than that of this particular candidate. Taking the minimum of the upper bound over admissible polynomials and dividing by the nonzero initial residual norm proves the spectral estimate. If the initial residual is zero, the system is already solved and no relative ratio is needed. □

6.5.2 Givens Rotations

Thm 6.17 **(Incremental QR Factorization of the Hessenberg Matrix)** At step k, applying the previously accumulated Givens rotations $\mathbf{G}_1, \ldots, \mathbf{G}_{k-1}$ transforms the first $k-1$ entries of the new column of $\bar{\mathbf{H}}_k$. The subdiagonal entry at position $(k+1, k)$ is then eliminated using a new Givens rotation $\mathbf{G}_k$ defined by

$$\mathbf{G}_k = \begin{pmatrix} c_k & s_k \\ -s_k & c_k \end{pmatrix}, \qquad c_k = \frac{h_{kk}^{(k-1)}}{\sqrt{(h_{kk}^{(k-1)})^2 + h_{k+1,k}^2}}, \quad s_k = \frac{h_{k+1,k}}{\sqrt{(h_{kk}^{(k-1)})^2 + h_{k+1,k}^2}}.$$

Applying $\mathbf{G}_k \dots \mathbf{G}_1$ transforms $\bar{\mathbf{H}}_k$ into an upper triangular matrix $\mathbf{R}_k$ with a zero final row, and transforms the right-hand side vector $\beta \boldsymbol{e}_1$ into $\boldsymbol{g}_{k+1} = (\gamma_1, \dots, \gamma_k, \gamma_{k+1})^T$. The residual norm at iteration k is obtained without forming $\boldsymbol{x}^{(k)}$:

$$\|\boldsymbol{r}^{(k)}\|_2 = |\gamma_{k+1}|.$$

Proof Each plane rotation acts as an orthogonal matrix on the full $(k+1)$-dimensional space by leaving the other coordinates unchanged. Their product $\mathbf{Q}_k = \mathbf{G}_k \cdots \mathbf{G}_1$ is therefore orthogonal and preserves Euclidean norms.

The first $k-1$ rotations are already known from the preceding step. Applying them to the new column leaves only one possible subdiagonal entry. The new rotation removes it. Thus

$$\mathbf{Q}_k \bar{\mathbf{H}}_k = \begin{pmatrix} \mathbf{R}_k \\ \mathbf{0}^T \end{pmatrix}, \qquad \mathbf{Q}_k (\beta \boldsymbol{e}_1) = \begin{pmatrix} \boldsymbol{g}_{1:k} \\ \gamma_{k+1} \end{pmatrix}.$$

Here $\mathbf{R}_k$ is $k \times k$. It is nonsingular because rotations preserve rank and $\bar{\mathbf{H}}_k$ has full column rank before breakdown.

For any coefficient vector $\boldsymbol{y}$, preserve the norm by applying $\mathbf{Q}_k$, then separate the last coordinate:

$$\begin{aligned} \|\beta \boldsymbol{e}_1 - \bar{\mathbf{H}}_k \boldsymbol{y}\|_2^2 &= \left\| \begin{pmatrix} \boldsymbol{g}_{1:k} - \mathbf{R}_k \boldsymbol{y} \\ \gamma_{k+1} \end{pmatrix} \right\|_2^2 \\ &= \|\boldsymbol{g}_{1:k} - \mathbf{R}_k \boldsymbol{y}\|_2^2 + |\gamma_{k+1}|^2. \end{aligned}$$

The last summand does not depend on $\boldsymbol{y}$, so it is a lower bound for the squared residual. Solve the triangular system $\mathbf{R}_k \boldsymbol{y} = \boldsymbol{g}_{1:k}$ to make the first summand zero. Hence that lower bound is attained, and the minimum residual norm is exactly $|\gamma_{k+1}|$.

Only the transformed right-hand side is needed to read this norm. Forming the full iterate $\boldsymbol{x}^{(0)} + \mathbf{V}_k \boldsymbol{y}$ can wait until the stopping test is satisfied. □

Remark 6.13 **(Happy breakdown vs. stagnation)** If the subdiagonal entry $h_{k+1,k}$ vanishes, Cor 4.27 shows that the Krylov subspace is invariant under $\mathbf{A}$. When $\mathbf{A}$ is nonsingular, its restriction to this invariant subspace is nonsingular, so the GMRES least-squares problem has residual zero and $\boldsymbol{x}^{(k)}$ is the exact solution. Termination for this reason is called happy breakdown. Stagnation is a different phenomenon. The Krylov space may continue to grow while its image under $\mathbf{A}$ contributes no useful correction to the residual. Non-normality, an unfavorable polynomial approximation problem, or a poor preconditioner can all cause this behavior, and the residual history alone does not identify which one is responsible.

6.5.3 Restarted GMRES and Practical Considerations

Def 6.15 **(Restarted GMRES(m))** Full GMRES requires storing k basis vectors of length n, incurring $O(kn)$ memory storage and $O(k^2 n)$ orthogonalization flops. To avoid memory exhaustion on large systems, the algorithm is restarted every m steps:

1. Run m steps of GMRES starting from $\boldsymbol{x}^{(0)}$ to produce $\boldsymbol{x}^{(m)}$.
2. Compute the residual $\boldsymbol{r}^{(m)} = \boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(m)}$.

3. Clear the Krylov basis vectors, set $\boldsymbol{x}^{(0)} \leftarrow \boldsymbol{x}^{(m)}$, and repeat.

Ex 6.33 **(Left versus right preconditioning)** Compare left preconditioning ($\mathbf{M}^{-1}\mathbf{A}\boldsymbol{x} = \mathbf{M}^{-1}\boldsymbol{b}$) and right preconditioning ($\mathbf{A}\mathbf{M}^{-1}\boldsymbol{y} = \boldsymbol{b}$, $\boldsymbol{x} = \mathbf{M}^{-1}\boldsymbol{y}$). Show that right-preconditioned GMRES minimizes the true residual $\|\boldsymbol{b} - \mathbf{A}\boldsymbol{x}\|_2$, whereas left-preconditioned GMRES minimizes $\|\mathbf{M}^{-1}(\boldsymbol{b} - \mathbf{A}\boldsymbol{x})\|_2$. Explain why a code that uses left preconditioning may need to monitor both quantities.

Remark 6.14 **(The trade-off of restarting)** Restarted GMRES(m) bounds memory usage to $O(mn)$ and bounds orthogonalization cost per cycle. However, restarting discards the accumulated Krylov subspace and destroys the global optimality property. If m is chosen too small, the algorithm may stagnate completely.

6.5.4 Further Exercises

Ex 6.34 **(Arnoldi factorization by hand)** For the matrix and initial residual

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}, \qquad \boldsymbol{r}^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

perform two steps of the Arnoldi iteration by hand.

1. Compute $\boldsymbol{v}_1$ and $\boldsymbol{w}_1 = \mathbf{A}\boldsymbol{v}_1$. Find h_{11}, h_{21} and $\boldsymbol{v}_2$.
2. Compute $\boldsymbol{w}_2 = \mathbf{A}\boldsymbol{v}_2$. Find h_{12}, h_{22}, h_{32} and $\boldsymbol{v}_3$.
3. Explicitly assemble the matrices $\mathbf{V}_2$ and the 3×2 upper Hessenberg matrix $\bar{\mathbf{H}}_2$.

Ex 6.35 **(Hessenberg QR and residual norm via Givens)** Using the 3×2 matrix $\bar{\mathbf{H}}_2$ from the preceding Arnoldi exercise:

1. Form the Givens rotation $\mathbf{G}_1$ that eliminates h_{21}, and the rotation $\mathbf{G}_2$ that eliminates h_{32}.
2. Apply these rotations to the right-hand side vector $\beta\boldsymbol{e}_1 = (\sqrt{2}, 0, 0)^T$.
3. Read off the residual norm $\|\boldsymbol{r}^{(2)}\|_2$ from the transformed vector without computing $\boldsymbol{x}^{(2)}$.

Ex 6.36 **(GMRES polynomial minimax on an ellipse)** Suppose the eigenvalues of $\mathbf{A}$ are enclosed within an ellipse in the complex plane centered at $c > 0$ with focal distance $d < c$ that does not contain the origin.

1. Using the scaled Chebyshev polynomial $p(z) = T_k((c - z)/d)/T_k(c/d)$, show that $p(0) = 1$.
2. Apply Thm 6.16 to derive an explicit upper bound for $\|\boldsymbol{r}^{(k)}\|_2/\|\boldsymbol{r}^{(0)}\|_2$ in terms of the semi-axes of the ellipse.

Ex 6.37 **(Exact stagnation for a cyclic permutation)** Let $\mathbf{A}$ be the cyclic permutation matrix defined by $\mathbf{A}\boldsymbol{e}_j = \boldsymbol{e}_{j+1}$ for $j < n$ and $\mathbf{A}\boldsymbol{e}_n = \boldsymbol{e}_1$. Take $\boldsymbol{x}^{(0)} = \mathbf{0}$ and $\boldsymbol{b} = \boldsymbol{e}_1$.

1. Prove that $\mathcal{K}_k(\mathbf{A}, \boldsymbol{b}) = \operatorname{span}\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_k\}$ for $k < n$.
2. Show that $\mathbf{A}\mathcal{K}_k = \operatorname{span}\{\boldsymbol{e}_2, \ldots, \boldsymbol{e}_{k+1}\}$ is orthogonal to $\boldsymbol{b}$. Conclude that $\|\boldsymbol{r}^{(k)}\|_2 = 1$ for every $k < n$.
3. Show that $\boldsymbol{x} = \boldsymbol{e}_n$ becomes available at step n and gives zero residual. Construct the degree-n residual polynomial that vanishes on the spectrum of $\mathbf{A}$, and explain why no polynomial of smaller degree can do so.

6.6 Multigrid Methods

A local iteration can quickly remove oscillations from an error while barely changing a smooth error. A coarse grid represents that smooth part with fewer unknowns. Multigrid combines the inexpensive local updates with corrections computed on successively coarser grids. We first analyze the updates, then derive the equation that the coarse correction must solve.

6.6.1 Oscillating Error and the Smoothing Property

Weighted Jacobi and Gauss–Seidel converge slowly as stand-alone solvers because smooth error components decay slowly. Their value as smoothers is that they reduce oscillatory error in a few sweeps. Unweighted Jacobi has a high-frequency exception, derived below. A Fourier analysis of the error makes this asymmetry precise, and multigrid is built on it. The smoother removes the oscillatory error components, and the remaining smooth error is passed to a coarser grid, where it becomes oscillatory and cheap to resolve.

Frequency-Dependent Error Decay in Jacobi Iteration For the 1D model problem, the Jacobi update reads

$$x_i^{(k+1)} = \tfrac{1}{2}(x_{i-1}^{(k)} + x_{i+1}^{(k)}) + \tfrac{1}{2}b_i.$$

Each new value is the *average of its two neighbors* plus a source term. Consider two extreme error profiles at some iteration:

- *Oscillatory error*: neighboring values often have opposite signs, so averaging damps many high-frequency modes rapidly. The exact alternating mode is an exception for unweighted Jacobi: its sign flips without shrinking. Weighted Jacobi or Gauss–Seidel supplies the smoothing used in multigrid.
- *Smooth error* ($e_i \propto \sin(ij\pi h)$ for small mode index j): neighbors of node i carry nearly the same value as e_i itself, so their average barely moves things. The error barely changes per sweep.

Jacobi acts as a local *averaging filter* that pulls each value toward the mean of its neighbors. Oscillations cancel in that average, whereas smooth trends do not. Starting from a random initial guess, the error spans all frequencies. With weighted Jacobi or Gauss–Seidel, the oscillatory error is reduced first. The remaining smooth error motivates a correction on a coarser grid. Error and residual have different spectral weights, so both should be interpreted through the operator.

Ex 6.38 **(Observing oscillating error)** Set $n = 63$, $\boldsymbol{b} = \mathbf{0}$, and $\boldsymbol{x}^* = \mathbf{0}$, so the error equals the iterate exactly. Run Jacobi and Gauss–Seidel from the same random initial vector $\boldsymbol{x}^{(0)} \sim \mathcal{N}(0, \mathbf{I})$.

1. Plot the error profile $\boldsymbol{e}^{(k)} = \boldsymbol{x}^{(k)}$ at iterations $k = 0, 1, 3, 10, 20$. Describe how the spectral composition of the error changes. Observe the behavior of the high-frequency components.
2. Decompose $\boldsymbol{e}^{(k)}$ into Fourier mode amplitudes $\hat{e}_j^{(k)} = \langle \boldsymbol{e}^{(k)}, \phi^{(j)} \rangle / \|\phi^{(j)}\|^2$ and plot $|\hat{e}_j^{(k)}|$ vs. j at the same iterations. Which modes decay fast, which persist?
3. The Jacobi iteration matrix has eigenvalues $\mu_j = \cos(j\pi h)$. Use this to predict the amplitude of mode j after k sweeps as a function of the initial amplitude $|\hat{e}_j^{(0)}|$. Does your plot match the prediction?

4. After 20 sweeps, what does the remaining error look like? Argue, using the eigenvalue formula, why no fine-grid iteration alone can efficiently eliminate it, necessitating a coarse-grid correction.

Def 6.16 **(Fourier Modes on a Uniform Grid)** Consider the 1D model problem with the $n \times n$ tridiagonal matrix $\mathbf{A}$ having 2 on the main diagonal and -1 on the off-diagonals (the finite-difference discretization of $-u'' = f$ on $[0,1]$ scaled by h^2). Let $h = 1/(n+1)$. For the nested grids below take $n = 2m+1$. The *Fourier modes* are the vectors

$$\phi_j^{(k)} = \sin(kj\pi h), \quad j = 1, \dots, n, \quad k = 1, \dots, n.$$

These form an orthogonal basis for $\mathbb{R}^n$ (by discrete sine orthogonality) and diagonalize $\mathbf{A}$:

$$\mathbf{A}\phi^{(k)} = \lambda_k \, \phi^{(k)}, \qquad \lambda_k = 2 - 2\cos(k\pi h) = 4\sin^2\Big(\frac{k\pi h}{2}\Big).$$

Small k yields long-wavelength (smooth) modes, while large k yields short-wavelength (oscillatory) modes.

Thm 6.18 **(Eigenvalues of the Jacobi Iteration Matrix)** For the 1D model problem, the Jacobi iteration matrix $\mathbf{T}_J = \mathbf{I} - \frac{1}{2}\mathbf{A}$ has eigenvalues

$$\mu_k = \cos(k\pi h), \quad k = 1, \dots, n,$$

with eigenvectors $\phi^{(k)}$. After ν Jacobi sweeps, mode k of the error is scaled by $\mu_k^\nu = \cos^\nu(k\pi h)$.

Proof Since $\mathbf{D} = 2\mathbf{I}$, the Jacobi iteration matrix is $\mathbf{T}_J = \mathbf{I} - \frac{1}{2}\mathbf{A}$. Applying $\mathbf{T}_J$ to the eigenvector $\phi^{(k)}$:

$$\begin{aligned}\mathbf{T}_J\phi^{(k)} &= \left(\mathbf{I} - \frac{1}{2}\mathbf{A}\right)\phi^{(k)} = \phi^{(k)} - \frac{1}{2}\mathbf{A}\phi^{(k)} \\ &= \phi^{(k)} - \frac{1}{2}\lambda_k\phi^{(k)} = \left(1 - \frac{\lambda_k}{2}\right)\phi^{(k)}.\end{aligned}$$

This calculation uses a general fact: if $\mathbf{A}\boldsymbol{v} = \lambda\boldsymbol{v}$, then $(\mathbf{I} - c\mathbf{A})\boldsymbol{v} = (1 - c\lambda)\boldsymbol{v}$. The eigenvector is unchanged, and only its eigenvalue changes. Here $c = 1/2$.

Substituting $\lambda_k = 2 - 2\cos(k\pi h)$ yields

$$1 - \frac{\lambda_k}{2} = 1 - (1 - \cos(k\pi h)) = \cos(k\pi h).$$

For an initial error $\boldsymbol{e}^{(0)} = \sum_k c_k\phi^{(k)}$, applying the iteration ν times gives $\boldsymbol{e}^{(\nu)} = \sum_k c_k\mu_k^\nu\phi^{(k)}$. This follows by applying the matrix to each basis vector separately and using linearity. It justifies the statement about each mode after repeated sweeps.

Therefore $\mathbf{T}_J\phi^{(k)} = \cos(k\pi h)\phi^{(k)}$. □

Def 6.17 **(Low- and High-Frequency Modes)** With respect to a grid of mesh size $h = 1/(n+1)$, partition the n Fourier modes by frequency:

- *Low-frequency (smooth) modes*: $k = 1, \dots, \lfloor n/2 \rfloor$, where $k\pi h \in (0, \pi/2)$. These satisfy $\mu_k = \cos(k\pi h) \in (0,1)$, and for small k, $\mu_k \approx 1 - (k\pi h)^2/2 \approx 1$, so these modes decay slowly.

- *High-frequency (oscillatory) modes*: $k = \lfloor n/2 \rfloor + 1, \ldots, n$, where $k\pi h \in [\pi/2, \pi)$.

Remark 6.15 Standard Jacobi ($\omega = 1$) is a poor smoother, because the highest-frequency mode ($k = n$) satisfies $\mu_n = \cos(n\pi h) = \cos(\pi n/(n+1)) \approx -1$, so the iterate oscillates in sign rather than decaying. *Weighted (damped) Jacobi* with $\omega = 2/3$ fixes this by controlling the entire high-frequency band.

Def 6.18 **(Weighted Jacobi Iteration)** The *weighted Jacobi* method introduces a damping parameter $\omega \in (0, 1)$:

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + \omega \mathbf{D}^{-1}(\boldsymbol{b} - \mathbf{A}\boldsymbol{x}^{(k)}),$$

with iteration matrix $\mathbf{T}_\omega = \mathbf{I} - \omega \mathbf{D}^{-1}\mathbf{A}$ and eigenvalues

$$\mu_k(\omega) = 1 - \omega(1 - \cos(k\pi h)).$$

The standard multigrid choice is $\omega = 2/3$.

Thm 6.19 **(Smoothing Property of Weighted Jacobi)** For the 1D model problem with $\omega = 2/3$, the *smoothing factor*

$$\bar{\mu} = \max_{\lfloor n/2 \rfloor < k \le n} |\mu_k(2/3)| \le \frac{1}{3}.$$

That is, every high-frequency component of the error is reduced by at least $2/3$ in a single weighted Jacobi step.

Proof The weighted Jacobi error matrix is

$$\mathbf{T}_\omega = \mathbf{I} - \omega \mathbf{D}^{-1}\mathbf{A}.$$

For this matrix $\mathbf{D} = 2\mathbf{I}$. A sine mode of index k has eigenvalue $2 - 2\cos(k\pi h)$ under $\mathbf{A}$. Substituting it into the error matrix gives its amplification factor

$$\begin{aligned} \mu_k(\omega) &= 1 - \tfrac{\omega}{2}[2 - 2\cos(k\pi h)] \\ &= 1 - \omega + \omega\cos(k\pi h). \end{aligned}$$

For $\omega = 2/3$, this is $1/3 + (2/3)\cos(k\pi h)$.

Because $n = 2m + 1$ and $h = 1/(n+1)$, a high-frequency index satisfies $k \ge m + 1$. Hence $k\pi h \in [\pi/2, \pi)$ and $-1 < \cos(k\pi h) \le 0$. It follows that

$$-\tfrac{1}{3} < \mu_k(2/3) \le \tfrac{1}{3}, \qquad |\mu_k(2/3)| \le \tfrac{1}{3}.$$

After ν sweeps, the coefficient of this mode is multiplied by $\mu_k(2/3)^\nu$, whose magnitude is at most $3^{-\nu}$. The sine modes form an orthogonal basis, so the same norm contraction holds for any combination restricted to this high-frequency band. No such bound is being asserted here for the low-frequency band. □

Remark 6.16 After ν weighted Jacobi sweeps, every high-frequency error component is reduced by $(1/3)^\nu$. The remaining error is dominated by *smooth* modes, which are nearly constant on the scale of h. This smooth, slowly-varying error is the target of the coarse-grid correction in multigrid.

Example **Eigenvalue spectrum of $\mathbf{T}_{2/3}$ for $n = 15$, $h = 1/16$.**
The eigenvalues $\mu_k = 1 - \frac{2}{3}(1 - \cos(k\pi/16))$ for selected k:

- $k = 1$ (smoothest): $\mu_1 \approx 0.981$ (minimal decay per sweep).
- $k = 4$: $\mu_4 = 1 - \frac{2}{3}(1 - \cos(\pi/4)) \approx 0.804$ (slow decay).
- $k = 8$: $\mu_8 = 1 - \frac{2}{3}(1 - \cos(\pi/2)) = 1/3 \approx 0.333$.
- $k = 12$: $\mu_{12} = 1 - \frac{2}{3}(1 - \cos(3\pi/4)) \approx -0.138$ (rapid decay).
- $k = 15$ (most oscillatory): $\mu_{15} \approx -0.330$ (magnitude bounded by $1/3$).

Modes $k \geq 8$ all satisfy $|\mu_k| \leq 1/3$, confirming the smoothing property. Modes $k \leq 7$ range from 0.63 to 0.98 and are essentially untouched by the smoother.

6.6.2 The Two-Grid Correction Scheme

A smooth error on a fine grid of mesh size h looks *oscillatory* on a coarser grid of mesh size $2h$. More precisely, the smooth fine-grid modes ($k < n/2$) oscillate at wavelength $> 2h$ and are well-represented on the coarse grid. Once the high-frequency components have been attenuated by the smoother, the remaining error lives in a low-dimensional smooth subspace that a direct solve on a coarse grid of size $n/2$ can eliminate efficiently.

Def 6.19 **(Prolongation Operator)** The *prolongation* (or interpolation) operator $\mathbf{P} : \mathbb{R}^m \to \mathbb{R}^n$ (where $m = (n-1)/2$ for grids satisfying $n = 2m+1$) transfers a coarse-grid vector to the fine grid by *linear interpolation*. For a coarse-grid vector $\boldsymbol{v} \in \mathbb{R}^m$, the fine-grid vector $\mathbf{P}\boldsymbol{v} \in \mathbb{R}^n$ is defined by

$$(\mathbf{P}\boldsymbol{v})_{2j} = v_j, \qquad (\mathbf{P}\boldsymbol{v})_{2j-1} = \tfrac{1}{2}(v_{j-1} + v_j), \quad j = 1, \dots, m,$$

with boundary conventions $v_0 = v_{m+1} = 0$ and the final midpoint value $(\mathbf{P}\boldsymbol{v})_{2m+1} = v_m/2$. In matrix form, $\mathbf{P}$ is $n \times m$ with entries 1 at coarse-grid points and $\frac{1}{2}$ weights at fine-grid midpoints.

Def 6.20 **(Restriction Operator)** The *restriction* operator $\mathbf{R} : \mathbb{R}^n \to \mathbb{R}^m$ transfers a fine-grid vector to the coarse grid. The standard *full-weighting restriction* averages each coarse point with its fine-grid neighbors:

$$(\mathbf{R}\boldsymbol{w})_j = \tfrac{1}{4}w_{2j-1} + \tfrac{1}{2}w_{2j} + \tfrac{1}{4}w_{2j+1}, \quad j = 1, \dots, m.$$

In matrix form, $\mathbf{R} = \frac{1}{2}\mathbf{P}^T$. This choice ensures that the Galerkin coarse-grid operator $\mathbf{A}_c = \mathbf{RAP}$ is SPD whenever $\mathbf{A}$ is SPD.

Remark 6.17 The relationship $\mathbf{R} = \frac{1}{2}\mathbf{P}^T$ is called the *variational principle* and preserves symmetry and positive definiteness of $\mathbf{A}_c$, since $\mathbf{P}$ has full column rank. Accuracy of the coarse space requires a separate approximation argument. For the physical scaling $\mathbf{A} = h^{-2}\,\mathrm{tridiag}(-1, 2, -1)$, $\mathbf{A}_c$ is the natural $2h$ finite-difference matrix. If the factor h^{-2} is omitted on the fine grid, the inherited coarse matrix is $\frac{1}{4}\,\mathrm{tridiag}(-1, 2, -1)$. This special property simplifies analysis but does not hold in general for variable-coefficient or higher-dimensional problems.

Let $\boldsymbol{\epsilon} = \boldsymbol{x}^* - \tilde{\boldsymbol{x}}$ be the correction we would like to make. Although it is unknown, its equation is known: $\mathbf{A}\boldsymbol{\epsilon} = \boldsymbol{r}$. Approximate the correction by a prolongated coarse vector $\mathbf{P}\boldsymbol{e}_c$, and require the restricted residual of this equation to vanish. This gives

$$\mathbf{R}(\boldsymbol{r} - \mathbf{AP}\boldsymbol{e}_c) = \mathbf{0}, \qquad (\mathbf{RAP})\boldsymbol{e}_c = \mathbf{R}\boldsymbol{r}.$$

This derives both the coarse matrix and its right-hand side. We transfer the residual because it supplies an equation for the unknown error.

Def 6.21 **(Two-Grid Correction Scheme)** Given $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ on the fine grid, one step of the *two-grid (TG) method* from an iterate $\boldsymbol{x}^{(0)}$ is:

1. **Pre-smoothing:** Apply $\nu_1 \geq 0$ smoother steps to get $\tilde{\boldsymbol{x}}$.
2. **Residual:** Compute $\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\tilde{\boldsymbol{x}}$.
3. **Restrict:** Form the coarse residual $\boldsymbol{r}_c = \mathbf{R}\boldsymbol{r} \in \mathbb{R}^m$.
4. **Coarse solve:** Solve $\mathbf{A}_c\boldsymbol{e}_c = \boldsymbol{r}_c$ exactly for $\boldsymbol{e}_c \in \mathbb{R}^m$.

5. **Prolong and correct:** Set $\hat{\boldsymbol{x}} = \tilde{\boldsymbol{x}} + \mathbf{P}\boldsymbol{e}_c$.

6. **Post-smoothing:** Apply $\nu_2 \geq 0$ smoother steps to $\hat{\boldsymbol{x}}$ to get $\boldsymbol{x}^{(1)}$.

The coarse-grid system $\mathbf{A}_c\boldsymbol{e}_c = \boldsymbol{r}_c$ solves the *residual equation*: if $\boldsymbol{x}^* - \tilde{\boldsymbol{x}}$ is smooth and lies in the range of $\mathbf{P}$, this recovers the error exactly.

Thm 6.20 **(Ideal Coarse-Grid Correction)** If the error $\boldsymbol{\epsilon} = \boldsymbol{x}^* - \tilde{\boldsymbol{x}}$ lies in the range of $\mathbf{P}$ (i.e., $\boldsymbol{\epsilon} = \mathbf{P}\boldsymbol{\delta}$ for some $\boldsymbol{\delta} \in \mathbb{R}^m$), then the two-grid correction recovers the exact solution: $\hat{\boldsymbol{x}} = \boldsymbol{x}^*$.

Proof Suppose the error before correction has the form $\boldsymbol{\epsilon} = \boldsymbol{x}^* - \tilde{\boldsymbol{x}} = \mathbf{P}\boldsymbol{\delta}$. The vector $\boldsymbol{\delta}$ need not be known. We show that the coarse equations recover it.

Use the exact equation $\mathbf{A}\boldsymbol{x}^* = \boldsymbol{b}$ to express the computable residual as

$$\boldsymbol{r} = \boldsymbol{b} - \mathbf{A}\tilde{\boldsymbol{x}} = \mathbf{A}(\boldsymbol{x}^* - \tilde{\boldsymbol{x}}) = \mathbf{A}\boldsymbol{\epsilon} = \mathbf{A}\mathbf{P}\boldsymbol{\delta}.$$

Restriction gives

$$\boldsymbol{r}_c = \mathbf{R}\boldsymbol{r} = \mathbf{R}\mathbf{A}\mathbf{P}\boldsymbol{\delta} = \mathbf{A}_c\boldsymbol{\delta}.$$

By definition the algorithm solves $\mathbf{A}_c\boldsymbol{e}_c = \boldsymbol{r}_c$. Thus $\boldsymbol{e}_c$ and $\boldsymbol{\delta}$ satisfy the same coarse linear system. The coarse matrix is nonsingular, so the solution is unique and $\boldsymbol{e}_c = \boldsymbol{\delta}$.

Now prolongate and add:

$$\hat{\boldsymbol{x}} = \tilde{\boldsymbol{x}} + \mathbf{P}\boldsymbol{e}_c = \tilde{\boldsymbol{x}} + \mathbf{P}\boldsymbol{\delta} = \tilde{\boldsymbol{x}} + (\boldsymbol{x}^* - \tilde{\boldsymbol{x}}) = \boldsymbol{x}^*.$$

This proves exact correction for errors in the range of $\mathbf{P}$. It does not claim that an arbitrary smooth error is exactly in that range. Approximation of other errors is a separate part of multigrid convergence theory (Briggs et al., 2000). □

Ex 6.39 Take $n = 7$, $\mathbf{A} = \text{tridiag}(-1, 2, -1)$, $\boldsymbol{b} = \mathbf{1}$, and $\boldsymbol{x}^{(0)} = \mathbf{0}$. Write out $\mathbf{P}$ and $\mathbf{R}$, and compute $\mathbf{A}_c = \mathbf{R}\mathbf{A}\mathbf{P}$. Compare the true residual norms after four weighted Jacobi sweeps and after a two-grid step with two pre- and two post-sweeps. Count the coarse solve as well as the smoothing work. Does a lower residual alone establish a lower total cost?

6.6.3 The Multigrid V-Cycle

The two-grid method replaces the fine-grid iteration matrix with one that damps all error components, at the cost of an exact coarse-grid solve. For large problems the coarse grid may itself be large. Multigrid applies the idea recursively. Instead of solving the coarse-grid problem exactly, it applies the two-grid scheme again at the coarser level, and so on, until the coarsest grid is trivially small (a single unknown). This gives the *V-cycle*.

Def 6.22 **(Multigrid Grid Hierarchy)** For $n = 2^L - 1$, define L nested grids with mesh sizes $h_\ell = 2^{-\ell}$, giving $n_\ell = 2^\ell - 1$ interior points at level $\ell = 1, \dots, L$. The finest grid is $\ell = L$ (mesh size h) and the coarsest is $\ell = 1$ (a single point). At each level ℓ we have a system matrix $\mathbf{A}_\ell$ (built via Galerkin: $\mathbf{A}_{\ell-1} = \mathbf{R}_\ell \mathbf{A}_\ell \mathbf{P}_\ell$), restriction $\mathbf{R}_\ell : \mathbb{R}^{n_\ell} \to \mathbb{R}^{n_{\ell-1}}$, and prolongation $\mathbf{P}_\ell : \mathbb{R}^{n_{\ell-1}} \to \mathbb{R}^{n_\ell}$.

Def 6.23 **(Multigrid V-Cycle)** $\texttt{MG}(\ell, \boldsymbol{x}, \boldsymbol{b})$: one multigrid V-cycle at level ℓ, returning an improved approximation to $\mathbf{A}_\ell \boldsymbol{x} = \boldsymbol{b}$:

1. **Base case** ($\ell = 1$): return $\mathbf{A}_1^{-1} \boldsymbol{b}$ (exact solve on a 1×1 system).
2. **Pre-smoothing:** $\boldsymbol{x} \leftarrow S_\ell^{\nu_1}(\boldsymbol{x}, \boldsymbol{b})$ (ν_1 steps of the smoother at level ℓ).
3. **Restrict residual:** $\boldsymbol{r}_{\ell-1} \leftarrow \mathbf{R}_\ell(\boldsymbol{b} - \mathbf{A}_\ell \boldsymbol{x})$.
4. **Coarse-grid correction:** $\boldsymbol{e}_{\ell-1} \leftarrow \texttt{MG}(\ell - 1, \mathbf{0}, \boldsymbol{r}_{\ell-1})$.
5. **Correct:** $\boldsymbol{x} \leftarrow \boldsymbol{x} + \mathbf{P}_\ell \boldsymbol{e}_{\ell-1}$.
6. **Post-smoothing:** $\boldsymbol{x} \leftarrow S_\ell^{\nu_2}(\boldsymbol{x}, \boldsymbol{b})$ (ν_2 steps of the smoother).
7. Return $\boldsymbol{x}$.

Calling MG once as the coarse-grid solver in step 4 gives the V-cycle. Calling it twice gives the *W-cycle*, which is more robust on difficult problems at higher cost.

Remark 6.18 A single V-cycle descends from the finest level L to the coarsest level 1 (the "down stroke") applying pre-smoothing and restriction, then ascends back (the "up stroke") applying prolongation and post-smoothing. Typical choices are $\nu_1 = \nu_2 = 1$ or 2. The W-cycle replaces the single recursive call in step 4 with two calls, more aggressively correcting smooth error with work governed by $W(n) = 2W(n/2) + O(n)$ in one dimension, hence $O(n \log n)$. In two or three dimensions, geometric coarsening gives a smaller recursion ratio and a W-cycle can still cost $O(n)$.

Remark 6.19 **(Mesh-independent convergence)** For the standard Dirichlet Poisson problem, suitable smoothing and coarse-grid approximation give a V-cycle contraction in the energy norm with a factor $q < 1$ independent of mesh size. This is a convergence theorem about the whole cycle, not a consequence of the smoothing estimate alone. Its proof is beyond the present development; see (Briggs et al., 2000). For a new discretization, transfer rule, or smoother, uniform convergence needs to be established again. The complexity estimate below is independent of that convergence question.

Ex 6.40 Let $\mathbf{C} = \mathbf{I} - \mathbf{P}(\mathbf{RAP})^{-1}\mathbf{RA}$. Prove that $\mathbf{CP} = \mathbf{0}$ and $\mathbf{C}^2 = \mathbf{C}$. For $\mathbf{R} = c\mathbf{P}^T$ with $c > 0$, show that $\mathbf{P}^T\mathbf{AC} = \mathbf{0}$ and that $\mathbf{C}$ is an orthogonal projection in the energy inner product. Deduce $\|\mathbf{C}\boldsymbol{e}\|_{\mathbf{A}} \le \|\boldsymbol{e}\|_{\mathbf{A}}$. Why does this only give a nonexpansive map, rather than a mesh-independent strict contraction?

Remark 6.20 For classical iterations on the 1D Poisson problem, $\rho(\mathbf{T}_J) = \cos(\pi h) \approx 1 - \pi^2 h^2/2 \to 1$ as $h \to 0$, requiring $O(h^{-2}) = O(n^2)$ Jacobi iterations for a fixed reduction of the slowest mode. With the V-cycle, the number of iterations to reach a fixed tolerance is bounded by a constant independent of n. The cost per V-cycle is $O(n)$ (geometric series over levels), making the total cost $O(n)$, which is the optimal asymptotic complexity for this class of problems.

Thm 6.21 **($O(n)$ Complexity of the V-Cycle)** Assume a fixed number of smoothing sweeps, $O(n_\ell)$ work for smoothing, residuals, and transfers at level ℓ, and geometric coarsening. For L levels with $n = n_L$ unknowns on the finest level, each V-cycle requires $O(n)$ floating-point operations, independent of the number of levels L.

Proof Let $n = n_L$ denote the number of unknowns on the finest level. The fixed number of sweeps means there is a constant C, independent of level and problem size, such that all local work at level ℓ is at most Cn_ℓ. This includes residuals and transfers as assumed in the statement.

In a V-cycle there is only one recursive call to the next coarser level. Thus every level is visited once on descent and ascent, with that work already included in C. The nested-grid sizes satisfy

$$n_\ell + 1 = 2^\ell = \frac{2^L}{2^{L-\ell}} = \frac{n+1}{2^{L-\ell}}.$$

Consequently $n_\ell \le (n+1)2^{-(L-\ell)}$. Summing the local bounds and the constant coarsest solve gives

$$\begin{aligned} W(n) &\le C \sum_{\ell=1}^{L} n_\ell + C_0 \\ &\le C(n+1) \sum_{j=0}^{L-1} 2^{-j} + C_0 \\ &\le 2C(n+1) + C_0. \end{aligned}$$

For $n \ge 1$, $n + 1 \le 2n$, so this is bounded by a constant times n. This proves the work bound for one V-cycle. A linear bound for the complete solution additionally requires a number of cycles bounded independently of the mesh. □

Example **Multigrid vs. classical iterations on the 2D Poisson problem.**
Discretize $-\Delta u = f$ on the unit square with an $n \times n$ grid ($h = 1/(n+1)$), giving $N = n^2$ unknowns. Approximate iteration counts to reduce the residual by 10^{-8}, and total work in units of N multiplications:

Method	Iterations	Work/iter	Total work
Jacobi	$O(n^2)$	$O(N)$	$O(N^2)$
Gauss-Seidel	$O(n^2)$	$O(N)$	$O(N^2)$
Optimal SOR	$O(n)$	$O(N)$	$O(N^{3/2})$
Multigrid V-cycle	$O(1)$	$O(N)$	$O(N)$

The table states asymptotic work for a fixed tolerance under the model assumptions. Actual cycle counts depend on the right-hand side, initial error, norm, and stopping test; measure them in the implementation exercises below.

Ex 6.41 **(Smoothing property)** For the $n = 15$ 1D model problem:

1. Compute the eigenvalues μ_k of the standard Jacobi ($\omega = 1$) and weighted Jacobi ($\omega = 2/3$) iteration matrices. Plot $|\mu_k|$ as a function of k on a single figure.

2. Starting from a random initial error $\boldsymbol{e}^{(0)} \in \mathbb{R}^{15}$, apply 3 steps of each method. Plot the initial and final errors, and describe qualitatively how each method reshapes the error.

3. Identify the "worst-case" mode (largest $|\mu_k|$) for each method over the high-frequency range $k > 7$. Why is the standard Jacobi smoother inadequate for multigrid?

Ex 6.42 **(Two-grid implementation)** Implement the two-grid correction scheme for the 1D model problem with $n = 2^m - 1$ interior points.

1. Write `restrict(r)` using the full-weighting formula and `prolongate(e)` using linear interpolation for transferring between a fine grid and the next coarser grid.

2. Implement `two_grid(A, b, x0, nu1, nu2)` with ν_1 pre-smoothing and ν_2 post-smoothing steps of weighted Jacobi ($\omega = 2/3$) and an exact coarse-grid solve via `np.linalg.solve`.

3. For $m = 6$ ($n = 63$), plot residual norm versus iteration count for: (a) weighted Jacobi alone, (b) repeated two-grid cycling. Use a semi-log scale. Estimate the per-cycle convergence factor ρ for the two-grid method.

Ex 6.43 **(Multigrid V-cycle)** Extend the two-grid implementation to a full V-cycle.

1. Implement `vcycle(x, b, A_list, level, nu1, nu2)` recursively. At level 1 (a single unknown), solve exactly. Test first on the two-grid case to verify it matches your two-grid implementation.

2. Run the V-cycle on the 1D Poisson problem with $n = 2^{10} - 1 = 1023$ and $\nu_1 = \nu_2 = 1$. Measure the empirical convergence factor $\rho_k = \|\boldsymbol{r}^{(k)}\|/\|\boldsymbol{r}^{(k-1)}\|$ for $k = 1, \ldots, 15$. Is it approximately constant (mesh-independent)?

3. Repeat for $n = 2^{12} - 1 = 4095$. Compare the convergence factors. Then vary $\nu_1 = \nu_2 \in \{1, 2, 3\}$ and report the convergence factor versus the extra cost per cycle.

Remark 6.21 **(Choosing a solver)** The structure of the coefficient matrix determines the method. The following summary collects the recommendations of the preceding sections.

1. **Dense symmetric positive definite.** Cholesky factorization (Thm 2.18), with $n^3/3$ flops and no pivoting.

2. **Large sparse symmetric positive definite.** Preconditioned conjugate gradients (Section 6.4) or multigrid, with $O(\mathrm{nnz}(\mathbf{A}))$ work per iteration.

3. **Dense general square.** Gaussian elimination with partial pivoting (Def 2.6), with $2n^3/3$ flops and the backward error of Thm 2.15.

4. **Large sparse general square.** Preconditioned GMRES (Section 6.5).

5. **Overdetermined with full column rank.** Householder QR (Thm 3.6), which avoids the squared condition number of the normal equations.

6. **Rank-deficient or nearly rank-deficient.** The SVD (Thm 4.22), which determines the numerical rank and the minimum-norm solution.

Read **Further reading.** Saad and Greenbaum give complementary treatments of Krylov methods and preconditioning (Saad, 2003; Greenbaum, 1997). The original conjugate-gradient and GMRES papers remain readable (Hestenes and Stiefel, 1952; Saad and Schultz, 1986). For multigrid, the tutorial by Briggs, Henson, and McCormick is a natural next step (Briggs et al., 2000).

6.7 Randomized Methods in Linear Algebra

A calculation may require only a sum of matrix entries or a few dominant directions. Computing an entire inverse or a full SVD can then do much more work than is needed. Random vectors let us estimate selected quantities through products and solves. We must specify what is random and how its error is measured.

We first estimate traces and diagonals from random probe vectors. We then compute low-rank approximations from the range of $\mathbf{A}\boldsymbol{\Omega}$ for a random matrix $\boldsymbol{\Omega}$. Finally, we show how random sampling limits fill during Gaussian elimination on graph Laplacians, which produces a preconditioner for the conjugate gradient method.

6.7.1 Trace and Diagonal Estimation

The trace of $\mathbf{A}^{-1}$ and the diagonal of $\mathbf{A}^{-1}$ appear as sums of variances in statistics, as effective resistances in graphs, and as sensitivities of linear models. Computing $\operatorname{diag}(\mathbf{A}^{-1})$ exactly requires solving $\mathbf{A}\boldsymbol{x} = \boldsymbol{e}_i$ for every i. The estimators below use only products $\boldsymbol{z} \mapsto \mathbf{A}\boldsymbol{z}$, or solves with $\mathbf{A}$, applied to a few random vectors $\boldsymbol{z}$.

For independent signs $z_i = \pm 1$ with equal probability, $z_i^2 = 1$ whereas $\mathbb{E}[z_i z_j] = 0$ for $i \neq j$. In a quadratic form, the diagonal contributions therefore survive averaging and the off-diagonal contributions cancel in expectation. A finite average still has random error. *Unbiased* means its expectation is correct, not that every sample is accurate.

Ex 6.44 For $\mathbf{A} = \begin{pmatrix} 2 & c \\ c & 3 \end{pmatrix}$, list the values of $\boldsymbol{z}^T\mathbf{A}\boldsymbol{z}$ for all four sign vectors. Compute their mean and variance. Explain why the mean is independent of c although the uncertainty is not.

Thm 6.22 **(Hutchinson's Trace Estimator)** Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be symmetric, and let $\boldsymbol{z} \in \mathbb{R}^n$ have independent entries with mean zero and variance one. Then $\mathbb{E}[\boldsymbol{z}^T\mathbf{A}\boldsymbol{z}] = \operatorname{tr}(\mathbf{A})$. If the entries of $\boldsymbol{z}$ are ± 1 with equal probability, then

$$\operatorname{Var}(\boldsymbol{z}^T\mathbf{A}\boldsymbol{z}) = 2\Big(\|\mathbf{A}\|_F^2 - \sum_{i=1}^{n} a_{ii}^2\Big).$$

If the entries of $\boldsymbol{z}$ are standard normal, then $\operatorname{Var}(\boldsymbol{z}^T\mathbf{A}\boldsymbol{z}) = 2\|\mathbf{A}\|_F^2$.

Proof Write the quadratic form as $\boldsymbol{z}^T\mathbf{A}\boldsymbol{z} = \sum_{i,j} a_{ij} z_i z_j$. Independence and zero means give $\mathbb{E}[z_i z_j] = 0$ for $i \neq j$, while unit variance gives $\mathbb{E}[z_i^2] = 1$. Linearity of expectation therefore gives

$$\mathbb{E}[\boldsymbol{z}^T\mathbf{A}\boldsymbol{z}] = \sum_{i} a_{ii} = \operatorname{tr}(\mathbf{A}).$$

This expectation argument needs no assumption about higher moments.

For independent signs, $z_i^2 = 1$ on every outcome. Symmetry of $\mathbf{A}$ pairs the off-diagonal terms, so the centered estimator is

$$Z = \boldsymbol{z}^T \mathbf{A} \boldsymbol{z} - \operatorname{tr}(\mathbf{A}) = 2 \sum_{i<j} a_{ij} z_i z_j.$$

Each product has mean zero and squared value one. Two different unordered pairs have zero cross expectation. If the pairs are disjoint, all four independent factors have mean zero. If they share one index, its square is one and the two remaining independent factors each have mean zero. Thus

$$\operatorname{Var}(Z) = \mathbb{E}[Z^2] = 4 \sum_{i<j} a_{ij}^2 = 2 \sum_{i \neq j} a_{ij}^2.$$

Subtracting the diagonal squares from $\|\mathbf{A}\|_F^2$ gives the first variance formula. The products need not be independent: the zero cross expectations are sufficient.

For Gaussian probes, diagonalize $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$ and put $\boldsymbol{g} = \mathbf{Q}^T \boldsymbol{z}$. The standard Gaussian density is proportional to $e^{-\|\boldsymbol{z}\|_2^2/2}$. Orthogonal changes preserve both the norm and volume, so $\boldsymbol{g}$ has the same independent standard normal coordinates. Hence

$$\boldsymbol{z}^T \mathbf{A} \boldsymbol{z} = \sum_i \lambda_i g_i^2.$$

A standard normal has second moment one and fourth moment three, obtained by integration by parts against its density. Thus $\operatorname{Var}(g_i^2) = 3 - 1 = 2$. Independence makes the covariances of distinct squared coordinates zero, giving

$$\operatorname{Var}(\boldsymbol{z}^T \mathbf{A} \boldsymbol{z}) = 2 \sum_i \lambda_i^2 = 2 \|\mathbf{A}\|_F^2.$$

The sign-probe estimator is attributed to Hutchinson (1990). □

Ex 6.45 Show that Hutchinson's estimator with ± 1 entries is exact for every diagonal matrix. Then show that, for every symmetric matrix, its variance is at most that of the Gaussian estimator, with equality exactly when the diagonal of $\mathbf{A}$ is zero.

Averaging k independent samples divides the variance by k. For a symmetric positive semidefinite matrix, $\|\mathbf{A}\|_F^2 = \sum_i \lambda_i^2 \leq (\sum_i \lambda_i)^2 = \operatorname{tr}(\mathbf{A})^2$, so with Gaussian probes the relative standard deviation of the averaged estimate is at most $\sqrt{2/k}$. The error decreases like $k^{-1/2}$, independently of n, which is the usual Monte Carlo rate. One hundred probes therefore give a relative standard deviation of at most about fourteen percent, and the estimator is most useful when this accuracy is sufficient or when $\|\mathbf{A}\|_F$ is much smaller than $\operatorname{tr}(\mathbf{A})$ (Hutchinson, 1990).

Prop 6.23 **(Diagonal Estimator)** Let $\boldsymbol{z}_1, \dots, \boldsymbol{z}_k$ be independent vectors with independent ± 1 entries, and let $\odot$ denote the entrywise product. Then

$$\boldsymbol{d}_k = \frac{1}{k} \sum_{l=1}^{k} \boldsymbol{z}_l \odot (\mathbf{A} \boldsymbol{z}_l)$$

satisfies $\mathbb{E}[\boldsymbol{d}_k] = \operatorname{diag}(\mathbf{A})$.

Ex 6.46 Expand entry i of $\boldsymbol{z} \odot (\mathbf{A} \boldsymbol{z})$. Separate the diagonal term, use $z_i^2 = 1$, and take expectations. Derive the variance of this entry and show that averaging k independent probes divides it by k.

To estimate $\operatorname{diag}(\mathbf{A}^{-1})$, we replace $\mathbf{A}\boldsymbol{z}_l$ by $\mathbf{A}^{-1}\boldsymbol{z}_l$ (Bekas et al., 2007). One sparse factorization of $\mathbf{A}$ is computed once, and each probe then costs one pair of triangular solves. For $k \ll n$, this replaces n solves by k solves. Bekas, Kokiopoulou, and Saad also use structured probe vectors that are orthogonal on blocks of the matrix, which removes the error from entries within those blocks.

6.7.2 Randomized Low-Rank Approximation

By Thm 4.21, the best rank-k approximation of $\mathbf{A}$ is given by its truncated SVD. Computing the full SVD of a large matrix in order to keep a few singular triplets wastes almost all of the work. If the singular values decay, the dominant left singular vectors span most of the range of $\mathbf{A}$. A product $\mathbf{A}\boldsymbol{\omega}$ with a random vector $\boldsymbol{\omega}$ is a random combination of the columns of $\mathbf{A}$, and it lies mostly in that dominant subspace. A few such products therefore span most of the dominant range.

Def 6.24 **(Randomized SVD)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$, let k be the target rank, and let $p \geq 2$ be an oversampling parameter with $k + p \leq \min(m, n)$. If the sample matrix is rank deficient, retain only its independent orthonormal columns and adjust the reduced dimensions below.

1. Draw $\boldsymbol{\Omega} \in \mathbb{R}^{n\times(k+p)}$ with independent standard normal entries.
2. Form $\mathbf{Y} = \mathbf{A}\boldsymbol{\Omega}$ and compute its thin QR factorization $\mathbf{Y} = \mathbf{Q}\mathbf{R}$.
3. Form $\mathbf{B} = \mathbf{Q}^T\mathbf{A} \in \mathbb{R}^{(k+p)\times n}$ and compute its SVD $\mathbf{B} = \tilde{\mathbf{U}}\boldsymbol{\Sigma}\mathbf{V}^T$.
4. Return the approximate SVD $\mathbf{A} \approx (\mathbf{Q}\tilde{\mathbf{U}})\boldsymbol{\Sigma}\mathbf{V}^T$, truncated to rank k if required.

Ex 6.47 Assume $\mathbf{A}$ has rank exactly k and the columns of $\mathbf{Y}$ span its range. Show that $\mathbf{Q}\mathbf{Q}^T\mathbf{A} = \mathbf{A}$, so the reduced SVD recovers the matrix exactly in exact arithmetic. For a matrix of larger rank, derive $\mathbf{A} - \mathbf{Q}\mathbf{B} = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^T)\mathbf{A}$. What error remains before any singular values of $\mathbf{B}$ are discarded?

The algorithm touches $\mathbf{A}$ only through the two products $\mathbf{A}\boldsymbol{\Omega}$ and $\mathbf{Q}^T\mathbf{A}$. For a dense matrix these cost $O(mn(k+p))$ flops, and for a sparse matrix $O(\operatorname{nnz}(\mathbf{A})(k+p))$. The QR factorization and the small SVD cost $O((m+n)(k+p)^2)$. Before the final truncation, the approximation error is $\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\|$, because $\mathbf{Q}\tilde{\mathbf{U}}\boldsymbol{\Sigma}\mathbf{V}^T = \mathbf{Q}\mathbf{Q}^T\mathbf{A}$. Halko, Martinsson, and Tropp prove that

$$\mathbb{E}\,\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\|_F \leq \Big(1 + \frac{k}{p-1}\Big)^{1/2}\Big(\sum_{j>k}\sigma_j^2\Big)^{1/2}$$

for Gaussian $\boldsymbol{\Omega}$ (Halko et al., 2011). The second factor is the error of the truncated SVD, so the randomized method loses at most the first factor in expectation. With $p = k + 1$, that factor is $\sqrt{2}$.

Remark 6.22 **(Slowly decaying singular values)** When the singular values decay slowly, the range of $\mathbf{A}\boldsymbol{\Omega}$ contains significant components from the trailing singular vectors. Replacing $\mathbf{Y} = \mathbf{A}\boldsymbol{\Omega}$ by $\mathbf{Y} = (\mathbf{A}\mathbf{A}^T)^q\mathbf{A}\boldsymbol{\Omega}$ multiplies the jth singular direction by σ_j^{2q+1} and suppresses the trailing directions, exactly as in power iteration (Def 4.5). In floating-point arithmetic, the columns are orthonormalized after each multiplication, because the powers otherwise become numerically dependent (Halko et al., 2011).

6.7.3 Random Sampling in Gaussian Elimination

Section 2.4 showed that eliminating a vertex joins all of its neighbors into a clique, and that the resulting fill makes sparse Cholesky expensive on large meshes. For the Laplacian of a weighted graph, the clique has an explicit form. Random sampling can replace it by a sparse graph that is correct on average.

Prop 6.24 **(Elimination in a Graph Laplacian)** Let $\mathbf{L} = \mathbf{D} - \mathbf{W}$ be the Laplacian of a connected graph with nonnegative edge weights w_{ij} and degrees $d_i = \sum_j w_{ij}$. Eliminating vertex v produces a Schur complement that is again a graph Laplacian. Its graph is the graph of $\mathbf{L}$ with v removed, plus an edge of weight $w_{iv}w_{jv}/d_v$ between each pair of neighbors i, j of v.

Proof Place the eliminated vertex first and write its Laplacian in blocks as

$$\mathbf{L} = \begin{pmatrix} d_v & \boldsymbol{r}^T \\ \boldsymbol{r} & \mathbf{B} \end{pmatrix}, \qquad r_i = -w_{iv}.$$

For a connected graph with more than one vertex, $d_v > 0$, so elimination is defined. Its Schur complement is $\mathbf{S} = \mathbf{B} - \boldsymbol{r}\boldsymbol{r}^T/d_v$.

For distinct remaining indices i, j, expand the entry:

$$s_{ij} = -w_{ij} - \frac{(-w_{iv})(-w_{jv})}{d_v} = -\left(w_{ij} + \frac{w_{iv}w_{jv}}{d_v}\right).$$

Thus a pair of neighbors acquires the additional edge weight in the statement. If either is not a neighbor, its contribution is zero. An existing edge receives that weight in addition to its previous weight.

We also check the diagonal through conservation of row sums. Removing column v from row i removes entry $-w_{iv}$, so the row sum of $\mathbf{B}$ is w_{iv}. On the other hand,

$$\sum_j \frac{r_i r_j}{d_v} = \frac{w_{iv}}{d_v} \sum_j w_{jv} = w_{iv}.$$

The row sum of $\mathbf{S}$ is therefore zero. Symmetry is preserved, and all off-diagonal entries are nonpositive. Its diagonal is consequently the sum of its incident edge weights, which proves that the whole Schur complement, not just its off-diagonal part, is the claimed graph Laplacian. □

Prop 6.25 **(An Unbiased Sparse Replacement for the Clique)** Let vertex v have neighbors $1, \ldots, \delta$ with weights $w_i = w_{iv}$ and $d = \sum_i w_i$. For each neighbor i, draw one neighbor j with probability w_j/d and, if $j \neq i$, add an edge between i and j with weight $w_i/2$. The Laplacian of the resulting graph of at most δ edges has expected value equal to the Laplacian of the clique in Prop 6.24.

Proof For a unit edge between distinct vertices i and j, let $\mathbf{L}_{ij} = (\boldsymbol{e}_i - \boldsymbol{e}_j)(\boldsymbol{e}_i - \boldsymbol{e}_j)^T$. The same matrix represents the edge in either order: $\mathbf{L}_{ij} = \mathbf{L}_{ji}$. Define $\mathbf{L}_{ii} = \mathbf{0}$, matching the rule that a self-selection adds no edge.

For the draw associated with vertex i, the event of selecting j has probability w_j/d. Its contributed Laplacian is $(w_i/2)\mathbf{L}_{ij}$. Taking its expectation gives

$$\sum_j \frac{w_j}{d}\frac{w_i}{2}\mathbf{L}_{ij}.$$

The Laplacian of a sum of edges is the sum of their Laplacians. Linearity of expectation therefore makes the expected total matrix

$$\sum_i \sum_j \frac{w_i w_j}{2d} \mathbf{L}_{ij}.$$

Terms with $i = j$ are zero. For each unordered pair $i < j$, there are two equal terms, from the draws associated with i and j. Their combined coefficient is $w_i w_j / d$. Thus the expected matrix is

$$\sum_{i<j} \frac{w_i w_j}{d} \mathbf{L}_{ij},$$

exactly the elimination clique. At most one edge is added per neighbor, so a realization has at most δ edges before merging duplicates. This calculation proves unbiasedness only, not a bound on the error of a realization. □

Ex 6.48 Verify Prop 6.25 for a vertex with three neighbors of weights 1, 2, and 3 by listing every outcome of the sampling and its probability.

An unbiased clique replacement does not by itself control the accumulated error or guarantee a useful preconditioner. A single sample may even disconnect the graph. A full approximate-elimination solver therefore needs concentration estimates and a carefully designed sampling scheme. For a connected Laplacian, the solve is restricted to the subspace orthogonal to $\mathbf{1}$, or one vertex is grounded, since the full matrix is singular. Kyng and Sachdeva use a refinement of this sampling, eliminate the vertices in random order, and prove that the resulting solver reaches relative accuracy ε in $O(m \log^3 n \log(1/\varepsilon))$ expected time for a graph with m edges (Kyng and Sachdeva, 2016). Practical implementations eliminate vertices in minimum-degree order, which reduces fill further but is not covered by the proof.

6.7.4 Further Exercises

Ex 6.49 Let $\mathbf{A}$ be the one-dimensional Poisson matrix with $n = 1000$. Estimate $\operatorname{tr}(\mathbf{A}^{-1})$ with $k = 10, 100, 1000$ probes using one Cholesky factorization. Plot the relative error against k on logarithmic axes and compare the slope with $-1/2$.

Ex 6.50 Construct a 1000×500 matrix with singular values $\sigma_j = 2^{-j}$ and random singular vectors. Apply Def 6.24 with $k = 20$ and $p = 0, 5, 20$, and compare $\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\|_2$ with σ_{21}. Repeat with $\sigma_j = j^{-1/2}$, with and without one step of the power iteration in the remark.

7 Ordinary Differential Equations

A differential equation specifies a rate of change rather than a finite update. We replace it by steps in time, estimate how the errors accumulate, and determine which step sizes preserve decay. The final construction concerns Hamiltonian equations, where preserving the geometry of the motion is also useful.

7.1 Numerical Methods for Ordinary Differential Equations

A differential equation gives the rate of change of a state. To compute a trajectory, we must turn that rate into a finite step. We begin with the simplest construction, then distinguish the error made in one step from the error accumulated over an interval.

Accuracy as the step size tends to zero does not settle how large a step we can use. A numerical solution can grow even when the exact solution decays. We study this restriction through a scalar test equation, then consider higher-order methods and the behavior of oscillating mechanical systems.

7.1.1 Initial Value Problems and Well-Posedness

Def 7.1 **(Initial Value Problem)** An initial value problem (IVP) for a system of ordinary differential equations has the form

$$\boldsymbol{y}'(t) = \boldsymbol{f}(t, \boldsymbol{y}(t)), \qquad \boldsymbol{y}(t_0) = \boldsymbol{y}_0,$$

where $t \in [t_0, T]$ and $\boldsymbol{y}(t) \in \mathbb{R}^m$. The function $\boldsymbol{f} : [t_0, T] \times \mathbb{R}^m \to \mathbb{R}^m$ is the vector field.

Example For $y' = -y$ and $y(0) = 1$, the exact solution is $y(t) = e^{-t}$. At time t_n, the derivative suggests the linear approximation $y(t_n + h) \approx y(t_n) + hy'(t_n)$. Replacing the exact value by the current approximation gives

$$y_{n+1} = y_n - hy_n = (1 - h)y_n.$$

This is forward Euler. Repeating the step gives $y_n = (1 - h)^n$. We can already ask two different questions: does it approach e^{-t_n} as h decreases, and does it decay for a fixed h?

Ex 7.1 Compute four Euler steps with $h = 1/2$ and compare with e^{-2}. Repeat with $h = 2$ and $h = 3$. When does the numerical sequence alternate in sign? When does its magnitude grow? Explain why the differential equation itself has none of this growth.

Before analyzing a method, we need an exact solution to compare it with. The next theorem gives a sufficient existence and uniqueness condition. Its global Lipschitz assumption is stronger than the local assumptions often used in ODE theory, but guarantees a solution throughout the stated finite interval.

Thm 7.1 **(Picard–Lindelöf Existence and Uniqueness)** Let $\boldsymbol{f}(t, \boldsymbol{y})$ be continuous in t and uniformly Lipschitz continuous in $\boldsymbol{y}$ on the domain $[t_0, T] \times \mathbb{R}^m$:

$$\|\boldsymbol{f}(t, \boldsymbol{u}) - \boldsymbol{f}(t, \boldsymbol{v})\| \le L\|\boldsymbol{u} - \boldsymbol{v}\| \qquad \text{for all } \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m,$$

where $L < \infty$ is the Lipschitz constant. Then the IVP has a unique continuous solution $\boldsymbol{y}(t)$ on $[t_0, T]$.

Proof Integrate the differential equation to obtain

$$\boldsymbol{y}(t) = \boldsymbol{y}_0 + \int_{t_0}^{t} \boldsymbol{f}(s, \boldsymbol{y}(s))\, ds.$$

On an interval of length Δ with $L\Delta < 1$, define $\mathcal{T}\boldsymbol{u}$ by the right-hand side with $\boldsymbol{u}$ in the integrand. In the supremum norm,

$$\|\mathcal{T}\boldsymbol{u} - \mathcal{T}\boldsymbol{v}\|_\infty \le L\Delta\, \|\boldsymbol{u} - \boldsymbol{v}\|_\infty.$$

Thus applying $\mathcal{T}$ repeatedly contracts differences by the factor $L\Delta$. Choose any initial continuous function $\boldsymbol{u}_0$ and define $\boldsymbol{u}_{j+1} = \mathcal{T}\boldsymbol{u}_j$. With $q = L\Delta < 1$, the contraction estimate implies

$$\|\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\|_\infty \le q^j \|\boldsymbol{u}_1 - \boldsymbol{u}_0\|_\infty.$$

For $r \geq 1$, the triangle inequality gives

$$\|\boldsymbol{u}_{j+r} - \boldsymbol{u}_j\|_\infty \leq \sum_{l=j}^{j+r-1} q^l \|\boldsymbol{u}_1 - \boldsymbol{u}_0\|_\infty \leq \frac{q^j}{1-q} \|\boldsymbol{u}_1 - \boldsymbol{u}_0\|_\infty.$$

This tends to zero independently of r. Completeness of the space of continuous functions in the supremum norm gives a uniform limit $\boldsymbol{y}$.

Starting with a continuous function, the successive differences are bounded by a geometric series. The iterates converge uniformly to a continuous function. Uniform convergence and the Lipschitz estimate bound the difference of the integrands uniformly by $L\|\boldsymbol{u}_j - \boldsymbol{y}\|_\infty$. The difference of their integrals is therefore at most $L\Delta\|\boldsymbol{u}_j - \boldsymbol{y}\|_\infty$, which tends to zero. Hence passing to the limit in the integral gives a fixed point and hence a solution. If there were two fixed points, the same inequality would force their distance to be zero.

Use the endpoint value as initial data on the next interval of length at most Δ. The global Lipschitz assumption gives the same contraction factor on every interval. Finitely many intervals cover $[t_0, T]$ and give the unique solution. If $L = 0$, the map is independent of its argument and the conclusion follows directly.

The integral of the continuous function $\boldsymbol{f}(s, \boldsymbol{y}(s))$ is continuously differentiable by the fundamental theorem of calculus. Thus the fixed point satisfies the original differential equation, not just its integral version. The finite-interval continuation is valid because the Lipschitz condition is global in the state variable. □

Ex 7.2 Let $q = L\Delta < 1$ and $\boldsymbol{u}_{j+1} = \mathcal{T}\boldsymbol{u}_j$. Prove $\|\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\|_\infty \leq q^j \|\boldsymbol{u}_1 - \boldsymbol{u}_0\|_\infty$ and bound $\|\boldsymbol{u}_{j+r} - \boldsymbol{u}_j\|_\infty$ by the tail of a geometric series. Explain how this establishes convergence and why the limit satisfies the integral equation. This is the contraction argument used in the proof.

7.1.2 Time-Stepping Methods

The exact increment satisfies $\boldsymbol{y}(t_{n+1}) - \boldsymbol{y}(t_n) = \int_{t_n}^{t_{n+1}} \boldsymbol{f}(t, \boldsymbol{y}(t))\, dt$. Approximating this integral by its left endpoint, right endpoint, or trapezoidal rule gives three basic methods. The right-endpoint methods contain the unknown new state on both sides, so each step requires solving equations.

Def 7.2 **(Forward Euler, Backward Euler, and Trapezoidal Methods)** For time step $h = t_{n+1} - t_n$:

1. **Forward Euler (explicit):** $\boldsymbol{y}_{n+1} = \boldsymbol{y}_n + h\boldsymbol{f}(t_n, \boldsymbol{y}_n)$.
2. **Backward Euler (implicit):** $\boldsymbol{y}_{n+1} = \boldsymbol{y}_n + h\boldsymbol{f}(t_{n+1}, \boldsymbol{y}_{n+1})$.
3. **Trapezoidal Rule (implicit Crank–Nicolson):**

$$\boldsymbol{y}_{n+1} = \boldsymbol{y}_n + \frac{h}{2}\left[\boldsymbol{f}(t_n, \boldsymbol{y}_n) + \boldsymbol{f}(t_{n+1}, \boldsymbol{y}_{n+1})\right].$$

7.1.3 Consistency, Stability, and the Discrete Gronwall Lemma

A numerical time-stepping method is convergent if the computed trajectory approaches the exact trajectory as the step size $h \to 0$. We quantify error at each step and track its accumulation over a finite time interval.

Def 7.3 **(Local Truncation Error and One-Step Defect)** Let $\boldsymbol{y}_{n+1} = \Phi_h(t_n, \boldsymbol{y}_n)$ denote an autonomous or non-autonomous one-step method. The local truncation error is the defect of the exact solution:

$$\boldsymbol{\tau}_{n+1} = \frac{\boldsymbol{y}(t_{n+1}) - \Phi_h(t_n, \boldsymbol{y}(t_n))}{h}.$$

The method is **consistent of order** p if $\|\boldsymbol{\tau}_{n+1}\| = O(h^p)$ as $h \to 0$ for all sufficiently smooth solutions.

There are two common error conventions. The unnormalized one-step defect is the numerator in this definition and is $h\boldsymbol{\tau}_{n+1}$. Our local truncation error divides it by h. For forward Euler, Taylor expansion gives a defect of order h^2 and a local truncation error of order h. Both statements describe the same calculation.

Lem 7.2 **(Discrete Gronwall Lemma)** Let $\{e_n\}_{n=0}^N$ be a sequence of non-negative real numbers satisfying the linear recurrence

$$e_{n+1} \le (1+\alpha)e_n + \beta, \qquad n = 0, 1, \ldots, N-1,$$

where $\alpha, \beta \ge 0$. For $\alpha > 0$,

$$e_n \le e^{n\alpha} e_0 + \beta \frac{e^{n\alpha} - 1}{\alpha}.$$

For $\alpha = 0$, the corresponding bound is $e_n \le e_0 + n\beta$.

Proof First let $\alpha > 0$. Apply the recurrence twice to see the pattern:

$$e_1 \le (1+\alpha)e_0 + \beta,$$
$$e_2 \le (1+\alpha)^2 e_0 + \beta[(1+\alpha) + 1].$$

Induction gives

$$e_n \le (1+\alpha)^n e_0 + \beta \sum_{j=0}^{n-1} (1+\alpha)^j.$$

Indeed, multiplying the bound at step n by $1+\alpha$ shifts all powers up by one, and adding β supplies the missing power zero. This is exactly the same formula for $n+1$.

The finite geometric sum is $[(1+\alpha)^n - 1]/\alpha$, because its ratio differs from one by α. The elementary inequality $1 + \alpha \le e^\alpha$ implies $(1+\alpha)^n \le e^{n\alpha}$. Since e_0, β are nonnegative, replacing the powers by this upper bound preserves the inequality:

$$e_n \le e^{n\alpha} e_0 + \beta \frac{e^{n\alpha} - 1}{\alpha}.$$

For $\alpha = 0$, no division by α is allowed. The recurrence reduces to $e_{n+1} \le e_n + \beta$. Applying it n times gives $e_n \le e_0 + n\beta$. Thus the separate zero case agrees with the limiting formula. □

Thm 7.3 **(Fundamental Theorem of One-Step Convergence)** Let Φ_h be a consistent one-step method of order p applied to an IVP whose vector field $\boldsymbol{f}$ is Lipschitz continuous with constant L. Suppose the numerical increment function satisfies $\|\Phi_h(t, \boldsymbol{u}) - \Phi_h(t, \boldsymbol{v})\| \le (1 + \Lambda h)\|\boldsymbol{u} - \boldsymbol{v}\|$

for some constant $\Lambda \geq 0$. Then on the fixed interval $[t_0, T]$ with $Nh = T - t_0$, the global error $\boldsymbol{e}_n = \boldsymbol{y}(t_n) - \boldsymbol{y}_n$ satisfies

$$\max_{0 \leq n \leq N} \|\boldsymbol{e}_n\| \leq e^{\Lambda(T-t_0)} \|\boldsymbol{e}_0\| + \frac{e^{\Lambda(T-t_0)} - 1}{\Lambda} \max_k \|\boldsymbol{\tau}_k\|,$$

where the quotient is interpreted as $T - t_0$ when $\Lambda = 0$. Consequently, when the numerical and exact solutions use the same initial value, a one-step defect of order $O(h^{p+1})$ yields a global error of order $O(h^p)$.

Proof Express the error at step $n+1$ by subtracting the numerical update from the exact solution:

$$\begin{aligned} \boldsymbol{e}_{n+1} &= \boldsymbol{y}(t_{n+1}) - \boldsymbol{y}_{n+1} \\ &= [\Phi_h(t_n, \boldsymbol{y}(t_n)) + h\boldsymbol{\tau}_{n+1}] - \Phi_h(t_n, \boldsymbol{y}_n) \\ &= [\Phi_h(t_n, \boldsymbol{y}(t_n)) - \Phi_h(t_n, \boldsymbol{y}_n)] + h\boldsymbol{\tau}_{n+1}. \end{aligned}$$

Taking norms and applying the triangle inequality yields

$$\|\boldsymbol{e}_{n+1}\| \leq (1 + \Lambda h)\|\boldsymbol{e}_n\| + h \max_k \|\boldsymbol{\tau}_k\|.$$

The first bracket compares two numerical steps starting from different states. The second term is the defect of a step starting from the exact state. Keeping these contributions separate explains why one is controlled by the step map's Lipschitz constant and the other by consistency.

Applying the Discrete Gronwall Lemma (Lem 7.2) with $\alpha = \Lambda h$ and $\beta = h \max_k \|\boldsymbol{\tau}_k\|$ gives

$$\|\boldsymbol{e}_n\| \leq e^{n\Lambda h}\|\boldsymbol{e}_0\| + h \max_k \|\boldsymbol{\tau}_k\| \frac{e^{n\Lambda h} - 1}{\Lambda h} = e^{\Lambda(t_n - t_0)}\|\boldsymbol{e}_0\| + \frac{e^{\Lambda(t_n - t_0)} - 1}{\Lambda} \max_k \|\boldsymbol{\tau}_k\|.$$

Since $t_n - t_0 \leq T - t_0$ and $\max_k \|\boldsymbol{\tau}_k\| = O(h^p)$, the global error is bounded by $O(h^p)$ when initial data are exact ($\boldsymbol{e}_0 = \boldsymbol{0}$).

If $\Lambda = 0$, use the zero-parameter case of the lemma directly: after n steps the added defect is at most $nh \max_k \|\boldsymbol{\tau}_k\| \leq (T - t_0) \max_k \|\boldsymbol{\tau}_k\|$. This justifies the quotient convention without dividing by zero. For fixed $T - t_0$, the exponential constants do not change as $h \to 0$, so a uniform $O(h^p)$ local truncation bound gives the stated global order. □

7.1.4 Euler Methods and Stability Functions

Thm 7.4 **(Stability Functions for Linear Test Equations)** Applied to the Dahlquist test equation $y' = \lambda y$ with $z = h\lambda \in \mathbb{C}$, the numerical update satisfies $y_{n+1} = R(z)y_n$. The stability functions are:

$$R_{\mathrm{FE}}(z) = 1 + z, \qquad R_{\mathrm{BE}}(z) = \frac{1}{1 - z}, \qquad R_{\mathrm{TR}}(z) = \frac{1 + z/2}{1 - z/2}.$$

The region of absolute stability is $\mathcal{S} = \{z \in \mathbb{C} : |R(z)| \leq 1\}$. Backward Euler and the Trapezoidal Rule are A-stable: $\mathbb{C}^- = \{z \in \mathbb{C} : \mathrm{Re}(z) \leq 0\} \subseteq \mathcal{S}$. Backward Euler is additionally L-stable: $\lim_{z \to -\infty} R_{\mathrm{BE}}(z) = 0$.

Proof Let $z = h\lambda$. Forward Euler substitutes the old value into the derivative:

$$y_{n+1} = y_n + h\lambda y_n = (1 + z)y_n.$$

Thus its multiplier is $R_{\mathrm{FE}}(z) = 1 + z$.

For backward Euler, the derivative uses the new value. Move its term to the left before dividing:

$$y_{n+1} = y_n + z y_{n+1}, \qquad (1 - z) y_{n+1} = y_n.$$

Whenever $1 - z \neq 0$, the multiplier is $1/(1 - z)$. Similarly the trapezoidal update gives

$$y_{n+1} = y_n + \tfrac{z}{2}(y_n + y_{n+1}), \qquad (1 - z/2) y_{n+1} = (1 + z/2) y_n,$$

so its multiplier is $(1 + z/2)/(1 - z/2)$ when defined.

To prove the stability assertions, write $z = x + iy$ with $x \leq 0$. Then

$$|1 - z|^2 = (1 - x)^2 + y^2 \geq 1.$$

The denominator of backward Euler is therefore nonzero and $|R_{\mathrm{BE}}(z)| \leq 1$. For the trapezoidal rule,

$$\begin{aligned} |1 - z/2|^2 - |1 + z/2|^2 &= [(1 - x/2)^2 + (y/2)^2] \\ &\quad - [(1 + x/2)^2 + (y/2)^2] = -2x \geq 0. \end{aligned}$$

Its denominator is also nonzero on this half-plane, so $|R_{\mathrm{TR}}(z)| \leq 1$. This proves A-stability for both implicit methods.

Along the negative real axis, $1/(1 - z) \to 0$ as $z \to -\infty$, whereas the trapezoidal multiplier tends to -1 after dividing numerator and denominator by z. Thus backward Euler has the additional L-stability property and the trapezoidal rule does not. See Hairer et al. (1993) for these stability notions. □

Ex 7.3 Derive $R_{\mathrm{TR}}(z)$ from the trapezoidal update. Write $z = x + iy$ and compute $|1 - z/2|^2 - |1 + z/2|^2$. Use its sign to prove A-stability and to locate the points where $|R_{\mathrm{TR}}(z)| = 1$. Take the limit along the negative real axis to show that the method is not L-stable. Finally, apply both implicit methods to $y' = -1000y$ with $h = 0.1$. Compare their first three values with the exact decay. Preservation of the magnitude of an oscillation does not imply exact phase: determine the trapezoidal phase change when $z = i\omega h$.

7.1.5 Runge–Kutta Methods

Runge–Kutta methods achieve high orders of accuracy without storing historical steps by evaluating intermediate stages within each time step.

Ex 7.4 A first improvement on Euler is to estimate the derivative halfway through the step. Form the provisional value $\boldsymbol{y}_{n+1/2} = \boldsymbol{y}_n + (h/2)\boldsymbol{f}(\boldsymbol{y}_n)$ and then set $\boldsymbol{y}_{n+1} = \boldsymbol{y}_n + h\boldsymbol{f}(\boldsymbol{y}_{n+1/2})$. For a smooth autonomous vector field, expand this update to order h^2 and compare it with the exact Taylor expansion. What is its one-step defect? This two-stage construction illustrates why intermediate stages can improve the order.

Def 7.4 **(Classical Fourth-Order Runge–Kutta Method (RK4))** For an autonomous ODE $\boldsymbol{y}' =$

$\boldsymbol{f}(\boldsymbol{y})$, the classical RK4 method computes four stages:

$$\begin{aligned}
\boldsymbol{k}_1 &= \boldsymbol{f}(\boldsymbol{y}_n),\\
\boldsymbol{k}_2 &= \boldsymbol{f}\left(\boldsymbol{y}_n + \frac{h}{2}\boldsymbol{k}_1\right),\\
\boldsymbol{k}_3 &= \boldsymbol{f}\left(\boldsymbol{y}_n + \frac{h}{2}\boldsymbol{k}_2\right),\\
\boldsymbol{k}_4 &= \boldsymbol{f}\left(\boldsymbol{y}_n + h\boldsymbol{k}_3\right),
\end{aligned}$$

and sets the update

$$\boldsymbol{y}_{n+1} = \boldsymbol{y}_n + \frac{h}{6}\left(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4\right).$$

The one-step defect is $O(h^5)$. With the normalization in Def 7.3, the local truncation error and the global error are both $O(h^4)$.

Thm 7.5 **(RK4 Stability Polynomial)** Applied to the scalar linear test equation $y' = \lambda y$, the stability function of RK4 is

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} = \sum_{k=0}^{4} \frac{z^k}{k!}, \qquad z = h\lambda.$$

Ex 7.5 Substitute $f(y) = \lambda y$ into the four stages. Express each stage as λy_n times a polynomial in $z = h\lambda$, and collect terms in the update. Why does matching the exponential on this scalar test problem alone not prove order four for a nonlinear ODE?

7.1.6 Symplectic Integrators and Geometric Mechanics

For the oscillator $q' = v$, $v' = -q$, the exact energy $H = (q^2 + v^2)/2$ is constant. Forward Euler instead multiplies this energy by $1 + h^2$ at every step. A small local error can therefore accumulate into a qualitatively wrong orbit over many periods. We now construct an update from simple motions that preserve the geometry of position and momentum. The matrix identity below defines precisely the property called symplecticity.

Def 7.5 **(Velocity Verlet Algorithm)** For a potential V with continuous second derivatives and a separable Hamiltonian $H(\boldsymbol{q}, \boldsymbol{v}) = \frac{1}{2}\|\boldsymbol{v}\|_2^2 + V(\boldsymbol{q})$ with equations of motion $\boldsymbol{q}' = \boldsymbol{v}$ and $\boldsymbol{v}' = \mathbf{F}(\boldsymbol{q}) = -\nabla V(\boldsymbol{q})$, the Velocity Verlet method is:

$$\begin{aligned}
\boldsymbol{v}_{n+1/2} &= \boldsymbol{v}_n + \frac{h}{2}\mathbf{F}(\boldsymbol{q}_n),\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\boldsymbol{v}_{n+1/2},\\
\boldsymbol{v}_{n+1} &= \boldsymbol{v}_{n+1/2} + \frac{h}{2}\mathbf{F}(\boldsymbol{q}_{n+1}).
\end{aligned}$$

Velocity Verlet first changes velocity with position held fixed, then changes position with velocity held fixed, and finally completes the velocity change. In one position dimension these are shears of the (q, v) plane. Each preserves area. In higher dimensions, symplecticity requires the stronger

matrix identity $\mathbf{J}^T\mathbf{\Omega}\mathbf{J} = \mathbf{\Omega}$, where $\mathbf{\Omega}$ pairs positions with their corresponding momenta. Volume preservation alone would not establish it.

Thm 7.6 **(Symplecticity and Area Preservation of Velocity Verlet)** The one-step Velocity Verlet map $\Psi_h : (\boldsymbol{q}_n, \boldsymbol{v}_n) \mapsto (\boldsymbol{q}_{n+1}, \boldsymbol{v}_{n+1})$ preserves the canonical symplectic structure. In particular, the Jacobian matrix $\mathbf{J} = \frac{\partial(\boldsymbol{q}_{n+1}, \boldsymbol{v}_{n+1})}{\partial(\boldsymbol{q}_n, \boldsymbol{v}_n)}$ satisfies

$$\det(\mathbf{J}) = 1.$$

The full symplectic assertion is that its Jacobian satisfies $\mathbf{J}^T\mathbf{\Omega}\mathbf{J} = \mathbf{\Omega}$ for the canonical symplectic matrix $\mathbf{\Omega}$. Symplecticity implies $\det(\mathbf{J}) = 1$, so Velocity Verlet also preserves phase-space volume.

Proof Decompose the Velocity Verlet step into the composition of three elementary shears:

$$\Psi_h = \psi_3 \circ \psi_2 \circ \psi_1,$$

where:

$$\begin{aligned}
\psi_1(\boldsymbol{q}, \boldsymbol{v}) &= \left(\boldsymbol{q},\, \boldsymbol{v} + \frac{h}{2}\mathbf{F}(\boldsymbol{q})\right), \\
\psi_2(\boldsymbol{q}, \boldsymbol{v}) &= (\boldsymbol{q} + h\boldsymbol{v},\, \boldsymbol{v}), \\
\psi_3(\boldsymbol{q}, \boldsymbol{v}) &= \left(\boldsymbol{q},\, \boldsymbol{v} + \frac{h}{2}\mathbf{F}(\boldsymbol{q})\right).
\end{aligned}$$

Evaluating the Jacobian of each shear map:

$$\mathbf{J}_1 = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \frac{h}{2}\nabla\mathbf{F}(\boldsymbol{q}) & \mathbf{I} \end{pmatrix}, \qquad \mathbf{J}_2 = \begin{pmatrix} \mathbf{I} & h\mathbf{I} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}, \qquad \mathbf{J}_3 = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \frac{h}{2}\nabla\mathbf{F}(\boldsymbol{q}_{n+1}) & \mathbf{I} \end{pmatrix}.$$

Here $\nabla\mathbf{F}$ denotes the Jacobian of the force, the matrix of its first partial derivatives. Since the force is the negative gradient of a potential with continuous second derivatives, that Jacobian is the negative Hessian of V. Equality of mixed partial derivatives makes it symmetric. This symmetry is essential for the symplectic calculation.

Because $\mathbf{F} = -\nabla V$, the Hessian block $\mathbf{S} = \frac{h}{2}\nabla\mathbf{F} = -\frac{h}{2}\nabla^2 V$ is symmetric ($\mathbf{S}^T = \mathbf{S}$). With canonical symplectic matrix $\mathbf{\Omega} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{pmatrix}$, evaluating the quadratic form for $\mathbf{J}_1$:

$$\begin{aligned}
\mathbf{J}_1^T\mathbf{\Omega}\mathbf{J}_1 &= \begin{pmatrix} \mathbf{I} & \mathbf{S} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{S} & \mathbf{I} \end{pmatrix} \\
&= \begin{pmatrix} -\mathbf{S} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{S} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} -\mathbf{S}+\mathbf{S} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{pmatrix} = \mathbf{\Omega}.
\end{aligned}$$

Similarly, for the translation shear $\mathbf{J}_2$:

$$\begin{aligned}
\mathbf{J}_2^T\mathbf{\Omega}\mathbf{J}_2 &= \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ h\mathbf{I} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{I} & h\mathbf{I} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \\
&= \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & h\mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{I} & h\mathbf{I} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & -h\mathbf{I}+h\mathbf{I} \end{pmatrix} = \mathbf{\Omega}.
\end{aligned}$$

The third shear $\mathbf{J}_3$ has the same form as $\mathbf{J}_1$ and satisfies $\mathbf{J}_3^T\boldsymbol{\Omega}\mathbf{J}_3 = \boldsymbol{\Omega}$. The canonical matrix $\boldsymbol{\Omega}$ records the pairing of each position coordinate with its corresponding momentum coordinate. The displayed calculations verify the full symplectic matrix identity for each individual shear, rather than checking its determinant alone.

By the chain rule, $\mathbf{J} = \mathbf{J}_3\mathbf{J}_2\mathbf{J}_1$, so

$$\mathbf{J}^T\boldsymbol{\Omega}\mathbf{J} = \mathbf{J}_1^T\mathbf{J}_2^T\left(\mathbf{J}_3^T\boldsymbol{\Omega}\mathbf{J}_3\right)\mathbf{J}_2\mathbf{J}_1 = \mathbf{J}_1^T\left(\mathbf{J}_2^T\boldsymbol{\Omega}\mathbf{J}_2\right)\mathbf{J}_1 = \mathbf{J}_1^T\boldsymbol{\Omega}\mathbf{J}_1 = \boldsymbol{\Omega}.$$

Each shear has unit determinant: $\det(\mathbf{J}_1) = \det(\mathbf{J}_2) = \det(\mathbf{J}_3) = 1$. Therefore

$$\det(\mathbf{J}) = \det(\mathbf{J}_3)\det(\mathbf{J}_2)\det(\mathbf{J}_1) = 1 \cdot 1 \cdot 1 = 1.$$

The product in the chain rule evaluates each Jacobian at the corresponding intermediate state. The identity for each factor holds at every such state, so the cancellations remain valid even for nonlinear forces. A unit determinant follows separately from their triangular block forms. This establishes both claims in the theorem. □

Ex 7.6 **Symplectic geometry of the harmonic oscillator.** Consider the harmonic oscillator $q' = v$, $v' = -q$ with exact energy $E(q, v) = \frac{1}{2}(q^2 + v^2)$.

1. Compute the one-step update matrix $\mathbf{M}_{\mathrm{FE}}$ for Forward Euler and show that $\det(\mathbf{M}_{\mathrm{FE}}) = 1 + h^2 > 1$.
2. Compute the one-step update matrix $\mathbf{M}_{\mathrm{Verlet}}$ for Velocity Verlet. Confirm that $\det(\mathbf{M}_{\mathrm{Verlet}}) = 1$.
3. For $0 < h < 2$, show that Velocity Verlet conserves the positive definite quadratic form $\tilde{H}(q, v) = \frac{1}{2}(q^2 + v^2) - \frac{h^2}{8}q^2 = H(q, v) + O(h^2)$.

7.1.7 Further Exercises

Ex 7.7 **The discrete Gronwall lemma and Forward Euler convergence.**

1. Let $\boldsymbol{y}(t)$ solve $\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y})$ with $\|\boldsymbol{f}(t, \boldsymbol{u}) - \boldsymbol{f}(t, \boldsymbol{v})\| \le L\|\boldsymbol{u} - \boldsymbol{v}\|$. Use Taylor's theorem to show that the one-step defect of Forward Euler satisfies $\|\boldsymbol{y}(t_{n+1}) - (\boldsymbol{y}(t_n) + h\boldsymbol{f}(t_n, \boldsymbol{y}(t_n)))\| \le \frac{h^2}{2}M_2$, where $M_2 = \max \|\boldsymbol{y}''(t)\|$.
2. Derive the error recurrence $\|\boldsymbol{e}_{n+1}\| \le (1 + Lh)\|\boldsymbol{e}_n\| + \frac{h^2}{2}M_2$.
3. Apply Lem 7.2 to establish the rigorous global error bound $\|\boldsymbol{e}_n\| \le \frac{M_2}{2L}(e^{L(t_n - t_0)} - 1)h$.

Ex 7.8 **Adaptive step size control via embedded Runge–Kutta pairs.**

1. An embedded pair computes a lower-order approximation $\boldsymbol{y}_{n+1}^{[p]}$ and a higher-order approximation $\boldsymbol{y}_{n+1}^{[p+1]}$ sharing stages. Show that the difference $\boldsymbol{e}_{n+1} = \boldsymbol{y}_{n+1}^{[p+1]} - \boldsymbol{y}_{n+1}^{[p]}$ satisfies $\|\boldsymbol{e}_{n+1}\| = Ch^{p+1} + O(h^{p+2})$.
2. Given error tolerance ε and current step size h_{old}, derive the optimal step size formula $h_{\mathrm{new}} = sh_{\mathrm{old}}(\varepsilon/\|\boldsymbol{e}_{n+1}\|)^{1/(p+1)}$ with safety factor $s \in (0, 1)$.
3. Explain why adaptive solvers reject a step and repeat with smaller h when $\|\boldsymbol{e}_{n+1}\| > \varepsilon$, and why an implementation may cap the growth factor, for example by $h_{\mathrm{new}} \le 5h_{\mathrm{old}}$. Explain why the numerical cap is a design choice.

Ex 7.9 **Stiffness in chemical cascades and BDF methods.** Consider the singularly perturbed initial value problem $\varepsilon y' = -y + \cos(t)$ with $y(0) = 0$ and $\varepsilon = 10^{-4}$.

1. Find the exact solution and identify the fast boundary layer transient $e^{-t/\varepsilon}$ and the slow quasi-steady state $\cos(t) + \varepsilon \sin(t)$.
2. Explain why explicit RK4 requires $h \lesssim 2.8 \times 10^{-4}$ throughout the entire simulation, even after the transient has decayed to zero.
3. Formulate the BDF2 update $\frac{3\boldsymbol{y}_{n+1} - 4\boldsymbol{y}_n + \boldsymbol{y}_{n-1}}{2h} = \boldsymbol{f}(t_{n+1}, \boldsymbol{y}_{n+1})$ and show that it allows time steps $h = 0.1$ in the smooth regime.

Read **Further reading.** Hairer, Nørsett, and Wanner give a careful account of ODE methods and their analysis (Hairer et al., 1993). Dahlquist's paper is the classical source for the stability barrier for linear multistep methods (Dahlquist, 1963). Quarteroni, Sacco, and Saleri provide a broader numerical-analysis reference (Quarteroni et al., 2007).

8 Applications

The decompositions and error estimates developed earlier now enter calculations with data, graphs, and evolving states. Each application introduces the meaning of its matrix before using its algebraic structure. The later sections on stochastic dynamics and observed evolution operators require the earlier material on spectra, least squares, and matrix exponentials.

8.1 Principal Component Analysis (PCA)

An observation with p measured features is a point in $\mathbb{R}^p$. A collection of observations may vary strongly along a few directions and very little along the others. Principal component analysis chooses orthogonal directions that retain as much of this variation as possible.

We first subtract the mean observation so that variation is measured about the origin. We then express the variance along a direction as a Rayleigh quotient. The SVD identifies the directions and gives the error caused by retaining only some of them.

8.1.1 Definitions and Preprocessing

Def 8.1 **(Data Centering)** Data matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ ($n \geq 2$ samples, p features) is first **centered**:

$$X_{ij} \leftarrow X_{ij} - \bar{x}_j, \quad \bar{x}_j = \frac{1}{n} \sum_{i=1}^{n} X_{ij}.$$

Throughout this section, $\mathbf{X}$ is assumed centered, and singular values beyond $\min(n, p)$ are understood to be zero. A variance ratio is defined only when $\|\mathbf{X}\|_F > 0$.

Def 8.2 **(Sample Covariance Matrix)** The matrix $\mathbf{S} = \frac{1}{n-1}\mathbf{X}^T\mathbf{X} \in \mathbb{R}^{p \times p}$.

- **Diagonal:** S_{jj} is the variance of feature j.
- **Off-diagonal:** S_{ij} is the covariance between features i and j.
- **Property:** $\mathbf{S}$ is symmetric positive semidefinite (SPSD).

Ex 8.1

1. Show that centering makes each column of $\mathbf{X}$ have mean zero.
2. Prove that the diagonal entries of $\mathbf{S}$ are sample variances.
3. Prove that $\mathbf{S}$ is symmetric positive semidefinite by computing $\boldsymbol{z}^T\mathbf{S}\boldsymbol{z}$.

8.1.2 Principal Directions and Scores

For a unit direction $\boldsymbol{v} \in \mathbb{R}^p$, the projected observations are the entries of $\mathbf{X}\boldsymbol{v}$. Their mean is zero because the columns of $\mathbf{X}$ are centered. Their sample variance is therefore

$$\frac{\|\mathbf{X}\boldsymbol{v}\|_2^2}{n-1} = \boldsymbol{v}^T\mathbf{S}\boldsymbol{v}.$$

The first principal direction maximizes this variance. By the spectral theorem, it is an eigenvector for the largest eigenvalue of $\mathbf{S}$. Subsequent directions maximize variance subject to orthogonality to the preceding ones.

Ex 8.2 Take centered observations $(2,0)$, $(-2,0)$, $(0,1)$, and $(0,-1)$ as rows of $\mathbf{X}$. Compute the variance along $\boldsymbol{v} = (\cos\theta, \sin\theta)^T$ and find its maximum. Identify the first and second principal directions. What changes if every second coordinate is multiplied by three?

Thm 8.1 **(Principal Directions and Scores)** Let $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ be the SVD of the centered data matrix.

- **Principal Directions (Loadings):** Columns of $\mathbf{V}$ (eigenvectors of $\mathbf{S}$).
- **Principal Components (Scores):** $\mathbf{Z} = \mathbf{X}\mathbf{V} = \mathbf{U}\mathbf{\Sigma}$.
- **Variances:** $\lambda_i = \mathrm{Var}(\boldsymbol{z}_i) = \frac{\sigma_i^2}{n-1}$.

Proof Substituting the SVD $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ into the sample covariance matrix $\mathbf{S}$:

$$\begin{aligned}
\mathbf{S} &= \frac{1}{n-1}\mathbf{X}^T\mathbf{X} \\
&= \frac{1}{n-1}\left(\mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^T\right)\left(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\right) \\
&= \frac{1}{n-1}\mathbf{V}\mathbf{\Sigma}^T(\mathbf{U}^T\mathbf{U})\mathbf{\Sigma}\mathbf{V}^T \\
&= \mathbf{V}\left(\frac{\mathbf{\Sigma}^T\mathbf{\Sigma}}{n-1}\right)\mathbf{V}^T.
\end{aligned}$$

Because $\mathbf{V}$ is orthogonal and $\frac{\mathbf{\Sigma}^T\mathbf{\Sigma}}{n-1}$ is diagonal with non-negative entries $\frac{\sigma_i^2}{n-1}$, this is the spectral decomposition of $\mathbf{S}$. The columns $\boldsymbol{v}_i$ are eigenvectors of $\mathbf{S}$ with eigenvalues $\lambda_i = \sigma_i^2/(n-1)$. The covariance eigenvalues are therefore squared singular values divided by $n-1$. This division is positive because the sample-size assumption is $n \geq 2$. If a singular value is zero, the corresponding direction has zero measured sample variance.

The principal component score matrix is

$$\mathbf{Z} = \mathbf{X}\mathbf{V} = \left(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\right)\mathbf{V} = \mathbf{U}\mathbf{\Sigma}(\mathbf{V}^T\mathbf{V}) = \mathbf{U}\mathbf{\Sigma}.$$

The sample covariance matrix of the scores is

$$\mathbf{S}_{\mathbf{Z}} = \frac{1}{n-1}\mathbf{Z}^T\mathbf{Z} = \frac{1}{n-1}(\mathbf{\Sigma}^T\mathbf{U}^T)(\mathbf{U}\mathbf{\Sigma}) = \frac{\mathbf{\Sigma}^T\mathbf{\Sigma}}{n-1} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p),$$

confirming that distinct score columns $\boldsymbol{z}_i, \boldsymbol{z}_j$ have sample covariance zero for $i \neq j$.

Because the centered columns of $\mathbf{X}$ have zero sums, $\mathbf{1}^T\mathbf{Z} = \mathbf{1}^T\mathbf{X}\mathbf{V} = \mathbf{0}^T$. Thus the score columns are centered too, making $\mathbf{Z}^T\mathbf{Z}/(n-1)$ their sample covariance without an additional centering step. Its diagonal form proves that distinct score columns are uncorrelated. It does not imply statistical independence, which is a stronger claim. See Jolliffe (2002) for the statistical interpretation. □

Remark 8.1 **(PCA stability)** Forming $\mathbf{X}^T\mathbf{X}$ squares the 2-norm condition number: $\kappa_2(\mathbf{X}^T\mathbf{X}) = \kappa_2(\mathbf{X})^2$ when $\mathbf{X}$ has full column rank. Compute the singular vectors directly from an SVD when small singular values or numerical rank matter.

Ex 8.3 Show that centering is left multiplication by $\mathbf{C} = \mathbf{I} - \mathbf{1}\mathbf{1}^T/n$. Prove that $\mathbf{C}^2 = \mathbf{C} = \mathbf{C}^T$, deduce $\operatorname{rank}(\mathbf{X}) \leq n-1$, and explain why at most $n-1$ principal components can have positive variance even when $p > n$.

8.1.3 Variance and Dimensionality Reduction

Def 8.3 **(Proportion of Variance Explained (PVE))** For the first k components:

$$\mathrm{PVE}_k = \frac{\sum_{i=1}^{k}\sigma_i^2}{\sum_{i=1}^{p}\sigma_i^2}.$$

Thm 8.2 **(Optimal Approximation)** The rank-k matrix

$$\mathbf{X}_k = \sum_{i=1}^{k}\sigma_i\boldsymbol{u}_i\boldsymbol{v}_i^T$$

is the optimal k-dimensional linear approximation of the data in the Frobenius norm, by the Eckart–Young theorem.

Ex 8.4

1. Use Thm 4.21 to prove Thm 8.2.
2. Show that $\|\mathbf{X} - \mathbf{X}_k\|_F^2 = \sum_{i>k}\sigma_i^2$.
3. Show that $\mathrm{PVE}_k = 1 - \|\mathbf{X} - \mathbf{X}_k\|_F^2/\|\mathbf{X}\|_F^2$.
4. Explain why a PVE threshold is a modeling choice, not a theorem.

8.1.4 Implementation Details

Remark 8.2 **(Standardization)** If features have different units, PCA will be dominated by large-scale features. When equal feature weighting is intended, standardize nonconstant columns by their standard deviations: $X_{ij} \leftarrow (X_{ij} - \bar{x}_j)/s_j$.

Remark 8.3 **(Interpretation)** Principal components are directions of large variance. A large loading identifies a direction of variation in the data matrix. It does not by itself explain why that variation occurs.

Def 8.4 **(Dual PCA for $p \gg n$)** When the number of features greatly exceeds the number of samples, we may diagonalize the $n \times n$ Gram matrix $\mathbf{K} = \mathbf{X}\mathbf{X}^T$ instead of the $p \times p$ covariance matrix. For each nonzero singular value, the feature-space principal direction is recovered from $\boldsymbol{v}_i = \mathbf{X}^T \boldsymbol{u}_i / \sigma_i$. Kernel PCA replaces this linear Gram matrix by a centered nonlinear kernel matrix.

8.1.5 Further Exercises

Ex 8.5 Center $\mathbf{X} = \begin{pmatrix} 2 & 1 \\ -1 & 3 \\ -1 & -4 \end{pmatrix}$ and compute $\mathbf{S}$. Find the PVE for $k = 1$.

Ex 8.6 If two largest singular values are equal, show that any orthonormal basis of their right singular subspace gives the same variance and reconstruction error. Explain which PCA subspace is determined and which individual directions are not uniquely determined.

Ex 8.7 Load the Iris dataset. Plot the data in the basis of the first two PCs. Do species cluster?

Ex 8.8 Verify that the sum of variances $\sum \lambda_i$ equals the total variance $\mathrm{Tr}(\mathbf{S})$.

Read **Further reading.** Jolliffe is a standard statistical reference for principal component analysis (Jolliffe, 2002). The Golub–Kahan paper gives the bidiagonalization behind modern large-scale SVD calculations (Golub and Kahan, 1965).

8.2 Spectral Graph Theory, Diffusion Maps, and Graph Laplacians

A graph records which pairs of vertices interact and how strongly. A scalar value at each vertex can describe a temperature, a signal, or a quantity attached to an observation. The graph Laplacian compares each value with those of its neighbors.

Its quadratic form adds the squared differences across edges. This connects eigenvalues to connectivity and suggests ways to partition a graph. We then normalize edge weights to obtain a random walk and use its probabilities to compare data points.

8.2.1 Graph Matrices and the Laplacian

Def 8.5 **(Adjacency and Degree Matrices)** For a weighted undirected graph $G = (V, E)$ with $n \geq 2$ vertices, take E to contain exactly the edges of positive weight:

- **Adjacency A:** $A_{ij} = w_{ij} \geq 0$ if $(i, j) \in E$, and $A_{ij} = 0$ otherwise. We assume $w_{ij} = w_{ji}$ and $A_{ii} = 0$.
- **Degree D:** Diagonal matrix with entries $D_{ii} = d_i = \sum_{j=1}^{n} w_{ij}$.

Example For the path 1–2–3 with unit weights, the middle vertex has two neighbors and the endpoints have one. Its Laplacian is

$$\mathbf{L} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

For values $\boldsymbol{x} = (a, b, c)^T$, its quadratic form is $(a-b)^2 + (b-c)^2$. A constant signal has zero energy. A jump across either edge has positive energy.

Ex 8.9 Derive this quadratic form by multiplying out $\boldsymbol{x}^T\mathbf{L}\boldsymbol{x}$. Replace the second edge weight by $\varepsilon > 0$ and repeat. Evaluate the form on $(1, 1, -2)^T$. How does the energy change as the second edge becomes weak?

Def 8.6 **(Unnormalized Graph Laplacian)** The unnormalized graph Laplacian is

$$\mathbf{L} = \mathbf{D} - \mathbf{A}.$$

For any signal $\boldsymbol{x} \in \mathbb{R}^n$ on the vertices, the action of $\mathbf{L}$ acts as a discrete Laplace operator:

$$(\mathbf{L}\boldsymbol{x})_i = \sum_{j \sim i} w_{ij}(x_i - x_j).$$

The associated quadratic form satisfies

$$\boldsymbol{x}^T\mathbf{L}\boldsymbol{x} = \frac{1}{2}\sum_{i,j=1}^{n} w_{ij}(x_i - x_j)^2 = \sum_{(i,j)\in E} w_{ij}(x_i - x_j)^2.$$

Prop 8.3 **(Laplacian Positive Semidefinite)** For an undirected graph with nonnegative weights, $\mathbf{L}$ is symmetric positive semidefinite, and $\mathbf{L}\mathbf{1} = \mathbf{0}$.

Proof Symmetry follows from $w_{ij} = w_{ji}$, so $\mathbf{A} = \mathbf{A}^T$ and $\mathbf{L} = \mathbf{D} - \mathbf{A} = \mathbf{L}^T$. Expanding the quadratic form algebraically from definitions:

$$\begin{aligned}
\boldsymbol{x}^T\mathbf{L}\boldsymbol{x} = \boldsymbol{x}^T(\mathbf{D} - \mathbf{A})\boldsymbol{x} &= \sum_{i=1}^{n} d_i x_i^2 - \sum_{i,j=1}^{n} w_{ij}x_i x_j \\
&= \sum_{i=1}^{n}\left(\sum_{j=1}^{n} w_{ij}\right) x_i^2 - \sum_{i,j=1}^{n} w_{ij}x_i x_j \\
&= \frac{1}{2}\sum_{i,j=1}^{n} w_{ij}x_i^2 + \frac{1}{2}\sum_{i,j=1}^{n} w_{ij}x_j^2 - \sum_{i,j=1}^{n} w_{ij}x_i x_j \\
&= \frac{1}{2}\sum_{i,j=1}^{n} w_{ij}(x_i - x_j)^2 = \sum_{(i,j)\in E} w_{ij}(x_i - x_j)^2.
\end{aligned}$$

To obtain the middle equality, interchange the indices in one copy of the degree sum and use $w_{ij} = w_{ji}$. The term $-2x_i x_j$ then combines with $x_i^2 + x_j^2$ to form a square. The factor $1/2$ removes the duplicate ordered pairs (i, j) and (j, i).

Because $w_{ij} \geq 0$ and $(x_i - x_j)^2 \geq 0$, each term is non-negative, so $\boldsymbol{x}^T\mathbf{L}\boldsymbol{x} \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$, proving positive semidefiniteness. For $\boldsymbol{x} = \mathbf{1}$, every coordinate satisfies $(\mathbf{L}\mathbf{1})_i = d_i - \sum_{j=1}^{n} w_{ij} = 0$, so $\mathbf{L}\mathbf{1} = \mathbf{0}$.

The quadratic-form inequality is the definition of positive semidefiniteness. It is not strict positive definiteness, because the nonzero constant vector has zero quadratic form and is annihilated by the matrix. □

Thm 8.4 **(Spectral Properties of the Graph Laplacian)** Let the eigenvalues of $\mathbf{L}$ be ordered as $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$.

1. $\lambda_1 = 0$ with eigenvector $\boldsymbol{v}_1 = \mathbf{1}$.
2. The multiplicity of eigenvalue 0 equals the number of connected components of G.
3. The second smallest eigenvalue λ_2 is the algebraic connectivity (Fiedler value). It satisfies $\lambda_2 > 0$ if and only if G is connected.

Proof Proposition 8.3 shows that all eigenvalues of $\mathbf{L}$ are nonnegative and that $\mathbf{1} \in \ker(\mathbf{L})$, so $\lambda_1 = 0$.

It remains to identify the entire nullspace. If $\boldsymbol{x} \in \ker(\mathbf{L})$, then

$$0 = \boldsymbol{x}^T \mathbf{L} \boldsymbol{x} = \sum_{(i,j)\in E} w_{ij}(x_i - x_j)^2.$$

Every term is nonnegative. Hence $x_i = x_j$ across every edge with positive weight, and therefore $\boldsymbol{x}$ is constant on each connected component. Conversely, a vector that is constant on each connected component has zero difference across every edge and belongs to $\ker(\mathbf{L})$. A constant on one connected component may be chosen independently of the constants on the other components, because there are no edges between them. The vector equal to one on a given component and zero elsewhere is therefore in the null space.

The indicator vectors of the connected components form a basis for this nullspace, so the multiplicity of eigenvalue zero equals the number of components.

In particular, G is connected exactly when $\ker(\mathbf{L}) = \operatorname{span}\{\mathbf{1}\}$. In that case zero is a simple eigenvalue, so $\lambda_2 > 0$. If G is disconnected, the nullspace has dimension at least two and $\lambda_2 = 0$.

These indicator vectors are independent because their supports are disjoint and nonempty. They span every vector constant on components by assigning its component-wise constants as coefficients. This supplies both halves of the basis assertion and the exact dimension count. □

Ex 8.10 **(Path graph Laplacian spectrum)** For a three-node unweighted path graph $1 - 2 - 3$, write down $\mathbf{A}$ and $\mathbf{L}$. Compute the eigenvalues and eigenvectors of $\mathbf{L}$ by hand. Verify that $\lambda_1 = 0$ with $\boldsymbol{v}_1 = (1, 1, 1)^T$, and compute the algebraic connectivity λ_2.

Cor 8.5 **(Rayleigh Characterization of Algebraic Connectivity)** For a connected undirected graph,

$$\lambda_2 = \min_{\boldsymbol{x}\perp\mathbf{1},\, \boldsymbol{x}\neq\mathbf{0}} \frac{\boldsymbol{x}^T\mathbf{L}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}} = \min_{\boldsymbol{x}\perp\mathbf{1},\, \boldsymbol{x}\neq\mathbf{0}} \frac{\sum_{(i,j)\in E} w_{ij}(x_i - x_j)^2}{\sum_i x_i^2}.$$

Proof For a connected graph, the null space is spanned by the constant vector. Normalize it as $\boldsymbol{q}_1 = \mathbf{1}/\sqrt{n}$ and complete an orthonormal eigenbasis $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$, with $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n$.

Expand a vector perpendicular to $\mathbf{1}$ in this basis. Its first coefficient is $\boldsymbol{q}_1^T\boldsymbol{x} = \mathbf{1}^T\boldsymbol{x}/\sqrt{n} = 0$, so $\boldsymbol{x} = \sum_{i=2}^n c_i \boldsymbol{q}_i$. For a nonzero such vector, at least one remaining coefficient is nonzero. Therefore

$$\begin{aligned}\frac{\boldsymbol{x}^T\mathbf{L}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}} &= \frac{\sum_{i=2}^n \lambda_i c_i^2}{\sum_{i=2}^n c_i^2} \\ &\geq \frac{\lambda_2 \sum_{i=2}^n c_i^2}{\sum_{i=2}^n c_i^2} = \lambda_2.\end{aligned}$$

The lower bound is attained by $\boldsymbol{x} = \boldsymbol{q}_2$, which is perpendicular to the constant vector. Hence it is the minimum over the entire allowed subspace.

Finally, substitute the graph energy identity $\boldsymbol{x}^T\mathbf{L}\boldsymbol{x} = \sum_{(i,j)\in E} w_{ij}(x_i - x_j)^2$ and $\boldsymbol{x}^T\boldsymbol{x} = \sum_i x_i^2$. This gives the second expression in the statement. The orthogonality constraint removes the constant direction, whose quotient would otherwise give the uninformative minimum zero. □

Ex 8.11 **(Rayleigh quotient for algebraic connectivity)** For the path graph in the preceding exercise, minimize the quotient in Cor 8.5 over vectors of the form $(a, b, -a-b)^T$. Verify that the minimum equals the computed value of λ_2. Then evaluate the numerator on a two-level vector that is constant on each side of a proposed cut and explain how weak cross-cut weights lower its energy.

Def 8.7 **(Normalized Laplacians)** Assume every vertex has positive degree. Normalized Laplacians account for high-degree vertices:

- **Symmetric Normalized Laplacian:** $\mathbf{L}_{\text{sym}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$. Its spectrum lies in the interval $[0, 2]$.
- **Random Walk Laplacian:** $\mathbf{L}_{\text{rw}} = \mathbf{D}^{-1}\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1}\mathbf{A}$. It is similar to $\mathbf{L}_{\text{sym}}$ and shares its eigenvalues.

Ex 8.12 Write $\boldsymbol{y} = \mathbf{D}^{-1/2}\boldsymbol{x}$ and use $(y_i - y_j)^2 \le 2y_i^2 + 2y_j^2$ to prove $0 \le \boldsymbol{x}^T\mathbf{L}_{\text{sym}}\boldsymbol{x} \le 2\|\boldsymbol{x}\|_2^2$. Deduce the spectral interval $[0, 2]$, and verify the similarity between the two normalized Laplacians. What definition must be added if isolated vertices are allowed?

8.2.2 Spectral Clustering

A cut divides the vertices into nonempty sets A and B. Let $\text{cut}(A, B) = \sum_{i\in A, j\in B} w_{ij}$ and $\text{vol}(A) = \sum_{i\in A} d_i$. The normalized cut divides the cross-edge weight by the volume on each side and adds the two results:

$$\text{Ncut}(A, B) = \text{cut}(A, B)\left(\frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)}\right).$$

The denominators discourage cutting off a set with tiny total degree. The discrete problem selects vertex sets. A relaxation permits real-valued assignments and gives a lower bound on the best discrete value. Assume positive degrees and a connected graph in the following construction.

Thm 8.6 **(Spectral Relaxation of Graph Cuts)** Partitioning a graph into two disjoint sets A and B to minimize the normalized cut is NP-hard. Relaxing the discrete indicator vector to a continuous vector subject to an orthogonality constraint yields the Rayleigh quotient problem

$$\min_{\boldsymbol{x}\perp\mathbf{D}\mathbf{1},\, \boldsymbol{x}\neq\mathbf{0}} \frac{\boldsymbol{x}^T\mathbf{L}\boldsymbol{x}}{\boldsymbol{x}^T\mathbf{D}\boldsymbol{x}}.$$

The optimal solution is the Fiedler eigenvector $\boldsymbol{v}_2$ of the generalized eigenvalue problem $\mathbf{L}\boldsymbol{v} = \lambda\mathbf{D}\boldsymbol{v}$, or equivalently the second eigenvector of $\mathbf{L}_{\text{rw}}$. Spectral clustering embeds vertices in $\mathbb{R}^k$ using the first k nontrivial eigenvectors, then partitions the coordinates using k-means.

Ex 8.13 For a cut A, B, let $v_A = \sum_{i\in A} d_i$ and $v_B = \sum_{i\in B} d_i$. Set $x_i = 1/v_A$ on A and $x_i = -1/v_B$ on B. Show that $\boldsymbol{x}^T\mathbf{D}\mathbf{1} = 0$ and that its generalized Rayleigh quotient is $\text{cut}(A, B)(1/v_A + 1/v_B)$. Then set $\boldsymbol{z} = \mathbf{D}^{1/2}\boldsymbol{x}$ and use the spectral theorem to derive the relaxed minimizer stated above. Explain why converting that vector back into a cut is a separate step.

8.2.3 Diffusion Maps and Nonlinear Manifolds

Remark 8.4 **(From graphs to manifolds)** When data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N \in \mathbb{R}^d$ lie on a low-dimensional manifold embedded in high-dimensional space, ambient Euclidean distances can obscure intrinsic connectivity or create shortcuts across a curved manifold. Diffusion maps define a random walk on a graph constructed over the point cloud and, under suitable sampling and bandwidth limits, approximate diffusion generated by a Laplace–Beltrami operator.

Def 8.8 **(Diffusion Operator and Random Walk)** Given point cloud $\{\boldsymbol{x}_i\}_{i=1}^N \subset \mathbb{R}^d$, construct a pairwise affinity matrix $\mathbf{K} \in \mathbb{R}^{N\times N}$ using a Gaussian kernel with bandwidth $\delta > 0$:

$$K_{ij} = \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}{2\delta^2}\right).$$

Let $d_i = \sum_{j=1}^N K_{ij}$ denote the row sums, and define $\mathbf{D} = \operatorname{diag}(d_1, \ldots, d_N)$. The row-stochastic diffusion matrix is

$$\mathbf{M} = \mathbf{D}^{-1}\mathbf{K}.$$

The entry $M_{ij} = K_{ij}/d_i$ defines the transition probability from point $\boldsymbol{x}_i$ to point $\boldsymbol{x}_j$ in a single diffusion step.

Thm 8.7 **(Spectral Decomposition of the Diffusion Operator)** The matrix $\mathbf{M} = \mathbf{D}^{-1}\mathbf{K}$ is similar to the real symmetric matrix

$$\mathbf{S} = \mathbf{D}^{-1/2}\mathbf{K}\mathbf{D}^{-1/2} = \mathbf{D}^{1/2}\mathbf{M}\mathbf{D}^{-1/2}.$$

Let $\mathbf{S} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^T$ be the eigendecomposition of $\mathbf{S}$ with orthonormal eigenvectors $\mathbf{V} = [\boldsymbol{v}_0, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_{N-1}]$ and eigenvalues

$$1 = \lambda_0 > \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{N-1} \geq 0.$$

Then $\mathbf{M}$ has real eigenvalues λ_k with right eigenvectors $\boldsymbol{\psi}_k = \mathbf{D}^{-1/2}\boldsymbol{v}_k$ and left eigenvectors $\boldsymbol{\phi}_k = \mathbf{D}^{1/2}\boldsymbol{v}_k$. The right eigenvectors satisfy $\mathbf{M}\boldsymbol{\psi}_k = \lambda_k\boldsymbol{\psi}_k$, with $\boldsymbol{\psi}_0 = \mathbf{1}/\sqrt{\sum_i d_i}$ when $\boldsymbol{v}_0$ is normalized as above.

Proof All degrees are positive because every Gaussian affinity is positive. Thus the diagonal square roots and inverse square roots of $\mathbf{D}$ are well defined. Since $\mathbf{K}$ is symmetric,

$$\mathbf{S}^T = (\mathbf{D}^{-1/2}\mathbf{K}\mathbf{D}^{-1/2})^T = \mathbf{D}^{-1/2}\mathbf{K}\mathbf{D}^{-1/2} = \mathbf{S}.$$

Also $\mathbf{D}^{1/2}\mathbf{M}\mathbf{D}^{-1/2} = \mathbf{S}$, so the two matrices are similar.

We justify positive semidefiniteness of the Gaussian kernel rather than inferring it merely from positive entries. Write

$$K_{ij} = a_i a_j \exp(\boldsymbol{x}_i^T\boldsymbol{x}_j/\delta^2), \qquad a_i = \exp(-\|\boldsymbol{x}_i\|_2^2/(2\delta^2)).$$

Expand the last exponential in powers. For every nonnegative integer l, the kernel $(\boldsymbol{x}_i^T\boldsymbol{x}_j)^l$ is a Gram kernel: expand the dot product power into sums indexed by l coordinate choices. For real coefficients c_i, the resulting quadratic form is a sum of squares of quantities $\sum_i c_i x_{i,s_1}\cdots x_{i,s_l}$. It is nonnegative. The exponential is a convergent sum of these kernels with positive weights $1/(l!\delta^{2l})$.

Multiplication by $a_i a_j$ replaces c_i by $a_i c_i$, preserving nonnegativity. Thus $\mathbf{K}$, and consequently $\mathbf{S}$, are positive semidefinite.

The symmetric spectral theorem now gives an orthonormal eigenbasis $\boldsymbol{v}_k$ of $\mathbf{S}$ with nonnegative eigenvalues. The row sums of $\mathbf{M}$ are one. For any eigenvector of $\mathbf{M}$, selecting an entry of maximal magnitude in its eigenvalue equation gives $|\lambda| \leq 1$ by the triangle inequality. Also $\mathbf{M1} = \mathbf{1}$, so one eigenvalue is one.

Its eigenspace is one-dimensional. For a real vector with $\mathbf{M}\boldsymbol{\psi} = \boldsymbol{\psi}$, a maximal entry is a strictly positive weighted average of all entries. It can equal their maximum only if every entry equals that maximum. Thus the vector is constant. Similarity to a symmetric matrix gives diagonalizability, so the eigenvalue one is algebraically simple as well. Hence all remaining eigenvalues are less than one.

To recover eigenvectors, use $\mathbf{M} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{1/2}$:

$$\mathbf{M}(\mathbf{D}^{-1/2}\boldsymbol{v}_k) = \mathbf{D}^{-1/2}\mathbf{S}\boldsymbol{v}_k = \lambda_k \mathbf{D}^{-1/2}\boldsymbol{v}_k,$$
$$(\mathbf{D}^{1/2}\boldsymbol{v}_k)^T\mathbf{M} = \boldsymbol{v}_k^T\mathbf{S}\mathbf{D}^{1/2} = \lambda_k \boldsymbol{v}_k^T\mathbf{D}^{1/2}.$$

These are the claimed right and left eigenvectors. Their pairing is $\boldsymbol{\phi}_j^T\boldsymbol{\psi}_k = \boldsymbol{v}_j^T\boldsymbol{v}_k = \delta_{jk}$. Finally the eigenvector of $\mathbf{S}$ for one is proportional to $\mathbf{D}^{1/2}\mathbf{1}$. Its squared norm is $\sum_i d_i$, yielding the stated normalization of $\boldsymbol{\psi}_0$. Diffusion-map normalization is discussed in Coifman and Lafon (2006). □

Def 8.9 **(Diffusion Distance and Diffusion Map Coordinates)** The t-step diffusion operator $\mathbf{M}^t$ represents running the random walk forward for time t. The diffusion distance between points $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ measures the difference between their conditional transition distributions weighted by inverse degrees. Here $\pi_i = d_i/\sum_j d_j$ is the stationary probability; using inverse stationary probabilities instead multiplies the following squared distance by $\sum_j d_j$:

$$D_t(\boldsymbol{x}_i, \boldsymbol{x}_j)^2 = \|\mathbf{M}^t(i,:) - \mathbf{M}^t(j,:)\|^2_{\mathbf{D}^{-1}} = \sum_{k=1}^{N-1} \lambda_k^{2t}\big(\psi_k(i) - \psi_k(j)\big)^2.$$

The diffusion map embedding of point $\boldsymbol{x}_i$ into an m-dimensional Euclidean space is

$$\Psi_t(\boldsymbol{x}_i) = \begin{pmatrix} \lambda_1^t\psi_1(i) \\ \lambda_2^t\psi_2(i) \\ \vdots \\ \lambda_m^t\psi_m(i) \end{pmatrix} \in \mathbb{R}^m.$$

Under this embedding, Euclidean distance in $\mathbb{R}^m$ equals the truncated diffusion distance:

$$\|\Psi_t(\boldsymbol{x}_i) - \Psi_t(\boldsymbol{x}_j)\|_2^2 \approx D_t(\boldsymbol{x}_i, \boldsymbol{x}_j)^2.$$

Remark 8.5 **(Diffusion time as a geometric low-pass filter)** The factor λ_k^{2t} in the diffusion distance acts as an exponential low-pass filter on manifold coordinates. Because $\lambda_k < 1$ for all $k \geq 1$, increasing t rapidly suppresses high-frequency modes associated with small eigenvalues, leaving only the dominant macroscopic geometric coordinates.

Remark 8.6 **(Density normalization)** The random walk of Def 8.8 depends on how densely the points sample the manifold as well as on its geometry. Coifman and Lafon separate the two effects

by renormalizing the kernel before forming the random walk (Coifman and Lafon, 2006). For a parameter $\alpha \in [0,1]$, they replace K_{ij} by $K_{ij}/(d_i^\alpha d_j^\alpha)$ and then compute new row sums and the row-stochastic matrix as before. In the limit of many points and small bandwidth, $\alpha = 0$ gives the classical normalized graph Laplacian, whose limit depends on the sampling density. The choice $\alpha = 1/2$ gives the generator of a Fokker–Planck diffusion, and $\alpha = 1$ recovers the Laplace–Beltrami operator independently of the density. The renormalized kernel is still symmetric, so Thm 8.7 applies with the new row sums.

8.2.4 Permutation Invariance and Graph Spectra

Remark 8.7 **(Comparing configurations of identical particles)** In molecular systems and particle clusters with N identical atoms, the state is specified by atomic positions in $\mathbb{R}^{3N}$. Comparing coordinate vectors requires deciding which atom in one configuration corresponds to which in the other, as well as accounting for rotations and translations. An exhaustive assignment search has $N!$ candidates and becomes expensive quickly. Distance-based graph quantities avoid choosing a coordinate alignment, although a graph spectrum does not uniquely identify every configuration.

Thm 8.8 **(Permutation Invariance of Graph Spectra)** Let $\mathbf{D} \in \mathbb{R}^{N\times N}$ be the pairwise interatomic distance matrix with entries $D_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|_2$. Under any atom permutation $\sigma \in S_N$, the distance matrix transforms by an orthogonal permutation matrix $\mathbf{P}_\sigma$:

$$\mathbf{D}_\sigma = \mathbf{P}_\sigma \mathbf{D} \mathbf{P}_\sigma^T.$$

Let $\mathbf{L}$ be any graph Laplacian constructed from edge weights $w_{ij} = f(D_{ij})$. Then the permuted Laplacian is

$$\mathbf{L}_\sigma = \mathbf{P}_\sigma \mathbf{L} \mathbf{P}_\sigma^T.$$

Because $\mathbf{P}_\sigma$ is orthogonal, $\mathbf{L}_\sigma$ is orthogonally similar to $\mathbf{L}$. Therefore, the eigenvalue spectrum

$$\operatorname{spec}(\mathbf{L}) = (\lambda_1, \lambda_2, \ldots, \lambda_N)$$

is invariant under all atomic permutations $\sigma \in S_N$ and all rigid Euclidean isometries SE(3).

Proof Because rigid rotations and translations preserve pairwise distances $\|\mathbf{r}_i - \mathbf{r}_j\|_2$, the matrix $\mathbf{D}$ is invariant under SE(3).

For a rigid transformation $\boldsymbol{r}_i \mapsto \mathbf{R}\boldsymbol{r}_i + \boldsymbol{t}$ with $\mathbf{R}^T\mathbf{R} = \mathbf{I}$, the difference transforms as $\mathbf{R}(\boldsymbol{r}_i - \boldsymbol{r}_j)$. Its squared length is $(\boldsymbol{r}_i - \boldsymbol{r}_j)^T\mathbf{R}^T\mathbf{R}(\boldsymbol{r}_i - \boldsymbol{r}_j)$, equal to its old squared length. Translation cancels in the difference. Any affinity built only from these distances is therefore unchanged.

Under a permutation $\sigma \in S_N$, vertex i maps to $\sigma(i)$, so the permuted affinity matrix has entries $W_{\sigma(i),\sigma(j)} = W_{ij}$. In matrix notation, $\mathbf{W}_\sigma = \mathbf{P}_\sigma \mathbf{W} \mathbf{P}_\sigma^T$. Similarly, the degree matrix transforms as $\mathbf{D}_\sigma = \mathbf{P}_\sigma \mathbf{D} \mathbf{P}_\sigma^T$. Thus

$$\mathbf{L}_\sigma = \mathbf{D}_\sigma - \mathbf{W}_\sigma = \mathbf{P}_\sigma(\mathbf{D} - \mathbf{W})\mathbf{P}_\sigma^T = \mathbf{P}_\sigma \mathbf{L} \mathbf{P}_\sigma^T.$$

Because permutation matrices are orthogonal ($\mathbf{P}_\sigma^{-1} = \mathbf{P}_\sigma^T$), the characteristic polynomial satisfies

$$\det(\mathbf{L}_\sigma - \lambda\mathbf{I}) = \det(\mathbf{P}_\sigma(\mathbf{L} - \lambda\mathbf{I})\mathbf{P}_\sigma^T) = \det(\mathbf{P}_\sigma)\det(\mathbf{L} - \lambda\mathbf{I})\det(\mathbf{P}_\sigma^T) = \det(\mathbf{L} - \lambda\mathbf{I}).$$

The roots of the characteristic polynomial are identical. Thus the eigenvalue multiset $\operatorname{spec}(\mathbf{L})$ is invariant under any relabeling of identical particles.

Similarity preserves the multiset of roots, including multiplicity. Sorting these eigenvalues therefore removes the arbitrary vertex ordering. This is an invariance statement, not a uniqueness statement: two different graphs or configurations can still have the same spectrum. □

8.2.5 Affinity Scaling and Bandwidth Selection

Remark 8.8 **(Sigmoidal behavior of the affinity sum)** The Gaussian affinity kernel $K_{ij} = \exp(-\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2/(2\delta^2))$ requires selecting the bandwidth parameter δ. If δ is too small, points are isolated and $K_{ij} \approx 0$ for $i \neq j$. If δ is too large, points are fully connected and $K_{ij} \approx 1$.
Define the total affinity sum

$$S(\delta) = \sum_{i=1}^{N} \sum_{j=1}^{N} \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}{2\delta^2}\right).$$

For distinct data points, the log-affinity sum passes between two limiting plateaus. Under adequate sampling of a smooth manifold, an intermediate range may exhibit the scaling below:

1. As $\delta \to 0$, $K_{ij} \to 0$ for $i \neq j$, so $S(\delta) \to N$ and $\ln S(\delta) \to \ln N$.
2. As $\delta \to \infty$, $K_{ij} \to 1$ for all i, j, so $S(\delta) \to N^2$ and $\ln S(\delta) \to 2 \ln N$.
3. In an intermediate range where the continuum approximation dominates the diagonal terms, $S(\delta)$ can scale approximately as δ^d, where d is the intrinsic manifold dimension. The slope $\frac{d \ln S(\delta)}{d \ln \delta} \approx d$ estimates dimension.

An intermediate scaling regime is a useful diagnostic for choosing δ, but does not guarantee recovery of manifold geometry. Sampling density, finite sample size, and connectivity must also be checked.

8.2.6 Further Exercises

Ex 8.14 **(Permutation invariance of graph spectra)** Consider a cluster of four identical atoms in $\mathbb{R}^3$.

1. Let the coordinates be arranged in a vector $\boldsymbol{x} \in \mathbb{R}^{12}$. Explain why a direct search over all assignments of identical atoms has $4! = 24$ candidate permutations. What information would be lost by comparing the coordinate vectors without allowing relabeling?
2. Construct the interatomic distance matrix $\mathbf{D}$ and form the graph Laplacian $\mathbf{L}$. Prove that for any permutation $\sigma \in S_4$, $\mathbf{L}_\sigma = \mathbf{P}_\sigma \mathbf{L} \mathbf{P}_\sigma^T$.
3. Explain how sorting the eigenvalues $\text{spec}(\mathbf{L}) = (\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ produces a coordinate-free, permutation-invariant fingerprint for molecular structure.

Ex 8.15 **(Diffusion time as a geometric low-pass filter)** Let a diffusion operator on a data manifold have leading eigenvalues $\lambda_1 = 0.98$ and $\lambda_2 = 0.80$.

1. Compute the ratio of modal contributions $(\lambda_2/\lambda_1)^{2t}$ for diffusion times $t = 1$, $t = 5$, and $t = 20$.
2. Explain why increasing t contracts local high-frequency fluctuations while preserving macroscopic manifold geometry.
3. What happens to the diffusion distance $D_t(\boldsymbol{x}_i, \boldsymbol{x}_j)$ as $t \to \infty$?

Ex 8.16 **(Sigmoidal bandwidth selection)** Let $N = 100$ data points be sampled uniformly from a circle of radius $R = 1$ in $\mathbb{R}^2$.

1. Write down the formula for the affinity sum $S(\delta) = \sum_{i,j} \exp(-\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2/(2\delta^2))$.
2. Evaluate the asymptotic limits of $\ln S(\delta)$ as $\delta \to 0$ and $\delta \to \infty$.
3. For $\delta \in [0.1, 0.5]$, estimate the slope numerically. Identify where it is near one and where diagonal contributions or the curvature of the circle affect it. Repeat for two sample sizes and explain why a finite interval of exact scaling is not guaranteed.

Read **Further reading.** Chung develops spectral graph theory from the graph-Laplacian side (Chung, 1997). Belkin and Niyogi introduce Laplacian eigenmaps, while Coifman and Lafon develop diffusion maps and the density normalization described above (Belkin and Niyogi, 2003; Coifman and Lafon, 2006).

8.3 Matrix Exponentials and Krylov Subspace Actions

The scalar equation $u' = au$ has solution $u(t) = e^{ta}u(0)$. For the system $\boldsymbol{u}' = \mathbf{A}\boldsymbol{u}$, we seek a matrix that advances every initial vector in the same way. Replacing scalar powers by matrix powers in the exponential series produces that matrix.

For a large sparse matrix, the full exponential can be dense even though applying $\mathbf{A}$ to a vector is cheap. If only one initial state $\boldsymbol{v}$ is given, our computational objective is the vector $e^{t\mathbf{A}}\boldsymbol{v}$. We construct an approximation in a Krylov subspace and use the defect in the differential equation to assess its error. The later sensitivity and logarithm constructions extend this calculation to estimating model parameters.

$$\boldsymbol{u}'(t) = \mathbf{A}\boldsymbol{u}(t), \qquad \boldsymbol{u}(0) = \boldsymbol{v}, \tag{2}$$

$$\boldsymbol{u}(t) = e^{t\mathbf{A}}\boldsymbol{v}. \tag{3}$$

8.3.1 The Exponential and Its Algebraic Properties

Def 8.10 **(The Matrix Exponential)** For any square matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$, the matrix exponential is defined by the convergent power series

$$e^{\mathbf{A}} = \sum_{k=0}^{\infty} \frac{\mathbf{A}^k}{k!} = \mathbf{I} + \mathbf{A} + \frac{1}{2}\mathbf{A}^2 + \frac{1}{6}\mathbf{A}^3 + \dots$$

For any scalar $t \in \mathbb{R}$, the matrix function $e^{t\mathbf{A}}$ satisfies $\frac{d}{dt}e^{t\mathbf{A}} = \mathbf{A}e^{t\mathbf{A}} = e^{t\mathbf{A}}\mathbf{A}$.

Ex 8.17 Use $\|\mathbf{A}^k\| \le \|\mathbf{A}\|^k$ to compare the defining series with a scalar exponential and prove absolute convergence. On a bounded interval of t, justify termwise differentiation to verify the differential equation and initial value. Compute the exponential of $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ directly from the series. Why does this example not require diagonalization?

Thm 8.9 **(Algebraic Properties of the Matrix Exponential)** Let $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{n\times n}$.

1. If $\mathbf{A}$ and $\mathbf{B}$ commute ($\mathbf{AB} = \mathbf{BA}$), then $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$. If they do not commute, this equality fails in general.
2. The matrix exponential is always nonsingular, with inverse $(e^{\mathbf{A}})^{-1} = e^{-\mathbf{A}}$.

3. For any nonsingular similarity transform $\mathbf{S}$, $e^{\mathbf{SAS}^{-1}} = \mathbf{S}e^{\mathbf{A}}\mathbf{S}^{-1}$. In particular, if $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}$ is diagonalizable, then $e^{\mathbf{A}} = \mathbf{Q}\,\mathrm{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n})\mathbf{Q}^{-1}$.

4. The determinant satisfies Jacobi's formula: $\det(e^{\mathbf{A}}) = e^{\mathrm{trace}(\mathbf{A})}$.

Proof The exponential series is absolutely convergent in a matrix norm, since $\|\mathbf{A}^k/k!\| \le \|\mathbf{A}\|^k/k!$ and the scalar exponential series converges. This permits the regrouping of products used below.

When $\mathbf{A}$ and $\mathbf{B}$ commute, the product of the two series can be grouped by total degree:

$$\begin{aligned} e^{\mathbf{A}}e^{\mathbf{B}} &= \sum_{k=0}^{\infty}\sum_{j=0}^{k}\frac{\mathbf{A}^j\mathbf{B}^{k-j}}{j!(k-j)!} \\ &= \sum_{k=0}^{\infty}\frac{1}{k!}\sum_{j=0}^{k}\binom{k}{j}\mathbf{A}^j\mathbf{B}^{k-j} \\ &= \sum_{k=0}^{\infty}\frac{(\mathbf{A}+\mathbf{B})^k}{k!} = e^{\mathbf{A}+\mathbf{B}}. \end{aligned}$$

Commutativity is needed for the matrix binomial theorem in the last step. Without it, terms such as $\mathbf{AB}$ and $\mathbf{BA}$ cannot be collected as though they were equal.

Set $\mathbf{B} = -\mathbf{A}$. These matrices commute, and $e^{\mathbf{0}} = \mathbf{I}$, so both $e^{\mathbf{A}}e^{-\mathbf{A}}$ and $e^{-\mathbf{A}}e^{\mathbf{A}}$ equal the identity. This proves nonsingularity and the inverse formula.

For similarity, induction gives $(\mathbf{SAS}^{-1})^k = \mathbf{SA}^k\mathbf{S}^{-1}$: the adjacent factors $\mathbf{S}^{-1}\mathbf{S}$ cancel when one more factor is multiplied. The case $k = 0$ also holds because $\mathbf{SIS}^{-1} = \mathbf{I}$. Substituting into the convergent series gives

$$e^{\mathbf{SAS}^{-1}} = \sum_{k=0}^{\infty}\frac{\mathbf{SA}^k\mathbf{S}^{-1}}{k!} = \mathbf{S}e^{\mathbf{A}}\mathbf{S}^{-1}.$$

For diagonal $\mathbf{A}$, each power and then each exponential entry is computed diagonally. This proves the diagonalizable-matrix formula.

For the determinant, use Schur form $\mathbf{A} = \mathbf{QTQ}^*$. Similarity invariance gives $e^{\mathbf{A}} = \mathbf{Q}e^{\mathbf{T}}\mathbf{Q}^*$. Every power of an upper triangular matrix is upper triangular, and its ith diagonal entry is t_{ii}^k. Therefore $e^{\mathbf{T}}$ is upper triangular with diagonal entries $e^{t_{ii}}$. Its determinant is their product:

$$\det(e^{\mathbf{A}}) = \det(e^{\mathbf{T}}) = \prod_i e^{t_{ii}} = e^{\sum_i t_{ii}} = e^{\mathrm{tr}(\mathbf{A})}.$$

Trace and determinant are unchanged by similarity. This establishes the final property without assuming diagonalizability. These are standard matrix-exponential identities (Higham, 2008). □

Ex 8.18 Use $\mathbf{B} = -\mathbf{A}$ to prove the inverse formula. Prove $(\mathbf{SAS}^{-1})^k = \mathbf{SA}^k\mathbf{S}^{-1}$ by induction and substitute it into the series to prove similarity invariance. Finally, use Schur form and the diagonal entries of powers of a triangular matrix to show $\det(e^{\mathbf{A}}) = e^{\mathrm{tr}(\mathbf{A})}$.

Remark 8.9 **(Storage and the exponential action)** Classical methods for computing the full $n \times n$ matrix $e^{\mathbf{A}}$ rely on scaling and squaring with rational Padé approximants. Evaluating an $[m/m]$ Padé approximant requires computing matrix powers and solving matrix linear systems, incurring $O(n^3)$ operations. More critically, even when $\mathbf{A}$ is sparse, the exponential $e^{t\mathbf{A}}$ is typically dense.

Storing $e^{t\mathbf{A}}$ for $n = 10^6$ requires eight terabytes of memory in double precision. Many evolution calculations require only the action $e^{t\mathbf{A}}\boldsymbol{v}$, which can be approximated without forming the full exponential.

8.3.2 Krylov Subspace Approximation of the Exponential Action

To compute the action $e^{t\mathbf{A}}\boldsymbol{v}$ for large sparse $\mathbf{A}$, we project the evolution onto an expanding Krylov subspace $\mathcal{K}_m(\mathbf{A}, \boldsymbol{v}) = \operatorname{span}\{\boldsymbol{v}, \mathbf{A}\boldsymbol{v}, \ldots, \mathbf{A}^{m-1}\boldsymbol{v}\}$.

Thm 8.10 **(Arnoldi Approximation of $e^{t\mathbf{A}}\boldsymbol{v}$)** Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ and let $\boldsymbol{v} \in \mathbb{C}^n$ be a nonzero vector with $\beta = \|\boldsymbol{v}\|_2$. After m steps of the Arnoldi iteration starting from $\boldsymbol{v}_1 = \boldsymbol{v}/\beta$, we obtain the Arnoldi relation

$$\mathbf{A}\mathbf{V}_m = \mathbf{V}_m\mathbf{H}_m + h_{m+1,m}\boldsymbol{v}_{m+1}\boldsymbol{e}_m^T,$$

where $\mathbf{V}_m \in \mathbb{C}^{n\times m}$ has orthonormal columns and $\mathbf{H}_m = \mathbf{V}_m^*\mathbf{A}\mathbf{V}_m \in \mathbb{C}^{m\times m}$ is upper Hessenberg. The Arnoldi approximation to the exponential action is

$$e^{t\mathbf{A}}\boldsymbol{v} \approx \beta\mathbf{V}_m e^{t\mathbf{H}_m}\boldsymbol{e}_1.$$

Proof We approximate the solution of $\boldsymbol{u}'(t) = \mathbf{A}\boldsymbol{u}(t)$ with $\boldsymbol{u}(0) = \boldsymbol{v}$ within the subspace $\mathcal{K}_m(\mathbf{A}, \boldsymbol{v})$ by setting $\boldsymbol{u}_m(t) = \mathbf{V}_m\boldsymbol{z}_m(t)$, where $\boldsymbol{z}_m(t) \in \mathbb{C}^m$. The residual defect of the ODE is $\boldsymbol{r}_m(t) = \boldsymbol{u}_m'(t) - \mathbf{A}\boldsymbol{u}_m(t) = \mathbf{V}_m\boldsymbol{z}_m'(t) - \mathbf{A}\mathbf{V}_m\boldsymbol{z}_m(t)$. The vector $\boldsymbol{z}_m(t)$ contains time-dependent coordinates in the fixed basis $\mathbf{V}_m$. We cannot generally make every component of the full differential defect zero with only these m coordinates. Instead we require zero components along all basis vectors. This is the meaning of the following Galerkin condition.

Imposing the Galerkin orthogonality condition $\mathbf{V}_m^*\boldsymbol{r}_m(t) = \mathbf{0}$ yields

$$\mathbf{V}_m^*\mathbf{V}_m\boldsymbol{z}_m'(t) - \mathbf{V}_m^*\mathbf{A}\mathbf{V}_m\boldsymbol{z}_m(t) = \mathbf{0} \implies \boldsymbol{z}_m'(t) = \mathbf{H}_m\boldsymbol{z}_m(t).$$

The initial condition $\boldsymbol{u}_m(0) = \boldsymbol{v} = \beta\boldsymbol{v}_1 = \beta\mathbf{V}_m\boldsymbol{e}_1$ requires $\boldsymbol{z}_m(0) = \beta\boldsymbol{e}_1$. Solving the reduced $m \times m$ linear system gives $\boldsymbol{z}_m(t) = \beta e^{t\mathbf{H}_m}\boldsymbol{e}_1$. Multiplying by $\mathbf{V}_m$ recovers $\boldsymbol{u}_m(t) = \beta\mathbf{V}_m e^{t\mathbf{H}_m}\boldsymbol{e}_1$.

The small exponential is $m \times m$, and $\boldsymbol{e}_1$ has length m. The final multiplication by $\mathbf{V}_m$ returns the vector to the original n-dimensional space. No $n \times n$ exponential was formed. At an invariant-subspace termination the same reduced equation is exact for this initial vector. □

An approximate trajectory can have the right initial value but fail to satisfy the differential equation. Its defect is the difference $\boldsymbol{u}_m' - \mathbf{A}\boldsymbol{u}_m$. This is computable from the projected solution. Subtracting the approximate equation from the exact one produces a forced linear equation for the error. Variation of constants then shows how past defects propagate to the final time.

Thm 8.11 **(Residual Representation and A Posteriori Error Bound)** The defect in the differential equation satisfies

$$\boldsymbol{u}_m'(t) - \mathbf{A}\boldsymbol{u}_m(t) = -\beta h_{m+1,m}[\boldsymbol{e}_m^T e^{t\mathbf{H}_m}\boldsymbol{e}_1]\boldsymbol{v}_{m+1}.$$

Integrating the defect via variation of constants yields the exact error identity

$$e^{t\mathbf{A}}\boldsymbol{v} - \beta\mathbf{V}_m e^{t\mathbf{H}_m}\boldsymbol{e}_1 = \beta h_{m+1,m}\int_0^t e^{(t-\tau)\mathbf{A}}\boldsymbol{v}_{m+1}[\boldsymbol{e}_m^T e^{\tau\mathbf{H}_m}\boldsymbol{e}_1]\,d\tau.$$

In the Euclidean norm, if the operator $\mathbf{A}$ generates a contractive semigroup satisfying $\|e^{\tau\mathbf{A}}\|_2 \le 1$ for all $\tau \ge 0$, the error is bounded by

$$\|e^{t\mathbf{A}}\boldsymbol{v} - \beta\mathbf{V}_m e^{t\mathbf{H}_m}\boldsymbol{e}_1\|_2 \le \beta|h_{m+1,m}| \int_0^t |\boldsymbol{e}_m^T e^{\tau\mathbf{H}_m}\boldsymbol{e}_1|\, d\tau.$$

Proof Using the Arnoldi relation $\mathbf{A}\mathbf{V}_m = \mathbf{V}_m\mathbf{H}_m + h_{m+1,m}\boldsymbol{v}_{m+1}\boldsymbol{e}_m^T$, the ODE defect is

$$\begin{aligned}
\boldsymbol{u}_m'(t) - \mathbf{A}\boldsymbol{u}_m(t) &= \mathbf{V}_m\boldsymbol{z}_m'(t) - \mathbf{A}\mathbf{V}_m\boldsymbol{z}_m(t) \\
&= \mathbf{V}_m\mathbf{H}_m\boldsymbol{z}_m(t) - \left(\mathbf{V}_m\mathbf{H}_m + h_{m+1,m}\boldsymbol{v}_{m+1}\boldsymbol{e}_m^T\right)\boldsymbol{z}_m(t) \\
&= -h_{m+1,m}\boldsymbol{v}_{m+1}\boldsymbol{e}_m^T\boldsymbol{z}_m(t) \\
&= -\beta h_{m+1,m}[\boldsymbol{e}_m^T e^{t\mathbf{H}_m}\boldsymbol{e}_1]\boldsymbol{v}_{m+1}.
\end{aligned}$$

The approximation error $\boldsymbol{e}(t) = \boldsymbol{u}(t) - \boldsymbol{u}_m(t)$ satisfies $\boldsymbol{e}'(t) = \mathbf{A}\boldsymbol{e}(t) - (\boldsymbol{u}_m'(t) - \mathbf{A}\boldsymbol{u}_m(t))$ with $\boldsymbol{e}(0) = \mathbf{0}$. The error equation can be integrated directly. Differentiate $e^{-t\mathbf{A}}\boldsymbol{e}(t)$, using the product rule and the fact that $\mathbf{A}$ commutes with its own exponential. The $\mathbf{A}\boldsymbol{e}$ terms cancel, leaving $e^{-t\mathbf{A}}[\mathbf{A}\boldsymbol{u}_m(t) - \boldsymbol{u}_m'(t)]$. Integrate from zero to t and multiply by $e^{t\mathbf{A}}$. The initial error is zero because both trajectories start at $\boldsymbol{v}$.

Integrating via variation of constants:

$$\boldsymbol{e}(t) = \int_0^t e^{(t-\tau)\mathbf{A}}\left[\mathbf{A}\boldsymbol{u}_m(\tau) - \boldsymbol{u}_m'(\tau)\right] d\tau = \beta h_{m+1,m}\int_0^t e^{(t-\tau)\mathbf{A}}\boldsymbol{v}_{m+1}[\boldsymbol{e}_m^T e^{\tau\mathbf{H}_m}\boldsymbol{e}_1]\, d\tau.$$

Taking norms with $\|e^{(t-\tau)\mathbf{A}}\| \le 1$ and $\|\boldsymbol{v}_{m+1}\|_2 = 1$ yields

$$\|\boldsymbol{e}(t)\| \le \beta|h_{m+1,m}| \int_0^t \|e^{(t-\tau)\mathbf{A}}\|_2\|\boldsymbol{v}_{m+1}\|_2|\boldsymbol{e}_m^T e^{\tau\mathbf{H}_m}\boldsymbol{e}_1|\, d\tau \le \beta|h_{m+1,m}| \int_0^t |\boldsymbol{e}_m^T e^{\tau\mathbf{H}_m}\boldsymbol{e}_1|\, d\tau.$$

The last inequality requires $t \ge 0$ and the stated bound on the propagator at every intervening time $t - \tau$. Without that contractivity assumption, the exact error identity remains valid, but its norm bound must retain $\|e^{(t-\tau)\mathbf{A}}\|_2$ under the integral. This distinguishes an exact representation from an estimate requiring an extra hypothesis. □

Ex 8.19 Repeat the norm estimate in the 1-norm. Explain why the right-hand side acquires the factor $\|\boldsymbol{v}_{m+1}\|_1$ when the Arnoldi basis is Euclidean-orthonormal. For a column-conservative Markov generator, use uniformization to show that $\|e^{t\mathbf{A}}\|_1 = 1$, and explain why this does not imply Euclidean contractivity.

Remark 8.10 **(Superlinear convergence of Krylov actions)** Hochbruck and Lubich bound the error of the Krylov approximation in terms of the spectrum and the time step (Hochbruck and Lubich, 1997). For a symmetric negative semidefinite matrix with eigenvalues in $[-4\rho, 0]$, the error in approximating $e^{\tau\mathbf{A}}\boldsymbol{v}$ from $\mathcal{K}_m(\mathbf{A}, \boldsymbol{v})$ is bounded by a constant multiple of $e^{-m^2/(5\rho\tau)}$ when $\sqrt{4\rho\tau} \le m \le 2\rho\tau$. For $m \ge 2\rho\tau$ the bound decays superlinearly in m. The relevant scale is therefore $\rho\tau$, not $\|\mathbf{A}\|_2$ alone, and in the first regime the required dimension grows only like the square root of $\rho\tau$. The same paper gives analogous bounds for non-Hermitian matrices whose numerical range lies in a known region of the left half-plane. The projected approximation also adapts to the spectrum seen from the starting vector rather than to the full matrix norm alone. Grid-independent convergence requires additional spectral or semigroup assumptions and should be checked for the discretization at hand.

8.3.3 The Continuous-Adjoint Sensitivity Formula

In parameter inference, we estimate unknown coefficients in an evolution model from discrete observations. Consider an affine generator family

$$\mathbf{A}(\boldsymbol{\theta}) = \mathbf{A}_0 + \sum_{j=1}^{M_{\text{lib}}} \theta_j \mathbf{A}_j, \tag{4}$$

where $\mathbf{A}_0$ is a baseline operator, the matrices $\mathbf{A}_j$ represent candidate physical mechanisms, and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_{M_{\text{lib}}})^T$ is a vector of unknown rate parameters.

Suppose the prediction $\boldsymbol{y}$ is compared with observed data through a real scalar function ℓ. To adjust a parameter, we need the derivative of that comparison with respect to the parameter. Differentiating the ODE gives one possible calculation for each parameter. An adjoint rearranges the chain rule so that all parameters share the same backward propagation. Parameters are real, and the initial vector is fixed. For complex states we use the real differential convention $d\ell = \operatorname{Re}[(\nabla_{\boldsymbol{y}}\ell)^* d\boldsymbol{y}]$.

Thm 8.12 **(Adjoint Gradient Formula for Exponential Actions)** Let $\boldsymbol{y} = e^{h\mathbf{A}(\boldsymbol{\theta})}\boldsymbol{v}$ denote the state propagated across an interval of length h. Let $\ell(\boldsymbol{y})$ be a differentiable scalar loss with endpoint gradient $\nabla_{\boldsymbol{y}}\ell$. Then the derivative of the loss with respect to coefficient θ_j is

$$\frac{\partial \ell}{\partial \theta_j} = h \operatorname{Re} \int_0^1 \boldsymbol{\lambda}(s)^* \mathbf{A}_j \boldsymbol{f}(s)\, ds,$$

where the forward path $\boldsymbol{f}(s)$ and adjoint path $\boldsymbol{\lambda}(s)$ are defined by

$$\boldsymbol{f}(s) = e^{sh\mathbf{A}}\boldsymbol{v}, \qquad \boldsymbol{\lambda}(s) = e^{(1-s)h\mathbf{A}^*}\nabla_{\boldsymbol{y}}\ell, \qquad 0 \le s \le 1.$$

Proof Let $\boldsymbol{f}(s)$ satisfy the ODE $\boldsymbol{f}'(s) = h\mathbf{A}(\boldsymbol{\theta})\boldsymbol{f}(s)$ with $\boldsymbol{f}(0) = \boldsymbol{v}$ and $\boldsymbol{y} = \boldsymbol{f}(1)$. Let $\boldsymbol{g}_j(s) = \partial \boldsymbol{f}(s)/\partial\theta_j$. The product rule applied to $\mathbf{A}(\boldsymbol{\theta})\boldsymbol{f}(s)$ gives two terms: one differentiates the matrix and the other the state. Since the family is affine, $\partial\mathbf{A}/\partial\theta_j = \mathbf{A}_j$. The initial state is fixed, so $\boldsymbol{g}_j(0) = \mathbf{0}$. Differentiating this initial-value problem with respect to θ_j yields the tangent equation

$$\frac{d}{ds}\left(\frac{\partial \boldsymbol{f}}{\partial \theta_j}\right) = h\mathbf{A}\left(\frac{\partial \boldsymbol{f}}{\partial \theta_j}\right) + h\mathbf{A}_j \boldsymbol{f}(s), \qquad \frac{\partial \boldsymbol{f}(0)}{\partial \theta_j} = \mathbf{0}.$$

Applying variation of constants at the endpoint $s = 1$ gives

$$\frac{\partial \boldsymbol{y}}{\partial \theta_j} = h \int_0^1 e^{(1-s)h\mathbf{A}} \mathbf{A}_j \boldsymbol{f}(s)\, ds.$$

Applying the chain rule to the scalar loss $\ell(\boldsymbol{y})$ and taking the real part produces

$$\begin{aligned}\frac{\partial \ell}{\partial \theta_j} &= \operatorname{Re}\left((\nabla_{\boldsymbol{y}}\ell)^* \frac{\partial \boldsymbol{y}}{\partial \theta_j}\right) = h \operatorname{Re} \int_0^1 (\nabla_{\boldsymbol{y}}\ell)^* e^{(1-s)h\mathbf{A}} \mathbf{A}_j \boldsymbol{f}(s)\, ds \\ &= h \operatorname{Re} \int_0^1 \left(e^{(1-s)h\mathbf{A}^*}\nabla_{\boldsymbol{y}}\ell\right)^* \mathbf{A}_j \boldsymbol{f}(s)\, ds = h \operatorname{Re} \int_0^1 \boldsymbol{\lambda}(s)^* \mathbf{A}_j \boldsymbol{f}(s)\, ds.\end{aligned}$$

The identity $(e^{(1-s)h\mathbf{A}^*})^* = e^{(1-s)h\mathbf{A}}$ follows by conjugate-transposing the power series term by term. This is the step that moves propagation from the state derivative to the loss gradient. In a real problem the conjugations and real-part operation reduce to ordinary transposes and the usual chain rule. □

Ex 8.20 Check the formula when $A(\theta) = \theta$ is scalar, $v = 1$, and $\ell(y) = \frac{1}{2}(y - y_{\text{obs}})^2$. Differentiate $y = e^{h\theta}$ directly, then evaluate the adjoint integral. Why does its integrand become independent of s in this example?

Remark 8.11 **(The adjoint sharing principle)** Evaluating a separate directional Fréchet derivative $L_{\exp}(h\mathbf{A}, h\mathbf{A}_j)$ for each candidate matrix $\mathbf{A}_j$ repeats the forward exponential propagation M_{lib} times. The adjoint identity separates propagation from the candidate directions. We compute the forward path $\boldsymbol{f}$ and adjoint path $\boldsymbol{\lambda}$ once per observation interval, and evaluate their contractions against $\mathbf{A}_j$ via numerical quadrature.

8.3.4 Shared Krylov Spaces and Quadrature

To evaluate the bilinear integral numerically, we approximate both paths using independent forward and adjoint Krylov spaces, and integrate the contraction using Gauss–Legendre quadrature.

Thm 8.13 **(Bilinear Quadrature Evaluation)** Let $\mathcal{K}_{m_f}(\mathbf{A}, \boldsymbol{v})$ have basis $\mathbf{V}_f$ and projected Hessenberg matrix $\mathbf{H}_f$, with $\beta_f = \|\boldsymbol{v}\|_2$. Let $\mathcal{K}_{m_a}(\mathbf{A}^*, \nabla_{\boldsymbol{y}}\ell)$ have basis $\mathbf{V}_a$ and projected matrix $\mathbf{H}_a$, with $\beta_a = \|\nabla_{\boldsymbol{y}}\ell\|_2$. At any Gauss–Legendre node $s_q \in [0, 1]$ with quadrature weight w_q, the path approximations are

$$\boldsymbol{f}^{(q)} = \beta_f \mathbf{V}_f e^{s_q h \mathbf{H}_f} \boldsymbol{e}_1, \qquad \boldsymbol{\lambda}^{(q)} = \beta_a \mathbf{V}_a e^{(1-s_q)h\mathbf{H}_a} \boldsymbol{e}_1.$$

The coefficient gradient evaluates to

$$\frac{\partial \ell}{\partial \theta_j} \approx h \sum_{q=1}^{n_q} w_q \operatorname{Re}\left[(\boldsymbol{\lambda}^{(q)})^* \mathbf{A}_j \boldsymbol{f}^{(q)}\right].$$

Ex 8.21 **One-Sided Termination:** Construct a 3×3 example where $\boldsymbol{v}$ is an exact eigenvector of $\mathbf{A}$, yielding forward residual $\rho_f = 0$ at step 1, while the adjoint starting vector is not an eigenvector. Explain why product heuristics fail in this setting.

Remark 8.12 **(One-sided termination in refinement heuristics)** A common stopping diagnostic tests the product of forward and adjoint residuals $\rho_f \rho_a$. If the forward path reaches an invariant subspace, $\rho_f = 0$, causing the product to vanish even when the adjoint path is at depth $m_a = 2$ and completely unresolved. Reliable stopping criteria require monitoring forward and adjoint convergence separately.

8.3.5 Matrix Logarithms and Generator Recovery

A discrete observation map records evolution over a fixed interval. In the scalar case, recovering a decay rate from a positive multiplier uses $a = h^{-1} \log k$. For oscillations, the exponential loses integer multiples of 2π in phase. The principal matrix logarithm chooses one branch, but recovering the actual generator also requires that its frequencies fall within that branch.

Def 8.11 **(The Matrix Logarithm)** Let $\mathbf{K} \in \mathbb{C}^{n \times n}$ have no eigenvalues on the closed negative real axis $(-\infty, 0]$. The principal matrix logarithm $\log(\mathbf{K})$ is the unique matrix $\mathbf{A}$ such that $e^{\mathbf{A}} = \mathbf{K}$ and all eigenvalues of $\mathbf{A}$ satisfy $|\operatorname{Im}(\lambda)| < \pi$.

Ex 8.22 **Logarithm Branch Cuts:** Let an evolution operator have oscillatory eigenvalue $\lambda = -\gamma + i\omega$ with $\omega = 50\pi$. Determine the maximum observation interval h that avoids branch-cut crossing in the principal matrix logarithm.

Remark 8.13 **(Projection and finite-time closure bias)** If P denotes the orthogonal projection onto a subspace of observables $\mathcal{F}$, the projected generator obtained from finite-time observations satisfies

$$\widehat{\mathbf{A}}_h = h^{-1} \log \left(P e^{h\mathcal{A}}|_{\mathcal{F}} \right) \neq P\mathcal{A}|_{\mathcal{F}}.$$

Unless $\mathcal{F}$ is invariant under $\mathcal{A}$, projection and exponentiation do not commute. The resulting finite-time closure bias can be expanded in powers of h, but its leading order depends on which terms leave the observable space. Any rate such as $O(h^r)$ therefore requires a separately defined closure depth and corresponding assumptions.

8.3.6 Further Exercises

Ex 8.23 **Dense vs. Action Complexity:** A sparse matrix $\mathbf{A} \in \mathbb{R}^{10^6 \times 10^6}$ has 10 nonzero entries per row. Compute the memory required to store $e^{h\mathbf{A}}$ as a dense array. Compare this with storing $m = 30$ Arnoldi basis vectors.

Ex 8.24 **Adjoint Sharing:** Let an overcomplete library contain $M_{\text{lib}} = 200$ candidate directions. Let each forward and adjoint Krylov construction use m products, and count products with $\mathbf{A}$ or $\mathbf{A}^*$ separately from contractions with candidate matrices. Compare rebuilding a forward and adjoint path separately for each of the 200 coefficients with sharing the two paths across all coefficients at five quadrature nodes. State the cost model for a candidate contraction, and explain why the number of candidates and nodes alone does not determine the cost of directional Fréchet derivatives.

Ex 8.25 **Taylor vs. Krylov Convergence:** For the 1D discrete Laplacian $\mathbf{A}$ with $\|\mathbf{A}\|_2 = 4 \times 10^4$ and $t = 10^{-2}$, compare the polynomial degrees suggested by a direct Taylor remainder bound and by Krylov projection. State every estimate you use, since neither degree is determined by $\|\mathbf{A}\|_2$ alone.

Read **Further reading.** Higham gives the general theory of matrix functions (Higham, 2008). Moler and Van Loan explain why apparently reasonable methods for the matrix exponential can fail, and Al-Mohy and Higham develop a modern action method that avoids forming the exponential (Moler and Van Loan, 2003; Al-Mohy and Higham, 2011).

8.4 Markov Chains, Continuous Generators, and the CME

A finite random system has a probability of occupying each possible state. A Markov model specifies how these probabilities change using only the current state. We first use a matrix of transition probabilities for a discrete step, then a matrix of transition rates for continuous time.

The matrix conventions matter: throughout this section, probabilities are columns, so column j describes transitions out of state j. We derive conservation and positivity, then assemble reaction models and examine the error introduced by retaining only a finite set of states.

8.4.1 Discrete-Time Markov Chains

Def 8.12 **(Probability Vector)** A vector $\boldsymbol{x} \in \mathbb{R}^n$ is a probability vector if

$$x_i \geq 0, \qquad \sum_{i=1}^{n} x_i = 1.$$

The entry x_i represents the probability of occupying state i.

Def 8.13 **(Column-Stochastic Matrix)** A matrix $\mathbf{P} \in \mathbb{R}^{n \times n}$ is column-stochastic if:

1. $P_{ij} \geq 0$ for all i, j.
2. $\sum_{i=1}^{n} P_{ij} = 1$ for each column j.

Example For two states, take $\mathbf{P} = \begin{pmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{pmatrix}$. Starting in state one gives distribution $(0.8, 0.2)^T$ after one step. Starting from probabilities $(0.6, 0.4)^T$ gives a mixture of the two columns, namely $(0.52, 0.48)^T$.

Ex 8.26 Compute the second step for each initial distribution. Prove that multiplying any probability vector by a column-stochastic matrix produces another probability vector. Identify separately where you use nonnegative entries and where you use the column sums.

Remark 8.14 **(Column convention)** Column j of $\mathbf{P}$ contains the transition probabilities from state j to all destination states i. The update rule for a probability distribution is $\boldsymbol{x}^{(k+1)} = \mathbf{P}\boldsymbol{x}^{(k)}$.

Thm 8.14 **(Spectrum of a Stochastic Matrix)** If $\mathbf{P}$ is column-stochastic, then $\lambda = 1$ is an eigenvalue, and every eigenvalue satisfies $|\lambda| \leq 1$.

Proof The column-sum condition can be written $\mathbf{1}^T\mathbf{P} = \mathbf{1}^T$. Taking transposes gives $\mathbf{P}^T\mathbf{1} = \mathbf{1}$, so one is an eigenvalue of $\mathbf{P}^T$. The characteristic polynomials of a matrix and its transpose agree, because

$$\det(\mathbf{P}^T - z\mathbf{I}) = \det((\mathbf{P} - z\mathbf{I})^T) = \det(\mathbf{P} - z\mathbf{I}).$$

Hence one is also an eigenvalue of $\mathbf{P}$. The constant vector need not be its right eigenvector under the column convention.

To bound any eigenvalue, use an eigenvector $\boldsymbol{v} \neq \mathbf{0}$ of $\mathbf{P}^T$ with eigenvalue λ. Choose j where $|v_j| = \max_i |v_i| > 0$. Its scalar eigenvalue equation is

$$\lambda v_j = \sum_i P_{ij} v_i.$$

Since the entries are nonnegative,

$$\begin{aligned} |\lambda||v_j| &\leq \sum_i P_{ij}|v_i| \\ &\leq \sum_i P_{ij}|v_j| = |v_j| \sum_i P_{ij} = |v_j|. \end{aligned}$$

Divide by the positive number $|v_j|$ to get $|\lambda| \leq 1$. The argument also covers complex eigenvectors because it uses absolute values. Equality of characteristic polynomials transfers this bound to every eigenvalue of $\mathbf{P}$. □

Def 8.14 **(Stationary Distribution)** A probability vector $\boldsymbol{\pi}$ is stationary if $\mathbf{P}\boldsymbol{\pi} = \boldsymbol{\pi}$. It is a right eigenvector of $\mathbf{P}$ associated with $\lambda = 1$, normalized so that $\mathbf{1}^T\boldsymbol{\pi} = 1$.

Ex 8.27 The matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ moves deterministically between two states. Find a stationary distribution, then start from $\boldsymbol{e}_1$ and show that the distributions do not converge. A stationary distribution can exist without attracting every initial distribution. Which hypothesis of the following theorem rules out this example?

Thm 8.15 **(Perron–Frobenius for Markov Chains)** If $\mathbf{P}$ is irreducible (every state can be reached from any other state in finitely many steps) and primitive (aperiodic, so $\mathbf{P}^k > 0$ elementwise for some $k \geq 1$), then:

1. $\lambda_1 = 1$ is an algebraically and geometrically simple eigenvalue.
2. $|\lambda_i| < 1$ for all other eigenvalues $\lambda_i \neq 1$.
3. The stationary distribution $\boldsymbol{\pi}$ is strictly positive in every entry.
4. For any initial probability vector $\boldsymbol{x}^{(0)}$, $\lim_{k\to\infty} \mathbf{P}^k\boldsymbol{x}^{(0)} = \boldsymbol{\pi}$.

Def 8.15 **(Spectral Gap and Mixing Time)** For a primitive chain with eigenvalues ordered as $1 = \lambda_1 > |\lambda_2| \geq \cdots \geq |\lambda_n|$, the spectral gap is

$$\delta = 1 - |\lambda_2|.$$

For a diagonalizable matrix, the error $\|\boldsymbol{x}^{(k)} - \boldsymbol{\pi}\|_2$ decays asymptotically as $\mathcal{O}(|\lambda_2|^k)$. The number of steps k_ε required to reach error tolerance ε satisfies

$$k_\varepsilon \approx \frac{\log(1/\varepsilon)}{\delta}.$$

Ex 8.28 **(Discrete spectrum and mixing)** Under the column-stochastic convention, let $\mathbf{P} = \begin{pmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{pmatrix}$. Find its stationary distribution $\boldsymbol{\pi}$, its second eigenvalue, and the spectral gap δ. Starting from $\boldsymbol{x}^{(0)} = \boldsymbol{e}_1$, derive the exact formula for $\|\mathbf{P}^k\boldsymbol{x}^{(0)} - \boldsymbol{\pi}\|_2$. Compare the first k for which this error is at most 10^{-3} with the estimate $\log(10^3)/\delta$, and explain why the latter is only an asymptotic rule.

Def 8.16 **(Detailed Balance (Reversibility))** A Markov chain satisfies detailed balance with respect to $\boldsymbol{\pi}$ if

$$P_{ij}\pi_j = P_{ji}\pi_i \qquad \text{for all } i, j.$$

A chain satisfying detailed balance is reversible.

Thm 8.16 **(Real Spectrum of Reversible Chains)** Let $\mathbf{P}$ satisfy detailed balance with stationary distribution $\boldsymbol{\pi} > 0$. Define $\mathbf{D}_{\boldsymbol{\pi}} = \operatorname{diag}(\boldsymbol{\pi})$. Then the matrix

$$\mathbf{S} = \mathbf{D}_{\boldsymbol{\pi}}^{-1/2}\mathbf{P}\mathbf{D}_{\boldsymbol{\pi}}^{1/2}$$

is symmetric. Consequently, all eigenvalues of $\mathbf{P}$ are real, and $\mathbf{P}$ has an orthogonal basis of eigenvectors in the inner product weighted by $\mathbf{D}_{\boldsymbol{\pi}}^{-1}$.

Proof Let $\mathbf{D}_{\boldsymbol{\pi}}$ have the positive stationary probabilities on its diagonal. Positivity makes its square root and inverse square root well defined. Entry (i, j) of the transformed matrix is

$$S_{ij} = \pi_i^{-1/2} P_{ij} \pi_j^{1/2} = \frac{P_{ij}\pi_j}{\sqrt{\pi_i \pi_j}}.$$

Detailed balance changes the numerator to $P_{ji}\pi_i$, giving

$$S_{ij} = \frac{P_{ji}\pi_i}{\sqrt{\pi_i \pi_j}} = \pi_j^{-1/2} P_{ji} \pi_i^{1/2} = S_{ji}.$$

Thus $\mathbf{S}$ is real symmetric and has a real orthonormal eigenbasis $\boldsymbol{q}_i$ with real eigenvalues λ_i.

Undo the similarity. The vectors $\boldsymbol{v}_i = \mathbf{D}_{\boldsymbol{\pi}}^{1/2} \boldsymbol{q}_i$ satisfy

$$\mathbf{P}\boldsymbol{v}_i = \mathbf{D}_{\boldsymbol{\pi}}^{1/2} \mathbf{S} \boldsymbol{q}_i = \lambda_i \boldsymbol{v}_i.$$

The invertible change of coordinates preserves their independence, so they form a basis. In the claimed weighted inner product their pairings are

$$\boldsymbol{v}_i^T \mathbf{D}_{\boldsymbol{\pi}}^{-1} \boldsymbol{v}_j = \boldsymbol{q}_i^T \mathbf{D}_{\boldsymbol{\pi}}^{1/2} \mathbf{D}_{\boldsymbol{\pi}}^{-1} \mathbf{D}_{\boldsymbol{\pi}}^{1/2} \boldsymbol{q}_j = \boldsymbol{q}_i^T \boldsymbol{q}_j = \delta_{ij}.$$

This proves the weighted orthonormality assertion in addition to reality of the spectrum. □

8.4.2 Continuous-Time Markov Chains and the CME

A rate differs from a probability. If the transition rate from j to i is A_{ij}, then over a small time Δt the corresponding transition probability is $A_{ij}\Delta t + o(\Delta t)$. The probability of remaining in j is one minus the total exit rate times Δt, to first order. Thus the transition matrix has expansion $\mathbf{I} + \Delta t \mathbf{A} + o(\Delta t)$, which explains the signs and column sums of a generator.

Ex 8.29 For two states with rates $1 \to 2$ equal to $a > 0$ and $2 \to 1$ equal to $b > 0$, write the generator and derive both equations for p_1' and p_2'. Show that $p_1 + p_2$ remains constant and find the stationary probabilities. Check the units of each matrix entry.

Def 8.17 **(Continuous-Time Infinitesimal Generator)** A continuous-time Markov jump process on a discrete state space evolves according to

$$\frac{d}{dt}\boldsymbol{p}(t) = \mathbf{A}\boldsymbol{p}(t),$$

where $\boldsymbol{p}(t)$ is the probability distribution vector and $\mathbf{A}$ is an infinitesimal generator matrix satisfying:

1. $A_{ij} \geq 0$ for all $i \neq j$ (transition rates between distinct states are nonnegative).
2. $\sum_i A_{ij} = 0$ for every column j (column sums vanish).
3. $A_{jj} = -\sum_{i \neq j} A_{ij} \leq 0$ (diagonal entries equal the negative total exit rate from state j).

Ex 8.30 Choose $\nu \geq \max_j(-A_{jj})$ with $\nu > 0$ and set $\mathbf{P} = \mathbf{I} + \mathbf{A}/\nu$. Prove that $\mathbf{P}$ is column-stochastic and that

$$e^{t\mathbf{A}} = e^{-\nu t} \sum_{k=0}^{\infty} \frac{(\nu t)^k}{k!} \mathbf{P}^k.$$

Deduce positivity, conservation of total mass, and $\|e^{t\mathbf{A}}\|_1 = 1$ for $t \geq 0$. Repeat the argument for a finite subgenerator whose column sums are nonpositive and prove that its exponential is substochastic. This construction is called uniformization.

A reaction model applies the same balance to molecule counts. Each term adds probability arriving from one neighboring count and subtracts probability leaving the current count. We take the propensity of an infeasible reaction to be zero, and terms involving states outside the nonnegative count space to be zero. For an infinite state space, conservation requires a nonexplosive process: only finitely many jumps occur in a finite time almost surely. The infinite-state statements below assume a well-defined conservative process and the summability needed for the displayed equations.

Def 8.18 **(Chemical Master Equation (CME))** Consider a chemically reacting system with S molecular species and R reaction channels. The discrete state $\boldsymbol{x} \in \mathbb{N}_0^S$ records the copy number of each species. Each reaction channel $k \in \{1, \ldots, R\}$ is characterized by a stoichiometric change vector $\boldsymbol{\nu}_k \in \mathbb{Z}^S$ and a propensity function $\alpha_k(\boldsymbol{x}; \boldsymbol{c}) \geq 0$ parameterized by rate constants $\boldsymbol{c}$.
The Chemical Master Equation governs the probability $p(\boldsymbol{x}, t)$ of occupying state $\boldsymbol{x}$ at time t:

$$\frac{\partial p(\boldsymbol{x}, t)}{\partial t} = \sum_{k=1}^{R} \Big[\alpha_k(\boldsymbol{x} - \boldsymbol{\nu}_k; \boldsymbol{c})\, p(\boldsymbol{x} - \boldsymbol{\nu}_k, t) - \alpha_k(\boldsymbol{x}; \boldsymbol{c})\, p(\boldsymbol{x}, t)\Big].$$

Enumerating the reachable states yields the linear system of ordinary differential equations

$$\dot{\boldsymbol{p}}(t) = \mathbf{A}(\boldsymbol{c})\boldsymbol{p}(t), \qquad \boldsymbol{p}(0) = \boldsymbol{p}_0,$$

whose exact formal solution is the matrix exponential action $\boldsymbol{p}(t) = \exp(t\mathbf{A}(\boldsymbol{c}))\boldsymbol{p}_0$.

Ex 8.31 For births at constant rate λ and deaths at rate μx, write the equation for $p_x'(t)$. Treat $x = 0$ separately. Identify which terms represent arrivals and departures, and explain why a death term from state zero must vanish. Write the first three columns of the infinite generator.

8.4.3 Generator Assembly and Stochastic Simulation

Two computations start from the list of reactions. The first builds the generator $\mathbf{A}(\boldsymbol{c})$ on a finite set of states and evolves the whole distribution. The second samples individual trajectories of the jump process and never enumerates the state space. The generator is needed for the finite state projection below. Sampled trajectories provide the data for the Koopman estimators of Section 8.5, and their histograms give an independent check on computed distributions.

Def 8.19 **(Generator Assembly by Forward Enumeration)** Let $\mathcal{S}$ be a finite set of n states, stored in a hash table that maps each state $\boldsymbol{x}$ to its index $\iota(\boldsymbol{x}) \in \{1, \ldots, n\}$. For each $\boldsymbol{x} \in \mathcal{S}$ and each reaction k with $\alpha_k(\boldsymbol{x}; \boldsymbol{c}) > 0$:

1. Compute the destination $\boldsymbol{y} = \boldsymbol{x} + \boldsymbol{\nu}_k$.
2. If $\boldsymbol{y} \in \mathcal{S}$, append the triplet $\big(\iota(\boldsymbol{y}), \iota(\boldsymbol{x}), \alpha_k(\boldsymbol{x}; \boldsymbol{c})\big)$.

Finally, append the diagonal triplet $(\iota(\boldsymbol{x}), \iota(\boldsymbol{x}), -\sum_k \alpha_k(\boldsymbol{x}; \boldsymbol{c}))$ for each $\boldsymbol{x}$, and convert the triplets to CSR format by Thm 1.26.

The diagonal entry contains the total exit rate, including the rates of reactions whose destination lies outside $\mathcal{S}$. Column $\iota(\boldsymbol{x})$ therefore sums to minus the rate at which probability leaves $\mathcal{S}$ from $\boldsymbol{x}$, which is the leakage measured by the finite state projection bound. Assembly visits each pair of a state and a reaction once and performs one expected constant-time hash lookup for each, so it costs $O(nR)$ operations and produces at most $n(R+1)$ nonzero entries. Searching all pairs of states for possible transitions would cost $O(n^2)$ instead.

Def 8.20 **(Stochastic Simulation Algorithm)** Let the process be in state $\boldsymbol{x}$ at time t. Gillespie's direct method advances it as follows (Gillespie, 1977).

1. Compute the propensities $\alpha_k(\boldsymbol{x}; \boldsymbol{c})$ and their sum $a_0(\boldsymbol{x}) = \sum_{k=1}^{R} \alpha_k(\boldsymbol{x}; \boldsymbol{c})$. If $a_0(\boldsymbol{x}) = 0$, stop.
2. Draw independent u_1, u_2 uniformly from $(0, 1)$.
3. Set $\tau = -\ln(u_1)/a_0(\boldsymbol{x})$, and choose the smallest k with $\sum_{l=1}^{k} \alpha_l(\boldsymbol{x}; \boldsymbol{c}) \geq u_2\, a_0(\boldsymbol{x})$.
4. Replace t by $t + \tau$ and $\boldsymbol{x}$ by $\boldsymbol{x} + \boldsymbol{\nu}_k$.

Thm 8.17 **(Exactness of the Stochastic Simulation Algorithm)** Assume $a_0(\boldsymbol{x}) > 0$, with zero-rate channels unable to fire. Suppose that from state $\boldsymbol{x}$ each positive-rate reaction k fires after an independent exponential waiting time with rate $\alpha_k(\boldsymbol{x}; \boldsymbol{c})$, and that the first reaction to fire determines the next state. Then the time to the next reaction is exponential with rate $a_0(\boldsymbol{x})$, the next reaction is k with probability $\alpha_k(\boldsymbol{x}; \boldsymbol{c})/a_0(\boldsymbol{x})$, and these two events are independent. The waiting time and reaction index produced by Def 8.20 have exactly this joint distribution.

Proof Fix the current state and abbreviate its positive reaction rates by α_k, with $a_0 = \sum_k \alpha_k > 0$. A rate-zero channel never fires and may be omitted. Let T_k be the independent exponential waiting time of channel k, let $T = \min_k T_k$, and let K be the first channel to fire. Ties have probability zero because these distributions are continuous.

For channel k to fire first at time s, its waiting-time density contributes $\alpha_k e^{-\alpha_k s}$, and every other waiting time must exceed s. Independence multiplies their survival probabilities. Thus the joint density in time is

$$\Pr(K = k,\ T \in ds)/ds = \alpha_k e^{-\alpha_k s} \prod_{l \neq k} e^{-\alpha_l s} = \alpha_k e^{-a_0 s}.$$

Integrate for $s > \tau$:

$$\Pr(K = k, T > \tau) = \int_{\tau}^{\infty} \alpha_k e^{-a_0 s}\, ds = \frac{\alpha_k}{a_0} e^{-a_0 \tau}.$$

Setting $\tau = 0$ gives $\Pr(K = k) = \alpha_k / a_0$. Summing over k gives $\Pr(T > \tau) = e^{-a_0 \tau}$. The joint expression is the product of these marginals, proving independence.

The algorithm reproduces both marginals. For u_1 uniform on $(0, 1)$,

$$\Pr(-\ln u_1 / a_0 > \tau) = \Pr(u_1 < e^{-a_0 \tau}) = e^{-a_0 \tau}.$$

The interval assigned to channel k by the cumulative sums has length α_k in an interval of length a_0. A uniform $u_2 a_0$ lands there with probability α_k/a_0. Independent uniforms preserve independence of the two selections. This proves exactness of this single jump, which can be repeated at each new state (Gillespie, 1977). □

Recomputing all propensities and selecting a reaction by linear search costs $O(R)$ per event. A tree reduces the selection cost to $O(\log R)$, but the total event cost also includes updating every propensity affected by the reaction. The number of steps up to time T is about $\int_0^T a_0(\mathbf{X}(t))\,dt$, which is large when some reactions are fast. An expected value $\mathbb{E}[f(\mathbf{X}(T))]$ is estimated by averaging f over N independent trajectories. By the central limit theorem, the error of this average has standard deviation $\sigma_f/\sqrt{N}$, where σ_f is the standard deviation of $f(\mathbf{X}(T))$. Each additional digit of accuracy therefore requires a hundred times more trajectories. The finite state projection below computes an approximation to the distribution instead. Its cost depends on the number of retained states and the accuracy required of both truncation and time propagation.

8.4.4 Finite State Projection (FSP)

Remark 8.15 **(Truncating the state space)** The CME generator $\mathbf{A}(\boldsymbol{c})$ is defined over an infinite or very large discrete state space $\mathbb{N}_0^S$. A finite computation therefore requires truncation, aggregation, or another representation that avoids enumerating the full state space.

Truncating the state space removes paths after they leave the retained set. It should not redirect an outgoing transition back into that set. We therefore keep the original negative diagonal entries while dropping the off-diagonal entries for omitted destinations. The resulting loss of total probability measures the discrepancy caused by truncation. Renormalizing the remaining probabilities would conceal that loss.

Def 8.21 **(Finite State Projection (FSP))** Let the discrete state space be partitioned into a finite retained index set $J \subset \mathbb{N}_0^S$ with $|J| = K$, and an infinite or omitted complement set J^c. Permuting states so that J appears first partitions the full generator:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_J & \mathbf{A}_{J,J^c} \\ \mathbf{A}_{J^c,J} & \mathbf{A}_{J^c} \end{pmatrix}.$$

The Finite State Projection of Munsky and Khammash (Munsky and Khammash, 2006) approximates the true marginal probability on J by solving the isolated subgenerator system

$$\dot{\boldsymbol{p}}_J(t) = \mathbf{A}_J \boldsymbol{p}_J(t), \qquad \boldsymbol{p}_J(0) = \boldsymbol{p}_0|_J.$$

Here $\mathbf{A}_J$ has nonpositive diagonal entries and nonnegative off-diagonal entries, but its column sums are nonpositive because transitions into J^c act as probability leaks.

Thm 8.18 **(FSP 1-Norm Leakage Bound)** Let $\boldsymbol{p}(t)$ be the exact probability distribution solving $\dot{\boldsymbol{p}} = \mathbf{A}\boldsymbol{p}$ with initial condition supported on J, so that $\mathbf{1}^T\boldsymbol{p}(t) = 1$. Let $\boldsymbol{p}_J(t) = \exp(t\mathbf{A}_J)\boldsymbol{p}_0|_J$ be the FSP solution. Then:

1. Pointwise underestimation: $\mathbf{0} \le \boldsymbol{p}_J(t) \le \boldsymbol{p}|_J(t)$ for all $t \ge 0$.
2. The full-space error of the zero-padded FSP approximation equals the leaked mass:

$$\|\boldsymbol{p}(t) - \widetilde{\boldsymbol{p}}_J(t)\|_1 = 1 - \mathbf{1}^T\boldsymbol{p}_J(t),$$

where $\widetilde{\boldsymbol{p}}_J(t)$ is $\boldsymbol{p}_J(t)$ padded with zeros on J^c. The retained-state error satisfies

$$\|\boldsymbol{p}|_J(t) - \boldsymbol{p}_J(t)\|_1 = \mathbf{1}^T\boldsymbol{p}|_J(t) - \mathbf{1}^T\boldsymbol{p}_J(t) \le 1 - \mathbf{1}^T\boldsymbol{p}_J(t).$$

Proof The exact retained dynamics satisfy the non-autonomous subgenerator ODE

$$\frac{d}{dt}\boldsymbol{p}|_J(t) = \mathbf{A}_J\boldsymbol{p}|_J(t) + \mathbf{A}_{J,J^c}\boldsymbol{p}|_{J^c}(t), \qquad \boldsymbol{p}|_J(0) = \boldsymbol{p}_0|_J.$$

The FSP approximation satisfies $\frac{d}{dt}\boldsymbol{p}_J(t) = \mathbf{A}_J\boldsymbol{p}_J(t)$ with the same initial condition $\boldsymbol{p}_J(0) = \boldsymbol{p}_0|_J$. Subtracting the two equations gives the defect ODE for the difference $\boldsymbol{\delta}_J(t) = \boldsymbol{p}|_J(t) - \boldsymbol{p}_J(t)$:

$$\frac{d}{dt}\boldsymbol{\delta}_J(t) = \mathbf{A}_J\boldsymbol{\delta}_J(t) + \mathbf{A}_{J,J^c}\boldsymbol{p}|_{J^c}(t), \qquad \boldsymbol{\delta}_J(0) = \mathbf{0}.$$

The two retained trajectories have exactly the same initial data. Their difference therefore starts at zero. The only forcing in its equation is probability returning from omitted states, represented by the nonnegative coupling term.

By Duhamel's formula (variation of constants):

$$\boldsymbol{\delta}_J(t) = \boldsymbol{p}|_J(t) - \boldsymbol{p}_J(t) = \int_0^t e^{(t-s)\mathbf{A}_J}\mathbf{A}_{J,J^c}\boldsymbol{p}|_{J^c}(s)\,ds.$$

For the finite retained matrix, nonnegativity of its exponential follows from the uniformization construction already given: shift its diagonal by a sufficiently large positive multiple of the identity and expand the resulting nonnegative matrix in powers. Every term of that exponential series is nonnegative.

Because $\mathbf{A}_J$ has non-negative off-diagonal entries, it is a Metzler matrix, and its matrix exponential $e^{\tau\mathbf{A}_J}$ is entrywise non-negative for all $\tau \ge 0$. Furthermore, the coupling matrix $\mathbf{A}_{J,J^c} \ge \mathbf{0}$ and exact probabilities $\boldsymbol{p}|_{J^c}(s) \ge \mathbf{0}$ are non-negative. Consequently, the integrand is entrywise non-negative, proving that

$$\boldsymbol{p}|_J(t) - \boldsymbol{p}_J(t) \ge \mathbf{0} \implies \mathbf{0} \le \boldsymbol{p}_J(t) \le \boldsymbol{p}|_J(t) \quad \text{for all } t \ge 0.$$

Because $\boldsymbol{p}_J(t) \le \boldsymbol{p}|_J(t)$, the marginal 1-norm difference on J simplifies without absolute value signs:

$$\|\boldsymbol{p}|_J(t) - \boldsymbol{p}_J(t)\|_1 = \sum_{i\in J}\left(p_i(t) - [\boldsymbol{p}_J(t)]_i\right) = \mathbf{1}^T\boldsymbol{p}|_J(t) - \mathbf{1}^T\boldsymbol{p}_J(t).$$

The total probability across the full space satisfies $\mathbf{1}^T\boldsymbol{p}|_J(t) + \mathbf{1}^T\boldsymbol{p}|_{J^c}(t) = 1$. The omitted probability mass on J^c is exactly $1 - \mathbf{1}^T\boldsymbol{p}|_J(t)$. Adding the unmodeled mass on J^c gives

$$\begin{aligned}\|\boldsymbol{p}(t) - \widetilde{\boldsymbol{p}}_J(t)\|_1 &= \|\boldsymbol{p}|_J(t) - \boldsymbol{p}_J(t)\|_1 + \|\boldsymbol{p}|_{J^c}(t)\|_1 \\ &= \left(\mathbf{1}^T\boldsymbol{p}|_J(t) - \mathbf{1}^T\boldsymbol{p}_J(t)\right) + \left(1 - \mathbf{1}^T\boldsymbol{p}|_J(t)\right) \\ &= 1 - \mathbf{1}^T\boldsymbol{p}_J(t).\end{aligned}$$

The zero-padded FSP approximation is a subprobability vector, so the familiar identity between total variation distance and half the 1-norm for two probability vectors does not apply directly. With the convention $d(\mu,\nu) = \sup_B|\mu(B) - \nu(B)|$, its discrepancy from the exact distribution is

the full leaked mass. A comparison of two probability distributions requires specifying how the missing mass is represented, for example by an absorbing sink.

The certificate concerns the exact solution of the finite subgenerator equation. A numerical approximation to that solution introduces an additional propagation error. If that error is bounded by ε_{num} in the 1-norm, the triangle inequality adds it to the truncation discrepancy. The truncation certificate should therefore not be presented as a complete error bound for an arbitrary numerical propagator (Munsky and Khammash, 2006). □

Ex 8.32 **(FSP 1-norm leakage bound derivation)** Consider a CTMC with generator $\mathbf{A}$ partitioned into retained set J and complement J^c.

1. Prove that $\mathbf{1}^T\mathbf{A}_J \leq \mathbf{0}^T$ elementwise, with strict inequality for at least one entry if transitions from J to J^c exist.
2. Show that $\frac{d}{dt}(\mathbf{1}^T\boldsymbol{p}_J(t)) \leq 0$, confirming that total modeled probability mass decreases monotonically over time.
3. Let $\varepsilon = 1 - \mathbf{1}^T\boldsymbol{p}_J(T)$. Verify from Thm 8.18 that the zero-padded FSP error is exactly ε in the 1-norm. Explain what additional identification of omitted states would be needed before comparing an absorbing sink with the exact distribution.

8.4.5 Truncation Strategy: Probability Flux versus Probability Mass

Remark 8.16 **(Failure of instantaneous mass truncation)** A naive heuristic for expanding or pruning states in the FSP retains states where the instantaneous probability $p_i(t)$ exceeds a threshold ε. In multiscale or stiff reaction networks, this heuristic can discard a low-mass intermediate that carries substantial probability flux and thereby produce an order-one error.

Def 8.22 **(Probability Flux)** For a transition from state j to state i with rate $A_{ij} > 0$, the instantaneous probability flux at time t is

$$\Phi_{ij}(t) = A_{ij}p_j(t).$$

The cumulative probability flux entering state i over the time interval $[0, T]$ is

$$F_i(T) = \int_0^T \sum_{j\neq i} A_{ij}p_j(s)\,ds.$$

Remark 8.17 **(Transient intermediate states)** Consider a reaction cascade $S_1 \xrightarrow{k_1} S_2 \xrightarrow{k_2} S_3$ with $k_2 \gg k_1$. State S_2 has a rapid exit rate, so its instantaneous probability is tiny: $p_2(t) \approx \frac{k_1}{k_2}p_1(t) \ll 1$. Pruning based on instantaneous mass $p_2(t) < \varepsilon$ deletes S_2 from the state space. However, every reaction producing S_3 must pass through S_2. Deleting S_2 severs the probability flow entirely, causing an $\mathcal{O}(1)$ error in $p_3(T)$.
Robust adaptive FSP algorithms monitor cumulative probability flux rather than instantaneous probability mass. A state may have negligible instantaneous mass yet carry substantial probability flux across the network.

8.4.6 Further Exercises

Ex 8.33 **(Simulation of a birth–death process)** Consider the reactions $\emptyset \to X$ with propensity k_1 and $X \to \emptyset$ with propensity $k_2 x$, started from $x = 0$. Implement Def 8.20, and compare the histogram of $X(T)$ over 10^4 trajectories with the Poisson distribution of mean $(k_1/k_2)(1 - e^{-k_2 T})$. Then assemble the generator on the states $0, \dots, n_{\max}$ by Def 8.19, compute $e^{T\mathbf{A}}\boldsymbol{e}_1$, and compare all three distributions.

Ex 8.34 **(Detailed balance)** Consider the column-stochastic matrix

$$\mathbf{P} = \begin{pmatrix} 0.6 & 0.2 & 0 \\ 0.4 & 0.5 & 0.1 \\ 0 & 0.3 & 0.9 \end{pmatrix}.$$

Find its stationary distribution $\boldsymbol{\pi}$, verify detailed balance on each edge, and compute $\mathbf{S} = \mathbf{D}_{\boldsymbol{\pi}}^{-1/2}\mathbf{P}\mathbf{D}_{\boldsymbol{\pi}}^{1/2}$. Verify directly that $\mathbf{S}$ is symmetric and that the eigenvalues of $\mathbf{P}$ are real.

Ex 8.35 **(Probability mass versus cumulative flux in stiff cascades)** Consider the three-state continuous-time Markov chain

$$X_1 \xrightarrow{k_1} X_2 \xrightarrow{k_2} X_3$$

with $k_1 = 1$, $k_2 = 10^4$, and initial condition $\boldsymbol{p}(0) = (1, 0, 0)^T$.

1. Solve the system analytically for $p_1(t), p_2(t), p_3(t)$.
2. Show that $\max_{t \geq 0} p_2(t) \approx 10^{-4}$. If a threshold $\varepsilon = 10^{-3}$ is used to prune states by instantaneous mass, state X_2 is discarded at all times.
3. Compute the cumulative flux $\int_0^T k_1 p_1(s)\, ds$ entering X_2 for $T = 3$. Show that over 95% of the total system probability passes through X_2 despite its negligible instantaneous mass.
4. Explain why adaptive truncation based on cumulative probability flux preserves state X_2 while mass-based thresholding disconnects the state space.

Ex 8.36 **(Birth–death process and infinite generator truncation)** For a simple birth–death process with birth rate λ and death rate μx, write down the tridiagonal subgenerator $\mathbf{A}_J$ for copy numbers $x \in \{0, 1, \dots, N\}$. Formulate the exit rate from boundary state N and derive the leakage rate $\frac{d}{dt}\mathbf{1}^T \boldsymbol{p}_J(t)$ as a function of $p_N(t)$.

Read **Further reading.** The finite-state projection method and its error certificate originate with Munsky and Khammash (Munsky and Khammash, 2006). Higham's treatment of matrix functions is useful background for the exponential calculations used here (Higham, 2008).

8.5 Koopman Theory, EDMD, and Continuous Generators

An observation of a dynamical system may be a function of its state, such as position, energy, or a molecule count. Instead of asking how the state changes, we can ask how each such function changes after one step. Composition with the state update defines the Koopman operator.

This operator is linear in the observed function even when the state update is nonlinear. Its function space may be infinite dimensional. We first examine that distinction, then fit matrices on selected functions using least squares. For reaction models, the same construction relates the evolution of expected observations to the master equation.

8.5.1 Koopman Operator Theory

Def 8.23 **(Nonlinear Dynamics and Observables)** Consider a discrete-time dynamical system

$$\boldsymbol{x}^{(k+1)} = \mathbf{F}(\boldsymbol{x}^{(k)}), \qquad \boldsymbol{x} \in \mathcal{M} \subseteq \mathbb{R}^n.$$

Choose a function space $\mathcal{F}$ invariant under composition with $\mathbf{F}$. An observable is a scalar function $g : \mathcal{M} \to \mathbb{R}$ belonging to $\mathcal{F}$. For an L^2 space, the measure and assumptions ensuring that composition is well defined must be specified. The Koopman operator $\mathcal{K} : \mathcal{F} \to \mathcal{F}$ is defined by

$$(\mathcal{K}g)(\boldsymbol{x}) = g(\mathbf{F}(\boldsymbol{x})).$$

The operator $\mathcal{K}$ is linear on $\mathcal{F}$, even when the underlying map $\mathbf{F}$ is nonlinear.

Prop 8.19 **(Linearity of the Koopman Operator)** The Koopman operator $\mathcal{K}$ satisfies

$$\mathcal{K}(\alpha g_1 + \beta g_2) = \alpha \mathcal{K} g_1 + \beta \mathcal{K} g_2$$

for all scalars $\alpha, \beta \in \mathbb{R}$ and observables $g_1, g_2 \in \mathcal{F}$.

Ex 8.37 Evaluate both sides at an arbitrary state and expand $(\alpha g_1 + \beta g_2)(\mathbf{F}(\boldsymbol{x}))$. Which operation is linear here, and which map may remain nonlinear?

Ex 8.38 For the state update $F(x) = x^3$, compute the images of $g_1(x) = x$ and $g_2(x) = x^3$. Show that $\operatorname{span}\{x, x^3\}$ is not invariant under composition with F. Which further monomials appear if you repeatedly add the missing images? Explain why linearity of the operator does not imply that these two observables give an exact matrix model.

Def 8.24 **(Koopman Eigenfunctions and Eigenvalues)** A nonzero function $\varphi \in \mathcal{F}$ is a Koopman eigenfunction with eigenvalue $\mu \in \mathbb{C}$ if

$$\mathcal{K}\varphi = \mu\varphi.$$

Along system trajectories, Koopman eigenfunctions evolve linearly:

$$\varphi(\boldsymbol{x}^{(k)}) = \mu^k \varphi(\boldsymbol{x}^{(0)}).$$

A sufficiently rich collection of Koopman eigenfunctions can provide coordinates in which part of the nonlinear dynamics evolves linearly. Such a collection need not exist globally, and finite dictionaries generally give only a projected approximation.

8.5.2 Dynamic Mode Decomposition (DMD) and EDMD

Suppose a scalar state is observed in pairs (x_m, y_m). Fitting one multiplier asks for $\min_a \sum_m (y_m - a x_m)^2$. Provided not all x_m vanish, the answer is $a = \sum_m x_m y_m / \sum_m x_m^2$. DMD applies this least-squares calculation to vectors, placing each observation in a column. This orientation differs from PCA, where observations were rows.

Def 8.25 **(Dynamic Mode Decomposition (DMD))** Given snapshot matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n \times M}$ where $\mathbf{Y}$ advances $\mathbf{X}$ by time step h:

$$\mathbf{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_M], \qquad \mathbf{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_M], \qquad \boldsymbol{y}_m = \mathbf{F}(\boldsymbol{x}_m),$$

DMD seeks the best-fit linear map $\mathbf{A} \approx \mathbf{Y}\mathbf{X}^\dagger$ in the Frobenius norm.
To compute the reduced model:

1. Compute the rank-r truncated SVD $\mathbf{X} \approx \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^T$.
2. Project onto the span of the retained left singular vectors (often called the proper orthogonal decomposition, or POD, subspace): $\widetilde{\mathbf{A}} = \mathbf{U}_r^T \mathbf{Y} \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \in \mathbb{R}^{r \times r}$.
3. Compute eigendecomposition $\widetilde{\mathbf{A}} \boldsymbol{w}_i = \mu_i \boldsymbol{w}_i$.
4. The DMD modes are $\boldsymbol{\phi}_i = \mathbf{Y} \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \boldsymbol{w}_i$.

Prop 8.20 **(DMD as a Least-Squares Fit)** The minimum-Frobenius-norm solution of

$$\min_{\mathbf{A}} \|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_F$$

is $\mathbf{A}_* = \mathbf{Y}\mathbf{X}^\dagger$. If $\mathbf{X} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^T$ is a rank-r SVD, the representation of this map on $\mathcal{R}(\mathbf{U}_r)$ is

$$\widetilde{\mathbf{A}} = \mathbf{U}_r^T \mathbf{A}_* \mathbf{U}_r = \mathbf{U}_r^T \mathbf{Y} \mathbf{V}_r \mathbf{\Sigma}_r^{-1}.$$

Proof Let row i of $\mathbf{A}$ be $\boldsymbol{a}_i^T$ and row i of $\mathbf{Y}$ be $\boldsymbol{y}_i^T$. The squared Frobenius residual is

$$\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_F^2 = \sum_i \|\boldsymbol{y}_i^T - \boldsymbol{a}_i^T \mathbf{X}\|_2^2 = \sum_i \|\boldsymbol{y}_i - \mathbf{X}^T \boldsymbol{a}_i\|_2^2.$$

Different rows of $\mathbf{A}$ appear in different summands. Hence we may minimize each summand independently. The pseudoinverse least-squares theorem gives its unique minimum-norm solution

$$\boldsymbol{a}_i = (\mathbf{X}^T)^\dagger \boldsymbol{y}_i = (\mathbf{X}^\dagger)^T \boldsymbol{y}_i.$$

Transposing back and collecting the rows gives $\mathbf{A}_* = \mathbf{Y}\mathbf{X}^\dagger$. Among all residual minimizers, the squared Frobenius norm is $\sum_i \|\boldsymbol{a}_i\|_2^2$. Choosing the minimum norm independently in every row therefore gives the minimum-Frobenius-norm matrix.

For a rank-r SVD, $\mathbf{X}^\dagger = \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{U}_r^T$. Its diagonal inverse is defined because the retained singular values are positive. Substitute into the reduced representation:

$$\begin{aligned} \mathbf{U}_r^T \mathbf{A}_* \mathbf{U}_r &= \mathbf{U}_r^T \mathbf{Y} \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{U}_r^T \mathbf{U}_r \\ &= \mathbf{U}_r^T \mathbf{Y} \mathbf{V}_r \mathbf{\Sigma}_r^{-1}. \end{aligned}$$

The last equality uses the orthonormal columns of $\mathbf{U}_r$. This proves the reduced formula as well as the least-squares assertion. □

Def 8.26 **(Extended Dynamic Mode Decomposition (EDMD))** Extended DMD (Williams et al., 2015) generalizes DMD by lifting state data through a dictionary of nonlinear observables $\boldsymbol{\psi}(\boldsymbol{x}) = (\psi_1(\boldsymbol{x}), \dots, \psi_K(\boldsymbol{x}))^T \in \mathbb{R}^K$.
Form data matrices

$$\mathbf{\Psi}_X = [\boldsymbol{\psi}(\boldsymbol{x}_1), \dots, \boldsymbol{\psi}(\boldsymbol{x}_M)] \in \mathbb{R}^{K \times M}, \qquad \mathbf{\Psi}_Y = [\boldsymbol{\psi}(\boldsymbol{y}_1), \dots, \boldsymbol{\psi}(\boldsymbol{y}_M)] \in \mathbb{R}^{K \times M}.$$

The finite-dimensional representation $\widehat{\mathbf{K}}_h \in \mathbb{R}^{K \times K}$ of the Koopman operator minimizes $\|\mathbf{\Psi}_Y - \mathbf{K}\mathbf{\Psi}_X\|_F^2$:

$$\widehat{\mathbf{K}}_h = \mathbf{\Psi}_Y \mathbf{\Psi}_X^\dagger = (\mathbf{\Psi}_Y \mathbf{\Psi}_X^T)(\mathbf{\Psi}_X \mathbf{\Psi}_X^T)^\dagger.$$

Left eigenvectors of $\widehat{\mathbf{K}}_h$ yield approximations to Koopman eigenfunctions.

Ex 8.39 Let a row vector $\boldsymbol{w}^T$ satisfy $\boldsymbol{w}^T\widehat{\mathbf{K}}_h = \mu\boldsymbol{w}^T$ and define $g(\boldsymbol{x}) = \boldsymbol{w}^T\boldsymbol{\psi}(\boldsymbol{x})$. If the fitted relation $\boldsymbol{\psi}(F(\boldsymbol{x})) = \widehat{\mathbf{K}}_h\boldsymbol{\psi}(\boldsymbol{x})$ were exact, prove $g(F(\boldsymbol{x})) = \mu g(\boldsymbol{x})$. With a nonzero fit residual, express the error in this identity. Why does a small training residual not guarantee the identity at unobserved states?

Remark 8.18 **(EDMD as a Galerkin method)** When the dictionary consists of the coordinate functions, $\boldsymbol{\psi}(\boldsymbol{x}) = \boldsymbol{x}$, the data matrices reduce to $\mathbf{X}$ and $\mathbf{Y}$, and EDMD reduces to the least-squares map of Prop 8.20. For a richer dictionary, the matrices $\boldsymbol{\Psi}_Y\boldsymbol{\Psi}_X^T/M$ and $\boldsymbol{\Psi}_X\boldsymbol{\Psi}_X^T/M$ are sample averages of inner products between dictionary functions. As the number of samples drawn from a measure μ grows, they converge to the corresponding inner-product matrices in $L^2(\mu)$, and $\widehat{\mathbf{K}}_h$ converges to the Galerkin projection of the Koopman operator onto the span of the dictionary (Williams et al., 2015). Enlarging the dictionary therefore raises a separate approximation question, namely how well that span represents the Koopman eigenfunctions of interest.

8.5.3 CME Koopman Generator and Polynomial Closure

Remark 8.19 **(Koopman generator for continuous Markov jump processes)** For a continuous-time Markov process with master equation $\dot{\boldsymbol{p}}(t) = \mathbf{A}(\boldsymbol{c})\boldsymbol{p}(t)$, the expectation of an observable $g(\boldsymbol{x})$ evolves according to $\frac{d}{dt}\mathbb{E}[g(\boldsymbol{x}(t))] = \frac{d}{dt}\sum_{\boldsymbol{x}} g(\boldsymbol{x})p(\boldsymbol{x},t) = \sum_{\boldsymbol{x}} g(\boldsymbol{x})[\mathbf{A}(\boldsymbol{c})\boldsymbol{p}(t)]_{\boldsymbol{x}} = \sum_{\boldsymbol{x}}[\mathbf{A}(\boldsymbol{c})^T g](\boldsymbol{x})p(\boldsymbol{x},t)$. The Koopman infinitesimal generator $\mathcal{A}$ is the formal adjoint of the master operator: $\mathcal{A} = \mathbf{A}(\boldsymbol{c})^T$.

Def 8.27 **(CME Koopman Generator)** For a chemical reaction network with stoichiometry $\boldsymbol{\nu}_k$ and propensities $\alpha_k(\boldsymbol{x};\boldsymbol{c}) = c_k h_k(\boldsymbol{x})$, the Koopman generator acts on observable g by

$$(\mathcal{A}g)(\boldsymbol{x}) = \sum_{k=1}^{R} \alpha_k(\boldsymbol{x};\boldsymbol{c})\Big(g(\boldsymbol{x}+\boldsymbol{\nu}_k) - g(\boldsymbol{x})\Big).$$

The generator contains a difference $g(\boldsymbol{x}+\boldsymbol{\nu}) - g(\boldsymbol{x})$, so it can reduce polynomial degree before multiplication by the propensity raises it. For instance, if $g(x) = x^2$, then $(x+1)^2 - x^2 = 2x+1$. Multiplying by a constant or linear propensity leaves degree at most two. A quadratic propensity can produce degree three. This degree count determines whether finitely many moments form a closed system.

Thm 8.21 **(Polynomial Closure in Chemical Kinetics)** Under mass-action kinetics, propensity functions are falling factorial polynomials:

$$h_k(\boldsymbol{x}) = \prod_{s=1}^{S} x_s^{(\nu_{ks}^-)}, \qquad x_s^{(m)} = x_s(x_s-1)\cdots(x_s-m+1).$$

Let $\mathcal{P}_d$ denote the space of polynomials in $\boldsymbol{x}$ of degree at most d.

1. **Unimolecular networks:** If all reactions have reactant order at most 1 (including immigration, conversion, and first-order degradation), then $\mathcal{A}\mathcal{P}_d \subseteq \mathcal{P}_d$ for all $d \geq 0$. The subspace $\mathcal{P}_d$ is invariant under $\mathcal{A}$, yielding exact finite-dimensional closure.

2. **Bimolecular networks:** For reactions of reactant order at most 2, $\mathcal{A}\mathcal{P}_d \subseteq \mathcal{P}_{d+1}$. A quadratic propensity can produce degree-$(d+1)$ terms, so the moment equations generally

fail to close at degree d unless additional cancellations, conservation laws, or invariant subspaces are present.

Proof If $d = 0$, every observable in the space is constant, so every difference $g(\boldsymbol{x}+\boldsymbol{\nu}_k) - g(\boldsymbol{x})$ is zero. Its image is the zero polynomial and both closure assertions hold. We may therefore assume $d \geq 1$ in the degree calculation below.

Let $g \in \mathcal{P}_d$ be a polynomial of degree d. For each stoichiometric change vector $\boldsymbol{\nu}_k$, the discrete difference operator expands via Taylor's formula on polynomials:

$$g(\boldsymbol{x}+\boldsymbol{\nu}_k) - g(\boldsymbol{x}) = \sum_{j=1}^{d} \frac{1}{j!}(\boldsymbol{\nu}_k \cdot \nabla)^j g(\boldsymbol{x}).$$

Because differentiation reduces polynomial degree by at least one, each term in the sum has degree at most $d-1$, so $g(\boldsymbol{x}+\boldsymbol{\nu}_k) - g(\boldsymbol{x}) \in \mathcal{P}_{d-1}$. Applying the Koopman generator (Def 8.27):

$$(\mathcal{A}g)(\boldsymbol{x}) = \sum_{k=1}^{R} \alpha_k(\boldsymbol{x};\boldsymbol{c})\big(g(\boldsymbol{x}+\boldsymbol{\nu}_k) - g(\boldsymbol{x})\big).$$

For unimolecular reactions, each propensity $\alpha_k(\boldsymbol{x};\boldsymbol{c})$ has reactant order at most 1, so $\alpha_k \in \mathcal{P}_1$. The product of a degree-1 polynomial and a degree-$(d-1)$ polynomial has degree at most $1 + (d-1) = d$, proving $\mathcal{A}\mathcal{P}_d \subseteq \mathcal{P}_d$. For bimolecular reactions, reactant order is at most 2, so $\alpha_k \in \mathcal{P}_2$. The product has degree at most $2 + (d-1) = d+1$, giving $\mathcal{A}\mathcal{P}_d \subseteq \mathcal{P}_{d+1}$.

The notation $\mathcal{A}\mathcal{P}_d \subseteq \mathcal{P}_d$ means that applying the generator to any member of this finite-dimensional polynomial space produces another member of the same space. Choosing a basis then gives a matrix for its action, so the corresponding expectations form a closed linear system. For the quadratic-propensity case, the inclusion in $\mathcal{P}_{d+1}$ only bounds the possible degree. It does not assert that a degree-$d+1$ term occurs for every polynomial or every network. □

Ex 8.40 **(Bimolecular moment hierarchy)** Consider the dimerization reaction $2A \xrightarrow{c} B$. The stoichiometry is $\boldsymbol{\nu} = (-2, 1)^T$ and the propensity is $\alpha(x) = cx(x-1)/2$.

1. Compute $(\mathcal{A}g)(x)$ for $g(x) = x$. Show that the result is quadratic in x.
2. Explain why computing the time derivative of the mean $\frac{d}{dt}\mathbb{E}[x]$ requires knowledge of the second moment $\mathbb{E}[x^2]$.

8.5.4 Continuous Generator Estimation

Remark 8.20 **(Discrete-to-continuous inversion)** Given an EDMD estimate of the discrete Koopman operator $\widehat{\mathbf{K}}_h \approx \exp(h\mathcal{A})$ over a sampling interval h, the continuous generator is recovered via the principal matrix logarithm:

$$\widehat{\mathbf{A}} = \frac{1}{h}\operatorname{logm}(\widehat{\mathbf{K}}_h).$$

If $\widehat{\mathbf{K}}_h$ is real and has no eigenvalues on the closed negative real axis $(-\infty, 0]$, then its principal matrix logarithm exists and is real. A matrix with eigenvalues on this branch cut may possess other logarithms, but it does not possess the principal logarithm used here.

Thm 8.22 **(Sensitivity of Recovered Decay Rates)** Let $\lambda_i < 0$ be a continuous eigenvalue of the generator $\mathcal{A}$, corresponding to a discrete Koopman eigenvalue $\mu_i = \exp(h\lambda_i) \in (0, 1)$. The relation between discrete and continuous eigenvalues is

$$\lambda_i = \frac{1}{h} \ln \mu_i.$$

The sensitivity of the continuous eigenvalue with respect to perturbations in the discrete eigenvalue is

$$\left| \frac{d\lambda_i}{d\mu_i} \right| = \frac{1}{h\mu_i} = \frac{1}{h \exp(h\lambda_i)}.$$

For fast decaying modes where $-\lambda_i \gg 1$, $\mu_i \to 0$ exponentially fast. Consequently, the noise amplification factor diverges:

$$\lim_{-\lambda_i \to \infty} \left| \frac{d\lambda_i}{d\mu_i} \right| = \infty.$$

A tiny statistical fluctuation $\Delta\mu_i$ in data-driven EDMD induces an exponentially magnified error $\Delta\lambda_i$ in the recovered rate.

Proof The scalar decay relation is $\mu = e^{h\lambda}$ with $h > 0$. For a positive multiplier it can be inverted as $\lambda = h^{-1} \ln \mu$. Differentiation gives

$$\frac{d\lambda}{d\mu} = \frac{1}{h\mu}.$$

Substitute the exponential expression for μ:

$$\left| \frac{d\lambda}{d\mu} \right| = \frac{e^{-h\lambda}}{h}.$$

When the negative decay rate becomes increasingly large in magnitude at fixed h, $-h\lambda \to +\infty$, so this derivative grows without bound.

The exact perturbation calculation also shows the range of the linear estimate. If $\mu + \Delta\mu > 0$, then

$$\Delta\lambda = \frac{1}{h}[\ln(\mu + \Delta\mu) - \ln \mu] = \frac{1}{h} \ln \left(1 + \frac{\Delta\mu}{\mu} \right).$$

For $|\Delta\mu|/\mu$ small, Taylor's formula $\ln(1 + s) = s + O(s^2)$ gives

$$\Delta\lambda = \frac{\Delta\mu}{h\mu} + O\left(\frac{(\Delta\mu)^2}{h\mu^2} \right).$$

Thus the derivative describes amplification of small absolute multiplier errors. It is not a valid linear approximation when the error is comparable to the multiplier. If the perturbed multiplier is nonpositive, the real scalar logarithm is unavailable, which is a separate obstruction. □

Ex 8.41 Differentiate $h^{-1} \ln \mu$ and substitute $\mu = e^{h\lambda}$ to obtain the stated amplification factor. Then compare the exact change $h^{-1} \ln(1 + \Delta\mu/\mu)$ with its linear approximation. What happens as $\Delta\mu \downarrow -\mu$? Distinguish this loss of a positive multiplier from a large but locally predictable perturbation.

8.5.5 Further Exercises

Ex 8.42 **(DMD residual)** Compute $\mathbf{A}_* = \mathbf{Y}\mathbf{X}^\dagger$ for a rank-one snapshot matrix and verify directly that the residual $\mathbf{Y} - \mathbf{A}_*\mathbf{X}$ is orthogonal to the row space of $\mathbf{X}$, as predicted by Prop 8.20.

Ex 8.43 **(Polynomial closure in unimolecular reaction networks)** Consider the reversible isomerization network $A \underset{c_2}{\overset{c_1}{\rightleftharpoons}} B$. Let x_1, x_2 denote the copy numbers of species A and B.

1. Write down the propensities $\alpha_1(\boldsymbol{x}) = c_1 x_1$ and $\alpha_2(\boldsymbol{x}) = c_2 x_2$, and the stoichiometry vectors $\boldsymbol{\nu}_1 = (-1, 1)^T$ and $\boldsymbol{\nu}_2 = (1, -1)^T$.
2. Apply the Koopman generator $\mathcal{A}$ to the linear observable $g(\boldsymbol{x}) = x_1$. Show that $(\mathcal{A}g)(\boldsymbol{x}) = -c_1 x_1 + c_2 x_2$, which is strictly linear.
3. Apply $\mathcal{A}$ to the second-order observable $g(\boldsymbol{x}) = x_1^2$. Show that $(\mathcal{A}g)(\boldsymbol{x})$ contains only polynomials of degree at most 2.
4. Explain why the subspace of polynomials of degree at most d is invariant under $\mathcal{A}$ for any unimolecular network.

Ex 8.44 **(Sensitivity of recovered decay rates)** Suppose a continuous dynamical system has decay rates $\lambda_1 = -1$ and $\lambda_2 = -50$. Data snapshots are collected at sampling interval $h = 0.1$.

1. Compute the true discrete eigenvalues $\mu_1 = \exp(h\lambda_1)$ and $\mu_2 = \exp(h\lambda_2)$.
2. Suppose statistical noise perturbs the discrete estimates by $\Delta\mu = 10^{-3}$. Compute the resulting perturbation $\Delta\lambda_1 \approx \frac{\Delta\mu}{h\mu_1}$.
3. Compute the resulting perturbation $\Delta\lambda_2 \approx \frac{\Delta\mu}{h\mu_2}$.
4. Explain why recovering continuous generators from discrete snapshots may benefit from regularized regression. Explain why regularization cannot recover a fast mode that the sampling interval has suppressed below the noise level.

Read **Further reading.** Koopman's original paper gives the operator viewpoint for Hamiltonian systems (Koopman, 1931). Schmid introduces dynamic mode decomposition in its fluid-mechanics form, and Williams, Kevrekidis, and Rowley connect a dictionary-based extension to the Koopman operator (Schmid, 2010; Williams et al., 2015).

9 Appendix: Notation and Software

These sections collect notation, proof conventions, and array operations used in the text. They can be consulted when a symbol or a NumPy expression is unfamiliar.

9.1 Mathematical Notation and Proof Techniques

This section establishes the logical, set-theoretic, and matrix notation used throughout these notes, together with the standard proof structures.

9.1.1 Logic and Set Notation

Def 9.1 **(Logical Quantifiers)**

- **Universal ($\forall$):** "For all."
- **Existential ($\exists$):** "There exists."
- **Example:** "$\forall \varepsilon > 0,\ \exists \delta > 0$" reads as "for every positive ε there is a positive δ."

Ex 9.1 Compare "for every real x there exists a real y with $y > x$" with "there exists a real y such that $y > x$ for every real x." Which statement is true? Explain why reversing the quantifiers changes the assertion. In an error estimate, what changes if its constant is allowed to depend on the problem size?

Def 9.2 **(Implication and Equivalence)**

- **Implication ($P \Rightarrow Q$):** P implies Q.
- **Equivalence ($P \Leftrightarrow Q$):** P if and only if Q (iff).
- **Contrapositive:** $\neg Q \Rightarrow \neg P$. Logically equivalent to $P \Rightarrow Q$.

Ex 9.2 For a square matrix, compare the statements "if $\mathbf{A}$ is invertible, then $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has a solution" and "if $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has a solution, then $\mathbf{A}$ is invertible." Give a counterexample to the second statement for one right-hand side. How must the solvability statement be strengthened to become equivalent to invertibility?

Def 9.3 **(Set Notation)**

- $x \in S$: x is an element of S.
- $S \subseteq T$: S is a subset of T.
- $S \cup T$ (union), $S \cap T$ (intersection), and $S \setminus T$ (set difference).
- $\{x \in S : P(x)\}$: Set builder notation.
- **Number Sets:** $\mathbb{Z}$ (integers), $\mathbb{R}$ (reals), $\mathbb{C}$ (complex), $\mathbb{R}^n$ (vectors), $\mathbb{R}^{m \times n}$ (matrices).

Ex 9.3

1. Write "every invertible matrix has a nonzero determinant" using quantifiers and implication.
2. State the contrapositive of: "if $\mathbf{A}$ is invertible, then $\mathbf{A}\boldsymbol{x} = \mathbf{0} \Rightarrow \boldsymbol{x} = \mathbf{0}$."
3. Let $S = \{1, 2, 3, 4\}$ and $T = \{3, 4, 5, 6\}$. Compute $S \cup T, S \cap T, S \setminus T$.

9.1.2 Proof Techniques

Def 9.4 **(Direct Proof)** Assume P is true and derive Q through a chain of valid logical steps.

Def 9.5 **(Proof by Contradiction)** Assume $\neg P$ and derive a logical contradiction ($R \wedge \neg R$). Proves P must hold.

Def 9.6 **(Proof by Induction)** To prove $P(n)$ holds for all integers $n \geq n_0$:

1. **Base case:** Verify $P(n_0)$ directly.
2. **Inductive step:** Assume $P(k)$ and prove $P(k+1)$.

Remark 9.1 **(Proof and numerical evidence)** A proof establishes a claim for every input satisfying its hypotheses. A finite experiment can test an implementation or expose a counterexample, but agreement on a sample does not prove a universal claim. In an error bound, state the norm, the limiting parameter, and which quantities the constant may depend on.

Ex 9.4

1. Prove: if $\mathbf{A}$ and $\mathbf{B}$ are symmetric, then $\mathbf{A} + \mathbf{B}$ is symmetric.
2. Prove by contradiction: if $\mathbf{A}\boldsymbol{x} = \mathbf{0} \Rightarrow \boldsymbol{x} = \mathbf{0}$, then $\mathbf{A}$ has no zero eigenvalue.
3. By induction, prove $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$.

9.1.3 Notation Conventions

Remark 9.2 **(Scalars, Vectors, and Matrices)**

- **Scalars:** Lowercase italic (a, b, x).
- **Vectors:** Lowercase bold $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{a})$. Default is column vector.
- **Matrices:** Uppercase bold $(\mathbf{A}, \mathbf{B}, \mathbf{Q})$.
- **Indexing:** Usually begins at one, unlike Python. The Fourier sections explicitly begin at zero. Here x_i denotes a vector entry and a_{ij} a matrix entry.

Remark 9.3 **(Transpose and Inverse)**

- $\mathbf{A}^T$: Transpose. The notation $\mathbf{A}^*$ denotes the conjugate transpose, which agrees with $\mathbf{A}^T$ for real entries.
- $\mathbf{A}^{-1}$: Matrix inverse (square, nonsingular matrices).
- $\mathbf{A}^{-T}$: Transpose of the inverse $(\mathbf{A}^{-1})^T$.
- $\mathbf{A}^k$: k-th power. $\mathbf{A}^0 = \mathbf{I}$.

Remark 9.4 **(Sums and Products)**

$$\sum_{i=1}^{n} a_i = a_1 + \cdots + a_n, \qquad \prod_{i=1}^{n} a_i = a_1 \dots a_n.$$

Remark 9.5 **(Functions)**

- $f : \mathbb{R}^n \to \mathbb{R}^m$: Maps n-vectors to m-vectors.
- $f \circ g$: Function composition $(f \circ g)(\boldsymbol{x}) = f(g(\boldsymbol{x}))$.

Remark 9.6 **(Asymptotic Notation)** Scaling of algorithms is described using Big-O notation ($O(n^2), O(n^3)$). Details in the Computational Complexity section.

9.2 Python and NumPy Primer

This section summarizes the NumPy array conventions, linear-algebra operations, and indexing idioms used throughout the computational examples in these notes.

9.2.1 Arrays and Basic Operations

Def 9.7 **(NumPy Arrays)** The `ndarray` is the fundamental structure.

- **Creation:** `np.array([1, 2])`, `np.zeros((m, n))`, `np.random.randn(m, n)`.
- **Indexing:** 0-indexed. `A[i, j]` for entries, `A[i, :]` for rows, and `A[:, j]` for columns.

Remark 9.7 **(Shapes and one-dimensional arrays)** `A.shape` returns `(m, n)`. A 1D array has shape `(n,)`, which is **not** equivalent to a column vector `(n, 1)`. These one-dimensional arrays can behave differently from column arrays under broadcasting. Use `x[:, None]` or `x.reshape(-1, 1)` to convert to 2D.

Ex 9.5

1. Create 4×4 identity matrix. Verify `shape` and `dtype`.
2. Extract row 2 and column 3 from a random 3×3 $\mathbf{A}$.
3. Compare `A @ x` for `x.shape == (3,)` and `x.shape == (3,1)`. Record the output shape in each case, then explain why the two representations behave differently under broadcasting.

9.2.2 Linear Algebra Operations

The core routines in `numpy.linalg` provide access to standard BLAS and LAPACK implementations:

Function	Computes
`A @ B`	Matrix product $\mathbf{AB}$
`np.linalg.solve(A, b)`	Solve $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ ($O(n^3)$)
`np.linalg.inv(A)`	Matrix inverse (prefer `solve` for linear systems)
`np.linalg.norm(x)`	Euclidean norm $\|\boldsymbol{x}\|_2$
`np.linalg.eig(A)`	Eigenvalues and eigenvectors
`np.linalg.svd(A)`	Singular Value Decomposition
`np.linalg.qr(A)`	QR Factorization
`np.linalg.cond(A)`	Condition number $\kappa_2(\mathbf{A})$

Remark 9.8 **(Vectorization and performance)** Explicit `for` loops in Python are slow. Push heavy lifting into NumPy/BLAS routines by **vectorizing** operations. For example, use `A @ B` instead of triple-nested loops.

Remark 9.9 **(Linear solve versus inversion)** `np.linalg.solve(A, b)` is faster and more stable than `np.linalg.inv(A) @ b`. Explicit inversion is almost never desirable in scientific computation.

Remark 9.10 **(Computed equality)** For computed floating-point quantities, compare the error with a tolerance appropriate to their scale. Functions such as `np.allclose` and `np.isclose` perform tolerance tests. Their default tolerances do not establish mathematical equality or certify accuracy for every problem. Exact equality is appropriate for integers, booleans, and deliberately exact structural checks.

Remark 9.11 **(Sparse matrices)** Large discretized PDE and graph problems should usually use `scipy.sparse`, not dense NumPy arrays. Use `scipy.sparse.linalg` for sparse solves, eigenvalue computations, CG, and GMRES.

Ex 9.6

1. Solve a 2×2 system and verify the residual $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|$.
2. Profile `solve` vs `inv @ b` for $n = 1000$.
3. Compare `np.linalg.eig` and `np.linalg.eigh` (optimized for symmetric matrices).

Ex 9.7 For a 3×3 matrix, form an approximation `x` with shape `(3,1)` and a right-hand side `b` with shape `(3,)`. Predict the shape of `b - A @ x` before evaluating it. Explain why that array is not the residual vector you intended. Make the shapes consistent and recompute the residual.

9.2.3 Key Idioms

- **Element-wise:** `*` and `**` are entry-wise. Matrix powers use `np.linalg.matrix_power`.
- **Broadcasting:** Automatic dimension expansion. For `A.shape == (m,n)` and `v.shape == (n,)`, `A * v` scales columns by the entries of `v`. To scale rows by a vector of length `m`, use `A * v[:,None]`.
- **Aliases, Views, Copies:** `B = A` makes a second name for the same array. Slices are usually views. Use `B = A.copy()` for independent data.
- **Sparse construction:** The `scipy.sparse` module provides `diags` for banded matrices and `csr_matrix` for compressed sparse row storage.

Ex 9.8

1. Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Compare `A * A` and `A @ A`.
2. Demonstrate the "view pitfall" by modifying a slice of a matrix.
3. Use broadcasting to perform diagonal scaling $\mathbf{AD}$ without constructing $\mathbf{D} = \mathrm{diag}(d)$.

Read **Further reading.** Heath gives a broad introduction to scientific computing at roughly the level of these notes (Heath, 2018). For the effects of floating-point arithmetic, Higham is the standard detailed reference (Higham, 2002). Meyer provides an accessible bridge from undergraduate linear algebra to matrix computation (Meyer, 2000).

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